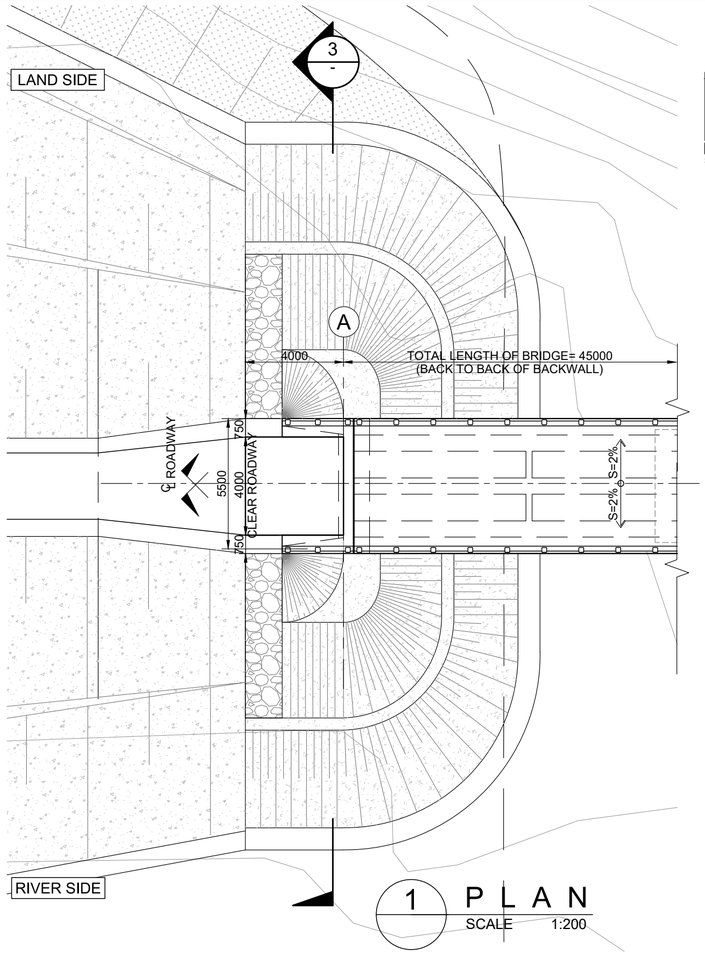
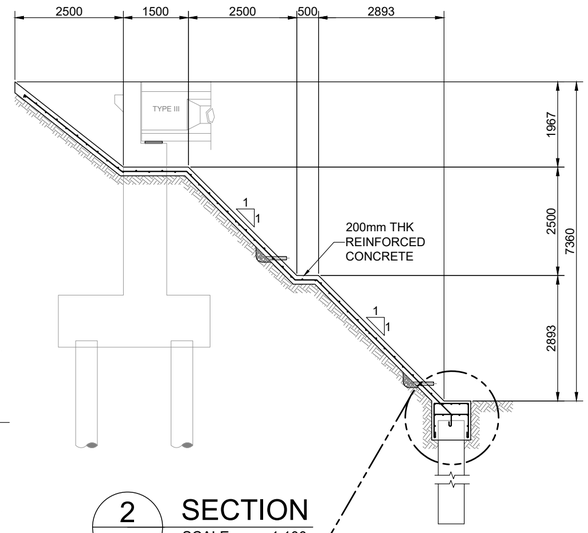
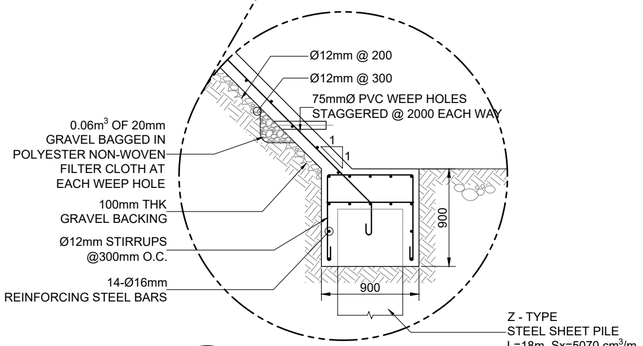


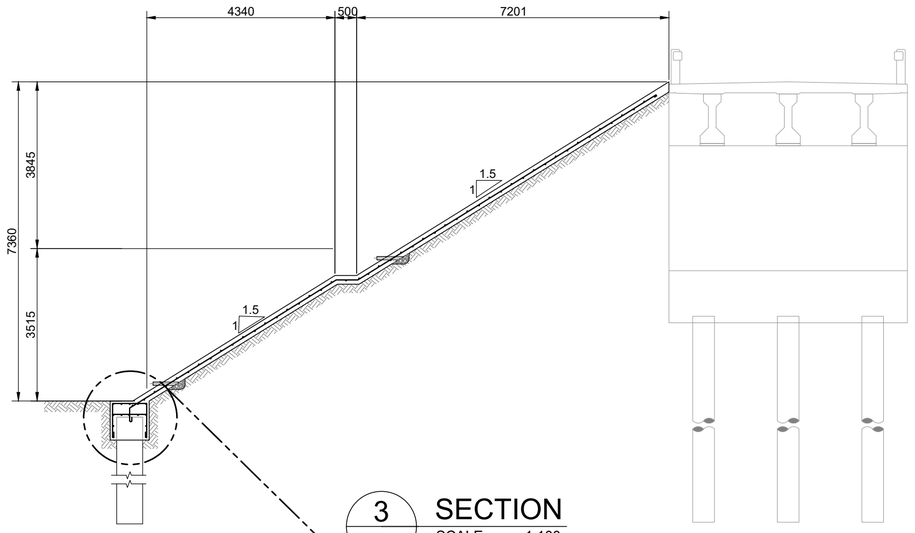
1 PLAN
SCALE 1:200



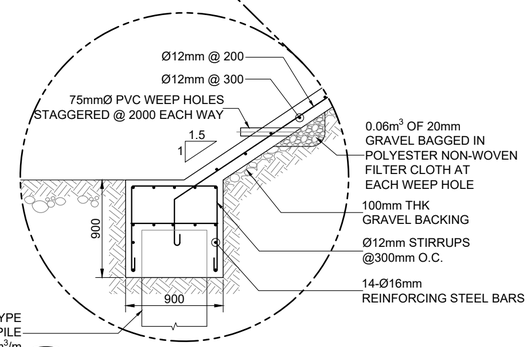
2 SECTION
SCALE 1:100



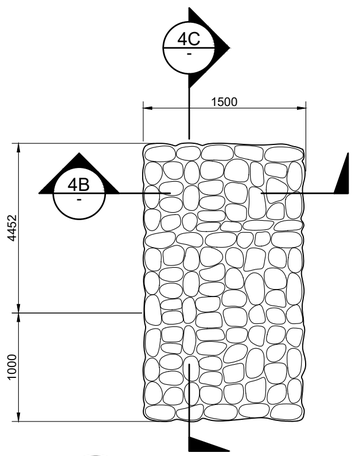
2A BLOW-UP DETAIL
SCALE 1:40



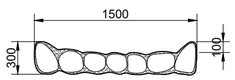
3 SECTION
SCALE 1:100



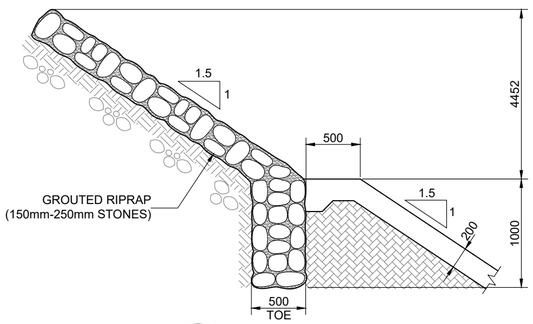
3A BLOW-UP DETAIL
SCALE 1:40



4A ELEVATION
SCALE 1:40

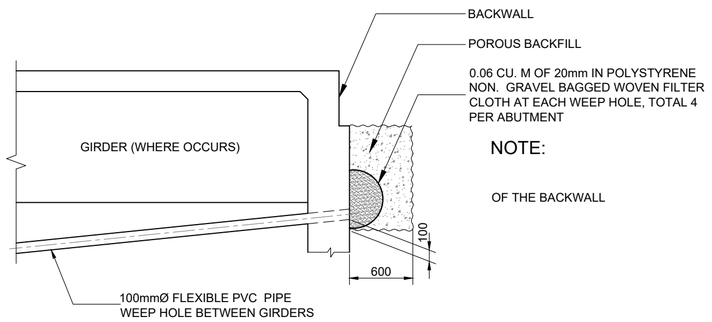


4B SECTION
SCALE 1:40



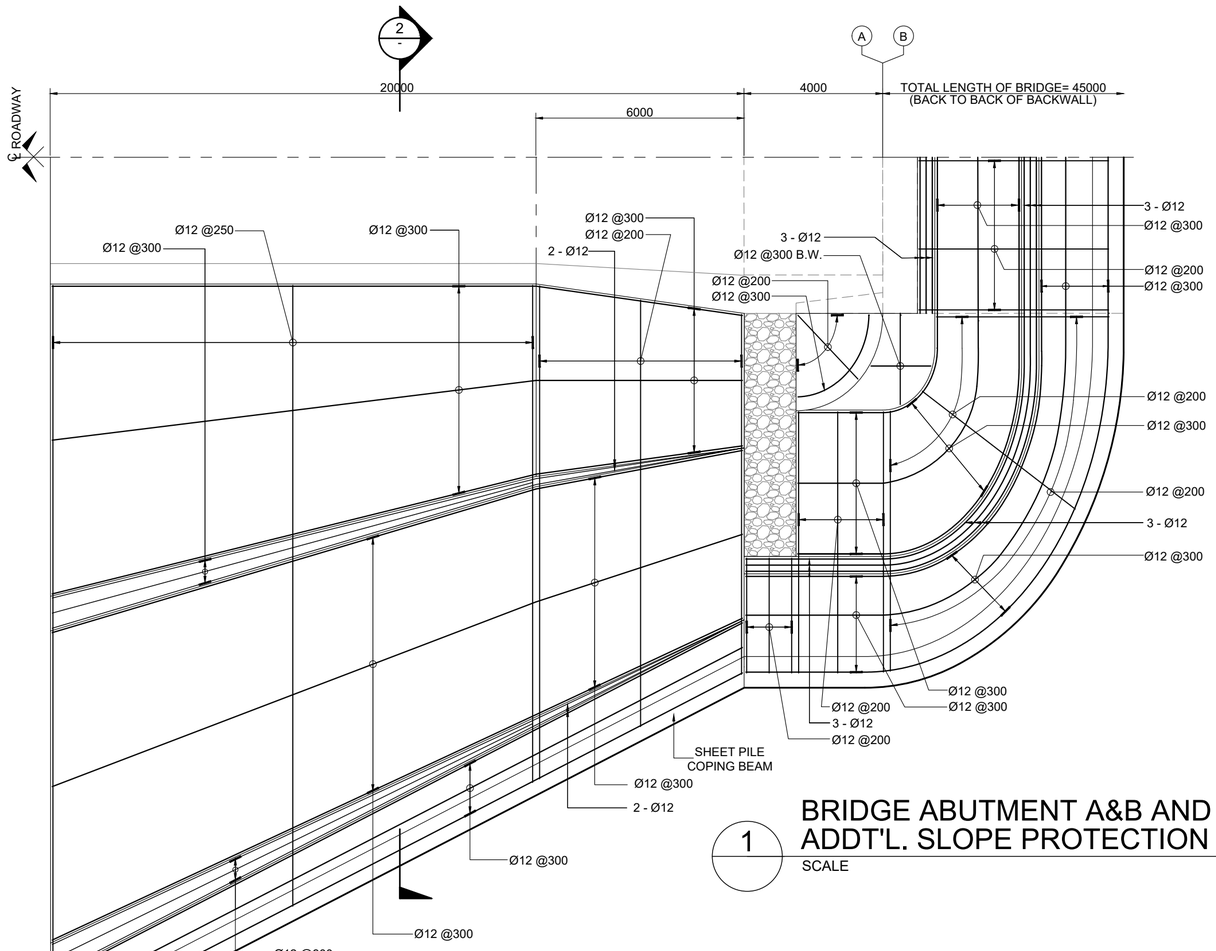
4C SECTION
SCALE 1:40

4 SIDE DRAIN DETAILS
SCALE AS SHOWN



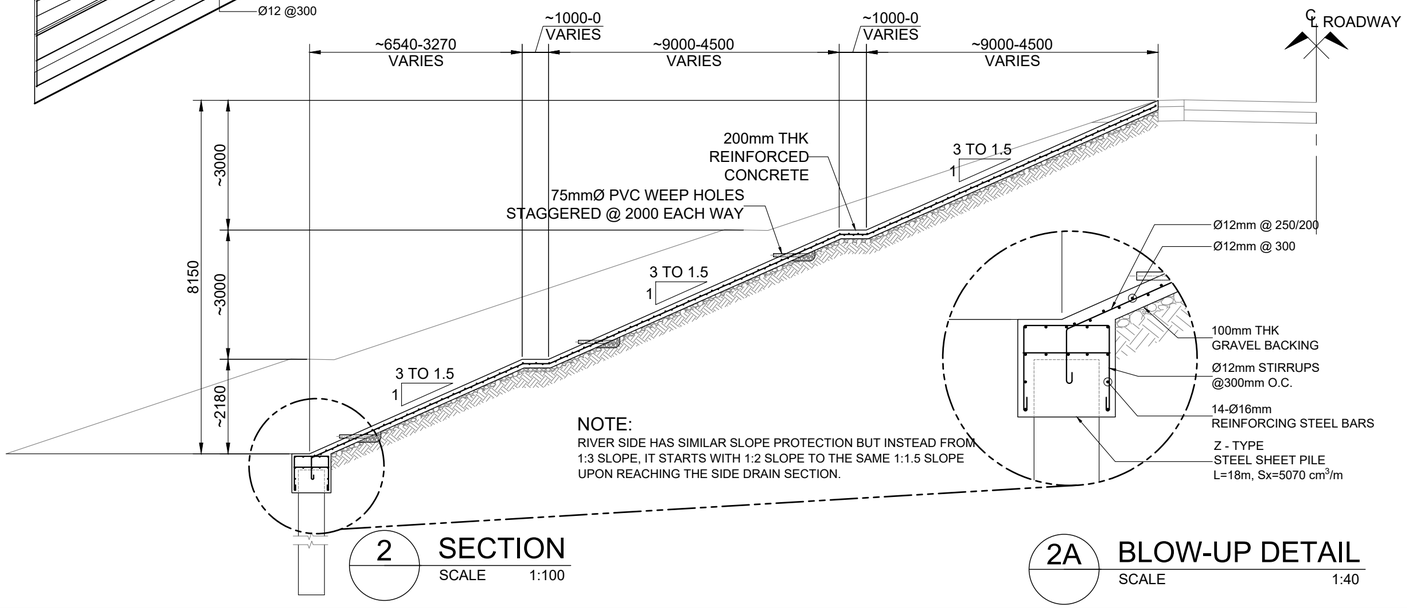
5 ABUTMENT A&B DRAINAGE
SCALE 1:25

	PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
	SUBMITTED:		ABUTMENT A&B SLOPE PROTECTION AND DRAINAGE DETAILS		DATE:	DATE:	DATE:	DATE: DATE:	DATE:	BP11 19 20	991 1154



BRIDGE ABUTMENT A&B AND LANDSIDE ADDT'L. SLOPE PROTECTION PLAN

1 SCALE 1:125



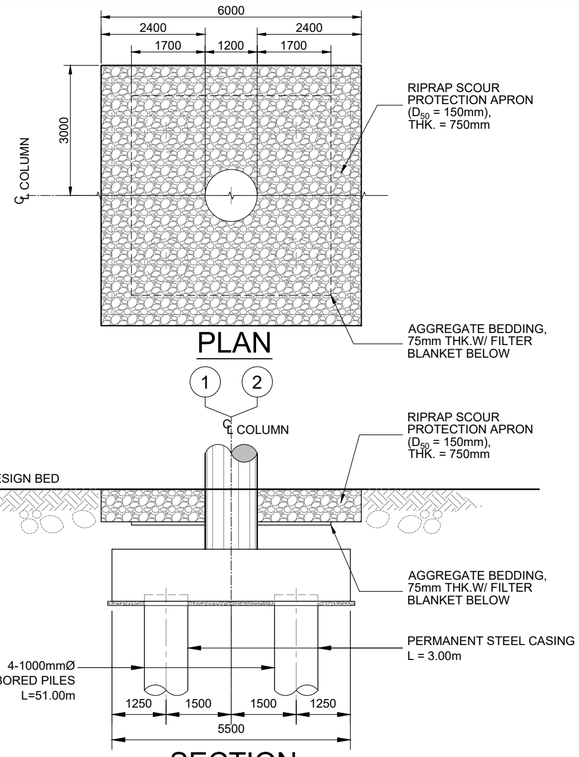
NOTE:
RIVER SIDE HAS SIMILAR SLOPE PROTECTION BUT INSTEAD FROM 1:3 SLOPE, IT STARTS WITH 1:2 SLOPE TO THE SAME 1:1.5 SLOPE UPON REACHING THE SIDE DRAIN SECTION.

SECTION

2 SCALE 1:100

BLOW-UP DETAIL

2A SCALE 1:40



SECTION

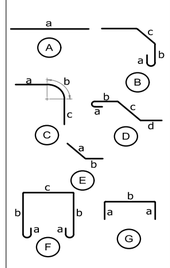
PIER 1&2 SCOUR PROTECTION APRON

3 SCALE 1:100

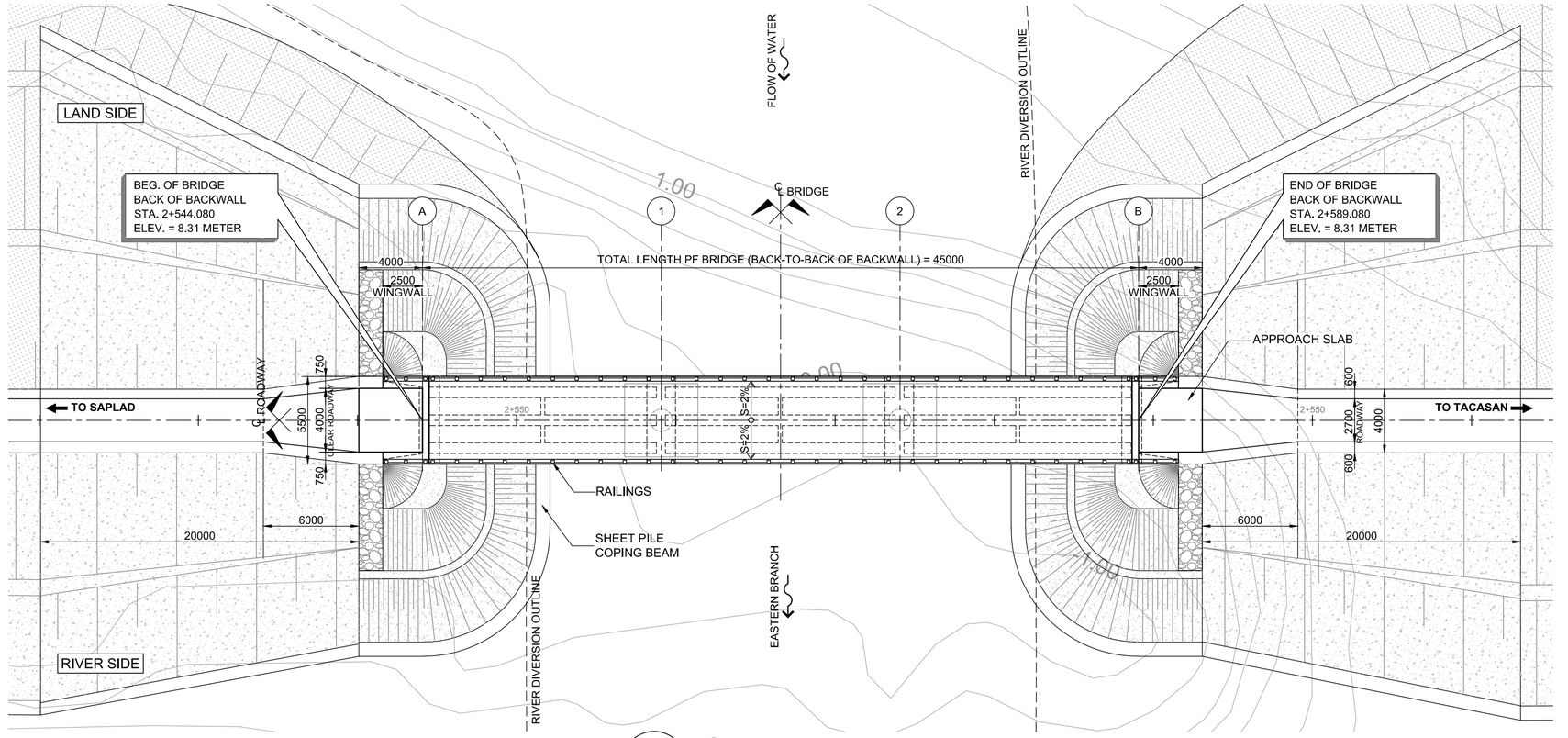
REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR SLOPE PROTECTION

REINFORCEMENT

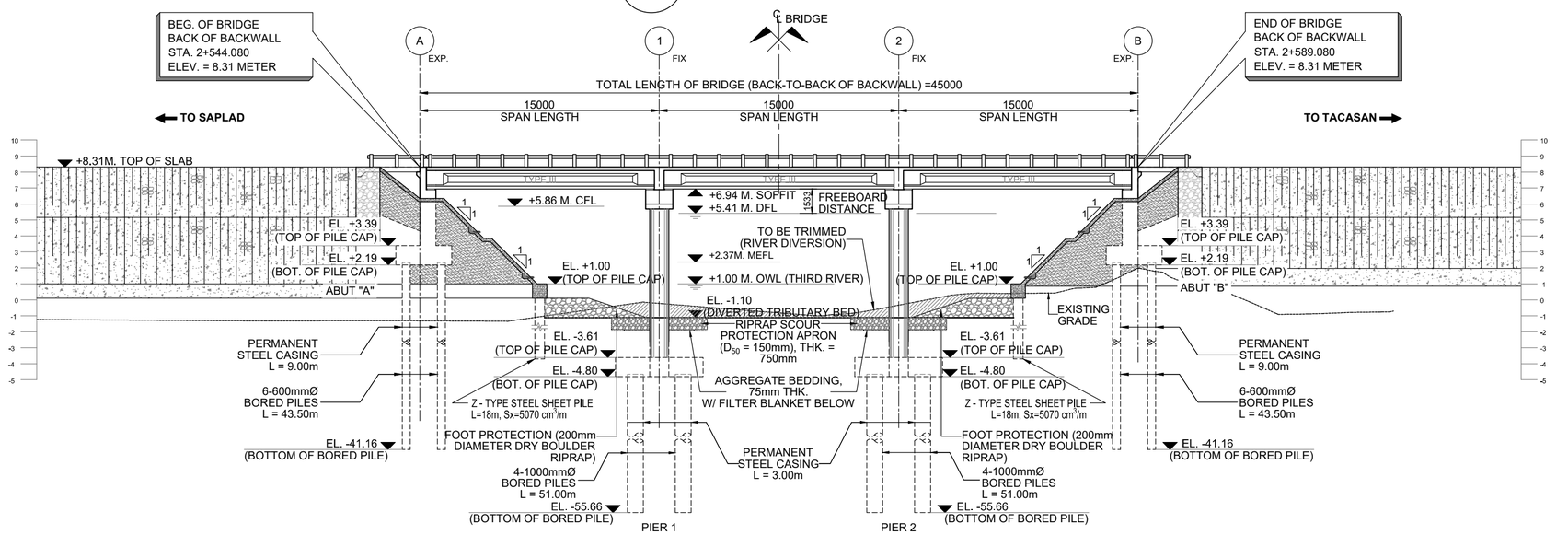
BENDING DIAGRAM	STRUCTURE COMPONENT	BAR SIZE (mm)	QUANTITY	SPACING	BAR SHAPE	BAR DIMENSIONS (m) a	b	c	d	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
	SLOPE PROTECTION @ AROUND ABUTMENT	12	91	300	A	45.106	-	-	-	45.11	4,104.65	0.89	3,653.13	117.34
		12	227	200	B	0.1	0.275	4.66	0.5	5.54	1,256.45	0.89	1,118.24	
		12	212	200	D	0.1	1.5	3.55	0.5	5.65	1,197.80	0.89	1,066.04	
		12	212	200	E	3.26	1.5	-	-	4.76	1,009.12	0.89	898.12	
	SLOPE PROTECTION @ LANDSIDE	12	121	300	A	22.51	-	-	-	22.51	2,723.71	0.89	2,424.10	64.16
		12	114	200	B	0.1	0.5	4.31	0.5	5.41	616.74	0.89	548.90	
		12	114	200	D	0.1	0.5	7.39	0.5	8.49	967.86	1.89	1,829.26	
		12	114	200	E	7.39	0.5	-	-	7.89	899.46	0.89	800.52	
	COPING BEAM	12	14	AS SHOWN	A	67.616	-	-	-	67.62	946.62	0.89	842.50	54.77
		12	227	300	F	0.15	0.75	0.75		2.55	578.85	0.89	515.18	
		12	227	300	G	0.45	0.75	-	-	1.65	374.55	0.89	333.35	
										GRAND TOTAL GRADE 60 =			14,029.33	
												SAY =	14,030.00	
												TOTAL =	14,030.00	237.00



PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:				DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 20 20	992 1154



1 GENERAL PLAN
SCALE 1:250



2 GENERAL ELEVATION
SCALE 1:250

HYDRAULIC DATA (BP11)			
A	=	184	m^2
Q_{50}	=	7.64	m^3/s
Q_{100}	=	8.28	m^3/s
O.W.L	=	1.00	m
D.F.L_{50}	=	5.41	m
C.F.L_{100}	=	5.86	m
V_{50}	=	0.05	m/s
V_{100}	=	0.06	m/s
SCOUR DEPTH			
Abut.A/B_{50}	=	4.72	m
Abut.A/B_{100}	=	5.94	m
Pier.1/2$_{50}$	=	2.64	m
Pier.1/2$_{100}$	=	3.40	m

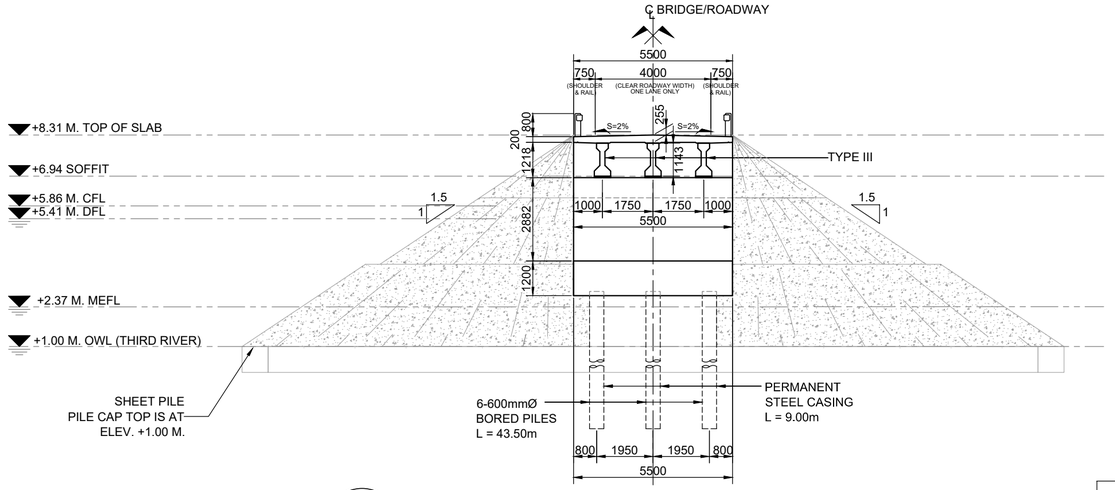
0 5.0 10.0 15.0 20.0 25.0m
SCALE 1: 250

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		GENERAL PLAN AND ELEVATION		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 05B 20	977 1154

SUMMARY OF QUANTITIES

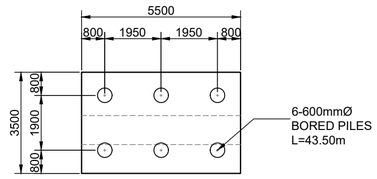
PAY ITEM NO.	DESCRIPTION	UNIT	SUPERSTRUCTURE	ABUTMENT		PIER		QUANTITY
				A	B	1	2	
PART C	**EARTHWORK**							
100 (1)	CLEARING AND GRUBBING	HA.	-	0.01	0.14	-	-	0.15
104 (7)	GRAVEL FILL	CU.M	-	117.84	117.84	-	-	235.68
PART E	**SURFACE COURSES**							
302 (2)	EMULSIFIED ASPHALT	SQ.M	292		-	-	-	292
310 (1) c	BITUMINOUS CONCRETE SURFACE WEARING COURSE, HOT-LAID, 50MM	SQ.M	292	-	-	-	-	292
PART F	**BRIDGE CONSTRUCTION**							
400 (17) aa	CONCRETE PILES CAST IN DRILLED HOLES, 0.60 M DIA.	M	-	261	261	-	-	522
400 (17) c	CONCRETE PILES CAST IN DRILLED HOLES, 1.0 M DIA.	M		-	-	204	204	408
400 (23) y	PERMANENT CASING, 0.60 M DIA., 6 MM THICK	M	-	54	54	-	-	108
400 (23) c1	PERMANENT CASING, 1.0 M DIA., 10 MM THICK	M	-	-	-	12	12	24
400 (26) b	PILE INTEGRITY TESTING, LOW STRAIN	EA.	-	6	6	4	4	20
400 (27)	HIGH STRAIN DYNAMIC TESTING (P.D.A)	EA.	-	6	6	4	4	20
401 (2) b	CONCRETE RAILING, BALUSTER	M	90	-	-	-	-	90
404 (1) a	REINFORCING STEEL, GRADE 40	KG	-	23	23	-	-	46
404 (1) b	REINFORCING STEEL, GRADE 60	KG	11899	6453	6453	10347	10347	45499
405 (1) b3	STRUCTURAL CONCRETE, 27.58 MPA, CLASS A, 28 DAYS	CU.M	76.25	360.02	360.02	68.95	68.95	934.48
406 (1) c1	PRESTRESSED STRUCTURAL CONCRETE MEMBERS, TYPE III	EA.	9	-	-	-	-	9
407 (8)	LEAN CONCRETE, CLASS B, 16.5 MPA	CU.M	-	1.05	1.05	3.25	3.25	8.61
404 (1) b	REINFORCING STEEL, GRADE 60 (ANCHOR BAR)	KG	-	52.16	-	-	-	52.16
412 (1) c	ELASTOMERIC BEARING PADS	EA.	18	-	-	-	-	18
413 (4) e6	EXPANSION JOINT, COMPRESSION SEAL, PREFABRICATED ASSEMBLY, 20MM GAP	M	12	-	-	-	-	12
PART G	**DRAINAGE AND SLOPE PROTECTION STRUCTURES**							
505 (1) a	RIPRAP, CLASS A	CU.M	-	-	-	27	27	54
505 (2) a	GROUTED RIPRAP, CLASS A	CU.M	-	10	10	-	-	20
509 (1) b1	STEEL SHEET PILES, SLOPE PROTECTION	M	-	2250	2250	-	-	4500
510 (2)	CONCRETE, SLOPE PROTECTION	CU.M	-	235.68	235.68	-	-	472
515 (3)	STRUCTURAL BACKFILL	CU.M	-	-	-	-	-	6046.24

*ALL QUANTITIES SHALL BE VERIFIED BEFORE CONSTRUCTION BEGINS.



1 CROSS SECTION AT ABUTMENT A&B

SCALE 1:150



2 ABUTMENT A&B FOUNDATION PLAN

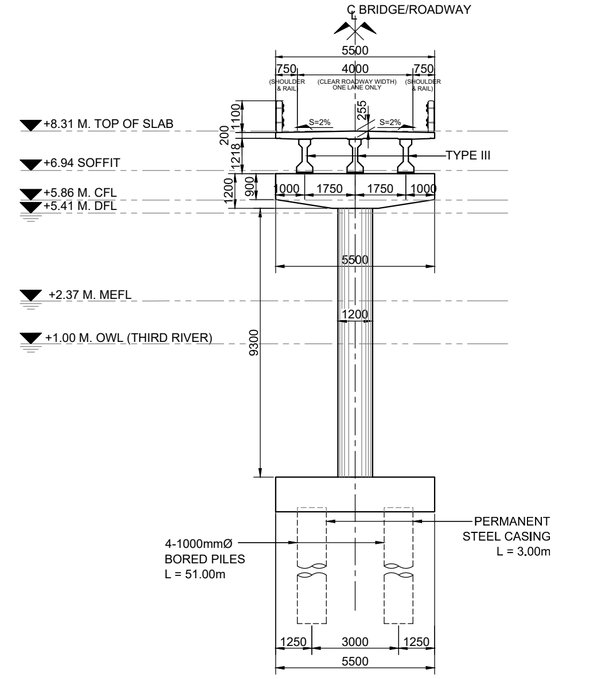
SCALE 1:150

A B U T M E N T - A & B						
	BORED PILE		STRENGTH DEMAND LOAD (kN)		EXTREME DEMAND LOAD (kN)	
	SIZE (mm)	LENGTH (m)	BEARING	PULL-OUT	BEARING	PULL-OUT
	600Ø	43.5	1650	COMP.	1579	COMP.
BOREHOLE NO.	BORED PILE		STRENGTH CAPACITY (kN)		EXTREME CAPACITY (kN)	
	SIZE (mm)	LENGTH (m)	BEARING	PULL-OUT	BEARING	PULL-OUT
* BH-34	600Ø	42.6	3557	2031	3211	1882

P I E R - 1 & 2						
	BORED PILE		STRENGTH DEMAND LOAD (kN)		EXTREME DEMAND LOAD (kN)	
	SIZE (mm)	LENGTH (m)	BEARING	PULL-OUT	BEARING	PULL-OUT
	1000Ø	51.0	2712	COMP.	2978	COMP.
BOREHOLE NO.	BORED PILE		STRENGTH CAPACITY (kN)		EXTREME CAPACITY (kN)	
	SIZE (mm)	LENGTH (m)	BEARING	PULL-OUT	BEARING	PULL-OUT
* BH-34	1000Ø	51.0	9435	5442	8344	4946

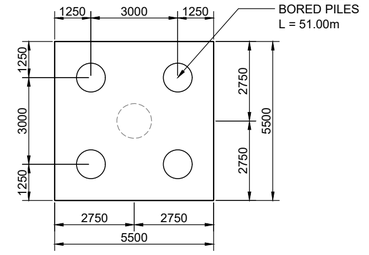
* SEE NEW GEOTECHNICAL INVESTIGATION REPORT.
DEMAND LOADS ARE CRITICAL REACTIONS AT BOTTOM OF PILES.

NOTE: CLOSEST BOREHOLE DATA (BH-34) FROM SAN ISIDRO - EB3 BRIDGE (SET NO.: 12) WAS UTILIZED IN THE BRIDGE DESIGN.



3 CROSS SECTION AT PIER 1&2

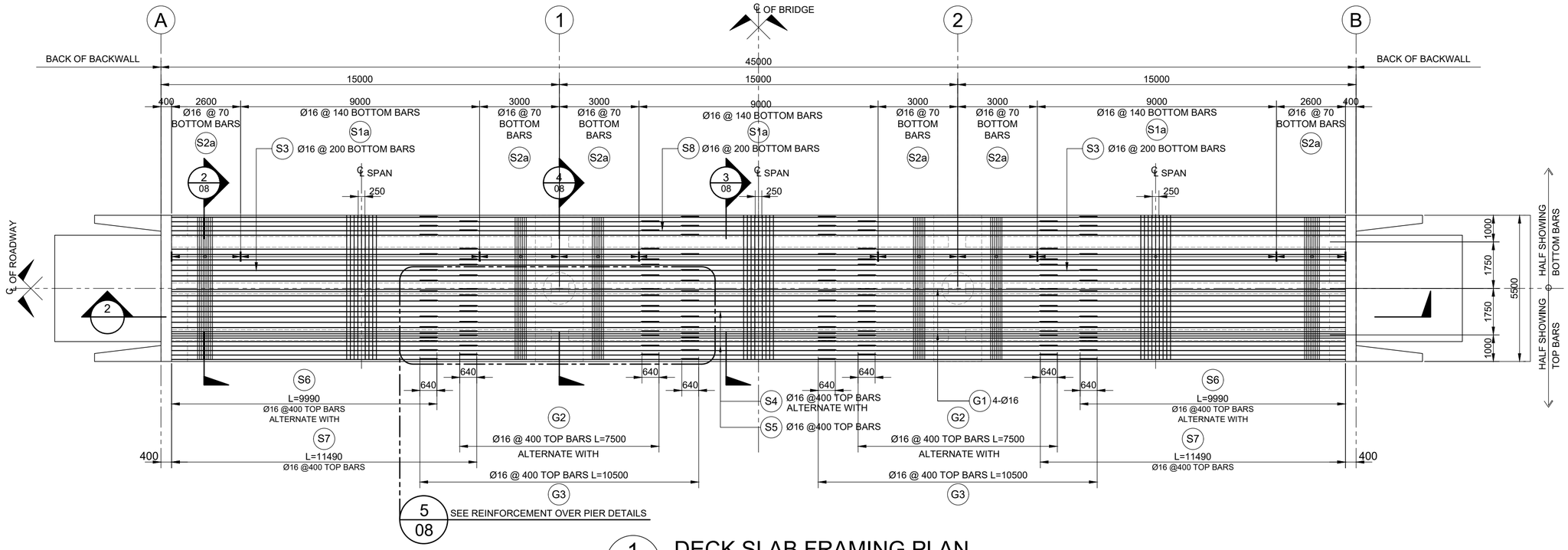
SCALE 1:150



4 PIER 1&2 FOUNDATION PLAN

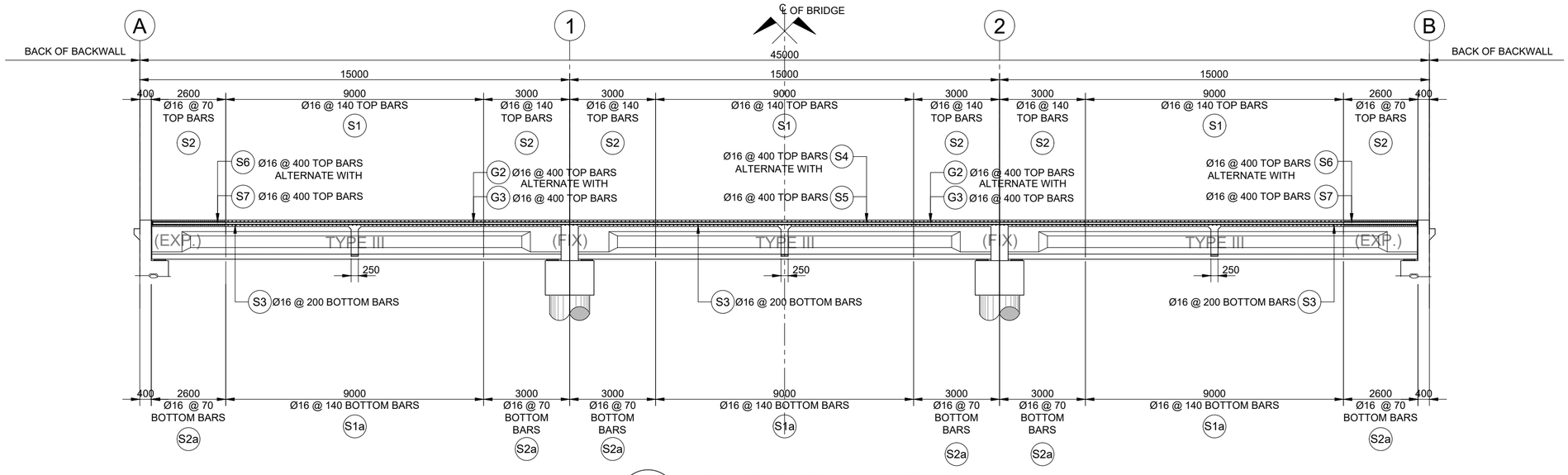
SCALE 1:150

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		SUMMARY OF QUANTITIES, ABUTMENT A&B AND PIER 1&2 CROSS SECTION AND FOUNDATION PLAN, AND BORED PILE DATA	DATE:	DATE:	DATE:	DATE: DATE:	DATE:	BP11 06 20	978 1154



1 DECK SLAB FRAMING PLAN

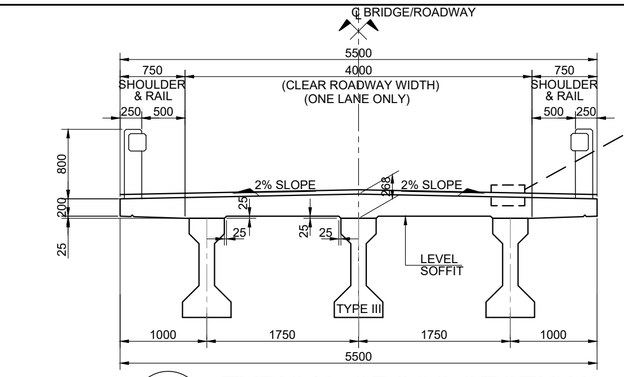
SCALE 1:125



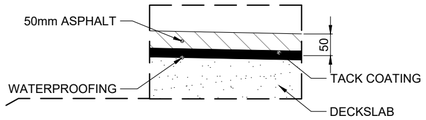
2 LONGITUDINAL SECTION

SCALE 1:125

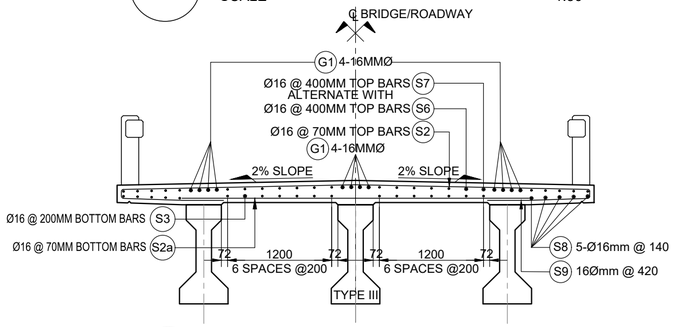
PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		DECK SLAB FRAMING PLAN AND LONGITUDINAL SECTION		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 07 20	979 1154



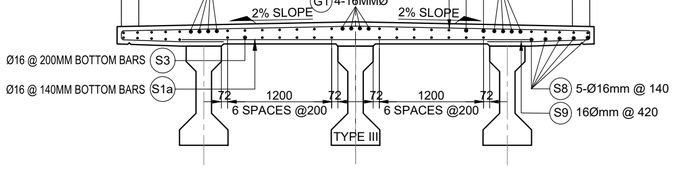
1 TYPICAL CROSS SECTION
SCALE 1:50



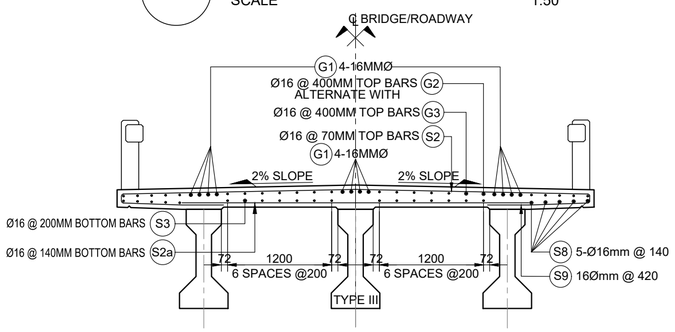
3A WEARING COURSE DET.
SCALE 1:10



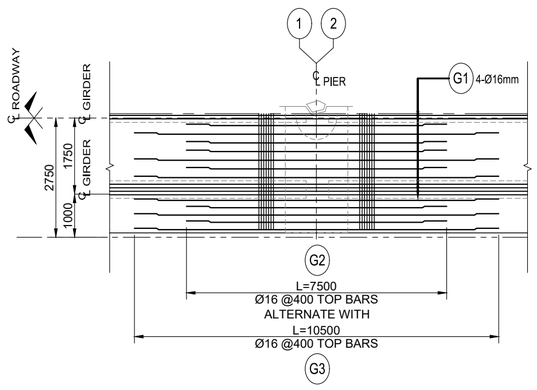
2 TRANSVERSE SECTION
SCALE 1:50



3 TRANSVERSE SECTION
SCALE 1:50



4 TRANSVERSE SECTION OVER PIER 1&2
SCALE 1:50



5 REINFORCEMENT OVER PIER 1&2
SCALE 1:100

REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR SUPERSTRUCTURE OF 3@15m SPAN

REINFORCEMENT

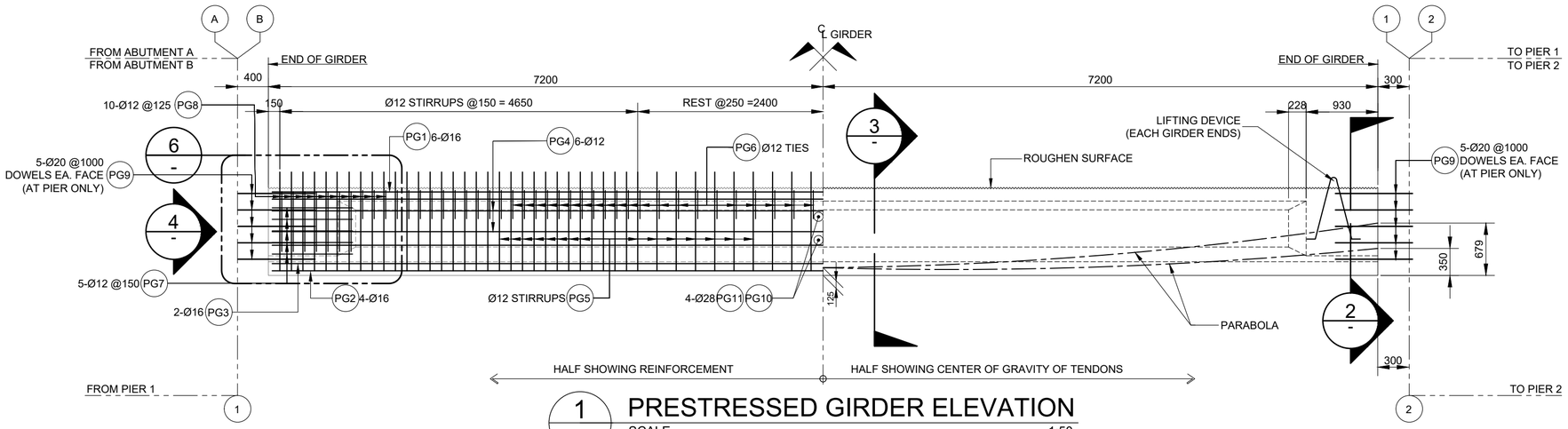
BENDING DIAGRAM	STRUCTURE COMPONENT	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	e	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
A	DECK SLAB	G1	16	12	AS SHOWN	C	1100	44900	-	-	47.10	565.20	1.58	893.02	41.39
		G2	16	12	AS SHOWN	A	7500	-	-	-	7.50	90.00	1.58	142.20	
		G3	16	12	AS SHOWN	A	10500	-	-	-	10.50	126.00	1.58	199.08	
		S1	16	195	140	E	120	2675	-	-	5.59	1,090.05	1.58	1,722.28	
B		S1a	16	195	140	F	2000	-	-	-	4.00	780.00	1.58	1,232.40	
		S2	16	172	100	E	120	2675	-	-	5.59	961.48	1.58	1,519.14	
		S2a	16	172	100	F	2000	-	-	-	4.00	688.00	1.58	1,087.04	
C		S3	16	14	200	A	44900	-	-	-	44.90	628.60	1.58	993.19	
		S4	16	12	400	A	9000	-	-	-	9.00	108.00	1.58	170.64	
		S5	16	12	400	A	12000	-	-	-	12.00	144.00	1.58	227.52	
D		S6	16	12	400	A	9990	-	-	-	9.99	119.88	1.58	189.41	
		S7	16	12	400	A	11490	-	-	-	11.49	137.88	1.58	217.85	
		S8	16	10	AS SHOWN	A	29900	-	-	-	29.90	299.00	1.58	472.42	
		S9	16	195	140	G	125	1130	-	-	1.26	244.73	1.58	386.67	
E		S9a	16	172	100	G	125	1130	-	-	1.26	215.86	1.58	341.06	
F	DIAPHRAGM	D1	28	2	AS SHOWN	F	2710	-	-	-	5.42	10.84	4.83	52.36	2.26
		D2	28	4	AS SHOWN	A	1150	-	-	-	1.15	4.60	4.83	22.22	
		D3	28	8	AS SHOWN	A	1500			-	1.50	12.00	4.83	57.96	
		D4	12	10	300	D	1260	220	150	-	3.26	32.60	0.89	29.01	
G		D5	28	2	AS SHOWN	F	2710	-	-	-	5.42	10.84	4.83	52.36	
		D6	28	4	AS SHOWN	A	1110	-	-	-	1.11	4.44	4.83	21.45	
		D7	28	16	AS SHOWN	A	1270	-	-	-	1.27	20.32	4.83	98.15	
H		D8	12	10	300	D	1350	220	150	-	3.44	34.40	0.89	30.62	
											GRAND TOTAL GRADE 60 =			10,158.02	
											SAY =			10,159.00	
											GRAND TOTAL GRADE 40 =			-	
											SAY =			-	
											TOTAL =	10,159.00		43.64	

DIMENSIONS, REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITES FOR ONE (1) PIER AND ABUTMENT BEARING SYSTEM - ANCHOR BAR

REINFORCEMENT

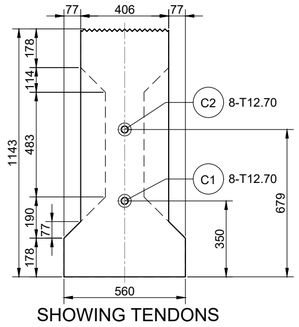
BENDING DIAGRAM	STRUCTURE COMPONENT	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	REMARKS
A	PIER	P6	28	13	300	A	2000	-	-	2.00	26.00	4.83	125.58	
	ABUTMENT	D9	28	6	300	A	1800		-	1.80	10.80	4.83	52.16	
										GRAND TOTAL GRADE 60 =			52.16	
										SAY =			53.00	
										GRAND TOTAL GRADE 40 =			-	
										SAY =			-	
										TOTAL =			53.00	

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		DECK SLAB TRANSVERSE SECTIONS, REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 08 20	980 1154

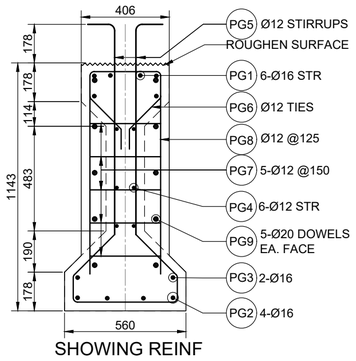


1 PRESTRESSED GIRDER ELEVATION

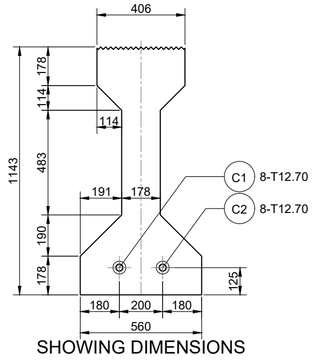
SCALE 1:50



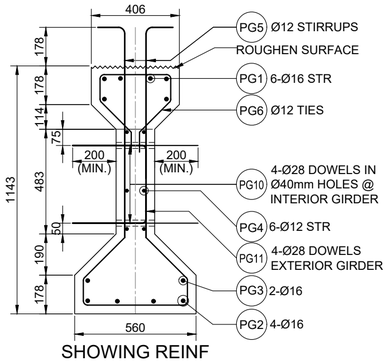
SHOWING TENDONS



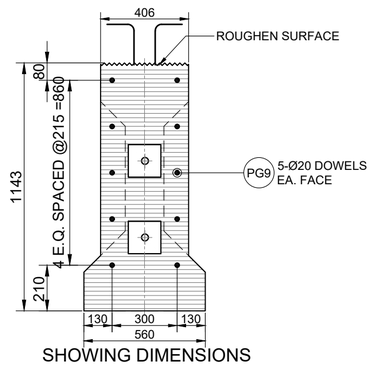
SHOWING REINF



SHOWING DIMENSIONS



SHOWING REINF



SHOWING DIMENSIONS

2 SECTION AT END BLOCK

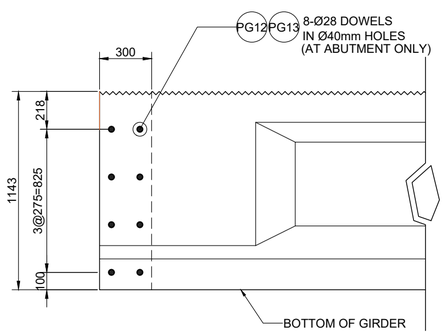
SCALE 1:20

3 SECTION AT MIDSPAN

SCALE 1:20

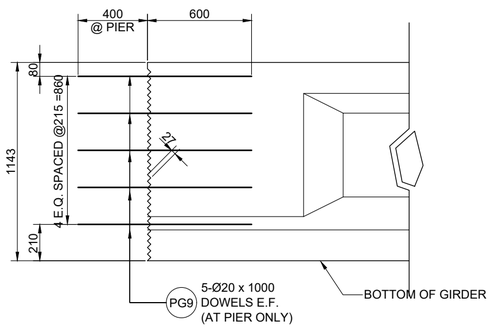
4 TOOTH END VIEW

SCALE 1:20



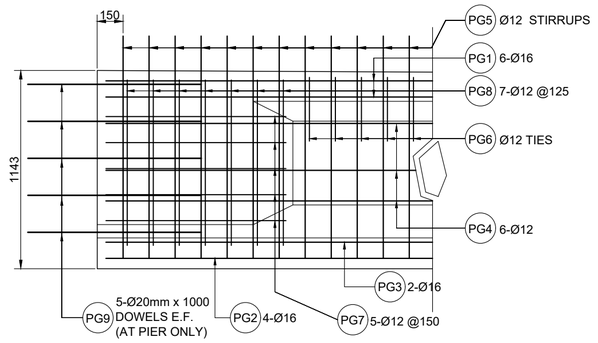
5A DOWELS AT END BLOCK (EXP.)

SCALE 1:25



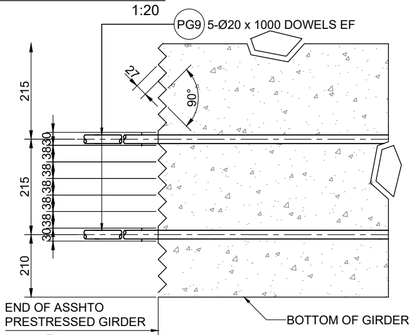
5B DOWELS AT END BLOCK (FIXED)

SCALE 1:25



6 END BLOCK REINFORCEMENT DETAILS

SCALE 1:25



7 TOOTH DETAILS

SCALE 1:10

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		GIRDER ELEVATION DETAILS AND SECTIONS (TYPE III)		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 09 20	981 1154

NOTE ON PRESTRESSING:

1. STRESSING SEQUENCE: C2-C1

2. PRESTRESSING FORCE
   C1 = 1102 kN
   C2 = 1102 kN

3. THE PROPOSED TYPE OF TENDONS WHICH WILL BE USED IN THE POST-TENSIONED DESIGNS. ALL NECESSARY ADDITIONAL DETAILS INCLUDING THOSE FOR END ANCHORAGES. METHODS TO BE EMPLOYED AND PROCEDURES TO BE FOLLOWED. SHALL BE AS APPROVED BY THE ENGINEER. THE TENDONS SHALL BE DRAPED LONGITUDINAL IN PARABOLIC POSITIONS. ALL TENDONS SHALL BE PLACED SO THAT THEIR CENTER OF GRAVITY WILL BE AT THE POSITION SHOWN ON THE DETAIL. THE REQUIRED FORCES AFTER LOSSES SHALL BE OBTAINED BY APPLYING INITIAL TENSILE FORCES OF SUFFICIENT MAGNITUDE TO ALLOW FOR ALL SUBSEQUENT LOSSES, INCLUDING THOSE FOR ELASTIC SHORTENING. SHRINKAGE, CREEP, RELAXATION, FRICTION AND EFFICIENCY OF END ANCHORAGE. AFTER SECURING THE END ANCHORAGES ALL TENDONS SHALL BE PRESSURE GROUTED IN THEIR CONDUITS IN ACCORDANCE WITH "SPECIFICATIONS".

4. STRANDS SHALL CONFORM TO AASHTO M 203 (ASTM A416) "GALVANIZED LOW-RELAXATION SEVEN-WIRE" STRANDS AND HAS THE FOLLOWING PROPERTIES:

5. PRE-STRESSING TENDON PROPERTIES
   NOMINAL DIAMETER = 12.7mm
   NOMINAL AREA = 98.70mm²
   TENSILE STRENGTH (fpu) = 1860 MPa
   0.1% PROOF LOAD (fpy) = 1670 MPa
   MODULUS OF ELASTICITY (Eps) = 197 GPa

6. PROPERTIES OF INTERNAL POST-TENSIONING CABLE:
   COEFFICIENT OF FRICTION (1/RAD). u=0.20
   WOBBLE FRICTION COEFFICIENT (1/M), K=0.00066
   ALLOWABLE WEDGE SLIDING = 9.5MM

7. CONCRETE STRENGTH OF GIRDER
   COMPRESSIVE STRENGTH @ 28TH DAY fc = 41.40 MPa
   COMPRESSIVE STRENGTH AT PRESTRESSING fci = 35.20 MPa

8. SHEATHING OR DUCTS SHALL BE APPROVED BY THE ENGINEER. THEY SHALL BE STRONG ENOUGH TO WITHSTAND THE VIBRATION OF THE CONCRETE WITHOUT SUFFERING ANY DAMAGE OR DEFORMATION AND SHALL BE MORTAR TIGHT, RIGIDLY SUPPORTED AT NO MORE THAN 500MM CENTER TO CENTER SPACING.

9. END ANCHORAGE DEVICE:

   a.) ANCHORAGE DEVICES SHALL WITHSTAND A FORCE OF NOT LESS THAN 95% OF THE SPECIFIED MINIMUM BREAKING STRENGTH OF THE TENDONS WITHOUT DAMAGE OR EXCESSIVE DEFORMATION OR DRAW-IN.

   b.) NO DAMAGED ANCHORAGE DEVICES SHALL BE USED. ALL STEEL PORTIONS SHALL BE PROTECTED FROM CORROSION AT ALL TIMES.

   c.) ALL TAPPED HOLES SHALL BE PROTECTED BY SUITABLE PLUGS UNTIL USED. THE ANCHORAGE DEVICES SHALL BE KEPT FREE FROM MORTAR, LOOSE RUST, GREASE, TAR, PAINT, OIL, MUD, OR ANY OTHER COATING.

10. GROUT ALL TENDONS IN SHEATHINGS AND THE RECESSES AT END ANCHORAGES. GROUT POINTS SHALL BE PROVIDED AT ALL CROWN POINTS. SAG POINTS AND ANCHORAGES.

11. IN CASE WHERE REINFORCING BARS INTERFERE WITH POST-TENSIONING DUCTS AND ANCHORAGES, REBAR REARRANGEMENT SHALL BE DONE WITH THE APPROVAL OF THE ENGINEER.

12. HANDLING IN PRESTRESSED CONCRETE BEAMS; THE BEAMS SHALL BE MAINTAINED IN AN UPRIGHT POSITION AND SHALL BE LIFTED BY SUITABLE DEVICES PROVIDED AT THE ENDS OF THE BEAMS. THE CONTRACTOR'S PROPOSED LIFTING DETAILS SHOULD BE GIVEN CAREFUL CONSIDERATION BEFORE BEING SUBMITTED ON SHOP DRAWINGS FOR APPROVAL. THE USE OF HOLES FOR LIFTING PURPOSES WILL NOT PERMITTED.

13. SHOP DRAWINGS SHALL BE SUBMITTED FOR REVIEW PRIOR TO FABRICATION. FABRICATION SHALL START ONLY AFTER APPROVAL OF SHOP DRAWINGS. SHOP DRAWINGS SHALL INCLUDE THE CALCULATED ELONGATION OF THE PRESTRESSING TENDONS CORRESPONDING TO THE REQUIRED JACKING FORCE.

14. PRIOR TO FABRICATION OF PRESTRESSED CONCRETE FIRDER. THE CONTRACTOR SHALL PREPARE AND SUBMIT FOR REVIEW AND APPROVAL OF THE ENGINEER COMPLETE PLANS, DETAIL DRAWINGS, STRUCTURAL DESIGN CALCULATION AND OTHER NECESSARY PRESENTATIONS COVERING THE UNDERTAKING OF THE CONTRACTOR AND IN CONSIDERATION OF THE FOLLOWING:

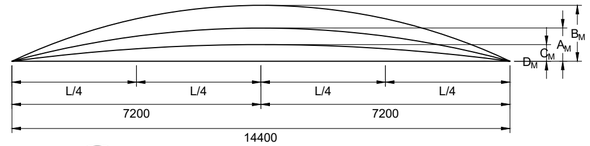
    METHODS AND EQUIPMENT TO BE USED IN PRESTRESSING OPERATIONS.

    MATERIALS AND APPURTENANCES TO BE PROVIDED AND INCORPORATED IN THE STRUCTURE.

    HANDLING AND TRANSPORT SYSTEM TO TRANSFER PRECAST PSC GIRDERS FROM CASTING YARD TO BRIDGE SITE

    ERECTION EQUIPMENT AND PROCEDURE TO BE UTILIZED TO INSTALL PSC GIRDER IN FINAL POSITION.

15. CABLES SHALL BE STRESSED AT BOTH ENDS



1 CAMBER DIAGRAM
SCALE NTS

NOTE ON CAMBER:

A = CAMBER OF CAST OF GIRDER, BEFORE GIRDER AND PRESTRESSING LOAD

B = CAMBER AFTER GIRDER AND PRESTRESSING LOAD

C = CAMBER AFTER SLAB, DIAPHRAGM

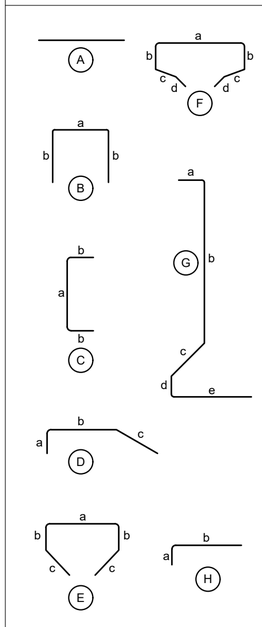
D = CAMBER FINAL AFTER SUPERIMPOSED DEAD LOAD

CAMBER ARE THEORETICAL AND MAY VARY WITH ACTUAL (AGE) CONCRETE STRENGTH, VARIOUS PRESTRESSING CONDITIONS, CREEP FACTOR AND PRESTRESS LOSSES. CONTRACTOR SHALL SURVEY TOP OF GIRDERS TO OBTAIN ACTUAL VALUE AND ADJUST PROFILE ACCORDINGLY.

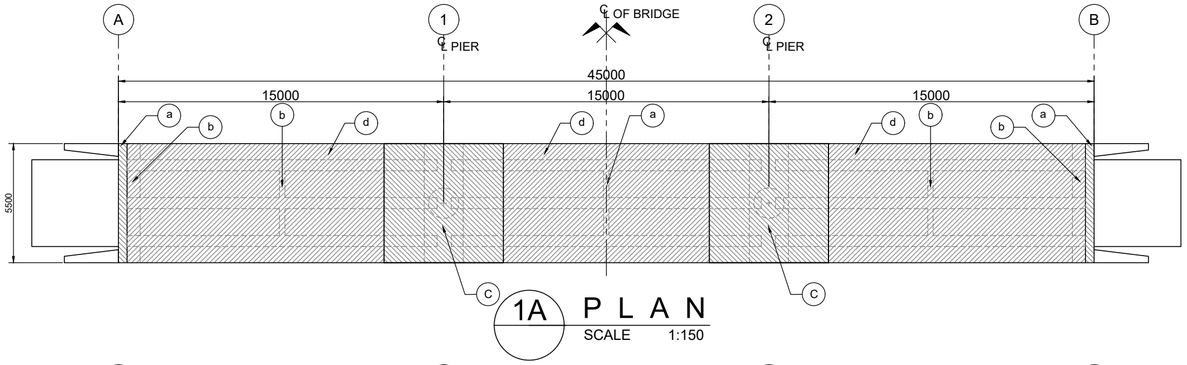
GIRDER LENGTH (m)	GIRDER TYPE	CAMBER (mm)				LOC.
		A	B	C	D	
14.400	III	3.0	14.0	7.0	5.0	MIDSPAN
		2.0	11.0	5.0	4.0	L/4 AND 3L/4

REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR PRESTRESSED GIRDER TYPE III (FOR 1 GIRDER ONLY)

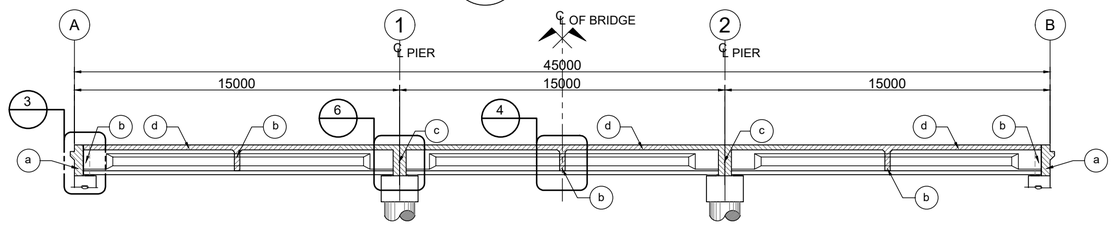
REINFORCEMENT

BENDING DIAGRAM	STRUCTURE COMPONENT	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (m)				LOCATION Lc	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
							a	b	c	d						
	TYPE III GIRDER	PG1	16	6	AS SHOWN	A	14.3	-	-	-	-	14.3	85.8	1.58	135.56	18.61
		PG2	16	4	AS SHOWN	A	14.3	-	-	-	-	14.3	57.2	1.58	90.38	
		PG3	16	2	AS SHOWN	A	14.3	-	-	-	-	14.3	28.6	1.58	45.19	
		PG4	12	4	AS SHOWN	A	14.3	-	-	-	-	14.3	57.2	0.89	50.91	
		PG5	12	352	AS SHOWN	G	0.125	0.975	0.275	0.13	0.5	2.005	705.76	0.89	628.13	
		PG6	12	176	AS SHOWN	E	0.33	0.37	0.2	-	-	1.47	258.72	0.89	230.26	
		PG7	12	20	150	D	0.33	0.86	0.23	-	-	1.54	30.8	0.89	27.41	
		PG8	12	14	125	B	0.43	0.94	-	-	-	3.11	43.54	0.89	38.75	
		PG9	20	20	AS SHOWN	H	0.15	1.1	-	-	-	1.25	25	2.47	61.75	
		PG10	28	4	AS SHOWN	A	0.58	-	-	-	-	0.58	2.32	4.83	11.21	
		PG11	28	4	AS SHOWN	H	0.155	0.35	-	-	-	0.505	2.02	4.83	9.76	
		PG12	28	2	AS SHOWN	A	0.603	-	-	-	-	0.603	4.824	4.83	23.30	
		PG13	28	8	AS SHOWN	H	0.155	0.35	-	-	-	0.505	4.04	4.83	19.51	
													GRAND TOTAL GRADE 60 =		1372.111	
														SAY =	1373	
													GRAND TOTAL GRADE 40 =		0.00	
														SAY =	0	
														TOTAL =	1373	18.61

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		GIRDER CAMBER DIAGRAM AND SCHEDULE OF REINFORCEMENT		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 10 20	982 1154



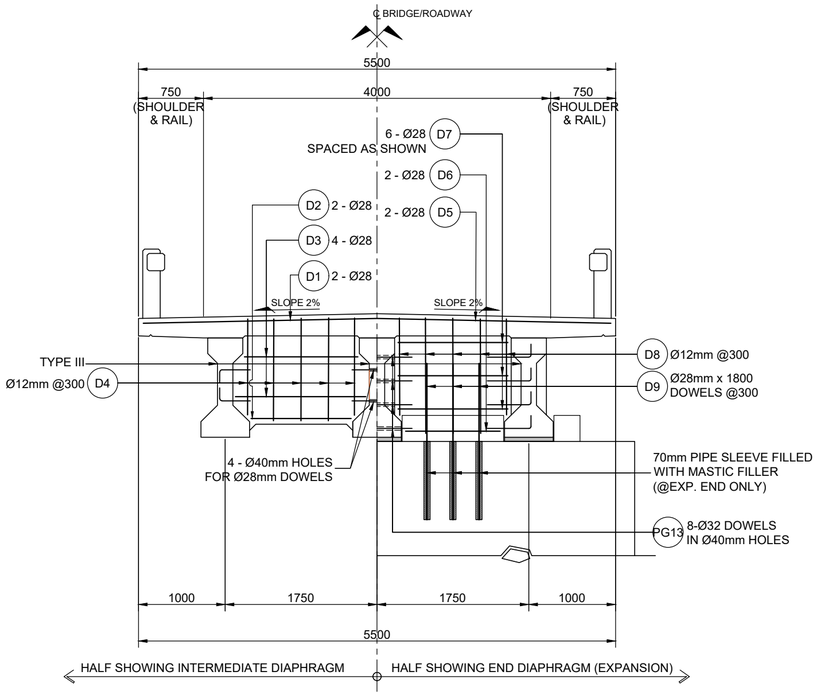
1A PLAN
SCALE 1:150



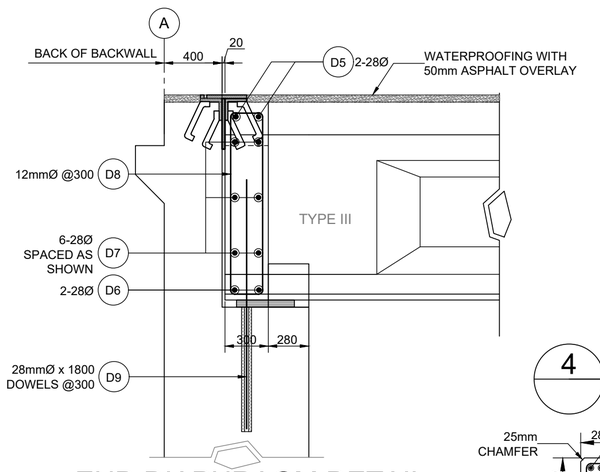
1B ELEVATION
SCALE 1:200

NOTES:

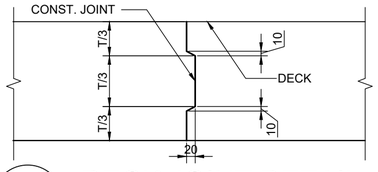
1. CONCRETE AT (a) AREAS SHALL BE PLACED AT LEAST ONE DAY AHEAD OF CONCRETE AT (b) AREAS.
2. CONCRETE AT (b) AREAS SHALL BE PLACED AT LEAST ONE DAY AHEAD OF CONCRETE AT (c) AREAS.
3. REINFORCEMENT SHALL BE CONTINUOUS AT CONSTRUCTION JOINTS



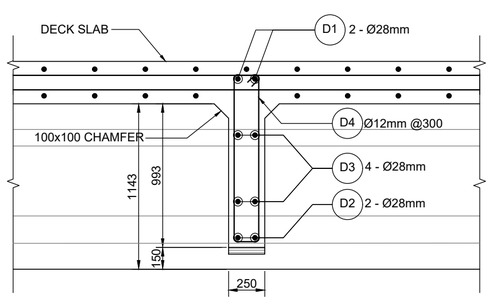
2 CROSS SECTION SHOWING DIAPHRAGM DETAIL
SCALE 1:50



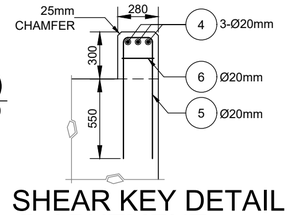
3 END DIAPHRAGM DETAIL @ ABUTMENT (EXPANSION)
SCALE 1:30



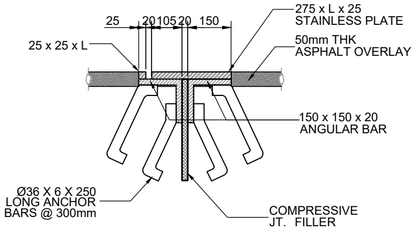
7 DECK JOINT DETAIL
SCALE 1:10



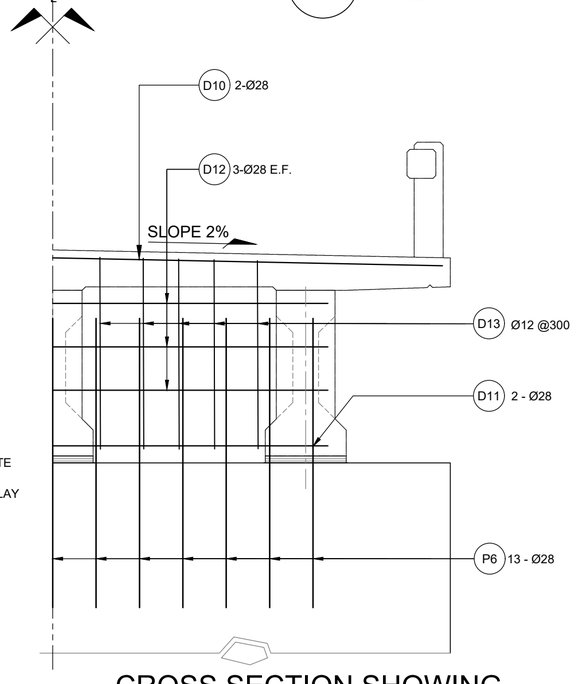
4 INTERMEDIATE DIAPHRAGM SECTION
SCALE 1:30



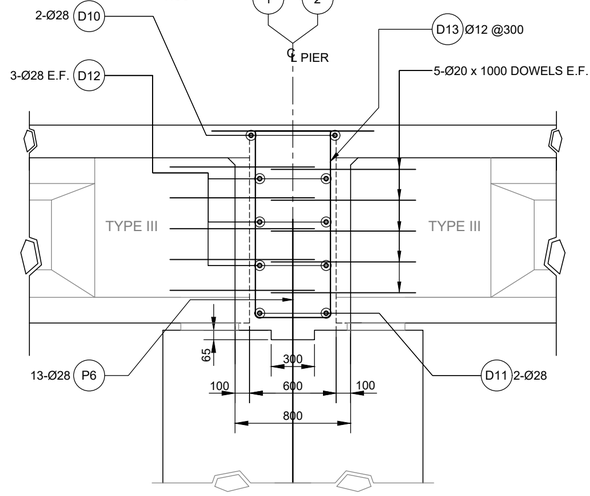
8 SHEAR KEY DETAIL @ ABUTMENT
SCALE 1:30



9 EXPANSION DAM DETAIL
SCALE 1:15

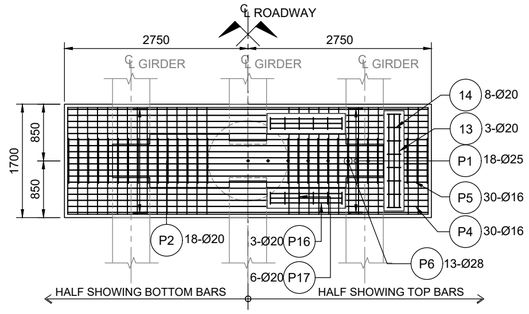


5 CROSS SECTION SHOWING DIAPHRAGM @ PIER 1&2
SCALE 1:30

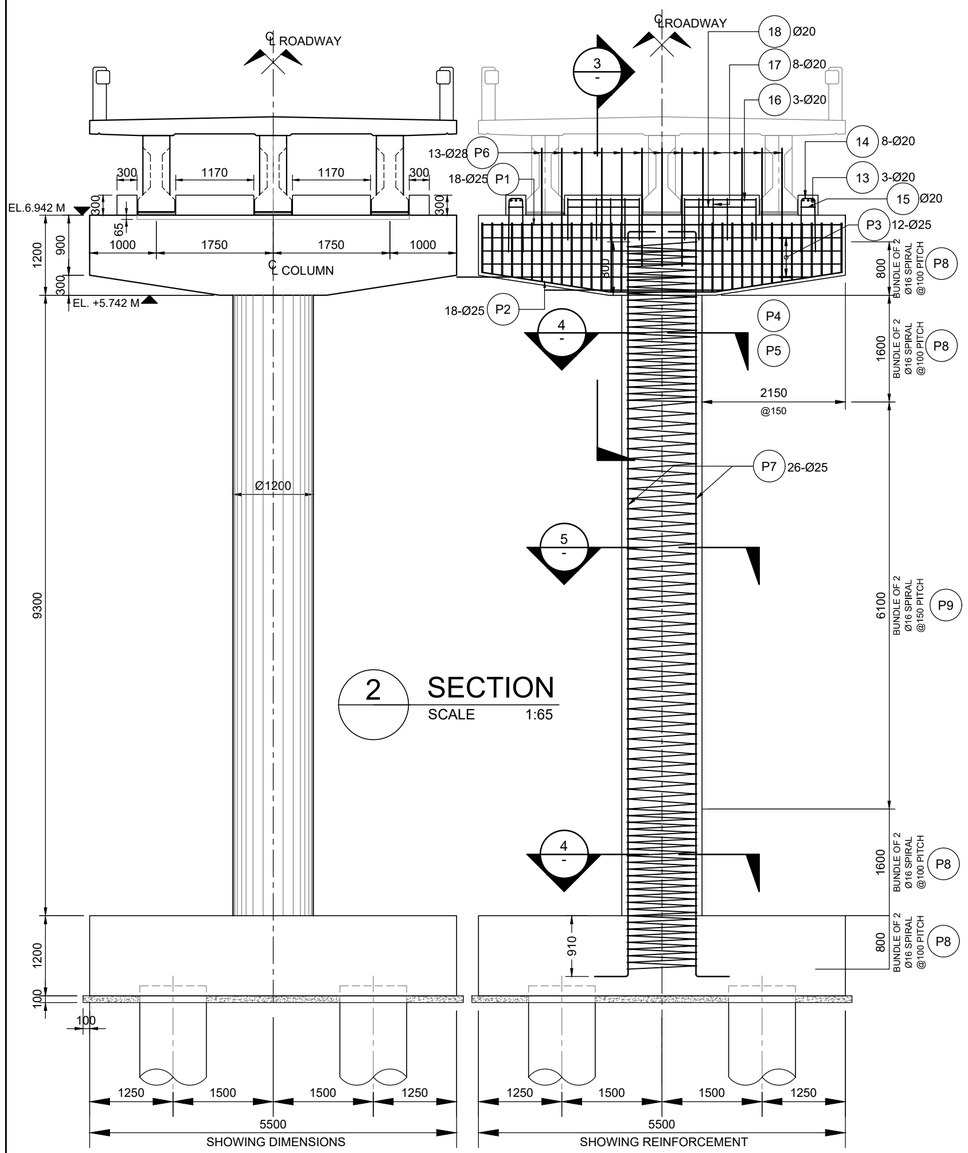


6 DIAPHRAGM DETAIL DETAIL @ PIER 1&2
SCALE 1:30

	PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:	APPROVED:	SET NO. / SHEET NO.
	SUBMITTED:		CONCRETE POURING SEQUENCE, DIAPHRAGM SECTIONS & DETAILS, SHEAR KEY & EXPANSION DAM DETAILS		DATE:	DATE:	DATE: DATE: DATE:	DATE:	BP11 11 20 / 983 1154



1 PLAN
SCALE 1:65

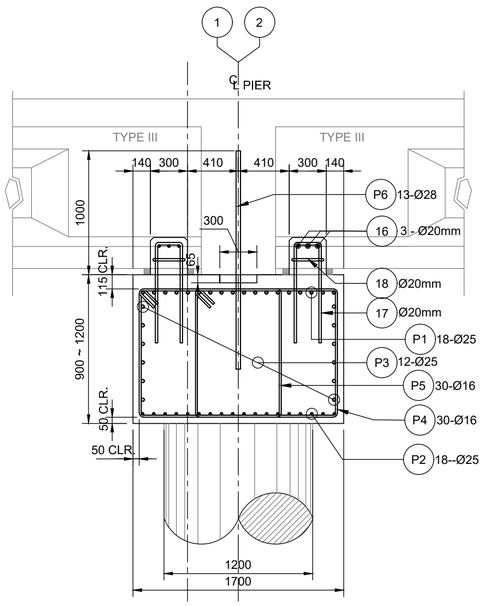


2 SECTION
SCALE 1:65

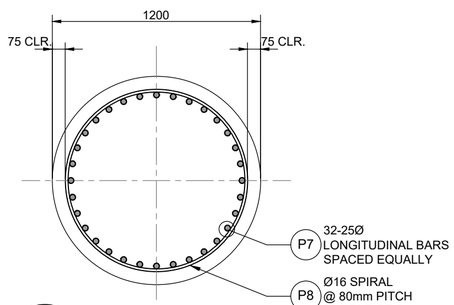
DIMENSIONS, REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITES FOR ONE (1) PIER

REINFORCEMENT

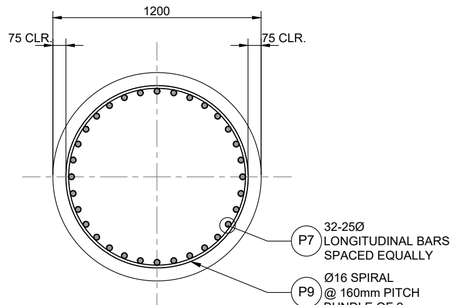
BENDING DIAGRAM	SPAN LENGTHS L1-L1-L1 (m)	COLUMN DIAMETER (m)	COPING BEAM W (m)	COPING BEAM H (m)	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)	REMARKS
	15-15-15	1.2	1.7	1.2	P1	25	18	AS SHOWN	B	5400	350	-	6.10	109.80	3.86	423.83	32.95	1. QUANTITIES ARE FOR THE TWO (2) COLUMNS OF ONE PIER ONLY.
					P2	25	18	AS SHOWN	B	5400	350	-	6.10	109.80	3.86	423.83		
					P3	25	12	AS SHOWN	A	5400	-	-	5.40	64.80	3.86	250.13		
					P4	16	30	AS SHOWN	C	1600	1035	195	5.66	169.80	1.58	268.28		
					P5	16	30	AS SHOWN	C	690	1035	195	3.84	115.20	1.58	182.02		
					P7	25	32	AS SHOWN	F	11200	500	-	12.20	390.40	3.86	1,506.94		
					P8	16	14	100	D	12000	-	-	12.00	168.00	1.58	265.44		
					P9	16	12	150	D	12000	-	-	12.00	144.00	1.58	227.52		
					13	20	6	AS SHOWN	A	1420	-	-	1.42	8.52	2.47	21.04		
					14	20	16	AS SHOWN	B	220	810	-	1.84	29.44	2.47	72.72		
					15	20	4	AS SHOWN	B	1420	220	-	1.86	7.44	2.47	18.38		
					16	20	12	AS SHOWN	A	1090	-	-	1.09	13.08	2.47	32.31		
					17	20	24	AS SHOWN	B	220	810	-	1.84	44.16	2.47	109.08		
					18	20	4	AS SHOWN	B	1090	220	-	1.53	6.12	2.47	15.12		
														GRAND TOTAL GRADE 60 =		3,816.63		
														SAY =		3,817.00		
														GRAND TOTAL GRADE 40 =		-		
														SAY =		-		
														TOTAL =		3,817.00	32.95	



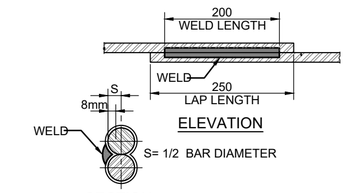
3 SECTION
SCALE 1:35



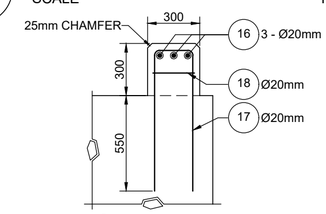
4 SECTION
SCALE 1:25



5 SECTION
SCALE 1:25

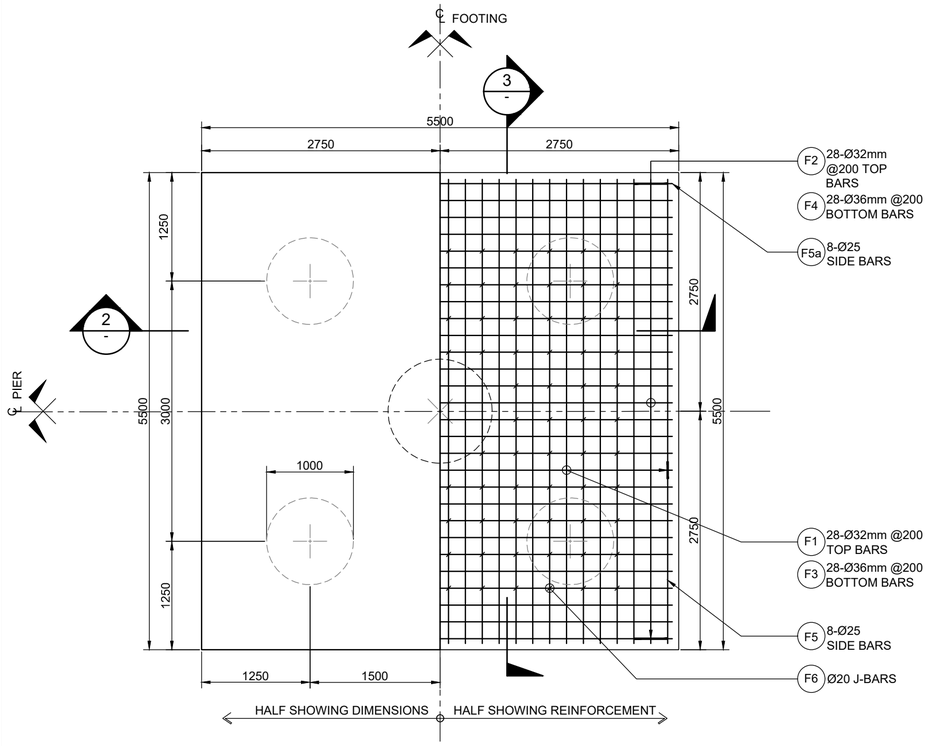


6 SPIRAL SPLICE DETAIL
SCALE NTS



7 SHEAR KEY DETAIL @ PIER
SCALE 1:25

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		PLAN, ELEVATION, SECTIONS AND SPIRAL SPLICE DETAIL OF PIER 1&2	DATE:	DATE:	DATE:	DATE: DATE:	DATE:	BP11 12 20	984 1154

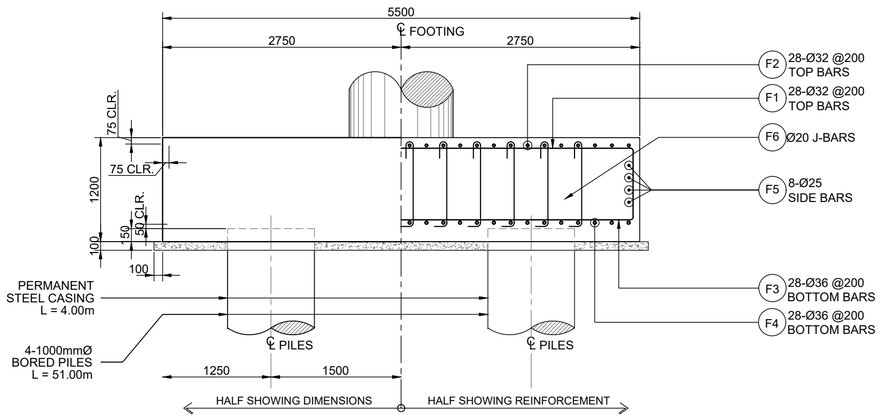


1 PLAN
SCALE 1:50

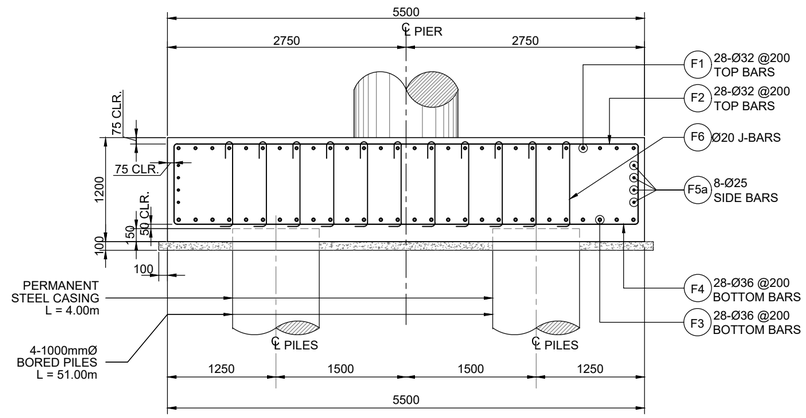
REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITES FOR FOOTING ON PILES

REINFORCEMENT

BENDING DIAGRAM	SPAN LENGTHS L1-L1-L1 (m)	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
	15-15-15	F1	32	28	200	B	5350	925	-	7.20	201.60	6.32	1,274.11	36.30
		F2	32	28	200	B	5350	925	-	7.20	201.60	6.32	1,274.11	
		F3	36	28	200	B	5350	925	-	7.20	201.60	7.99	1,610.78	
		F4	36	28	200	B	5350	925	-	7.20	201.60	7.99	1,610.78	
		F5	25	8	AS SHOWN	A	5350	-	-	5.35	42.80	3.86	165.21	
		F5a	25	8	AS SHOWN	A	5350	-	-	5.35	42.80	3.86	165.21	
		F6	20	122	AS SHOWN	D	200	925	300	1.43	173.85	2.47	429.41	
										GRAND TOTAL GRADE 60 =			6,529.62	
										SAY =			6,530.00	
										GRAND TOTAL GRADE 40 =			-	
										SAY =			-	
										TOTAL =			6,530.00	37.00



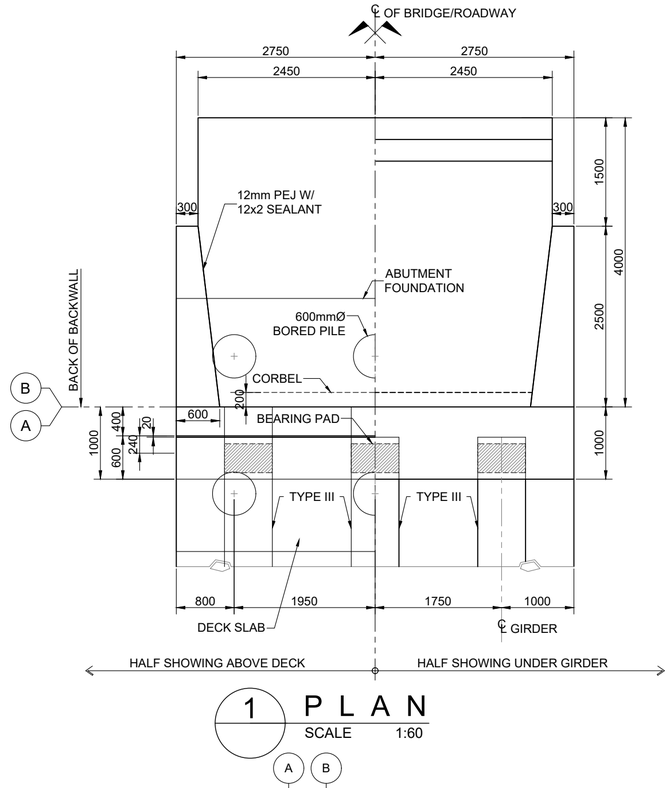
2 TRANSVERSE SECTION
SCALE 1:50



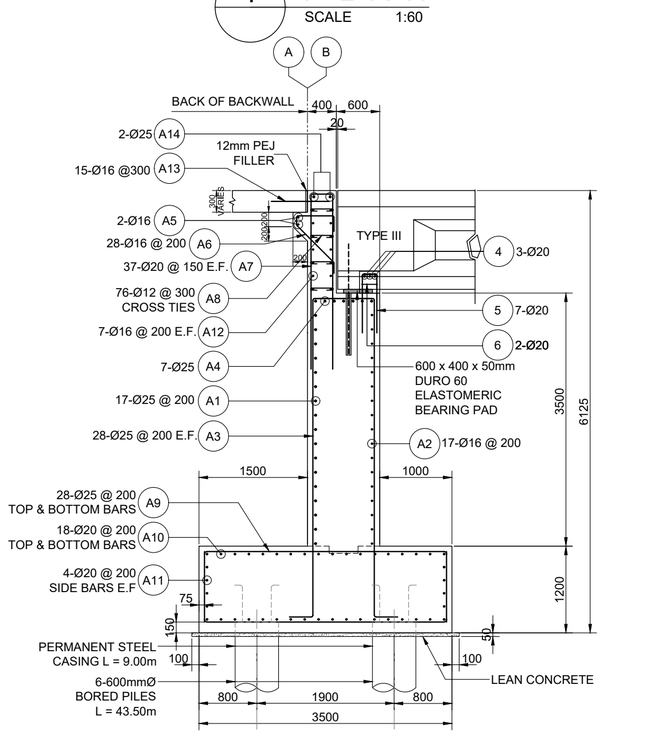
3 LONGITUDINAL SECTION
SCALE 1:50

A PIER 1&2 FOOTING ON PILES
SCALE AS SHOWN

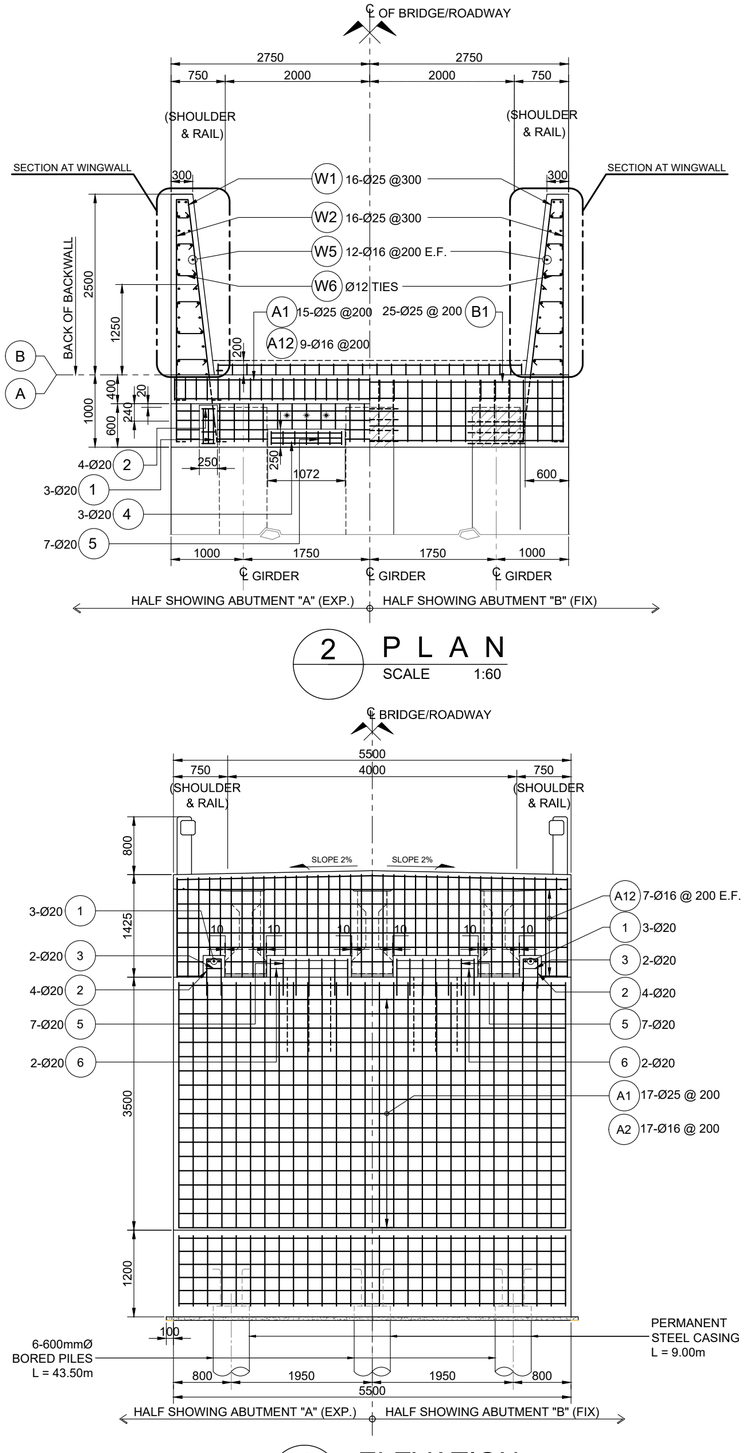
PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		DETAIL OF PIER 1&2 FOOTING ON PILES	DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 13 20	985 1154



1 PLAN
SCALE 1:60

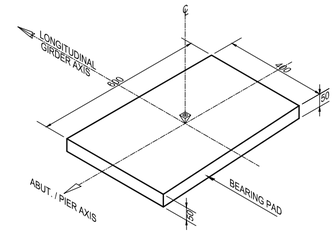


3 ELEVATION (EXP.)
SCALE 1:60

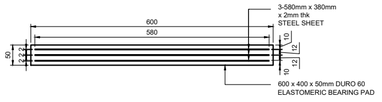


2 PLAN
SCALE 1:60

4 ELEVATION
SCALE 1:60



ISOMETRIC



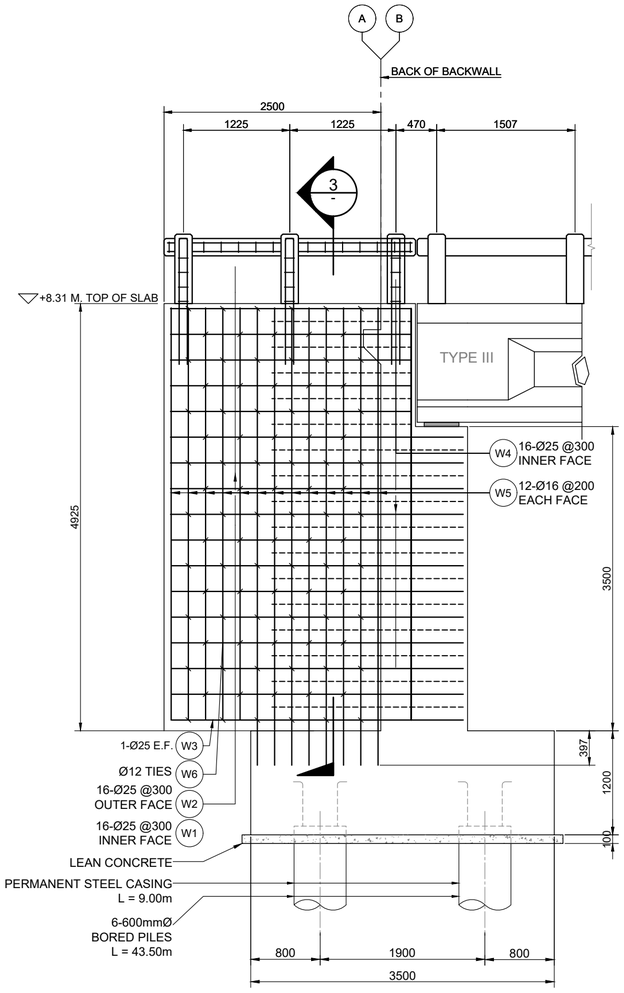
5 BEARING PAD DETAILS
SCALE NTS

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		ABUTMENT A&B PLAN AND ELEVATION DETAILS		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 14 20	986 1154

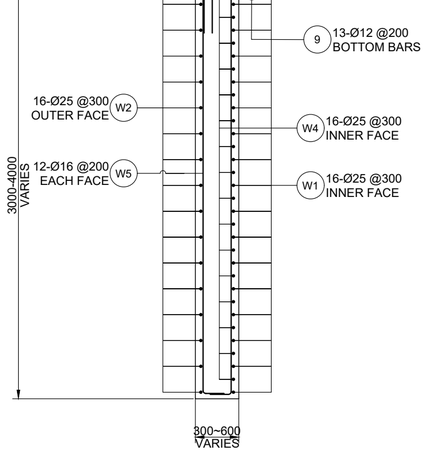
REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR ABUTMENT (MOVABLE)

REINFORCEMENT

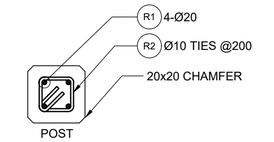
BENDING DIAGRAM	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	d	e	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
	A1	25	17	200	C	700	5400	-	-	-	6.80	115.60	3.86	446.22	
	A2	16	17	200	C	700	5400	-	-	-	6.80	115.60	1.58	182.65	
	A3	25	28	200	D	350	4400	850	-	-	10.35	289.80	3.86	1,118.63	
	A4	25	7	AS SHOWN	C	700	5400	-	-	-	6.80	47.60	3.86	183.74	
	A5	16	2	AS SHOWN	A	5400	-	-	-	-	5.40	10.80	1.58	17.06	
	A6	16	28	200	E	225	470	132	750	150	1.73	48.36	1.58	76.40	
a (A)	A7	20	37	150	C	2430	300	-	-	-	5.16	190.92	2.47	471.57	
a b (B)	A8	12	76	300	J	96	285	96	-	-	0.48	36.25	0.89	32.26	
	A9	25	56	200	C	600	3350	-	-	-	4.55	254.80	3.86	983.53	
	A10	20	36	200	B	110	5350	-	-	-	5.46	196.56	2.47	485.50	
a b a (C)	A11	20	8	200	B	110	5350	-	-	-	5.46	43.68	2.47	107.89	
	A12	16	14	200	C	230	5400	-	-	-	5.86	82.04	1.58	129.62	
c b b a a (D)	A13	16	19	300	A	800	-	-	-	-	0.80	15.20	1.58	24.02	
	A14	25	2	AS SHOWN	C	230	5400	-	-	-	5.86	11.72	3.86	45.24	
	W1	25	32	300	B	300	3148	-	-	-	3.45	110.35	3.86	425.96	
	W2	25	32	300	B	300	3125	-	-	-	3.43	109.60	3.86	423.06	
a b c (E)	W3	25	4	AS SHOWN	B	4775	3350	-	-	-	8.13	32.50	3.86	125.45	45.81
	W4	25	32	300	B	450	1950	-	-	-	2.40	76.80	3.86	296.45	
	W5	16	52	200	C	300	5090	-	-	-	5.69	295.86	1.58	467.46	
b c a (F)	W6	12	188	AS SHOWN	G	200	300	120	-	-	0.62	116.56	0.89	103.74	
	1	20	6	AS SHOWN	A	500	-	-	-	-	0.50	3.00	2.47	7.41	
b a c (G)	2	20	8	AS SHOWN	C	500	200	-	-	-	1.20	9.60	2.47	23.71	
	3	20	4	AS SHOWN	C	220	500	-	-	-	0.94	3.76	2.47	9.29	
	4	20	6	AS SHOWN	A	1070	-	-	-	-	1.07	6.42	2.47	15.86	
c d b e=200 a f=VARIES (H)	5	20	14	AS SHOWN	C	500	200	-	-	-	1.20	16.80	2.47	41.50	
	6	20	4	AS SHOWN	C	200	1070	-	-	-	1.47	5.88	2.47	14.52	
	7	12	12	200	A	2400	-	-	-	-	2.40	28.80	0.89	25.63	
a c c b b (I) a	8	16	26	200	G	150	650	100	-	-	0.90	23.40	1.58	36.97	
	9	12	26	200	B	525	100	-	-	-	0.63	16.25	0.89	14.46	
	R1	20	24	AS SHOWN	A	1460	-	-	-	-	1.46	35.04	2.47	86.55	
	R2	10	30	200	I	120	120	80	-	-	0.64	19.20	0.62	11.85	
	R3	16	8	AS SHOWN	A	2420	-	-	-	-	2.42	19.36	1.58	30.59	
b a c (J)	R4	10	28	200	I	120	120	80	-	-	0.64	17.92	0.62	11.06	
												GRAND TOTAL GRADE 60 =		6,452.93	
													SAY =	6,453.00	
												GRAND TOTAL GRADE 40 =		22.90	
													SAY =	23.00	
													TOTAL =	6,476.00	46.00

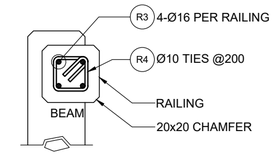


2 ELEVATION
SCALE 1:50



3 SECTION
SCALE 1:50

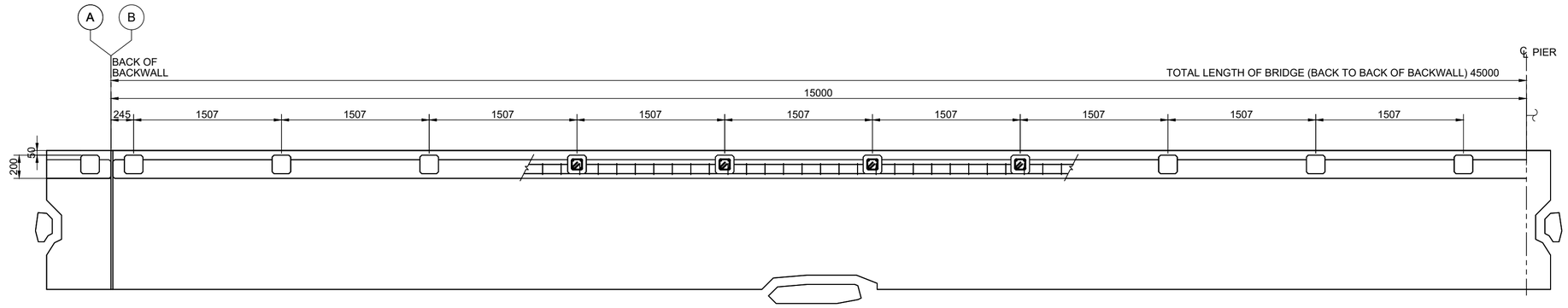




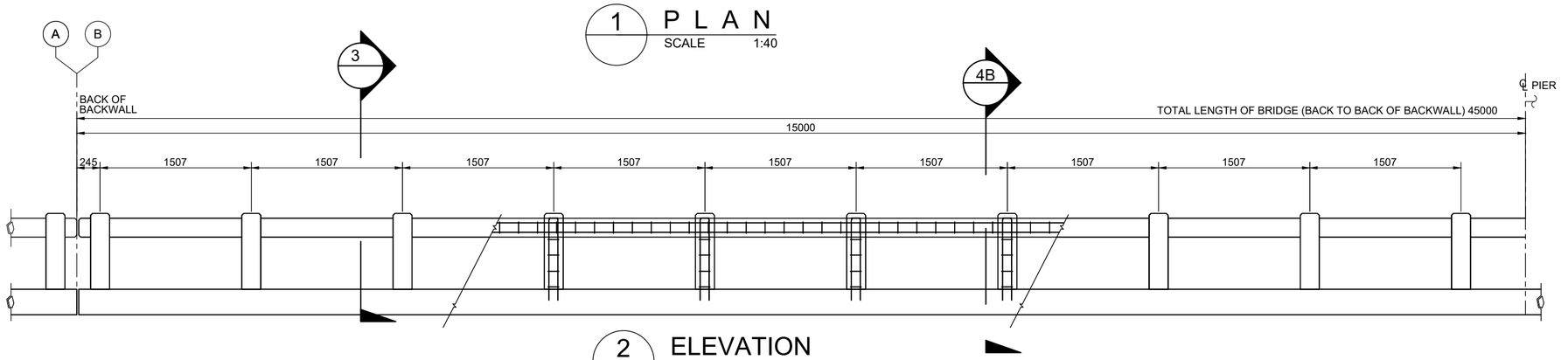
4 RAILING POST DETAILS
SCALE 1:30

1 WINGWALL DETAILS
SCALE AS SHOWN

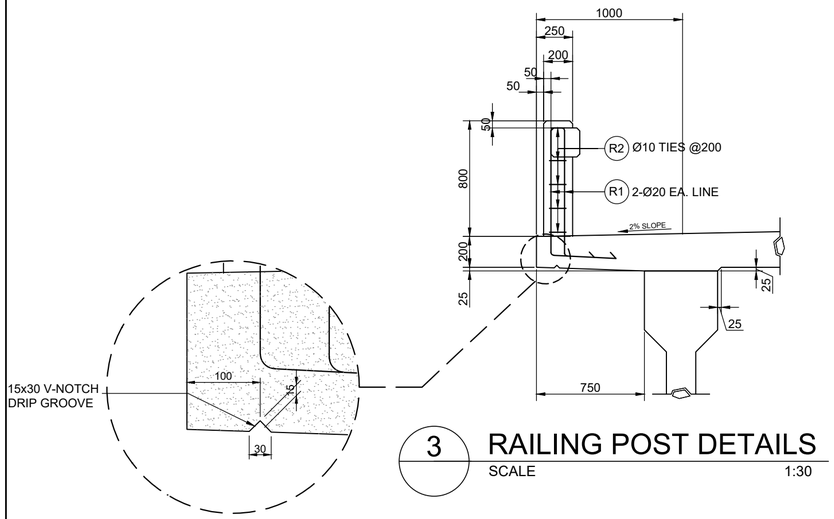
PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		WINGWALL DETAILS AND ABUTMENT A&B REINFORCEMENT SCHEDULE		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 15 20	987 1154



1 PLAN
SCALE 1:40

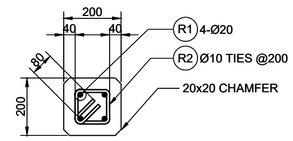


2 ELEVATION
SCALE 1:40

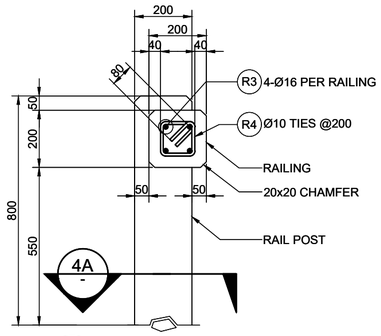


3 RAILING POST DETAILS
SCALE 1:30

3A DRIP GROOVE DETAILS
SCALE 1:5



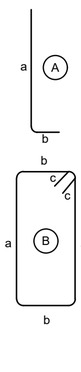
4A PLAN
SCALE 1:15



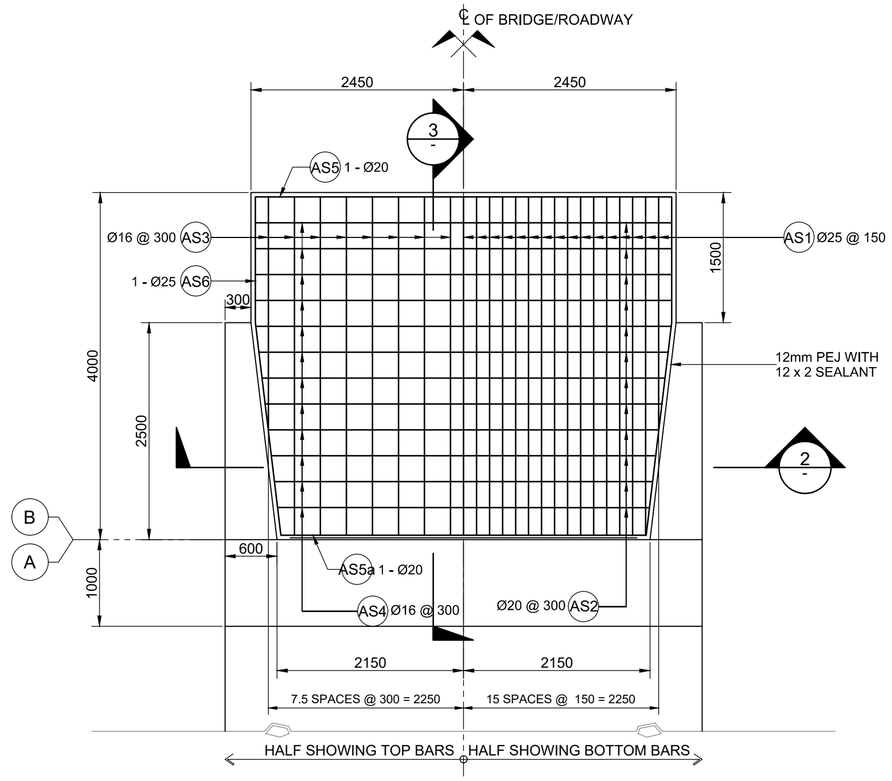
4B SECTION
SCALE 1:15

REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR RAILPOST

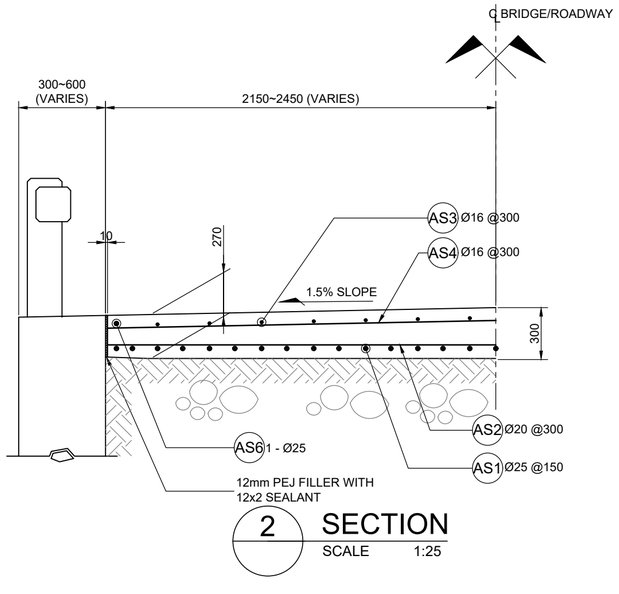
REINFORCEMENT

BENDING DIAGRAM	STRUCTURE COMPONENT	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
	RAILING POST	R1	20	240	AS SHOWN	A	1240	150	-	1.39	333.60	2.47	823.99	1.64
		R2	10	300	200	B	170	170	100	0.88	264.00	0.62	162.89	
	RAILING	R3	16	8	AS SHOWN	A	44160	-	-	44.16	353.28	1.58	558.18	3.53
		R4	10	443.6	200	C	120	120	80	0.64	283.90	0.62	175.17	
										GRAND TOTAL GRADE 60 =			1,382.17	
										SAY =			1,383.00	
										GRAND TOTAL GRADE 40=			338.06	
										SAY =			339.00	
										TOTAL =			1,722.00	6.00

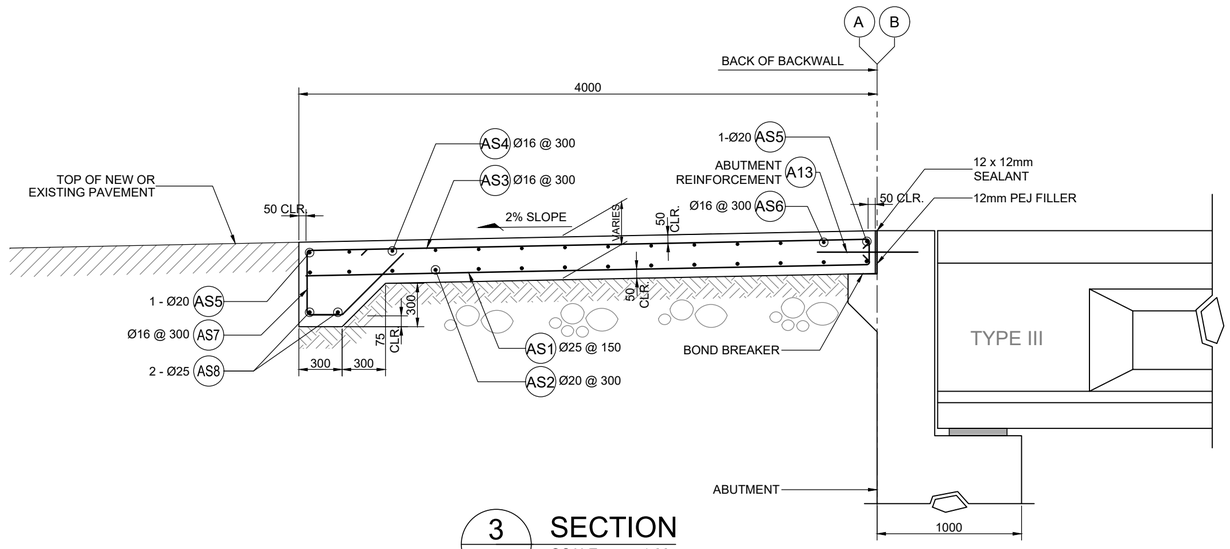
PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		RAILING PLAN, ELEVATION, RAIL POST DETAILS AND REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES	DATE:	DATE:	DATE:	DATE: DATE:	DATE:	BP11 16 20	988 1154



1 PLAN
SCALE 1:50



2 SECTION
SCALE 1:25



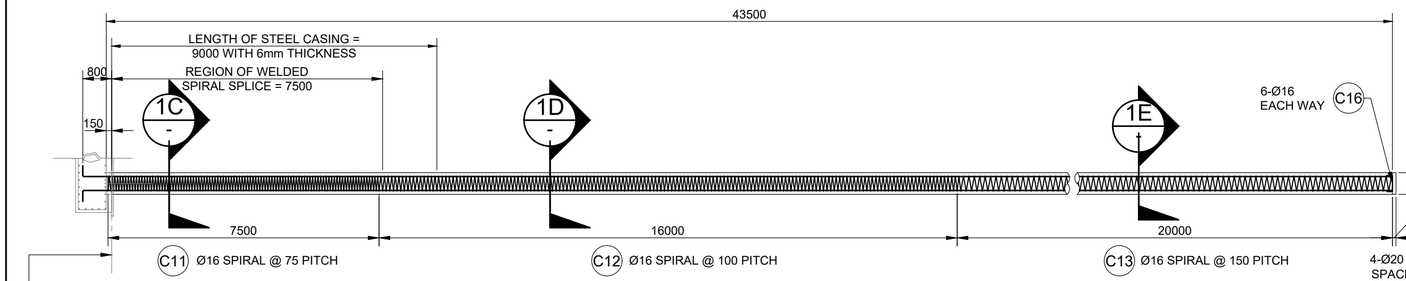
3 SECTION
SCALE 1:30

REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR APPROACH SLAB

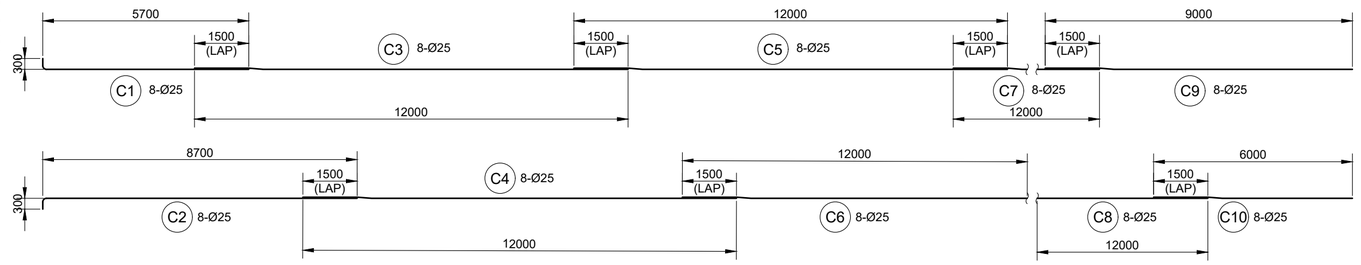
REINFORCEMENT

BENDING DIAGRAM	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	LOCATION Lc	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
a A	AS1	25	30	150	B	3900	150	-	-	4.05	121.50	3.86	468.99	4.80
	AS2	20	12	300	A	3900	-	-	-	3.90	46.80	2.47	115.60	
	AS3	16	13	300	A	3900	-	-	-	3.90	50.70	1.58	80.11	
	AS4	16	13	300	A	3900	-	-	-	3.90	50.70	1.58	80.11	
b B a	AS5	20	2	AS SHOWN	A	4800	-	-	-	4.80	9.60	2.47	23.71	
	AS6	25	2	AS SHOWN	C	1447	2471	-	-	3.92	7.84	3.86	30.25	
a C b	AS7	16	15	300	D	450	250	600	-	1.70	25.50	1.58	40.29	
400 a c b D	AS8	25	2	AS SHOWN	A	3900	-	-	-	3.90	7.80	3.86	30.11	
										GRAND TOTAL GRADE 60 =			869.15	
										SAY =			870.00	
										GRAND TOTAL GRADE 40 =			-	
										SAY =			-	
										TOTAL =			870.00	4.80

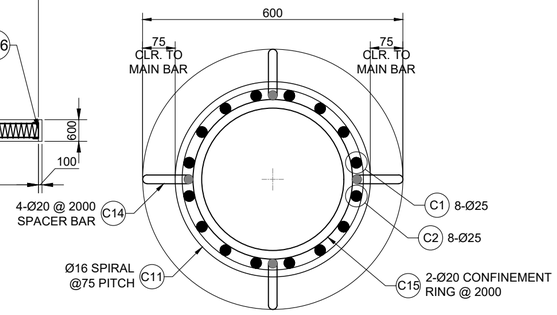
PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		APPROACH SLAB DETAIL		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 17 20	989 1154



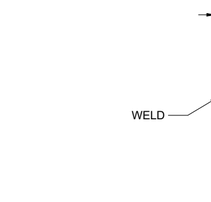
1A PILE ELEVATION
SCALE 1:150



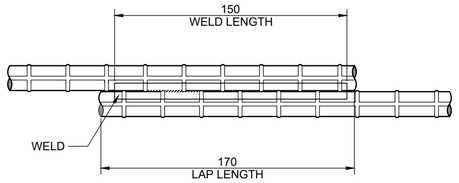
1B BAR SCHEMATIC DIAGRAM
SCALE 1:120



1C SECTION
SCALE 1:10

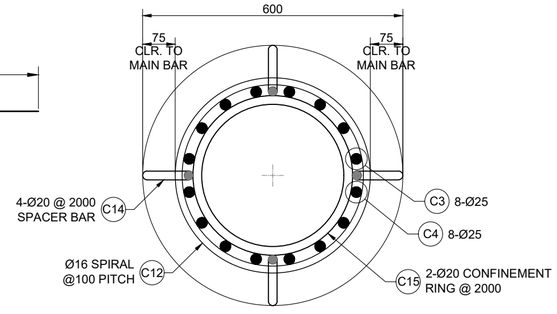


SECTION

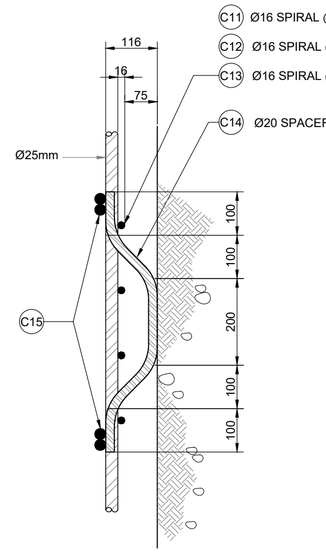


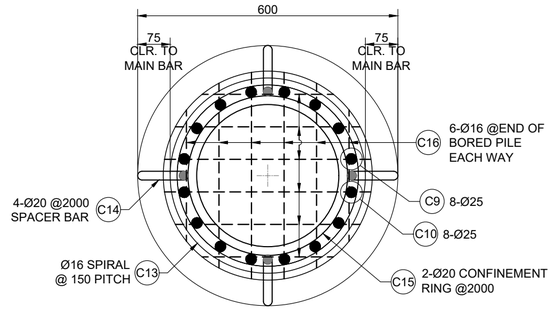
ELEVATION

1G SPIRAL SPLICE DETAIL
SCALE NTS



1D SECTION
SCALE 1:10





1E SECTION
SCALE 1:10

1F SPACER DETAIL
SCALE 1:10

REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR ONE (1) BORED PILE

REINFORCEMENT

BENDING DIAGRAM	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm) a	b	c	d	e	LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
A	C1	25	8	AS SHOWN	B	5700	300	-	-	-	6	48	3.86	185.28	12.30
B	C2	25	8	AS SHOWN	B	8700	300	-	-	-	9	72	3.86	277.92	
C	C3	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
D	C4	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
E	C5	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
F	C6	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
	C7	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
	C8	25	8	AS SHOWN	A	12000	-	-	-	-	12	96	3.86	370.56	
	C9	25	8	AS SHOWN	A	9000	-	-	-	-	9	72	3.86	277.92	
	C10	25	8	AS SHOWN	A	6000	-	-	-	-	6	48	3.86	185.28	
	C11	16	101	75	C	1416	-	-	-	-	1.416	144	1.58	227.52	
	C12	16	161	100	C	1418	-	-	-	-	1.418	229	1.58	361.82	
	C13	16	135	150	C	1422	-	-	-	-	1.422	192	1.58	303.36	
	C14	20	92	2000	E	100	140	200	140	100	0.68	62.56	2.47	154.52	
	C15	20	46	2000	D	2311	-	-	-	-	2.311	106.306	2.47	262.58	
	C16	16	12	AS SHOWN	F	450	150	-	-	-	0.75	9	1.58	14.22	
												GRAND TOTAL GRADE 60 =		4473.78	
												SAY =		4474	
												GRAND TOTAL GRADE 40 =		0.00	
												SAY =		0	
												TOTAL =		4474	13.00

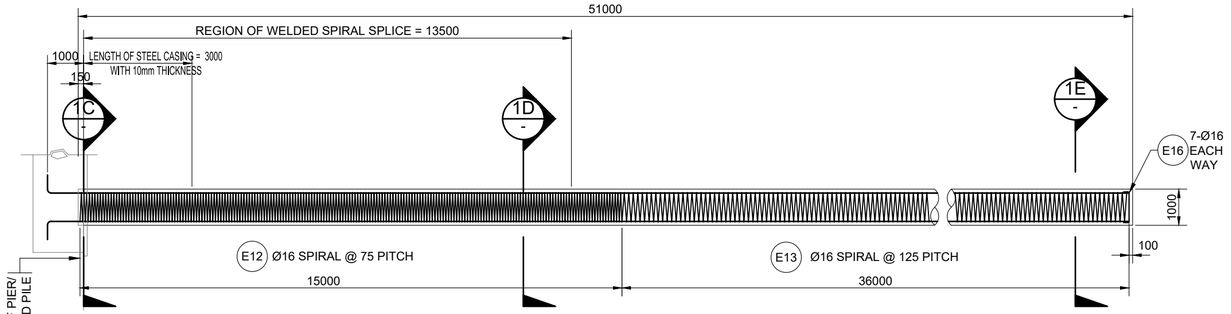
1 ABUTMENT A&B BORED PILE DETAILS
SCALE AS SHOWN

NOTE:
- CONCRETE SHALL BE CAST IN ONE POURING FOR THE FULL LENGTH OF EACH PILE BY THE USE OF TREMIE. BOTTOM OF DRILLED HOLE SHALL BE CLEANED AND FREE OF DEBRIS OR ANY LOOSE MATERIALS BEFORE POURING OF CONCRETE.
- PROVIDE 4-40mm (INTERNAL DIAMETER) PIPE FOR FULL LENGTH OF BORED PILE FOR ULTRASONIC TEST. THE PIPE SHALL HAVE SCREW CAPS AT BOTH ENDS.

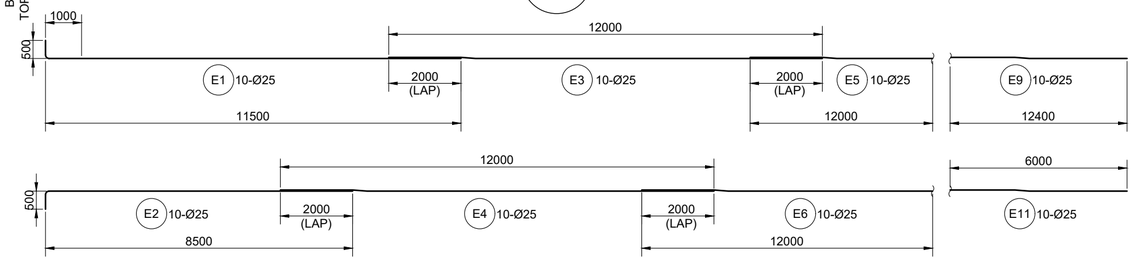
NOTES:
- CONFINEMENT RING AND SPACERS EVERY 2000mm
- CUT-OFF LENGTH : 1000mm
- LAP SPLICE FOR SPIRAL MIN. 700 mm
- SPACERS AND CONFINEMENT RINGS TO BE WELDED

MATERIALS:
- CONCRETE SHALL BE CLASS "D" WITH fc'=28MPa (4000 psi) AT 28 DAYS AND WITH 19mm MAXIMUM COARSE AGGREGATE
- REINFORCING STEEL SHALL CONFORM TO ASTM A615, GRADE 60 (fy=414MPa), DEFORMED TYPE

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:			APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		ABUTMENT A&B BORED PILE DETAILS		DATE:	DATE:	DATE:	DATE:	DATE:	DATE:	BP11 18A 20	990 1154

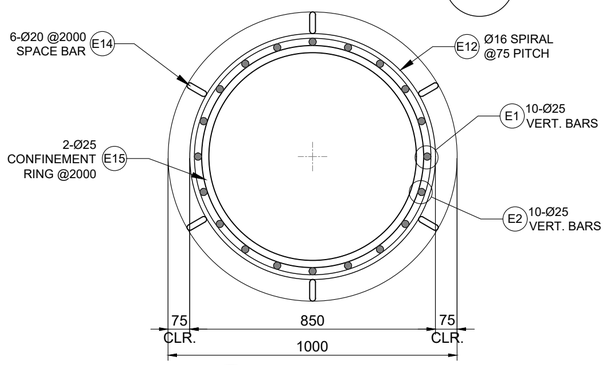


1A PILE ELEVATION
SCALE 1:120

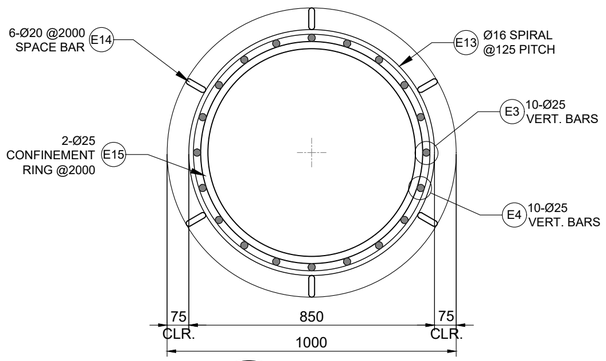


1B BAR SCHEMATIC DIAGRAM
SCALE 1:120

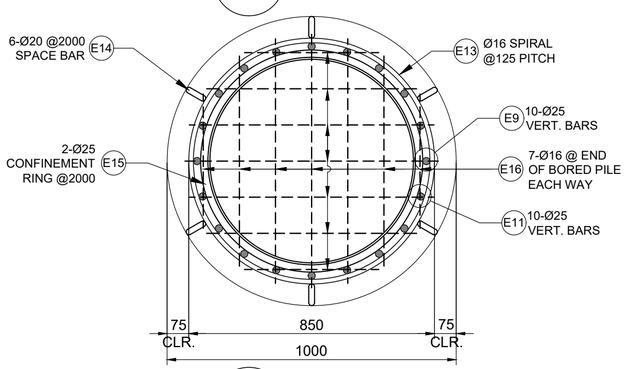
REINFORCEMENT SCHEDULE AND ESTIMATED QUANTITIES FOR ONE (1) BORED PILE															
REINFORCEMENT															
BENDING DIAGRAM	BAR MARK	BAR SIZE (mm)	QUANTITY	SPACING (mm)	BAR SHAPE	BAR DIMENSIONS (mm)					LENGTH PER BAR (m)	TOTAL LENGTH (m)	UNIT WEIGHT (kg/m)	TOTAL WEIGHT (kg)	CONCRETE VOLUME (m3)
						a	b	c	d	e					
A, B, C, D, E, F	E1	25	10	AS SHOWN	B	11500	500	-	-	-	12	120	3.86	463.20	40.06
	E2	25	10	AS SHOWN	B	8500	500	-	-	-	9	90	3.86	347.40	
	E3	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E4	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E5	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E6	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E7	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E8	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E9	25	10	AS SHOWN	A	12400	-	-	-	-	12.4	124	3.86	478.64	
	E10	25	10	AS SHOWN	A	12000	-	-	-	-	12	120	3.86	463.20	
	E11	25	10	AS SHOWN	A	6000	-	-	-	-	6	60	3.86	231.60	
	E12	16	201	75	C	1416	-	-	-	-	1.416	285	1.58	450.30	
	E13	16	289	125	C	1420	-	-	-	-	1.42	411	1.58	649.38	
	E14	20	108	2000	E	100	140	200	140	100	0.68	73.44	2.47	181.40	
	E15	25	54	2000	D	2311	-	-	-	-	2.311	124.794	3.86	481.70	
	E16	16	14	AS SHOWN	F	450	150	-	-	-	0.75	10.5	1.58	16.59	
												GRAND TOTAL GRADE 60 =		6542.61	
												SAY =		6543	
												GRAND TOTAL GRADE 40 =		0.00	
												SAY =		0	
												TOTAL =		6543	41.00



1C SECTION
SCALE 1:15



1D SECTION
SCALE 1:15



1E SECTION
SCALE 1:15

NOTE:

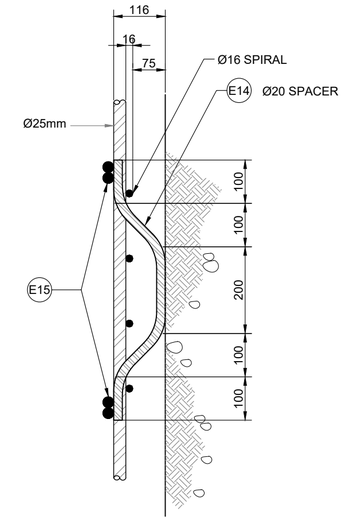
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- PROVIDE 4-40mm (INTERNAL DIAMETER) PIPE FOR FULL LENGTH OF BORED PILE FOR ULTRASONIC TEST. THE PIPE SHALL HAVE SCREW CAPS AT BOTH ENDS.

NOTES:

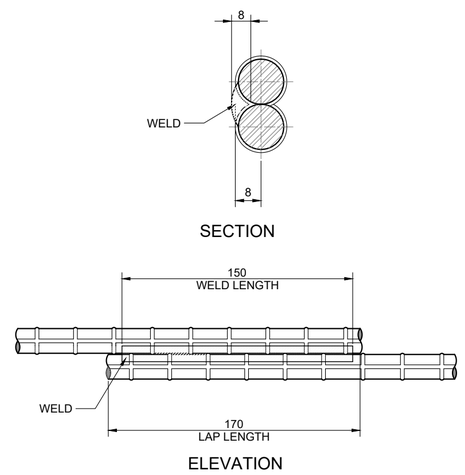
- CONFINEMENT RING AND SPACERS EVERY 2000mm
- CUT-OFF LENGTH : 1200mm
- LAP SPLICE FOR SPIRAL MIN. 700 mm
- SPACERS AND CONFINEMENT RINGS TO BE WELDED

MATERIALS:

- CONCRETE SHALL BE CLASS "D" WITH fc'=28MPa (4000 psi) AT 28 DAYS AND WITH 19mm MAXIMUM COARSE AGGREGATE
- REINFORCING STEEL SHALL CONFORM TO ASTM A615, GRADE 60 (fy=414MPa), DEFORMED TYPE



1G SPACER DETAIL
SCALE 1:10



1H SPIRAL SPLICE DETAIL
SCALE NTS

1 PIER 1&2 BORED PILE DETAILS
SCALE AS SHOWN

PREPARED:	PROJECT NAME AND LOCATION:	SHEET CONTENTS:		CHECKED:	REVIEWED:	RECOMMENDED:		APPROVED:	SET NO.	SHEET NO.
SUBMITTED:		PIER 1&2 BORED PILE DETAILS		DATE:	DATE:	DATE:	DATE:	DATE:	BP11 18 B 20	990 1154

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER (14.40m TYPE III)

Designed by/Date: KH - RCDG 05/04/26

Conversion Factor: $kN \equiv 1000 \cdot newton$ $Pa \equiv \frac{newton}{m^2}$ $kPa \equiv \frac{kN}{m^2}$ $MPa \equiv \frac{newton}{mm^2}$

DESIGN THEORY:

1. Use concrete stress limits from AASHTO LRFD 2012 Sec 5.9.3 and 4 at service limit states.
2. Use load resistance factor design to check ultimate moment and shear capacity at the factored level considering (HL-93) truck and lane loads, design tandem and lane loads, or permit loads, whichever is greater.
3. Gross properties are used as preliminary design.
4. Net/Transformed section properties will be calculated and used for the final design in the analysis.

1. MATERIAL DATA and SPECIFICATIONS

1.1 Bridge Geometry

Girder Span Length: $L := 14.40 \cdot m$

No. of Girder: $N_g := 3$

Slab Thickness: $t_s := 220 \cdot mm$

Roadway Width: $RW := 4 \cdot m$

Haunch Thickness: $t_{haunch} := 25 \cdot mm$

Sidedwalk Width: $SW := 0 \cdot m$

Girder Spacing: $S_g := 1.75 \cdot m$

No. of Diaphragm (interior diaphragm): $N_{diap} := 1$

Center of Girder to Edge of Slab: $L_c := 1.0 \cdot m$ (at critical curved area)

1.2 Prestressed Concrete

Concrete Compressive Strength @ 28 days: $f'_c := 41.4 \cdot MPa$

Concrete Compressive Strength @ 28 days (for cast-in-place deck): $f'_{c_slab} := 27.6 \cdot MPa$

Concrete Compressive Strength @ transfer: $f'_{ci} := 0.85 \cdot f'_c$

$f'_{ci} = 35.19 \cdot MPa$

Unit Weight of Concrete: $_{conc} := 24.00 \cdot \frac{kN}{m^3}$

Modulus of Rupture of normal wt. conc.: $f_r := 0.63 \cdot \sqrt{f'_c \cdot MPa}$

$f_r = 4.05 \cdot MPa$

Unit Weight of Wearing coarse: $_{fws} := 22 \cdot \frac{kN}{m^3} \cdot (80 \cdot mm)$

$_{fws} = 1.76 \cdot kPa$

Modulus of Elasticity of concrete @ transfer: $E_{ci} := 0.043 \cdot \left(\frac{_{conc} \cdot m^3}{kgf} \right)^{1.5} \cdot \sqrt{f'_{ci} \cdot MPa}$

$E_{ci} = 30883 \cdot MPa$

1.3 Prestressing Steel - *Stress Relieved Strands*

Ultimate Strength of prestressing steel: $f'_s := 1860 \cdot MPa$

From AASHTO 5.9.5.2.2

Yield Strength of prestressing steel: $f^*_y := 0.90 \cdot f'_s$

$f^*_y = 1674 \cdot MPa$

Friction Curvature Coefficient: $:= 0.20$

Friction Wobble Coefficient: $k := \frac{0.00066}{m}$

Modulus of Elasticity of prestressing steel: $E_p := 197000 \cdot MPa$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

1.4 Non-Prestressing Steel : *Mild Steel Reinforcement (Intermediate Grade Steel)*

Yield Strength of intermediate grade steel $f'_{sy} := 414 \cdot MPa$

Modulus of Elasticity of intermediate grade steel $E_s := 200000 \cdot MPa$

2. STRESS LIMITS FOR CONCRETE (AASHTO 5.9.4)

2.1 At Transfer Stage

Compression Fiber: $f'_{ci_allow} := 0.60 \cdot f'_{ci}$ $f'_{ci_allow} = 21.11 \cdot MPa$

Tension Fiber: $f'_{ti_allow} := -0.63 \cdot \sqrt{f'_{ci} \cdot MPa}$ $f'_{ti_allow} = -3.737 \cdot MPa$

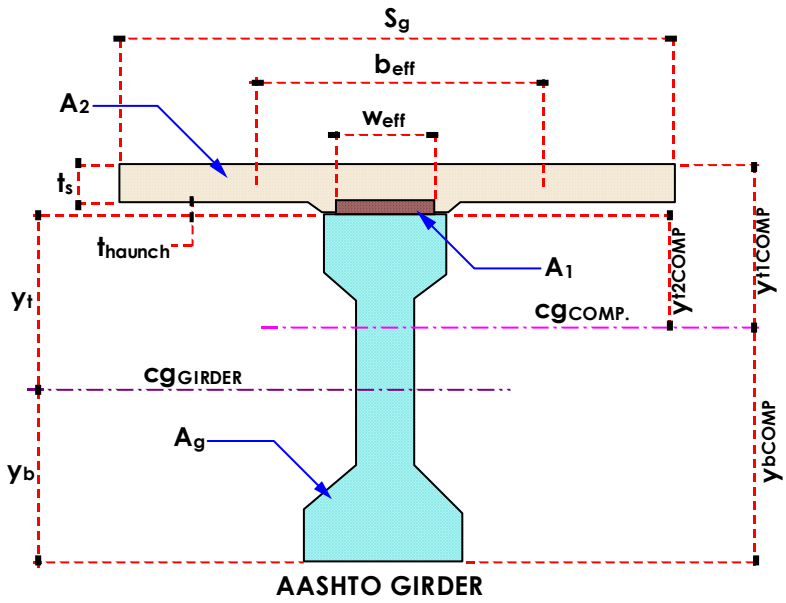
2.2 At Service Stage

Compression Fiber: $f'_{c_allow} := 0.45 \cdot f'_{c}$ $f'_{c_allow} = 18.63 \cdot MPa$

*** Tension Fiber:** $f'_{t_allow} := -0.50 \cdot \sqrt{f'_{c} \cdot MPa}$ $f'_{t_allow} = -3.217 \cdot MPa$

** If site is located at severe corrosive condition, $f'_{t_allow} = -0.25(f'_c)^{0.5}$*

3. CROSS SECTIONAL PROPERTIES OF AASHTO I-GIRDER



Type := 3 **<<< Input AASHTO Girder Type**

Note: Types 7 & 8 are actually Types IV-A & IV-B respectively.

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Table of AASHTO Girder Properties:

$j := 1..8$

j =	D_{pc_j} :=	B_{bf_j} :=	B_{tf_j} :=	T_{w_j} :=	Z_{pc_j} :=	A_{pc_j} :=	$I_{y_pc_j}$:=	$I_{z_pc_j}$:=	$I_{x_pc_j}$:=
1.00	0.711	0.406	0.305	0.152	0.3198	0.1778	0.00945	0.001384	0.001303
2.00	0.914	0.457	0.305	0.152	0.4020	0.2374	0.02115	0.002193	0.002407
3.00	1.143	0.559	0.406	0.178	0.5146	0.3613	0.05216	0.005055	0.005232
4.00	1.372	0.660	0.508	0.203	0.6283	0.5087	0.10845	0.010039	0.009901
5.00	1.600	0.711	1.067	0.203	0.8096	0.6532	0.21493	0.02384	0.011063
6.00	1.829	0.711	1.067	0.203	0.9223	0.6997	0.30303	0.024003	0.011702
7.00	1.473	0.660	0.508	0.203	0.7012	0.5604	0.13795	0.01115	0.01459
8.00	1.575	0.660	0.508	0.203	0.7707	0.6119	0.17007	0.012257	0.022886

$D_{pc} := D_{pc} \cdot m \quad B_{bf} := B_{bf} \cdot m \quad B_{tf} := B_{tf} \cdot m \quad T_w := T_w \cdot m \quad Z_{pc} := Z_{pc} \cdot m$

$A_{pc} := A_{pc} \cdot m^2 \quad I_{y_pc} := I_{y_pc} \cdot m^4 \quad I_{z_pc} := I_{z_pc} \cdot m^4 \quad I_{x_pc} := I_{x_pc} \cdot m^4$

Girder Height: $h_g := D_{pc_{Type}}$ → $h_g = 1.143\,m$

Moment of Inertia: $I_g := I_{y_pc_{Type}}$ → $I_g = 0.0522\,m^4$

Girder Area: $A_g := A_{pc_{Type}}$ → $A_g = 0.3613\,m^2$

Dist. of C.G. from bottom of girder : $y_b := Z_{pc_{Type}}$ → $y_b = 0.5146\,m$

Dist. of C.G. from top of girder : $y_t := h_g - y_b$ → $y_t = 0.6284\,m$

Section Modulus at top of girder : $S_t := \frac{I_g}{y_t}$ → $S_t = 0.083 \cdot m^3$

Section Modulus at bottom of girder : $S_b := \frac{I_g}{y_b}$ → $S_b = 0.1014 \cdot m^3$

Web Thickness: $T_w := T_{w_{Type}}$ → $T_w = 0.178\,m$

4. COMPOSITE CROSS SECTIONAL PROPERTIES OF AASHTO I-GIRDER

4.1 Effective Width of Slab

Width of slab should not be greater than the following:

a) $S_1 := \frac{L}{8}$ → $S_1 = 1800 \cdot mm$

b) $S_2 := 6 \cdot t_s + \frac{B_{tf_{Type}}}{4}$ → $S_2 = 1421.5 \cdot mm$

c) $S_3 := \frac{S_g}{2}$ → $S_3 = 875 \cdot mm$

$S := L_c + \min((S_1 \;\; S_2 \;\; S_3))$ → $S = 1875 \cdot mm$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Modular ratio:

$$E_{c_slab} := 0.043 \cdot \left(\frac{conc \cdot m^3}{kgf} \right)^{1.5} \cdot \sqrt{f'_{c_slab} \cdot MPa} \qquad E_{c_slab} = 27350 \cdot MPa$$

$$E_{c_girder} := 0.043 \cdot \left(\frac{conc \cdot m^3}{kgf} \right)^{1.5} \cdot \sqrt{f'_{c} \cdot MPa} \qquad E_{c_girder} = 33497 \cdot MPa$$

$$n := \frac{E_{c_slab}}{E_{c_girder}} \qquad n = 0.8165$$

Effective width of Flange, b_{eff} $\quad b_{eff} := n \cdot S \qquad b_{eff} = 1.53\,m$

Effective width of haunch, w_{eff} $\quad w_{eff} := n \cdot B_{tf_{Type}} \qquad w_{eff} = 0.33\,m$

Determination c.g. with respect to bottom:

Composite haunch area $\quad A_1 := w_{eff} \cdot t_{haunch} \qquad A_1 = 0.008\,m^2$

Composite deck slab area $\quad A_2 := b_{eff} \cdot t_s \qquad A_2 = 0.34\,m^2$

Total Composite Area $\quad A_{comp} := A_g + A_1 + A_2 \qquad A_{comp} = 0.71\,m^2$

$$y_{bcomp} := \frac{A_g \cdot y_b + A_1 \cdot \left(h_g + \frac{1}{2} \cdot t_{haunch} \right) + A_2 \cdot \left(h_g + t_{haunch} + \frac{1}{2} \cdot t_s \right)}{A_{comp}} \qquad y_{bcomp} = 886.1 \cdot mm$$

$$y_{t1comp} := \left(h_g + t_{haunch} + t_s \right) - y_{bcomp} \qquad y_{t1comp} = 501.9 \cdot mm$$

$$y_{t2comp} := h_g - y_{bcomp} \qquad y_{t2comp} = 256.9 \cdot mm$$

Moment of Inertia for the composite section:

$$I_{comp} := I_g + A_g \cdot \left(y_{bcomp} - y_b \right)^2 + \left[\frac{w_{eff} \cdot t_{haunch}^3}{12} + A_1 \cdot \left(y_{t1comp} - t_s - \frac{t_{haunch}}{2} \right)^2 \right] + \left[\frac{b_{eff} \cdot t_s^3}{12} + A_2 \cdot \left(y_{t1comp} - \frac{t_s}{2} \right)^2 \right] \qquad I_{comp} = 0.1557\,m^4$$

Section Modulus of Composite section:

$$S_{bcomp} := \frac{I_{comp}}{y_{bcomp}} \qquad S_{bcomp} = 0.1757 \cdot m^3$$

$$S_{t2comp} := \frac{I_{comp}}{y_{t2comp}} \qquad S_{t2comp} = 0.6061 \cdot m^3$$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

5. LOAD ANALYSIS PER GIRDER (DEAD LOAD)

Dead loads due to girder, slab, and intermediate diaphragm (DC):

$w_g := A_g \cdot \text{conc}$ $\quad w_g = 8.67 \cdot \frac{kN}{m}$

$w_{slab} := \left(t_s \cdot S + t_{haunch} \cdot B_{tf_{Type}}\right) \cdot \text{conc}$ $\quad w_{slab} = 10.14 \cdot \frac{kN}{m}$

$P_{diaph} := \left(\frac{N_{diap} \cdot 0.25m \cdot 1.60m \cdot 1.0m}{2}\right) \cdot \text{conc}$ $\quad P_{diaph} = 4.8 \cdot kN$

Dead loads due to superimposed loads:

$N_{post} := 10$

$w_{railings} := 0.20m \cdot 0.20m \cdot \text{conc} \cdot 1$ $\quad w_{railings} = 0.96 \cdot \frac{kN}{m}$

$w_{rail_post} := \frac{0.8m \cdot 0.20m \cdot 0.20m \cdot \text{conc} \cdot N_{post}}{L}$ $\quad w_{rail_post} = 0.53 \cdot \frac{kN}{m}$

$w_{sidewalk} := SW \cdot 0.26m \cdot \text{conc}$ $\quad w_{sidewalk} = 0 \cdot \frac{kN}{m}$

$ws_{trib_width.ext} := S - 0.25m$ $\quad ws_{trib_width.ext} = 1.625\,m$

$w_{fws} := ws_{trib_width.ext} \cdot \text{fws}$ $\quad w_{fws} = 2.86 \cdot \frac{kN}{m}$

$w_{utilities1} := \left(w_{railings} + w_{rail_post} + w_{sidewalk} + w_{fws}\right) \cdot 0.10$ $\quad w_{utilities1} = 0.44 \cdot \frac{kN}{m}$

$w_{utilities2} := 1.00 \frac{kN}{m}$ *(est. minimum; 1 kN/m)*

$w_{utilities} := \max\left(w_{utilities1}, w_{utilities2}\right)$ $\quad w_{utilities} = 1.00 \cdot \frac{kN}{m}$

$w_{sup} := w_{railings} + w_{rail_post} + w_{sidewalk} + w_{fws} + w_{utilities}$ $\quad w_{sup} = 5.35 \cdot \frac{kN}{m}$

6. DESIGN MOMENT FORCES

6.1 Moment at Midspan Due to Dead Loads

$L_b := 13.8m$ Bearing-to-bearing distance

(From StaadPro Analysis of Continuous Girders)

Girder

$M_g := \frac{w_g \cdot L_b^2}{8}$

$M_g = 206.42 \cdot kN \cdot m$

Slab

$M_{slab} := \frac{w_{slab} \cdot L_b^2}{8}$

$M_{slab} = 241.47 \cdot kN \cdot m$

Superimposed loads

$M_{sup} := 23.843 \cdot kN \cdot m + 61.769 \cdot kN \cdot m$

$M_{sup} = 85.61 \cdot kN \cdot m$

$$M_{diaph} := \begin{cases} \frac{P_{diaph} \cdot L_b}{4} & \text{if } N_{diap} = 1 \\ \frac{P_{diaph} \cdot L_b}{3} & \text{if } N_{diap} = 2 \\ \frac{P_{diaph} \cdot L_b}{2} & \text{if } N_{diap} = 3 \end{cases}$$

Intermediate diaphragm

$M_{diaph} = 16.56 \cdot kN \cdot m$

Note: If there is no interior diaphragm, M_{diaph} = 0.00 kN-m.

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

6.2 Moment at Midspan Due to Live Loads

6.2.1 Distribution Factor of Live Loads Per Lane for Moment in Exterior Beams

Dynamic Load Allowance Factor $IM := 1.33$

Multiple Presence factor (One Lane) $MF_1 := 1.2$

Multiple Presence factor (Two Lane) $MF_2 := 1.0$

Table 11.3.2.2-6 Dist. of Live Loads per Lane for Moment in Exterior Longitudinal Beams, DGCS (Vol.5)

*Use lever rule, since **Ng = 3**:*

(First wheel is at 600m away from railings, wheels spacing is 1.8m)

$$DF_1 := \left(0.5 + \frac{50mm}{S_g}\cdot 0.5\right)\cdot MF_1 \qquad g_1 := DF_1 = 0.617$$

$$DF_2 := \left(0.5 + \frac{50mm}{S_g}\cdot 0.5\right)\cdot MF_2 \qquad g_2 := DF_2 = 0.514$$

6.2.2 Truck Loading

Note: Live Load Moments are from StaadPro Analysis for Continuous Span.

Design Truck $M_{Tr} := 620\cdot kN\cdot m$

Design Tandem $M_T := 578\cdot kN\cdot m$

Unfactored Design Truck/Tandem + IM

$$M_{DT} := \max(M_{Tr}, M_T)\cdot IM \qquad M_{DT} = 824.60\cdot kN\cdot m$$

6.2.3 Lane Loading

$$w_{lane} := 9.34\cdot\frac{kN}{m}$$

$$M_{lane} := 194\cdot kN\cdot m \qquad M_{lane} = 194.00\cdot kN\cdot m$$

6.2.4 Unfactored Moment (Due to Truck + Lane Load) x Distribution Factor DF

$$M_{LL} := (M_{DT} + M_{lane})\cdot g_1 \qquad \textit{(lever arm rule)}$$

$$M_{LL} = 628.62\cdot kN\cdot m$$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

6.2.5 Permit Truck Loading

Caltrans P-7

Note: Live Load Moments are from StaadPro Analysis for Continuous Span.

$M_{PT} := 765 \cdot kN \cdot m$

Unfactored Permit Load + IM

Dynamic Load Allowance Factor for Permit Load

$IM_P := 1.25$ Dynamic Load Allowance Factor for Permit Load

$M_{PTim} := M_{PT} \cdot IM_P$

$M_{PTim} = 956.25 \cdot kN \cdot m$

Final force applied to girder (AASHTO 4.6.2.2.5)

$Z := 1.0$ Factor *(1.0 if lever rule is used)*

$$M_P := M_{PTim} \cdot \left(\frac{g_1}{Z} \right) + M_{LL} \cdot \left(g_2 - \frac{g_1}{Z} \right) \qquad M_P = 525.48 \cdot kN \cdot m$$

6.2.6 Strength Limit States Check

Strength I Limit State (Design Truck):

$M_{uL} := 1.75 \cdot M_{LL}$

$M_{uL} = 1100.09 \cdot kN \cdot m$

Strength II Limit State (Permit Truck):

$M_{uP} := 1.35 \cdot M_P$

$M_{uP} = 709.40 \cdot kN \cdot m$

$$Strength_check := \begin{cases} \text{"Design Truck governs!"} & \text{if } M_{uL} \geq M_{uP} \\ \text{"Permit Truck governs!"} & \text{otherwise} \end{cases}$$

Strength_check = "Design Truck governs!"

6.2.7 Pedestrian / Bicycle Loading

$SW_C := 0.75 \cdot m$ *Assumed pedestrian/bicycle lane width (beside railings, both sides, 0.75mx2=1.5m)*

$w_{PL} := 4.30 \cdot kPa \cdot SW_C = 3.225 \cdot \frac{kN}{m}$ (Pedestrian/Bicycle LL = 90 psf)

(from STAAD analysis for Continuous)

$M_{PL} := 52 \cdot kN \cdot m$ $M_{PL} = 52.00 \cdot kN \cdot m$

6.2.8 Final Design Moment (Due to Design Truck + Lane Load + Pedestrian Load) x DF

$M_{LLf} := (M_{DT} + M_{lane} + M_{PL}) \cdot g_1$ *(lever arm rule)*

$M_{LLf} = 660.71 \cdot kN \cdot m$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

7. STRESSES DUE TO EXTERNAL LOADS AT MIDSPAN

Note: *1. Positive values denotes compression*
2. Negative values denotes tension

7.1 Due to Beam/Girder Acting Alone

	TOP FIBER	BOTTOM FIBER
Due to Girder:	$f_{tg} := \frac{M_g}{S_t}$	$f_{bg} := \frac{M_g}{S_b}\cdot(-1)$
	$f_{tg} = 2.49\cdot MPa$	$f_{bg} = -2.04\cdot MPa$
Due to Slab:	$f_{t_slab} := \frac{M_{slab}}{S_t}$	$f_{b_slab} := \frac{M_{slab}}{S_b}\cdot(-1)$
	$f_{t_slab} = 2.91\cdot MPa$	$f_{b_slab} = -2.38\cdot MPa$
Due to Diaphragm:	$f_{t_diaph} := \frac{M_{diaph}}{S_t}$	$f_{b_diaph} := \frac{M_{diaph}}{S_b}\cdot(-1)$
	$f_{t_diaph} = 0.2\cdot MPa$	$f_{b_diaph} = -0.16\cdot MPa$

7.2 Due to Beam/Girder Acting as Composite

	TOP FIBER	BOTTOM FIBER
Due to Superimposed Dead Loads :	$f_{t_sup} := \frac{M_{sup}}{S_{t2comp}}$	$f_{b_sup} := \frac{M_{sup}}{S_{bcomp}}\cdot(-1)$
	$f_{t_sup} = 0.14\cdot MPa$	$f_{b_sup} = -0.49\cdot MPa$
Due to Live Loads : (Service III on Tension)	$f_{tLL} := \frac{M_{LLf}}{S_{t2comp}}$	$f_{bLL} := \frac{0.80\cdot M_{LLf}}{S_{bcomp}}\cdot(-1)$
	$f_{tLL} = 1.09\cdot MPa$	$f_{bLL} = -3.01\cdot MPa$

7.3 Total Stresses

Total Stresses at Top Fiber:

$ft_{total} := f_{tg} + f_{t_slab} + f_{t_diaph} + f_{t_sup} + f_{tLL}$ $\quad ft_{total} = 6.83\cdot MPa$

Total Stresses at Bottom Fiber:

$fb_{total} := f_{bg} + f_{b_slab} + f_{b_diaph} + f_{b_sup} + f_{bLL}$ $\quad fb_{total} = -8.08\cdot MPa$

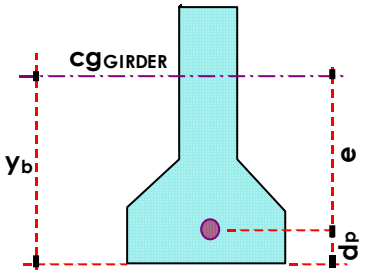
Note: *These stresses are used to simplify calculations. Girder will be tensioned so these stresses never occur.*

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

8. REQUIRED PRESTRESSING STEEL

Eccentricity of prestressing steel:



$d_p := 125 \cdot mm$ distance of prestressing steel from c.g. to the bottom of girder

$e_c := y_b - d_p$ $e_c = 389.60 \cdot mm$

$r_1 := \frac{I_g}{A_g}$ $r_1 = 0.144\,m^2$

$c_b := e_c \cdot \frac{y_b}{r_1}$ $c_b = 1.39$

From General Equation,

$f'_{t_allow} = f_{prestress} + f_{total}$

where: $f_{prestress} = P_f / A_g + P_f (e/Sb)$

$f_{total} = f_{b_total}$

Solving for Prestressing Force Required,

$P_f := \left(f_{t_allow} - fb_{total}\right) \cdot \frac{A_g}{c_b + 1}$ $P_f = 735.1 \cdot kN$

Maximum jacking stress (AASHTO 5.9.3): $f_{pi} := 0.75 \cdot f'_s = 1395.00 \cdot MPa$

Estimated total post tension loss: $Loss := 20\% \cdot f_{pi} = 279.00 \cdot MPa$

Working prestress required: $W_f := f_{pi} - Loss$ $W_f = 1116.00 \cdot MPa$

REQUIRED PRESTRESSING STEEL

Required prestressing steel: $A1^*_s := \frac{P_f}{W_f}$ $A1^*_s = 659 \cdot mm^2$

Note: *Area of 1- strand = 98.70 mm²* $A_{str} := 98.70 \cdot mm^2$

No. of prestressing steel: $N_o := \frac{A1^*_s}{A_{str}}$ $N_o = 6.67$ ***say,*** $N_o := 16$

Try 2-ducts - 8T12.7 strands per duct $N_{duct} := 2$

Total Area of Prestressing Steel: $A^*_s := N_o \cdot A_{str}$ $A^*_s = 1579 \cdot mm^2$

JACKING FORCE REQUIRED:

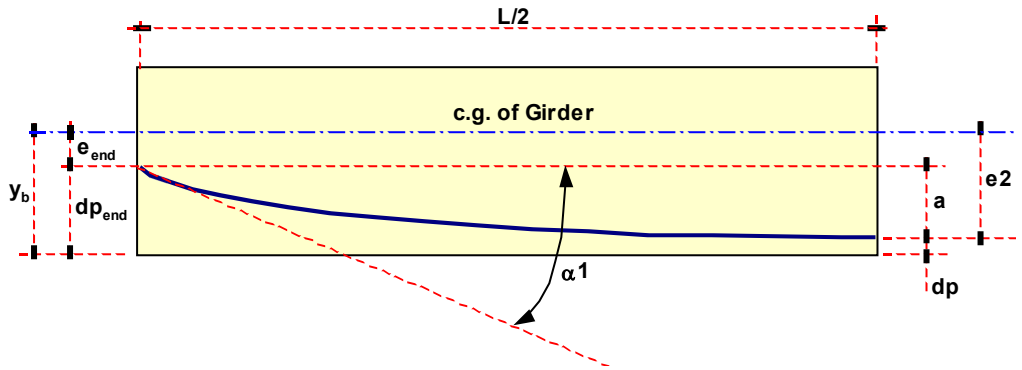
Note: AASHTO allows **0.90 f*$_y$** for short time prior to seating *(AASHTO 5.9.3)*
CALTRANS allows **0.75 f'$_s$**

Required Jacking Force: $P_j := 0.75 \cdot f'_s \cdot A^*_s$ $P_j = 2202.98 \cdot kN$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

9. PRESTRESS LOSSES

9.1 Friction Losses (AASHTO 5.9.5.2.2b)



Determine c.g. of Prestressing Steel from bottom of Girder (at end):
Note: *Y = Distance from c.g. of duct to the bottom of girder.*

$n := N_{duct}$ $i := 1 .. n$

$Duct_i :=$

1
2

$N_{strand_i} :=$

8
8

$Y_i :=$

350
679

$N_{strand_i} \cdot Y_i =$

2800.00
5432.00

$Y := Y \cdot mm$

$$dp_{end} := \frac{\sum_{i=1}^{n} \left(N_{strand_i} \cdot Y_i\right)}{N_o} \qquad dp_{end} = 514.50 \cdot mm$$

$a := dp_{end} - d_p$ $a = 389.50 \cdot mm$

Steel stress at jacking end: $T_o := \frac{P_j}{A^*_s}$ $T_o = 1395 \cdot MPa$

AT MIDSPAN:

Total angular change of prestressing steel profile from jacking to midspan

$$1 := L \cdot \frac{a}{\left(\frac{L}{2}\right)^2} \qquad 1 = 0.108$$

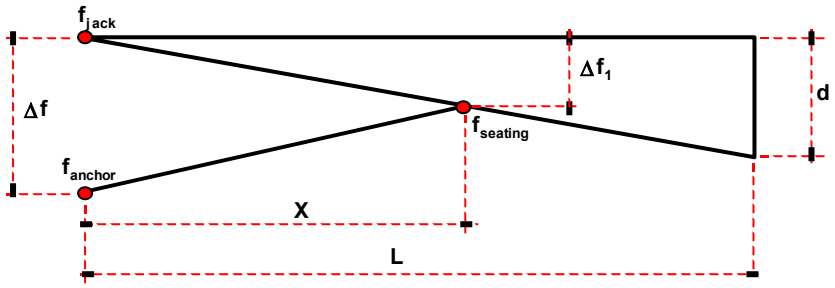
Friction Loss $FR := f_{pi} \cdot \left[1 - e^{-\left(k \cdot \frac{L}{2} + \cdot \ 1\right)}\right]$ $FR = 36.33 \cdot MPa$

Steel stress at midspan: $T_{x1} := f_{pi} - FR$ $T_{x1} = 1358.67 \cdot MPa$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

9.2 Anchorage Slip (Anchor Set Loss)

Note: Anchorage slip effect normally doesn't affect force at midspan when using Stress Relieved Strands, but must be checked and verified.



Jacking Stress at Jacking end before seating losses occur : $f_{jack} := T_o$ $f_{jack} = 1395.00 \cdot MPa$

Friction Loss in length L : $d := T_o - T_{x1}$ $d = 36.33 \cdot MPa$

Anchor set (normally 9.5mm) $L := 9.5 \cdot mm$ <<< *From CALTRANS*

Length influenced by anchor set: $X := \sqrt{\frac{E_p \cdot L \cdot L}{d}}$ $X = 27.23 \cdot m$

Change in stress due to anchor set: $f := \frac{2 \cdot d \cdot X}{L}$ $f = 137.44 \cdot MPa$

$f1 := \frac{d \cdot X}{L}$ $f1 = 68.72 \cdot MPa$

Anchorage Slip Loss at midspan

$$AN := \begin{cases} 0.00 \cdot MPa & \text{if } X \le \frac{L}{2} \\ \frac{f \cdot \left(X - \frac{L}{2}\right)}{X} & \text{otherwise} \end{cases}$$

$AN = 101.10 \cdot MPa$

9.3 Elastic Shortening Loss *(AASHTO 5.9.5.2.3b)*

Maximum moment due to girder dead load: $M_g = 206.4 \cdot kN \cdot m$

Concrete stress at the c.g. of prestressing steel due to prestressing force and girder immediately after release of anchorage(At this stage, initial stress in the tendons has been reduced by elastic shortening of concrete and tendon friction).

$P_{jcir} := 0.70 \cdot f'_s \cdot A^*_s$

$P_{jcir} = 2056.12 \cdot kN$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Number of identical prestressing tendons:

$N_i := 2$

Sum of concrete stress at center of gravity of tendons:

$$f_{cgp} := \left(\frac{P_{jcir}}{A_g} + P_{jcir}\cdot\frac{e_c^2}{I_g}\right) - M_g\cdot\frac{e_c}{I_g}$$

$f_{cgp} = 10.13\cdot MPa$

Elastic Shortening Loss:

$$ES := \frac{N_i - 1}{2\cdot N_i}\cdot\frac{E_p}{E_{ci}}\cdot f_{cgp}$$

$ES = 16.16\cdot MPa$

9.4 Time-Dependent Losses *(AASHTO 5.9.5.3)*

Note: Long-term prestress loss due to creep and shrinkage of concrete, and relaxation of steel

Average annual ambient relative humudity (%):

$H_u := 79$

Estimate relaxation loss for low relaxation strand:

$f_{pR} := 17\cdot MPa$

Correction factor for relative humidity of the ambient air:

$_h := 1.7 - 0.01\cdot H_u = 0.91$

Correction factor for specified concrete strength at time of prestress transfer to the concrete member:

$$_{st} := \frac{35}{7 + \frac{f'_{ci}}{MPa}} = 0.83$$

Initial prestressing steel stress immediately prior to transfer:

$f_{pi} = 1395.00\cdot MPa$

Total long-term prestress loss:

$$f_{pLT} := 10\cdot f_{pi}\cdot\frac{A^*_s}{A_g}\cdot {}_h\cdot {}_{st} + 83MPa\cdot {}_h\cdot {}_{st} + f_{pR}$$

$f_{pLT} = 125.69\cdot MPa$

9.5 TOTAL LOSSES:

9.5.1 At Transfer Stage $\quad Loss_t := FR + AN + ES$

$Loss_t = 153.59\cdot MPa$

9.5.2 At Service Stage $\quad Loss_s := FR + AN + ES + f_{pLT}$

$Loss_s = 279.28\cdot MPa$

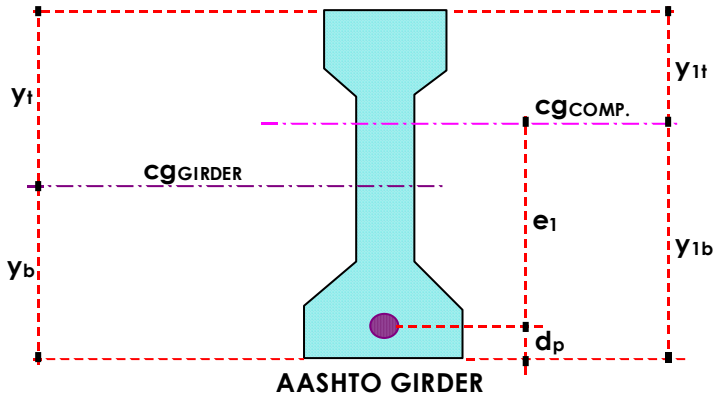
DESIGN OF PRESTRESSED EXTERIOR GIRDER

10. CHECK STRESSES OF GIRDER AT DIFFERENT STAGES

10.1 STAGE I - *Tension Girder (Net Section)*

Ungrouted Tendons:Girder Dead Load resist uplift from tensioning

Deduct holes for the ducts.



10.1.1 **STAGE I** : Net Section Properties

Duct Diameter: $D_{duct} := \left(2 + \frac{5}{8}\right)\cdot in$ $D_{duct} = 0.067\,m$

Duct Moment of Inertia: $I_{duct} := \frac{\cdot D_{duct}^{4}}{64}\cdot N_{duct}$ $I_{duct} = 0.00000194\,m^{4}$

Duct Area: $A_{duct} := \frac{\cdot D_{duct}^{2}}{4}\cdot N_{duct}$ $A_{duct} = 0.007\cdot m^{2}$

Note: (Using only 50% of mild steel)

Area of non-prestressing steel (bottom bars) $A_{stL} := 402\cdot mm^{2}$

Modulus of Elasticity of non-prestressing steel $E_{stL} := 200000\cdot MPa$

Depth of mild steel from girder bottom $d_{stL} := 60\cdot mm$

Modular ratio: $n_{stL} := \frac{E_{stL}}{E_{ci}}$ $n_{stL} = 6.476$

Equivalent concrete area of steel: $A_{equiv_stL} := n_{stL}\cdot A_{stL}$ $A_{equiv_stL} = 0.00\cdot m^{2}$

Total concrete net area : $A_{net1} := A_g - A_{duct} + A_{equiv_stL}$ $A_{net1} = 0.3569\cdot m^{2}$

Determine y_{1b} about bottom of girder:

$$y_{1b} := \frac{A_g\cdot y_b - A_{duct}\cdot d_p + A_{equiv_stL}\cdot d_{stL}}{A_{net1}}$$ $y_{1b} = 518.91\cdot mm$

$y_{1t} := h_g - y_{1b}$ $y_{1t} = 624.09\cdot mm$

Determine Moment of Inertia:

$I_{1bottom} := \left(I_g + A_g\cdot y_b^{2}\right) - \left(I_{duct} + A_{duct}\cdot d_p^{2}\right) + A_{equiv_stL}\cdot d_{stL}^{2}$ $I_{1bottom} = 0.1477\,m^{4}$

$I_{1cg} := I_{1bottom} - A_{net1}\cdot y_{1b}^{2}$ $I_{1cg} = 0.0516\,m^{4}$

Eccentricity e_1: $e_1 := y_{1b} - d_p$ $e_1 = 393.91\cdot mm$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

10.1.2 **STAGE I** : Stresses at Midspan due to External Loads

Note: *At this Stage, considered only the stress due to Girder weight.*

	TOP FIBER	**BOTTOM FIBER**
Due to Girder:	$f_{tg} := \frac{M_g \cdot y_{1t}}{I_{1cg}}$	$f_{bg} := \frac{M_g \cdot y_{1b}}{I_{1cg}} \cdot (-1)$
	$f_{tg} = 2.5 \cdot MPa$	$f_{bg} = -2.07 \cdot MPa$

10.1.3 **STAGE I** : Effective Prestress Force

Note: Deduct friction loses from jacking force. Some elastic shortening can be included when jacking multiple tendons separately.

$$P_1 := P_j - (Loss_t) \cdot A^*_s \qquad P_1 = 1960.43 \cdot kN$$

10.1.4 **STAGE I** : Final Stresses at Midspan - Girder Only

Stress @ top

$$f_{t1} := \left(\frac{P_1}{A_{net1}} - P_1 \cdot \frac{e_1 \cdot y_{1t}}{I_{1cg}} \right) + f_{tg} \qquad f_{t1} = -1.35 \cdot MPa$$

$$CHECK_f_{t1} := \begin{cases} \text{"OK"} & \text{if } (f_{t1} > 0) \cdot (f_{t1} \le f'_{ci_allow}) \\ \text{"OK"} & \text{if } (f_{t1} < 0) \cdot (f_{t1} \ge f'_{ti_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{t1} = \text{"OK"}$$

Stress @ bottom

$$f_{b1} := \left(\frac{P_1}{A_{net1}} + P_1 \cdot \frac{e_1 \cdot y_{1b}}{I_{1cg}} \right) + f_{bg} \qquad f_{b1} = 11.18 \cdot MPa$$

$$CHECK_f_{b1} := \begin{cases} \text{"OK"} & \text{if } (f_{b1} > 0) \cdot (f_{b1} \le f'_{ci_allow}) \\ \text{"OK"} & \text{if } (f_{b1} < 0) \cdot (f_{b1} \ge f'_{ti_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{b1} = \text{"OK"}$$

10.2 STAGE II - *Grout Tendons, Set Girder and pour Deckslab & Diaphragm*

10.2.1 **STAGE II** : Transformed Section Properties(Grouted Tendons)

Equivalent concrete area of prestressing steel, A_{eq}

$$n_1 := \frac{E_p}{E_{ci}} \qquad n_1 = 6.379$$

$$A_{eq} := A^*_s \cdot (n_1 - 1) \qquad A_{eq} = 0.01 \cdot m^2$$

Total transformed area:

$$A_{net2} := A_g + A_{eq} \qquad A_{net2} = 0.37 \cdot m^2$$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Determine y_{2b} about bottom of girder

$$y_{2b} := \frac{A_g \cdot y_b + A_{eq} \cdot d_p}{A_{net2}} \qquad y_{2b} = 506 \cdot mm$$

$$y_{2t} := h_g - y_{2b} \qquad y_{2t} = 637 \cdot mm$$

Eccetricity e_2:

$$e_2 := y_{2b} - d_p \qquad e_2 = 381 \cdot mm$$

Determine Moment of Inertia

$$I_{2bot} := I_g + A_g \cdot y_b^{\,2} + A_{eq} \cdot d_p^{\,2} \qquad I_{2bot} = 0.148\,m^4$$

$$I_{2cg} := I_{2bot} - A_{net2} \cdot y_{2b}^{\,2} \qquad I_{2cg} = 0.0534\,m^4$$

10.2.2 **STAGE II** : Stresses at Midspan due to External Loads

Note: *At this Stage, considered only the stress due to Girder weight, Deck slab, and Diaphragm.*

	TOP FIBER	**BOTTOM FIBER**
Due to Deck slab	$f_{t_slab} := \frac{M_{slab} \cdot y_{2t}}{I_{2cg}}$	$f_{b_slab} := \frac{M_{slab} \cdot y_{2b}}{I_{2cg}} \cdot (-1)$
	$f_{t_slab} = 2.88 \cdot MPa$	$f_{b_slab} = -2.29 \cdot MPa$
Due to Diaphragm	$f_{t_diaph} := \frac{M_{diaph} \cdot y_{2t}}{I_{2cg}}$	$f_{b_diaph} := \frac{M_{diaph} \cdot y_{2b}}{I_{2cg}} \cdot (-1)$
	$f_{t_diaph} = 0.2 \cdot MPa$	$f_{b_diaph} = -0.16 \cdot MPa$

10.2.3 **STAGE II** : Effective Prestress Force

Note: At this Stage, 40% losses is assumed, plus friction losses and elastic shortenig and anchorage slip. This value should be based on actual constuction schedule.

$$P_2 := P_j - \left(Loss_t + 40\% \cdot f_{pLT}\right) \cdot A^*_s \qquad P_2 = 1881.03 \cdot kN$$

10.2.4 **STAGE II** : Final Stresses at Midspan - Grouted Tendons, Poured Deckslab and Diaphragm

Stress @ top

$$f_{t2} := \left(\frac{P_2}{A_{net2}} - P_2 \cdot \frac{e_2 \cdot y_{2t}}{I_{2cg}}\right) + f_{tg} + f_{t_slab} + f_{t_diaph} \qquad f_{t2} = 2.12 \cdot MPa$$

$$CHECK_f_{t2} := \begin{cases} \text{"OK"} & \text{if } (f_{t2} > 0) \cdot (f_{t2} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{t2} < 0) \cdot (f_{t2} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{t2} = \text{"OK"}$$

Stress @ bottom

$$f_{b2} := \left(\frac{P_2}{A_{net2}} + P_2 \cdot \frac{e_2 \cdot y_{2b}}{I_{2cg}}\right) + f_{bg} + f_{b_slab} + f_{b_diaph} \qquad f_{b2} = 7.35 \cdot MPa$$

$$CHECK_f_{b2} := \begin{cases} \text{"OK"} & \text{if } (f_{b2} > 0) \cdot (f_{b2} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{b2} < 0) \cdot (f_{b2} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{b2} = \text{"OK"}$$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

10.3 STAGE III - *Superstructure Construction Complete (Superimposed Dead Load plus Live Loads)*

10.3.1 STAGE III : Composite Transformed Section Properties

Total transformed area:

$A_{net2} = 0.37 \cdot m^2$

Deck Slab properties:

$A_{slab} := t_s \cdot b_{eff}$ $\quad A_{slab} = 0.34 \cdot m^2$

$I_{slab} := \frac{b_{eff} \cdot t_s^3}{12}$ $\quad I_{slab} = 1.36 \times 10^{-3}\, m^4$

Haunch properties:

$A_{haunch} := w_{eff} \cdot t_{haunch}$ $\quad A_{haunch} = 0.008\, m^2$

$I_{haunch} := \frac{w_{eff} \cdot t_{haunch}^3}{12}$ $\quad I_{haunch} = 4.3164 \times 10^{-7} \cdot m^4$

Total Composite Transformed Area:

$A_{net3} := A_{net2} + A_{slab} + A_{haunch}$ $\quad A_{net3} = 0.71 \cdot m^2$

Determine y_{3b} about bottom of girder:

$$y_{3b} := \frac{\left[\left(A_{net2}\right) \cdot y_{2b} + A_{slab} \cdot \left(h_g + t_{haunch} + \frac{1}{2} \cdot t_s\right)\right] \ldots + A_{haunch} \cdot \left(h_g + \frac{t_{haunch}}{2}\right)}{A_{net3}} \qquad y_{3b} = 0.877\, m$$

$y_{3t} := h_g - y_{3b}$ $\quad y_{3t} = 0.27\, m$

Detemine Moment of Inertia:

$$I_{3bottom} := \left(I_{2cg} + A_{net2} \cdot y_{2b}^2\right) \ldots + \left[I_{slab} + A_{slab} \cdot \left(h_g + t_{haunch} + \frac{1}{2} \cdot t_s\right)^2\right] \ldots + \left[I_{haunch} + A_{haunch} \cdot \left(h_g + \frac{t_{haunch}}{2}\right)^2\right] \qquad I_{3bottom} = 0.7105\, m^4$$

$I_{3cg} := I_{3bottom} - A_{net3} \cdot y_{3b}^2$ $\quad I_{3cg} = 0.1606\, m^4$

10.3.2 STAGE III : Stresses at Midspan due to External Loads

Note: *At this Stage, the external loads are Girder, Deck slab, Diaphragm, Superimposed dead load, Live load and Pedestrian Load.*

	TOP FIBER	BOTTOM FIBER
Due to LL + Sup. DL (Service III on Tension)	$f_{tLL_sup} := \frac{\left(M_{LLf} + M_{sup}\right) \cdot y_{3t}}{I_{3cg}}$	$f_{bLL_sup} := \frac{\left(0.80 \cdot M_{LLf} + M_{sup}\right) \cdot y_{3b}}{I_{3cg}} \cdot (-1)$
	$f_{tLL_sup} = 1.24 \cdot MPa$	$f_{bLL_sup} = -3.35 \cdot MPa$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

10.3.3 **STAGE III** : Effective Prestress Force

Note: At this Stage (Final Stage), Final Prestress Force after all losses.

$P_3 := P_j - Loss_s \cdot A^*_s$ $\quad P_3 = 1761.94 \cdot kN$

10.3.4 **STAGE III** : Final Stresses at Midspan - Service Condition

Stress @ top

$$f_{t3} := \left(\frac{P_3}{A_{net2}} - P_3 \cdot \frac{e_2 \cdot y_{2t}}{I_{2cg}}\right) + f_{tg} + f_{t_slab} \ldots + f_{t_diaph} + f_{tLL_sup}$$

$f_{t3} = 3.57 \cdot MPa$

$f'_{c_allow} = 18.63 \cdot MPa$

$$CHECK_f_{t3} := \begin{cases} \text{"OK"} & \text{if } (f_{t3} > 0) \cdot (f_{t3} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{t3} < 0) \cdot (f_{t3} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$CHECK_f_{t3} = \text{"OK"}$

Stress @ bottom

$$f_{b3} := \left(\frac{P_3}{A_{net2}} + P_3 \cdot \frac{e_2 \cdot y_{2b}}{I_{2cg}}\right) + f_{bg} + f_{b_slab} \ldots + f_{b_diaph} + f_{bLL_sup}$$

$f_{b3} = 3.24 \cdot MPa$

$f'_{t_allow} = -3.22 \cdot MPa$

$$CHECK_f_{b3} := \begin{cases} \text{"OK"} & \text{if } (f_{b3} > 0) \cdot (f_{b3} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{b3} < 0) \cdot (f_{b3} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$CHECK_f_{b3} = \text{"OK"}$

11. CHECK FLEXURAL RESISTANCE VS FACTORED MOMENT

Resistance factor (AASHTO 5.5.4.2): $:= 0.95$

Total prestressing steel area $A^*_s = 1579.20 \cdot mm^2$

Ultimate Strength of prestressing steel $f'_s = 1860.00 \cdot MPa$

Yield Strength of prestressing steel $f^*_y = 1674.00 \cdot MPa$

Pretressing steel type factor (AASHTO 5.5.4.2):

$$k_f := 2 \cdot \left(1.04 - \frac{f^*_y}{f'_s}\right) = 0.28$$

Distance of c.g. of prestressing steel to top of slab:

$d_{pt} := h_g + t_{haunch} + t_s - d_p$

$d_{pt} = 1.26\,m$

Area of mild steel tension reinforcement:

$$A_{stL} := 4 \cdot \frac{\cdot (16mm)^2}{4} = 804.25 \cdot mm^2$$

Yield Strength of intermediate grade steel $f'_{sy} = 414.00 \cdot MPa$

Prestressed Concrete Compressive Strength $f'_c = 41.40 \cdot MPa$

$$_1 := \begin{cases} 0.85 & \text{if } f'_c \le 28 \cdot MPa \\ 0.85 - \frac{0.05}{7 \cdot MPa} \cdot (f'_c - 28MPa) & \end{cases}$$

(β_1 is not less than 0.65)

$_1 = 0.75$

Top slab (flange) thickness: $t_s = 220.00 \cdot mm$

Top slab (flange) effective width: $b_{eff} = 1.53\,m$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Distance from extreme compression fiber to neutral axis (assumed rectangular section behavior):
Note: compression reinforcement conservatively ignored

$$c_r := \frac{A^*_s \cdot f'_s + A_{stL} \cdot f'_{sy}}{0.85 \cdot f'_c \cdot \beta_1 \cdot b_{eff} + k_f \cdot A^*_s \cdot \frac{f'_s}{d_{pt}}}$$

$c_r = 79.21 \cdot mm$

$$CHECK := \begin{cases} \text{"N.A. is inside the flange, Analyze as Rectangular Section"} & \text{if } c_r \le t_s \\ \text{"N.A. is outside the flange, Analyze as T-Beam Section"} & \text{otherwise} \end{cases}$$

CHECK = "N.A. is inside the flange, Analyze as Rectangular Section"

Depth of equivalent stress block:

$a := c_r \cdot \beta_1 = 59.75 \cdot mm$

Distance from extreme compression fiber to centroid of mild tension reinforcement:

$d_s := h_g + t_{haunch} + t_s - d_{stL}$

$d_s = 1.33 \cdot m$

Stress in prestressing steel (AASHTO 5.7.3.1.1-1):

$$f_{ps} := f'_s \cdot \left(1 - k_f \cdot \frac{c_r}{d_{pt}}\right)$$

$f_{ps} = 1827.34 \cdot MPa$

Norminal resistance (rectangular section behavior):

$$M_{nr} := A^*_s \cdot f_{ps} \cdot \left(d_{pt} - \frac{a}{2}\right) + A_{stL} \cdot f'_{sy} \cdot \left(d_s - \frac{a}{2}\right)$$

$M_{nr} = 3990.70 \cdot kN \cdot m$

FLEXURAL RESISTANCE (AASHTO 5.7.3.2):

$M_r := \phi \cdot M_{nr}$

$M_r = 3791.16 \cdot kN \cdot m$

Actual Ultimate Factored Moment Applied: (STRENGTH I)

$$M_u := 1.05\left[1.25 \cdot \left(M_g + M_{slab} + M_{diaph}\right) + 1.50 \cdot M_{sup} + 1.75 \cdot M_{LLf}\right]$$

$M_u = 1958.49 \cdot kN \cdot m$

$Flexure_check := \text{if}\left(M_r \ge M_u, \text{"OK"}, \text{"FAIL"}\right)$

Flexure_check = "OK"

MINIMUM PRESTRESSED AND NON-PRESTRESSED TENSILE STEEL REQUIREMENT (AASHTO 5.7.3.3.2)

Section Modulus at bottom of girder, composite section:

$$S_{bc} := \frac{I_{3cg}}{y_{3b}}$$

$S_{bc} = 0.1831 \cdot m^3$

Section Modulus at bottom of girder, non-composite section:

$S_b = 0.1014 \cdot m^3$

Modulus of rupture of concrete:

$f_r = 4.05 \cdot MPa$

Flexural cracking variability factor:

$\gamma_1 := 1.6$

Prestress variability factor:

$\gamma_2 := 1.1$

Ratio of fy/fu factor (1.0 for prestress concrete structures):

$\gamma_3 := 1.0$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Compressive stress in concrete due to effective prestress forces only (after all loses):

$$f_{cpe} := \frac{P_3}{A_g} + \frac{P_3 \cdot e_c}{S_b} \qquad f_{cpe} = 11.65 \cdot MPa$$

Total unfactored DL moment acting on non-composite section:

$$M_{dnc} := M_g + M_{slab} + M_{diaph}$$

$$M_{dnc} = 464.45 \cdot kN \cdot m$$

Unless otherwise specified, at any section of a flexural component, the amount of prestressed and non-prestressed tensile reinforcement shall be adequate to develop a factored flexural resistance, Mr, at least equal to the lesser of:

$$M_{r1} := 1.33 \cdot M_u = 2604.79 \cdot kN \cdot m$$

or the cracking strength:

$$M_{cr} := {}_3\cdot\left[\left({}_1\cdot f_r + {}_2\cdot f_{cpe}\right)\cdot S_{bc} - M_{dnc}\cdot\left(\frac{S_{bc}}{S_b} - 1\right)\right] \qquad M_{cr} = 3158.98 \cdot kN \cdot m$$

$$Mmin := \text{if}\left(M_{r1} > M_{cr}, \text{"Mcr!"}, \text{"1.33Mu!"}\right) \qquad Mmin = \text{"1.33Mu!"}$$

Minimum flexural resistance:

$$M_{rmin} := \min\left(M_{r1}, M_{cr}\right) \qquad M_{rmin} = 2604.79 \cdot kN \cdot m$$

$$Min_check := \text{if}\left(M_r \geq M_{rmin}, \text{"OK"}, \text{"FAIL"}\right) \qquad Min_check = \text{"OK"}$$

12. DESIGN SHEAR FORCES AT SUPPORT

12.1 Shear Due to Dead Loads

Due to Girder:

$$V_g := w_g \cdot \frac{L}{2} \qquad V_g = 62.43 \cdot kN$$

Due to Slab:

$$V_{slab} := w_{slab} \cdot \frac{L}{2} \qquad V_{slab} = 73.03 \cdot kN$$

Due to Int. Diaphragm:

$$V_{diaph} := \frac{P_{diaph}}{2} \qquad V_{diaph} = 2.40 \cdot kN$$

Due to End Diaphragm:

$$V_{ediaph} := \frac{0.30m \cdot 1.25m \cdot 1.60m \cdot {}_{conc}}{2} \qquad V_{ediaph} = 7.20 \cdot kN$$

Due to Sidewalk and Railings (Superimposed DC):

$$w_{siderail} := w_{railings} + w_{rail_post} + w_{sidewalk} \qquad w_{siderail} = 1.49 \cdot \frac{kN}{m}$$

(From StaadPro Analysis of Continuous Girders)

$$V_{siderail} := 13 \cdot kN \qquad V_{siderail} = 13.00 \cdot kN$$

Due to wearing surface and utilities (DW):

$$w_{fwsg} := w_{fws}$$

$$w_{DW} := w_{fwsg} + w_{utilities} \qquad w_{DW} = 3.86 \cdot \frac{kN}{m}$$

$$V_{DW} := 33.5 \cdot kN \qquad V_{DW} = 33.50 \cdot kN$$

Total Shear due to Dead Loads:

$$V_{dl} := V_g + V_{slab} + V_{diaph} + V_{ediaph} + V_{siderail} + V_{DW} \qquad V_{dl} = 191.57 \cdot kN$$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

12.2 Shear Due to Live Loads

12.2.1 Distribution Factor of Live Loads Per Lane for Shear in Exterior Beams

Table 11.3.2.2-10 Dist. of Live Loads per Lane for Shear in Exterior Longitudinal Beams, DGCS (Vol.5)

*Use lever rule, since **Ng = 3**:*

(First wheel is at 600m away from railings, wheels spacing is 1.8m)

$$DF_{S1} := \left(0.5 + \frac{50mm}{S_g}\cdot 0.5\right)\cdot MF_1 \qquad DF_{S2} := \left(0.5 + \frac{50mm}{S_g}\cdot 0.5\right)\cdot MF_2$$

$g_{s1} := DF_{S1} = 0.617$ $g_{s2} := DF_{S2} = 0.514$

12.2.2 Truck Loading

Note: Live Load Shears are from StaadPro Analysis for Continuous Span.

Design Truck *Design Tandem*

$V_{Tr} := 276.5\cdot kN$ $V_T := 211\cdot kN$

Unfactored Design Truck/Tandem + IM

$V_{DT} := \max(V_{Tr}, V_T)\cdot IM$ $V_{DT} = 367.75\cdot kN$

12.2.3 Lane Loading

Note: Live Load Shears are from StaadPro Analysis of Continuous Girders.

$V_{lane} := 83\cdot kN$ $V_{lane} = 83.00\cdot kN$

12.2.4 Unfactored Shear (Due to Design Truck + Lane Load) x DF

$V_{LL} := (V_{DT} + V_{lane})\cdot g_{s1}$ *(lever arm rule)*

$V_{LL} = 278.17\cdot kN$

12.2.5 Permit Truck Loading

Note: Live Load Shears are from StaadPro Analysis of Continuous Girders.

Caltrans P-7

$V_{PT} := 401\cdot kN$

Unfactored Permit Load + IM

$V_{PTim} := V_{PT}\cdot IM_P$ $V_{PTim} = 501.25\cdot kN$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Final force applied to girder (AASHTO 4.6.2.2.5)

$Z = 1.00$ Factor *(1.0 if lever rule is used)*

$$V_P := V_{PTim}\cdot\left(\frac{g_{s1}}{Z}\right) + V_{LL}\cdot\left(g_{s2} - \frac{g_{s1}}{Z}\right)$$

$V_P = 280.73\cdot kN$

12.2.6 Strength Limit States Check

Strength I Limit State (Design Truck):

$V_{uL} := 1.75\cdot V_{LL}$

$V_{uL} = 486.80\cdot kN$

Strength II Limit State (Permit Truck):

$V_{uP} := 1.35\cdot V_P$

$V_{uP} = 378.99\cdot kN$

$$Strength_check_v := \begin{cases} \text{"Design Truck governs!"} & \text{if } V_{uL} \geq V_{uP} \\ \text{"Permit Truck governs!"} & \text{otherwise} \end{cases}$$

$Strength_check_v = \text{"Design Truck governs!"}$

12.2.7 Pedestrian and Bicycle Loading

Note: Live Load Shears are from StaadPro Analysis of Continuous Girders.

$V_{PL} := 28\cdot kN$ $V_{PL} = 28.00\cdot kN$

12.2.8 Final Design Shear (Due to Design Truck + Lane Load + Pedestrian Load) x DF

$V_{LLf} := (V_{DT} + V_{lane} + V_{PL})\cdot g_{s1}$

$V_{LLf} = 295.45\cdot kN$

12.3 Total Factored Shear Force (STRENGTH I):

$$V_u := 1.05\left[1.25\cdot\left(V_g + V_{slab} + V_{diaph} + V_{ediaph} + V_{siderail}\right) + 1.50\cdot V_{DW} + 1.75\cdot V_{LLf}\right]$$

$V_u = 803.12\cdot kN$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

13. DESIGN SHEAR RESISTANCE

Shear resistance factor (AASHTO 5.5.4.2): $\phi_s := 0.90$

Effective web width: $b_v := T_w = 178\cdot mm$

Effective shear depth dv (greater of 0.9de or 0.72h):

$$d_e := \frac{A^*_s\cdot f_{ps}\cdot d_{pt} + A_{stL}\cdot f'_{sy}\cdot d_s}{A^*_s\cdot f_{ps} + A_{stL}\cdot f'_{sy}}$$

$d_e = 1.27\cdot m$

$0.9\cdot d_e = 1.143\,m$

Height of composite member:

$h := h_g + t_s + t_{haunch}$

$h = 1.39\,m$

$0.72\cdot h = 0.999\,m$

$d_v := \max(0.9\cdot d_e, 0.72\cdot h)$ $\quad d_v = 1.143\,m$

Unfactored shear force due to dead load: $V_{dl} = 192\cdot kN$

Factored shear force at section due to external load occuring simultaneously with M_{max}: $V_u = 803.12\cdot kN$

Total unfactored DL moment acting on non-composite section at support: $M_{dncs} := 0$

Multipy by -(4/5) to approximate support moment reaction for continuous beam: -(4/5) x (1/8)wL² = -(1/10)wL²

Max. factored moment at section due to externally applied loads: $M_{max} := -(1.50\cdot M_{sup} + 1.75\cdot M_{LLf}) = -1284.67\cdot kN\cdot m$

Eccentricity between center of girder to center of prestress at end support: $e_{end} := y_b - dp_{end}$ $\quad e_{end} = 0.10\cdot mm$

Compressive stress in concrete due to effective prestress forces only (after all loses) at bottom of girder: $f_{cpe} := \frac{P_3}{A_g} + \frac{P_3\cdot e_{end}}{S_b}$ $\quad f_{cpe} = 4.88\cdot MPa$

Modulus of rupture when used to calculate the cracking moment of a member in Artide 5.8.3.4.3: $f_{rc} := 0.52\cdot\sqrt{f'_c\cdot MPa}$ $\quad f_{rc} = 3.35\cdot MPa$

Moment causing flexural cracking due to externally applied loads (AASHTO 5.8.3.4.3-2) at support: $M_{cre} := S_{bc}\cdot\left(f_{rc} + f_{cpe} - \frac{M_{dncs}}{S_b}\right)$ $\quad M_{cre} = 1505.72\cdot kN\cdot m$

Nominal shear resistance provided by concrete when inclined cracking results from combined shear and moment (AASHTO 5.8.3.4.3-1):

$$V_{ci1} := \begin{cases} \left(0.0525\cdot\sqrt{f'_c\cdot MPa}\cdot b_v\cdot d_v + V_{dl}\right) & \text{if } M_{max} \le 0 \\ \left(0.0525\cdot\sqrt{f'_c\cdot MPa}\cdot b_v\cdot d_v + V_{dl} + \frac{V_u\cdot M_{cre}}{M_{max}}\right) & \text{otherwise} \end{cases}$$

$V_{ci1} = 260.28\cdot kN$ $\quad V_{ci2} := 0.16\cdot\sqrt{f'_c\cdot MPa}\cdot b_v\cdot d_v = 209.41\cdot kN$

$V_{ci} := \max(V_{ci1}, V_{ci2})$

$V_{ci} = 260.28\cdot kN$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Compressive stress in concrete (after allowance for all prestress losses) at centroid of cross section resisting externally applied load

$$f_{pc} := \frac{P_3}{A_g}$$

$$f_{pc} = 4.88 \cdot MPa$$

Vertical component of prestressing force, Vp is taken as zero when AASHTO 5.8.3.4.3 is applied:

$$V_p := 0 \cdot kN$$

Nominal shear resistance provided by concrete when inclined cracking results from excessive principal tensions in web (AASHTO 5.8.3.4.3-3):

$$V_{cw} := \left(0.16 \cdot \sqrt{f'_c \cdot MPa} + 0.30 \cdot f_{pc}\right) \cdot b_v \cdot d_v + V_p$$

$$V_{cw} = 507.00 \cdot kN$$

Controlling shear resistance provided by concrete:

$$V_c := \min\left(V_{ci}, V_{cw}\right)$$

$$V_c = 260.28 \cdot kN$$

Design shear resistance needed by transverse reinforcement:

$$V_s := \frac{V_u}{s} - V_c$$

$$V_s = 632.08 \cdot kN$$

DESIGN STIRRUP SPACING:

Diameter of Stirrups: $d_s := 12 \cdot mm$

Area of Stirrups: $A_v := \frac{\cdot d_s^2}{4}$

$$A_v = 113.10 \cdot mm^2$$

No. of legs of stirrups: $N_{legs} := 2$

$$s_1 := \frac{N_{legs} \cdot A_v \cdot f'_{sy} \cdot d_v}{V_s}$$

$$s_1 = 169.30 \cdot mm$$

say, $s_1 := 150 \cdot mm$

SHEAR RESISTANCE (AASHTO 5.8.3.3):

Shear resistance provided by transverse reinforcement:

$$V_s := \frac{N_{legs} \cdot A_v \cdot f'_{sy} \cdot d_v}{s_1} \qquad V_s = 713.42 \cdot kN$$

Nominal shear resistance:

$$V_{n1} := V_c + V_s \qquad V_{n2} := 0.25 \cdot f'_c \cdot b_v \cdot d_v$$

$$V_{n1} = 973.70 \cdot kN \qquad V_{n2} = 2105.29 \cdot kN$$

$$V_n := \min\left(V_{n1}, V_{n2}\right) \qquad V_n = 973.70 \cdot kN$$

Shear resistance:

$$V_r := \ _s \cdot V_n = 876.33 \cdot kN$$

Factored shear force:

$$V_u = 803.12 \cdot kN$$

$$Shear_check := \text{if}\left(V_r \geq V_u, \text{"OK"}, \text{"FAIL"}\right)$$

$$Shear_check = \text{"OK"}$$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

14. DOWELS AT GIRDER ENDS

The required dowels in girder ends to transfer girder reactions into diaphragms. Use shear friction, AASHTO 5.8.4

Concrete Area resisting shear transfer: $A_{cv} := A_g$ $A_{cv} = 0.36 \cdot m^2$

Diaphragm concrete strength: $f'_{c_diaph} := 27.6 \cdot MPa$

Shear reduction factor: $_v := 0.85$

Diaphragm reinforcement strength: $f'_{sy} := 414 \cdot MPa$

Friction factor (surface roughened): $_{} := 1.00$

Reinforcement diameter: $d_b := 20 \cdot mm$ $V_u = 803.12 \cdot kN$

Nominal Shear Force: $V_n := \frac{V_u}{v}$ $V_n = 944.85 \cdot kN$

$$V_{n_max} := \begin{cases} 0.20 \cdot f'_{c_diaph} \cdot A_{cv} & \text{if } \left(0.20 \cdot f'_{c_diaph} \cdot A_{cv}\right) \le 0.80ksi \cdot A_{cv} \\ 0.80ksi \cdot A_{cv} & \text{otherwise} \end{cases} \qquad V_{n_max} = 1992.86 \cdot kN$$

$$CHECK_V_n := \begin{cases} \text{"OK"} & \text{if } V_{n_max} > V_n \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_V_n = \text{"OK"}$$

Required steel area of dowel reinforcement:

$$A_{dowel} := \frac{\cdot d_b^2}{4} \qquad A_v := \frac{V_n}{f'_{sy} \cdot}$$

$A_{dowel} = 314.16 \cdot mm^2$ $A_v = 2282.24 \cdot mm^2$

$N_dowels := \frac{A_v}{A_{dowel}}$ $N_dowels = 7.26$ **Say 5 each face, 10 total**

Determine embedment length:

$Min := 0.4 \cdot d_b \cdot \frac{f'_{sy}}{ksi}$ $A_b := A_{dowel}$

$Min = 480.36 \cdot mm$ $factor1 := 1.40$ *(AASHTO 5.11.2.1.2)*

$factor2 := 0.80$ *(AASHTO 5.11.2.1.3)*

$$Basic := \frac{1.25 \cdot A_b \cdot f'_{sy}}{\sqrt{f'_{c_diaph} \cdot ksi} \cdot in}$$

$Basic = 463.99 \cdot mm$

$Embedment := factor1 \cdot factor2 \cdot Basic$ $Embedment = 520 \cdot mm$

Dowel length = embedment length

NOTE: Bars spaced at 150 mm min and 75 mm min cover. Distance between girder should be greater than dowel length plus clearance, otherwise hooks will be needed.

DESIGN OF PRESTRESSED EXTERIOR GIRDER

15. ESTIMATED CAMBER AT MIDSPAN

NOTE: + means upward camber
- means downward camber

The multiplier values were taken from PCI Bridge Design Manual, 3rd Ed., 2014 (8.7.1 Multiplier Method):

Initial Prestressing Force: $P_1 = 1960.43 \cdot kN$ *Final Prestressing Force:* $P_3 = 1761.94 \cdot kN$

$y_{2b} = 505.651 \cdot mm$ $E_c := E_{c_girder}$ $w_{diaph} := \frac{P_{diaph}}{L}$

$dp_{end} = 514.50 \cdot mm$ $e_2 = 380.65 \cdot mm$ $w_{diaph} = 0.333 \cdot \frac{kN}{m}$

$e_{end} := y_{2b} - dp_{end}$ $e_{end} = -8.85 \cdot mm$

$e_{mid} := e_2 - e_{end}$ $e_{mid} = 389.50 \cdot mm$

15.1 CAMBER AT TRANSFER

15.1.1 Due to Prestress including End Moment

$$pc := \frac{P_1 \cdot L^2}{48 \cdot E_{ci} \cdot I_{2cg}} \cdot \left(5 \cdot e_{mid} + 6 e_{end}\right) \qquad pc = 9.73 \cdot mm$$

15.1.2 Due to Girder Weight

$$_{gdr} := \left(\frac{5 \cdot w_g \cdot L^4}{384 \cdot E_{ci} \cdot I_{2cg}} \right) \cdot (-1) \qquad _{gdr} = -2.94 \cdot mm$$

15.1.2 Initial Camber at Transfer

$Camber_{Transfer} := pc + {}_{gdr}$ $Camber_{Transfer} = 6.78 \cdot mm$

15.2 CAMBER AT ERECTION

15.2.1 Due to Prestress including End Moment

$m_1 := 1.80$

$$'pc := \frac{m_1 \cdot P_1 \cdot L^2}{48 \cdot E_c \cdot I_{2cg}} \cdot \left(5 \cdot e_{mid} + 6 e_{end}\right) \qquad 'pc = 16.14 \cdot mm$$

15.2.2 Due to Girder Weight

$m_2 := 1.85$

$$'_{gdr} := \left(\frac{m_2 \cdot 5 \cdot w_g \cdot L^4}{384 \cdot E_c \cdot I_{2cg}} \right) \cdot (-1) \qquad '_{gdr} = -5.02 \cdot mm$$

15.2.3 Camber at Erection

$Camber_{Erection} := 'pc + {}'_{gdr}$ $Camber_{Erection} = 11.12 \cdot mm$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

15.3 CAMBER AT FINAL (SERVICE STAGE)

15.3.1 Due to Slab

$$\Delta_{slab} := \left(\frac{5 \cdot w_{slab} \cdot L^4}{384 \cdot E_c \cdot I_{2cg}}\right) \cdot (-1) \qquad \Delta_{slab} = -3.17 \cdot mm$$

15.3.2 Due to Diaphragm

$$\Delta_{diaph} := \left(\frac{5 \cdot w_{diaph} \cdot L^4}{384 \cdot E_c \cdot I_{2cg}}\right) \cdot (-1) \qquad \Delta_{diaph} = -0.10 \cdot mm$$

15.3.3 Due to Superimposed Dead Loads

$$\Delta_{sup} := \left(\frac{5 \cdot w_{sup} \cdot L^4}{384 \cdot E_c \cdot I_{3cg}}\right) \cdot (-1) \qquad \Delta_{sup} = -0.56 \cdot mm$$

15.3.4 Final Camber at Service Stage (Long Term Deflection)

$m_3 := 2.2$

Due to P/S:
$$\Delta PS_F := \frac{m_3 \cdot P_3 \cdot L^2}{48 \cdot E_c \cdot I_{2cg}} \cdot (5 \cdot e_{mid} + 6 e_{end}) \qquad \Delta PS_F = 17.73 \cdot mm$$

$m_4 := 2.4$

Due to Girder:
$$\Delta Gdr_F := \left(\frac{m_4 \cdot 5 \cdot w_g \cdot L^4}{384 \cdot E_c \cdot I_{2cg}}\right) \cdot (-1) \qquad \Delta Gdr_F = -6.51 \cdot mm$$

$m_5 := 2.3$

Due to Composite DL:
$$\Delta Slab_F := m_5 \cdot (\Delta_{slab} + \Delta_{diaph}) \qquad \Delta Slab_F = -7.54 \cdot mm$$

$m_6 := 3.0$

Due to SDL:
$$\Delta SDL_F := m_6 \cdot \Delta_{sup} \qquad \Delta SDL_F = -1.67 \cdot mm$$

$$Camber_{Final} := \Delta PS_F + \Delta Gdr_F + \Delta Slab_F + \Delta SDL_F$$

$$Camber_{Final} = 2.01 \cdot mm$$

15.4 CAMBER SCHEME

A) Camber of cast of girder, before girder and prestressing load

$$A_c := |Camber_{Final}| \qquad A_c = 2.01 \cdot mm$$

B) Camber after girder and prestressing load

Deflection due to Girder and P/S

$$\Delta GdrPS_F := \Delta Gdr_F + \Delta PS_F \qquad \Delta GdrPS_F = 11.22 \cdot mm$$

Resulting camber

$$B_c := A_c + \Delta GdrPS_F \qquad B_c = 13.22 \cdot mm$$

C) Camber after slab, diaphragm

$$C_c := B_c + \Delta Slab_F \qquad C_c = 5.68 \cdot mm$$

D) Camber final after Superimposed DL

$$D_c := C_c + \Delta SDL_F \qquad D_c = 4.01 \cdot mm$$

DESIGN OF PRESTRESSED EXTERIOR GIRDER

16. DEFLECTION DUE TO LIVE LOAD AND IMPACT

Allowable deflection due to live load *(LRFD 2.5.2.6.2)*

$L = 14.40\,m$

$$_Liveload_{allow} := \frac{L}{800} \qquad _Liveload_{allow} = 18.00\cdot mm$$

Live load deflection criteria

If the owner invokes the optional live load deflection criteria specified in LRFD Art. 2.5.2.6.2, the deflection is the greater of:

- that resulting from the design truck plus impact Δ_{LT}, or [LRFD Art. 3.6.1.3.2]
- that resulting from 25% of the design truck plus impact Δ_{LT}, taken together with the design lane load, Δ_{LL}.

Note: LRFD Article 2.5.2.6.2 states that the dynamic load allowance be included in the calculation of live load deflection.

The *LRFD Specifications* states that all the beams should be assumed to deflect equally under the applied live load and impact. [LRFD Art. 2.5.2.6.2]

Deflection due to Design Lane Load

Lane load: $w_{lane} = 9.34\cdot\frac{kN}{m}$

Distribution factor: $g_1 = 0.617$

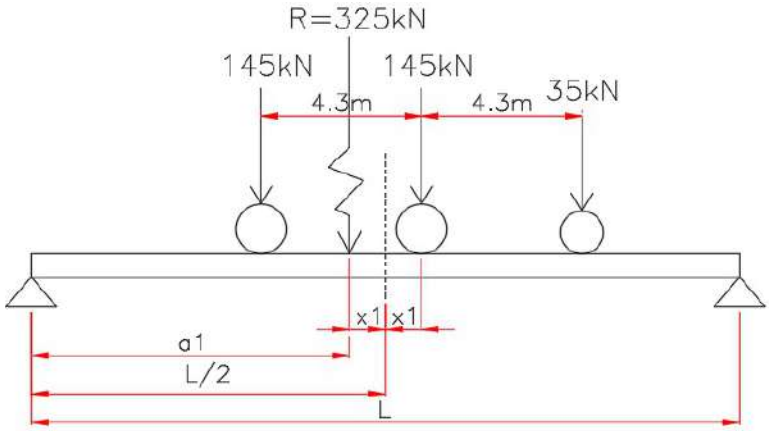
Modulus of elasticity of composite beam: $E_{c_girder} = 33497\cdot MPa$

Moment of inertia of composite beam: $I_{3cg} = 0.1606\,m^4$

$$_LL := \left(\frac{5\cdot w_{lane}\cdot L^4}{384\cdot E_{c_girder}\cdot I_{3cg}}\right)\cdot(g_1) \qquad _LL = 0.60\cdot mm$$

Deflection due to Design Truck Load + Impact

To calculate the maximum deflection due to truck load, set the spacing between the rear axles to 4.3m. And have the centerline of the beam coincide with the middle point of the distance between the inner 145kN axle and the resultant of the truck load (325kN).



DESIGN OF PRESTRESSED EXTERIOR GIRDER

where:

Point load resultant of the loads in wheels

$R_t := (145 + 145 + 35)\cdot kN$ $\quad R_t = 325.00\cdot kN$

Location of deflection to be determined

$x_{mid} := \frac{L}{2}$ $\quad x_{mid} = 7.200\,m$

Distance from R to centerline of beam

$x_1 := \frac{1}{2}\cdot\frac{(145\cdot kN\cdot 4.3\cdot m - 35\cdot kN\cdot 4.3\cdot m)}{R_t}$ $\quad x_1 = 727.69\cdot mm$

Distance from left end of beam to point load R

$a_1 := x_{mid} - x_1$ $\quad a_1 = 6.472\,m$

Distance from point load R to right end of beam

$b_1 := L - a_1$ $\quad b_1 = 7.928\,m$

Minimum of a or b

$c_1 := \min(a_1, b_1)$

Impact factor

$IM = 1.33$

$$_LT := \frac{R_t\cdot c_1}{48\cdot E_{c_girder}\cdot I_{3cg}}\cdot\left(3\cdot L^2 - 4\cdot c_1^{\,2}\right)\cdot IM\cdot(g_1)$$

$_LT = 3.04\cdot mm$

Deflection due to Live Load and Check vs Allowable

$_Liveload_{Def} := \max(\ _LT, 0.25\cdot\ _LT + \ _LL)$ $\quad _Liveload_{Def} = 3.04\cdot mm$

$_Liveload_{allow} = 18.00\cdot mm$

$LLDeflection_check := if(\ _Liveload_{allow} \geq \ _Liveload_{Def}, "OK", "FAIL")$

$LLDeflection_check = "OK"$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

17. END ANCHORAGE DESIGN

17.1. Compressive Stress (Local Zone Check)

Member thickness

$t_{tf} := B_{tf_{Type}}$

$t_{tf} = 406.00 \cdot mm$

Lateral dimension of the effective bearing area measured parallel to the larger dimension of the cross-section

$a_{ef} := 150mm$

Lateral dimension of the effective bearing area measured parallel to the smaller dimension of the cross-section

$b_{ef} := 150mm$

Area of bearing plate

$A_{plate} := a_{ef} \cdot b_{ef}$

$A_{plate} = 22500.0 \cdot mm^2$

Diameter of confinement

$D_s := 150mm$

Area of confinement

$A_{conf} := \frac{\pi}{4} \cdot D_s^2$

$A_{conf} = 17671.5 \cdot mm^2$

Equivalent dimensions of confined area:

$a_{conf} := \sqrt{A_{conf}}$

$a_{conf} = 132.9 \cdot mm$

$b_{conf} := \sqrt{A_{conf}}$

$b_{conf} = 132.9 \cdot mm$

Center-to-center spacing of anchorages

$s_a := 353mm$

Number of anchorages in a row

$n_a := 2.0$

Diameter of duct

$D := D_{duct} = 0.067\,m$

No. of duct

$N_a := \frac{A^*_s}{12 \cdot A_{str}}$

$N_a = 1.33$

Factored tendon force

$P_{ulz} := \frac{1.25 \cdot P_j}{N_a}$

$P_{ulz} = 2065.30 \cdot kN$

Longitudinal extent of confining reinf. of the local zone

$l_c := \max(1.15a_{ef}, 1.15b_{ef})$

$l_c = 172.5 \cdot mm$

Correction factor for closely spaced anchorages

$$K_a := \begin{cases} 1 + \left(2 - \frac{s_a}{a_{ef}}\right)\left(0.3 + \frac{n_a}{15}\right) & \text{if } a_{ef} \le s_a < 2a_{ef} \\ 1 & \text{if } s_a \ge 2a_{ef} \end{cases}$$

$K_a = 1.00$

Note: If a group of anchorages is closely spaced in two directions, the product of the correction factors, k_a, for each direction shall be used.

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

The effective bearing area, A_{eb}, shall be taken as the larger of the anchor bearing plate area; A_{plate}, or the bearing area of the confined concrete in the local zone; A_{conf}, with the following limitations:

$CHECK_A_{eb} := \text{if}\left(A_{plate} > A_{conf}, \text{"A.plate controls"}, \text{"A.conf controls"}\right)$ $\quad CHECK_A_{eb} = \text{"A.plate controls"}$

$$A_{eb} := \begin{cases} \min\left(A_{plate}, \frac{4}{}\cdot A_{conf} - \frac{}{4}\cdot D^2\right) & \text{if } \left(A_{plate} > A_{conf}\right)\left(A_{plate} \le \frac{4}{}\cdot A_{conf}\right) \\ A_{conf} - \frac{}{4}\cdot D^2 & \text{if } \left(A_{plate} < A_{conf}\right)\left(\max\left(a_{conf}, b_{conf}\right) < 2\max\left(a_{ef}, b_{ef}\right)\right)\left(\max\left(a_{conf}, b_{conf}\right) < 3\min\left(a_{ef}, b_{ef}\right)\right) \\ A_{plate} & \text{otherwise} \end{cases}$$

$A_{plate} = 22500.00\cdot mm^2$

$A_{eb} = 19008.47\cdot mm^2$

$\frac{4}{}\cdot A_{conf} - \frac{}{4}\cdot D^2 = 19008.47\cdot mm^2$

Concrete compressive stress ahead of the achorage devices

$$f_{ca} := \frac{0.6\,P_{ulz}\,K_a}{A_{eb}\cdot\left[1 + l_c\cdot\left(\frac{1}{b_{ef}} - \frac{1}{t_{tf}}\right)\right]}$$

$$A_{eb}\cdot\left[1 + l_c\cdot\left(\frac{1}{b_{eff}} - \frac{1}{t_{tf}}\right)\right] = 13074.02\cdot mm^2$$

$f_{ca} = 37.79\cdot MPa$

17.2. Bursting Force (General Zone Check)

Factored tendon force

$P_{ugz} := 1.25\cdot P_j$

$P_{ugz} = 2753.73\cdot kN$

Lateral dimension of the anchorage device or group of devices in the direction considered

$a_b := \min\left(N_a\cdot a_{ef}, t_{tf}\right)$

$a_b = 200.0\cdot mm$

Lateral dimension of the cross-section in the direction considered

$h_b := h_g$

$h_b = 1143.00\cdot mm$

Reduction factor for tension in steel in anchorages zones

$_b := 1.0$

Angle of inclination of a tendon force with respect to the centerline of the member, positive for concentric tendons or if the anchor force points toward the centroid of the section; negative if the anchor force points away from the centroid of the section

$1 = 0.108\cdot rad$

Eccentricity of the anchorage device or group of devices with respect to the centroid of the cross-section; always taken as positive

$e_b := y_b - dp_{end}$

$e_b = 0.10\cdot mm$

Reduction factor for compression in anchorages zones

$_{ca} := 0.8$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

Max. factored concrete compressive strength for the general zone

$$\frac{P_{ugz}}{a_b \cdot h_b} = 12.05 \cdot MPa$$

$f_{camax} := 0.7 \cdot {}_{ca} \cdot f'_{ci}$ $\quad f_{camax} = 19.71 \cdot MPa$

$$CHECK_f_{camax} := \begin{cases} "OK" & \text{if } \frac{P_{ugz}}{a_b \cdot h_b} < f_{camax} \\ "NG" & \text{otherwise} \end{cases}$$

$CHECK_f_{camax} = "OK"$

Distance from anchorage device to the centroid of the bursting force

$$d_{burst} := \frac{0.5\left(h_b - 2e_b\right) + 5 \cdot e_b \cdot \sin(\ 1)}{N_a}$$

$d_{burst} = 428.59 \cdot mm$

Tensile force in the anchorage zone acting ahead of the anchorage device and transverse to the tendon axis

$$T_{burst} := 0.25 P_{ugz} \cdot \left(1 - \frac{a_b}{h_b}\right) + 0.5 \left| P_{ugz} \cdot \sin(\ 1) \right|$$

$T_{burst} = 716.65 \cdot kN$

Required area of bursting bars

$$A_{rburst} := \frac{T_{burst}}{{}_b \cdot N_a \cdot f'_{sy}}$$

$A_{rburst} = 1298.28 \cdot mm^2$

Provided area of bursting bars

diameter bursting bars $\quad d_{bburst} := 12mm$

no. bursting bars $\quad N_{bburst} := 8$

$$A_{pburst} := \frac{\pi}{4} \cdot d_{bburst}^{\ 2} \cdot 2 \cdot N_{bburst}$$

$A_{pburst} = 1809.56 \cdot mm^2$

$$CHECK_A_{burst} := \begin{cases} "OK" & \text{if } A_{rburst} < A_{pburst} \\ "NG" & \text{otherwise} \end{cases}$$

$CHECK_A_{burst} = "OK"$

CALCULATION SHEET

DESIGN OF PRESTRESSED EXTERIOR GIRDER

18. TENDONS LAYOUT

Number of ducts $N_{duct} = 2$

Cable number | No. of strands | Distance from bottom of girder at ends

$Duct_i =$

1
2

$N_{strand_i} =$

8
8

$Y_i =$

350	·mm
679	

Girder Span Length

$L = 14400 \cdot mm$

Distance from bottom of girder at midspan

$d_p = 125 \cdot mm$

General equation of parabola: y = a(x-h)^2 + k

a = (y - k) / (x-h)^2

where vertex of paraboloa is V(h,k) = V(L/2,dp)

$$a_i := \frac{Y_i - d_p}{(0 - 0.5 \cdot L)^2}$$

$a_i =$

0.00434	$\cdot m^{-1}$
0.01069	

Horizontal distance from end to midspan of girder

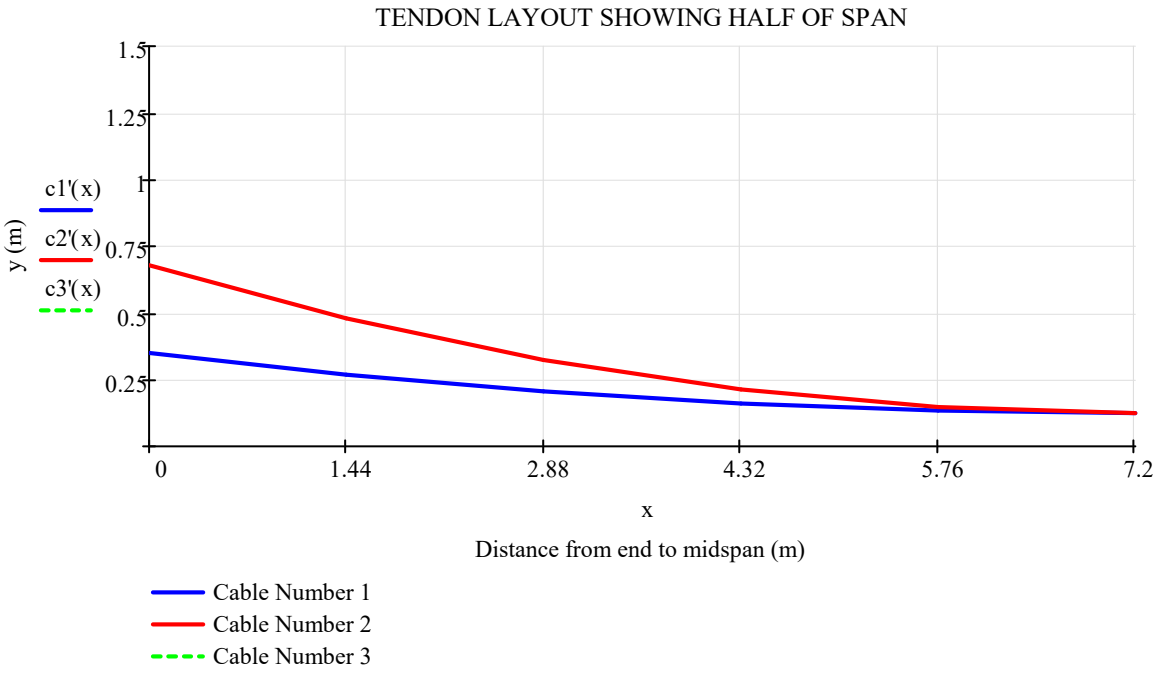
$x := 0, 0.1 \cdot L .. 0.5L$

Parabolic equations of per cable duct number (vertical distance from bottom of girder)

$$c1'(x) := a_1 \cdot (x - 0.5 \cdot L)^2 + d_p \qquad c2'(x) := a_2 \cdot (x - 0.5 \cdot L)^2 + d_p \qquad c3'(x) := a_3 \cdot (x - 0.5 \cdot L)^2 + d_p$$

x =		c1'(x) =		c2'(x) =	
0	·mm	350	·mm	679	·mm
1440		269		479.56	
2880		206		324.44	
4320		161		213.64	
5760		134		147.16	
7200		125		125	

DESIGN OF PRESTRESSED EXTERIOR GIRDER



19. REQUIRED JACKING FORCE PER DUCT

Total jacking force: $P_j = 2202.98 \cdot kN$

Say: $P_j := 2204 \cdot kN$

$$Pj_i := \frac{N_{strand_i}}{N_o} \cdot P_j$$

$Duct_i$ =	N_{strand_i} =	Pj_i =	
1	8	1102	·kN
2	8	1102	

per duct

$N_o = 16$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER (14.40m TYPE III)

Designed by/Date: KH - RCDG 05/04/26

Conversion Factor: $kN \equiv 1000 \cdot newton$ $Pa \equiv \frac{newton}{m^2}$ $kPa \equiv \frac{kN}{m^2}$ $MPa \equiv \frac{newton}{mm^2}$

DESIGN THEORY:

1. Use concrete stress limits from AASHTO LRFD 2012 Sec 5.9.3 and 4 at service limit states.
2. Use load resistance factor design to check ultimate moment and shear capacity at the factored level considering (HL-93) truck and lane loads, design tandem and lane loads, or permit loads, whichever is greater.
3. Gross properties are used as preliminary design.
4. Net/Transformed section properties will be calculated and used for the final design in the analysis.

1. MATERIAL DATA and SPECIFICATIONS

1.1 Bridge Geometry

Girder Span Length	$L := 14.40 \cdot m$	No. of Girder	$N_g := 3$
Slab Thickness	$t_s := 220 \cdot mm$	Roadway Width	$RW := 4 \cdot m$
Haunch Thickness	$t_{haunch} := 25 \cdot mm$	Sidedwalk Width	$SW := 0 \cdot m$
Girder Spacing	$S_g := 1.75 \cdot m$	No. of Diaphragm (interior diaphragm)	$N_{diap} := 1$

1.2 Prestressed Concrete

Concrete Compressive Strength @ 28 days	$f'_c := 41.4 \cdot MPa$	Concrete Compressive Strength @ 28 days (for cast-in-place deck)	$f'_{c_slab} := 27.6 \cdot MPa$
Concrete Compressive Strength @ transfer	$f'_{ci} := 0.85 \cdot f'_c$	Unit Weight of Concrete	$_{conc} := 24.00 \cdot \frac{kN}{m^3}$
Modulus of Elasticity of concrete @ transfer	$E_{ci} := 0.043 \cdot \left(\frac{_{conc} \cdot m^3}{kgf}\right)^{1.5} \cdot \sqrt{f'_{ci} \cdot MPa}$		
	$E_{ci} = 30883 \cdot MPa$	Unit Weight of Wearing coarse	$_{fws} := 22 \cdot \frac{kN}{m^3} (80mm)$
		Modulus of Rupture of normal wt. conc.	$f_r := 0.63 \cdot \sqrt{f'_c \cdot MPa}$
			$f_r = 4.05 \cdot MPa$

1.3 Prestressing Steel - *Stress Relieved Strands*

Ultimate Strength of prestressing steel	$f'_s := 1860 \cdot MPa$	*From AASHTO 5.9.5.2*	
Yield Strength of prestressing steel	$f^*_y := 0.90 \cdot f'_s$	Friction Curvature Coefficient	$:= 0.20$
	$f^*_y = 1674 \cdot MPa$	Friction Wobble Coefficient	$k := \frac{0.00066}{m}$
Modulus of Elasticity of prestressing steel	$E_p := 197000 \cdot MPa$		

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

1.4 Non-Prestressing Steel : *Mild Steel Reinforcement (Intermediate Grade Steel)*

Yield Strength of intermediate grade steel $f'_{sy} := 414 \cdot MPa$

Modulus of Elasticity of intermediate grade steel $E_s := 200000 \cdot MPa$

2. STRESS LIMITS FOR CONCRETE (AASHTO 5.9.4)

2.1 At Transfer Stage

Compression Fiber: $f'_{ci_allow} := 0.60 \cdot f'_{ci}$ $f'_{ci_allow} = 21.11 \cdot MPa$

Tension Fiber: $f'_{ti_allow} := -0.63 \cdot \sqrt{f'_{ci} \cdot MPa}$ $f'_{ti_allow} = -3.737 \cdot MPa$

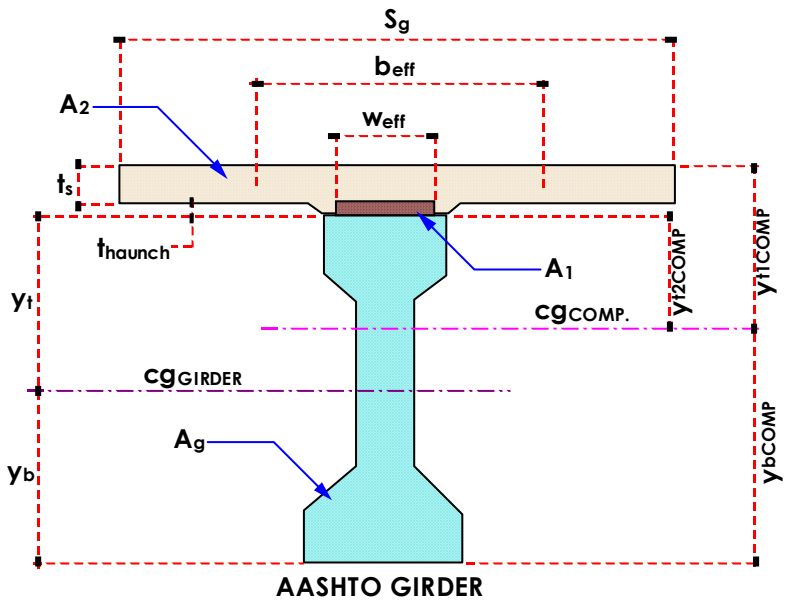
2.2 At Service Stage

Compression Fiber: $f'_{c_allow} := 0.45 \cdot f'_{c}$ $f'_{c_allow} = 18.63 \cdot MPa$

*** Tension Fiber:** $f'_{t_allow} := -0.50 \cdot \sqrt{f'_{c} \cdot MPa}$ $f'_{t_allow} = -3.217 \cdot MPa$

** If site is located at severe corrosive condition, $f'_{t_allow} = -0.25(f'_c)^{0.5}$*

3. CROSS SECTIONAL PROPERTIES OF AASHTO I-GIRDER



Type := 3 **<<< Input AASHTO Girder Type**

Note: Types 7 & 8 are actually Types IV-A & IV-B respectively.

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Table of AASHTO Girder Properties:

$j := 1..8$

j =	D_{pc_j} :=	B_{bf_j} :=	B_{tf_j} :=	T_{w_j} :=	Z_{pc_j} :=	A_{pc_j} :=	$I_{y_pc_j}$:=	$I_{z_pc_j}$:=	$I_{x_pc_j}$:=
1.00	0.711	0.406	0.305	0.152	0.3198	0.1778	0.00945	0.001384	0.001303
2.00	0.914	0.457	0.305	0.152	0.4020	0.2374	0.02115	0.002193	0.002407
3.00	1.143	0.559	0.406	0.178	0.5146	0.3613	0.05216	0.005055	0.005232
4.00	1.372	0.660	0.508	0.203	0.6283	0.5087	0.10845	0.010039	0.009901
5.00	1.600	0.711	1.067	0.203	0.8096	0.6532	0.21493	0.02384	0.011063
6.00	1.829	0.711	1.067	0.203	0.9223	0.6997	0.30303	0.024003	0.011702
7.00	1.473	0.660	0.508	0.203	0.7012	0.5604	0.13795	0.01115	0.01459
8.00	1.575	0.660	0.508	0.203	0.7707	0.6119	0.17007	0.012257	0.022886

$D_{pc} := D_{pc} \cdot m \quad B_{bf} := B_{bf} \cdot m \quad B_{tf} := B_{tf} \cdot m \quad T_w := T_w \cdot m \quad Z_{pc} := Z_{pc} \cdot m$

$A_{pc} := A_{pc} \cdot m^2 \quad I_{y_pc} := I_{y_pc} \cdot m^4 \quad I_{z_pc} := I_{z_pc} \cdot m^4 \quad I_{x_pc} := I_{x_pc} \cdot m^4$

Girder Height: $h_g := D_{pc_{Type}}$ → $h_g = 1.143\,m$

Moment of Inertia: $I_g := I_{y_pc_{Type}}$ → $I_g = 0.0522\,m^4$

Girder Area: $A_g := A_{pc_{Type}}$ → $A_g = 0.3613\,m^2$

Dist. of C.G. from bottom of girder : $y_b := Z_{pc_{Type}}$ → $y_b = 0.5146\,m$

Dist. of C.G. from top of girder : $y_t := h_g - y_b$ → $y_t = 0.6284\,m$

Section Modulus at top of girder : $S_t := \frac{I_g}{y_t}$ → $S_t = 0.083 \cdot m^3$

Section Modulus at bottom of girder : $S_b := \frac{I_g}{y_b}$ → $S_b = 0.1014 \cdot m^3$

Web Thickness: $T_w := T_{w_{Type}}$ → $T_w = 0.178\,m$

4. COMPOSITE CROSS SECTIONAL PROPERTIES OF AASHTO I-GIRDER

4.1 Effective Width of Slab

Width of slab should not be greater than the following:

a) $S_1 := \frac{L}{4}$ → $S_1 = 3600 \cdot mm$

b) $S_2 := S_g$ → $S_2 = 1750 \cdot mm$

c) $S_3 := 12 \cdot t_s + T_w$ → $S_3 = 2818 \cdot mm$

$S := \min((S_1 \; S_2 \; S_3))$ → $S = 1750.00 \cdot mm$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Modular ratio:

$$E_{c_slab} := 0.043 \cdot \left(\frac{conc \cdot m^3}{kgf}\right)^{1.5} \cdot \sqrt{f'_{c_slab} \cdot MPa} \qquad E_{c_slab} = 27350 \cdot MPa$$

$$E_{c_girder} := 0.043 \cdot \left(\frac{conc \cdot m^3}{kgf}\right)^{1.5} \cdot \sqrt{f'_{c} \cdot MPa} \qquad E_{c_girder} = 33497 \cdot MPa$$

$$n := \frac{E_{c_slab}}{E_{c_girder}} \qquad n = 0.8165$$

Effective width of Flange, b_{eff} $\quad b_{eff} := n \cdot S \qquad b_{eff} = 1.43\,m$

Effective width of haunch, w_{eff} $\quad w_{eff} := n \cdot B_{tf_{Type}} \qquad w_{eff} = 0.33\,m$

Determination c.g. with respect to bottom:

Composite haunch area $\quad A_1 := w_{eff} \cdot t_{haunch} \qquad A_1 = 0.008\,m^2$

Composite deck slab area $\quad A_2 := b_{eff} \cdot t_s \qquad A_2 = 0.31\,m^2$

Total Composite Area $\quad A_{comp} := A_g + A_1 + A_2 \qquad A_{comp} = 0.68\,m^2$

$$y_{bcomp} := \frac{A_g \cdot y_b + A_1 \cdot \left(h_g + \frac{1}{2} \cdot t_{haunch}\right) + A_2 \cdot \left(h_g + t_{haunch} + \frac{1}{2} \cdot t_s\right)}{A_{comp}} \qquad y_{bcomp} = 873.24 \cdot mm$$

$$y_{t1comp} := \left(h_g + t_{haunch} + t_s\right) - y_{bcomp} \qquad y_{t1comp} = 514.76 \cdot mm$$

$$y_{t2comp} := h_g - y_{bcomp} \qquad y_{t2comp} = 269.76 \cdot mm$$

Moment of Inertia for the composite section:

$$I_{comp} := I_g + A_g \cdot \left(y_{bcomp} - y_b\right)^2 \ldots + \left[\frac{w_{eff} \cdot t_{haunch}^3}{12} + A_1 \cdot \left(y_{t1comp} - t_s - \frac{t_{haunch}}{2}\right)^2\right] \ldots + \left[\frac{b_{eff} \cdot t_s^3}{12} + A_2 \cdot \left(y_{t1comp} - \frac{t_s}{2}\right)^2\right] \qquad I_{comp} = 0.1521\,m^4$$

Section Modulus of Composite section:

$$S_{bcomp} := \frac{I_{comp}}{y_{bcomp}} \qquad S_{bcomp} = 0.1741 \cdot m^3$$

$$S_{t2comp} := \frac{I_{comp}}{y_{t2comp}} \qquad S_{t2comp} = 0.5637 \cdot m^3$$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

5. LOAD ANALYSIS PER GIRDER (DEAD LOAD)

Dead loads due to girder, slab, and intermediate diaphragm (DC):

$w_g := A_g \cdot \text{conc}$ $\quad w_g = 8.67 \cdot \frac{kN}{m}$

$w_{slab} := \left(t_s \cdot S + t_{haunch} \cdot B_{tf_{Type}}\right) \cdot \text{conc}$ $\quad w_{slab} = 9.48 \cdot \frac{kN}{m}$

$P_{diaph} := N_{diap} \cdot 0.25m \cdot 1.60m \cdot 1.00m \cdot \text{conc}$ $\quad P_{diaph} = 9.6 \cdot kN$

Dead loads due to superimposed loads:

$w_{railings} := 0.20 \cdot 0.20 \cdot m^2 \cdot 0 \cdot \text{conc}$ $\quad w_{railings} = 0 \cdot \frac{kN}{m}$

$w_{rail_post} := \frac{0.80m \cdot 0.20m \cdot 0.20m \cdot 0 \cdot \text{conc}}{1.50m}$ $\quad w_{rail_post} = 0 \cdot \frac{kN}{m}$

$w_{sidewalk} := SW \cdot 0.26m \cdot \text{conc} \cdot 0$ $\quad w_{sidewalk} = 0 \cdot \frac{kN}{m}$

$w_{fws} := S \cdot \text{fws}$ $\quad w_{fws} = 3.08 \cdot \frac{kN}{m}$

$w_{utilities1} := \left(w_{railings} + w_{rail_post} + w_{sidewalk} + w_{fws}\right) \cdot 0.10$

$w_{utilities2} := 1.00 \frac{kN}{m}$ *(est. minimum; 1 kN/m)*

$w_{utilities} := \max\left(w_{utilities1}, w_{utilities2}\right)$ $\quad w_{utilities} = 1.00 \cdot \frac{kN}{m}$

$w_{sup} := w_{railings} + w_{rail_post} + w_{sidewalk} + w_{fws} + w_{utilities}$ $\quad w_{sup} = 4.08 \cdot \frac{kN}{m}$

6. DESIGN MOMENT FORCES

6.1 Moment at Midspan Due to Dead Loads

$L_b := 13.80m$ Bearing-to-bearing distance

(From StaadPro Analysis of Continuous Girders)

Girder

$M_g := \frac{w_g \cdot L_b^2}{8}$

$M_g = 206.42 \cdot kN \cdot m$

Slab

$M_{slab} := \frac{w_{slab} \cdot L_b^2}{8}$

$M_{slab} = 225.76 \cdot kN \cdot m$

Superimposed loads

$M_{sup} := 0 \cdot kN \cdot m + 65.30 \cdot kN \cdot m$

$M_{sup} = 65.3 \cdot kN \cdot m$

$$M_{diaph} := \begin{cases} \frac{P_{diaph} \cdot L_b}{4} & \text{if } N_{diap} = 1 \\ \frac{P_{diaph} \cdot L_b}{3} & \text{if } N_{diap} = 2 \\ \frac{P_{diaph} \cdot L_b}{2} & \text{if } N_{diap} = 3 \end{cases}$$

Intermediate diaphragm

$M_{diaph} = 33.12 \cdot kN \cdot m$

Note: If there is no interior diaphragm, M_{diaph} = 0.00 kN-m.

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

6.2 Moment at Midspan Due to Live Loads

6.2.1 Distribution Factor of Live Loads Per Lane for Moment in Interior Beams

Dynamic Load Allowance Factor $IM := 1.33$

Modular ratio of beam to deck $n_d := \frac{E_{c_girder}}{E_{c\ slab}} = 1.22$

Distance bet. centers of gravity of beam and deck $e_g := y_t + t_{haunch} + \frac{t_s}{2} = 763.4 \cdot mm$

Longitudinal Stiffness Parameter $K_g := n_d \cdot \left(I_g + A_g \cdot e_g^{\,2} \right)$ $K_g = 3.22 \times 10^{11} \cdot mm^4$

Distribution Factor for One Design Lane Loaded:

$$DF_{M1} := 0.06 + \left(\frac{S_g}{4300 \cdot mm} \right)^{0.4} \cdot \left(\frac{S_g}{L_b} \right)^{0.3} \cdot \left(\frac{K_g}{L_b \cdot t_s^{\,3}} \right)^{0.1}$$

$DF_{M1} = 0.466$

Distribution Factor for Two or More Design Lane Loaded:

$$DF_{M2} := 0.075 + \left(\frac{S_g}{2900 \cdot mm} \right)^{0.6} \cdot \left(\frac{S_g}{L_b} \right)^{0.2} \cdot \left(\frac{K_g}{L_b \cdot t_s^{\,3}} \right)^{0.1}$$

$DF_{M2} = 0.604$

Note: If Ng=3, use lesser of the values above or the lever rule

Using lever rule, since Ng = 3:

multiple presence factor $mf_1 := 1.2$ (for one lane) $mf_2 := 1.0$ (for multiple lane)

(first wheel is at 600m away from railings, wheels spacing is 1.8m)

$DF_{M3.1} := (0.5) \cdot mf_1$ $DF_{M3.1} = 0.600$

Using lever arm rule value (more conservative)

$DF_{M3.2} := (0.5) \cdot mf_2$ $DF_{M3.2} = 0.500$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

6.2.2 Truck Loading

Note: Live Load Moments are from StaadPro Analysis of Continuous Girders.

Design Truck *Design Tandem*

$M_{Tr} := 620 \cdot kN \cdot m$ $M_T := 578 \cdot kN \cdot m$

Unfactored Design Truck/Tandem + IM

$M_{DT} := \max(M_{Tr}, M_T) \cdot IM$ $M_{DT} = 824.60 \cdot kN \cdot m$

6.2.3 Lane Loading

$w_{lane} := 9.34 \cdot \frac{kN}{m}$

$M_{lane} := 194 \cdot kN \cdot m$ $M_{lane} = 194.00 \cdot kN \cdot m$

6.2.4 Unfactored Moment (Due to Truck + Lane Load) x Distribution Factor DF

$M_{LL} := (M_{DT} + M_{lane}) \cdot DF_{M3.1}$ *(lever arm rule)*

$M_{LL} = 611.16 \cdot kN \cdot m$

6.2.5 Permit Truck Loading

Caltrans P-7 *Note:* Live Load Moments are from StaadPro Analysis of Continuous Girders.

$M_{PT} := 765 \cdot kN \cdot m$

Unfactored Permit Load + IM

$IM_P := 1.25$ Dynamic Load Allowance Factor for Permit Load

$M_{PTim} := M_{PT} \cdot IM_P$

$M_{PTim} = 956.25 \cdot kN \cdot m$

DESIGN OF PRESTRESSED INTERIOR GIRDER

Final force applied to girder (AASHTO 4.6.2.2.5)

$Z := 1.00$ Factor *(1.0 if lever rule is used)*

$$M_P := M_{PTim}\cdot\left(\frac{DF_{M3.1}}{Z}\right) + M_{LL}\cdot\left(DF_{M3.2} - \frac{DF_{M3.1}}{Z}\right)$$

$M_P = 512.63\cdot kN\cdot m$

6.2.6 Strength Limit States Check

Strength I Limit State (Design Truck):

$M_{uL} := 1.75\cdot M_{LL}$

$M_{uL} = 1069.53\cdot kN\cdot m$

Strength II Limit State (Permit Truck):

$M_{uP} := 1.35\cdot M_P$

$M_{uP} = 692.06\cdot kN\cdot m$

$$Strength_check := \begin{cases} \text{"Design Truck governs!"} & \text{if } M_{uL} \geq M_{uP} \\ \text{"Permit Truck governs!"} & \text{otherwise} \end{cases}$$

$Strength_check = \text{"Design Truck governs!"}$

6.2.7 Pedestrian / Bicycle Loading

$SW_C := 0\cdot m$ Clear sidewalk width; 0 if interior

$w_{PL} := 4.30\cdot kPa\cdot SW_C = 0.000\cdot \frac{kN}{m}$ (Pedestrian/Bicycle LL = 90 psf)

$M_{PL} := 0$

$M_{PL} = 0.00\cdot kN\cdot m$

6.2.8 Final Design Moment (Due to Design Truck + Lane Load + Pedestrian Load) x DF

$M_{LLf} := (M_{DT} + M_{lane} + M_{PL})\cdot DF_{M3.1}$

$M_{LLf} = 611.16\cdot kN\cdot m$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

7. STRESSES DUE TO EXTERNAL LOADS AT MIDSPAN

Note: *1. Positive values denotes compression*
2. Negative values denotes tension

7.1 Due to Beam/Girder Acting Alone

	TOP FIBER	BOTTOM FIBER
Due to Girder:	$f_{tg} := \frac{M_g}{S_t}$	$f_{bg} := \frac{M_g}{S_b}\cdot(-1)$
	$f_{tg} = 2.49\cdot MPa$	$f_{bg} = -2.04\cdot MPa$
Due to Slab:	$f_{t_slab} := \frac{M_{slab}}{S_t}$	$f_{b_slab} := \frac{M_{slab}}{S_b}\cdot(-1)$
	$f_{t_slab} = 2.72\cdot MPa$	$f_{b_slab} = -2.23\cdot MPa$
Due to Diaphragm:	$f_{t_diaph} := \frac{M_{diaph}}{S_t}$	$f_{b_diaph} := \frac{M_{diaph}}{S_b}\cdot(-1)$
	$f_{t_diaph} = 0.4\cdot MPa$	$f_{b_diaph} = -0.33\cdot MPa$

7.2 Due to Beam/Girder Acting as Composite

	TOP FIBER	BOTTOM FIBER
Due to Superimposed Dead Loads :	$f_{t_sup} := \frac{M_{sup}}{S_{t2comp}}$	$f_{b_sup} := \frac{M_{sup}}{S_{bcomp}}\cdot(-1)$
	$f_{t_sup} = 0.12\cdot MPa$	$f_{b_sup} = -0.37\cdot MPa$
Due to Live Loads : (Service III on Tension)	$f_{tLL} := \frac{M_{LLf}}{S_{t2comp}}$	$f_{bLL} := \frac{0.80\cdot M_{LLf}}{S_{bcomp}}\cdot(-1)$
	$f_{tLL} = 1.08\cdot MPa$	$f_{bLL} = -2.81\cdot MPa$

7.3 Total Stresses

Total Stresses at Top Fiber:

$ft_{total} := f_{tg} + f_{t_slab} + f_{t_diaph} + f_{t_sup} + f_{tLL}$ $\quad ft_{total} = 6.81\cdot MPa$

Total Stresses at Bottom Fiber:

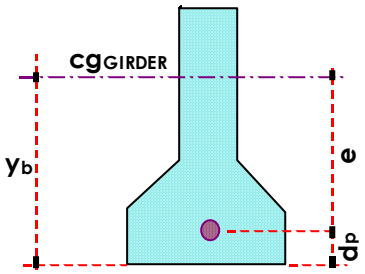
$fb_{total} := f_{bg} + f_{b_slab} + f_{b_diaph} + f_{b_sup} + f_{bLL}$ $\quad fb_{total} = -7.77\cdot MPa$

Note: *These stresses are used to simplify calculations. Girder will be tensioned so these stresses never occur.*

DESIGN OF PRESTRESSED INTERIOR GIRDER

8. REQUIRED PRESTRESSING STEEL

Eccentricity of prestressing steel:



$d_p := 125 \cdot mm$ distance of prestressing steel from c.g. to the bottom of girder

$e_c := y_b - d_p$ $\quad e_c = 389.60 \cdot mm$

$r_l := \frac{I_g}{A_g}$ $\quad r_l = 0.144\,m^2$

$c_b := e_c \cdot \frac{y_b}{r_l}$ $\quad c_b = 1.39$

From General Equation,

$f'_{t_allow} = f_{prestress} + f_{total}$ where: $f_{prestress} = P_f / A_g + P_f (e/Sb)$

$f_{total} = f_{b_total}$

Solving for Prestressing Force Required,

$P_f := \left(f'_{t_allow} - fb_{total}\right) \cdot \frac{A_g}{c_b + 1}$ $\quad P_f = 689.12 \cdot kN$

Maximum jacking stress (AASHTO 5.9.3): $f_{pi} := 0.75 \cdot f'_s = 1395.00 \cdot MPa$

Estimated total post tension loss: $Loss := 20\% \cdot f_{pi} = 279.00 \cdot MPa$

Working prestress required: $W_f := f_{pi} - Loss$ $\quad W_f = 1116.00 \cdot MPa$

REQUIRED PRESTRESSING STEEL

Required prestressing steel: $A1^*_s := \frac{P_f}{W_f}$ $\quad A1^*_s = 617 \cdot mm^2$

Note: *Area of 1- strand = 98.7 mm²* $\quad A_{str} := 98.7 \cdot mm^2$

No. of prestressing steel: $N_o := \frac{A1^*_s}{A_{str}}$ $\quad N_o = 6.26$ ***say,*** $N_o := 16$

Try 2-ducts - 8T12.7 strands per duct $\quad N_{duct} := 2$

Total Area of Prestressing Steel: $A^*_s := N_o \cdot A_{str}$ $\quad A^*_s = 1579 \cdot mm^2$

JACKING FORCE REQUIRED:

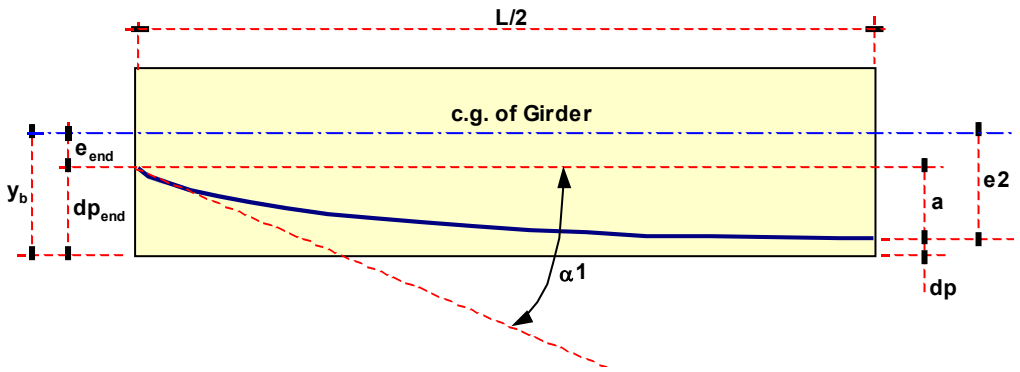
Note: AASHTO allows **0.90 f*$_y$** for short time prior to seating *(AASHTO 5.9.3)*
CALTRANS allows **0.75 f'$_s$**

Required Jacking Force: $P_j := 0.75 \cdot f'_s \cdot A^*_s$ $\quad P_j = 2202.98 \cdot kN$

DESIGN OF PRESTRESSED INTERIOR GIRDER

9. PRESTRESS LOSSES

9.1 Friction Losses (AASHTO 5.9.5.2.2b)



Determine c.g. of Prestressing Steel from bottom of Girder (at end):
Note: *Y = Distance from c.g. of duct to the bottom of girder.*

$n := N_{duct}$ $i := 1 .. n$

$Duct_i :=$

1
2

$N_{strand_i} :=$

8
8

$Y_i :=$

350
679

$N_{strand_i} \cdot Y_i =$

2800.00
5432.00

$Y := Y \cdot mm$

$$dp_{end} := \frac{\sum_{i=1}^{n} \left(N_{strand_i} \cdot Y_i\right)}{N_o}$$

$dp_{end} = 514.50 \cdot mm$

$a := dp_{end} - d_p$ $a = 389.50 \cdot mm$

Steel stress at jacking end: $T_o := \frac{P_j}{A^*_s}$ $T_o = 1395 \cdot MPa$

AT MIDSPAN:

Total angular change of prestressing steel profile from jacking to midspan

$$1 := L \cdot \frac{a}{\left(\frac{L}{2}\right)^2}$$ $1 = 0.108$

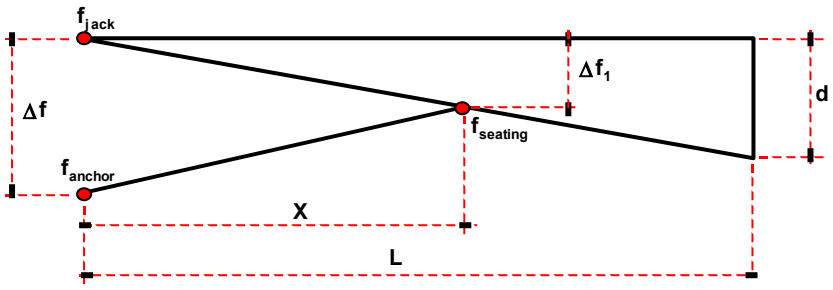
Friction Loss $FR := f_{pi} \cdot \left[1 - e^{-\left(k \cdot \frac{L}{2} + \cdot \ 1\right)}\right]$ $FR = 36.33 \cdot MPa$

Steel stress at midspan: $T_{x1} := f_{pi} - FR$ $T_{x1} = 1358.67 \cdot MPa$

DESIGN OF PRESTRESSED INTERIOR GIRDER

9.2 Anchorage Slip (Anchor Set Loss)

Note: Anchorage slip effect normally doesn't affect force at midspan when using Stress Relieved Strands, but must be checked and verified.



Jacking Stress at Jacking end before seating losses occur :

$f_{jack} := T_o$ $\quad f_{jack} = 1395.00 \cdot MPa$

Friction Loss in length L :

$d := T_o - T_{x1}$ $\quad d = 36.33 \cdot MPa$

Anchor set (normally 9.5mm)

$L := 9.5 \cdot mm$ <<< *From CALTRANS*

Length influenced by anchor set:

$$X := \sqrt{\frac{E_p \cdot L \cdot L}{d}} \qquad X = 27.23 \cdot m$$

Change in stress due to anchor set:

$$f := \frac{2 \cdot d \cdot X}{L} \qquad f = 137.44 \cdot MPa$$

$$f1 := \frac{d \cdot X}{L} \qquad f1 = 68.72 \cdot MPa$$

Anchorage Slip Loss at midspan

$$AN := \begin{cases} 0.00 \cdot MPa & \text{if } X \le \frac{L}{2} \\ \frac{f \cdot \left(X - \frac{L}{2}\right)}{X} & \text{otherwise} \end{cases} \qquad AN = 101.10 \cdot MPa$$

9.3 Elastic Shortening Loss *(AASHTO 5.9.5.2.3b)*

Maximum moment due to girder dead load: $M_g = 206.4 \cdot kN \cdot m$

Concrete stress at the c.g. of prestressing steel due to prestressing force and girder immediately after release of anchorage(At this stage, initial stress in the tendons has been reduced by elastic shortening of concrete and tendon friction).

$P_{jcir} := 0.70 \cdot f'_s \cdot A^*_s$

$P_{jcir} = 2056.12 \cdot kN$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Number of identical prestressing tendons:

$N_i := 2$

Sum of concrete stress at center of gravity of tendons:

$$f_{cgp} := \left(\frac{P_{jcir}}{A_g} + P_{jcir} \cdot \frac{e_c^2}{I_g} \right) - M_g \cdot \frac{e_c}{I_g}$$

$f_{cgp} = 10.13 \cdot MPa$

Elastic Shortening Loss:

$$ES := \frac{N_i - 1}{2 \cdot N_i} \cdot \frac{E_p}{E_{ci}} \cdot f_{cgp}$$

$ES = 16.16 \cdot MPa$

9.4 Time-Dependent Losses *(AASHTO 5.9.5.3)*

Note: Long-term prestress loss due to creep and shrinkage of concrete, and relaxation of steel

Average annual ambient relative humidity (%):

$H_u := 79$

Estimate relaxation loss for low relaxation strand:

$f_{pR} := 17 \cdot MPa$

Correction factor for relative humidity of the ambient air:

$_h := 1.7 - 0.01 \cdot H_u = 0.91$

Correction factor for specified concrete strength at time of prestress transfer to the concrete member:

$$_{st} := \frac{35}{7 + \frac{f'_{ci}}{MPa}} = 0.83$$

Initial prestressing steel stress immediately prior to transfer:

$f_{pi} = 1395.00 \cdot MPa$

Total long-term prestress loss:

$$f_{pLT} := 10 \cdot f_{pi} \cdot \frac{A^*_s}{A_g} \cdot {}_h \cdot {}_{st} + 83MPa \cdot {}_h \cdot {}_{st} + f_{pR}$$

$f_{pLT} = 125.69 \cdot MPa$

9.5 TOTAL LOSSES:

9.5.1 At Transfer Stage

$Loss_t := FR + AN + ES$

$Loss_t = 153.59 \cdot MPa$

9.5.2 At Service Stage

$Loss_s := FR + AN + ES + f_{pLT}$

$Loss_s = 279.28 \cdot MPa$

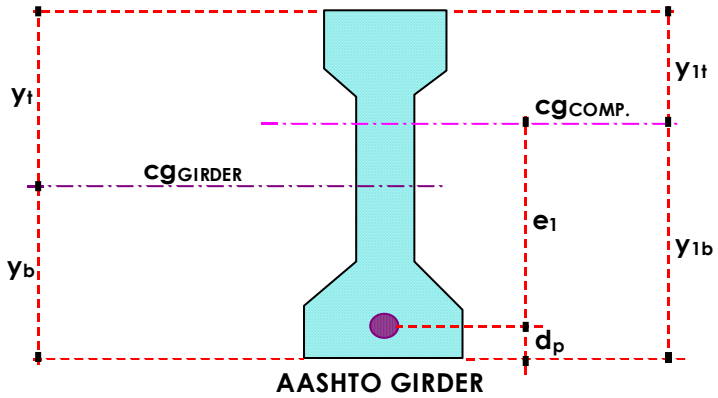
DESIGN OF PRESTRESSED INTERIOR GIRDER

10. CHECK STRESSES OF GIRDER AT DIFFERENT STAGES

10.1 STAGE I - *Tension Girder (Net Section)*

Ungrouted Tendons:Girder Dead Load resist uplift from tensioning

Deduct holes for the ducts.



10.1.1 **STAGE I** : Net Section Properties

Duct Diameter:

$$D_{duct} := \left(2 + \frac{5}{8}\right)\cdot in \qquad D_{duct} = 0.067\,m$$

Duct Moment of Inertia:

$$I_{duct} := \frac{\cdot D_{duct}^{4}}{64}\cdot N_{duct} \qquad I_{duct} = 0.00000194\,m^{4}$$

Duct Area:

$$A_{duct} := \frac{\cdot D_{duct}^{2}}{4}\cdot N_{duct} \qquad A_{duct} = 0.007\cdot m^{2}$$

Note: (Using only 50% of mild steel)

Area of non-prestressing steel (bottom bars) $A_{stL} := 402\cdot mm^{2}$

Modulus of Elasticity of non-prestressing steel $E_{stL} := 200000\cdot MPa$

Depth of mild steel from girder bottom $d_{stL} := 60\cdot mm$

Modular ratio:

$$n_{stL} := \frac{E_{stL}}{E_{ci}} \qquad n_{stL} = 6.476$$

Equivalent concrete area of steel:

$$A_{equiv_stL} := n_{stL}\cdot A_{stL} \qquad A_{equiv_stL} = 0.00\cdot m^{2}$$

Total concrete net area :

$$A_{net1} := A_g - A_{duct} + A_{equiv_stL} \qquad A_{net1} = 0.3569\cdot m^{2}$$

Determine y_{1b} about bottom of girder:

$$y_{1b} := \frac{A_g\cdot y_b - A_{duct}\cdot d_p + A_{equiv_stL}\cdot d_{stL}}{A_{net1}} \qquad y_{1b} = 518.91\cdot mm$$

$$y_{1t} := h_g - y_{1b} \qquad y_{1t} = 624.09\cdot mm$$

DESIGN OF PRESTRESSED INTERIOR GIRDER

Determine Moment of Inertia:

$$I_{1bottom} := \left(I_g + A_g \cdot y_b^{\,2}\right) - \left(I_{duct} + A_{duct} \cdot d_p^{\,2}\right) + A_{equiv_stL} \cdot d_{stL}^{\,2} \qquad I_{1bottom} = 0.1477\,m^4$$

$$I_{1cg} := I_{1bottom} - A_{net1} \cdot y_{1b}^{\,2} \qquad I_{1cg} = 0.0516\,m^4$$

Eccentricity e_1: $\quad e_1 := y_{1b} - d_p \qquad e_1 = 393.91 \cdot mm$

10.1.2 **STAGE I** : Stresses at Midspan due to External Loads

Note: *At this Stage, considered only the stress due to Girder weight.*

	TOP FIBER	**BOTTOM FIBER**
Due to Girder:	$f_{tg} := \dfrac{M_g \cdot y_{1t}}{I_{1cg}}$	$f_{bg} := \dfrac{M_g \cdot y_{1b}}{I_{1cg}} \cdot (-1)$
	$f_{tg} = 2.5 \cdot MPa$	$f_{bg} = -2.07 \cdot MPa$

10.1.3 **STAGE I** : Effective Prestress Force

Note: Deduct friction loses from jacking force. Some elastic shortening can be included when jacking multiple tendons separately.

$$P_1 := P_j - \left(Loss_t\right) \cdot A^*_s \qquad P_1 = 1960.43 \cdot kN$$

10.1.4 **STAGE I** : Final Stresses at Midspan - Girder Only

Stress @ top

$$f_{t1} := \left(\frac{P_1}{A_{net1}} - P_1 \cdot \frac{e_1 \cdot y_{1t}}{I_{1cg}}\right) + f_{tg} \qquad f_{t1} = -1.35 \cdot MPa$$

$$CHECK_f_{t1} := \begin{cases} \text{"OK"} & \text{if } \left(f_{t1} > 0\right) \cdot \left(f_{t1} \le f'_{ci_allow}\right) \\ \text{"OK"} & \text{if } \left(f_{t1} < 0\right) \cdot \left(f_{t1} \ge f'_{ti_allow}\right) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{t1} = \text{"OK"}$$

Stress @ bottom

$$f_{b1} := \left(\frac{P_1}{A_{net1}} + P_1 \cdot \frac{e_1 \cdot y_{1b}}{I_{1cg}}\right) + f_{bg} \qquad f_{b1} = 11.18 \cdot MPa$$

$$CHECK_f_{b1} := \begin{cases} \text{"OK"} & \text{if } \left(f_{b1} > 0\right) \cdot \left(f_{b1} \le f'_{ci_allow}\right) \\ \text{"OK"} & \text{if } \left(f_{b1} < 0\right) \cdot \left(f_{b1} \ge f'_{ti_allow}\right) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{b1} = \text{"OK"}$$

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10.2 STAGE II - *Grout Tendons, Set Girder and pour Deckslab & Diaphragm*

10.2.1 **STAGE II** : Transformed Section Properties(Grouted Tendons)

Equivalent concrete area of prestressing steel, A_{eq}

$$n_1 := \frac{E_p}{E_{ci}} \qquad n_1 = 6.379$$

$$A_{eq} := A^*_s \cdot (n_1 - 1) \qquad A_{eq} = 0.01 \cdot m^2$$

Total transformed area:

$$A_{net2} := A_g + A_{eq} \qquad A_{net2} = 0.37 \cdot m^2$$

Determine y_{2b} about bottom of girder

$$y_{2b} := \frac{A_g \cdot y_b + A_{eq} \cdot d_p}{A_{net2}} \qquad y_{2b} = 506 \cdot mm$$

$$y_{2t} := h_g - y_{2b} \qquad y_{2t} = 637 \cdot mm$$

Eccetricity e_2:

$$e_2 := y_{2b} - d_p \qquad e_2 = 381 \cdot mm$$

Determine Moment of Inertia

$$I_{2bot} := I_g + A_g \cdot y_b^2 + A_{eq} \cdot d_p^2 \qquad I_{2bot} = 0.148\,m^4$$

$$I_{2cg} := I_{2bot} - A_{net2} \cdot y_{2b}^2 \qquad I_{2cg} = 0.0534\,m^4$$

10.2.2 **STAGE II** : Stresses at Midspan due to External Loads

Note: *At this Stage, considered only the stress due to Girder weight, Deck slab, and Diaphragm.*

	TOP FIBER	**BOTTOM FIBER**
Due to Deck slab	$f_{t_slab} := \frac{M_{slab} \cdot y_{2t}}{I_{2cg}}$	$f_{b_slab} := \frac{M_{slab} \cdot y_{2b}}{I_{2cg}} \cdot (-1)$
	$f_{t_slab} = 2.69 \cdot MPa$	$f_{b_slab} = -2.14 \cdot MPa$
Due to Diaphragm	$f_{t_diaph} := \frac{M_{diaph} \cdot y_{2t}}{I_{2cg}}$	$f_{b_diaph} := \frac{M_{diaph} \cdot y_{2b}}{I_{2cg}} \cdot (-1)$
	$f_{t_diaph} = 0.4 \cdot MPa$	$f_{b_diaph} = -0.31 \cdot MPa$

10.2.3 **STAGE II** : Effective Prestress Force

Note: At this Stage, 40% losses is assumed, plus friction losses and elastic shortening and anchorage slip. This value should be based on actual constuction schedule.

$$P_2 := P_j - (Loss_t + 40\% \cdot f_{pLT}) \cdot A^*_s \qquad P_2 = 1881.03 \cdot kN$$

DESIGN OF PRESTRESSED INTERIOR GIRDER

10.2.4 **STAGE II** : Final Stresses at Midspan - Grouted Tendons, Poured Deckslab and Diaphragm

Stress @ top

$$f_{t2} := \left(\frac{P_2}{A_{net2}} - P_2 \cdot \frac{e_2 \cdot y_{2t}}{I_{2cg}}\right) + f_{tg} + f_{t_slab} + f_{t_diaph} \qquad f_{t2} = 2.13 \cdot MPa$$

$$CHECK_f_{t2} := \begin{cases} \text{"OK"} & \text{if } (f_{t2} > 0) \cdot (f_{t2} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{t2} < 0) \cdot (f_{t2} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{t2} = \text{"OK"}$$

Stress @ bottom

$$f_{b2} := \left(\frac{P_2}{A_{net2}} + P_2 \cdot \frac{e_2 \cdot y_{2b}}{I_{2cg}}\right) + f_{bg} + f_{b_slab} + f_{b_diaph} \qquad f_{b2} = 7.34 \cdot MPa$$

$$CHECK_f_{b2} := \begin{cases} \text{"OK"} & \text{if } (f_{b2} > 0) \cdot (f_{b2} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{b2} < 0) \cdot (f_{b2} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{b2} = \text{"OK"}$$

10.3 STAGE III - *Superstructure Construction Complete (Superimposed Dead Load plus Live Loads)*

10.3.1 **STAGE III** : Composite Transformed Section Properties

Total transformed area:

$$A_{net2} = 0.37 \cdot m^2$$

Deck Slab properties:

$$A_{slab} := t_s \cdot b_{eff} \qquad A_{slab} = 0.31 \cdot m^2$$

$$I_{slab} := \frac{b_{eff} \cdot t_s^3}{12} \qquad I_{slab} = 1.27 \times 10^{-3}\, m^4$$

Haunch properties:

$$A_{haunch} := w_{eff} \cdot t_{haunch} \qquad A_{haunch} = 0.008\, m^2$$

$$I_{haunch} := \frac{w_{eff} \cdot t_{haunch}^3}{12} \qquad I_{haunch} = 4.3164 \times 10^{-7} \cdot m^4$$

Total Composite Transformed Area:

$$A_{net3} := A_{net2} + A_{slab} + A_{haunch} \qquad A_{net3} = 0.69 \cdot m^2$$

Determine y_{3b} about bottom of girder:

$$y_{3b} := \frac{\left[(A_{net2}) \cdot y_{2b} + A_{slab} \cdot \left(h_g + t_{haunch} + \frac{1}{2} \cdot t_s\right)\right] \ldots + A_{haunch} \cdot \left(h_g + \frac{t_{haunch}}{2}\right)}{A_{net3}} \qquad y_{3b} = 0.864\, m$$

$$y_{3t} := h_g - y_{3b} \qquad y_{3t} = 0.28\, m$$

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Detemine Moment of Inertia:

$$I_{3bottom} := \left(I_{2cg} + A_{net2}\cdot y_{2b}^{2}\right) \ldots + \left[I_{slab} + A_{slab}\cdot\left(h_g + t_{haunch} + \frac{1}{2}\cdot t_s\right)^{2}\right] \ldots + \left[I_{haunch} + A_{haunch}\cdot\left(h_g + \frac{t_{haunch}}{2}\right)^{2}\right] \qquad I_{3bottom} = 0.6737\,m^4$$

$$I_{3cg} := I_{3bottom} - A_{net3}\cdot y_{3b}^{2} \qquad I_{3cg} = 0.1568\,m^4$$

10.3.2 **STAGE III** : Stresses at Midspan due to External Loads

Note: *At this Stage, the external loads are Girder, Deck slab, Diaphragm, Superimposed dead load, Live load and Pedestrian Load.*

	TOP FIBER	**BOTTOM FIBER**
Due to LL + Sup. DL (Service III on Tension)	$f_{tLL_sup} := \frac{(M_{LLf} + M_{sup})\cdot y_{3t}}{I_{3cg}}$	$f_{bLL_sup} := \frac{(0.80\cdot M_{LLf} + M_{sup})\cdot y_{3b}}{I_{3cg}}\cdot(-1)$
	$f_{tLL_sup} = 1.2\cdot MPa$	$f_{bLL_sup} = -3.05\cdot MPa$

10.3.3 **STAGE III** : Effective Prestress Force

Note: At this Stage (Final Stage), Final Prestress Force after all losses.

$$P_3 := P_j - Loss_s\cdot A^*_s \qquad P_3 = 1761.94\cdot kN$$

10.3.4 **STAGE III** : Final Stresses at Midspan - Service Condition

Stress @ top

$$f_{t3} := \left(\frac{P_3}{A_{net2}} - P_3\cdot\frac{e_2\cdot y_{2t}}{I_{2cg}}\right) + f_{tg} + f_{t_slab} \ldots + f_{t_diaph} + f_{tLL_sup} \qquad f_{t3} = 3.55\cdot MPa$$

$$f'_{c_allow} = 18.63\cdot MPa$$

$$CHECK_f_{t3} := \begin{cases} \text{"OK"} & \text{if } (f_{t3} > 0)\cdot(f_{t3} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{t3} < 0)\cdot(f_{t3} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{t3} = \text{"OK"}$$

Stress @ bottom

$$f_{b3} := \left(\frac{P_3}{A_{net2}} + P_3\cdot\frac{e_2\cdot y_{2b}}{I_{2cg}}\right) + f_{bg} + f_{b_slab} \ldots + f_{b_diaph} + f_{bLL_sup} \qquad f_{b3} = 3.53\cdot MPa$$

$$f'_{t_allow} = -3.22\cdot MPa$$

$$CHECK_f_{b3} := \begin{cases} \text{"OK"} & \text{if } (f_{b3} > 0)\cdot(f_{b3} \le f'_{c_allow}) \\ \text{"OK"} & \text{if } (f_{b3} < 0)\cdot(f_{b3} \ge f'_{t_allow}) \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_f_{b3} = \text{"OK"}$$

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11. CHECK FLEXURAL RESISTANCE VS FACTORED MOMENT

Resistance factor (AASHTO 5.5.4.2): $:= 0.95$

Total prestressing steel area $A^*_s = 1579.20\cdot mm^2$

Ultimate Strength of prestressing steel $f'_s = 1860.00\cdot MPa$

Yield Strength of prestressing steel $f^*_y = 1674.00\cdot MPa$

Area of mild steel tension reinforcement:

$$A_{stL} := 4\cdot\frac{\cdot(16mm)^2}{4} = 804.25\cdot mm^2$$

Yield Strength of intermediate grade steel $f'_{sy} = 414.00\cdot MPa$

Prestressed Concrete Compressive Strength $f'_c = 41.40\cdot MPa$

Pretressing steel type factor (AASHTO 5.5.4.2):

$$k_f := 2\cdot\left(1.04 - \frac{f^*_y}{f'_s}\right) = 0.28$$

$$_1 := \begin{cases} 0.85 & \text{if } f'_c \le 28\cdot MPa \\ 0.85 - \frac{0.05}{7\cdot MPa}\cdot\left(f'_c - 28MPa\right) & \end{cases}$$

(β_1 is not less than 0.65) $_1 = 0.75$

Distance of c.g. of prestressing steel to top of slab:

$$d_{pt} := h_g + t_{haunch} + t_s - d_p$$

$d_{pt} = 1.26\,m$

Top slab (flange) thickness: $t_s = 220.00\cdot mm$

Top slab (flange) effective width: $b_{eff} = 1.43\,m$

Distance from extreme compression fiber to neutral axis (assumed rectangular section behavior):
Note: compression reinforcement conservatively ignored

$$c_r := \frac{A^*_s\cdot f'_s + A_{stL}\cdot f'_{sy}}{0.85\cdot f'_c\cdot\ _1\cdot b_{eff} + k_f\cdot A^*_s\cdot\frac{f'_s}{d_{pt}}}$$

$c_r = 84.77\cdot mm$

$$CHECK := \begin{cases} \text{"N.A. is inside the flange, Analyze as Rectangular Section"} & \text{if } c_r \le t_s \\ \text{"N.A. is outside the flange, Analyze as T-Beam Section"} & \text{otherwise} \end{cases}$$

CHECK = "N.A. is inside the flange, Analyze as Rectangular Section"

Depth of equivalent stress block: $:= c_r\cdot\ _1 = 63.94\cdot mm$

Distance from extreme compression fiber to centroid of mild tension reinforcement:

$d_s := h_g + t_{haunch} + t_s - d_{stL}$

$d_s = 1.33\cdot m$

Stress in prestressing steel (AASHTO 5.7.3.1.1-1):

$$f_{ps} := f'_s\cdot\left(1 - k_f\cdot\frac{c_r}{d_{pt}}\right) \qquad f_{ps} = 1825.04\cdot MPa$$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Norminal resistance (rectangular section behavior):

$$M_{nr} := A^*_s \cdot f_{ps} \cdot \left(d_{pt} - \frac{}{2} \right) + A_{stL} \cdot f'_{sy} \cdot \left(d_s - \frac{}{2} \right) \qquad M_{nr} = 3979.49 \cdot kN \cdot m$$

FLEXURAL RESISTANCE (AASHTO 5.7.3.2):

$$M_r := \cdot M_{nr} \qquad M_r = 3780.51 \cdot kN \cdot m$$

Actual Ultimate Factored Moment Applied: (STRENGTH I)

$$M_u := 1.05\left[1.25 \cdot \left(M_g + M_{slab} + M_{diaph}\right) + 1.50 \cdot M_{sup} + 1.75 \cdot M_{LLf}\right] \qquad M_u = 1836.55 \cdot kN \cdot m$$

$$\text{Flexure_check} := \text{if}\left(M_r \geq M_u, \text{"OK"}, \text{"FAIL"}\right) \qquad \text{Flexure_check} = \text{"OK"}$$

MINIMUM PRESTRESSED AND NON-PRESTRESSED TENSILE STEEL REQUIREMENT (AASHTO 5.7.3.3.2)

Section Modulus at bottom of girder, composite section:

$$S_{bc} := \frac{I_{3cg}}{y_{3b}}$$

$$S_{bc} = 0.1814 \cdot m^3$$

Section Modulus at bottom of girder, non-composite section:

$$S_b = 0.1014 \cdot m^3$$

Modulus of rupture of concrete: $f_r = 4.05 \cdot MPa$

Flexural cracking variability factor: $_1 := 1.6$

Prestress variability factor: $_2 := 1.1$

Ratio of fy/fu factor (1.0 for prestress concrete structures): $_3 := 1.0$

Compressive stress in concrete due to effective prestress forces only (after all loses):

$$f_{cpe} := \frac{P_3}{A_g} + \frac{P_3 \cdot e_c}{S_b} \qquad f_{cpe} = 11.65 \cdot MPa$$

Total unfactored DL moment acting on non-composite section:

$$M_{dnc} := M_g + M_{slab} + M_{diaph}$$

$$M_{dnc} = 465.30 \cdot kN \cdot m$$

Unless otherwise specified, at any section of a flexural component, the amount of prestressed and non-prestressed tensile reinforcement shall be adequate to develop a factored flexural resistance, Mr, at least equal to the lesser of:

$$M_{r1} := 1.33 \cdot M_u = 2442.62 \cdot kN \cdot m$$

or the cracking strength:

$$M_{cr} := {}_3 \cdot \left[\left({}_1 \cdot f_r + {}_2 \cdot f_{cpe} \right) \cdot S_{bc} - M_{dnc} \cdot \left(\frac{S_{bc}}{S_b} - 1 \right) \right] \qquad M_{cr} = 3133.84 \cdot kN \cdot m$$

$$\text{Mmin} := \text{if}\left(M_{r1} > M_{cr}, \text{"Mcr!"}, \text{"1.33Mu!"}\right) \qquad \text{Mmin} = \text{"1.33Mu!"}$$

Minimum flexural resistance:

$$M_{rmin} := \min\left(M_{r1}, M_{cr}\right) \qquad M_{rmin} = 2442.62 \cdot kN \cdot m$$

$$\text{Min_check} := \text{if}\left(M_r \geq M_{rmin}, \text{"OK"}, \text{"FAIL"}\right) \qquad \text{Min_check} = \text{"OK"}$$

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12. DESIGN SHEAR FORCES AT SUPPORT

12.1 Shear Due to Dead Loads

Due to Girder: $V_g := w_g \cdot \frac{L}{2}$ $V_g = 62.43 \cdot kN$

Due to Slab: $V_{slab} := w_{slab} \cdot \frac{L}{2}$ $V_{slab} = 68.28 \cdot kN$

Due to Int. Diaphragm: $V_{diaph} := \frac{P_{diaph}}{2}$ $V_{diaph} = 4.80 \cdot kN$

Due to End Diaphragm: $V_{ediaph} := 0.30m \cdot 1.25m \cdot 1.60m \cdot \gamma_{conc}$ $V_{ediaph} = 14.40 \cdot kN$

Due to Sidewalk and Railings (Superimposed DC): *(From StaadPro Analysis of Continuous Girders)*

$V_{siderail} := 0 \cdot kN$

Due to wearing surface and utilities (DW): *(From StaadPro Analysis of Continuous Girders)*

$w_{fwsg} := w_{fws}$ $w_{utilities} = 1.00 \cdot \frac{kN}{m}$

$w_{DW} := w_{fwsg} + w_{utilities}$ $w_{DW} = 4.08 \cdot \frac{kN}{m}$

$V_{DW} := 35.2 \cdot kN$ $V_{DW} = 35.20 \cdot kN$

Total Shear due to Dead Loads: $V_{dl} := V_g + V_{slab} + V_{diaph} + V_{ediaph} + V_{siderail} + V_{DW}$ $V_{dl} = 185.11 \cdot kN$

12.2 Shear Due to Live Loads

12.2.1 Distribution Factor of Live Loads Per Lane for Shear in Interior Beams

Dynamic Load Allowance Factor $IM = 1.33$

Girder Spacing $S_g = 1.75\,m$

Distribution Factor for One Design Lane Loaded:

$DF_{S1} := 0.36 + \frac{S_g}{7600 \cdot mm}$ $DF_{S1} = 0.590$

Distribution Factor for Two or More Design Lane Loaded:

$DF_{S2} := 0.2 + \frac{S_g}{3600 \cdot mm} - \left(\frac{S_g}{10700mm}\right)^2$ $DF_{S2} = 0.659$

Note: If Ng=3, use lever rule

DESIGN OF PRESTRESSED INTERIOR GIRDER

Using lever rule, since Ng = 3:

$DF_{S3.1} := DF_{M3.1} = 0.60$

$DF_{S3.2} := DF_{M3.2} = 0.50$

12.2.2 Truck Loading

Note: Live Load Shears are from StaadPro Analysis for Simple Span.

Design Truck *Design Tandem*

$V_{Tr} := 276.5 \cdot kN$ $V_T := 211 \cdot kN$

Unfactored Design Truck/Tandem + IM

$V_{DT} := \max(V_{Tr}, V_T) \cdot IM$ $V_{DT} = 367.75 \cdot kN$

12.2.3 Lane Loading

$V_{lane} := 83\ kN$ $V_{lane} = 83.00 \cdot kN$

12.2.4 Unfactored Shear (Due to Design Truck + Lane Load) x DF

$V_{LL} := (V_{DT} + V_{lane}) \cdot DF_{S3.1}$ *(lever rule)*

$V_{LL} = 270.45 \cdot kN$

12.2.5 Permit Truck Loading

Caltrans P-7 *Note:* Live Load Shears are from StaadPro Analysis for Continuous Girders

$V_{PT} := 401 \cdot kN$

Unfactored Permit Load + IM

$V_{PTim} := V_{PT} \cdot IM_P$

$V_{PTim} = 501.25 \cdot kN$

Final force applied to girder (AASHTO 4.6.2.2.5)

$Z = 1.00$ Factor *(1.0 if lever rule is used)*

$$V_P := V_{PTim} \cdot \left(\frac{DF_{S3.1}}{Z} \right) + V_{LL} \cdot \left(DF_{S3.2} - \frac{DF_{S3.1}}{Z} \right)$$

$V_P = 273.71 \cdot kN$

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12.2.6 Strength Limit States Check

Strength I Limit State (Design Truck):

$V_{uL} := 1.75 \cdot V_{LL}$

$V_{uL} = 473.28 \cdot kN$

Strength II Limit State (Permit Truck):

$V_{uP} := 1.35 \cdot V_P$

$V_{uP} = 369.50 \cdot kN$

$$Strength_check_v := \begin{cases} \text{"Design Truck governs!"} & \text{if } V_{uL} \geq V_{uP} \\ \text{"Permit Truck governs!"} & \text{otherwise} \end{cases}$$

$Strength_check_v = \text{"Design Truck governs!"}$

12.2.7 Pedestrian and Bicycle Loading

$V_{PL} := 0$ $\quad V_{PL} = 0.00 \cdot kN$

12.2.8 Final Design Shear (Due to Design Truck + Lane Load + Pedestrian Load) x DF

$V_{LLf} := (V_{DT} + V_{lane} + V_{PL}) \cdot DF_{S3.1}$ (lever rule)

$V_{LLf} = 270.45 \cdot kN$

12.3 Total Factored Shear Force (STRENGTH I):

$V_u := 1.05\left[1.25 \cdot (V_g + V_{slab} + V_{diaph} + V_{ediaph} + V_{siderail}) + 1.50 \cdot V_{DW} + 1.75 \cdot V_{LLf}\right]$

$V_u = 749.15 \cdot kN$

13. DESIGN SHEAR RESISTANCE

Shear Resistance factor (AASHTO 5.5.4.2): $_s := 0.90$

Effective web width: $b_v := T_w = 178 \cdot mm$

Effective shear depth dv (greater of 0.9de or 0.72h):

$$d_e := \frac{A^*_s \cdot f_{ps} \cdot d_{pt} + A_{stL} \cdot f'_{sy} \cdot d_s}{A^*_s \cdot f_{ps} + A_{stL} \cdot f'_{sy}}$$

$d_e = 1.27 \cdot m$

$0.9 \cdot d_e = 1.143\,m$

Height of composite member:

$h := h_g + t_s + t_{haunch}$

$h = 1.39\,m$

$0.72 \cdot h = 0.999\,m$

$d_v := \max(0.9 \cdot d_e, 0.72 \cdot h)$ $\quad d_v = 1.143\,m$

Unfactored shear force due to dead load: $V_{dl} = 185 \cdot kN$

Factored shear force at section due to external load occuring simultaneously with M_{max}. $V_u = 749.15 \cdot kN$

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$V_u = 749.15 \cdot kN$

Unfactored shear force due to dead load: $V_{dl} = 185 \cdot kN$

Factored shear force at section due to external load occuring simultaneously with M_{max}.

Total unfactored DL moment acting on non-composite section at support:

$$M_{dncs} := 0$$

Max. factored moment at section due to externally applied loads:

$$M_{max} := -\left(1.50 \cdot M_{sup} + 1.75 \cdot M_{LLf}\right) = -1167.48 \cdot kN \cdot m$$

Eccentricity between center of girder to center of prestress at end support:

$$e_{end} := y_b - dp_{end} \qquad e_{end} = 0.10 \cdot mm$$

Compressive stress in concrete due to effective prestress forces only (after all loses) at bottom of girder:

$$f_{cpe} := \frac{P_3}{A_g} + \frac{P_3 \cdot e_{end}}{S_b} \qquad f_{cpe} = 4.88 \cdot MPa$$

Modulus of rupture when used to calculate the cracking moment of a member in Artide 5.8.3.4.3:

$$f_{rc} := 0.52 \cdot \sqrt{f'_c \cdot MPa} \qquad f_{rc} = 3.35 \cdot MPa$$

Moment causing flexural cracking due to externally applied loads (AASHTO 5.8.3.4.3-2) at support:

$$M_{cre} := S_{bc} \cdot \left(f_{rc} + f_{cpe} - \frac{M_{dncs}}{S_b} \right) \qquad M_{cre} = 1492.04 \cdot kN \cdot m$$

Nominal shear resistance provided by concrete when inclined cracking results from combined shear and moment (AASHTO 5.8.3.4.3-1):

$$V_{ci1} := \begin{cases} \left(0.0525 \cdot \sqrt{f'_c \cdot MPa} \cdot b_v \cdot d_v + V_{dl}\right) & \text{if } M_{max} \le 0 \\ \left(0.0525 \cdot \sqrt{f'_c \cdot MPa} \cdot b_v \cdot d_v + V_{dl} + \frac{V_u \cdot M_{cre}}{M_{max}}\right) & \text{otherwise} \end{cases}$$

$V_{ci1} = 253.83 \cdot kN$

$$V_{ci2} := 0.16 \cdot \sqrt{f'_c \cdot MPa} \cdot b_v \cdot d_v = 209.41 \cdot kN$$

$$V_{ci} := \max\left(V_{ci1}, V_{ci2}\right)$$

$V_{ci} = 253.83 \cdot kN$

Compressive stress in concrete (after allowance for all prestress losses) at centroid of cross section resisting externally applied load

$$f_{pc} := \frac{P_3}{A_g}$$

$f_{pc} = 4.88 \cdot MPa$

Vertical component of prestressing force, Vp is taken as zero when AASHTO 5.8.3.4.3 is applied:

$$V_p := 0 \cdot kN$$

Nominal shear resistance provided by concrete when inclined cracking results from excessive principal tensions in web (AASHTO 5.8.3.4.3-3):

$$V_{cw} := \left(0.16 \cdot \sqrt{f'_c \cdot MPa} + 0.30 \cdot f_{pc}\right) \cdot b_v \cdot d_v + V_p$$

$V_{cw} = 507.00 \cdot kN$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Controlling shear resistance provided by concrete:

$V_c := \min(V_{ci}, V_{cw})$

$V_c = 253.83 \cdot kN$

Design shear resistance needed by transverse reinforcement:

$V_s := \frac{V_u}{s} - V_c$

$V_s = 578.56 \cdot kN$

DESIGN STIRRUP SPACING:

Diameter of Stirrups: $d_s := 12 \cdot mm$

Area of Stirrups: $A_v := \frac{\cdot d_s^{\,2}}{4}$

$A_v = 113.10 \cdot mm^2$

No. of legs of stirrups: $N_{legs} := 2$

$s_1 := \frac{N_{legs} \cdot A_v \cdot f'_{sy} \cdot d_v}{V_s}$

$s_1 = 184.96 \cdot mm$

say, $s_1 := 150 \cdot mm$

SHEAR RESISTANCE (AASHTO 5.8.3.3):

Shear resistance provided by transverse reinforcement:

$V_s := \frac{N_{legs} \cdot A_v \cdot f'_{sy} \cdot d_v}{s_1}$ $\quad V_s = 713.42 \cdot kN$

Nominal shear resistance:

$V_{n1} := V_c + V_s$ $\quad V_{n1} = 967.25 \cdot kN$

$V_{n2} := 0.25 \cdot f'_c \cdot b_v \cdot d_v$ $\quad V_{n2} = 2105.30 \cdot kN$

$V_n := \min(V_{n1}, V_{n2})$ $\quad V_n = 967.25 \cdot kN$

Shear resistance:

$V_r := \ _s \cdot V_n = 870.52 \cdot kN$

Factored shear force:

$V_u = 749.15 \cdot kN$

$Shear_check := \text{if}(V_r \geq V_u, \text{"OK"}, \text{"FAIL"})$

$Shear_check = \text{"OK"}$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

14. DOWELS AT GIRDER ENDS

The required dowels in girder ends to transfer girder reactions into diaphragms. Use shear friction, AASHTO 5.8.4

Concrete Area resisting shear transfer: $A_{cv} := A_g$ $A_{cv} = 0.36 \cdot m^2$

Diaphragm concrete strength: $f'_{c_diaph} := 27.6 \cdot MPa$

Shear reduction factor: $_v := 0.85$

Diaphragm reinforcement strength: $f'_{sy} := 414 \cdot MPa$

Friction factor (surface roughened): $_{} := 1.00$

Reinforcement diameter: $d_b := 20 \cdot mm$ $V_u = 749.15 \cdot kN$

Nominal Shear Force: $V_n := \frac{V_u}{_v}$ $V_n = 881.35 \cdot kN$

$$V_{n_max} := \begin{cases} 0.20 \cdot f'_{c_diaph} \cdot A_{cv} & \text{if } \left(0.20 \cdot f'_{c_diaph} \cdot A_{cv}\right) \le 0.80ksi \cdot A_{cv} \\ 0.80ksi \cdot A_{cv} & \text{otherwise} \end{cases} \qquad V_{n_max} = 1992.86 \cdot kN$$

$$CHECK_V_n := \begin{cases} \text{"OK"} & \text{if } V_{n_max} > V_n \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad CHECK_V_n = \text{"OK"}$$

Required steel area of dowel reinforcement:

$A_{dowel} := \frac{\cdot d_b^2}{4}$ $A_{dowel} = 314.16 \cdot mm^2$

$A_v := \frac{V_n}{f'_{sy} \cdot}$ $A_v = 2128.87 \cdot mm^2$

$N_dowels := \frac{A_v}{A_{dowel}}$ $N_dowels = 6.78$ **Say 5 each face, 10 total**

Determine embedment length:

$Min := 0.4 \cdot d_b \cdot \frac{f'_{sy}}{ksi}$ $Min = 480.36 \cdot mm$

$A_b := A_{dowel}$

$factor1 := 1.40$ *(AASHTO 5.11.2.1.2)*

$factor2 := 0.80$ *(AASHTO 5.11.2.1.3)*

$Basic := \frac{1.25 \cdot A_b \cdot f'_{sy}}{\sqrt{f'_{c_diaph} \cdot ksi \cdot in}}$ $Basic = 463.99 \cdot mm$

$Embedment := factor1 \cdot factor2 \cdot Basic$ $Embedment = 520 \cdot mm$

Dowel length = embedment length

NOTE: Bars spaced at 150 mm min and 75 mm min cover. Distance between girder should be greater than dowel length plus clearance, otherwise hooks will be needed.

DESIGN OF PRESTRESSED INTERIOR GIRDER

15. ESTIMATED CAMBER AT MIDSPAN

NOTE: + means upward camber
- means downward camber

The multiplier values were taken from PCI Bridge Design Manual, 3rd Ed., 2014 (8.7.1 Multiplier Method):

Initial Prestressing Force: $P_1 = 1960.43 \cdot kN$ *Final Prestressing Force:* $P_3 = 1761.94 \cdot kN$

$y_{2b} = 505.651 \cdot mm$ $E_c := E_{c_girder}$ $w_{diaph} := \frac{P_{diaph}}{L}$

$dp_{end} = 514.50 \cdot mm$ $e_2 = 380.65 \cdot mm$ $w_{diaph} = 0.667 \cdot \frac{kN}{m}$

$e_{end} := y_{2b} - dp_{end}$ $e_{end} = -8.85 \cdot mm$

$e_{mid} := e_2 - e_{end}$ $e_{mid} = 389.50 \cdot mm$

15.1 CAMBER AT TRANSFER

15.1.1 Due to Prestress including End Moment

$$pc := \frac{P_1 \cdot L^2}{48 \cdot E_{ci} \cdot I_{2cg}} \cdot (5 \cdot e_{mid} + 6e_{end}) \qquad pc = 9.73 \cdot mm$$

15.1.2 Due to Girder Weight

$$gdr := \left(\frac{5 \cdot w_g \cdot L^4}{384 \cdot E_{ci} \cdot I_{2cg}} \right) \cdot (-1) \qquad gdr = -2.94 \cdot mm$$

15.1.2 Initial Camber at Transfer

$Camber_{Transfer} := pc + gdr$ $Camber_{Transfer} = 6.78 \cdot mm$

15.2 CAMBER AT ERECTION

15.2.1 Due to Prestress including End Moment

$m_1 := 1.80$

$$'pc := \frac{m_1 \cdot P_1 \cdot L^2}{48 \cdot E_c \cdot I_{2cg}} \cdot (5 \cdot e_{mid} + 6e_{end}) \qquad 'pc = 16.14 \cdot mm$$

15.2.2 Due to Girder Weight

$m_2 := 1.85$

$$'_{gdr} := \left(\frac{m_2 \cdot 5 \cdot w_g \cdot L^4}{384 \cdot E_c \cdot I_{2cg}} \right) \cdot (-1) \qquad '_{gdr} = -5.02 \cdot mm$$

15.2.3 Camber at Erection

$Camber_{Erection} := 'pc + '_{gdr}$ $Camber_{Erection} = 11.12 \cdot mm$

DESIGN OF PRESTRESSED INTERIOR GIRDER

15.3 CAMBER AT FINAL (SERVICE STAGE)

15.3.1 Due to Slab

$$\text{slab} := \left(\frac{5\cdot w_{slab}\cdot L^4}{384\cdot E_c\cdot I_{2cg}}\right)\cdot(-1) \qquad \text{slab} = -2.97\cdot\text{mm}$$

15.3.2 Due to Diaphragm

$$\text{diaph} := \left(\frac{5\cdot w_{diaph}\cdot L^4}{384\cdot E_c\cdot I_{2cg}}\right)\cdot(-1) \qquad \text{diaph} = -0.21\cdot\text{mm}$$

15.3.3 Due to Superimposed Dead Loads

$$\text{sup} := \left(\frac{5\cdot w_{sup}\cdot L^4}{384\cdot E_c\cdot I_{3cg}}\right)\cdot(-1) \qquad \text{sup} = -0.44\cdot\text{mm}$$

15.3.4 Final Camber at Service Stage (Long Term Deflection)

Due to P/S:

$$_PS_F := \frac{m_3\cdot P_3\cdot L^2}{48\cdot E_c\cdot I_{2cg}}\cdot\left(5\cdot e_{mid} + 6e_{end}\right) \qquad m_3 := 2.2 \qquad _PS_F = 17.73\cdot\text{mm}$$

Due to Girder:

$$_Gdr_F := \left(\frac{m_4\cdot 5\cdot w_g\cdot L^4}{384\cdot E_c\cdot I_{2cg}}\right)\cdot(-1) \qquad m_4 := 2.4 \qquad _Gdr_F = -6.51\cdot\text{mm}$$

Due to Composite DL:

$$_Slab_F := m_5\cdot\left(\text{slab} + \text{diaph}\right) \qquad m_5 := 2.3 \qquad _Slab_F = -7.30\cdot\text{mm}$$

Due to SDL:

$$_SDL_F := m_6\cdot \text{sup} \qquad m_6 := 3.0 \qquad _SDL_F = -1.31\cdot\text{mm}$$

$$\text{Camber}_{Final} := _PS_F + _Gdr_F + _Slab_F + _SDL_F \qquad \text{Camber}_{Final} = 2.61\cdot\text{mm}$$

15.4 CAMBER SCHEME

A) Camber of cast of girder, before girder and prestressing load

$$A_c := \left|\text{Camber}_{Final}\right| \qquad A_c = 2.61\cdot\text{mm}$$

B) Camber after girder and prestressing load

Deflection due to Girder and P/S

$$_GdrPS_F := _Gdr_F + _PS_F \qquad _GdrPS_F = 11.22\cdot\text{mm}$$

Resulting camber

$$B_c := A_c + _GdrPS_F \qquad B_c = 13.82\cdot\text{mm}$$

C) Camber after slab, diaphragm

$$C_c := B_c + _Slab_F \qquad C_c = 6.52\cdot\text{mm}$$

D) Camber final after Superimposed DL

$$D_c := C_c + _SDL_F \qquad D_c = 5.21\cdot\text{mm}$$

DESIGN OF PRESTRESSED INTERIOR GIRDER

16. DEFLECTION DUE TO LIVE LOAD AND IMPACT

Allowable deflection due to live load *(LRFD 2.5.2.6.2)*

$L = 14.40\,m$

$$_Liveload_{allow} := \frac{L}{800}$$

$_Liveload_{allow} = 18.00 \cdot mm$

Live load deflection criteria

If the owner invokes the optional live load deflection criteria specified in LRFD Art. 2.5.2.6.2, the deflection is the greater of:

- that resulting from the design truck plus impact Δ_{LT}, or [LRFD Art. 3.6.1.3.2]
- that resulting from 25% of the design truck plus impact Δ_{LT}, taken together with the design lane load, Δ_{LL}.

Note: LRFD Article 2.5.2.6.2 states that the dynamic load allowance be included in the calculation of live load deflection.

The *LRFD Specifications* states that all the beams should be assumed to deflect equally under the applied live load and impact. [LRFD Art. 2.5.2.6.2]

Deflection due to Design Lane Load

Lane load: $w_{lane} = 9.34 \cdot \frac{kN}{m}$

Distribution factor: $DF_{M2} = 0.604$

Modulus of elasticity of composite beam: $E_{c_girder} = 33497 \cdot MPa$

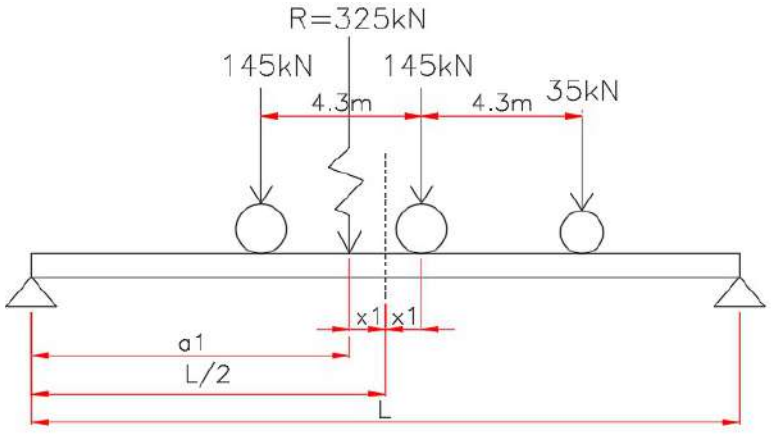
Moment of inertia of composite beam: $I_{3cg} = 0.1568\,m^4$

$$_LL := \left(\frac{5 \cdot w_{lane} \cdot L^4}{384 \cdot E_{c_girder} \cdot I_{3cg}} \right) \cdot DF_{M2}$$

$_LL = 0.60 \cdot mm$

Deflection due to Design Truck Load + Impact

To calculate the maximum deflection due to truck load, set the spacing between the rear axles to 4.3m. And have the centerline of the beam coincide with the middle point of the distance between the inner 145kN axle and the resultant of the truck load (325kN).



DESIGN OF PRESTRESSED INTERIOR GIRDER

where:

Point load resultant of the loads in wheels

$R_t := (145 + 145 + 35)\cdot kN$ $\quad R_t = 325.00\cdot kN$

Location of deflection to be determined

$x_{mid} := \frac{L}{2}$ $\quad x_{mid} = 7.200\,m$

Distance from R to centerline of beam

$x_1 := \frac{1}{2}\cdot\frac{(145\cdot kN\cdot 4.3\cdot m - 35\cdot kN\cdot 4.3\cdot m)}{R_t}$ $\quad x_1 = 727.69\cdot mm$

Distance from left end of beam to point load R

$a_1 := x_{mid} - x_1$ $\quad a_1 = 6.472\,m$

Distance from point load R to right end of beam

$b_1 := L - a_1$ $\quad b_1 = 7.928\,m$

Minimum of a or b

$c_1 := \min(a_1, b_1)$

Impact factor

$IM = 1.33$

$$_LT := \frac{R_t\cdot c_1}{48\cdot E_{c_girder}\cdot I_{3cg}}\cdot\left(3\cdot L^2 - 4\cdot c_1^{\,2}\right)\cdot IM\cdot DF_{M2} \qquad _LT = 3.04\cdot mm$$

Deflection due to Live Load and Check vs Allowable

$_Liveload_{Def} := \max(\ _LT, 0.25\cdot\ _LT + \ _LL)$ $\quad _Liveload_{Def} = 3.04\cdot mm$

$_Liveload_{allow} = 18.00\cdot mm$

$LLDeflection_check := if(\ _Liveload_{allow} \geq \ _Liveload_{Def}, "OK", "FAIL")$

$LLDeflection_check = "OK"$

DESIGN OF PRESTRESSED INTERIOR GIRDER

17. END ANCHORAGE DESIGN

17.1. Compressive Stress (Local Zone Check)

Member thickness

$t_{tf} := B_{tf_{Type}}$

$t_{tf} = 406.00 \cdot mm$

Lateral dimension of the effective bearing area measured parallel to the larger dimension of the cross-section

$a_{ef} := 150mm$

Lateral dimension of the effective bearing area measured parallel to the smaller dimension of the cross-section

$b_{ef} := 150mm$

Area of bearing plate

$A_{plate} := a_{ef} \cdot b_{ef}$

$A_{plate} = 22500.0 \cdot mm^2$

Diameter of confinement

$D_s := 150mm$

Area of confinement

$A_{conf} := \frac{\pi}{4} \cdot D_s^2$

$A_{conf} = 17671.5 \cdot mm^2$

Equivalent dimensions of confined area:

$a_{conf} := \sqrt{A_{conf}}$

$a_{conf} = 132.9 \cdot mm$

$b_{conf} := \sqrt{A_{conf}}$

$b_{conf} = 132.9 \cdot mm$

Center-to-center spacing of anchorages

$s_a := 353mm$

Number of anchorages in a row

$n_a := 2$

Diameter of duct

$D := 67.0mm$

No. of duct

$N_a := \frac{A^*_s}{12 \cdot A_{str}}$

$N_a = 1.33$

Factored tendon force

$P_{ulz} := \frac{1.25 \cdot P_j}{N_a}$

$P_{ulz} = 2065.30 \cdot kN$

Longitudinal extent of confining reinf. of the local zone

$l_c := \max(1.15a_{ef}, 1.15b_{ef})$

$l_c = 172.5 \cdot mm$

Correction factor for closely spaced anchorages

$$K_a := \begin{cases} 1 + \left(2 - \frac{s_a}{a_{ef}}\right)\left(0.3 + \frac{n_a}{15}\right) & \text{if } a_{ef} \le s_a < 2a_{ef} \\ 1 & \text{if } s_a \ge 2a_{ef} \end{cases}$$

$K_a = 1.00$

Note: If a group of anchorages is closely spaced in two directions, the product of the correction factors, k_a, for each direction shall be used.

DESIGN OF PRESTRESSED INTERIOR GIRDER

The effective bearing area, A_{eb}, shall be taken as the larger of the anchor bearing plate area; A_{plate}, or the bearing area of the confined concrete in the local zone; A_{conf}, with the following limitations:

$CHECK_A_{eb} := \text{if}\left(A_{plate} > A_{conf}, \text{"A.plate controls"}, \text{"A.conf controls"}\right)$ $CHECK_A_{eb} = \text{"A.plate controls"}$

$$A_{eb} := \begin{cases} \min\left(A_{plate}, \frac{4}{}\cdot A_{conf} - \frac{}{4}\cdot D^2\right) & \text{if } \left(A_{plate} > A_{conf}\right)\left(A_{plate} \le \frac{4}{}\cdot A_{conf}\right) \\ A_{conf} - \frac{}{4}\cdot D^2 & \text{if } \left(A_{plate} < A_{conf}\right)\left(\max\left(a_{conf}, b_{conf}\right) < 2\max\left(a_{ef}, b_{ef}\right)\right)\left(\max\left(a_{conf}, b_{conf}\right) < 3\min\left(a_{ef}, b_{ef}\right)\right) \\ A_{plate} & \text{otherwise} \end{cases}$$

$A_{plate} = 22500.00\cdot mm^2$

$A_{eb} = 18974.35\cdot mm^2$

$\frac{4}{}\cdot A_{conf} - \frac{}{4}\cdot D^2 = 18974.35\cdot mm^2$

Concrete compressive stress ahead of the achorage devices

$$f_{ca} := \frac{0.6\,P_{ulz}\,K_a}{A_{eb}\cdot\left[1 + l_c\cdot\left(\frac{1}{b_{ef}} - \frac{1}{t_{tf}}\right)\right]}$$

$$A_{eb}\cdot\left[1 + l_c\cdot\left(\frac{1}{b_{eff}} - \frac{1}{t_{tf}}\right)\right] = 13203.26\cdot mm^2$$

$f_{ca} = 37.86\cdot MPa$

17.2. Bursting Force (General Zone Check)

Factored tendon force

$P_{ugz} := 1.25\cdot P_j$

$P_{ugz} = 2753.73\cdot kN$

Lateral dimension of the anchorage device or group of devices in the direction considered

$a_b := \min\left(N_a\cdot a_{ef}, t_{tf}\right)$

$a_b = 200.0\cdot mm$

Lateral dimension of the cross-section in the direction considered

$h_b := h_g$

$h_b = 1143.00\cdot mm$

Reduction factor for tension in steel in anchorages zones

$_b := 1.0$

Angle of inclination of a tendon force with respect to the centerline of the member, positive for concentric tendons or if the anchor force points toward the centroid of the section; negative if the anchor force points away from the centroid of the section

$1 = 0.108\cdot rad$

Eccentricity of the anchorage device or group of devices with respect to the centroid of the cross-section; always taken as positive

$e_b := y_b - dp_{end}$

$e_b = 0.10\cdot mm$

Reduction factor for compression in anchorages zones

$_{ca} := 0.8$

CALCULATION SHEET

DESIGN OF PRESTRESSED INTERIOR GIRDER

Max. factored concrete compressive strength for the general zone

$$\frac{P_{ugz}}{a_b \cdot h_b} = 12.05 \cdot MPa$$

$f_{camax} := 0.7 \cdot {}_{ca} \cdot f'_{ci}$ $\quad f_{camax} = 19.71 \cdot MPa$

$$CHECK_f_{camax} := \begin{cases} "OK" & \text{if } \frac{P_{ugz}}{a_b \cdot h_b} < f_{camax} \\ "NG" & \text{otherwise} \end{cases}$$

$CHECK_f_{camax} = "OK"$

Distance from anchorage device to the centroid of the bursting force

$$d_{burst} := \frac{0.5(h_b - 2e_b) + 5 \cdot e_b \cdot \sin(\ 1)}{N_a}$$

$d_{burst} = 428.59 \cdot mm$

Tensile force in the anchorage zone acting ahead of the anchorage device and transverse to the tendon axis

$$T_{burst} := 0.25 P_{ugz} \cdot \left(1 - \frac{a_b}{h_b}\right) + 0.5 \left| P_{ugz} \cdot \sin(\ 1) \right|$$

$T_{burst} = 716.65 \cdot kN$

Required area of bursting bars

$$A_{rburst} := \frac{T_{burst}}{{}_b \cdot N_a \cdot f'_{sy}}$$

$A_{rburst} = 1298.28 \cdot mm^2$

Provided area of bursting bars

diameter bursting bars $\quad d_{bburst} := 12mm$

no. bursting bars $\quad N_{bburst} := 10$

$$A_{pburst} := \frac{\pi}{4} \cdot d_{bburst}^2 \cdot 2 \cdot N_{bburst}$$

$A_{pburst} = 2261.95 \cdot mm^2$

$$CHECK_A_{burst} := \begin{cases} "OK" & \text{if } A_{rburst} < A_{pburst} \\ "NG" & \text{otherwise} \end{cases}$$

$CHECK_A_{burst} = "OK"$

DESIGN OF PRESTRESSED INTERIOR GIRDER

18. TENDONS LAYOUT

Number of ducts $N_{duct} = 2$

Cable number | No. of strands | Distance from bottom of girder at ends

$Duct_i$ =	N_{strand_i} =	Y_i =	
1	8	350	·mm
2	8	679	

Girder Span Length $L = 14400 \cdot mm$

Distance from bottom of girder at midspan $d_p = 125 \cdot mm$

General equation of parabola: y = a(x-h)^2 + k

a = (y - k) / (x-h)^2

where vertex of paraboloa is V(h,k) = V(L/2,dp)

$$a_i := \frac{Y_i - d_p}{(0 - 0.5 \cdot L)^2}$$

a_i =

0.00434	$\cdot m^{-1}$
0.01069	

Horizontal distance from end to midspan of girder

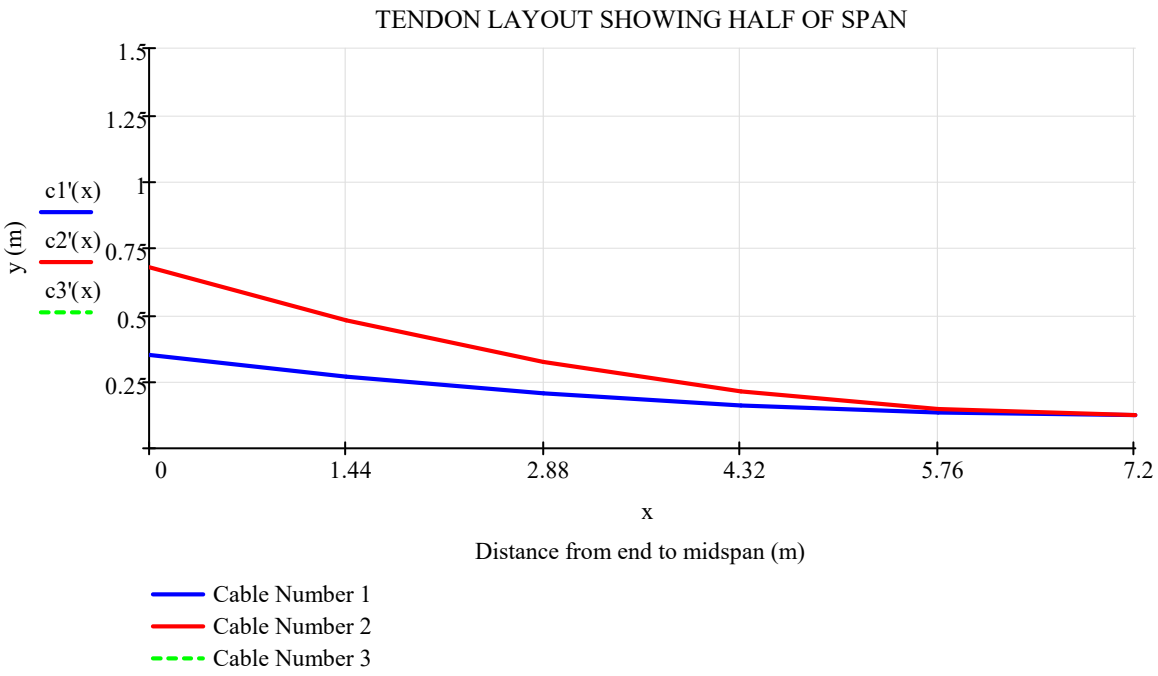
$x := 0, 0.1 \cdot L .. 0.5L$

Parabolic equations of per cable duct number (vertical distance from bottom of girder)

$c1'(x) := a_1 \cdot (x - 0.5 \cdot L)^2 + d_p$ $c2'(x) := a_2 \cdot (x - 0.5 \cdot L)^2 + d_p$ $c3'(x) := a_3 \cdot (x - 0.5 \cdot L)^2 + d_p$

x =		c1'(x) =		c2'(x) =	
0	·mm	350	·mm	679	·mm
1440		269		479.56	
2880		206		324.44	
4320		161		213.64	
5760		134		147.16	
7200		125		125	

DESIGN OF PRESTRESSED INTERIOR GIRDER



19. REQUIRED JACKING FORCE PER DUCT

Total jacking force: $P_j = 2202.98 \cdot kN$

Say: $P_j := 2204 \cdot kN$

$$Pj_i := \frac{N_{strand_i}}{N_o} \cdot P_j$$

$Duct_i$ =	N_{strand_i} =	Pj_i =	
1	8	1102	·kN
2	8	1102	

per duct

$N_o = 16$

CALCULATION SHEET

DESIGN FOR NEGATIVE MOMENT AT PIER

Designed by/Date: KH - RCDG 05/15/26

Conversion Factor: $kN \equiv 1000 \cdot newton$ $Pa \equiv \frac{newton}{m^2}$ $kPa \equiv \frac{kN}{m^2}$ $MPa \equiv \frac{newton}{mm^2}$

MATERIAL DATA and SPECIFICATIONS:

Slab Thickness $t_s := 220 \cdot mm$

Haunch Thickness $t_{haunch} := 25 \cdot mm$

Girder Spacing $S_g := 1.75 \cdot m$

Width of bottom flange of girder $b := 559 \cdot mm$

Yield Strength of reinforcing steel $f'_{sy} := 414 \cdot MPa$

Diameter of bar $d_b := 16 \cdot mm$

Area of one bar $A_b := \frac{\cdot d_b^2}{4} = 201.06 \cdot mm^2$

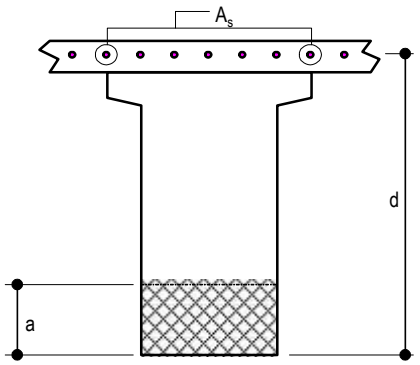
Depth from c.g of steel bar to bottom of girder $d := 1.143m + t_{haunch} + t_s - 0.074m$

$d = 1.314\,m$

Concrete Compressive Strength @ 28 days $f'_c := 27.6 \cdot MPa$

Modulus of Rupture of normal wt. conc. $f_r := 0.63 \cdot \sqrt{f'_c \cdot MPa}$

$f_r = 3.31 \cdot MPa$



DESIGN NEGATIVE MOMENT:

(From StaadPro Analysis of Continuous Girders)

Moment from superimposed dead load DC: $M_{DC} := 21.10 \cdot kN \cdot m$

Note: Int. and Ext. Girder values were averaged as the governing value

Moment from superimposed dead load DW: $M_{DW} := 83.53 \cdot kN \cdot m$

Dynamic Load Allowance Factor $IM := 1.33$

Moment due to Truck Load: $M_T := 382.07 \cdot kN \cdot m \cdot IM = 508.15 \cdot kN \cdot m$

Moment due to Lane Load: $M_{Lane} := 227.84 \cdot kN \cdot m$

Moment due to Pedestrian Load: $M_{PL} := 45.66 \cdot kN \cdot m$

Distribution Factor One Lane: $DF_{M3} := 0.611$

Total Moment due to Live Loads per Girder: $M_{LL} := (M_T + M_{Lane} + M_{PL}) \cdot DF_{M3}$

$M_{LL} = 477.59 \cdot kN \cdot m$

Total Factored Design Moment: $M_u := 1.25 \cdot M_{DC} + 1.5 \cdot M_{DW} + 1.75 \cdot M_{LL}$

$M_u = 987.45 \cdot kN \cdot m$

CALCULATION SHEET

DESIGN FOR NEGATIVE MOMENT AT PIER

DESIGN FLEXURAL REINFORCEMENT:

Let: $:= 0.90$ Resistance Factor

$m_1 := 0.85 \cdot \frac{f'_c}{f'_{sy}}$ $\quad m_2 := \frac{2}{0.85 \cdot f'_c}$ $\quad R_n := \frac{M_u}{\cdot b \cdot d^2} = 1.14 \cdot MPa$

Required steel ratio: $:= m_1 \cdot \left(1 - \sqrt{1 - m_2 \cdot R_n}\right) = 0.0028$

Required toal area of reinf: $A_s := \cdot b \cdot d$ $\quad A_s = 2068.26 \cdot mm^2$

Required number of bars: $n := \frac{A_s}{A_b}$ $\quad n = 10.29$

Say, $n := 12$

Total area of reinf: $A_s := n \cdot A_b$ $\quad A_s = 2412.74 \cdot mm^2$

Depth of rectangular stress block: $a := \frac{A_s \cdot f'_{sy}}{0.85 \cdot f'_c \cdot b}$ $\quad a = 76.17 \cdot mm$

Flexural Resistance: $M_r := \cdot A_s \cdot f'_{sy} \cdot \left(d - \frac{a}{2}\right)$ $\quad M_r = 1147.03 \cdot kN \cdot m$

$Flexure_check := if\left(M_r \geq M_u, "OK", "FAIL"\right)$ $\quad Flexure_check = "OK"$

Provide 12 - ϕ16mm additional bars at slab per girder or
4 - ϕ16mm @100mm at top of girder then rest @200mm o.c.

CHECK MINIMUM REINFORCEMENT:

Flexural cracking variability factor: $_1 := 1.6$

Ratio of fy/fu factor (A615 Grade 60 reinforcement): $_3 := 0.67$

Composite Moment of Inertia: $I_{3cg} := 0.1606\ m^4$

Distance from neutral axis to extreme tension fiber: $y_{tc} := 0.27\ m + t_{haunch} + t_s$ $\quad y_{tc} = 0.515 \cdot m$

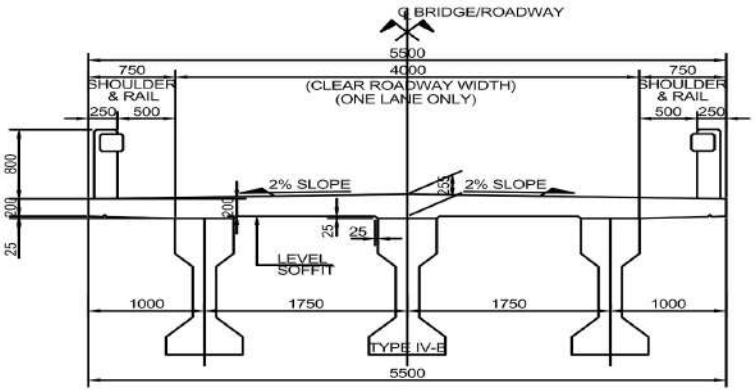
Cracking Moment: $M_{cr} := {}_1 \cdot {}_3 \cdot f_r \cdot \frac{I_{3cg}}{y_{tc}}$ $\quad M_{cr} = 1106.44 \cdot kN \cdot m$

Minimum Flexural Resistance: $M_{min} := \min\left(M_{cr}, 1.33 \cdot M_u\right)$ $\quad M_{min} = 1106.44 \cdot kN \cdot m$

$Minimum_check := if\left(M_r \geq M_{min}, "OK", "FAIL"\right)$ $\quad Minimum_check = "OK"$

INTERIOR DECK SLAB

1.0 GEOMETRY AND DESIGN PARAMETERS



1.1 Bridge Geometry

Bridge Span Length	=	45	m
Girder Spacing, S	=	1.75	m
No. of Girder Supports	=	3	pcs.
Top Flange Width	=	500	mm
Curb - Curb Distance	=	5000	mm
Deck Width	=	5500	mm
Sidewalk Thickness	=	0	mm
Sidewalk Width	=	0	mm
Square Post Dimension	=	200	mm
Square Post Height	=	800	mm
Total Number of Post (each side)	=	30	pcs.
Square Railing Dimension	=	200	mm
Total Number of Square Railing	=	1	pcs.
Future Wearing Coarse Thickness, t_{FWS}	=	50	mm
Deck Slab Thickness, t_{slab}	=	220	mm
Deck type	=	k	type
Number of Lanes, m	=	1	lane/s
Type of Girder	=	IV-B	type

1.2 Design Parameters

Design width, b	=	1000	mm
Concrete Cover, cc	=	50	mm
Unit Weight of Concrete, γ_{conc}	=	24	kN/m^3
Unit Weight of Future Wearing Coarse, γ_{FWS}	=	22	kN/m^3
Concrete Compressive Strength, fc'	=	28	MPa
Steel Reinforcement Yield Strength, fy	=	414	MPa
Main Bar Diameter	=	16	mm
Longitudinal Bar Diameter	=	16	mm
Strength Resistance Factor for Flexure, Φ_f	=	0.9	-

2.0 DEAD LOAD ANALYSIS

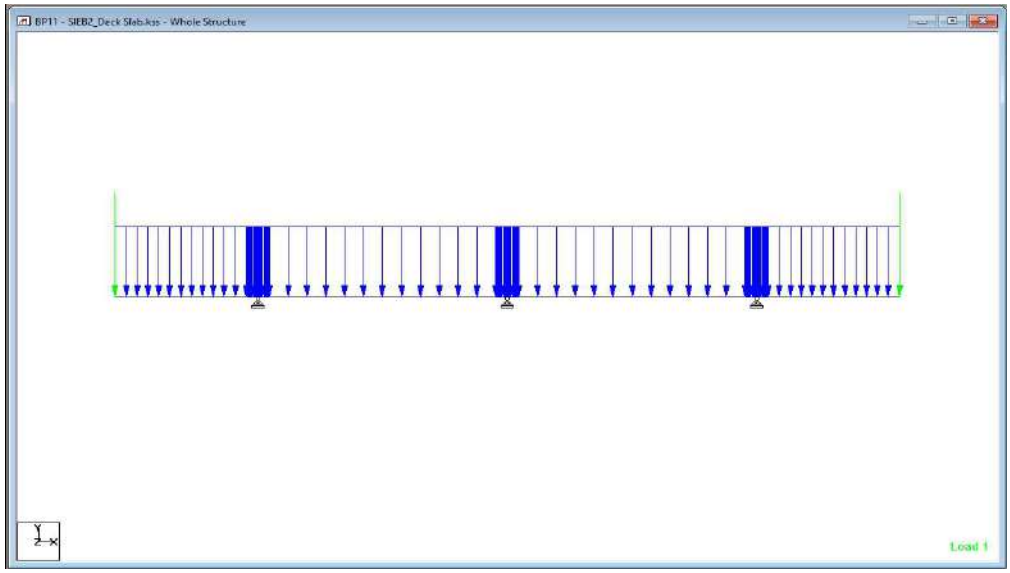
2.1 Uniform Dead Load for Interior Slabs

Deck Slab Self Weight	=	5.28 kN/m
Future Wearing Surface Self Weight	=	1.1 kN/m

2.2 Concentrated Force on Cantilever:

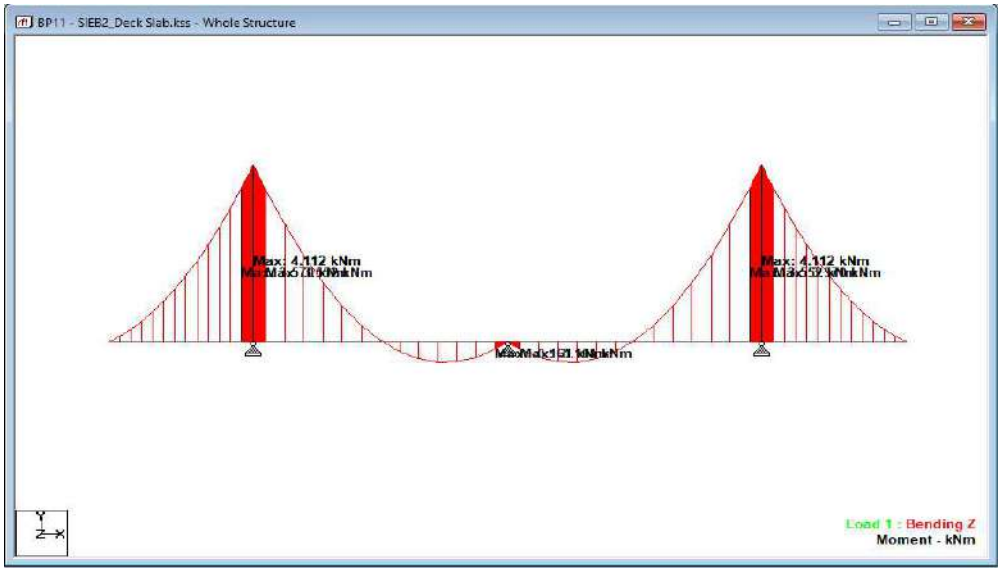
Sidewalk Weight	=	0 kN
Post Weight	=	0.512 kN
Railing Weight	=	0.96 kN

2.3 STAAD.Pro Load and Definition

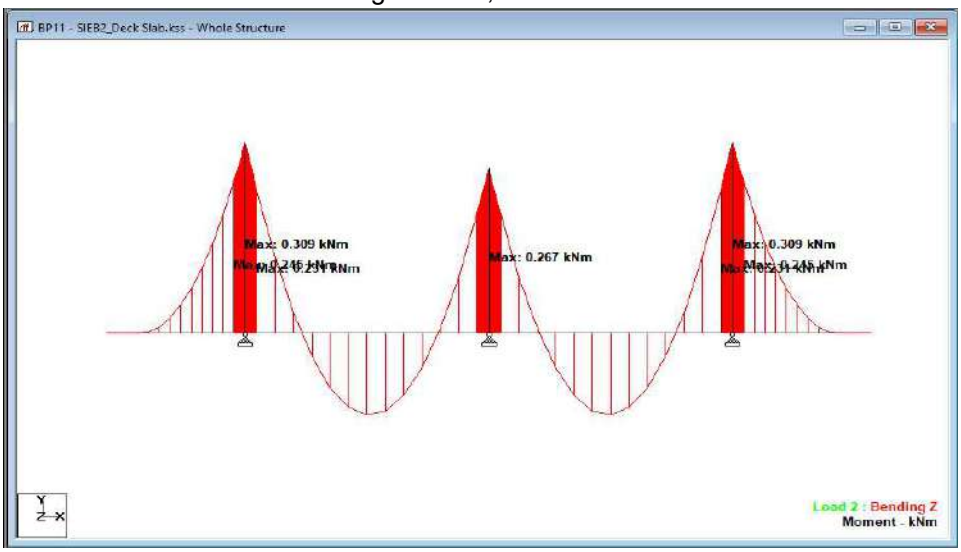


2.4 STAAD Pro. Results (Positive and Negative Moments)

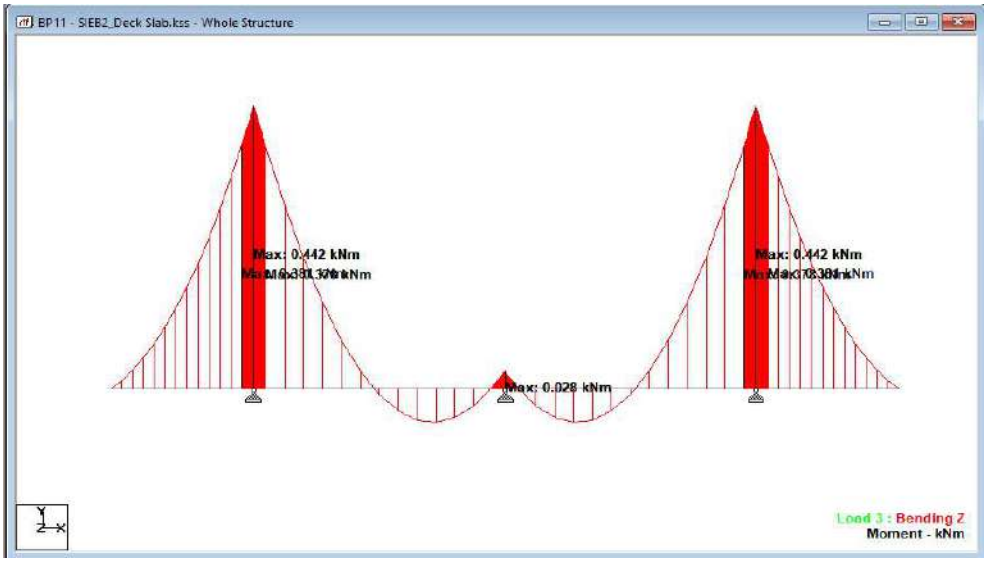
2.4.1 Dead Load, DC



2.4.2 Future Wearing Surface, DW



2.4.3 Utilities



3.0 LIVE LOAD ANALYSIS

For decks covered by Table A4-1, the total moment acting on the edge beam, including the multiple presence factor and the dynamic load allowance, may be calculated by multiplying the moment per unit width, taken from Table A4-1, by the corresponding full strip width specified in Article 4.6.2.1.3.

Table A4-1 may be used in determining the design moments for different girder arrangements. The following assumptions and limitations were used in developing this table and should be considered when using the listed values for design:

- The moments are calculated using the equivalent strip method as applied to concrete slabs supported on parallel girders.
- Multiple presence factors and the dynamic load allowance are included in the tabulated values.
- See Article 4.6.2.1.6 for the distance between the center of the girders to the location of the design sections for negative moments in the deck. Interpolation between the listed values may be used for distances other than those listed in Table A4-1.
- The moments represent the upper bound for the moments in the interior regions of the slab and, for any specific girder spacing, were taken as the maximum value calculated, assuming different numbers of girders in the bridge cross-section. For each combination of girder spacing and number of girders, the following two cases of overhang width were considered:
  - (a.) Minimum total overhang width of 530mm measured from the center of the exterior girder, and
  - (b.) Maximum total overhang width equal to the smaller of 0.625 times the girder spacing and 1800mm.

  A railing system width of 530mm was used to determine the clear overhang width. For other widths of railing systems, the difference in the moments in the interior regions of the deck is expected to be within the acceptable limits for practical design.
- The moments do not apply to the deck overhangs and the adjacent regions of the deck that need to be designed taking into account the provisions of Article A13.4.1.
- It was found that the effect of two 108kN axles of the tandem, placed at 1200mm from each other, produced maximum effects under each of the tires approximately equal to the effect of the 145kN truck axle. The tandem produces a larger total moment, but this moment is spread ocver a larger width. It was concluded that repeating calculations with a different strip width for the tandem would not result in a significant difference.

AASHTO Table A4-1 Maximum Live Load Moments Per Unit Width, (kNm/m)								
s (m)	Positive Moment	Negative Moment						
		0 mm	75 mm	150 mm	225 mm	300 mm	450 mm	600 mm
1.3	21.13	11.72	10.27	8.94	7.95	7.15	6.06	5.47
1.4	21.01	14.14	12.21	10.34	8.94	7.67	5.96	5.12
1.5	21.05	16.32	14.03	11.72	9.98	8.24	5.82	5.25
1.6	21.19	18.4	15.78	13.16	11.03	8.97	5.91	4.29
1.7	21.44	20.14	17.29	14.45	12.01	9.71	6.06	4.51
1.8	21.79	21.69	18.66	15.63	12.93	10.44	6.27	4.79
1.9	22.24	23.05	19.88	16.71	13.78	11.13	6.65	5.13
2	22.78	24.26	20.96	17.67	14.55	11.77	7.03	5.57
2.1	23.38	26.78	23.19	19.58	16.06	12.87	7.41	6.08

2.2	24.04	27.67	24.02	20.37	16.74	13.49	7.36	6.73
2.3	24.75	28.45	24.76	21.07	17.38	14.57	9.08	8.05
2.4	25.5	29.14	25.42	21.7	17.98	15.41	10.87	9.34
2.5	26.31	29.72	25.99	22.25	18.51	16.05	12.4	10.63
2.6	27.22	30.22	26.47	22.73	18.98	16.48	13.66	11.88
2.7	28.12	30.68	26.92	23.17	19.42	16.76	14.71	13.11
2.8	29.02	31.05	27.3	23.55	19.99	17.41	15.54	14.31
2.9	29.91	32.49	28.72	24.94	21.26	18.41	16.8	15.48
3	30.8	34.63	30.79	26.96	23.12	19.46	18.03	16.62
3.1	31.66	36.63	32.77	28.89	23.97	21.15	19.23	17.78
3.2	32.5	38.57	34.67	30.77	26.88	22.98	20.38	18.91
3.3	33.36	40.44	36.52	32.6	28.68	24.77	21.5	20.01
3.4	34.21	42.25	38.34	34.43	30.52	26.61	22.6	21.09
3.5	35.05	43.97	40.03	36.09	32.15	28.21	23.67	22.13
3.6	35.87	45.65	41.7	37.76	33.81	29.87	24.7	23.15
3.7	36.67	47.25	43.31	39.37	35.43	31.49	25.79	24.14
3.8	37.45	48.82	44.88	40.94	37.01	33.07	27.08	25.1
3.9	38.23	50.32	46.39	42.46	38.54	34.6	28.33	25.55
4	38.97	51.79	47.87	43.95	40.03	36.11	29.57	26.41
4.1	39.71	53.19	49.28	45.37	41.47	37.57	30.77	27.85
4.2	40.42	54.56	50.67	46.77	42.88	38.99	31.96	28.73
4.3	41.12	55.88	52	48.13	44.25	40.38	33.13	29.57
4.4	41.8	57.15	53.29	49.44	45.58	41.72	34.25	30.4
4.5	42.46	58.42	54.58	50.74	46.9	43.06	35.38	31.29
4.6	43.11	59.62	55.8	51.98	48.16	44.34	36.7	32.36

Given Girder Spacing, S: 1.75 m
Flange Width: 500 mm
Based on DGCS Vol. 5, 11.3.2.1.5
Negative Moment = 17.6489 kN-m/m
Positive Moment = 21.615 kN-m/m

Minimum Reinforcement:

Mcr = $y_1 y_3 f_r S$ *AASHTO LRFD 10th Ed. Art. 5.6.3.3*

f_r	=	$0.63\sqrt{Fc'}$	=	3.33365	MPa
y_1	=	Other Concrete Structures	=	1.6	
y_2	=	No Tendons	=	0	
y_3	=	A615 - G80	=	0.76	
S	=	$bt^2 / 6$	=	8.1E+06	mm^3

Therefore, Mcr = 32.70 kN-m

4.0 DESIGN FOR FLEXURE (TOP REINFORCEMENT)

LOAD CASES	NEGATIVE MOMENT (kN-m)						LOAD FACTOR
	SPAN 1			SPAN 2			
	LEFT	MID	RIGHT	LEFT	MID	RIGHT	STR - I
DC	3.55	0.00	0.00	0.00	0.00	3.55	1.25
FWS	0.23	0.00	0.19	0.19	0.00	0.23	1.5
UTLY	0.38	0.00	0.00	0.00	0.00	0.38	1.5
LL	17.65	0.00	17.65	17.65	0.00	17.65	1.75
LL*IF*m (Beyond Deck Joint)	28.168	0.000	28.168	28.168	0.000	28.168	1.75
LL*IF*m (At Deck Joint)	37.063	0.000	37.063	37.063	0.000	37.063	1.75
SERVICE (Beyond Deck Joint)	32.329	0.000	28.363	28.363	0.000	32.329	kN-m
SERVICE (Deck Joint)	41.224	0.000	37.258	37.258	0.000	41.224	kN-m
STRENGTH I (Beyond Deck Joint)	54.647	0.000	49.586	49.586	0.000	54.647	kN-m
STRENGTH I (Deck Joint)	70.213	0.000	65.152	65.152	0.000	70.213	kN-m

4.1 For Dynamic Allowance of 33% (Beyond Deck Joints)

Design moment, Max[Mu, Min(Mcr, 1.33Mu)]	=	54.65	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	2.31	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.01	-
Required As = ρbd	=	954.22	mm
Therefore, use spacing	=	210.709	mm
Use	**16 mm ⌀**	**at**	**140 mm**

4.2 For Dynamic Allowance of 75% (At Deck Joints)

Design moment, Max[Mu, Min(Mcr, 1.75Mu)]	=	70.21	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	2.97	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.01	-
Required As = ρbd	=	1246.66	mm
Therefore, use spacing	=	161.281	mm
Use	**16 mm ⌀**	**at**	**100 mm**

5.0 FLEXURAL CAPACITY CHECK (TOP REINFORCEMENT)

5.1 Assumed Spacing			
Beyond Deck Joints	=	140	mm
At Deck Joints	=	100	mm
5.2 Assumed Area, $A_{provided} = A_b b/S_{provided}$			
Beyond Deck Joints	=	1436.16	mm^2
At Deck Joints	=	2010.62	mm^2
5.3 Depth of Compression Block, a = Asfy/0.85fc'b			
Beyond Deck Joints	=	24.98	mm
At Deck Joints	=	34.97	mm
5.4 Nominal Moment Capacity, Mn = AsFy(d-a/2)			
Beyond Deck Joints	=	88.89	kN-m
At Deck Joints	=	120.29	kN-m
5.5 Design Moment, ΦMn			
Beyond Deck Joints	=	80.00	kN-m
At Deck Joints	=	108.26	kN-m
5.6 Capacity-Demand Ratio, ΦMn/Mu > 1.0			
Beyond Deck Joints	=	1.46	Adequate
At Deck Joints	=	1.54	Adequate

6.0 CRACK CONTROL CHECK(DGCS Art. 12.4.3.4)

Modulus of Elasticity of Steel, Es	=	200000	MPa
Modulus of Elasticity of Concrete, Ec = 4730√fc'	=	25028.8	MPa
Modular Ratio, n = Es/Ec	=	7.99079	-
Exposure Factor, γ_e (AASHTO LRFD 10th Ed., Art. 5.6.7-1)	=	0.75	-
Concrete Cover from Centroid of Closest Steel, dc	=	58	mm

Transformed Section

6.1 Moment at Service Limit State, M_{SVC}			
Beyond Deck Joints	=	32.329	kN-m
At Deck Joints	=	41.224	kN-m
6.2 Ratio of Concrete Width to Transformed Area, B = b / (nAs)			
Beyond Deck Joints	=	0.12448	
At Deck Joints	=	0.09336	
6.3 Depth of Neutral Axis, kd = [√(2dB+1) - 1] / B			
Beyond Deck Joints	=	43.6127	mm
At Deck Joints	=	49.1645	mm
6.4 Neutral Axis Ratio, k = kd / d			
Beyond Deck Joints	=	0.26921	
At Deck Joints	=	0.30348	

6.5 Lever Arm Factor, $j = 1 - k / 3$

Beyond Deck Joints	=	0.91026
At Deck Joints	=	0.89884

6.6 Tensile Stress at Service Limit of Steel, $f_{ss} = M / (j\ d\ As)$

Beyond Deck Joints	=	218.075 MPa
At Deck Joints	=	211.209 MPa

Required Spacing

6.7 Ratio of Strain Factor, $\beta_s = 1 + de / 0.7(h - dc)$

Beyond Deck Joints	=	1.43739
At Deck Joints	=	1.43739

6.8 Maximum Reinforcement Spacing, $Smax = (123000\gamma_e/\beta_s f_{ss}) - 2dc$

Beyond Deck Joints				=	178.297 mm
Smax =	178.297	>	140.000 = Spr	Sufficient reinforcement.	
At Deck Joints				=	187.865 mm
Smax =	187.865	>	100.000 = Spr	Sufficient reinforcement.	

7.0 DESIGN FOR FLEXURE (BOTTOM REINFORCEMENT)

LOAD CASES	POSITIVE MOMENT (kN-m)						LOAD FACTOR
	SPAN 1			SPAN 2			
	LEFT	MID	RIGHT	LEFT	MID	RIGHT	STR - I
DC	0.00	0.48	0.00	0.00	0.48	0.00	1.25
FWS	0.00	0.14	0.00	0.00	0.11	0.00	1.5
UTLY	0.00	0.06	0.00	0.00	0.05	0.00	1.5
LL	0.00	21.62	0.00	0.00	21.62	0.00	1.75
LL*IF*m (Beyond Deck Joint)	0.000	34.498	0.000	0.000	34.498	0.000	1.75
LL*IF*m (At Deck Joint)	0.000	45.392	0.000	0.000	45.392	0.000	1.75
SERVICE (Beyond Deck Joint)	0.000	35.176	0.000	0.000	35.139	0.000	kN-m
SERVICE (Deck Joint)	0.000	46.070	0.000	0.000	46.033	0.000	kN-m
STRENGTH I (Beyond Deck Joint)	0.000	61.267	0.000	0.000	61.212	0.000	kN-m
STRENGTH I (Deck Joint)	0.000	80.332	0.000	0.000	80.276	0.000	kN-m

7.1 For Dynamic Allowance of 33% (Beyond Deck Joints)

Design moment, Max[Mu, Min(Mcr, 1.33Mu)]	=	61.27	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	2.59	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.01	MPa
Required As = ρbd	=	1077.33	mm
Therefore, use spacing	=	186.631	mm

Use 16 mm ⌀ at 140 mm

Adopt tighter spacing between top and bottom reinforcement.

7.2 For Dynamic Allowance of 75% (At Deck Joints)

Design moment, Max[Mu, Min(Mcr, 1.75Mu)]	=	80.33	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	3.40	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.01	MPa
Required As = ρbd	=	1442.58	mm
Therefore, use spacing	=	139.377	mm

Use 16 mm ⌀ at 100 mm

8.0 FLEXURAL CAPACITY CHECK (BOTTOM REINFORCEMENT)

8.1 Assumed Spacing

Beyond Deck Joints	=	140	mm
At Deck Joints	=	100	mm

8.2 Assumed Area, $A_{provided} = A_b b/S_{provided}$

Beyond Deck Joints	=	1436.16	mm^2
At Deck Joints	=	2010.62	mm^2

8.3 Depth of Compression Block, a = Asfy/0.85fc'b

Beyond Deck Joints	=	24.98	mm
At Deck Joints	=	34.97	mm

8.4 Nominal Moment Capacity, Mn = AsFy(d-a/2)

Beyond Deck Joints	=	88.89	kN-m
At Deck Joints	=	120.29	kN-m

8.5 Design Moment, ΦMn

Beyond Deck Joints	=	80.00	kN-m
At Deck Joints	=	108.26	kN-m

8.6 Capacity-Demand Ratio, ΦMn/Mu > 1.0

Beyond Deck Joints	=	1.31	Adequate
At Deck Joints	=	1.35	Adequate

9.0 CRACK CONTROL CHECK *(DGCS Art. 12.4.3.4)*

Modulus of Elasticity of Steel, Es	=	200000	MPa
Modulus of Elasticity of Concrete, Ec = 4730√fc'	=	25028.8	MPa
Modular Ratio, n = Es/Ec	=	7.99079	-
Exposure Factor, γ_e (AASHTO LRFD 10th Ed., Art. 5.6.7-1)	=	0.75	-
Concrete Cover from Centroid of Closest Steel, dc	=	58	mm

Transformed Section

9.1 Moment at Service Limit State, M_{SVC}

Beyond Deck Joints	=	35.176	kN-m
At Deck Joints	=	46.070	kN-m

9.2 Ratio of Concrete Width to Transformed Area, B = b / (nAs)

Beyond Deck Joints	=	0.11203
At Deck Joints	=	0.08091

9.3 Depth of Neutral Axis, kd = [√(2dB+1) - 1] / B

Beyond Deck Joints	=	45.5869	mm
At Deck Joints	=	52.1159	mm

9.4 Neutral Axis Ratio, k = kd / d

Beyond Deck Joints	=	0.2814

At Deck Joints = 0.3217

9.5 Lever Arm Factor, j = 1 - k / 3

Beyond Deck Joints = 0.9062

At Deck Joints = 0.89277

9.6 Tensile Stress at Service Limit of Steel, f_{ss} = M / (j d As)

Beyond Deck Joints = 214.508 MPa

At Deck Joints = 205.956 MPa

Required Spacing

9.7 Ratio of Strain Factor, β_s = 1 + de / 0.7(h - dc)

Beyond Deck Joints = 1.43739

At Deck Joints = 1.43739

Maximum Reinforcement Spacing, Smax = $(123000\gamma_e/\beta_s f_{ss})$ - 2dc

Beyond Deck Joints = 183.19 mm

Smax = 183.19 > 160.000 = Spr — Sufficient reinforcement.

At Deck Joints = 195.614 mm

Smax = 195.614 > 120.000 = Spr — Sufficient reinforcement.

10.0 DISTRIBUTION REINFORCEMENT

The distribution reinforcement shall be placed transverse to the main steel reinforcement in the bottom of all slabs to provide for the lateral distribution of the concentrated live loads. DGCS Vol. 5, Sec. 14.4.3.1.

10.1 Effective Span of Slab

S = if (B_{tf}/T_{tf})<4.0 use S_g-T_w , otherwise S_g-$B_{tf}/2$ = 1547 mm

%As, 3840/√S = 97.6306 %

Area of main reinforcement, $As_{provided}$ = 1436.16 mm^2

Area of Secondary Steel, As_{dr} = if(%As<=67%,%As,67%)*Asp

Asdr = 962.225 mm^2

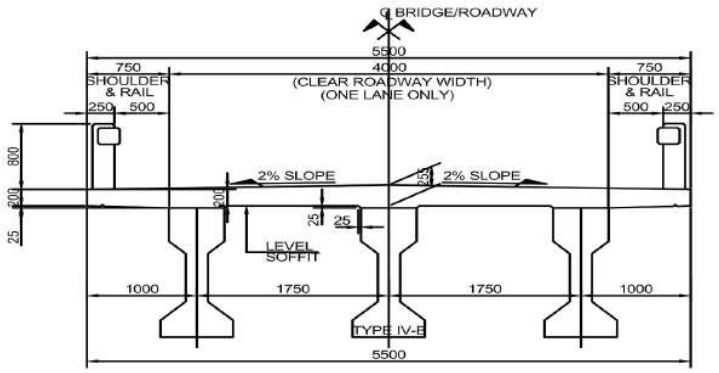
Required Spacing of Secondary Steel, Sdr = $A_b b/As_{dr}$

Sdr = 208.955 mm

Therefore: Use **16 mmΦ** at **200 mm O.C.**

CANTILEVER DECK SLAB DESIGN

1.0 GEOMETRY AND DESIGN PARAMETERS



1.1 Bridge Geometry

Bridge Span Length	=	30	m
Girder Spacing, S	=	1.75	m
Cantilever Thickness	=	220	mm
Cantilever Width	=	750	mm
Sidewalk Thickness	=	0	mm
Sidewalk Width	=	0	mm
Square Post Dimension	=	200	mm
Square Post Height	=	800	mm
Total Number of Post (each side)	=	30	pcs.
Square Railing Dimension	=	200	mm
Total Number of Square Railing	=	1	pcs.
Future Wearing Coarse Thickness, t_{FWS}	=	50	mm
Deck Slab Thickness, t_{slab}	=	220	mm
Number of Lanes, m	=	1	lane/s
Type of Girder	=	IV-B	type

1.2 Design Parameters

Design width, b	=	1000	mm
Concrete Cover, cc	=	50	mm
Unit Weight of Concrete, γ_{conc}	=	24	kN/m^3
Unit Weight of Future Wearing Coarse, γ_{FWS}	=	22	kN/m^3
Concrete Compressive Strength, fc'	=	28	MPa
Steel Reinforcement Yield Strength, fy	=	414	MPa
Main Bar Diameter	=	16	mm
Longitudinal Bar Diameter	=	16	mm
Strength Resistance Factor for Flexure, Φ_f	=	0.9	-

2.0 DESIGN LOADS

2.1 Dead Load

Component	Weight [kN]	Moment Arm [m]	Moment [kN-m]
Sidewalk	0.00	0.00	0.00
Wearing Surface	0.61	0.50	0.30
Slab	3.96	0.38	1.49
Railing Post	0.77	0.15	0.12
Railing	0.96	0.15	0.14
Misc.	2.00	0.38	0.75
TOTAL	8.29		2.80

2.2 Live Load

Truck Load (HL - 93)	=	72.00	kN
Location of Load from the Face of the Railing	=	304.80	mm

DGCS 3.6.1.3.4 - Deck Overhang Load

Distance from Load to Point of Support, x	=	195.20	mm
Axle Load Distribution Width for Overhang, E	=	1302.60	mm

DGCS 11.3.2.1.2, Table 11.3.2.1-1

Live Load Moment	=	12.95	kN-m

2.3 Design Moments, Mu = $1.25M_{DL} + 1.75M_{LL}$

Beyond Deck Joints	=	33.63	kN-m
At Deck Joints	=	43.15	kN-m

3.0 DESIGN FOR FLEXURE (TOP AND BOTTOM)

3.1 Minimum Reinforcement

Mcr = $y_1 y_3 f_r S$ *AASHTO LRFD 10th Ed. Art. 5.6.3.3*

f_r	=	$0.63\sqrt{Fc'}$	=	3.33	MPa
y_1	=	Other Concrete Structures	=	1.60	-
y_2	=	No Tendons	=	0.00	-
y_3	=	A615 - G60	=	0.67	-
S	=	$bt^2 / 6$	=	0.01	m^3

Therefore, Mcr	=	28.83	kN-m

3.2 **For Dynamic Allowance of 33% (Beyond Deck Joints)**

Design moment, Max[Mu, Min(Mcr, 1.33Mu)]	=	33.63	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	1.4E-06	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.00	MPa
Required As = ρbd	=	557.16	mm
Therefore, spacing	=	360.869	mm

Use 16 mm ⌀ at 140 mm

3.2 **For Dynamic Allowance of 75% (At Deck Joints)**

Design moment, Max[Mu, Min(Mcr, 1.33Mu)]	=	43.15	kN-m
Effective depth, d	=	162	mm
$Rn = Mu / \Phi bd^2$	=	1.8E-06	MPa
$\rho = 0.85fc' / fy (1 - \sqrt{(1-2Rn / 0.85fc')})$	=	0.00	MPa
Required As = ρbd	=	714.82	mm
Therefore, use spacing	=	281.277	mm

Use 16 mm ⌀ at 100 mm

4.0 FLEXURAL CAPACITY CHECK

4.1 Spacing

Beyond Deck Joints	=	140	mm
At Deck Joints	=	100	mm

4.2 Assumed Area, $A_{provided} = A_b b / S_{provided}$

Beyond Deck Joints	=	1436.16	mm^2
At Deck Joints	=	2010.62	mm^2

4.3 Depth of Compression Block, a = Asfy/0.85fc'b

Beyond Deck Joints	=	24.98	mm
At Deck Joints	=	34.97	mm

4.4 Nominal Moment Capacity, Mn = AsFy(d-a/2)

Beyond Deck Joints	=	88.89	kN-m
At Deck Joints	=	120.29	kN-m

4.5 Design Moment, ΦMn

Beyond Deck Joints	=	80.00	kN-m
At Deck Joints	=	108.26	kN-m

4.6 Capacity-Demand Ratio, ΦMn/Mu > 1.0

Beyond Deck Joints	=	2.38	Adequate
At Deck Joints	=	2.51	Adequate

5.0 CRACK CONTROL CHECK (DGCS Art. 12.4.3.4)

Modulus of Elasticity of Steel, Es	=	200000	MPa
Modulus of Elasticity of Concrete, Ec = 4730√fc'	=	25028.8	MPa
Modular Ratio, n = Es/Ec	=	7.99079	-
Exposure Factor, γ_e (AASHTO LRFD 10th Ed., Art. 5.6.7-1)	=	0.75	-
Concrete Cover from Centroid of Closest Steel, dc	=	58	mm

Transformed Section

5.1 Moment at Service Limit State, M_{SVC}

Beyond Deck Joints	=	16.67	
At Deck Joints	=	17.84	

5.2 Ratio of Concrete Width to Transformed Area, B = b / (nAs)

Beyond Deck Joints	=	0.09	
At Deck Joints	=	0.06	

5.3 Depth of Neutral Axis, kd = [√(2dB+1) - 1] / B

Beyond Deck Joints	=	50.57	mm
At Deck Joints	=	57.85	mm

5.4 Neutral Axis Ratio, k = kd / d

Beyond Deck Joints	=	0.31	
At Deck Joints	=	0.36	

5.5 Lever Arm Factor, j = 1 - k / 3

Beyond Deck Joints	=	0.90	
At Deck Joints	=	0.88	

5.6 Tensile Stress at Service Limit of Steel, f_{ss} = M / (j d As)

Beyond Deck Joints	=	79.96	MPa
At Deck Joints	=	62.18	MPa

5.7 Ratio of Strain Factor, β_s = 1 + de / 0.7(h - dc)

Beyond Deck Joints	=	1.44	
At Deck Joints	=	1.44	

Maximum Reinforcement Spacing, Smax = $(123000\gamma_e/\beta_s f_{ss})$ - 2dc

Beyond Deck Joints				=	686.66
Smax =	686.66	>	140 = Spr		Sufficient reinforcement.
At Deck Joints				=	916.20
Smax =	916.20	>	100 = Spr		Sufficient reinforcement.

6.0 DISTRIBUTION REINFORCEMENT

The distribution reinforcement shall be placed transverse to the main steel reinforcement in the bottom of all slabs to provide for the lateral distribution of the concentrated live loads. DGCS Vol. 5, Sec. 14.4.3.1.

6.1 Effective Span of Slab

S = if (B_{tf}/T_{tf})<4.0 use S_g-T_w , otherwise S_g-$B_{tf}/2$	=	1547	mm
%As, 3840/√S	=	97.63	%
Area of main reinforcement, $As_{provided}$	=	1436.16	mm^2
Area of Secondary Steel, As_{dr} = if(%As<=67%,%As,67%)*Asp			
Asdr	=	962.22	mm^2
Required Spacing of Secondary Steel, Sdr = $A_b b/As_{dr}$			
Sdr	=	208.96	mm

Therefore: Use **16 mmΦ** at **200 mm O.C.**

CALCULATION SHEET

DESIGN OF ABUTMENT

Project: Prop San Isidro-EB2 Bridge (Revised) Proj. Location: Minalin, Pampanga

Designed by/Date: KH-RCDG 05/06/26

1.1 General Design Conditions & Criteria

1.1.1 Structural Conditions

One lane carriageways; total width= 5.50 m

15-15-15m single span AASHTO girders-Type III

Regular Bridge (non-skewed bridge)

Abutment type: cantilever type on concrete bored piles

1.1.2 Seismic Design Requirements and Ground Conditions

Table 7.2-1 Seismic Design Requirements and Ground Conditions

Bridge Operational Classification=	OC-III
Earthquake Ground Motion=	Level 2
Ground Type=	3
Seismic Performace Level=	3
Seismic Performance Zone=	4
Peak Ground Accelaration=	0.6g

1.1.3 Site Factors

Site Factors	
Fpga=	0.783
Fa=	0.783
Fv=	2.267
As=	0.47

1.1.4 Hydrology & Hydraulic Data

50 years Return Period

Discharge:	4.38	m3/s
Water Level, DFL:	5.60	m
Velocity:	4.00	m/s
Freeboard :	1.50	m
Drainage Area:	-	sqm
Computed scour depth:	-	m

1.1.5 Design Loads

1. Permanent Loads	
DC=	Dead load pertaining to structural and non structural components
DW=	Dead Load pertaining to future wearing surface
EH=	Horizontal earth pressure
ES=	Earth surcharge
EV=	Vertical pressure from earth fill

2.Transient Loads	
FQ=	Earthquake Load
LL/IM=	Vehicular Load/Impact load
LS=	Liveload surcharge
WA=	Water Load
FR=	Friction Load
BF=	Braking Force Load

For Seismic load analysis: Refer to BSDS

Load Combinations and factors: Refer to DGCS 10.0

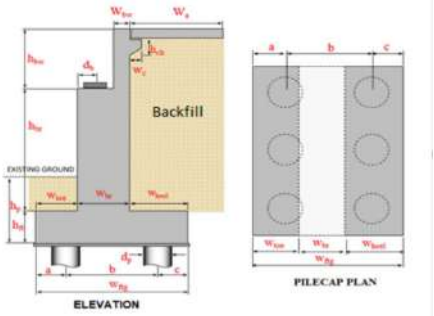
1.1.6 Material and Soil Property

Conc. Compressive strength @ 28 days, f'c	27.6	MPa
Reinforcing steel (ASTM 615),fy	414	MPa
Unit Weight of concrete,	24	kN/m^3
UnitWeight of soil,Ysoil	19	kN/m^3

Angle of internal friction of soil for granular soil,Ø			
Estimated value Ø between,30°-35°		30	deg
Angle of friction between soil and wall(JRA 2002)			
JRA TableC.22.5	seismic=	0	deg
	static=	10	deg

1.2 Geometry and Load Calculations

1.2.1 Geometry of Abutment

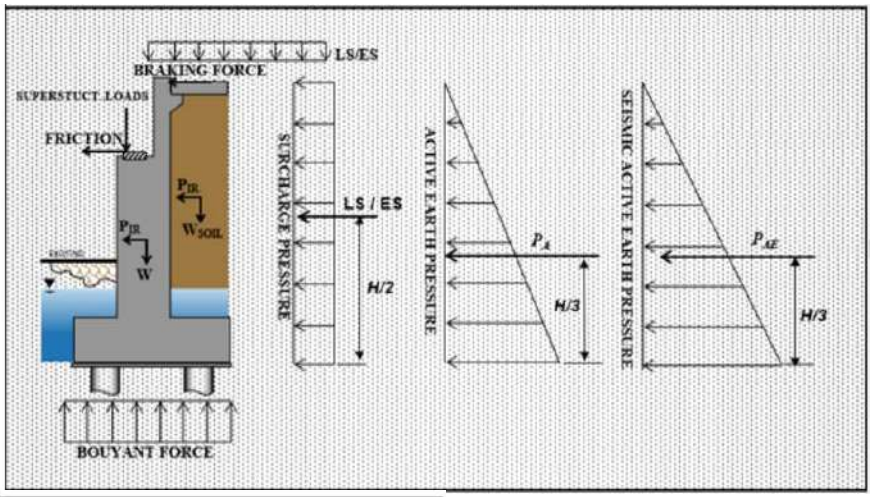


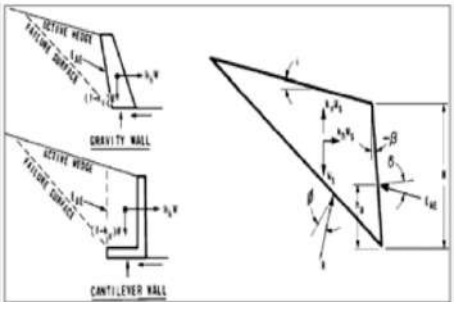
Railings			
height (h_{rl})	=	0.80 m	
thickness (t_{rl})	=	0.20 m	
length (L_{rl})	=	2.90 m	
Sidewalk			
thickness (t_{sw})	=	0.00 m	
width (w_{sw})	=	0.00 m	
length (L_{sw})	=	0.00 m	
Approach Slab			
height (h_{ap})	=	0.30 m	
width (w_{ap})	=	4.00 m	
length (L_{ap})	=	4.00 m	
Corbel			
height (h_{cb})	=	0.40 m	
width (w_{cb})	=	0.20 m	
length (L_{cb})	=	4.00 m	
Backwall			
height (h_{bw})	=	1.425 m	
width (w_{bw})	=	0.40 m	
length (L_{bw})	=	5.50 m	
Breast wall			
height (h_{br})	=	3.50 m	
width (w_{br})	=	1.00 m	
length (L_{br})	=	5.50 m	

Footing		
height (h_{ftg})	=	1.20 m
width (w_{ftg})	=	3.50 m
width (w_{toe})	=	1.00 m
width (w_{heel})	=	1.50 m
length (L_{ftg})	=	5.50 m
Wingwall		
height (h_{ww-A})	=	4.925 m
thickness (t_{ww-A})	=	0.60 m
width (w_{ww})	=	2.50 m
height (h_{ww-B})	=	4.925 m
thickness (t_{ww-B})	=	0.30 m

Bearing		
d_b	=	0.30 m
Piles size and location		
d_p	=	0.60 m
a	=	0.80 m
b	=	1.90 m
c	=	0.80 m
No. of piles, (N)	=	6
Height of soil or backfill		
Active Soil (h_a)	=	6.125 m
Passive Soil (h_p)	=	1.20 m
		(neglect passive)

1.2.2 Diagram of forces acting to Abutment





Slope of wall to vertical, β	0	deg.
Slope angle of backfill, *i*	0	deg.
Wall slope to horizontal, α	90	deg.
Backslope angle , θ	13.22	deg.

Mononobe-Okabe Method to Consider Cohesion

a. Horizontal forces from earth pressure

a.1 Active lateral earth pressure PA =1/2 YsoilH2KA

Where :

$$K_A := \frac{(\sin(\alpha+\phi))^2}{(\sin(\alpha))^2 \cdot \left(\sin(\alpha-\delta_{static})\right) \cdot \left(1+\sqrt{\frac{\sin(\phi+\delta_{static})\cdot\sin(\phi-i)}{\sin(\alpha-\delta_{static})\cdot\sin(\alpha+i)}}\right)^2}$$

Active lateral earth forces			
	P_A	P_A	P_A
K_A	Abutment (full length)	Backwall (per meter)	Breast wall (per meter)
0.308	604.65 kN	5.95 kN/m	71.08 kN/m

a.2 Seismic lateral earth pressure $P_{AE} = 1/2\ Y_{soil} H^2 K_{AE}$

Where:

$$K_{AE} := \frac{(\cos(\phi-\theta-\beta))^2}{\cos(\theta)\cdot(\cos(\beta))^2\cdot\cos(\delta_{seismic}+\beta+\theta)\left(1+\sqrt{\frac{\sin(\phi+\delta_{seismic})\cdot\sin(\phi-\theta-i)}{\cos(\delta_{seismic}+\beta+\theta)\cdot\cos((i-\beta))}}\right)^2}$$

Seismic active earth forces			
	P_{AE}	P_{AE}	P_{AE}
K_{AE}	Abutment (full length)	Backwall (per meter)	Breast wall (per meter)
0.504	988.45 kN	9.73 kN/m	116.20 kN/m

b. Horizontal forces from inertial mass of abutment, P_{IR}

Calculation of seismic acceleration coefficient of wall, k_h

Effective peak ground acc. Coefficient, A_s =		0.47	From site specific factor analysis
Seismic horizontal coefficient, $k_{ho} = A_s$ =		0.47	for other foundations
Seismic horizontal coefficient, $k_{ho} = 1.2\ A_s$		0.564	for walls founded on hard or soft rock
Vertical acceleration coefficient, k_v		0	Assume 0 for vertical acceleration

Note: in this exercise, the abutment is not founded on rock
Therefore,kho = As = 0.47

The final hor'l acceleration coefficient, kh = 1/2 kho = **0.235** for seismic coefficient of wall

---

However, the peak ground acceleration and backslope of angle shall be verified and satisfied:

verify peak ground acceleration: $(1-kv)\tan(\phi - i)$ = 0.577 $\theta = \arctan(kh/(1-kv))$ = 13.225

0.577 >k_h OK

$i + \arctan(kh/(1-kv))$ = 13.225

verify backslope angle:

ϕ > 13.225 OK

therefore, k_h = 0.235

Horizontal forces due to inertial mass of abutment, P_{IR}				
COMPONENT	Weight (kN/m)	kh*Weight (kN/m)	W* (kN)	kh*W* (kN)
Railings			22.27	5.23
Sidewalks			0.00	0.00
Wingwalls			265.95	62.50
Approach slab	28.80	6.77	115.20	27.07
Backwall	13.68	3.21	75.24	17.68
Corbel	1.92	0.45	7.68	1.80
Breast wall	84.00	19.74	462.00	108.57
Footing	100.80	23.69	554.40	130.28
Soil at heel	131.81	30.98	724.97	170.37

W* = full weight of component

c. Horizontal forces from live load surcharge pressures, LS

A live load shall be applied where vehicular load is expected to act on the surface of the backfill within a distance equal to the wall height behind the back face of the wall

DGCS Table 10.15.5.4-1

Abutment Height (m)	heq (m)
1.5	1.2
3.0	0.9
greater than 6.0	0.61

Equivalent height of soil for traffic perpendicular to wall

Labut = 5.50 m total length of Abutment wall

Habut = 6.13 m total height of Abutment wall heq = 0.61 m

Horizontal pressure due to live load surcharge = LS =	$K_A Y_{soil} h_{eq}$	=	3.58	kPa
Live load surcharge acting on abutment (whole) = LS* Habut * Labut		=	120.44	kN
Live load surcharge acting on backwall (meter strip) = LS*hbw*1m		=	5.09	kN
Live load surcharge acting on breast wall (meter strip) = LS*(hbw+hbr)*1m		=	17.61	kN

d. Horizontal forces from uniform surcharge pressures, ES

Where a uniform surcharge is present, a constant horizontal earth pressure shall be added to the basic earth pressure.

This constant earth pressure may be taken as: $\Delta_p = k_s q_s$

where:

Δ_p = *Constant horizontal earth pressure due to uniform surcharge*
k_s = *Coefficient of earth pressure due to surcharge or* K_A
q_s = *Uniform surcharge applied to the upper surface of the active earth wedge*

Horizontal pressure due to uniform surcharge = ES=	$K_A \delta_c h_{ap}$	=	2.22	kPa
Uniform surcharge acting on abutment (whole) = ES* Habut * Labut		=	74.82	kN
Uniform surcharge acting on backwall (meter strip) = ES*hbw*1m		=	3.16	kN
Uniform surcharge acting on breast wall (meter strip) = ES*(hbw+hbr)*1m		=	10.94	kN

e. Horizontal force due to movement of superstructure from via friction effect of bearing pad , FR

Dead load reaction force of the superstructure , DC	466.05	kN
Dead load of wearing surfaces and utilities, DW	102.2	kN
Live load reaction force, LL	343.797	kN
Coefficient of friction of bearing pad, μf	0.15	*(Assumption)*
Dynamic load allowance, IM	n/a	

SERVICE 1

Load Factor			Unfactored Load	Factored (kN)	
Maximum	Minimum	Type of load	(kN)	max	min
1	n/a	DC	466.05	466.05	n/a
1	n/a	DW	102.2	102.20	n/a
1	n/a	LL	343.797	343.80	n/a
		V =		912.05	
		FR = V x μ_f		136.81	

STRENGTH 1 *(Load Factor for FR = 1.0 max)*

Load Factor, Y_p		Type of load	Unfactored Load (kN)	Factored (kN)	
Maximum	Minimum			max	min
1	0.9	DC	466.05	466.05	419.45
1	0.65	DW	102.2	102.20	66.43
1		LL	343.797	343.80	0.00
			V =	912.05	485.88
			FR = V x μ_f	136.81	72.88

EXTREME EVENT 1 *(Load Factor for FR = 1.0 max, EQLL = 0.5 max)*

Load Factor, Y_p		Type of load	Unfactored Load (kN)	Factored (kN)	
Maximum	Minimum			max	min
1	0.9	DC	466.05	466.05	419.45
1	0.65	DW	102.2	102.20	66.43
0.5		LL	343.797	171.90	0.00
			V =	740.15	485.88
			FR = V x μ_f	111.02	72.88

f. Horizontal force due to braking force at approach slab, BF

The breaking force shall be taken as the greater of:

a. 25% of the axle weights of the design truck or design tandem

b. 5% of the design truck plus lane load or 5% of the design tandem plus lane load



Design lane load	9.34 kN/m	Multiple presence		
Number of lanes	1	factor, m	1.20	
Length of approach slab	4.00 m			
Length of abutment	5.50 m			

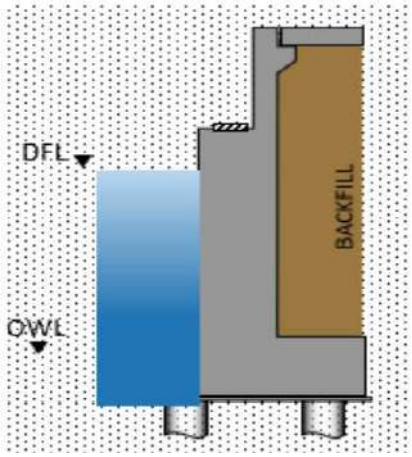
Tandem:	25% x 2 x (110kN) x no. of lanes x m /Labut =	12.00 kN/m	**Governs!**
Lane:	(5%x2x(110kN)+lane load x length of slab) x no. of lanes x m /Labut =	10.55 kN/m	
Truck:	25% x 35kN x no. of lanes x m / Labut=	1.91 kN/m	
Lane:	(5%x35kN+lane load x length of slab) x no. of lanes x m/Labut =	8.53 kN/m	

BF =	12.00	kN (considering 1m strip)
BF =	66.00	kN (considering full length)

g. Uplift force due to buoyant force, WA

Bouyancy shall be considered to be an uplift force, taken as the sum of the vertical components of static pressures acting on all components below the water.

Unit weight of water	10	kN/m^3		
Height of ordinary water level	0	m	(from bottom of ftg)	
Height of design flood level	2.72	m	(from bottom of ftg)	
Height of passive soil, hp	1.20	m		



Weight of water during ordinary water level, OWL
WA = weight of water x (volume of submerged including backfill/passive soil)

WA	DIST.
0	1.75

Buoyant force acting on footing, WA 0 kN during OWL
horizontal distance from toe 1.75 m

Weight of water during design flood level, DFL
WA = weight of water x (volume of submerged including backfill/passive soil)

WA	DIST.
523.60	1.75

Buoyant force acting on footing, WA 523.60 kN during OWL
horizontal distance from toe 1.75 m

1.2.3 Load modifiers, factors, and combinations

General load equation:

The total factored force effect shall be taken as:

$$Q = \sum \eta_i \gamma_i Q_i$$

where:

For loads for which a maximum value of is appropriate:

$$\eta_i = \eta_D \eta_R \eta_I \geq 0.95$$

For loads for which a minimum value of γi is appropriate:

$$\eta_i = \frac{1}{\eta_D \eta_R \eta_I} \leq 1.0$$

where:

γi	=	load factor
η_i	=	load modifier
η_D	=	a factor relating to ductility
η_R	=	a factor relating to redundancy
η_I	=	a factor relating to operational importance
Q_i	=	force effect

Load modifiers:

• Ductility

For strength limit state

for non-ductile components and connections	**ηD**		1.05
for conventional designs complying with AASHTO	**ηD**	=	1.00
for components and connections	**ηD**		0.95
For all other limit states	**ηD**		1.00

• Redundancy

For strength limit state

for non-redundant members	**ηR**		1.05
for conventional levels of redundancy	**ηR**	=	1.00
for exceptional level of redundancy	**ηR**		0.95
For all other limit states	**ηR**		1.00

• Operational Importance

For strength limit state

for critical and essential bridges	**ηI**		1.05
for typical bridges	**ηI**	=	1.00
for relatively less important bridges	**ηI**		0.95
For all other limit states	**ηI**		1.00

Load factors:

Table 10.3-1 Load Combination and Load Factors

Load Combination / Limit State	DC DD DW EH EV ES EL PS CR SH	LL IM CE BR PL LS	WA	WS	FR	TU	TG	SE	EQ	Use one of these at a time: BL	CT	CV
STRENGTH-I (Unless noted)	γ_p	1.75	1.00	-	1.00	0.50/1.20	0.0	γ_{se}	-	-	-	-
STRENGTH-II	γ_p	1.35	1.00	-	1.00	0.50/1.20	0.0	γ_{se}	-	-	-	-
STRENGTH-III	γ_p	1.35	1.00	-	1.00	0.50/1.20	0.0	γ_{se}	-	-	-	-
STRENGTH-III	γ_p	-	1.00	1.4	1.00	0.50/1.20	0.0	γ_{se}	-	-	-	-
STRENGTH-IV	γ_p	-	1.00	-	1.00	0.50/1.20	-	-	-	-	-	-
EH, EV,ES,DW,	1.5											
DC ONLY												
STRENGTH-V	γ_p	1.35	1.00	0.40	1.00	0.50/1.20	0.0	γ_{se}	-	-	-	-
EXTREME EVENT - I	γ_p	γ_{EQ}	1.00	-	1.00	-	-	-	1.00	-	-	-
EVENT - I												
EXTREME EVENT - II	γ_p	0.5	1.00	-	1.00	-	-	-	-	1.00	1.00	1.00
EVENT - II												
SERVICE - I	1.00	1.00	1.00	0.30	1.00	1.00/1.20	0.0	γ_{se}	-	-	-	-
SERVICE - II	1.00	1.3	1.00	-	1.00	1.00/1.20	-	-	-		-	-
SERVICE - III	1.00	0.8	1.00	-	1.00	1.00/1.20	0.0	γ_{se}	-	-	-	-
SERVICE - IV	1.00	-	1.00	0.70	1.00	1.00/1.20	-	1.0	-	-	-	-
FATIGUE – I LL, IM, & CE ONLY	-	1.50	-	-	-	-	-	-	-	-	-	-
FATIGUE – II LL, IM, & CE ONLY	-	0.75	-	-	-	-	-	-	-	-	-	-

Table 10.3-2 Load Factors for Permanent Loads, γ_p

Type of Load	Load Factor	
	Max	Min
DC: Component and Attachments	1.25	0.90
DD: Downdrag	1.80	0.45
DW: Wearing Surfaces and Utilities	1.50	0.65
EH: Horizontal Earth Pressure		
Active	1.50	0.90
At-Rest	1.35	0.90
EL: Locked-in Erection Stresses	1.00	1.00
EV: Vertical Earth Pressure		
Retaining Walls and Abutments	1.35	1.00
Rigid Buried Structure	1.30	0.90
Rigid Frames	1.35	0.90
Flexible Buried Structures other than Metal Box Culverts	1.95	0.90
Flexible Metal Box Culverts	1.50	0.90
ES: Earth Surcharge	1.50	0.75

1.3 Design of Backwall

1.3.1 Determine the applicable loads calculated from section 1.2 Geometry and Load Calculations

1.) Permanent loads

Dead load from the superstructure, DC 1	N/A	kN	
Dead load from self weight, DC 2			
a) Weight of backwall, DC 2.1	13.68	kN	Note: loads per meter strip
b) Weight of corbel, DC 2.2	1.92	kN	
c) Weight of approach slab, DC 2.3	28.80	kN	
Deadload of future wearing surface and utilities, DW	N/A	kN	
Horizontal earth pressure load, EH (=P_A)	5.95	kN	
Earth surcharge load, ES	3.16	kN	
2.) Braking force, BF	12.00	kN	
3.) Earthquake force, EQ			
Seismic earth force, *PAE*	9.73	kN	
Seismic inertial force, P_{IR}			
a) kh * Backwall	3.21	kN	
b) kh * Corbel	0.45	kN	
c) kh * Approach slab	6.77	kN	
d) kh * Soil	7.53	kN	
4.) Vehicular live load	N/A	kN	
5.) Live load surcharge, LS	5.09	kN	
6.) Friction load, FR	N/A	kN	
7.) Water load and stream pressure, WA	N/A	kN	

1.3.2 Determine the load combinations with applied load modifiers and load factors

Load modifier for maximum values, ηi: 1.05
Load modifier for minimum values, ηi: 0.95

a. Load Combination: STRENGTH 1

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min		Hor'l	Vert'l		max	min	max	min	max	min
1.25	0.90	DC 2.1	0.00	13.68	0.00	17.10	12.312	0	0	0	0
1.25	0.90	DC 2.2	0.00	1.92	-0.30	2.40	1.728	0	0	-0.72	-0.52
1.25	0.90	DC 2.3	0.00	28.80	-0.30	36.00	25.92	0	0	-10.8	-7.78
1.50	0.90	EH	5.95	0.00	0.48	0	0	8.93	5.36	4.24	2.54
1.50	0.75	ES	3.16	0.00	0.71	0	0	4.75	2.37	3.38	1.69
1.75		BF	12.00	0.00	1.43	0	0	21.00	0	29.93	0
1.75		LS	5.09	0.00	0.71	0	0	8.92	0	6.35	0
					(Strength I) Design load:	**58.28**	**37.96**	**45.77**	**7.34**	**34.00**	**-3.86**

b. Load Combination: EXTREME EVENT 1

Note: the lateral force to be applied to the wall due to seismic and earth pressure loading should be determined considering the combined effects of PAE and PIR considering and them not to be concurrent. Two cases should be investigated:

CASE 1: Combined 100% of PAE plus 50% of PIR
CASE 2: Combined 50% of PAE but no less than static active earth pressure with 100% if PIR.
The most conservative result of the two cases shall be used in the design of the backwall

b1. Load Combination: EXTREME EVENT 1 (CASE 1) (Case1: 100% PAE + 50% PIR)

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min		Hor'l	Vert'l		max	min	max	min	max	min
1.25	0.90	DC 2.1	0	13.68	0.00	17.10	12.312	0	0	0	0
1.25	0.90	DC 2.2	0	1.92	-0.30	2.4	1.728	0	0	-0.72	-0.52
1.25	0.90	DC 2.3	0	28.80	-0.30	36.00	25.92	0	0	-10.8	-7.78
1.5	0.75	ES	3.16	0	0.71	0	0	4.75	2.37	3.38	1.69
0.5		BF	12.00	0	1.43	0	0	6.00	0	8.55	0
0.5		LS	5.09	0	0.71	0	0	2.55	0	1.81	0
1.00		50%PIR_a	1.61	0	0.71	0	0	1.61	0	1.15	0
1.00		50%PIR_b	0.23	0	0.93	0	0	0.23	0	0.21	0
1.00		50%PIR_c	3.38	0	1.28	0	0	3.38	0	4.31	0
1.00		50%PIR_d	3.77	0	0.56	0	0	3.77	0	2.12	0
1.00		PAE	9.73	0	0.48	0	0	9.73	0	4.62	0
					(Extreme Event 1-Case1) Design load:	**55.50**	**39.96**	**32.01**	**2.37**	**14.64**	**-6.60**

b2. Load Combination: EXTREME EVENT 1 (CASE 2) (Case2: 50% PAE + 100% PIR)

However, if 50% PAE < PA, use PA, else use 50% PAE
Verification:
50% PAE = 4.86
P_A = EH = 5.95 > 50% PAE
therefore use EH

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
			Hor'l	Vert'l		Axial force, kN		Shear force, kN		Moment, kN-m	
max	min					max	min	max	min	max	min
1.25	0.90	DC 2.1	0	13.68	0.00	17.10	12.31	0	0	0	0
1.25	0.90	DC 2.2	0	1.92	-0.30	2.40	1.73	0	0	-0.72	-0.52
1.25	0.90	DC 2.3	0	28.80	-0.30	36.00	25.92	0	0	-10.80	-7.78
1.50	0.75	ES	3.16	0	0.71	0	0	4.75	2.37	3.38	1.69
0.50		BF	12.00	0	1.43	0	0	6.00	0	8.55	0
0.50		LS	5.09	0	0.71	0	0	2.55	0	1.81	0
1.00		PIR_a	3.21	0	0.71	0	0	3.21	0	2.29	0
1.00		PIR_b	0.45	0	0.93	0	0	0.45	0	0.42	0
1.00		PIR_c	6.77	0	1.28	0	0	6.77	0	8.63	0
1.00		PIR_d	7.53	0	0.56	0	0	7.53	0	4.24	0
1.00		EH	5.95	0	0.48	0	0	5.95	0	2.83	0
					(Extreme Event 1-Case2) Design load:	**55.50**	**39.96**	**37.21**	**2.37**	**20.63**	**-6.60**

c. Load Combination: SERVICE 1

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
			Hor'l	Vert'l		Axial force, kN		Shear force, kN		Moment, kN-m	
max	min					max	min	max	min	max	min
1.00		DC 2.1	0	13.68	0.00	13.68	0	0	0	0	0
1.00		DC 2.2	0	1.92	-0.30	1.92	0	0	0	-0.58	0
1.00		DC 2.3	0	28.80	-0.30	28.80	0	0	0	-8.64	0
1.00		EH	5.95	0	0.48	0	0	5.95	0	2.83	0
1.00		ES	3.16	0	0.71	0	0	3.16	0	2.25	0
1.00		BF	12.00	0	1.43	0	0	12.00	0	17.10	0
1.00		LS	5.09	0	0.71	0	0	5.09	0	3.63	0
					(Strength I) Design load:	**44.40**	**0**	**26.21**	**0**	**16.60**	**0**

b3. Summary of Load Combinations:

STRENGTH 1						**EXTREME EVENT 1 (CASE 1)**					
Axial force, kN		Shear force, kN		Moment, kN-m		Axial force, kN		Shear force, kN		Moment, kN-m	
max	min	max	min	max	min	max	min	max	min	max	min
58.28	**37.96**	**45.77**	**7.34**	**34.00**	**-3.86**	**55.50**	**39.96**	**32.01**	**2.37**	**14.64**	**-6.60**

EXTREME EVENT 1 (CASE 2)						**SERVICE 1**					
Axial force, kN		Shear force, kN		Moment, kN-m		Axial force, kN		Shear force, kN		Moment, kN-m	
max	min	max	min	max	min	max	min	max	min	max	min
55.50	**39.96**	**37.21**	**2.37**	**20.63**	**-6.60**	**44.40**	**0.00**	**26.21**	**0.00**	**16.60**	**0.00**

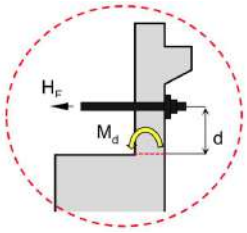
1.3.3 Determine the governing design forces

Load Combination:	**STRENGTH 1**	
Axial force =	**58.28**	kN
Shear force =	**45.77**	kN
Moment =	**34.00**	kN-m

Note: The governing design forces shall be verified from : Verification of demand forces for unseating prevention device (or if the bridge superstructure has fixed connection) from Design of Breast Wall.

The design forces required for the unseating prevention device, or if the superstructure has fixed connection, as determined from Breast wall design are as follows: **- NOT APPLICABLE**

H_F = Vd = **127** kN
Md = **76** kN-m



1.3.4 Verification of flexural resistance

Demand moment, *Md*	**34**	kN-m
Concrete cover	50	mm
Diameter of reinforcing bar	20	mm
Area of reinforcing bar, A_b	314.16	mm^2
Diameter of shrinkage bar	16	mm
Diameter of cross ties	12	mm
Effective depth of concrete, *de*	340	mm
Width to be considered, *b*	1000	mm
Overall thickness of component, *h*	400	mm

Minimum reinforcement

Flexural cracking variability factor, γ_1	1.6	*for all other concrete*	
Ratio of specified min. to ult. Tensile strength of steel, γ_3	0.67	*for A6165, 414MPa steel*	
Modulus of rupture, f_r	3.310	MPa	
Section modulus, S_c	2.7E+07	mm^3	
Cracking moment, M_{cr}			
$M_{cr} = \gamma_3 (\gamma_1 \cdot f_r) Sc$	94.61	kN-m	
Mu_{min} 1.33*Md =	45.22	kN-m	
Condition: if Md > min(M_{cr}, Mu_{min}), Md, min(M_{cr}, Mu_{min})			
Therefore, Design moment for backwall	**45**	kN-m	**Governs!!!**

Steel ratio

β1 = Coefficient Criterion: = 0.85
the factor β1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β1 shall be reduced *at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65*

For required steel ratio, ρ

m_1 =	0.0567	
m_2 =	0.0853	mm^2/N
R_n =	0.435	MPa
ρ =	0.0011	

Computation for main reinforcement

Required steel area, *As*				360.30	mm^2	
Required spacing, *s*				871.93	say: 150	mm
Provided steel for backwall, *As*				2094.40	mm^2	
Depth of compression block, *a*				36.96	mm	
Compression fiber to neutral axis, *c*				43.48	mm	
Stress Check:	fs=600((d-c)/c)=	4091.57	MPa	fs > fy	tension steel yields	
Nominal moment capacity of section, *Mn*				278.78	kN-m	
Check net tensile strain:	εt = 0.003((de/c)-1) =	0.020		> 0.005	tension is controlled	
Resistance factor, *Φ*				0.9		
0.75 ≤ φ = 0.65 + 0.15(deff/c-1) ≤ 0.9	φ = 1.67	*(φ is between 0.75 and 0.9 only)*				
Ultimate moment capacity of section, *Φ Mn*				**250.91**	kN-m	**OK!**
				c/d =	5.55	

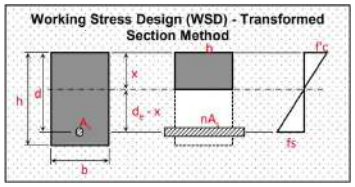
Using of 20mm □ main bars spaced at 150mm O.C. for backwall is adequate

Control of cracking by distribution of reinforcement

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1			16.60	kN-m
80% of Modulus of rupture, f_r	80% x 0.52√f'c =		2.185	MPa
Tensile stress in steel at the service limit, f_{ss}	Ms/Sc =		0.622	MPa

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does NOT need to be satisfied.



Extreme tension fiber to center of flexural reinforcement, d_c	-	mm	
Overall thickness of component, *h*	-	mm	
Compression fiber to the centroid of extreme tension steel, d_e	-	mm	
Neutral axis to extreme compression fiber, *x*	-	mm	
Modulus of elasticity of steel, E_s	-	GPa	
Modulus of elasticity of concrete, E_c	-	GPa	
Modular ration, *n*	-		
Cracked section moment of inertia of section, I_{NA}	-	mm^4	
Exposure factor, Y_e	-		
Exposure condition: Class 1			
Tensile stress in steel reinforcement at the service limit, f_{ss}	-	MPa	
βs = 1 + d_c/(0.7(h-d_c))	-		
The spacing shall satisfy: s≤123000Y_e/$\beta_s f_{ss}$-2dc	minimum spacing, s =	-	mm

1.3.5 Verification of shear resistance

Effective shear depth, d_v		321.52	mm
Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h	0.9*de =	306	mm
	0.72*h =	288	mm

Factor indicating ability of diagonally cracked concrete to transmit tension, β	3.91		
Solution for β: **GENERAL PROCEDURE**			
Area of prestressing steel on tension side, A_{ps}	0	mm^2	
Area of non-prestressing steel, A_s	2094.40	mm^2	
Maximum aggregate size, a_g	20	mm	
Modulus of elasticity of prestressing tendons, f_{po}	0	MPa	
Factored axial force, N_u	-58275	N	
*Positive for tension; Negative for compression			
Factored shear force, V_u	45768.06	N	
Absolute value of the factored moment, $\|M_u\|$	3.40E+07	N-mm	
*But not less than $\|V_u - V_p\| d_v$			
Modulus of elasticity of prestressing steel, E_p	0	GPa	
Modulus of elasticity of steel, *Es*	200	GPa	
Net longitudinal tensile strain, εs	0.000		
Crack spacing parameter, S_{xe}	313.04	mm	
Shear resistance from steel (cross ties), *Vs*	-	kN	
Effective prestressing force, V_p	0	kN	
Shear resistance provided by concrete, V_c	548.51	kN	
The nominal shear resistance, V_n	548.51	kN	
*shall be determined as the lesser of:			
$V_n=V_c+V_s+V_p$	548.51	kN	
$V_n=0.25f'c\ b_v d_v+V_p$	2218.49	kN	
Factored ultimate shear demand, Vu = Vd =	**46**	kN	
Resistance factor for normal weight concrete, *Φ*	0.9		
Ultimate shear capacity of section, *Φ* Vn	**494**	kN	**OK!**
	c/d =	10.79	

1.3.5.1 Check Shear Reinforcement Requirement

If Vu > Φ(Vc + Vp) provide shear reinforcement, Vs

Vu = 46 kN < 493.66 kN = Φ(Vc + Vp)

∴ Section without shear reinfrocement is adequate.

Required Vs = (Vu/Φ) - (Vc + Vp)

Req'd Vs = - kN

Shear resistance from provided cross ties, Vs = (b/splength)*Av*fsy*dv / spheight

splength = - mm

spheight = - mm

Diameter of cross ties - mm

Area of cross ties, Av = - mm2

Provided Vs = - kN -

No need to provide shear reinforcement.

1.3.6 Verification of interface shear resistance

Interface shear transfer shall be considered across a given plane at:
a) An existing or potential crack
b) An interface between dissimilar materials
c) An interface between two concrete cast at different times
d) The interface between different elements of the cross-section

Number of bars provided (for both faces per meter strip), *N*	say	13.33	pcs	
Area of shear reinforcement crossing the shear plane, A_{vf}		4,188.79	mm^2	
*Minimum area of shear interface shall satisfy:				
$A_{vf} \geq (0.35A_{cv})/f_y$		192.75	mm^2	**OK!**
Interface length considered to be engaged inshear transfer, L_{vi}		1000	mm	
Interface width considered to be engaged inshear transfer, b_{vi}		228	mm	
Area of concrete considered to be engaged in interface shear transfer, A_{cv}		228000	mm^2	
Permanent net compressive force normal to the shear plane, P_c		58275	N	
Factored interface shear force due to total load, V_{ui}		**46**	kN	

Cohesion and friction factors

**for concrete placed against a clean concrete surface, free of laitance, with surface intentionally roughened to an amplitude of 6mm:*

Cohesion factor, *c*	1.7	MPa
Friction factor, *m*	1.00	
Fraction of concrete to resist interface shear, K_1	0.25	
Limiting interface shear resistance, K_2	10.3	MPa

The nominal shear resistance of the interface plane shall be taken as:

Vni=cAcv+ μ(Avffy+Pc) 2180.03 kN

The nominal shear resistance, V_{ni} shall not be greater than the lesser of:

a) K1f'cAcv 1,573.20 kN
b) K_2A_{cv} 2348.40 kN

Nominal shear resistance of the interface plane, V_{ni} 1573 kN
Resistance factor for normal weight concrete, Φ 0.9
Factored interface shear resistance of the section. V_{ri} **1416** kN **OK!**
c/d = 30.94

1.3.7 Verification of shrinkage and temperature reinforcement

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete.

Diameter of shrinkage and temperature bar 16 mm
Assumed spacing, S 200 mm
Assumed shrinkage and temperature reinforcement, A_s 1.005 mm^2/mm

Shrinkage and temperature reinforcement shall satisfy:
a) As≥ (0.75bh)/(2(b+h)fy) = 0.259 1.005 mm^2/mm
b) 0.233 ≤ As ≤ 1.27 1.00531 mm^2/mm

Spacing shall not exceed:
a) 3.0 times the component thickness, or 450mm
b) 300mm for walls and footings greater than 450mm thick
c) 300mm for other components greater than 900mm thick

Final shrinkage and temperature reinforcement, A_s 1005.31 mm^2 per meter
Final spacing to be used, S 200 mm
say: 200 mm 200

Use 16mm for temperature and shrinkage bar spaced at 200mm O.C. each face

1.3.8 Development of reinforcement

Diameter of main bars, d_b 20 mm
Area of main bars, A_b 314.16 mm^2
Basic tension development length, l_{db} 495.14 mm
Minimum development length *(only for d_b 36mm and lesser)* 496.80 mm

Modification.
Modification factor 1 1.00
Modification factor 2 0.17
Final development length, *ld* 496.80 say: 550 mm

1.4 DESIGN OF BREAST WALL

1.4.1 Determine the applicable loads calculated from section 1.2 Geometry and Load Calculations

1. Permanent Loads			• loads per meter strip
Dead load reaction force of the superstructure, DC 1	84.74	kN	
Dead load from self-weight, DC 2			
a) Weight of backwall, DC 2.1	13.68	kN	
b) Weight of corbel, DC 2.2	1.92	kN	
c) Weight of breast wall, DC 2.3	84.00	kN	
d) Weight of approach slab, DC 2.4	28.80	kN	
Dead load of wearing surfaces and utilities,DW	18.58	kN	
Horizontal Earth pressure load, $_{E}$H(= PA)	71.08	kN	
Earth surcharge load, ES	10.94	kN	
2. Braking force, BF	12.00	kN	• BF is the breaking from approach slab.
3. Earthquake force, EQ			BF from
Seismic earth force, PAE	116.20	kN	superstructure in N/A
Seismic inertia force, PIR			
a) kh* Backwall	3.21	kN	• Note: FR loads are
b) kh*Corbel	0.45	kN	already factored
c) kh*Breast wall	19.74	kN	
d) kh* Approach slab	6.77	kN	
e) kh* Soil	30.98	kN	
f) kh*(DC+DW+0.5LL)	31.62	kN	• DL from superstructure
4. Vehicular Live Load	62.51	kN	
5. Live load surcharge, LS	17.61	kN	
6. Friction load, FR			
a) Strength 1			
-at maximum condition	24.87	kN	
-at minimum condition	13.25	kN	
b) Extreme event 1			
-at maximum condition	20.19	kN	
-at minimum condition	13.25	kN	
c) Service 1			
-at maximum condition	24.87	kN	
-at minimum condition	0	kN	
7. Water load and stream pressure,WA	N/A	kN	

1.4.2 Determine the load combination with applied load modifiers and load factors

Load modifier for maximum values, n_i = 1.05
Load modifier for minimum values, n_i = 0.95

• Applications of modifier are similar to backwall

a. Load Combination: STRENGTH 1

Load Factor			Unfactored load,kN		Lever arm	Factored					
						Axial force,kN		Shear force,kN		Moment,kN-m	
max	min	Load Type	Hor'l	Ver'l	(m)	max	min	max	min	max	min
1.25	0.90	DC1	0	84.74	0.20	105.92	76.26	0	0	21.18	15.25
1.25	0.90	DC2.1	0	13.68	-0.30	17.1	12.31	0	0	-5.13	-3.69
1.25	0.90	DC2.2	0	1.92	-0.60	2.4	1.73	0	0	-1.44	-1.04
1.25	0.90	DC2.3	0	84.00	0	105	75.60	0	0	0.00	0.00
1.25	0.90	DC2.4	0	28.80	-0.60	36	25.92	0	0	-21.60	-15.55
1.50	0.65	DW	0	18.58	0.20	27.87	12.08	0	0	5.57	2.42
1.50	0.90	EH	71.08	0	1.64	0	0	106.62	63.97	175.03	105.02
1.50	0.75	ES	10.94	0	2.4625	0	0	16.41	8.20	40.40	20.20
		FR	0	0	3.50	0	0	24.87	13.25	87.06	46.38
1.75		LL	0	62.51	0.20	109.39	0	0	0	21.88	0.00
1.75		BF	12.00	0	4.93	0	0	21.00	0	103.43	0.00
1.75		LS	17.61	0	2.4625	0	0	30.81	0	75.88	0.00
		(Strength 1) Design load:				**423.87**	**193.71**	**209.70**	**81.15**	**527.38**	**160.54**

• Modifier, n_i
$n_D \geq$ 1.05
$n_R \geq$ 1.00
$n_i \geq$ 1.00
for max. values
ni=(1.05x1.0x1.0)
n_i=1.05
for min. values
n_i=1/(1.05x1.0x1.0)
n_i=0.95

• Note: Friction forces are already factored on section Geometry and Load

b. Load Combination:EXTREME EVENT I

b.1 Load Combination: EXTREME EVENT I (CASE 1)(Case 1: 100% P_{AE} +50% P_{IE})

Load Factor			Unfactored load,kN		Lever arm	Factored					
						Axial force,kN		Shear force,kN		Moment,kN-m	
max	min	Load Type	Hor'l	Ver'l	(m)	max	min	max	min	max	min
1.25	0.90	DC 1	0	84.74	0.20	105.92	76.26	0.00	0.00	21.18	15.25
1.25	0.90	DC 2.1	0	13.68	-0.30	17.10	12.31	0.00	0.00	-5.13	-3.69
1.25	0.90	DC 2.2	0	1.92	-0.60	2.4	1.73	0.00	0.00	-1.44	-1.04
1.25	0.90	DC 2.3	0	84.00	0.00	105.00	75.60	0.00	0.00	0.00	0.00
1.25	0.90	DC 2.4	0	28.80	-0.60	36.00	25.92	0.00	0.00	-21.60	-15.55
1.50	0.65	DW	0	18.58	0.20	27.87	12.08	0.00	0.00	5.57	2.42
1.50	0.75	ES	10.94	0	2.46	0.00	0.00	16.41	8.20	40.40	20.20
0.50		LL	0	62.51	0.20	31.25	0.00	0.00	0.00	6.25	0.00
0.50		BF	12.00	0	4.93	0.00	0.00	6.00	0.00	29.55	0.00
0.50		LS	17.61	0	2.46	0.00	0.00	8.80	0.00	21.68	0.00
		FR	0	0	3.50	0.00	0.00	20.19	13.25	70.65	46.38
1.00		PAE	116.20	0	1.64	0.00	0.00	116.20	0.00	190.76	0.00
1.00		50%P_{IE}_a	1.61	0	4.21	0.00	0.00	1.61	0.00	6.77	0.00
1.00		50%P_{IE}_b	0.23	0	4.43	0.00	0.00	0.23	0.00	1.00	0.00
1.00		50%P_{IE}_c	9.87	0	1.75	0.00	0.00	9.87	0.00	17.27	0.00
1.00		50%P_{IE}_d	3.39	0	4.78	0.00	0.00	3.39	0.00	16.16	0.00
1.00		50%P_{IE}_e	15.49	0	2.31	0.00	0.00	15.49	0.00	35.82	0.00
1.00		50%P_{IE}_f	15.81	0	3.50	0.00	0.00	15.81	0.00	55.34	0.00
		(Extreme Event 1-Case1) Design load:				**325.55**	**203.90**	**213.98**	**21.45**	**490.24**	**63.97**

• Modifier, n_i
ni=1.00

• Note: Friction forces are already factored on section Geometry and Load

b.2 Load Combination: EXTREME EVENT I (CASE 2)(Case 2: 50% P_{AE} +100% P_{IR})

However, if 50% P_{AE} <P_A , use P_A, else use 50% P_{AE}

Verification:

50%P_{AE} = 58.10

P_A = EH = 71.08 > 50% PAE

therefore use EH

Load Factor		Load Type	Unfactored load,kN		Lever arm	Factored					
						Axial force,kN		Shear force,kN		Moment,kN-m	
max	min		Hor'l	Ver'l	(m)	max	min	max	min	max	min
1.25	0.90	DC 1	0	84.74	0.20	105.92	76.26	0.00	0.00	21.18	15.25
1.25	0.90	DC 2.1	0	13.68	-0.30	17.10	12.31	0.00	0.00	-5.13	-3.69
1.25	0.90	DC 2.2	0	1.92	-0.60	2.40	1.73	0.00	0.00	-1.44	-1.04
1.25	0.90	DC 2.3	0	84.00	0.00	105.00	75.60	0.00	0.00	0.00	0.00
1.25	0.90	DC 2.4	0	28.80	-0.60	36.00	25.92	0.00	0.00	-21.60	-15.55
1.50	0.65	DW	0	18.58	0.20	27.87	12.08	0.00	0.00	5.57	2.42
1.50	0.75	ES	10.94	0	2.46	0.00	0.00	16.41	8.20	40.40	20.20
0.50		LL	0	62.51	0.20	31.25	0.00	0.00	0.00	6.25	0.00
0.50		BF	12.00	0	4.93	0.00	0.00	6.00	0.00	29.55	0.00
0.50		LS	17.61	0	2.46	0.00	0.00	8.80	0.00	21.68	0.00
		FR	0	0	3.50	0.00	0.00	20.19	13.25	70.65	46.38
1.00		EH	71.08	0	1.64	0.00	0.00	71.08	0.00	116.57	0.00
1.00		P_{IR}_a	3.21	0	4.21	0.00	0.00	3.21	0.00	13.54	0.00
1.00		P_{IR}_b	0.45	0	4.43	0.00	0.00	0.45	0.00	2.00	0.00
1.00		P_{IR}_c	19.74	0	1.75	0.00	0.00	19.74	0.00	34.55	0.00
1.00		P_{IR}_d	6.77	0	4.78	0.00	0.00	6.77	0.00	32.33	0.00
1.00		P_{IR}_e	30.98	0	2.31	0.00	0.00	30.98	0.00	71.63	0.00
1.00		P_{IR}_f	31.62	0	3.50	0.00	0.00	31.62	0.00	110.69	0.00
		(Extreme Event 1-Case2) Design load:				**325.55**	**203.90**	**215.25**	**21.45**	**548.42**	**63.97**

- Modifier, n_i ni= 1.00
- Note: Friction forces are already factored on section Geometry and Load

c. Load CombinationsL SERVICE 1

Load Factor		Load Type	Unfactored load,kN		Lever arm	Factored					
						Axial force,kN		Shear force,kN		Moment,kN-m	
max	min		Hor'l	Ver'l	(m)	max	min	max	min	max	min
1		DC 1	0	84.74	0.2	84.74	0.00	0.00	0.00	16.95	0.00
1		DC 2.1	0	13.68	-0.3	13.68	0.00	0.00	0.00	-4.10	0.00
1		DC 2.2	0	1.92	-0.60	1.92	0.00	0.00	0.00	-1.15	0.00
1		DC 2.3	0	84.00	0	84.00	0.00	0.00	0.00	0.00	0.00
1		DC 2.4	0	28.80	-0.6	28.80	0.00	0.00	0.00	-17.28	0.00
1		DW	0	18.58	0.2	18.58	0.00	0.00	0.00	3.72	0.00
1		EH	71.08	0	1.64	0.00	0.00	71.08	0.00	116.57	0.00
1		ES	10.94	0	2.4625	0.00	0.00	10.94	0.00	26.94	0.00
1		FR	0	0	3.5	0.00	0.00	24.87	0.00	87.06	0.00
1		LL	0	62.51	0.2	62.51	0.00	0.00	0.00	12.50	0.00
1		LS	17.61	0	2.4625	0.00	0.00	17.61	0.00	43.36	0.00
1		BF	12.00	0	4.925	0.00	0.00	12.00	0.00	59.10	0.00
		(Service 1) Design load:				**294.23**	**0**	**136.50**	**0**	**343.65**	**0**

- Modifier, n_i ni= 1.00

Summary of Load Combinations:

STRENGTH I						EXTREME EVENT I (CASE 1)					
Axial force,kN		Shear force,kN		Moment,kN-m		Axial force,kN		Shear force,kN		Moment,kN-m	
max	min	max	min	max	min	max	min	max	min	max	min
423.87	**193.71**	**209.70**	**81.15**	**527.38**	**160.54**	**325.55**	**203.90**	**213.98**	**21.45**	**490.24**	**63.97**

EXTREME EVENT I (CASE 2)						SERVICE I					
Axial force,kN		Shear force,kN		Moment,kN-m		Axial force,kN		Shear force,kN		Moment,kN-m	
max	min	max	min	max	min	max	min	max	min	max	min
325.55	**203.90**	**215.25**	**21.45**	**548.42**	**63.97**	**294.23**	**0**	**136.50**	**0**	**343.65**	**0**

1.4.3 Determine the governing design forces:

Load Combination:	**EXTREME EVENT I (CASE 2)**
Axial Force=	**325.55** kN
Shear Force=	**215.25** kN
Moment=	**548.42** kN

- The summary of load combinations show that EXTREME EVENT I (CASE 2) is critical

1.4.4 Verification of Flexural Resistance

Demand moment	**548.42**	kN-m		
Concrete cover	75	mm		
Diameter of reinforcing bar	25	mm	Ab = 490.87 mm2	
Diameter of shrinkage bar	16	mm		
Diameter of cross ties	12	mm		
Effective depth of concrete, de	912.5	mm		
Width to be considered, b	1000	mm		
Overall thickness of component, h	1000	mm		
Minimum |Reinforcements				
Flexural cracking variability factor, $Y1$			1.6	*for all other concrete*
Ratio of specified min. to ult. tensile strength of steel, $Y3$			0.67	*for A615, 414MPa steel*
Modulus of rupture, f_r			3.310 MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c			1.67E+08 mm^3	Sc=I/c
Cracking Moment, M_{cr}				
M_{cr}=Y_3(Y_1*fr) S_c			591.34 kN-m	
$M_{u_}$min 1.33*Md=			729.40 kN-m	
Condition: if Md > min (Mcr, Mu_min), Md, min (Mcr,Mu_min)				
Design moment for breast wall			**591.34** kN-m	**Governs!!!**

- per 1m-width design
- DGCS 12.4.3.3

Steel ratio

β_1 Coefficient Criterion: = 0.85 • DGCS 12.4.2.1

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ

m1 =	0.0567		m1=0.85*f'c/fy
m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)
Rn =	0.789	MPa	
ρ=	0.0019		ρ=m1*(1-sqrt(1-m2Rn))

Computation for main reinforcement

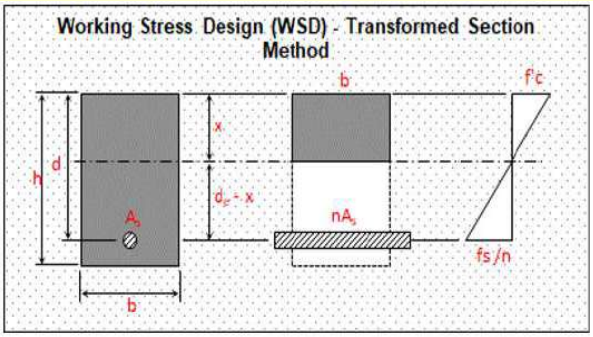
Required steel area, As	1769.531	mm^2		
Required spacing	277.40		say: 200	mm
Provided steel for breast wall, As	2454.37	mm^2		
Compression fiber to neutral axis, c	50.96	mm		
Depth of compression block, a	43.31	mm		
Nominal moment capacity of section, M_n	905.19	kN-m		
Resistance factor, Ø	0.9	tension is controlled		
0.75 ≤ Ø =0.65+0.15 (deff/c −1) ≤0.9	φ = 3.19	*(φ is between 0.75 and 0.9 only)*		
Ultimate moment capacity of section, ØM_n	**814.67**	kN-m	**OK!**	
	c/d = 1.38			

Control of cracking by distribution of reinforcement • DGCS 12.4.3.4

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1	343.65	kN-m	• DGCS 12.1.1.6
80% of Modulus of rupture, f_r	2.185	MPa	
Tension in the cross section	2.062	MPa	

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does NOT need to be satisfied.



Extreme tension fiber to center of flexural reinforcement, d_c	-	mm	• by quadratic equation
Overall thickness of component, h	-	mm	determine x:
Compression fiber to the centroid of extreme tension steel, d_e	-	mm	a= -
Neutral axis to extreme compression fiber, x	-	mm	b= -
Modulus elasticity of steel, E_s	-	Gpa	c= -
Modulus elasticity of concrete, E_c	-	GPa	
Modular ratio, n	-		
Cracked section moment of inertia of section, I NA	-	mm^4	
Exposure factor, Y_e	-		
Exposure condition: Class 1			
Tensile stress in steel reinforcement at the service limit, fss	-	MPa	
$\beta s = 1 + {}_dc / (0.7 (h - dc))$	-		
The spacing shall satisfy: $s \le 123000 y_e/\beta s f_{ss} - 2dc$	-	mm	

Initial spacing: - mm -

Using of 25mm ⌀ main bars spaced at 200mm O.C. for breast wall is adequate and safe

1.4.5 Verification of shear resistance

• DGCS 12.5.3.2

Effective shear depth, d_v		890.84	mm
Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h	=0.9*de	821.25	mm
	=0.72*h	720	mm

Factor indicating ability of diagonally cracked concrete to transmit tension,	1.66		
Solution for β: **GENERAL PROCEDURE**			• General procedure is basically applicable to design of walls,slab and footings,with thickness > 400mm
Area of prestressing steel on tension side, A_{ps}	0	mm^2	
Area of non-prestressing steel, A_s	2454.37	mm^2	
Maximum aggregate size, a_g	20	mm	
Modulus of elasticity of prestressing tendons, f_{po}	0	MPa	
Factored axial force, N_u	-325547.5	N	• DGCS 12.5.3.3.2
*Positive for tension; Negative for compression			

Factored shear force, V_u	**215252.6**	N		
Absolute value of the factored moment, $\|M_u\|$	5.48E+08	N-mm		
*But not less than $\|V_u - V_p\| d_v$				
Modulus of elasticity of prestressing steel, E_p	0	GPa		
Modulus of elasticity of steel, E_s	200	GPa		
Net longitudinal tensile strain, e_s	0.0014			
Crack spacing parameter, S_{xe}	867.34	mm		
Shear resistance from steel, V_s	0	kN		
Effective prestressing force, V_p	0	kN		
Shear resistance provided by concrete, V_c	643.32	kN		
The nominal shear resistance, V_n	643.32	kN		
*shall be determined as the lesser of:				
$V_n=V_c+V_s+V_p$	643.32	kN		
$V_n=0.25fc'b_vd_v+V_p$	6146.82	kN		
Factored ultimate shear demand, Vu = Vd =	215.25	kN		
Resistance factor for normal weight concrete, Ø	0.9			
Ultimate shear capacity of section, ØV n	**578.99**	kN	**OK!**	
	c/d=	2.69		
Section without shear reinforcement is adequate and safe.				

1.4.5.1 Check Shear Reinforcement Requirement

If Vu > Φ(Vc + Vp) provide shear reinforcement, Vs

Vu = 215.25 kN < 578.99 kN = Φ(Vc + Vp)

∴ Section without shear reinfrocement is adequate.

Required Vs = (Vu/Φ) - (Vc + Vp)

Req'd Vs = - kN

Shear resistance from provided cross ties, Vs = (b/splength)*Av*fsy*dv / spheight

splength = - mm

spheight = - mm

Diameter of cross ties = - mm

Area of cross ties, Av = - mm^2

Provided Vs = - kN -

No need to provide shear reinforcement.

1.4.6 Verification of interface shear resistance

Interface shear transfer shall be considered across a given plane at:
a) An existing or potential crack
b) An interface between dessimilar materials
c) An interface between two concrete cast at different times
d) The interface between different elements of the cross-section

• DGCS 12.5.5

Number of bars provided (for both faces per meter strip), N	10.00	pcs		• N=(b/s)x 2sides
Area of shear reinforcement crossing the shear plane, Avf	4,908.74	mm^2		• DGCS 12.5.5.3
*Minimum area of shear interface shall satisfy:				
$A_{vf}\geq(0.35A_{cv})/fy$	649.28	mm^2	**SATISFIED!**	

Interface length considered to be engaged inshear transfer, L_{vi}	1000	mm	
Interface width considered to be engaged inshear transfer, b_{vi}	768	mm	
Area of concrete considered to be engaged in interface shear transfer, A_{cv}	768000	mm^2	
Permanent net compressive force normal to the shear plan, P_c	325547	N	
Factored interface shear force due to total load, V_{ui}	215.25	kN	
Cohesion and friction factors			• DGCS 12.5.5.2
**for concrete placed against a clean concrete surface, free of laitance, with surface intentionally roughened to an amplitude of 6mm:*			
Cohesion factor, c	1.7	Mpa	
Friction factor, m	1		
Fraction of concrete to resist interface shear, K_1	0.25		
Limiting interface shear resistance, K_2	10.3	MPa	
The nominal shear resistance of the interface plane shall be taken as:			
Vni= cAcv+ μ(Avffy+Pc)	3663.37	kN	
The nominal shear resistance, V_{ni} shall not be greater than the lesser of:		kN	
a) $k_1f'cA_{cv}$	5,299.20	kN	
b) k_2A_{cv}	7,910.40	kN	
Nominal shear resistance of the interface plane, V_{ni}	3663.37	kN	
Resistance factor for normal weight concrete, $_f$	0.9	kN	
Factored interace shear resistance of the section, V_{ri}	**3297.03**	kN	**OK!**

Section is adequate for interface shear transfer.

1.4.7 Verification of shrinkage and temperature reinforcement

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete.

• DGCS 12.7.8

Diameter of shrinkage and temperature bar	16	mm	
Assumed spacing, S	200	mm	
Assumed shrinkage and temperature reinforcement, As	1.005	mm^2/mm	
Shrinkage and temperature reinforcement shall satisfy:			
a) As ≥ (0.75bh)/(2(b+h)fy) = 0.453	1.005	mm^2/mm	
b) 0.233 ≤ As ≤1.27	1.00531	mm^2/mm	
Spacing shall not exceed:			
a)3.0 times the component thickness, or 450 mm			
b)300 mm for walls and footings greater than 450 mm thick			
c)300 mm for other components greater than 900 mm thick			
Final shrinkage and temperature reinforcement, A_s	1,005.31	mm^2 per meter	
Final spacing to be used	200	mm	
say:	200	mm	

Use 16mm for temperature and shrinkage bar spaced at 200mm O.C. each face

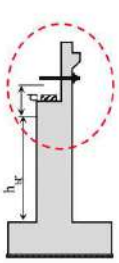
1.4.8 Development of reinforcement

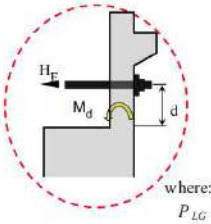
Diameter of main bars, d_b	25	mm		• DGCS 12.8.2.1
Area of main bars, A_b	490.87	mm^2		
Basic tension development length, l_{db}	773.65	mm		
Minimum development length *(only for d_b 36 mm and lesser)*	621.00	mm		
Modification factor that decrease l db				
Modification factor 1	0.8			
Modification factor 2	0.72			
Final development length, l_d	621.00		say: 700 mm	
Note: For effective anchorage the rebars should rest on pilecap bottom bars				

1.4.9 Verification of demand forces for unseating prevention device for backwall - NOT APPLICABLE

The ultimate strength of an unseating prevention device shall not be less than the design seismic force .

• BSDS 7.3





When the unseating prevention device directly connects the superstructure, the design seismic force shall be:

$$H_F = PL_G$$

However, H_F shall not exceed 1.5 R_D

where:

P_{LG} = The lesser value corresponding to lateral (hor'l) capacity of the breast wall calculated from its nominal flexural resistance, or the nominal shear resistance of the breast wall.

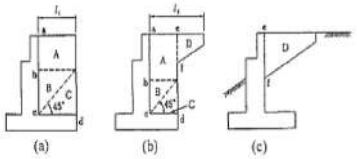
Distance of unseating prevention device from the base of backwall, d	0.60	m		*(assumed height if with unseating)*
Nominal flexural resistance of the breast wall, M_n	905.19	kN-m		*or half of girder (if fixed connection)*
Deadload reaction from superstructure (per meter strip), R_D	84.74	kN		
a. Lateral capacity of breast wall from its nominal flexural resistance	258.63	kN	=Mn/hbr	
b. Nominal shear resistance of the breast wall, V_n	643.32	kN		
c. 1.5 times the deadload reaction of superstructure	127.10	kN		
Therefore, the design seismic force of the unseating prevention device	**127**	kN	$=H_F$	
Design moment to be considered in designing the backwall, M_d	**76**	kN-m	$=H_F$ x d	

1.5 DESIGN OF WING WALLS

Reference: JRA, 2002-Part IV substructure Art. 8.4.4

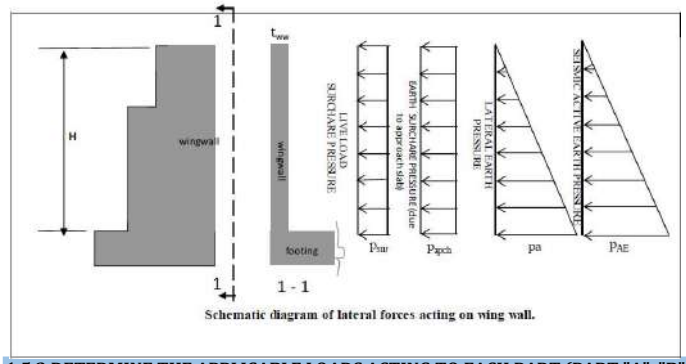
(1) The wing walls shall be designed as slabs to receive superimposed loads due to live loads and the earth pressure.

(2) The slabs in this case shall be cantilevers fixed to a wall or slabs fixed on two sides to a wall and footing.



Note:
When l1 or l2 are greater than 8m, design the wingwall fixed on 2 sides (vertical wall and footing). Otherwise design part A, B, and C as cantilever wall fixed on sides a-b, b-c, and c-d, respectively. While part D fixed on side e-f.

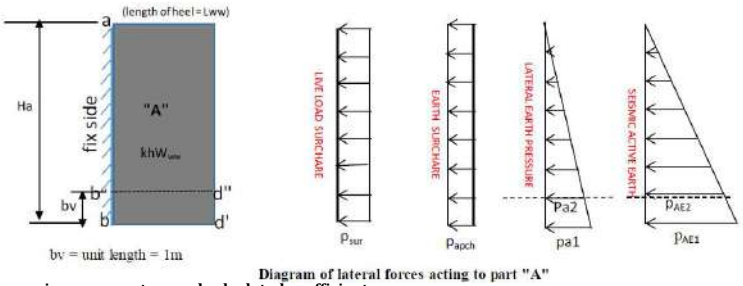
1.5.1 DIAGRAM OF FORCES ACTING TO WALL



Schematic diagram of lateral forces acting on wing wall.

1.5.2 DETERMINE THE APPLICABLE LOADS ACTING TO EACH PART (PART "A", "B" AND "C") AS SHOWN IN THE SHAPE OF WING WALL (a).

PART "A"



bv = unit length = 1m

Diagram of lateral forces acting to part "A"

From given parameters and calculated coefficients :

	kh=	0.235	
	ka=	0.308	
	k_{AE} =	0.504	
	δ_c =	31.20	kN/m^3
(let ww constant sect.)	***Y soil* =**	19.00	kN/m^3
WINGWALL DIMENSIONS	**H=**	4.93	m
	t_{ww} =	0.6 ~ 0.3	m
	L_{ww} =	1.50	m
	height (h_{ww-A})	4.93	m
	thickness (t_{ww-A})	0.60	m
	height (h_{ww-B})	4.93	m
	thickness (t_{ww-B})	0.30	m
Surcharge=	**h_{sur} =**	0.61	m
Approach slab=	**h_{apch} =**	0.30	m

*(unit wt. of conc. is increased by **30%** to account for railing and sidewalk)*

Hb = (tan 45) x Lww =	1.50
Ha = (H - Hb) =	3.43

Calculation of horizontal pressure:

Loads	pressure at bd' (for Ha)		pressure at b"d" (for Ha -1.0m)	
Due to lateral pressure	kPa		kPa	
pa1 = ka*Y_s	pa1 =			14.21
p_{AE1} = k_{AE}*Y_s*Ha	p_{AE1} =	32.81	p = k *Ys*(Ha - 1.0m) =	23.23
p = ka * Y *h	p_{sur} =	3.58		
p = ka * Yc*h	p_{apch} =	2.89		
Due to inertial mass, p_{IR}				
khW_{ww} = kh*(L_{ww}*Ha*t_{ww}*Y_c)	khW_{ww} =	19.21	kN	

Calculation of forces per unit length: bv= 1.00 m

Loads	Force, kN		Lever arm, m		Moment, kN*m
Earth Pressure	(pa1 +pa2)/2 *L_{ww}*bv =	25.71	L_{ww}/2 =	0.75	19.29
Seismic earth pressure (ΔpAE)	(PAE1 + PAE2)/2 *Lww*bv =	42.04	L_{ww}/2 =	0.75	31.53
Liveload surcharge(LS)	p_{sur} *L_{ww}*bv =	5.36	L_{ww}/2 =	0.75	4.02
Earth surcharge(ES)	p_{apch} *L_{ww}*bv =	4.33	L_{ww}/2 =	0.75	3.25
Inertial force mass (pIR)	khW_{ww}/Ha*bv =	5.61	L_{ww}/2 =	0.75	4.21

where : Moment = Force x lever arm

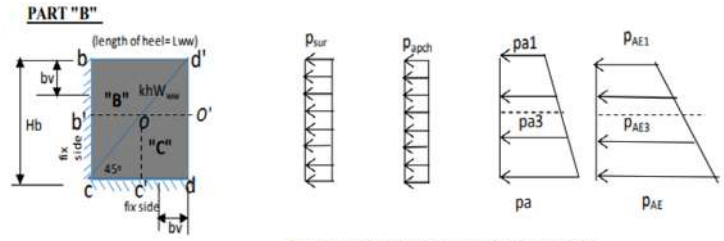


Diagram of lateral forces of part "B" and "C"

Calculation of horizontal pressure:

Loads	pressure at bd' (for Ha)		pressure at b' O' (for Ha +Hb/2)	
Due to lateral pressure	kPa		kPa	
pa1 = ka*Y_s*Ha	pa1 =	20.07	pa2 = ka*Ys*(Ha + Hb/2) =	24.47
p_{AE1} = k_{AE}*Y_s*Ha	p_{AE1} =	32.81	p_{AE2} = k_{AE}*Ys*(Ha + Hb/2) =	40.00
p_{sur} = ka * Y_s*h_{sur}	p_{sur} =	3.58		
p_{apch} = ka * Yc*h_{apch}	p_{apch} =	2.89		
Due to inertial mass, p_{IR}				
khW_{ww} = kh*(1/2*L_{ww}*Hb*t_{ww}*Y_c)	khW_{ww} =	4.21	kN	

Calculation of forces per unit length: bv= 1.00 m

Loads	Force, kN		Lever arm, m		Moment, kN*m
Earth Pressure	(pa1 +pa2)/2 *1/2*Hb*3/4*L_{ww} =	18.79	3/4*L_{ww}/2 =	0.56	10.57
Seismic earth pressure (ΔpAE)	(PAE1 + PAE2)/2 *1/2*Hb*3/4*L_{ww} =	30.72	3/4*L_{ww}/2 =	0.56	17.28
Liveload surcharge(LS)	p_{sur}*1/2*Hb*3/4*L_{ww} =	3.02	3/4*L_{ww}/2 =	0.56	1.70
Earth surcharge(ES)	p_{apch}*1/2*Hb*3/4*L_{ww} =	2.44	3/4*L_{ww}/2 =	0.56	1.37
Inertial force mass (pIR)	khW_{ww}/Hb =	2.80	3/4*L_{ww}/2 =	0.56	1.58

Note: ¾ Lww is average length between b-b'

where : Moment = Force x lever arm

PART "C" (refer to figure of PART "B")

Calculation of horizontal pressure:

Loads	pressure at bd' (for Ha)		
Due to lateral pressure	kPa		
pa = ka*Y_s*H	pa =	28.86	
p_{AE} = k_{AE}*Y_s*H	p_{AE} =	47.19	
p_{sur} = ka * Y_s*h_{sur}	p_{sur} =	3.58	
p_{apch} = ka * Yc*h_{apch}	p_{apch} =	2.89	
Due to inertial mass, p_{IR}			
khW_{ww} = kh*(1/2*L_{ww}*Hb*t_{ww}*Y_c)	khW_{ww} =	4.21	kN

Calculation of forces per unit length: bv= 1.00 m

Loads	Force, kN		Lever arm, m		Moment, kN*m
Earth Pressure	pa*3/4*Hb*bv =	32.47	3/4Hb/2 =	0.56	18.27
Seismic earth pressure (ΔpAE)	p_{AE}*3/4*Hb*bv=	53.08	3/4Hb/2 =	0.56	29.86
Liveload surcharge(LS)	p_{sur}*3/4*Hb*bv =	4.02	3/4Hb/2 =	0.56	2.26
Earth surcharge(ES)	p_{apch}*3/4*Hb*bv =	3.25	3/4Hb/2 =	0.56	1.83
Inertial force mass (pIR)	khW_{ww}/Lww*bv =	2.80	3/4Hb/2 =	0.56	1.58

Note: 3/4 Hb is average height between c'-d

where : Moment = Force x lever arm

PART "D"

Calculation of horizontal pressure:

Loads	pressure at bottom		pressure at 1m strip from bott. (for H -1.0m)	
Due to lateral pressure	kPa		kPa	
pa = $ka*Y_s*H$	pa =	28.86	pa2 = $ka*Ys*(Ha - 1.0m)$ =	23.00
$p_{AE} = k_{AE}*Y_s*H$	p_{AE} =	47.19	$p_{AE2} = k_{AE}*Ys*(Ha - 1.0m)$ =	37.61
$p_{sur} = ka * Y_s*h_{sur}$	p_{sur} =	3.58		
$p_{apch} = ka * Yc*h_{apch}$	p_{apch} =	2.89		
Due to inertial mass, p_{IR}				
$khW_{ww} = kh*(*L*H*t*Y_c)$	khW_{ww} =	15.17	kN	

Calculation of forces per unit length: bv= 1.00 m

Loads	Force, kN		Lever arm, m		Moment, kN*m
Earth Pressure	(pa+pa2)/2 *L*bv =	25.93	L/2 =	0.50	12.97
Seismic earth pressure (ΔpAE)	$(p_{AE}+p_{AE2})/2$ *L*bv=	42.40	L/2 =	0.50	21.20
Liveload surcharge(LS)	p_{sur}*L*bv =	3.58	L/2 =	0.50	1.79
Earth surcharge(ES)	p_{apch}*L*bv =	2.89	L/2 =	0.50	1.44
Inertial force mass (pIR)	khW_{ww}/L*bv=	3.08	L/2 =	0.50	1.54

where : Moment = Force x lever arm

1.5.3a DESIGN PART "A"

SUMMARY OF UNFACTORED LOADS		
Loads	**Force, kN**	**Moment, kN*m**
Earth Pressure (PA)	51.65	71.15
Seismic earth pressure (PAE)	84.43	116.32
Liveload surcharge(LS)	8.94	11.17
Earth surcharge(ES)	7.22	9.02
Inertial force mass (p_{IR})	8.69	10.37

η = 1.0 for other bridges hence modifier for wingwalls is assumed 1.0

a. Load Combinations:

Strength -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.50	0.90	EH	77.47	46.48	106.73	64.04
1.75	0.00	LS	15.64	0.00	19.55	0.00
1.50	0.75	ES	10.83	5.41	13.53	6.77
		Total:	**103.94**	**51.90**	**139.82**	**70.81**

EXTREME EVENT - 1 : CASE 1 (100% P_{AE} + 50% P_{IR})

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	PAE	84.43	0.00	116.32	0.00
0.50	0.00	LS	4.47	0.00	5.59	0.00
1.50	0.75	ES	10.83	5.41	13.53	6.77
1.00	0.00	50%p_{IR}	4.34	0.00	5.18	0.00
		Total:	**104.07**	**5.41**	**140.62**	**6.77**

EXTREME EVENT - 1 : CASE 2 (50% P_{AE} +100% P_{IR})

Verification: if 50%P_{AE} < Pa, use Pa, else use 50% P_{AE} (both for Force and Moment)

Force, kN	
50% P_{AE} =	42.22
Pa =EH =	51.65

Moment, kN*m	
50% P_{AE} =	58.16
Pa =EH =	71.15

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	51.65	0.00	71.15	0.00
0.50	0.00	LS	4.47	0.00	5.59	0.00
1.50	0.75	ES	10.83	5.41	13.53	6.77
1.00	0.00	p_{IR}	8.69	0.00	10.37	0.00
		Total:	**75.63**	**5.41**	**100.64**	**6.77**

Service -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	51.65	0.00	71.15	0.00
1.00	0.00	LS	8.94	0.00	11.17	0.00
1.00	0.00	ES	7.22	0.00	9.02	0.00
		Total:	**67.80**	**0.00**	**91.35**	**0.00**

b. Determine the governing design forces:

Load Combination:	EXTREME EVENT - 1 : CASE 1 (100% PAE + 50% PIR)	
Shear Force=	104.07	kN
Moment=	140.62	kN-m

1.5.3b DESIGN PART "B"

SUMMARY OF UNFACTORED LOADS		
Loads	Force, kN	Moment, kN*m
Earth Pressure (PA)	44.73	62.44
Seismic earth pressure (PAE)	73.12	102.07
Liveload surcharge(LS)	6.59	8.85
Earth surcharge(ES)	5.32	7.14
Inertial force mass (p_{IR})	5.88	7.74

η = 1.0 for other bridges hence modifier for wingwalls is assumed 1.0

a. Load Combinations:

Strength -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.50	0.90	EH	67.09	40.25	93.66	56.19
1.75	0.00	LS	11.54	0.00	15.48	0.00
1.50	0.75	ES	7.99	3.99	10.72	5.36
		Total:	86.61	44.25	119.86	61.55

EXTREME EVENT - 1 : CASE 1 (100% P_{AE} + 50% P_{IR})

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	PAE	73.12	0.00	102.07	0.00
0.50	0.00	LS	3.30	0.00	4.42	0.00
1.50	0.75	ES	7.99	3.99	10.72	5.36
1.00	0.00	50%p_{IR}	2.94	0.00	3.87	0.00
		Total:	87.34	3.99	121.08	5.36

EXTREME EVENT - 1 : CASE 2 (50% P_{AE} +100% P_{IR})

Verification: if 50%P_{AE} < Pa, use Pa, else use 50% P_{AE} (both for Force and Moment)

Force, kN	
50% P_{AE} =	36.56
Pa =EH =	44.73

Moment, kN*m	
50% P_{AE} =	51.04
Pa =EH =	62.44

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	44.73	0.00	62.44	0.00
0.50	0.00	LS	3.30	0.00	4.42	0.00
1.50	0.75	ES	7.99	3.99	10.72	5.36
1.00	0.00	p_{IR}	5.88	0.00	7.74	0.00
		Total:	61.89	3.99	85.32	5.36

Service -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	44.73	0.00	62.44	0.00
1.00	0.00	LS	6.59	0.00	8.85	0.00
1.00	0.00	ES	5.32	0.00	7.14	0.00
		Total:	56.64	0.00	78.43	0.00

b. Determine the governing design forces:

Load Combination:	EXTREME EVENT - 1 : CASE 1 (100% PAE + 50% PIR)	
Shear Force=	87.34	kN
Moment=	121.08	kN-m

1.5.3c DESIGN PART "C"

SUMMARY OF UNFACTORED LOADS		
Loads	**Force, kN**	**Moment, kN*m**
Earth Pressure (PA)	32.47	18.27
Seismic earth pressure (PAE)	53.08	29.86
Liveload surcharge(LS)	4.02	2.26
Earth surcharge(ES)	3.25	1.83
Inertial force mass (p_{IR})	2.80	1.58

η = 1.0 for other bridges hence modifier for wingwalls is assumed 1.0

a. Load Combinations:

Strength -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	**min**		**max**	**min**	**max**	**min**
1.50	0.90	EH	48.71	29.23	27.40	16.44
1.75	0.00	LS	7.04	0.00	3.96	0.00
1.50	0.75	ES	4.87	2.44	2.74	1.37
		Total:	**60.62**	**31.66**	**34.10**	**17.81**

EXTREME EVENT - 1 : CASE 1 (100% P_{AE} + 50% P_{IR})

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	**min**		**max**	**min**	**max**	**min**
1.00	0.00	PAE	53.08	0.00	29.86	0.00
0.50	0.00	LS	2.01	0.00	1.13	0.00
1.50	0.75	ES	4.87	2.44	2.74	1.37
1.00	0.00	50%p_{IR}	1.40	0.00	0.79	0.00
		Total:	**61.37**	**2.44**	**34.52**	**1.37**

EXTREME EVENT - 1 : CASE 2 (50% P_{AE} +100% P_{IR})

Verification: if 50%P_{AE} < Pa, use Pa, else use 50% P_{AE} (both for Force and Moment)

Force, kN	
50% P_{AE} =	26.54
Pa =EH =	32.47

Moment, kN*m	
50% P_{AE} =	14.93
Pa =EH =	18.27

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	**min**		**max**	**min**	**max**	**min**
1.00	0.00	EH	32.47	0.00	18.27	0.00
0.50	0.00	LS	2.01	0.00	1.13	0.00
1.50	0.75	ES	4.87	2.44	2.74	1.37
1.00	0.00	p_{IR}	2.80	0.00	1.58	0.00
		Total:	**42.16**	**2.44**	**23.72**	**1.37**

Service -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	**min**		**max**	**min**	**max**	**min**
1.00	0.00	EH	32.47	0.00	18.27	0.00
1.00	0.00	LS	4.02	0.00	2.26	0.00
1.00	0.00	ES	3.25	0.00	1.83	0.00
		Total:	**39.74**	**0.00**	**22.36**	**0.00**

b. Determine the governing design forces:

Load Combination: EXTREME EVENT - 1 : CASE 1 (100% PAE + 50% PIR)

Shear Force= **61.37** kN

Moment= **34.52** kN-m

1.5.3d DESIGN PART "D"

SUMMARY OF UNFACTORED LOADS		
Loads	**Force, kN**	**Moment, kN*m**
Earth Pressure (PA)	25.93	12.97
Seismic earth pressure (PAE)	42.40	21.20
Liveload surcharge(LS)	3.58	1.79
Earth surcharge(ES)	2.89	1.44
Inertial force mass (p_{IR})	3.08	1.54

η = 1.0 for other bridges hence modifier for wingwalls is assumed 1.0

a. Load Combinations:

Strength -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.50	0.90	EH	38.90	23.34	19.45	11.67
1.75	0.00	LS	6.26	0.00	3.13	0.00
1.50	0.75	ES	4.33	2.17	2.17	1.08
		Total:	**49.49**	**25.51**	**24.74**	**12.75**

EXTREME EVENT - 1 : CASE 1 (100% P_{AE} + 50% P_{IR})

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	PAE	42.40	0.00	21.20	0.00
0.50	0.00	LS	1.79	0.00	0.89	0.00
1.50	0.75	ES	4.33	2.17	2.17	1.08
1.00	0.00	50%p_{IR}	1.54	0.00	0.77	0.00
		Total:	**50.05**	**2.17**	**25.03**	**1.08**

EXTREME EVENT - 1 : CASE 2 (50% P_{AE} +100% P_{IR})

Verification: if 50%P_{AE} < Pa, use Pa, else use 50% P_{AE} (both for Force and Moment)

Force, kN	
50% P_{AE} =	21.20
Pa =EH =	25.93

Moment, kN*m	
50% P_{AE} =	10.60
Pa =EH =	12.97

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	25.93	0.00	12.97	0.00
0.50	0.00	LS	1.79	0.00	0.89	0.00
1.50	0.75	ES	4.33	2.17	2.17	1.08
1.00	0.00	p_{IR}	3.08	0.00	1.54	0.00
		Total:	**35.13**	**2.17**	**17.57**	**1.08**

Service -1

Load Factor		Loads	Factored Loads			
			Force, kN		Moment, kN*m	
max	min		max	min	max	min
1.00	0.00	EH	25.93	0.00	12.97	0.00
1.00	0.00	LS	3.58	0.00	1.79	0.00
1.00	0.00	ES	2.89	0.00	1.44	0.00
		Total:	**32.40**	**0.00**	**16.20**	**0.00**

b. Determine the governing design forces:

Load Combination: EXTREME EVENT - 1 : CASE 1 (100% PAE + 50% PIR)

Shear Force= **50.05** kN

Moment= **25.03** kN-m

c. Verification of flexural resistance

c.1 Part "A" Flexural Design

Moment Demand, Md=					140.62 kN-m
Main reinf. (inner face/horizontal), db1	Ab1=	490.87	mm^2		25 mm
Secondary reinf. (outer face/vertical), db2	Ab2=	201.06	mm^2		16 mm
design cover, dc.					75 mm
concrete compressive strength, f'c					27.6 MPa
yield strength of steel, fy					414 MPa
thickness of wall, t_{ww}					600 mm
unit width of wall, bv					1000 mm
Height of wall, Ha					3425 mm
Length of wall, L_{ww}					1500 mm

Minimum reinforcement

$Mcr = Y_3 (Y_1 f_r) S_c$

where:

fr=	3.31	Mpa	(fr = 0.63*√f'c)
$Y_{1\,=}$	1.6		*for all other concrete
$Y_{3\,=}$	0.67		*for A615, 414MPa steel
S_c = 1/6(bt^2)			Section modulus
S_c = (bv*tww^2)/6			
S_c =	60,000,000.00	mm^3	

$Mcr = Y_3 (Y_1 f_r) S_c$ =	212.88	kN-m
Mu_min=1.33*Md=	187.03	kN-m

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Therefore, Md= 187.03 kN-m

Computation for main reinforcement

effective d_e = tww-dc-(db1/2) — d_e = 512.5 mm

$m_1 = 0.85 * \frac{f'_c}{fy} =$ — m1= 0.057

$m_2 = \frac{2}{0.85 * fc} =$ — m2= 0.085 mm^2/N

$Rn = \frac{Md}{\emptyset bv * (d_e)^2}$ — Rn= 0.791 MPa

$\rho = m_1 * (1 - \sqrt{1 - m_2 * Rn})$ — ρ = 0.001944

$A_s = \rho * bv * d_e$ — As= 996.52 mm^2

$S = Ab1 * \frac{bv}{As}$ — S= 492.59 mm

Try: say S_prov= — S_prov= 150 mm

$A_{s_prov} = Ab1 * \frac{bv}{S_prov}$ — As_prov= 3272.492 mm^2

β= — β= 0.853

$c = \frac{As_{prov} * fy}{0.85 * f'c * \beta * bv}$ — c= 67.713 mm

$a = c * \beta$ — a= 57.750 mm

$Mn = (As_{prov} * fy) * d_e - \frac{a}{2} =$ — Mn= 655.221 kN-m

Check net tensile strain, εt

$\varepsilon t = 0.003 * \left(\frac{d_e}{c} - 1\right) =$ — εt= 0.020 > 0.005

Tension Controlled!, Reduction factor = ***0.9***

Ultimate moment capacity of section, $\emptyset M_n$ — ØMn= 589.70 kN-m

c/d= 3.15 ***Section is safe in flexure!***

Therefore: **Use 25 mm dia. @ 150 mm O.C**

c. Verification of flexural resistance

c.2 Part "B" Flexural Design

Moment Demand, Md=				121.08	kN-m
Main reinf. (inner face/horizontal), db1	Ab1=	490.87	mm^2	25	mm
Secondary reinf. (outer face/vertical), db2	Ab2=	201.06	mm^2	16	mm
design cover, dc.				75	mm
concrete compressive strength, f'c				27.6	MPa
yield strength of steel, fy				414	MPa
thickness of wall, t_{ww}				600	mm
unit width of wall, bv				1000	mm
Height of wall, Hb				1500	mm
Length of wall, L_{ww}				1500	mm

Minimum reinforcement

$Mcr = Y_3 (Y_1 f_r) S_c$

where:

fr=	3.31	Mpa	(fr = 0.63*√f'c)
$Y_{1\,=}$	1.6		*for all other concrete
$Y_{3\,=}$	0.67		*for A615, 414MPa steel
S_c = 1/6(bt^2)			Section modulus
S_c = (bv*tww^2)/6			
S_c =	60,000,000.00	mm^3	

$Mcr = Y_3 (Y_1 f_r) S_c$ =	212.88	kN-m
Mu_min=1.33*Md=	161.04	kN-m

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Therefore, Md= 161.04 kN-m

Computation for main reinforcement

effective d_e = tww-dc-(db1/2) — d_e = 512.5 mm

$m_1 = 0.85 * \frac{f'c}{fy} =$ — m1= 0.057

$m_2 = \frac{2}{0.85 * fc} =$ — m2= 0.085 mm^2/N

$Rn = \frac{Md}{\emptyset bv * (d_e)^2}$ — Rn= 0.681 MPa

$\rho = m_1 * (1 - \sqrt{1 - m_2 * Rn})$ — ρ = 0.001670

$A_s = \rho * bv * d_e$ — As= 855.93 mm^2

$S = Ab1 * \frac{bv}{As}$ — S= 573.50 mm

Try: say S_prov= — S_prov= 150 mm

$A_{s_prov} = Ab1 * \frac{bv}{S_prov}$ — As_prov= 3272.492 mm^2

β= — β= 0.853

$c = \frac{As_{prov} * fy}{0.85 * f'c * \beta * bv}$ — c= 67.713 mm

$a = c * \beta$ — a= 57.750 mm

$Mn = (As_{prov} * fy) * d_e - \frac{a}{2} =$ — Mn= 655.221 kN-m

Check net tensile strain, εt

$\varepsilon t = 0.003 * \left(\frac{d_e}{c} - 1\right) =$ — εt= 0.020 > 0.005

Tension Controlled!, Reduction factor = ***0.9***

Ultimate moment capacity of section, $\emptyset M_n$ — ØMn= 589.70 kN-m

c/d= 3.66 ***Section is safe in flexure!***

Therefore: **Use 25 mm dia. @ 150 mm O.C**

c. Verification of flexural resistance

c.3 Part "C" Flexural Design

Moment Demand, Md=					34.52 kN-m
Main reinf. (vertical), db1	Ab1=	201.06	mm^2	16	mm
Secondary reinf. (horizontal), db2	Ab2=	490.87	mm^2	25	mm
design cover, dc.				75	mm
concrete compressive strength, f'c				27.6	MPa
yield strength of steel, fy				414	MPa
thickness of wall, t_{ww}				510	mm
unit width of wall, bv				1000	mm
Height of wall, Hb				1500	mm
Length of wall, L_{ww}				1500	mm

Minimum reinforcement

$Mcr = Y_3 (Y_1 f_r) S_c$

where:

fr= 3.31 Mpa (fr = 0.63*√f'c)

$Y_{1\,=}$ 1.6 **for all other concrete*

$Y_{3\,=}$ 0.67 **for A615, 414MPa steel*

S_c = 1/6(bt^2) Section modulus

S_c = (bv*tww^2)/6

S_c = 43,350,000.00 mm^3

Mcr = $Y_3 (Y_1 f_r) S_c$ = 153.81 kN-m

Mu_min=1.33*Md= 45.91 kN-m

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Therefore, Md= 45.91 kN-m

Computation for main reinforcement

effective d_e = tww-dc-db2-(db1/2) — d_e = 402 mm

$m_1 = 0.85 * \frac{f'c}{fy} =$ — m1= 0.057

$m_2 = \frac{2}{0.85 * fc} =$ — m2= 0.085 mm^2/N

$Rn = \frac{Md}{\emptyset bv * (d_e)^2}$ — Rn= 0.316 MPa

$\rho = m_1 * (1 - \sqrt{1 - m_2 * Rn})$ — ρ = 0.000768

$A_s = \rho * bv * d_e$ — As= 308.61 mm^2

$S = Ab1 * \frac{bv}{As}$ — S= 651.50 mm

Try: say S_prov= — S_prov= 200 mm

$A_{s_prov} = Ab1 * \frac{bv}{S_prov}$ — As_prov= 1005.310 mm^2

β= — β= 0.853

$c = \frac{As_{prov} * fy}{0.85 * f'c * \beta * bv}$ — c= 20.802 mm

$a = c * \beta$ — a= 17.741 mm

$Mn = (As_{prov} * fy) * d_e - \frac{a}{2}$ = — Mn= 163.620 kN-m

Check net tensile strain, εt

$\varepsilon t = 0.003 * \left(\frac{d_e}{c} - 1\right)$ = — εt= 0.055 > 0.005

Tension Controlled!, Reduction factor = 0.9

Ultimate moment capacity of section, $ØM_n$ — ØMn= 147.26 kN-m

c/d= 3.21 ***Section is safe in flexure!***

Therefore: **Use 16 mm dia. @ 200 mm O.C**

c. Verification of flexural resistance

c.4 Part "D" Flexural Design

Moment Demand, Md=				25.03	kN-m
Main reinf. (inner face/horizontal), db1	Ab1=	490.87	mm²	25	mm
Secondary reinf. (outer face/vertical), db2	Ab2=	201.06	mm²	16	mm
design cover, dc.				75	mm
concrete compressive strength, f'c				27.6	MPa
yield strength of steel, fy				414	MPa
thickness of wall, t				420	mm
unit width of wall, bv				1000	mm
Height of wall, H				4925	mm
Length of wall, L				1000	mm

Minimum reinforcement

$Mcr = Y_3 (Y_1 f_r) S_c$

where:

fr= 3.31 Mpa (fr = 0.63*√f'c)

$Y_{1\,=}$ 1.6 *for all other concrete

$Y_{3\,=}$ 0.67 *for A615, 414MPa steel

S_c = 1/6(bt^2) Section modulus

S_c = (bv*tww^2)/6

S_c = 29,400,000.00 mm^3

$Mcr = Y_3 (Y_1 f_r) S_c$ = 104.31 kN-m

Mu_min=1.33*Md= 33.29 kN-m

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Therefore, Md= 33.29 kN-m

Computation for main reinforcement

effective d_e = tww-dc-(db1/2) — d_e = 332.50 mm

$m_1 = 0.85 * \frac{f'_c}{fy}$ = — m1= 0.057

$m_2 = \frac{2}{0.85 * fc}$ = — m2= 0.085 mm^2/N

$Rn = \frac{Md}{Øbv * (d_e)^2}$ — Rn= 0.335 MPa

$\rho = m_1 * (1 - \sqrt{1 - m_2 * Rn})$ — ρ = 0.000814

$A_s = \rho * bv * d_e$ — As= 270.62 mm^2

$S = Ab1 * \frac{bv}{As}$ — S= 1813.90 mm

Try: say S_prov= — S_prov= 300 mm

$A_{s_prov} = Ab1 * \frac{bv}{S_prov}$ — As_prov= 1636.246 mm^2

β= — β= 0.853

$c = \frac{As_{prov} * fy}{0.85 * f'c * \beta * bv}$ — c= 33.857 mm

$a = c * \beta$ — a= 28.875 mm

$Mn = (As_{prov} * fy) * d_e - \frac{a}{2}$ = — Mn= 215.457 kN-m

Check net tensile strain, εt

$\varepsilon t = 0.003 * \left(\frac{d_e}{c} - 1\right)$ = — εt= 0.026 > 0.005

Tension Controlled!, Reduction factor = 0.9

Ultimate moment capacity of section, $ØM_n$ — ØMn= 193.91 kN-m

c/d= 5.83 ***Section is safe in flexure!***

Therefore: **Use 25 mm dia. @ 300 mm O.C**

Control of cracking by distribution of reinforcement

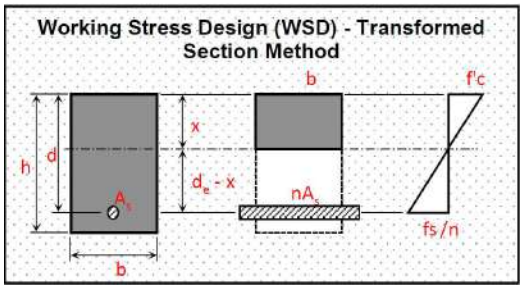
Note: This provision applies to all members when tension in the cross section exceeds the 80% of the modulus rupture @ applicable service limit load combination.

Part "A" control of Cracking by Distribution of Reinforcement

Moment demand at Service 1	**Ms=**	**91.35**	**kN-m**
fr = 0.52*√f'c	fr=	2.732	Mpa
	80%fr =	2.185	Mpa
fss = Ms/Sc	fss=	1.522	Mpa

Provision does not need to be satisfied.

Es = Modulus of elasticity of reinf.	Es=	200000.00	MPa
Ec = Modulus of elasticity of concrete	Ec=	29089.30	MPa
n= modular ratio		6.88	
n*As= transformed area of reinforcement	n*As=	22500	mm^2
d_e= effective d_e	d_e=	513	mm
bv = unit width	bv =	1000	mm



Determine N.A.

Gen. Eq.: $1/2 * bv * x^2 = nAs*(de-x)$

x= 131.02 mm

j*de= de - (x/3) = 468.83 mm

Actual fss = Mservice 1 /(As j*de) = 59.54 Mpa

where

Y_e = 1 for class 1 exposure	Y_e =	1	
d_c = conc cover to centroid of main bars	d_c =	75	mm
h=thickness of wall, t_{ww}	h =	600	mm
f_{ss} = tensile stress of reinf at service limit		Mservice 1/Asjd	

$$\beta s = 1 + \frac{d_c}{(0.7(h - d_c))}$$

β_s= 1.204

The spacing shall satisfy: Eq 12.4.3.4-1

$$s \leq \frac{123000\gamma_e}{\beta_s f_{ss}} - 2d_c$$

s= 1565.68 mm

S_prov= 150 mm

Section 12.4.3.4 satisfied!!!

The following shall be satisfied:

Allowable f_{ss} = $123000*Y_e/(\beta_s(S_prov + 2*d_c))$ = 340.51 >Actual f_{ss} = 59.54

Section 12.4.3.4 satisfied!!!

Therefore using 25mm diam @ 150 is adequate

Check for minimum spacing of reinforcement.

For cast in place concrete, clear distance between paraller bars in a layer shall not be less than:

1.5 x nominal diam of bars=	37.5	mm	***satisfied the required min. spacing!!!***
1.5 x maximum size of aggregates=	30	mm	***satisfied the required min. spacing!!!***
38mm=	38	mm	***satisfied the required min. spacing!!!***

Check for maximum spacing of reinforcement (for walls and slabs)

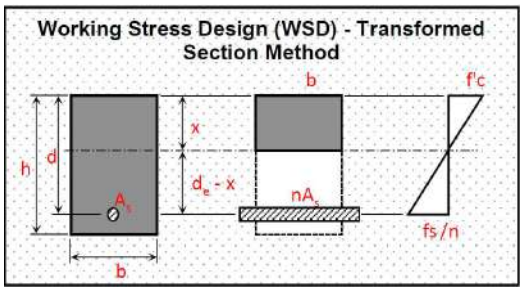
s < 1.5 x t=	900	mm	***satisfied the required max. spacing!!!***
450mm=	450	mm	***satisfied the required max. spacing!!!***

Part "B" control of Cracking by Distribution of Reinforcement

Moment demand at Service 1	**Ms=**	**78.43**	**kN-m**
fr = 0.52*√f'c	fr=	2.732	Mpa
	80%fr =	2.185	Mpa
fss = Ms/Sc	fss=	1.307	Mpa

Provision does not need to be satisfied.

Es = Modulus of elasticity of reinf.		Es=	200000.00	MPa
Ec = Modulus of elasticity of concrete		Ec=	29089.30	MPa
n= modular ratio			6.88	
n*As= transformed area of reinforcement		n*As=	22500	mm^2
d_e=	effective d_e	d_e=	513	mm
bv =	unit width	bv =	1000	mm



Determine N.A.

Gen. Eq.: $1/2 * bv * x^2 = nAs*(de-x)$

x= 131.02 mm

j*de= de - (x/3) = 468.83 mm

Actual fss = Mservice 1 /(As j*de) = 51.12 Mpa

where

Y_e = 1 for class 1 exposure — Y_e = 1

d_c = conc cover to centroid of main bars — d_c = 75 mm

h=thickness of wall, t_{ww} — h = 600 mm

f_{ss} = tensile stress of reinf at service limit — Mservice 1/Asjd

$$\beta s = 1 + \frac{d_c}{(0.7(h - d_c))}$$

β_s= 1.204

The spacing shall satisfy: Eq 12.4.3.4-1

$$s \leq \frac{123000\gamma_e}{\beta_s f_{ss}} - 2d_c$$

s= 1848.27 mm

S_prov= 150 mm

Section 12.4.3.4 satisfied!!!

The following shall be satisfied:

Allowable f_{ss} = $123000*Y_e/(\beta_s(S_prov + 2*d_c))$ = 340.51 >Actual f_{ss} = 51.12

Section 12.4.3.4 satisfied!!!

Therefore using 25mm diam @ 150 is adequate

Check for minimum spacing of reinforcement.

For cast in place concrete, clear distance between paraller bars in a layer shall not be less than:

1.5 x nominal diam of bars=	37.5	mm	***satisfied the required min. spacing!!!***
1.5 x maximum size of aggregates=	30	mm	***satisfied the required min. spacing!!!***
38mm=	38	mm	***satisfied the required min. spacing!!!***

Check for maximum spacing of reinforcement (for walls and slabs)

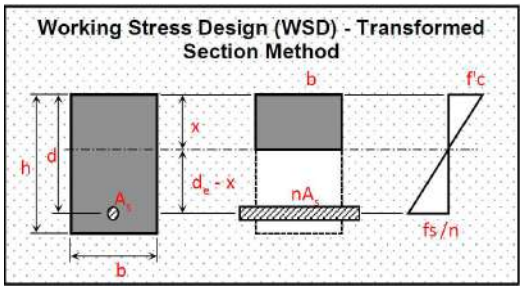
s < 1.5 x t=	900	mm	***satisfied the required max. spacing!!!***
450mm=	450	mm	***satisfied the required max. spacing!!!***

Part "C" control of Cracking by Distribution of Reinforcement

Moment demand at Service 1		**Ms=**	**22.36**	**kN-m**
fr = 0.52*√f'c		fr=	2.732	Mpa
		80%fr =	2.185	Mpa
fss = Ms/Sc		fss=	0.516	Mpa

Provision does not need to be satisfied.

Es = Modulus of elasticity of reinf.		Es=	200000.00	MPa
Ec = Modulus of elasticity of concrete		Ec=	29089.30	MPa
n= modular ratio			6.88	
n*As= transformed area of reinforcement		n*As=	6912	mm^2
d_e=	effective d_e	d_e=	402	mm
bv =	unit width	bv =	1000	mm



Determine N.A.

Gen. Eq.: $1/2 * bv * x^2 = nAs*(de-x)$

x= 67.95 mm

j*de= de - (x/3) = 379.35 mm

Actual fss = Mservice 1 /(As j*de) = 58.62 Mpa

where

Y_e = 1 for class 1 exposure — Y_e = 1

d_c = conc cover to centroid of main bars — d_c = 75 mm

h=thickness of wall, t_{ww} — h = 510 mm

f_{ss} = tensile stress of reinf at service limit — Mservice 1/Asjd

$$\beta s = 1 + \frac{d_c}{(0.7(h - d_c))}$$

β_s= 1.246

The spacing shall satisfy: Eq 12.4.3.4-1

$$s \leq \frac{123000\gamma_e}{\beta_s f_{ss}} - 2d_c$$

s= 1533.59 mm

S_prov= 200 mm

Section 12.4.3.4 satisfied!!!

The following shall be satisfied:

Allowable f_{ss} = $123000*Y_e/(\beta_s(S_prov + 2*d_c))$ = 281.98 >Actual f_{ss} = 58.62

Section 12.4.3.4 satisfied!!!

Therefore using 16mm diam @ 200 is adequate

Check for minimum spacing of reinforcement.

For cast in place concrete, clear distance between paraller bars in a layer shall not be less than:

1.5 x nominal diam of bars=	24	mm	***satisfied the required min. spacing!!!***
1.5 x maximum size of aggregates=	30	mm	***satisfied the required min. spacing!!!***
38mm=	38	mm	***satisfied the required min. spacing!!!***

Check for maximum spacing of reinforcement (for walls and slabs)

s < 1.5 x t=	765	mm	***satisfied the required max. spacing!!!***
450mm=	450	mm	***satisfied the required max. spacing!!!***

Part "D" control of Cracking by Distribution of Reinforcement

Moment demand at Service 1 — **Ms=** **16.20** **kN-m**

fr = $0.52*\sqrt{f'c}$ — fr= 2.732 Mpa

80%fr = 2.185 Mpa

fss = Ms/Sc — fss= 0.551 Mpa

Provision does not need to be satisfied.

Es = Modulus of elasticity of reinf. — Es= 200000.00 MPa

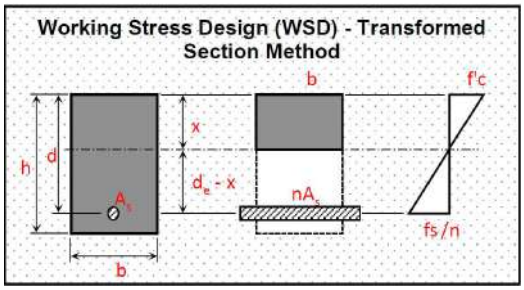
Ec = Modulus of elasticity of concrete — Ec= 29089.30 MPa

n= modular ratio — 6.88

n*As= transformed area of reinforcement — n*As= 11250 mm^2

d_e= effective d_e — d_e= 333 mm

bv = unit width — bv = 1000 mm



Determine N.A.

Gen. Eq.: $1/2 * bv * x^2 = nAs*(de-x)$

x= 75.97 mm

j*de= de - (x/3) = 307.18 mm

Actual fss = Mservice 1 /(As j*de) = 32.23 Mpa

where

Y_e = 1 for class 1 exposure — Y_e = 1

d_c = conc cover to centroid of main bars — d_c = 75 mm

h=thickness of wall, t_{ww} — h = 420 mm

f_{ss} = tensile stress of reinf at service limit — Mservice 1/Asjd

$$\beta s = 1 + \frac{d_c}{(0.7(h - d_c))}$$

β_s= 1.311

The spacing shall satisfy: Eq 12.4.3.4-1

$$s \leq \frac{123000\gamma_e}{\beta_s f_{ss}} - 2d_c$$

s= 2762.16 mm

S_prov= 300 mm

Section 12.4.3.4 satisfied!!!

The following shall be satisfied:

Allowable f_{ss} = $123000*Y_e/(\beta_s(S_prov + 2*d_c))$ = 208.56 >Actual f_{ss} = 32.23

Section 12.4.3.4 satisfied!!!

Therefore using 25mm diam @ 300 is adequate

d.1 Part "A" Verification of shear resistance

3 procedures of determining shear resistance:

Simplified procedure for non-prestressed sections

General procedure <<<< this procedure is applicable for design of wingwall

Simplied procedure for prestressed and prestressed sections

Shear demand, Vd = **104.07** kN

Nominal shear resistance, Vn= min (Vn1, Vn2)

where :

Vn1= Vc + Vs

Vc = 0.083*β(√f'c)bv dv

Vs = [Av*fy*dv*(cot ɵ+ cot α)*sin α] / s

Vn2 = 0.25*f'c*bv*dv

d_e =	512.5	mm
dv1 = (de - a/2)	483.6	mm
dv2= 0.9d_e	461.3	mm
dv3= 0.72tww	432.0	mm
dv=	483.6	mm
bv=	1000	mm

Calculate for β:

$$\beta = \frac{4.8}{(1 + 750\varepsilon_s)} \frac{51}{(39 + S_{xe})}$$

DGCS Eq 12.5.3-2 β in english units

Factor indicating ability of diagonally cracked concrete to transmit tension

where :

$$\varepsilon_s = \frac{\left(\frac{|M_u|}{d_v} + 0.5N_u + |V_u - V_p| - A_{ps}f_{po}\right)}{(E_s A_s + E_p A_{ps})}$$

Net longitudinal tensile strain

Mu =	140.62	kN-m
dv=	0.484	m
Nu =	0	kN
Vu=	104.07	kN
Es=	2.00E+08	kN/m²
As=	0.003272	m²
Mu > Vu*dv	OK!	

EpAps =	0				
Apsfpo=	0				
Vp =	0				

Sxe = Sx *[35/(ag + 16)]

where :

ag =size of aggregates	20	mm		
Sx =dv	483.6	mm		
Sxe =	470.2	mm	300mm≤Sxe≤2025mm	limit to min.: OK! limit to max.: OK!
ε_s =	0.0006			
β=	2.931			

Therefore:

Vc= 0.083*β(√f'c)bv dv =	618.01	kN	Ø= 0.9
Vs=	0	kN	*(Assume no transverse reinf.)*
Vn1 = Vc + Vs=	618.01	kN	
Vn2 = 0.25*f'c*bv*dv =	3,337.01	kN	
Vn = min (Vn1, Vn2) =	618.01	kN	
Ultimate shear resistance, ØVn =	**556.21**	kN	
Shear demand, Vd =	**104.07**	kN	
c/d=	5.34		

Section is safe in shear!

d.2 Part "B" Verification of shear resistance

3 procedures of determining shear resistance:

Simplified procedure for non-prestressed sections
General procedure <<<< this procedure is applicable for design of wingwall
Simplied procedure for prestressed and prestressed sections

Shear demand, Vd =	**87.34**	
Nominal shear resistance,		Vn= min (Vn1, Vn2)

where :

Vn1= Vc + Vs
Vc = 0.083*β(√f'c)bv dv
Vs = [Av*fy*dv*(cot ɵ+ cot α)*sin α] / s

Vn2 = 0.25*f'c*bv*dv

d_e =	512.5	mm
dv1 = (de - a/2)	483.6	mm
dv2= 0.9d_e	461.3	mm
dv3= 0.72tww	432.0	mm
dv=	483.6	mm
bv=	1000	mm

Calculate for β:

$$\beta = \frac{4.8}{(1+750\varepsilon_s)}\frac{51}{(39+S_{xe})}$$

DGCS Eq 12.5.3-2 β in english units

Factor indicating ability of diagonally cracked concrete to transmit tension

where :

$$\varepsilon_s = \frac{\left(\frac{|M_u|}{d_v}+0.5N_u+|V_u-V_p|-A_{ps}f_{po}\right)}{\left(E_sA_s+E_pA_{ps}\right)}$$

Net longitudinal tensile strain

Mu =	121.08	kN-m
dv=	0.484	m
Nu =	0	kN
Vu=	87.34	kN
Es=	2.00E+08	kN/m²
As=	0.003272	m²
Mu > Vu*dv	OK!	
EpAps =	0	
Apsfpo=	0	
Vp =	0	

Sxe = Sx *[35/(ag + 16)]

where :

ag =size of aggregates	20	mm		
Sx =dv	483.6	mm		
Sxe =	470.2	mm	300mm≤Sxe≤2025mm	limit to min.: OK! limit to max.: OK!
ε_s =	0.0005			
β=	3.069			

Therefore:

Vc= 0.083*β(√f'c)bv dv =	647.19	kN	Ø= 0.9
Vs=	0	kN	*(Assume no transverse reinf.)*
Vn1 = Vc + Vs=	647.19	kN	
Vn2 = 0.25*f'c*bv*dv =	3,022.66	kN	
Vn = min (Vn1, Vn2) =	647.19	kN	
Ultimate shear resistance, ØVn =	**582.47**	kN	
Shear demand, Vd =	**87.34**	kN	
c/d=	6.67		

Section is safe in shear!

d.3 Part "C" Verification of shear resistance

3 procedures of determining shear resistance:

Simplified procedure for non-prestressed sections
General procedure <<<< this procedure is applicable for design of wingwall
Simplied procedure for prestressed and prestressed sections

Shear demand, Vd =	**61.37**	
Nominal shear resistance,		Vn= min (Vn1, Vn2)

where :

Vn1= Vc + Vs
Vc = 0.083*β(√f'c)bv dv
Vs = [Av*fy*dv*(cot ɵ+ cot α)*sin α] / s

Vn2 = 0.25*f'c*bv*dv

d_e =	402.0	mm
dv1 = (de - a/2)	393.1	mm
dv2= 0.9d_e	361.8	mm
dv3= 0.72tww	367.2	mm
dv=	393.1	mm
bv=	1000	mm

Calculate for β:

$$\beta = \frac{4.8}{(1+750\varepsilon_s)}\frac{51}{(39+S_{xe})}$$

DGCS Eq 12.5.3-2 β in english units

Factor indicating ability of diagonally cracked concrete to transmit tension

where :

$$\varepsilon_s = \frac{\left(\frac{|M_u|}{d_v}+0.5N_u+|V_u-V_p|-A_{ps}f_{po}\right)}{\left(E_sA_s+E_pA_{ps}\right)}$$

Net longitudinal tensile strain

Mu =	34.52	kN-m
dv=	0.393	m
Nu =	0	kN
Vu=	61.37	kN
Es=	2.00E+08	kN/m²
As=	0.001005	m²
Mu > Vu*dv	OK!	
EpAps =	0	
Apsfpo=	0	
Vp =	0	

Sxe = Sx *[35/(ag + 16)]

where :

ag =size of aggregates	20	mm		
Sx =dv	393.1	mm		
Sxe =	382.2	mm	300mm≤Sxe≤2025mm	limit to min.: OK! limit to max.: OK!
ε_s =	0.0007			
β=	2.910			

Therefore:

Vc= 0.083*β(√f'c)bv dv =	498.84	kN	Ø= 0.9
Vs=	0	kN	*(Assume no transverse reinf.)*
Vn1 = Vc + Vs=	498.84	kN	
Vn2 = 0.25*f'c*bv*dv =	2,712.59	kN	
Vn = min (Vn1, Vn2) =	498.84	kN	
Ultimate shear resistance, ØVn =	**448.96**	kN	
Shear demand, Vd =	**61.37**	kN	
c/d=	7.32		

Section is safe in shear!

d.4 Part "D" Verification of shear resistance

3 procedures of determining shear resistance:

Simplified procedure for non-prestressed sections
General procedure <<<< this procedure is applicable for design of wingwall
Simplied procedure for prestressed and prestressed sections

Shear demand, Vd =	**50.05**	
Nominal shear resistance,		Vn= min (Vn1, Vn2)

where :

Vn1= Vc + Vs
Vc = 0.083*β(√f'c)bv dv
Vs = [Av*fy*dv*(cot ɵ+ cot α)*sin α] / s

Vn2 = 0.25*f'c*bv*dv

d_e =	332.5	mm
dv1 = (de - a/2)	318.1	mm
dv2= 0.9d_e	299.3	mm

dv3= 0.72tww	302.4	mm
dv=	318.1	mm
bv=	1000	mm

Calculate for β:

$$\beta = \frac{4.8}{(1+750\varepsilon_s)}\,\frac{51}{(39+S_{xe})}$$

DGCS Eq 12.5.3-2 β in english units

Factor indicating ability of diagonally cracked concrete to transmit tension

where :

$$\varepsilon_s = \frac{\left(\frac{|M_u|}{d_v}+0.5N_u+|V_u-V_p|-A_{ps}f_{po}\right)}{\left(E_sA_s+E_pA_{ps}\right)}$$

Net longitudinal tensile strain

Mu =	25.03	kN-m
dv=	0.318	m
Nu =	0	kN
Vu=	50.05	kN
Es=	2.00E+08	kN/m^2
As=	0.001636	m^2
Mu > Vu*dv	OK!	
EpAps =	0	
Apsfpo=	0	
Vp =	0	

Sxe = Sx *[35/(ag + 16)]

where :

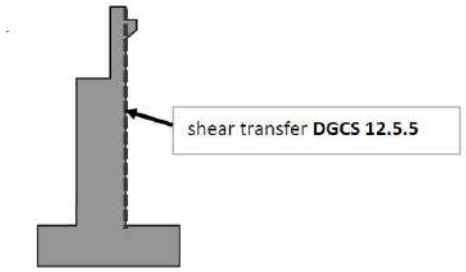
ag =size of aggregates	20	mm		
Sx =dv	318.1	mm		
Sxe =	309.2	mm	300mm≤Sxe≤2025mm	limit to min.: OK! limit to max.: OK!
ε_s =	0.0004			
β=	3.694			

Therefore:

Vc= 0.083*β(√f'c)bv dv =	512.29	kN	Ø= 0.9
Vs=	0	kN	*(Assume no transverse reinf.)*
Vn1 = Vc + Vs=	512.29	kN	
Vn2 = 0.25*f'c*bv*dv =	2,194.63	kN	
Vn = min (Vn1, Vn2) =	512.29	kN	
Ultimate shear resistance, ØVn =	**461.06**	kN	
Shear demand, Vd =	**50.05**	kN	
c/d=	9.21		

Section is safe in shear!

e.1 Part "A" Verification of interface shear resistance



Shear demand, Vd = Vui Vui = **104.07** kN

Nominal interface shear resistance, Vni

Factored interface resistance, Vri = ØVni

Vri ≥ Vui

where :

Vni = c A_{cv} + μ (A_{vf} fy + P_c)

Vni ≤ min (K_1f'$_c$ A_{cv}, K_2 A_{cv})

A_{cv} = area of concrete cosidered to be engaged in interface shear transfer, mm^2 — 393000 mm^2

bvi Lvi

b_{vi} = interface width considered to be engaged in shear transfer, mm — 393 mm

tww - 2*cc-2(db1)-2(db2)

L_{vi} = interface length considered to be engaged in shear transfer, = bv, mm — 1000 mm

Avf = area of shear reinf. crossing the shear plane within the area of A_{cv}, mm^2 — 3943 mm^2

inner face = A_prov =	3272.49	mm^2
outer face = Ab2*bv/S_outerface	670.21	mm^2

Note: area of outer face to be confirmed, see Shrinkage /Temp bars computations.

try S_outerface = 300 mm

c =	cohesion factor, Mpa	1.7	MPa
μ =	friciton factor, Mpa	1	
fy =	yield strength of reinf. not to exceed 60ksi	414	MPa
P_c =	permanent net compressive normal to the shear plane; if force is tensile, P_c=0.0kN	0	kN
f'_c =	compressive strength of the weaker concrete either side of the interface, Mpa	27.6	MPa
K_1 =	fraction of concrete available to resist interface transfer shear	0.25	
K_2 =	limiting interface shear resistance	10.3	MPa

$Vni = c A_{cv} + \mu (A_{vf} fy + P_c)$ 2300 kN

$Vn1 = K_1 f'_c A_{cv}$ 2712 kN

$Vn2 = K_2 A_{cv}$ 4048 kN

Therefore, Vni = 2300 kN

Factored interface resistance, Vri = ØVni = **2070.34** kN Ø= 0.9

Shear demand, Vd = Vui= **104.07** kN

c/d = 19.89

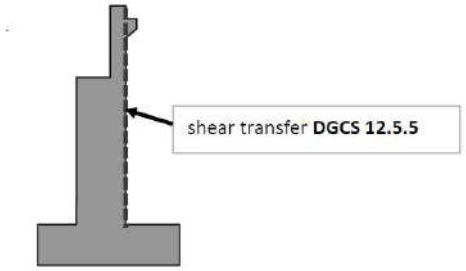
Section is safe in interface shear!

Minimum area of interface shear

$A_{vf} \geq 0.35 A_{cv}/fy$ = 332.25 mm^2

satisfied the required min. area!

e.2 Part "B" Verification of interface shear resistance



Shear demand, Vd = Vui Vui = **87.34** kN

Nominal interface shear resistance, Vni

Factored interface resistance, Vri = ØVni

Vri ≥ Vui

where :

$Vni = c A_{cv} + \mu (A_{vf} fy + P_c)$

$Vni \leq \min (K_1 f'_c A_{cv}, K_2 A_{cv})$

A_{cv} =	area of concrete cosidered to be engaged in interface shear transfer, mm^2	393000	mm^2
b_{vi} =	bvi Lvi interface width considered to be engaged in shear transfer, mm	393	mm
	tww - 2*cc-2(db1)-2(db2)		
L_{vi} =	interface length considered to be engaged in shear transfer, = bv, mm	1000	mm
Avf =	area of shear reinf. crossing the shear plane within the area of A_{cv}, mm^2	3943	mm^2

inner face = A_prov = 3272.49 mm^2

outer face = Ab2*bv/S_outerface 670.21 mm^2

Note: area of outer face to be confirmed, see Shrinkage /Temp bars computations.

try S_outerface = 300 mm

c =	cohesion factor, Mpa	1.7	MPa
μ =	friciton factor, Mpa	1	
fy =	yield strength of reinf. not to exceed 60ksi	414	MPa
P_c =	permanent net compressive normal to the shear plane; if force is tensile, P_c=0.0kN	0	kN
f'_c =	compressive strength of the weaker concrete either side of the interface, Mpa	27.6	MPa
K_1 =	fraction of concrete available to resist interface transfer shear	0.25	
K_2 =	limiting interface shear resistance	10.3	MPa

$V_{ni} = c A_{cv} + \mu (A_{vf} fy + P_c)$	2300	kN	
$V_{n1} = K_1 f'_c A_{cv}$	2712	kN	
$V_{n2} = K_2 A_{cv}$	4048	kN	
Therefore, V_{ni} =	2300	kN	
Factored interface resistance, $V_{ri} = \emptyset V_{ni}$ =	**2070.34**	kN	Ø= 0.9
Shear demand, $V_d = V_{ui}$ =	**87.34**	kN	
c/d =	23.70		

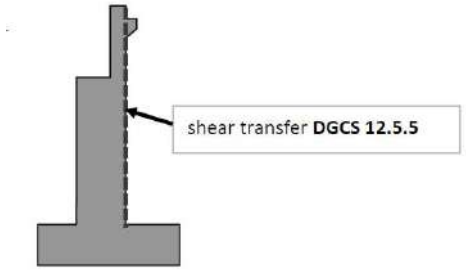
Section is safe in interface shear!

Minimum area of interface shear

$A_{vf} \geq 0.35 A_{cv}/fy$ = 332.25 mm^2

satisfied the required min. area!

e.3 Part "C" Verification of interface shear resistance



Shear demand, Vd = Vui V_{ui} = **61.37** kN

Nominal interface shear resistance, Vni

Factored interface resistance, Vri = ØVni

$V_{ri} \geq V_{ui}$

where :

$V_{ni} = c A_{cv} + \mu (A_{vf} fy + P_c)$

$V_{ni} \leq \min (K_1 f'_c A_{cv}, K_2 A_{cv})$

A_{cv} =	area of concrete cosidered to be engaged in interface shear transfer, mm^2	294000	mm^2
	bvi Lvi		
b_{vi} =	interface width considered to be engaged in shear transfer, mm	294	mm
	tww - 2*cc-2(db1)-2(db2)		
L_{vi} =	interface length considered to be engaged in shear transfer, = bv, mm	1000	mm
Avf =	area of shear reinf. crossing the shear plane within the area of A_{cv}, mm^2	2232	mm^2
	inner face = A_prov =	1005.31	mm^2
	outer face = Ab2*bv/S_outerface	1227.18	mm^2

Note: area of outer face to be confirmed, see Shrinkage /Temp bars computations.

try S_outerface = 400 mm

c =	cohesion factor, Mpa	1.7	MPa
μ =	friciton factor, Mpa	1	
fy =	yield strength of reinf. not to exceed 60ksi	414	MPa
P_c =	permanent net compressive normal to the shear plane; if force is tensile, P_c=0.0kN	0	kN
f'_c =	compressive strength of the weaker concrete either side of the interface, Mpa	27.6	MPa
K_1 =	fraction of concrete available to resist interface transfer shear	0.25	
K_2 =	limiting interface shear resistance	10.3	MPa

$V_{ni} = c A_{cv} + \mu (A_{vf} fy + P_c)$	1424	kN
$V_{n1} = K_1 f'_c A_{cv}$	2029	kN
$V_{n2} = K_2 A_{cv}$	3028	kN
Therefore, V_{ni} =	1424	kN
Factored interface resistance, $V_{ri} = \emptyset V_{ni}$ =	**1281.65**	kN
Shear demand, $V_d = V_{ui}$ =	**61.37**	kN
c/d =	20.88	

Ø= 0.9

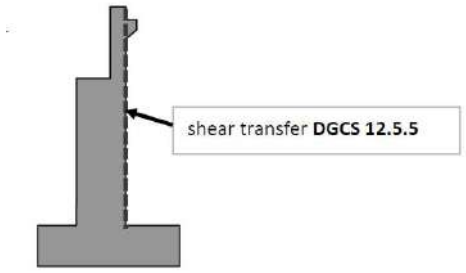
Section is safe in interface shear!

Minimum area of interface shear

$A_{vf} \geq 0.35A_{cv}/fy =$ 248.55 mm^2

satisfied the required min. area!

e.4 Part "D" Verification of interface shear resistance



Shear demand, Vd = Vui Vui = **50.05** kN

Nominal interface shear resistance, Vni

Factored interface resistance, Vri = ØVni

$Vri \geq Vui$

where :

$Vni = c A_{cv} + \mu (A_{vf} fy + P_c)$

$Vni \leq \min (K_1 f'_c A_{cv}, K_2 A_{cv})$

A_{cv} = area of concrete cosidered to be engaged in interface shear transfer, mm^2 — 213000 mm^2

bvi Lvi

b_{vi} = interface width considered to be engaged in shear transfer, mm — 213 mm

tww - 2*cc-2(db1)-2(db2)

L_{vi} = interface length considered to be engaged in shear transfer, = bv, mm — 1000 mm

A_{vf} = area of shear reinf. crossing the shear plane within the area of A_{cv}, mm^2 — 2306 mm^2

inner face = A_prov = 1636.25 mm^2

outer face = Ab2*bv/S_outerface 670.21 mm^2

Note: area of outer face to be confirmed, see Shrinkage /Temp bars computations.

try S_outerface = 300 mm

c = cohesion factor, Mpa — 1.7 MPa

μ = friciton factor, Mpa — 1

fy = yield strength of reinf. not to exceed 60ksi — 414 MPa

P_c = permanent net compressive normal to the shear plane; if force is tensile, P_c=0.0kN — 0 kN

f'_c = compressive strength of the weaker concrete either side of the interface, Mpa — 27.6 MPa

K_1 = fraction of concrete available to resist interface transfer shear — 0.25

K_2 = limiting interface shear resistance — 10.3 MPa

$Vni = c A_{cv} + \mu (A_{vf} fy + P_c)$ — 1317 kN

$Vn1 = K_1 f'_c A_{cv}$ — 1470 kN

$Vn2 = K_2 A_{cv}$ — 2194 kN

Therefore, Vni = 1317 kN

Factored interface resistance, Vri = ØVni = **1185.27** kN Ø= 0.9

Shear demand, Vd = Vui= **50.05** kN

c/d = 23.68

Section is safe in interface shear!

Minimum area of interface shear

$A_{vf} \geq 0.35A_{cv}/fy =$ 180.07 mm^2

satisfied the required min. area!

f. Verification of shrinkage and temperature reinforcement

$Ast = 0.75(bxh)/2(b+h)fy$

Limitations: $0.223 < Ast < 1.27$

where:

b = least width of component section, mm	3425	mm
h = least thickness of component section, mm	600	mm

Ast=	0.462	mm^2/mm			
Limitations:	0.223<Ast<1.27				
therefore,					
Ast=	0.462	mm^2/mm each face/each direction			
db1 =	16	mm	S_assumed =	200	mm
Ab1=	201	mm^2	Ast=	1.005	mm^2/mm
checking:					
Ast=	Ab1/S_final =		1.005	mm^2/mm	
S_final =	Ab1/ Ast				
S_final =	200	mm say	200	mm	

Use : 16mm dia. @ 200mm each face temperature bars

where Spacing S is:

- S should not be greater than 3 X thickness, or 450mm
- S should not be greater than 300 for walls and footings > 450mm
- S should be 300mm for other components > 900mm thick.

satisfied the required max. spacing!!!
satisfied the required max. spacing!!!

g. Development length of the reinforcement.

sizes	Ab	l_{db}		Modification factors(MF) that increase ld	Modification factors(MF) that decrease ld	$l_d = l_{db} * MF$	l_{min}
	mm^2	mm				mm	mm
32	803.84		1267	Note: factors are not applicable in this exercise	see below factors	1,014	795
28	615.44	0.02*Ab*fy/√f'c =	970			776	696
25	490.63		773			619	621
20	314.00	(l_{min} = .06dbfy)	495			396	497
16	200.96		317			253	397

fy=	414	MPa
f'c=	27.6	MPa
ld=	300mm minimum	

Note:
(a) Reinforcement being developed in the length under consideration space laterally not less than 150mm c to c and not less than 75mm clear cover measured in the direction of the spacing. , MF =0.8
(b) Anchorage or development for the full yield strength of reinf. is not required or where reinf. in flexural is in excess of that required by analysis. MF = As_req'd/As_prov
(c) Reinf. is enclosed within a spiral composed of bars of not less than 6mm in diam and spaced at not more than a 100 mm pitch. MF = 0.75

1.6 DESIGN OF PILE CAP

1.6.1 Determine the applicable loads calculated from section 1.2 Geometry and Load Calculations

1. Permanent loads

Dead load rection force of the superstructure, *DC 1*		466.05	kN
Dead load from self weight, *DC 2*			
a) Weight of backwall, *DC 2.1*		75.24	kN
b) Weight of corbel, *DC 2.2*		7.68	kN
c) Weight of breast wall, *DC 2.3*		462.00	kN
d) Weight of footing, *DC 2.4*		554.40	kN
e) Weight of wingwalls, *DC 2.5*	*(increased by 30% to account railings & sidewalk)*	345.74	kN
f) Weight of approach slab, *DC 2.6*		115.20	kN
Vertical pressure from dead load of earth fill, *EV*		724.97	kN
Deadload of wearing surfaces and utilities, *DW*		102.20	kN
Horizontal earth pressure load, *EH (=P $_A$)*		604.65	kN
Earth surcharge load, *ES*		74.82	kN

Note:
Loads Considering full length of abutment

2. Braking force, BF — 66.00 kN

3. Earthquake force, EQ

Seismic earth force, *PAE*		988.45	kN
Seismic inertial force, *P $_{IR}$*			
a) kh * Backwall		17.68	kN
b) kh * Corbel		1.80	kN
c) kh * Breast wall		108.57	kN
d) kh * Footing		130.28	kN
e) kh * Wingwalls		81.25	kN
f) kh * Approach slab		27.07	kN
g) kh * Soil (EV)		170.37	kN
h) kh* DL(DC+DW+0.5LL)		173.93	kN

kh= 0.235

4. Vehicular live load — 343.80 kN
5. Live load surcharge, LS — 120.44 kN
6. Friction load, FR

a) Strength 1		
- at maximum condition	136.81	kN
- at minimum condition	72.88	kN
b) Extreme event 1		
- at maximum condition	111.02	kN
- at minimum condition	72.88	kN
c) Service 1		
- at maximum condition	136.81	kN
- at minimum condition	0.00	kN

7. Stream Flow, WA

Water load and stream pressure, *WA*		
a) Water load due to OWL, *WA 1*	0.00	kN
b) Water load due to DFL, *WA 2*	523.60	kN

1.6.2 Determine the load combinations with applied load modifiers and load factors.

(Moment lever arm is from center of footing)

Load modifier for maximum values, η_i — 1.05
Load modifier for minimum values, η_i — 0.95

Load Combination: STRENGTH 1

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min		Hor'l	Vert'l		max	min	max	min	max	min
1.25	0.90	DC 1	0.00	466.05	0.45	582.56	419.45	0.00	0.00	262.15	188.75
1.25	0.90	DC 2.1	0.00	75.24	-0.05	94.05	67.72	0.00	0.00	-4.70	-3.39
1.25	0.90	DC 2.2	0.00	7.68	-0.35	9.60	6.91	0.00	0.00	-3.36	-2.42
1.25	0.90	DC 2.3	0.00	462.00	0.25	577.50	415.80	0.00	0.00	144.38	103.95
1.25	0.90	DC 2.4	0.00	554.40	0.00	693.00	498.96	0.00	0.00	0.00	0.00
1.25	0.90	DC 2.5	0.00	345.74	-1.43	432.17	311.16	0.00	0.00	-618.24	-445.13
1.25	0.90	DC 2.6	0.00	115.20	-0.35	144.00	103.68	0.00	0.00	-50.40	-36.29
1.35	1.00	EV	0.00	724.97	-1.00	978.71	724.97	0.00	0.00	-978.71	-724.97
1.50	0.65	DW	0.00	102.20	0.45	153.30	66.43	0.00	0.00	68.99	29.89
1.50	0.90	EH	604.65	0.00	2.04	0.00	0.00	906.98	544.19	1851.75	1111.05
1.50	0.75	ES	74.82	0.00	3.06	0.00	0.00	112.23	56.11	343.70	171.85
		FR	0.00	0.00	4.70	0.00	0.00	136.81	72.88	642.99	342.54
1.75	0	LL	0.00	343.80	0.45	601.64	0.00	0.00	0.00	270.74	0.00
1.75	0	BF	66.00	0.00	6.13	0.00	0.00	115.50	0.00	707.44	0.00
1.75	0	LS	120.44	0.00	3.06	0.00	0.00	210.76	0.00	645.47	0.00
1.00	0	WA 2	0.00	-523.60	0.00	-523.60	0.00	0.00	0.00	0.00	0.00
(Strength I) Design load:						**3930.08**	**2484.32**	**1556.39**	**639.52**	**3446.29**	**699.04**

•Modifier, ηi
ηD ≥ 1.05
ηR ≥ 1
ηi ≥ 1

for max. values
ηi = (1.05x1.0x1.0)
ηi = **1.05**

for min values
ηi = 1/(1.05x1.0x1.0)
ηi = **0.95**

• Note: Friction forces are already factored on section Geometry and Load Calculation

Load Combination: EXTREME EVENT I

Load Combination: EXTREME EVENT I (CASE 1) (Case1: 100% PAE + 50% PIR)

Load Factor		Load Type	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min		Hor'l	Vert'l		max	min	max	min	max	min
1.25	0.90	DC 1	0.00	466.05	0.45	582.56	419.45	0.00	0.00	262.15	188.75
1.25	0.90	DC 2.1	0.00	75.24	-0.05	94.05	67.72	0.00	0.00	-4.70	-3.39
1.25	0.90	DC 2.2	0.00	7.68	-0.35	9.60	6.91	0.00	0.00	-3.36	-2.42
1.25	0.90	DC 2.3	0.00	462.00	0.25	577.50	415.80	0.00	0.00	144.38	103.95
1.25	0.90	DC 2.4	0.00	554.40	0.00	693.00	498.96	0.00	0.00	0.00	0.00
1.25	0.90	DC 2.5	0.00	345.74	-1.43	432.17	311.16	0.00	0.00	-618.24	-445.13
1.25	0.90	DC 2.6	0.00	115.20	-0.35	144.00	103.68	0.00	0.00	-50.40	-36.29
1.35	1.00	EV	0.00	724.97	-1.00	978.71	724.97	0.00	0.00	-978.71	-724.97
1.50	0.65	DW	0.00	102.20	0.45	153.30	66.43	0.00	0.00	68.99	29.89
1.50	0.75	ES	74.82	0.00	3.06	0.00	0.00	112.23	56.11	343.70	171.85
0.50	0	LL	0.00	343.80	0.45	171.90	0.00	0.00	0.00	77.35	0.00
0.50	0	BF	66.00	0.00	6.13	0.00	0.00	33.00	0.00	202.13	0.00
0.50	0	LS	120.44	0.00	3.06	0.00	0.00	60.22	0.00	184.42	0.00
		FR	0.00	0.00	4.70	0.00	0.00	111.02	72.88	521.80	342.54
1.00	0	WA 1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1.00	0	PAE	988.45	0.00	2.04	0.00	0.00	988.45	0.00	2018.09	0.00
1.00	0	50%P_{IR}_a	8.84	0.00	5.41	0.00	0.00	8.84	0.00	47.85	0.00
1.00	0	50%P_{IR}_b	0.90	0.00	5.63	0.00	0.00	0.90	0.00	5.08	0.00
1.00	0	50%P_{IR}_c	54.29	0.00	2.95	0.00	0.00	54.29	0.00	160.14	0.00
1.00	0	50%P_{IR}_d	65.14	0.00	0.60	0.00	0.00	65.14	0.00	39.09	0.00
1.00	0	50%P_{IR}_e	40.62	0.00	3.66	0.00	0.00	40.62	0.00	148.78	0.00
1.00	0	50%P_{IR}_f	13.54	0.00	5.98	0.00	0.00	13.54	0.00	80.88	0.00
1.00	0	50%P_{IR}_g	85.18	0.00	3.51	0.00	0.00	85.18	0.00	299.21	0.00
1.00	0	50%P_{IR}_h	86.97	0.00	4.70	0.00	0.00	86.97	0.00	408.75	0.00
(Extreme Event 1-Case1) Design load:						**3836.79**	**2615.07**	**1660.40**	**129.00**	**3357.36**	**-375.21**

• Note: Friction forces are already factored on section Geometry and Load Calculation

•Modifier, ηi

ηi ≥ 1

Load Combination: EXTREME EVENT I (CASE 2) **(Case2: 50% P_{AE} + 100% P_{IR})**

However, if 50% P_{AE} < P_A, use P_A, else use 50%P_{AE}

P_A =EH = Verification: 604.65 >50% PAE

50% P_{AE}= 494.23 therefore use EH

Load Factor		Load	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min	Type	Hor'l	Vert'l		max	min	max	min	max	min
1.25	0.90	DC 1	0.00	466.05	0.45	582.56	419.45	0.00	0.00	262.15	188.75
1.25	0.90	DC 2.1	0.00	75.24	-0.05	94.05	67.72	0.00	0.00	-4.70	-3.39
1.25	0.90	DC 2.2	0.00	7.68	-0.35	9.60	6.91	0.00	0.00	-3.36	-2.42
1.25	0.90	DC 2.3	0.00	462.00	0.25	577.50	415.80	0.00	0.00	144.38	103.95
1.25	0.90	DC 2.4	0.00	554.40	0.00	693.00	498.96	0.00	0.00	0.00	0.00
1.25	0.90	DC 2.5	0.00	345.74	-1.43	432.17	311.16	0.00	0.00	-618.24	-445.13
1.25	0.90	DC 2.6	0.00	115.20	-0.35	144.00	103.68	0.00	0.00	-50.40	-36.29
1.35	1.00	EV	0.00	724.97	-1.00	978.71	724.97	0.00	0.00	-978.71	-724.97
1.50	0.65	DW	0.00	102.20	0.45	153.30	66.43	0.00	0.00	68.99	29.89
1.50	0.75	ES	74.82	0.00	3.06	0.00	0.00	112.23	56.11	343.70	171.85
0.50	0	LL	0.00	343.80	0.45	171.90	0.00	0.00	0.00	77.35	0.00
0.50	0	BF	66.00	0.00	6.13	0.00	0.00	33.00	0.00	202.13	0.00
0.50	0	LS	120.44	0.00	3.06	0.00	0.00	60.22	0.00	184.42	0.00
		FR	0.00	0.00	4.70	0.00	0.00	111.02	72.88	521.80	342.54
1.00	0	WA 1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1.00	0	EH	604.65	0.00	2.04	0.00	0.00	604.65	0.00	1234.50	0.00
1.00	0	P_{IR}_a	17.68	0.00	5.41	0.00	0.00	17.68	0.00	95.70	0.00
1.00	0	P_{IR}_b	1.80	0.00	5.63	0.00	0.00	1.80	0.00	10.15	0.00
1.00	0	P_{IR}_c	108.57	0.00	2.95	0.00	0.00	108.57	0.00	320.28	0.00
1.00	0	P_{IR}_d	130.28	0.00	0.60	0.00	0.00	130.28	0.00	78.17	0.00
1.00	0	P_{IR}_e	81.25	0.00	3.66	0.00	0.00	81.25	0.00	297.57	0.00
1.00	0	P_{IR}_f	27.07	0.00	5.98	0.00	0.00	27.07	0.00	161.76	0.00
1.00	0	P_{IR}_g	170.37	0.00	3.51	0.00	0.00	170.37	0.00	598.42	0.00
1.00	0	P_{IR}_h	173.93	0.00	4.70	0.00	0.00	173.93	0.00	817.49	0.00
(Extreme Event 1-Case 2) Design load:						**3836.79**	**2615.07**	**1632.08**	**129.00**	**3763.54**	**-375.21**

• Note: Friction forces are already factored on section Geometry and Load Calculation

•Modifier, ηi

ηi ≥ 1

Load Combination: SERVICE 1

Load Factor		Load	Unfactored load, kN		Lever arm (m)	Factored					
						Axial force, kN		Shear force, kN		Moment, kN-m	
max	min	Type	Hor'l	Vert'l		max	min	max	min	max	min
1	0	DC 1	0.00	466.05	0.45	466.05	0.00	0.00	0.00	209.72	0.00
1	0	DC 2.1	0.00	75.24	-0.05	75.24	0.00	0.00	0.00	-3.76	0.00
1	0	DC 2.2	0.00	7.68	-0.35	7.68	0.00	0.00	0.00	-2.69	0.00
1	0	DC 2.3	0.00	462.00	0.25	462.00	0.00	0.00	0.00	115.50	0.00
1	0	DC 2.4	0.00	554.40	0.00	554.40	0.00	0.00	0.00	0.00	0.00
1	0	DC 2.5	0.00	345.74	-1.43	345.74	0.00	0.00	0.00	-494.59	0.00
1	0	DC 2.6	0.00	115.20	-0.35	115.20	0.00	0.00	0.00	-40.32	0.00
1	0	EV	0.00	724.97	-1.00	724.97	0.00	0.00	0.00	-724.97	0.00
1	0	DW	0.00	102.20	0.45	102.20	0.00	0.00	0.00	45.99	0.00
1	0	EH	604.65	0.00	2.04	0.00	0.00	604.65	0.00	1234.50	0.00
1	0	ES	74.82	0.00	3.06	0.00	0.00	74.82	0.00	229.13	0.00
		FR	0.00	0.00	4.70	0.00	0.00	136.81	0.00	642.99	0.00
1	0	LL	0.00	343.80	0.45	343.80	0.00	0.00	0.00	154.71	0.00
1	0	BF	66.00	0.00	6.13	0.00	0.00	66.00	0.00	404.25	0.00
1	0	LS	120.44	0.00	3.06	0.00	0.00	120.44	0.00	368.84	0.00
1	0	WA 1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(Service 1) Design load:						**3197.27**	**0.00**	**1002.71**	**0.00**	**2139.30**	**0.00**

Summary of Load Combinations:

STRENGTH I						EXTREME EVENT 1 (CASE 1)					
Axial force (kN)		Shear Force (kN)		Moment (kN-m)		Axial force (kN)		Shear Force (kN)		Moment (kN-m)	
max	min	max	min	max	min	max	min	max	min	max	min
3930.08	**2484.32**	**1556.39**	**639.52**	**3446.29**	**699.04**	**3836.79**	**2615.07**	**1660.40**	**129.00**	**3357.36**	**-375.21**

EXTREME EVENT 1 (CASE 2)						SERVICE 1					
Axial force (kN)		Shear Force (kN)		Moment (kN-m)		Axial force (kN)		Shear Force (kN)		Moment (kN-m)	
max	min	max	min	max	min	max	min	max	min	max	min
3836.79	**2615.07**	**1632.08**	**129.00**	**3763.54**	**-375.21**	**3197.27**	**0.00**	**1002.71**	**0.00**	**2139.30**	**0.00**

1.6.3 Determine the governing design forces

Load Combination: **EXTREME EVENT 1 (CASE 2)**

Axial force(min) =	2615.07	kN
Axial force(max)=	3836.79	kN
Shear Force =	1632.08	kN
Max. Moment =	3763.54	kN-m

Determination of pile reactions

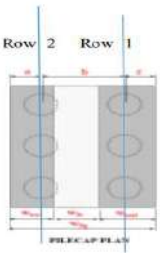
	No. of piles/row	Mom. of Inertia	Dist. from N.A. (m)
Row 1	3	2.7075	0.95
Row 2	3	2.7075	0.95
	I =	5.415	m2

**Modified stress formula for piles (multiplying/canceling same area of piles on both sides of equation)*

F=P/N ± (Mc)/I (+) *For Compression*

(-) *For Tension*

Design moment, *Mu*	**3763.54**	kN-m			
Distance of neutral axis to piles, *c*					
For row 1, C_1	0.95	m	a=c=	0.80	m
For row 2, C_2	0.95	m	wftg =	3.50	m
Moment of Inertia, *I*	5.415	m2	L=	5.50	m
Number of piles, *N*	6				



Reaction per pile:

(@Axial min.)for piles at row 1, (F_1)	**-224.42**	kN	***GOVERNS!***
(@Axial max.)for piles at row 1, (F_1)	**-20.81**	kN	
(@Axial min.)for piles at row 2, (F2)	**1096.12**	kN	
(@Axial max.)for piles at row 2, (F_2)	**1299.73**	kN	***GOVERNS!***

NOTE: For verification, the reactions from pile group analysis may be compared to the reactions obtained from section 7.8 Design of Piles analysis and use the conservative values for design of pile cap.

1.6.4 Verification of flexural resistance

1.6.4.1 Determine design moment for compression side (to design bottom bars)

Passive soil above footing	0.00	m	(neglect, conservative)
Weight of soil above toe	0.00	kN	
Weight of toe	158.40	kN	
Number of piles at row 2	3		
Total reaction force from piles in row 2	3899.20	kN	(@ compression)
Distance of piles from face of column	0.20	m	
Design moment for bottom bars	700.64	kN-m	

Design for bottom bars

Demand moment for bottom bars	700.64	kN-m	
Concrete cover	200	mm	*Concrete cover for bottom bars = Pile embedment + 50mm*
Pile embedment	150	mm	
Diameter of reinforcing bar	25	mm	Ab = 490.87 mm2
Diameter of shrinkage bar	20	mm	
Diameter of cross ties	0	mm	
Effective depth of concrete	987.5	mm	
Length to be considered	5500	mm	
Overall thickness of component	1200	mm	

Minimum reinforcement

Flexural cracking variability factor, Y_1	1.6		**for all other concrete*
Ratio of specified min. to ult. tensile strength of steel, Y_3	0.67		**for A615, 414MPa steel*
Modulus of rupture, f_r	3.310	Mpa	(fr = 0.63*√f'c)
Section modulus, S_c	1.32E+09	mm^3	
Cracking moment, M_{cr}			
•$M_{cr}=Y_3(Y_1 fr)\,Sc$	4683.43	kN-m	
*Mu_min 1,33*Md =*	931.85	kN-m	
Design moment for bottom bars	**931.85**	kN-m	

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Steel ratio

β_1 Coefficient Criterion : 0.85

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ

m_1 =	0.057		concrete compressive strength, f'c	27.6	MPa
m_2 =	0.085	mm^2/N	yield strength of steel, fy	414	MPa
R_n =	0.193	Mpa			
ρ =	0.0005				

Computation for main reinforcement

Required steel area	2543.11	mm^2		
Required spacing	1061.62		say: 200	mm
Provided steel for bottom bars, A_s	13499.03	mm^2		
Compression fiber to neutral axis, c	50.96	mm		
Depth of compression block, a	43.31	mm		
Nominal moment capacity of section, M_n	5,397.71	kN-m		
Resistance factor, Ø	0.9	tension is controlled		
$0.75 \le Ø = 0.65+0.15\,(\,deff/c - 1) \le 0.9$	3.41			
Ultimate moment capacity of section, $ØM_n$	**4857.94**	kN-m	**OK!**	
	c/d =	5.21		

Determination of pile reactions @ Service 1

Axial force=	3197.27	kN
Shear Force =	1002.71	kN
Max. Moment =	2139.30	kN-m

Reaction per pile:

Axial for piles at row 1, (F1)	**157.56**	kN
Axial for piles at row 2, (F2)	**908.19**	kN

Determine design moment for compression side (to check design bottom bars from cracking)

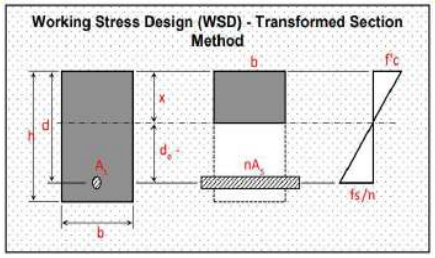
Passive soil above footing	0.00	m	(neglect, conservative)
Weight of soil above toe	0.00	kN	
Weight of toe	158.40	kN	
Number of piles at row 2	3		
Total reaction force from piles in row 2	2724.58	kN	(@ compression)
Distance of piles from face of column	0.20	m	
Design moment for bottom bars	465.72	kN-m	

Control of cracking by distribution of reinforcement

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs

Ultimate moment from service loads, M_s	465.72	kN-m	
80% of Modulus of rupture, f_r	2.185	MPa	fr = 0.52*√f'c
Tension in the cross section, fss	0.3528	MPa	fss = Ms/Sc

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



Extreme tension fiber to center of flexural reinforcement, d_c	-	mm					
Overall thickness of component, h	-	mm					
Compression fiber to the centroid of extreme tension steel, d_e	-	mm	b*x*x/2 = nAs*(de-x)				
Neutral axis to extreme compression fiber, x	-	mm	bx^2/2=nAsde-nAsx				
Modulus elasticity of steel, E_s	-	GPa	bx^2/2+nAsx-nAsde=0				
Modulus elasticity of concrete, E_c	-	GPa	A = b/2	-			
Modular ratio, n	-		B = nAs	-			
Cracked section moment of inertia of section, I_{NA}	-	mm^4	C = -nAs de	-			
Exposure factor, Y_e	-						
Exposure condition:	Class 1		j*de=	de - (x/3) =	-	mm	
Tensile stress in steel reinforcement at the service limit, f_s	-	Mpa	Actual fss =	Mservice 1 /(As j*de) =	-	Mpa	
$\beta s = 1+d_c/(0.7(h-d_c))$	-						
The spacing shall satisfy: $s \le 123000Y_e/\beta_s f_{ss} - 2dc$	-	mm					

Initial spacing: - mm -

Using of 25mm ⌀ main bars spaced at 200mm O.C. for breast wall is adequate and safe

1.6.4.2 Determine design moment for tension side (to design top bars)

Passive soil above footing	4.63	m
Weight of soil above heel	724.97	kN
Weight of heel	237.60	kN
Number of piles at row 1	3	
Total reaction force from piles in row 1	-673.27	kN (@ tension)
Distance of piles from face of column	0.70	m
Design moment for top bars	-1193.22	kN-m

Design for top bars

Demand moment for top bars	1193.22	kN-m	
Concrete cover	75	mm	
Diameter of reinforcing bar	25	mm	Ab = 490.87 mm2
Diameter of shrinkage bar	20	mm	
Diameter of cross ties	0	mm	
Effective depth of concrete	1112.5	mm	
Length to be considered	5500	mm	
Overall thickness of component	1200	mm	

Minimum reinforcement

Flexural cracking variability factor, Y_1	1.6	*for all other concrete	concrete compressive strength, f'c	27.6	MPa
Ratio of specified min. to ult. tensile strength of steel, Y_3	0.67	*for A615, 414MPa steel	yield strength of steel, fy	414	MPa
Modulus of rupture, F_r	3.310	Mpa (fr = 0.63*√f'c)	S_c = 1/6(bt^2)		
Section modulus, S_c	1.32E+09	mm^3	S_c = (bv*tww^2)/6		
Cracking moment, M_{cr}					
• $M_{cr}=Y_3(Y_1 fr)\,Sc$	4683.43	kN-m			
*Mu_min 1,33*Md =*	1586.98	kN-m			
Design moment for top bars	**1586.98**	kN-m			

Condition: if Md > (min (Mcr, Mu_min), Md, (min (Mcr,Mu_min))

Steel ratio

β_1 Coefficient Criterion : 0.85

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ	m_1 =	0.0567	
	m_2 =	0.0853	mm^2/N
	R_n =	0.2590	Mpa
	ρ =	0.0006	

Computation for main reinforcement

Required steel area	3849.87	mm^2		
Required spacing	701.27	say:	200	mm
Provided steel for top bars, *A s*	13499.03	mm^2		
Compression fiber to neutral axis, *c*	50.96	mm		
Depth of compression block, *a*	43.31	mm		
Nominal moment capacity of section, M_n	6096.29	kN-m		
Resistance factor, Ø	0.9	tension is controlled		
0.75 ≤ Ø =0.65+0.15 (deff/c −1) ≤0.9	3.77			
Ultimate moment capacity of section, ØM_n	**5486.66**	kN-m	**OK!**	
	c/d =	3.46		

Reaction per pile (at Service 1):

Axial for piles at row 1, (F1) **157.56** kN *Reaction is compression still not in tension!*

Determine design moment for tension side (to check design top bars from cracking)

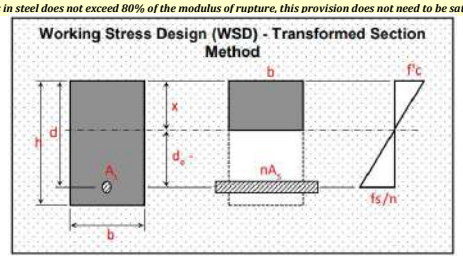
Design moment for top bars (Factored from tension side of piles)	-1193.22	kN-m
Design moment for top bars for cracking check	-897.16	kN-m
(Let Factored Moment / 1.33 = Service Moment)		

Control of cracking by distribution of reinforcement

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs

Ultimate moment from service loads, M_s	897.16	kN-m	
80% of Modulus of rupture, f_r	2.185	MPa	fr = 0.52*√f'c
Tension in the cross section, fss	0.680	MPa	fss = Ms/Sc

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



Extreme tension fiber to center of flexural reinforcement, d_c	-	mm	b*x*x/2 = nAs*(de-x)				
Overall thickness of component, *h*	-	mm	bx^2/2=nAsde-nAsx				
Compression fiber to the centroid of extreme tension steel, d_e	-	mm	bx^2/2+nAsx-nAsde=0				
Neutral axis to extreme compression fiber, *x*	-	mm	A = b/2	-			
Modulus elasticity of steel, E_s	-	GPa	B = nAs	-			
Modulus elasticity of concrete, E_c	-	GPa	C = -nAs de	-			
Modular ratio, *n*	-						
Cracked section moment of inertia of section, I_{NA}	-	mm^4	j*de=	de - (x/3) =	-	mm	
Exposure factor, Y_e	-		Actual fss =	Mservice 1 /(As j*de) =	-	Mpa	
Exposure condition:	Class 1				-		
Tensile stress in steel reinforcement at the service limit, f_s	-	Mpa					
βs=1+d_c/(0.7(h-d_c))	-						
The spacing shall satisfy: s ≤ 123000Y_e/$\beta_s f_{ss}$-2dc	-	mm					

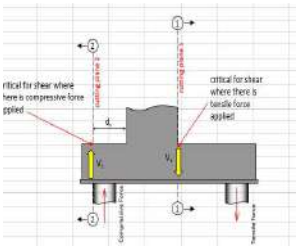
Initial spacing: - mm -

• for detailing purposes, it is recommended to adopt the conservative spacings of bars for top and bottom bars.

Using of 25mm ⌀ main bars spaced at 200mm O.C. for breast wall is adequate and safe

1.6.5 Verification of shear resistance

a. Nominal resistance for one-way action



For section 1 - 1

Reaction force of single pile in row 1, F_1	-224.42	kN	
Total force acting on tension side	-1635.84	kN	
Tension governs; critical for shear is at face of column			

distance of compressive force to face of breast wall= 0.2 m

For section 2 - 2

Reaction force of single pile in row 2, F_2	1299.73	kN	• by inspection, compression side is not critical because shear action falls outside the critical shear.
Total force acting on compression side	3740.80	kN	
Compression does not govern because critical for shear is at distance dv			
Ultimate shear based on demand, V_u	**1635.84**	kN	***Governs!!!***
Effective shear depth, d_v	965.84	mm	

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2) 965.84
dv2= 0.9d_e 888.75
dv3= 0.72tww 864

Factor indicating ability of diagonally cracked concrete to transmit tension	1.69		
Solution for β: **GENERAL PROCEDURE**			
Area of prestressing steel on tension side, A_{ps}	0	mm^2	
Area of non-prestressing steel, A_s	13499.03	mm^2	
Maximum aggregate size, a_u	20	mm	
Modulus of elasticity of prestressing tendons, f_{po}	0	MPa	
Factored axial force, N_u	0.00	N	
Positive for tension; Negative for compression			
Factored shear force, V_u	1,635,842.06	N	
Absolute value of the factored moment, $\lvert M_u \rvert$	1.59E+09	N-mm	
*But not less than $\lvert V_u - V_p \rvert d_v$	*controls as /Mu/ =* 1.58E+09	N-mm	
Modulus of elasticity of prestressing steel, E_p	0	GPa	
Modulus of elasticity of steel, E_s	200	GPa	
Net longitudinal tensile strain, e_x	0.0012		
Crack spacing parameter, S_{xe}	940.36	mm	Sxe = Sx *(35/ag + 16) Sx =dv
Shear resistance from steel, V_s	0	kN	
Effective prestressing force, V_p	0	kN	
Shear resistance provided by concrete, V_c	3,903.38	kN	Vc= 0.083*β(f'c^0.5)bv dv
The nominal shear resistance, V_n	3,903.38	kN	
*shall be determined as the lesser of:			
V_n=Vc+Vs+Vp	3,903.38	kN	
V_n=0.25fc'$b_v d_v$+Vp	36,653.77	kN	
Resistance factor for normal weight concrete,Ø	0.9		
Ultimate shear capacity of section, ØV_n	**3,513.04**	kN	**OK!**
	c/d =	2.15	

$$\beta = \frac{4.8}{(1+750\varepsilon_x)}\frac{51}{(39+S_{xe})}$$

$$\varepsilon_x = \frac{\left(\frac{|M_u|}{d_v}+0.5N_u+|V_u-V_p|-A_{ps}f_{po}\right)}{(E_sA_s+E_pA_{ps})}$$

Section without shear reinforcement is adequate and safe.

1.6.5.1 Check Shear Reinforcement Requirement

If Vu > Φ(Vc + Vp) provide shear reinforcement, Vs

Vu = 1636 kN < 3,513.04 kN = Φ(Vc + Vp)

∴ Section without shear reinfrocement is adequate.

Required Vs = (Vu/Φ) - (Vc + Vp)

Req'd Vs = - kN

Shear resistance from provided cross ties, Vs = (b/splength)*Av*fsy*dv / spheight

splength = - mm

Diameter of cross ties - mm spheight = - mm

Area of cross ties, Av = - mm2

Provided Vs = - kN -

No need to provide shear reinforcement.

b. Nominal shear resistance for two-way action

For two-way action for sections without transverse reinforcement, the nominal shear resistance, V_n of the concrete shall be taken as:

$$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv \le 0.33\sqrt{(fc')}\,bodv$$

Effective shear depth of footing, d_v	966	mm	
Ratio of long to short direction of pile, βc	1		
Perimeter of the critical section, b_o	4919	mm	
Maximum reaction of single pile, V_u	**1299.73**	kN	
Nominal shear capacity of concrete, V_n	8237.11	kN	
a)$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv$	12480.47	kN	
b) $Vn= 0.33\sqrt{(fc')}\,bodv$	8237.11	kN	
Resistance factor for normal weight concrete,Ø	0.9		
Ultimate shear capacity of section, ØV_n	**7413.40**	kN	**OK!**

Pile cap is adequate for shear due to maximum reaction of pile

1.6.6 Verification of shrinkage and temperature reinforcement

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete.

Diameter of shrinkage and temperature bar	20	mm	Ab= 314.16
Assumed spacing, S	200	mm	
Assumed shrinkage and temperature reinforcement, A_s	1.57	mm^2/mm	
Shrinkage and temperature reinforcement shall satisfy:			
a) As ≥ (0.75 bh)/(2(b+h)fy) = 0.89	1.57	mm^2/mm	

b) 0.233 ≤As ≤ 1.27 — 1.270 mm^2/mm

Spacing shall not exceed:
a) 3.0 times the component thickness, or 450 mm
b) 300 mm for walls and footings greater than 450 mm thick
c) 300 mm for other components greather than 900 mm thick

Final shrinkage and temperature reinforcement, A_s		1270.00	mm^2 per meter
Final spacing to be used		247.37	mm
	say:	200.00	mm

Use 20mm for temperature and shrinkage bar spaced at 200mm O.C. (Side Bars)

1.7 A DESIGN OF PILES - ABUT. A

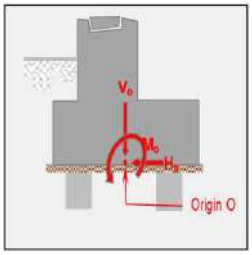
1.7.1 Determine the pile springs and geometric properties

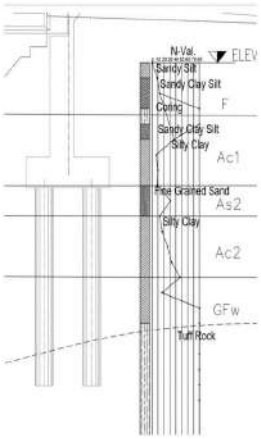
Design forces in Longitudinal Direction			
Governing Load Case			
EXTREME EVENT 1			
Forces acting at the bottom of pile cap/footing			
	Vertical Load	Lateral Load	Moment
Location	V_o	H_o	M_o
Origin O	kN	kN	kN-m
	2615.07	1660.40	3357.36
Pmax =	3836.79		

Note: Piles shall be evaluated both Pmax and Pmin. Pmin is critical in the pull out action.

STRENGTH 1			
Forces acting at the bottom of pile cap/footing			
	Vertical Load	Lateral Load	Moment
Location	V_o	H_o	M_o
Origin O	kN	kN	kN-m
	2484.32	1556.39	3446.29
Pmax =	3930.08		

Service I			
	Vertical Load	Lateral Load	Moment
Location	Vs	Hs	Ms
Origin O	kN	kN	kN-m
	3197.27	1002.71	2139.30





BH - ABUTMENT (See Plans For Actual Soil Profile)

a. Horizontal pile spring constant (K_h)

Horizontal Modulus of Subgrade Reaction
(See Geotechnical Report)

Borehole No., BH - 34
Pile diameter, D = 0.600 m

STRENGTH EVENT

BH Depth (m)	Thickness of soil (m)	Depth of pile z, (m) Embankment Up to Bottom of Pile Cap	Kh (kN/m³)	ks = Kh*D*z Soil Spring Stiffness, ks (kN/m) Embankment Up to Bottom of Pile Cap
	1.5	1.5	5609*	5048
0.000	1.5	3.0	5609*	5048
1.500	1.5	4.5	5609	5048
3.000	1.5	6.0	10401	9361
4.500	1.5	7.5	8774	7897
6.000	1.5	9.0	5609	5048
7.500	1.5	10.5	7175	6458
9.000	1.5	12.0	7175	6458
10.500	1.5	13.5	7175	9686
13.500	3.0	16.5	4083	7349
16.500	3.0	19.5	12052	21694
19.500	3.0	22.5	22348	40226
22.500	3.0	25.5	24116	54261
27.000	4.5	30.0	71176	192175
31.500	4.5	34.5	95075	256703
36.000	4.5	39.0	109292	295088
40.500	4.5	43.5	129903	175369

EXTREME EVENT

BH Depth (m)	Thickness (m)	Depth of pile z, (m) Embankment Up to Bottom of Pile Cap	Kh (kN/m³)	ks = Kh*D*z Soil Spring Stiffness, ks (kN/m) Embankment Up to Bottom of Pile Cap
	1.5	1.5	65527*	58974
0.000	1.5	3.0	65527*	58974
1.500	1.5	4.5	65527	58974
3.000	1.5	6.0	110229	99206
4.500	1.5	7.5	98412	88571
6.000	1.5	9.0	41572	37415
7.500	1.5	10.5	86060	77454
9.000	1.5	12.0	86060	77454
10.500	1.5	13.5	52593	71001
13.500	3.0	16.5	55050	99090
16.500	3.0	19.5	74317	133771
19.500	3.0	22.5	191796	345233
22.500	3.0	25.5	201782	454010
27.000	4.5	30.0	443906	1198546
31.500	4.5	34.5	329034	888392
36.000	4.5	39.0	361068	974884
40.500	4.5	43.5	405145	546946

* Embankment assumed value

Pile end vertical spring	
Kv (kN/m³)	Soil Axial Spring, kv (kN/m)
191,462	54135

Pile end vertical spring	
Kv (kN/m³)	Soil Axial Spring, kv (kN/m)
536,737	151759

b. Geometric properties of piles

Select Pile Section :
-- Circular Section
Square Section
Select Pile Installation Method :
Driven Piles (Blow Method)
Driven Piles (Vibro-Hammer Method)
Cast-in-place RC Piles
-- Bored Piles
Pre-Boring Piles
Steel Pile Soil Cement Piles

Input Pile Dimension : Diameter	0.6	m
Input Number of Piles	6	piles
Input Pile Length :	43.5	m
Calculate Section Properties :		
Cross-section Area	0.283	m^2
Perimeter of Pile :	1.885	m
Pile Moment of Inertia :	0.00636	m^4
Pile Flexural Stiffness :	1.74E+05	$kN-m^2$
Concrete Material Properties :		
Design Compressive Strength at 28th day	27.6	N/mm^2
Unit Density for Concrete	2400	kg/m^3
Unit weight for Reinforced Concrete	24	kN/m^3
Young's Modulus of Elasticity	2.735E+07	kN/m^2
Reinforcement Material Properties :		
Minimum Yield Strength	414	N/mm^2
Ultimate Tensile Strength	620	N/mm^2
Young's Modulus of Elasticity	2.00E+08	kN/m^2

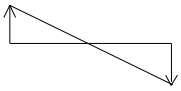
1.7.2 Determine the pile displacement and reaction force

a. Converted pressure loadings to pile cap with piles:

Footing			Piles size and location		
height (h_{ftg})	=	1.20 m	d_p =		0.60 m
width (w_{ftg})	=	3.50 m	a =		0.80 m
width (w_{toe})	=	1.00 m	b =		1.90 m
width (w_{heel})	=	1.50 m	c =		0.80 m
length (L_{ftg})	=	5.50 m	No. of piles, (N)	=	6

Governing Load Case: Extreme (Longitudinal Direction)

Vertical Load (uniform)			Lateral Load (uniform)			Moment (triangular, coupled)		
$V_{o\,min}$ =	135.85	kPa	H_o =	86.25	kPa	$+M_o$ =	298.99	kPa
$V_{o\,max}$ =	199.31	kPa				$-M_o$ =	-298.99	kPa



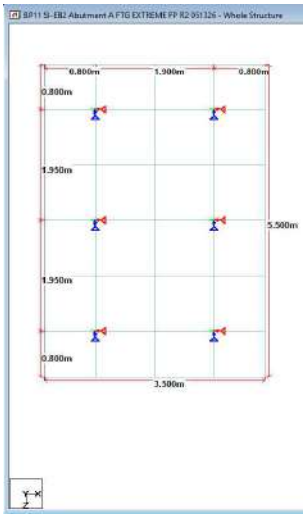
Strength Load Case: (Longitudinal Direction)

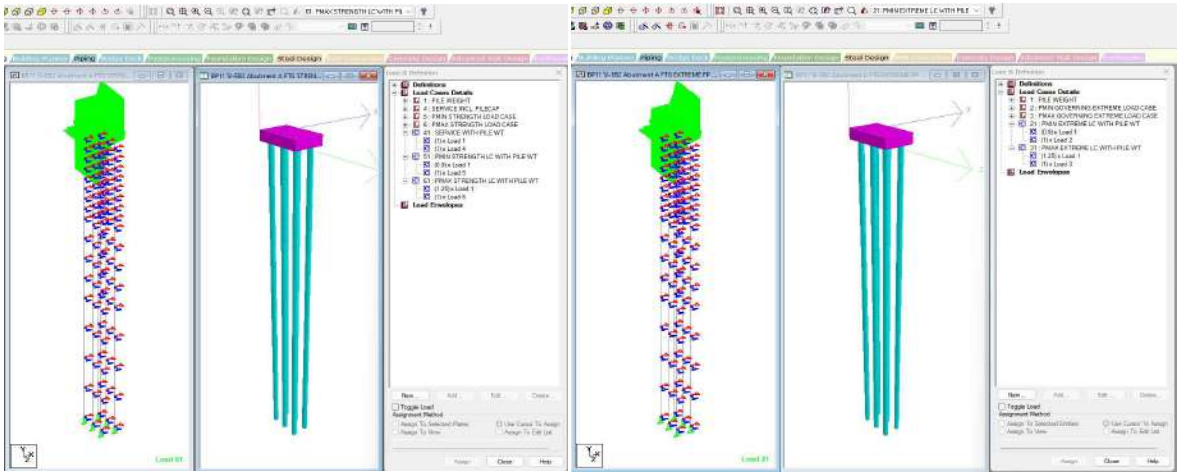
$V_{o\,min}$ =	129.06	kPa	H_o =	80.85	kPa	$+M_o$ =	306.91	kPa
$V_{o\,max}$ =	204.16	kPa				$-M_o$ =	-306.91	kPa

Service Load Case: (Longitudinal Direction)

V_s =	166.09	kPa	Hs =	52.09	kPa	$+M_s$ =	190.51	kPa
						$-Ms$ =	-190.51	kPa

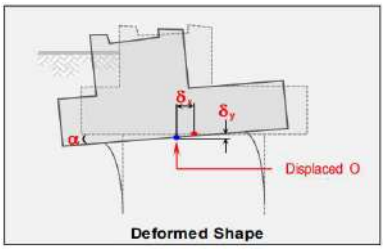
b. StaadPro modeling, loading and analysis:

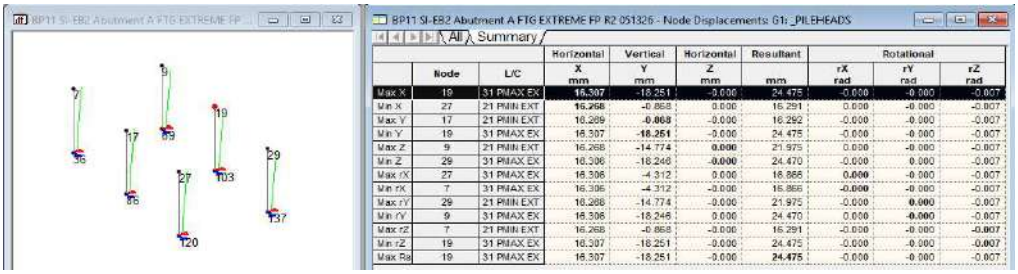




c. Displacement:

Longitudinal Displacement			
For Governing Load Case			
EXTREME EVENT 1			
Location	Displacement		
	Lateral	Vertical	Rotational
	d_x	d_y	a
	mm	mm	rad
Pile Head	**16.307**	**18.251**	**0.007**





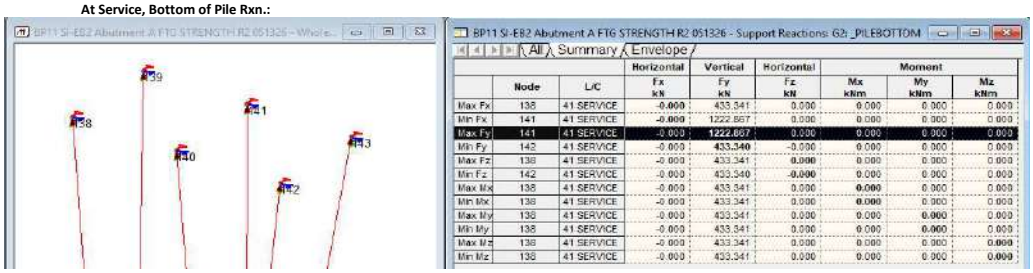
BP11 SI-EB2 Abutment A FTG EXTREME EP R2 051326 - Node Displacements: G1: _PILEHEADS

All / Summary

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	rY rad	rZ rad
Max X	19	31 PMAX EX	16.307	-18.251	-0.000	24.475	-0.000	-0.000	-0.007
Min X	27	21 PMIN EXT	16.268	-0.868	0.000	16.291	0.000	-0.000	-0.007
Max Y	17	21 PMIN EXT	16.289	-0.868	-0.000	16.292	-0.000	-0.000	-0.007
Min Y	19	31 PMAX EX	16.307	-18.251	-0.000	24.475	-0.000	-0.000	-0.007
Max Z	9	21 PMIN EXT	16.268	-14.774	0.000	21.975	0.000	-0.000	-0.007
Min Z	29	31 PMAX EX	16.306	-18.246	-0.000	24.470	-0.000	-0.000	-0.007
Max rX	27	31 PMAX EX	16.306	-4.312	0.000	16.866	0.000	-0.000	-0.007
Min rX	7	31 PMAX EX	16.306	-4.312	-0.000	16.866	-0.000	-0.000	-0.007
Max rY	29	21 PMIN EXT	16.268	-14.774	-0.000	21.975	-0.000	0.000	-0.007
Min rY	9	31 PMAX EX	16.306	-18.246	0.000	24.470	0.000	-0.000	-0.007
Max rZ	7	21 PMIN EXT	16.268	-0.868	-0.000	16.291	-0.000	-0.000	-0.007
Min rZ	19	31 PMAX EX	16.307	-18.251	-0.000	24.475	-0.000	-0.000	-0.007
Max Re	19	31 PMAX EX	16.307	-18.251	-0.000	24.475	-0.000	-0.000	-0.007

d. Reaction Forces:

At Service, Bottom of Pile Rxn.:

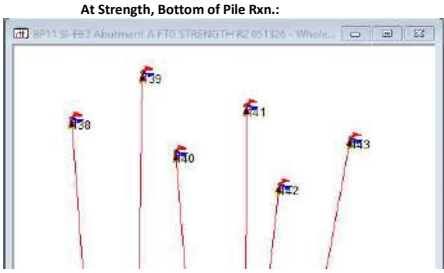


BP11 SI-EB2 Abutment A FTG STRENGTH R2 051326 - Support Reactions: G2: _PILEBOTTOM

All / Summary / Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Min Fx	141	41 SERVICE	-0.000	1222.867	0.000	0.000	0.000	0.000
Max Fy	141	41 SERVICE	-0.000	1222.867	0.000	0.000	0.000	0.000
Min Fy	142	41 SERVICE	-0.000	433.340	-0.000	0.000	0.000	0.000
Max Fz	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Min Fz	142	41 SERVICE	-0.000	433.340	-0.000	0.000	0.000	0.000
Max Mx	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Min Mx	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Max My	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Min My	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Max Mz	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000
Min Mz	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	0.000

At Strength, Bottom of Pile Rxn.:

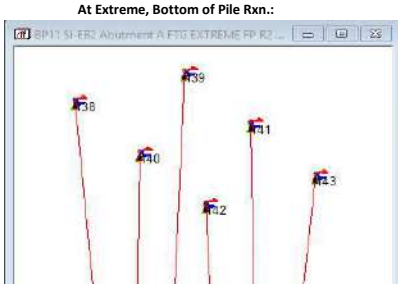


BP11 SI-EB2 Abutment A FTG STRENGTH R2 051326 - Support Reactions: G2: _PILEBOTTOM

All / Summary / Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	Moment My kNm	Moment Mz kNm
Max Fx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Fx	141	61 PMAX ST	-0.000	1650.052	0.000	0.000	0.000	0.000
Max Fy	141	61 PMAX ST	-0.000	1650.052	0.000	0.000	0.000	0.000
Min Fy	140	51 PMIN STR	-0.000	58.105	-0.000	0.000	0.000	0.000
Max Fz	143	51 PMIN STR	-0.000	1361.293	0.000	0.000	0.000	0.000
Min Fz	139	51 PMIN STR	-0.000	1361.284	-0.000	0.000	0.000	0.000
Max Mx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Mx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Max My	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min My	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Max Mz	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Mz	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000

At Extreme, Bottom of Pile Rxn.:

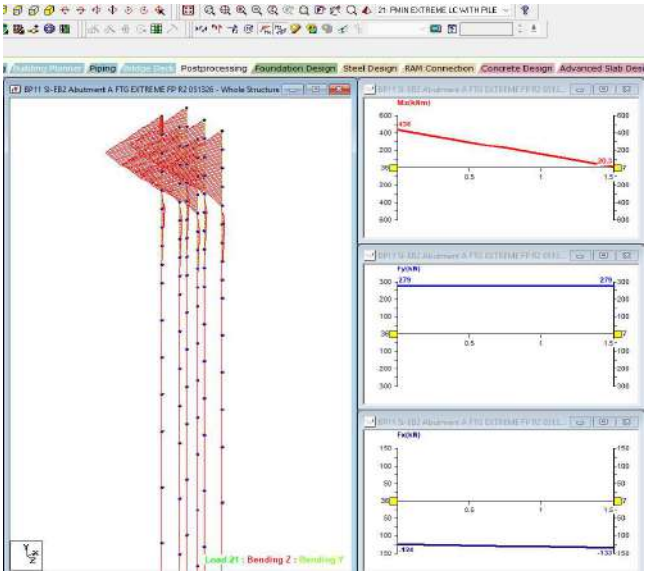


BP11 SI-EB2 Abutment A FTG EXTREME FP R2 051326 - Support Reactions: G2: _PILEBOTTO...

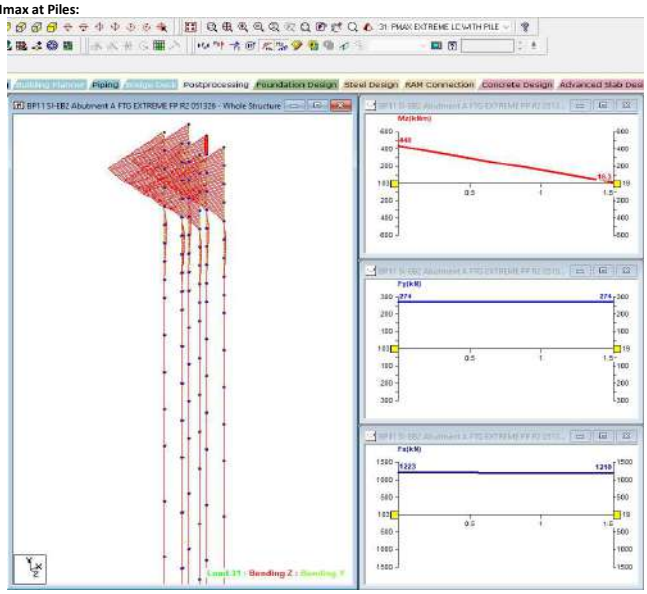
All / Summary / Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	Moment My kNm	Moment Mz kNm
Max Fx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Fx	141	31 PMAX EX	-0.000	1579.167	0.000	0.000	0.000	0.000
Max Fy	141	31 PMAX EX	-0.000	1579.167	0.000	0.000	0.000	0.000
Min Fy	140	21 PMIN EXT	-0.000	132.200	0.000	0.000	0.000	0.000
Max Fz	138	31 PMAX EX	-0.000	437.985	0.000	0.000	0.000	0.000
Min Fz	142	31 PMAX EX	-0.000	437.964	-0.000	0.000	0.000	0.000
Max Mx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Mx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Max My	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min My	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Max Mz	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Mz	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000

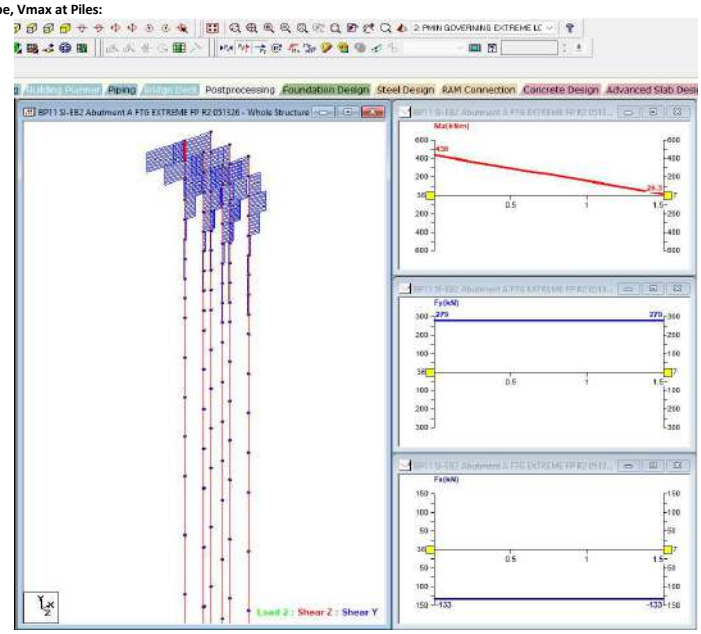
At Ultimate, Pmin with Mmax at Piles:



At Ultimate, Pmax with Mmax at Piles:



At Ultimate Envelope, Vmax at Piles:



1.7.3 Verification of Pile Stability

a. The factored resistance of piles shall be taken as: *(If comparing demand at top of pile)*

$$R_R = \Upsilon(\emptyset R_n - Ws) + W_s - W$$

where:

R_R	factored resitance of pile, kN	
R_n	nominal resistance of pile, kN	
W_s	effective weight of soil replaced by pile, kN	
W	effective weight of pile and soil inside pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state 0.65-BSDS Article 5.4.1(5)	0.65
Υ	modification coefficient depending on nominal bearing resistance 1.00-BSDS Table 5.4.3.3-1	

b. The nominal bearing capacity can be obtained from the empirical bearing capacity estimation formula:

$$R_n = q_d A_p + U\Sigma L_i f_i$$

where:

R_n	nominal bearing capacity of pile, kN
A_p	area of pile tip
q_d	nominal end bearing resistance intensity per unit area, kN/m^2
U	perimeter of pile
L_i	thickness of soil layer considering shaft resistance, m
f_i	maximum shaft resistance of soil layer considering pile shaft resistance, kN/m^2

c. The factored axial pull-out resistance of a single pile shall be obtained considering soil conditions and construction methods:

(If comparing demand at top of pile)

$$P_R = \emptyset P_n + W$$

where:

P_R	factored axial pull-out resistance of pile, kN	
P_n	nominal axial pull-out resistance, kN	
W	effective weight of pile, kN	
Ø	resistance factore for pile under extreme event limit state	0.50

-BSDS Article 5.4.1(5)

Pile Bearing and Pull-Out Capacity for 600mm dia. BORED PILE (Est. From Geotechnical Report)

BH-34 L = 43.50 m

Allowable Pile Bearing Capacity of Pile, Ra :	**(vs Service)**	3557	kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Bearing Capacity of Pile, Rn :	**(vs Strength)**	3557	kN	
Extreme (Already Factored) Resistance of Pile, RR :	**(vs Extreme)**	3211	kN	
Allowable Uplift Capacity of Pile, Pa :	**(vs Service)**	2031	kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Pull-out Capacity of Pile, Pn :	**(vs Strength)**	2031	kN	
Extreme (Already Factored) Pull-out Resistance of Pile, PR	**(vs Extreme)**	1882	kN	

Verification for Max. Axial Resistance (Service)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1222.87	3557	2.91	OK

Verification for Max. Axial Pull-out Resistance (Service)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
433.34	2031	OK	OK	*(No pull-out demand at bottom)*

Verification for Max. Axial Resistance (Strength)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1650.05	3557	2.16	OK

Verification for Max. Axial Pull-out Resistance (Strength)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
58.11	2031	OK	OK	*(No pull-out demand at bottom)*

Verification for Maximum Axial Resistance (Extreme)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1579.17	3211	2.03	OK

Verification for Maximum Axial Pull-out Resistance (Extreme)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
132.20	1882	OK	OK	*(No pull-out demand at bottom)*

Verification for Lateral Displacement

Displacement		C/D	Verification
Demand	Capacity		
mm	mm		
16.31	30	1.84	OK

Capacity =5% x Dpile. (See commentary BSDS pp 4-15)

under earthquake loading this value ... may reach as much as 5%

Verification for Tip Displacement/Settlement

Displacement		C/D	Verification
Demand	Allowable		
mm	mm		
10.41	25	2.40	OK

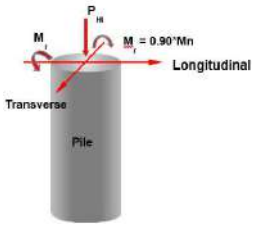
1.7.4 Verification of Pile Structural Resistance

Pmin with Mmax	
Rigid Pile Head	
P_u	M_u
kN	kN-m
-124	438

Note: The connection is rigid.

Pmax with Mmax	
Rigid Pile Head	
P_u	M_u
kN	kN-m
1223	440

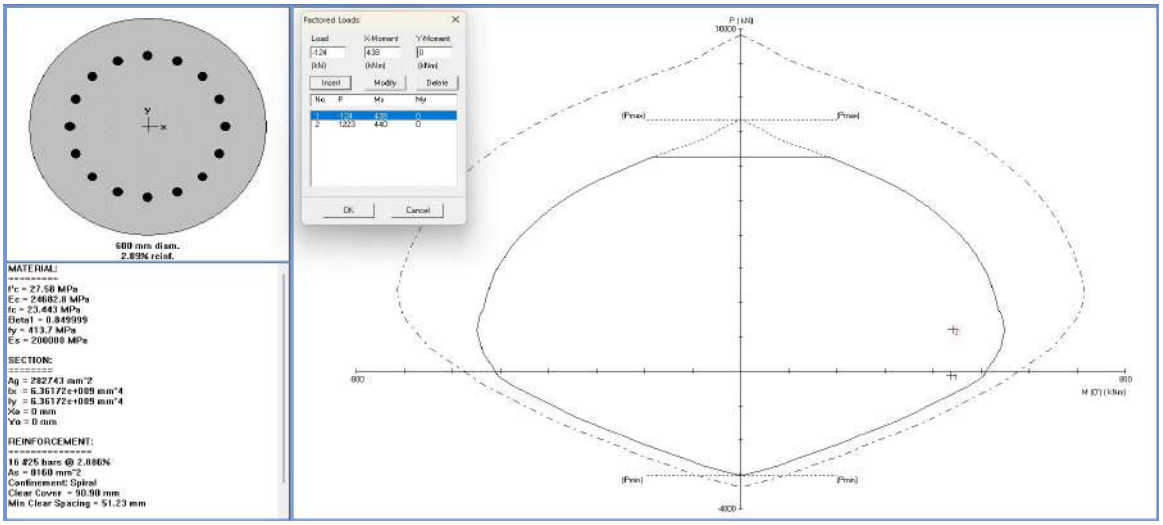
Note: The connection is rigid.



Define Reinforcement of Pile :

Diameter of Longitudinal Bars	25	mm
Diameter of Hoops/Spiral	16	mm
Reinforcement concrete cover to spiral/hoops	75	mm

SPColumn Axial and Flexure (P-M) Design of Pile



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	-124.00	438.00	0.00	505.25	0.00	1.154	188	496	0.00494	0.897
2	1223.00	440.00	0.00	549.25	0.00	1.248	260	496	0.00274	0.764

Verification of Flexure with Axial Load Resistance

P = -124 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
438.00	505.25	1.15	OK

P = 1223 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
440.00	549.25	1.25	OK

Calculate Pile Shear Design and Resistance

Shear Demand, Vu = 279 kN

The nominal shear resistance, Vn is lesser of the following:

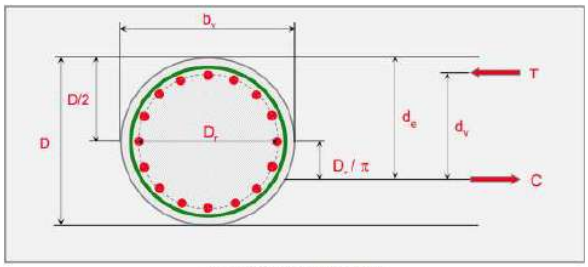
Vn = Vc + Vs + *Vp* OR Vn = 0.25f'c bv dv + *Vp*

where: Vc = 0.083*β*Sqrt(f'c)*bv*dv
Vs = [Av*fy*dv*(cotθ+cotα) sinα] / s
Vp = 0.00 for non-prestressed concrete
Vp = component in the direction of applied shear of effective prestressing force

Ultimate shear resistance, Vr = ØVn: Vr = ØVn > Vd_EQ where Ø= 0.9 for shear

Shear design parameters:

D	diameter of pile (out to out dimension)	600	mm
cc	concrete cover (to spiral or hoops)	75	mm
db	main bars or longitudinal bars	25	mm
ds	size of spirals	16	mm
f'c	concrete compressive strength	27.6	MPa
fy	yield strength of reinforcing steel	414	MPa
bv	effective web width, mm = diameter of column	600	mm
de	de = (D / 2) + (Dr / π)	417	mm
dv	effective shear depth = 0.90 x de	375	mm
β	factor indicating ability of diagonally cracked concrete to transmit tension	2.0	
θ	angle of inclination of diagonal compressive stresses	45	deg
α	angle of inclination of transverse reinforcement	90	deg
Dr	diameter of circle passing thru centers of the longitudinal reinf.	368	mm
Dc	diameter of circle measured ourside the diameter of spiral/hoops	450	mm
Ac	Area of core measured outside diameter of spiral/hoops	159043	mm^2
Ag	Gross area of section	282743.339	mm^2



: For definition of terms:

From above parameters:

ds	size of spirals	16	mm
Asp	area of shear reinf.	201	mm
s	Try spacing	75	mm
No	Try no. of legs	2	No. legs
Total area of shear reinf. Av = Av = No*(Asp)		402	mm^2

Definition of no. of legs

leg ⬭ leg

no. of spirals	No. of legs
single spiral	2 legs
bundle spiral	4 legs

cot θ = 1/tan 45° = 1
cot α = 1/tan 90° = 0
sin α = sin90° = 1

Vc = 0.00 kN *For conservative approach, assume Vc = 0.0, i.e. concrete has no shear strength.; If Pmin = negative value (tension), Vc = 0.0.*
Vp = 0.00 kN *For non prestressed concrete*
Vs = [Av*fy*dv*(cot θ + cot α)sin α]/s = **833.3** kN

Vn is lesser of the following:

Vn = Vc + Vs + Vp =	833.3	kN	**Governing shear capacity!**
Vn = 0.25f'cbvdv+Vp=	1554.3	kN	
Vd_EQ =	279	kN	Shear Demand
Vr =	**750**	**kN**	Ultimate Shear Resistance, ØVn
c/d =	2.69		Shear resistance is OK!

c. Verification of shear/transverse reinforcement for confinement at plastic hinges

DGCS 12.7.11.2

Length of column end region will be the largest of the following :

a.) 3D + Max M Loc. *(assume 5D)* = 4.80 m *Conservative, see pile moment diagram above*
b.)450 mm (18 inches) = 0.45 m
therefore, **Length** = **4.80 m**

d. Verification of minimum required transverse reinforcement

1. The greater of the following:
ρs1 = 0.12*(f'c/fy) = 0.00800 *AASHTO 5.11.4.1.4 @ Plastic Hinge Region*
Note: for circular shape:
ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) = 0.02333 **Governs!** *AASHTO 5.6.4.6 @ Column General Region*

2. Checking from provided confinement, where Av represent spiral leg on one (1) side

$$\rho_{sprovided} = \frac{4*Av}{Dc*s} = 0.02383$$ **OK!!**

Maximum spacing

1. S should not be greater than 1/4 min dimension of member (=D/4) **OK!!**
2. S should not be greater than 100mm **OK!!**

Minimum clear spacing

1. Sc should not be less than 25mm **OK!!**
2. Sc should not be less than 1.33 x aggregate size (1.33 x 20mm) **OK!!**

e. Transverse reinforcement for outside region

ds	size of spirals	16	mm
Asp	area of shear reinforcement	201	mm^2
s	Try spacing of transverse steel	150	mm
No	Try number of legs	2	No. legs
	Total area of shear reinf. Av = Av = No*(Asp)	402.12	mm^2

DGCS 12.5.2.4

Min area of transverse reinf. spacing :

Av = 0.083 x sqrt(f'c) x bv x s/fy = 94.79 mm^2 **OK!!**

f. Summary of shear/transverse reinforcement

Location	Range, m	Spacing/Pitch of spirals, mm	Diameter, mm	No. of legs
End regions	5.00	75	16	2
Outside region	38.50	150	16	2

Pile Details Summary at Abutment:

Use 6 - 600mm Dia. Bored Piles, L=43.5m, with 16-25mm dia. main bars, 16mm dia. spiral hoops spaced @ 75mm for the first 5m; then 16mm dia. spiral hoops spaced @ 150mm all throughout the rest of total length.

1.7 B DESIGN OF PILES - ABUT. B

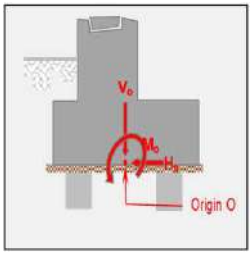
1.7.1 Determine the pile springs and geometric properties

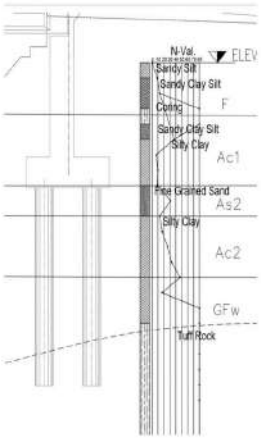
Design forces in Longitudinal Direction			
Governing Load Case			
EXTREME EVENT 1			
Forces acting at the bottom of pile cap/footing			
	Vertical Load	Lateral Load	Moment
Location	V_o	H_o	M_o
Origin O	kN	kN	kN-m
	2615.07	1660.40	3357.36
Pmax =	3836.79		

Note: Piles shall be evaluated both Pmax and Pmin. Pmin is critical in the pull out action.

STRENGTH 1			
Forces acting at the bottom of pile cap/footing			
	Vertical Load	Lateral Load	Moment
Location	V_o	H_o	M_o
Origin O	kN	kN	kN-m
	2484.32	1556.39	3446.29
Pmax =	3930.08		

Service I			
	Vertical Load	Lateral Load	Moment
Location	Vs	Hs	Ms
Origin O	kN	kN	kN-m
	3197.27	1002.71	2139.30





BH - ABUTMENT (See Plans For Actual Soil Profile)

a. Horizontal pile spring constant (K_h)

Horizontal Modulus of Subgrade Reaction
(See Geotechnical Report)

Borehole No., BH - 34
Pile diameter, D = 0.600 m

STRENGTH EVENT

BH Depth (m)	Thickness of soil (m)	Depth of pile z, (m) Embankment Up to Bottom of Pile Cap	Kh (kN/m³)	ks = Kh*D*z Soil Spring Stiffness, ks (kN/m) Embankment Up to Bottom of Pile Cap
	1.5	1.5	5609*	5048
0.000	1.5	3.0	5609*	5048
1.500	1.5	4.5	5609	5048
3.000	1.5	6.0	10401	9361
4.500	1.5	7.5	8774	7897
6.000	1.5	9.0	5609	5048
7.500	1.5	10.5	7175	6458
9.000	1.5	12.0	7175	6458
10.500	1.5	13.5	7175	9686
13.500	3.0	16.5	4083	7349
16.500	3.0	19.5	12052	21694
19.500	3.0	22.5	22348	40226
22.500	3.0	25.5	24116	54261
27.000	4.5	30.0	71176	192175
31.500	4.5	34.5	95075	256703
36.000	4.5	39.0	109292	295088
40.500	4.5	43.5	129903	175369

EXTREME EVENT

BH Depth (m)	Thickness (m)	Depth of pile z, (m) Embankment Up to Bottom of Pile Cap	Kh (kN/m³)	ks = Kh*D*z Soil Spring Stiffness, ks (kN/m) Embankment Up to Bottom of Pile Cap
	1.5	1.5	65527*	58974
0.000	1.5	3.0	65527*	58974
1.500	1.5	4.5	65527	58974
3.000	1.5	6.0	110229	99206
4.500	1.5	7.5	98412	88571
6.000	1.5	9.0	41572	37415
7.500	1.5	10.5	86060	77454
9.000	1.5	12.0	86060	77454
10.500	1.5	13.5	52593	71001
13.500	3.0	16.5	55050	99090
16.500	3.0	19.5	74317	133771
19.500	3.0	22.5	191796	345233
22.500	3.0	25.5	201782	454010
27.000	4.5	30.0	443906	1198546
31.500	4.5	34.5	329034	888392
36.000	4.5	39.0	361068	974884
40.500	4.5	43.5	405145	546946

* Embankment assumed value

Pile end vertical spring	
Kv (kN/m³)	Soil Axial Spring, kv (kN/m)
191,462	54135

Pile end vertical spring	
Kv (kN/m³)	Soil Axial Spring, kv (kN/m)
536,737	151759

b. Geometric properties of piles

Select Pile Section :
-- Circular Section
Square Section
Select Pile Installation Method :
Driven Piles (Blow Method)
Driven Piles (Vibro-Hammer Method)
Cast-in-place RC Piles
-- Bored Piles
Pre-Boring Piles
Steel Pile Soil Cement Piles

Input Pile Dimension : Diameter	0.6	m
Input Number of Piles	6	piles
Input Pile Length :	43.5	m
Calculate Section Properties :		
Cross-section Area	0.283	m^2
Perimeter of Pile :	1.885	m
Pile Moment of Inertia :	0.00636	m^4
Pile Flexural Stiffness :	1.74E+05	$kN-m^2$
Concrete Material Properties :		
Design Compressive Strength at 28th day	27.6	N/mm^2
Unit Density for Concrete	2400	kg/m^3
Unit weight for Reinforced Concrete	24	kN/m^3
Young's Modulus of Elasticity	2.735E+07	kN/m^2
Reinforcement Material Properties :		
Minimum Yield Strength	414	N/mm^2
Ultimate Tensile Strength	620	N/mm^2
Young's Modulus of Elasticity	2.00E+08	kN/m^2

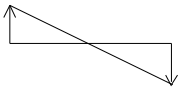
1.7.2 Determine the pile displacement and reaction force

a. Converted pressure loadings to pile cap with piles:

Footing			Piles size and location		
height (h_{ftg})	=	1.20 m	d_p =		0.60 m
width (w_{ftg})	=	3.50 m	a =		0.80 m
width (w_{toe})	=	1.00 m	b =		1.90 m
width (w_{heel})	=	1.50 m	c =		0.80 m
length (L_{ftg})	=	5.50 m	No. of piles, (N)	=	6

Governing Load Case: Extreme (Longitudinal Direction)

Vertical Load (uniform)			Lateral Load (uniform)			Moment (triangular, coupled)		
$V_{o\,min}$ =	135.85	kPa	H_o =	86.25	kPa	$+M_o$ =	298.99	kPa
$V_{o\,max}$ =	199.31	kPa				$-M_o$ =	-298.99	kPa



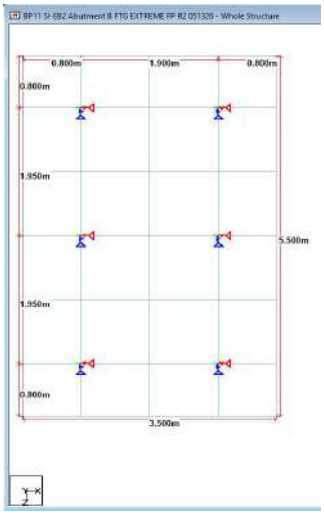
Strength Load Case: (Longitudinal Direction)

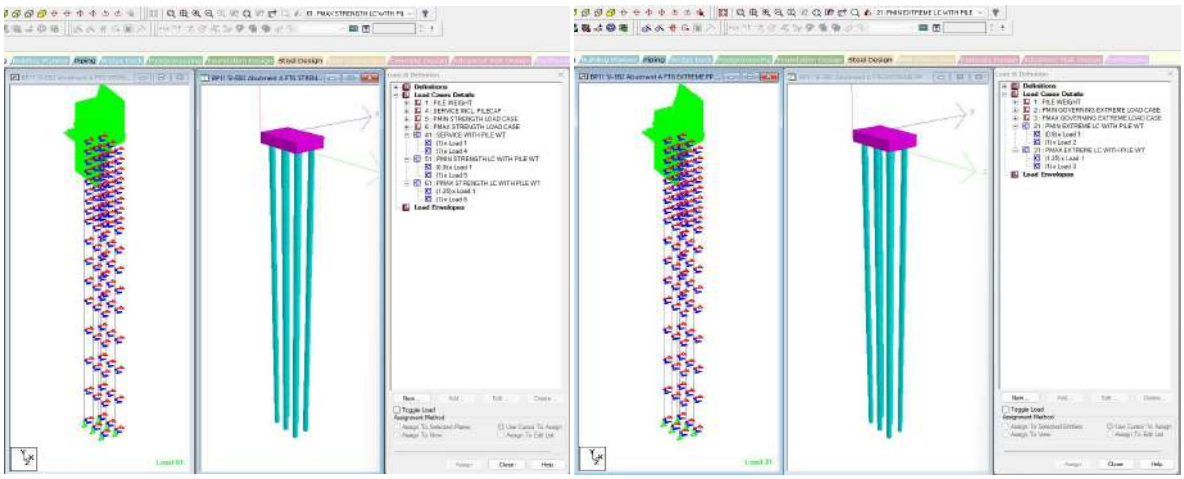
$V_{o\,min}$ =	129.06	kPa	H_o =	80.85	kPa	$+M_o$ =	306.91	kPa
$V_{o\,max}$ =	204.16	kPa				$-M_o$ =	-306.91	kPa

Service Load Case: (Longitudinal Direction)

V_s =	166.09	kPa	Hs =	52.09	kPa	$+M_s$ =	190.51	kPa
						$-Ms$ =	-190.51	kPa

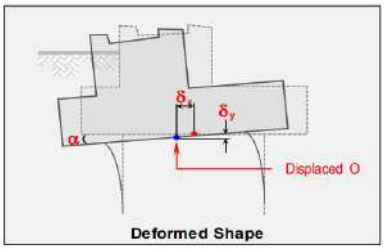
b. StaadPro modeling, loading and analysis:

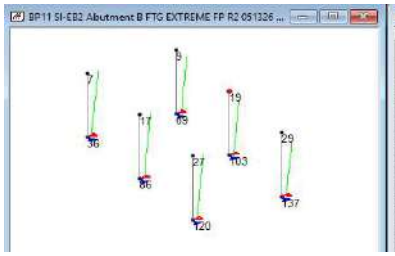


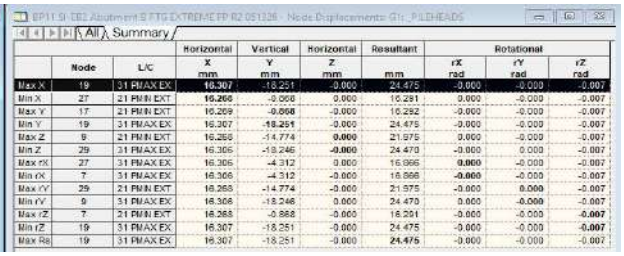


c. Displacement:

Longitudinal Displacement			
For Governing Load Case			
EXTREME EVENT 1			
Location	Displacement		
	Lateral	Vertical	Rotational
	d_x	d_y	a
	mm	mm	rad
Pile Head	**16.307**	**18.251**	**0.007**



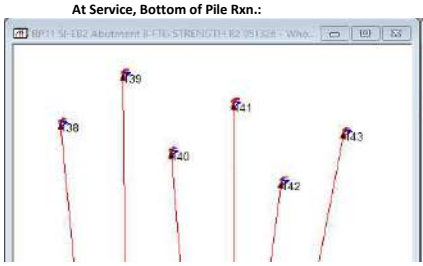


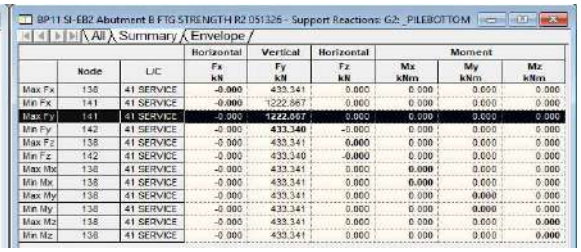


	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	rY rad	rZ rad
Max X	19	31 PMAX EX	**16.307**	-18.251	-0.000	24.475	-0.000	-0.000	-0.007
Min X	27	21 PMIN EXT	**16.266**	-0.068	0.000	16.291	0.000	-0.000	-0.007
Max Y	17	21 PMIN EXT	16.269	**-0.068**	-0.000	16.292	-0.000	-0.000	-0.007
Min Y	19	31 PMAX EX	16.307	**-18.251**	-0.000	24.475	-0.000	-0.000	-0.007
Max Z	9	21 PMIN EXT	16.268	-14.774	**0.000**	21.575	0.000	-0.000	-0.007
Min Z	29	31 PMAX EX	16.306	-18.246	**-0.000**	24.470	-0.000	0.000	-0.007
Max rX	27	31 PMAX EX	16.306	-4.312	0.000	16.866	**0.000**	-0.000	-0.007
Min rX	7	31 PMAX EX	16.306	-4.312	-0.000	16.866	**-0.000**	-0.000	-0.007
Max rY	29	21 PMIN EXT	16.268	-14.774	-0.000	21.575	-0.000	**0.000**	-0.007
Min rY	9	31 PMAX EX	16.306	-18.246	0.000	24.470	0.000	**-0.000**	-0.007
Max rZ	7	21 PMIN EXT	16.268	-0.868	-0.000	16.291	-0.000	-0.000	**-0.007**
Min rZ	19	31 PMAX EX	16.307	-18.251	-0.000	24.475	-0.000	-0.000	**-0.007**
Max Rs	19	31 PMAX EX	16.307	-18.251	-0.000	**24.475**	-0.000	-0.000	-0.007

d. Reaction Forces:

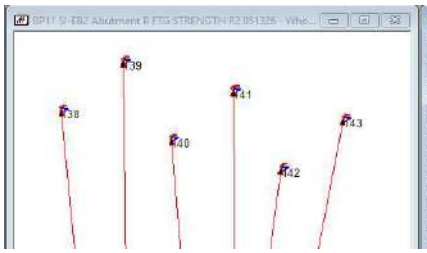
At Service, Bottom of Pile Rxn.:





	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	138	41 SERVICE	**-0.000**	433.341	0.000	0.000	0.000	0.000
Min Fx	141	41 SERVICE	**-0.000**	1222.867	0.000	0.000	0.000	0.000
Max Fy	141	41 SERVICE	-0.000	**1222.867**	0.000	0.000	0.000	0.000
Min Fy	142	41 SERVICE	-0.000	**433.340**	-0.000	0.000	0.000	0.000
Max Fz	138	41 SERVICE	-0.000	433.341	**0.000**	0.000	0.000	0.000
Min Fz	142	41 SERVICE	-0.000	433.340	**-0.000**	0.000	0.000	0.000
Max Mx	138	41 SERVICE	-0.000	433.341	0.000	**0.000**	0.000	0.000
Min Mx	138	41 SERVICE	-0.000	433.341	0.000	**0.000**	0.000	0.000
Max My	138	41 SERVICE	-0.000	433.341	0.000	0.000	**0.000**	0.000
Min My	138	41 SERVICE	-0.000	433.341	0.000	0.000	**0.000**	0.000
Max Mz	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	**0.000**
Min Mz	138	41 SERVICE	-0.000	433.341	0.000	0.000	0.000	**0.000**

At Strength, Bottom of Pile Rxn.:

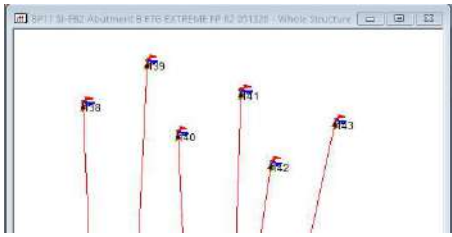


BP11 SI-EB2 Abutment B FTG STRENGTH R2 051326 - Support Reactions: G2: _PILEBOTTOM

All \ Summary \ Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	Moment My kNm	Moment Mz kNm
Max Fx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Fx	141	61 PMAX ST	-0.000	1650.052	0.000	0.000	0.000	0.000
Max Fy	141	61 PMAX ST	-0.000	1650.052	0.000	0.000	0.000	0.000
Min Fy	140	51 PMIN STR	-0.000	58.116	-0.000	0.000	0.000	0.000
Max Fz	143	51 PMIN STR	-0.000	1301.293	0.000	0.000	0.000	0.000
Min Fz	138	51 PMIN STR	-0.000	1301.294	-0.000	0.000	0.000	0.000
Max Mx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Mx	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Max My	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min My	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Max Mz	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000
Min Mz	138	51 PMIN STR	-0.000	58.116	0.000	0.000	0.000	0.000

At Extreme, Bottom of Pile Rxn.:

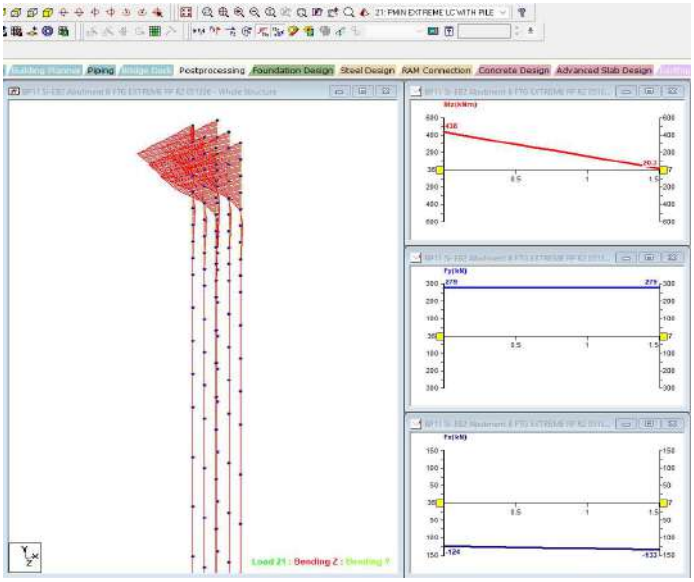


BP11 SI-EB2 Abutment B FTG EXTREME FP R2 051326 - Support Reactions: G2: _PILEBOTT...

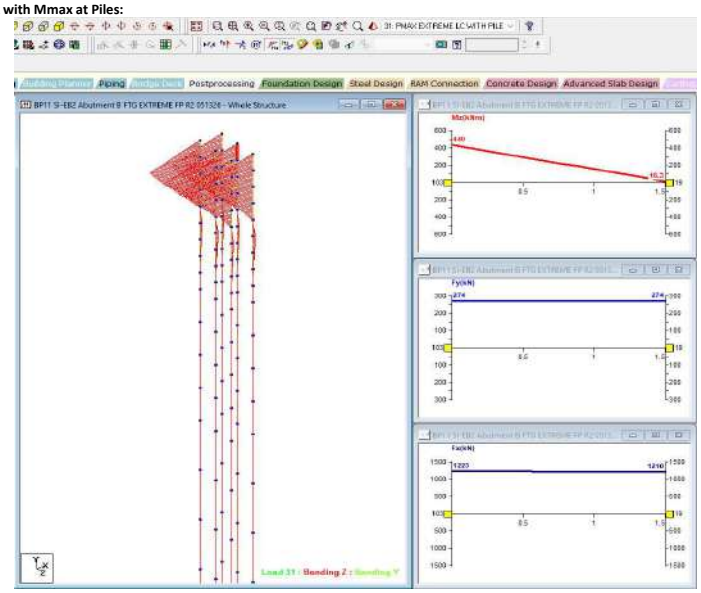
All \ Summary \ Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	Moment My kNm	Moment Mz kNm
Max Fx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Fx	141	31 PMAX EX	-0.000	1579.167	0.000	0.000	0.000	0.000
Max Fy	141	31 PMAX EX	-0.000	1579.167	0.000	0.000	0.000	0.000
Min Fy	140	21 PMIN EXT	-0.000	132.200	0.000	0.000	0.000	0.000
Max Fz	138	31 PMAX EX	-0.000	437.965	0.000	0.000	0.000	0.000
Min Fz	142	31 PMAX EX	-0.000	437.964	-0.000	0.000	0.000	0.000
Max Mx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Mx	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Max My	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min My	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Max Mz	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000
Min Mz	138	21 PMIN EXT	-0.000	132.219	0.000	0.000	0.000	0.000

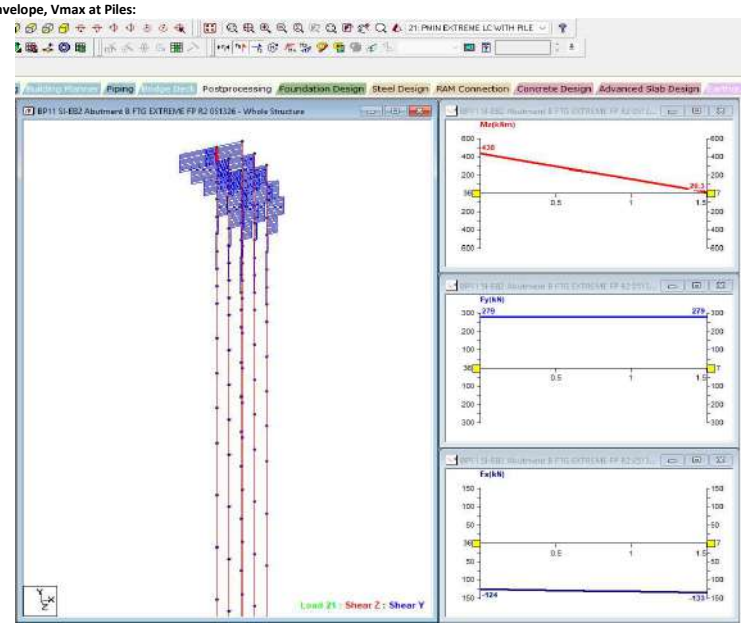
At Ultimate, Pmin with Mmax at Piles:



At Ultimate, Pmax with Mmax at Piles:



At Ultimate Envelope, Vmax at Piles:



1.7.3 Verification of Pile Stability

a. The factored resistance of piles shall be taken as: *(If comparing demand at top of pile)*

$$R_R = \Upsilon(\emptyset R_n - Ws) + W_s - W$$

where:

R_R	factored resitance of pile, kN	
R_n	nominal resistance of pile, kN	
W_s	effective weight of soil replaced by pile, kN	
W	effective weight of pile and soil inside pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state 0.65-BSDS Article 5.4.1(5)	0.65
Υ	modification coefficient depending on nominal bearing resistance 1.00-BSDS Table 5.4.3.3-1	

b. The nominal bearing capacity can be obtained from the empirical bearing capacity estimation formula:

$$R_n = q_d A_p + U\Sigma L_i f_i$$

where:

R_n	nominal bearing capacity of pile, kN
A_p	area of pile tip
q_d	nominal end bearing resistance intensity per unit area, kN/m^2
U	perimeter of pile
L_i	thickness of soil layer considering shaft resistance, m
f_i	maximum shaft resistance of soil layer considering pile shaft resistance, kN/m^2

c. The factored axial pull-out resistance of a single pile shall be obtained considering soil conditions and construction methods:

(If comparing demand at top of pile)

$$P_R = \emptyset P_n + W$$

where:

P_R	factored axial pull-out resistance of pile, kN	
P_n	nominal axial pull-out resistance, kN	
W	effective weight of pile, kN	
Ø	resistance factore for pile under extreme event limit state	0.50

-BSDS Article 5.4.1(5)

Pile Bearing and Pull-Out Capacity for 600mm dia. BORED PILE (Est. From Geotechnical Report)

BH-34 L = 43.50 m

Allowable Pile Bearing Capacity of Pile, Ra :	**(vs Service)**	2292	kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Bearing Capacity of Pile, Rn :	**(vs Strength)**	2292	kN	
Extreme (Already Factored) Resistance of Pile, RR :	**(vs Extreme)**	3126	kN	
Allowable Uplift Capacity of Pile, Pa :	**(vs Service)**	1395	kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Pull-out Capacity of Pile, Pn :	**(vs Strength)**	1395	kN	
Extreme (Already Factored) Pull-out Resistance of Pile, PR	**(vs Extreme)**	1853	kN	

Verification for Max. Axial Resistance (Service)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1222.87	2292	1.87	OK

Verification for Max. Axial Pull-out Resistance (Service)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
433.34	1395	OK	OK	*(No pull-out demand at bottom)*

Verification for Max. Axial Resistance (Strength)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1650.05	2292	1.39	OK

Verification for Max. Axial Pull-out Resistance (Strength)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
58.11	1395	OK	OK	*(No pull-out demand at bottom)*

Verification for Maximum Axial Resistance (Extreme)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1579.17	3126	1.98	OK

Verification for Maximum Axial Pull-out Resistance (Extreme)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
132.20	1853	OK	OK	*(No pull-out demand at bottom)*

Verification for Lateral Displacement

Displacement		C/D	Verification
Demand	Capacity		
mm	mm		
16.31	30	1.84	OK

Capacity =5% x Dpile. (See commentary BSDS pp 4-15)

under earthquake loading this value ... may reach as much as 5%

Verification for Tip Displacement/Settlement

Displacement		C/D	Verification
Demand	Allowable		
mm	mm		
10.41	25	2.40	OK

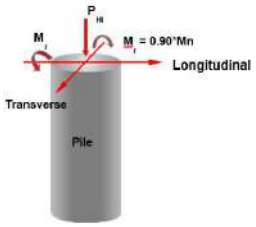
1.7.4 Verification of Pile Structural Resistance

Pmin with Mmax	
Rigid Pile Head	
P_u	M_u
kN	kN-m
-124	438

Note: The connection is rigid.

Pmax with Mmax	
Rigid Pile Head	
P_u	M_u
kN	kN-m
1223	440

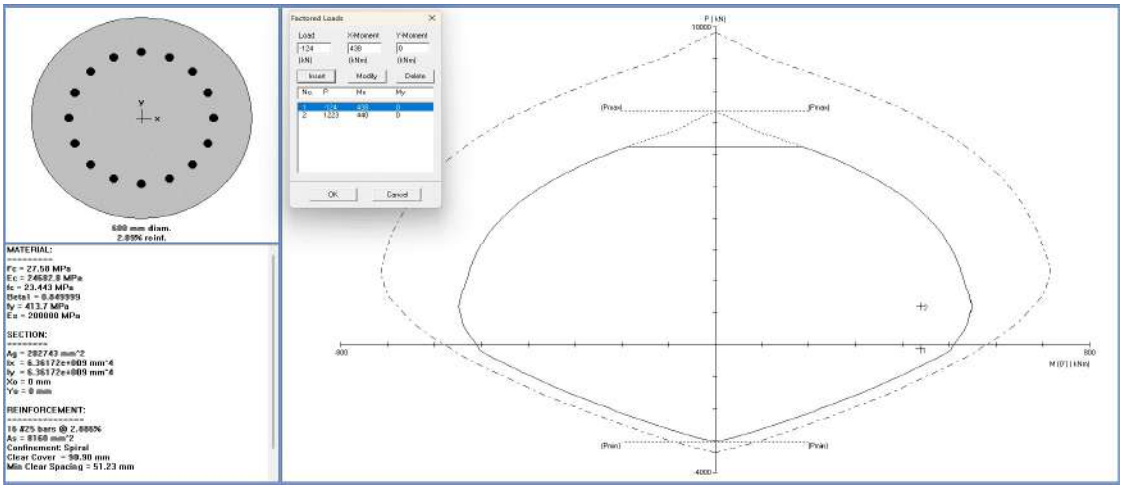
Note: The connection is rigid.



Define Reinforcement of Pile :

Diameter of Longitudinal Bars	25	mm
Diameter of Hoops/Spiral	16	mm
Reinforcement concrete cover to spiral/hoops	75	mm

SPColumn Axial and Flexure (P-M) Design of Pile



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	-124.00	438.00	0.00	505.25	0.00	1.154	188	496	0.00484	0.897
2	1223.00	440.00	0.00	548.25	0.00	1.246	260	496	0.00274	0.784

Verification of Flexure with Axial Load Resistance

P = -124 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
438.00	505.25	1.15	OK

P = 1223 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
440.00	549.25	1.25	OK

Calculate Pile Shear Design and Resistance

Shear Demand, Vu = 279 kN

The nominal shear resistance, Vn is lesser of the following:

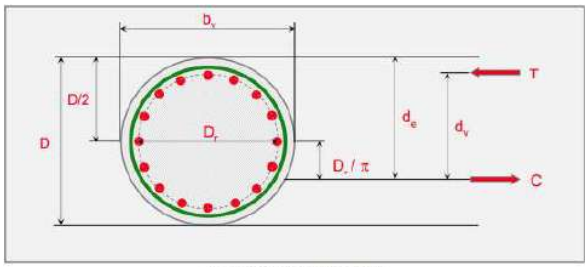
Vn = Vc + Vs + *Vp* OR Vn = 0.25f'c bv dv + *Vp*

where: Vc = 0.083*β*Sqrt(f'c)*bv*dv
Vs = [Av*fy*dv*(cotθ+cotα) sinα] / s
Vp = 0.00 for non-prestressed concrete
Vp = component in the direction of applied shear of effective prestressing force

Ultimate shear resistance, Vr = ØVn: Vr = ØVn > Vd_EQ where Ø= 0.9 for shear

Shear design parameters:

D	diameter of pile (out to out dimension)	600	mm
cc	concrete cover (to spiral or hoops)	75	mm
db	main bars or longitudinal bars	25	mm
ds	size of spirals	16	mm
f'c	concrete compressive strength	27.6	MPa
fy	yield strength of reinforcing steel	414	MPa
bv	effective web width, mm = diameter of column	600	mm
de	de = (D / 2) + (Dr / π)	417	mm
dv	effective shear depth = 0.90 x de	375	mm
β	factor indicating ability of diagonally cracked concrete to transmit tension	2.0	
θ	angle of inclination of diagonal compressive stresses	45	deg
α	angle of inclination of transverse reinforcement	90	deg
Dr	diameter of circle passing thru centers of the longitudinal reinf.	368	mm
Dc	diameter of circle measured ourside the diameter of spiral/hoops	450	mm
Ac	Area of core measured outside diameter of spiral/hoops	159043	mm^2
Ag	Gross area of section	282743.339	mm^2



: For definition of terms:

From above parameters:

ds	size of spirals	16	mm
Asp	area of shear reinf.	201	mm
s	Try spacing	75	mm
No	Try no. of legs	2	No. legs
Total area of shear reinf. Av = Av = No*(Asp)		402	mm^2

Definition of no. of legs

leg ⬭ leg

no. of spirals	No. of legs
single spiral	2 legs
bundle spiral	4 legs

cot θ = 1/tan 45° = 1
cot α = 1/tan 90° = 0
sin α = sin90° = 1

Vc = 0.00 kN *For conservative approach, assume Vc = 0.0, i.e. concrete has no shear strength.; If Pmin = negative value (tension), Vc = 0.0.*
Vp = 0.00 kN *For non prestressed concrete*
Vs = [Av*fy*dv*(cot θ + cot α)sin α]/s = **833.3** kN

Vn is lesser of the following:

Vn = Vc + Vs + Vp =	833.3	kN	**Governing shear capacity!**
Vn = 0.25f'cbvdv+Vp=	1554.3	kN	
Vd_EQ =	279	kN	Shear Demand
Vr =	**750**	**kN**	Ultimate Shear Resistance, ØVn
c/d =	2.69		Shear resistance is OK!

c. Verification of shear/transverse reinforcement for confinement at plastic hinges

DGCS 12.7.11.2

Length of column end region will be the largest of the following :

a.) 3D + Max M Loc. *(assume 5D)* = 4.80 m *Conservative, see pile moment diagram above*
b.)450 mm (18 inches) = 0.45 m
therefore, **Length** = **4.80 m**

d. Verification of minimum required transverse reinforcement

1. The greater of the following:
ρs1 = 0.12*(f'c/fy) = 0.00800 *AASHTO 5.11.4.1.4 @ Plastic Hinge Region*
Note: for circular shape:
ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) = 0.02333 **Governs!** *AASHTO 5.6.4.6 @ Column General Region*

2. Checking from provided confinement, where Av represent spiral leg on one (1) side

$$\rho_{sprovided} = \frac{4*Av}{Dc*s} = 0.02383$$ **OK!!**

Maximum spacing

1. S should not be greater than 1/4 min dimension of member (=D/4) **OK!!**
2. S should not be greater than 100mm **OK!!**

Minimum clear spacing

1. Sc should not be less than 25mm **OK!!**
2. Sc should not be less than 1.33 x aggregate size (1.33 x 20mm) **OK!!**

e. Transverse reinforcement for outside region

ds	size of spirals	16	mm
Asp	area of shear reinforcement	201	mm^2
s	Try spacing of transverse steel	150	mm
No	Try number of legs	2	No. legs
	Total area of shear reinf. Av = Av = No*(Asp)	402.12	mm^2

DGCS 12.5.2.4

Min area of transverse reinf. spacing :

Av = 0.083 x sqrt(f'c) x bv x s/fy = 94.79 mm^2 **OK!!**

f. Summary of shear/transverse reinforcement

Location	Range, m	Spacing/Pitch of spirals, mm	Diameter, mm	No. of legs
End regions	5.00	75	16	2
Outside region	39.00	150	16	2

Pile Details Summary at Abutment:

Use 6 - 600mm Dia. Bored Piles, L=43.5m, with 16-25mm dia. main bars, 16mm dia. spiral hoops spaced @ 75mm for the first 5m; then 16mm dia. spiral hoops spaced @ 150mm all throughout the rest of total length.

CALCULATION SHEET

DESIGN OF PIER 1 & 2

Project: Prop San Isidro - Eastern Branch 2 (SI-EB2) Bridge Proj. Location: Minalin, Pampanga
Designed by/Date: KH-RCDG 05/28/26

5.1 GENERAL DESIGN CONDITIONS & CRITERIA

5.1.1 Structural Conditions

One lane carriageway; total width= 5.50 m
3-15m continuous AASHTO girders-Type III
Bearing restraints: M-F-F-M (F=fixed, M=movable)
Regular Bridge (non-skewed bridge)
Pier type: one column on concrete bored piles foundation with pile cap
Abutment type: cantilever type on concrete bored piles

5.1.2 Seismic Design Requirements and Ground Conditions

Table 7.2-1 Seismic Design Requirements and Ground Conditions

Bridge Operational Classification=	OC-III
Earthquake Ground Motion=	Level 2
Ground Type=	3
Seismic Performace Level=	3
Seismic Performance Zone=	4
Peak Ground Accelaration=	0.6g

5.1.3 Site Factors

Site Factors	
Fpga=	0.783
Fa=	0.783
Fv=	2.267
As=	0.47

5.1.4 Hydrology & Hydraulic Data

50 years Return Period

Discharge:	4.38	m3/s	
Water Level, DFL:	5.60	m	
Velocity:	4.00	m/s	*(assumed, conservative)*
Freeboard :	1.50	m	
Drainage Area:	-	sqm	
Computed scour depth:	-	m	

5.1.5 Design Loads

1. Permanent Loads	
DC=	Dead load pertaining to structural and non structural components
DW=	Dead Load pertaining to future wearing surface
EH=	Horizontal earth pressure
ES=	Earth surcharge
EV=	Vertical pressure from earth fill

2.Transient Loads	
EQ=	Earthquake Load
LL/IM=	Vehicular Load/Impact load
LS=	Liveload surcharge
WA=	Water Load
FR=	Friction Load
BF=	Braking Force Load

For Seismic load analysis: Refer to BSDS
Load Combinations and factors: Refer to DGCS 10.0

5.1.6 Material and Soil Property

Conc. Compressive strength @ 28 days, f'c	27.6	MPa
Reinforcing steel (ASTM 615), fy	414	MPa
Unit Weight of concrete, Yc	24	kN/m^3
Unit Weight of soil, Ysoil	19	kN/m^3
Unit Weight of rock, Yr	20	kN/m^3

5.2 LOAD CALCULATIONS

5.2.1 Seismic Loads

Obtain the elastic seismic forces of column at its base due to earthquake loadings in longitudinal (Long'l EQ) and transverse (Trans. EQ) directions from the bridge seismic analysis.

Loads = Long'l EQ + 50% LL
Trans. EQ + 50% LL

LC1 = 1.0*Long'l EQ + 0.3*Trans. EQ
LC2 = 1.0*Trans. EQ + 0.3*Long'l EQ

Calculate the orthogonal forces and inelastic hinging effect with Modification Factor, R.

Modified elastic forces(LC moment /R)

STAADPro Default Notations

Orthogonal Direction	Global Axis	**Column** Principal Local Axis Forces		**Beam** Principal Local Axis Forces		Column/Beam Local Axis Force
		Shear	Moment	Shear	Moment	Axial
Longitudinal	X-Dir.	Fy	Mz	Fz	My	Fx
Transverse	Z-Dir.	Fz	My	Fy	Mz	
Vertical	Y-Dir.					

(From Staad Bridge EQ Analysis)

Pier 1/2 - Column 1 *Shear and Axial Forces unit is kN; Moment unit is kN-m*

Loadings	**COLUMN BOTTOM**				
	Fx	**Fy**	**Fz**	**My**	**Mz**
Longitudinal EQx	103.2	544.2	0.0	0.0	5033.5
Transverse EQz	0.0	0.0	658.4	5032.2	0.0
EQ1 = 1.0EQx + 0.30EQz	103.2	544.2	197.5	1509.7	5033.5
EQ2 = 1.0EQz + 0.30EQx	31.0	163.3	658.4	5032.2	1510.1
EQ1 moment / R	**103.2**	**544.2**	**197.5**	**503.2**	**1677.8**
EQ2 moment / R	**31.0**	**163.3**	**658.4**	**1677.4**	**503.4**

Response Modification Factor, R = 3

Note: For regular/normal bridges, forces are absolute values that is without regards to its sign, beacause the seismic load can act in either direction.

BSDS 3.8: RESPONSE MODIFICATION FACTOR, R

Table 3.8.1-1 Response Modification Factors – Substructures

Substructure	**Operational Category**		
	OC-I (Critical)	**OC-II (Essential)**	**OC-III (Others)**
Wall-type piers – larger dimension	1.5	1.5	2.0
Reinforced concrete pile bents			
• Vertical piles only	1.5	2.0	3.0
• With batter piles	1.5	1.5	2.0
Single columns	1.5	2.0	3.0
Steel or composite steel and concrete pile bents			
• Vertical piles only	1.5	3.5	5.0
• With batter piles	1.5	2.0	3.0
Multiple column bents	1.5	3.5	5.0

5.2.2 Permanent Loads

a. Superstructure

DC_super = R_{DL} = Deadload reaction to Pier 1 (top of pier cap/coping beam) from superstructure permanent loads (including diaphragms, sidewalks, railings, slab, girders, etc., except future wearing surface).

DC_G1 = 394.35 kN *(From rxn of DC_DL from bridge StaadPro model)*
DC_G2 = 349.88 kN *(at link below girders)*
DC_G3 = 394.35 kN

DC_super = **1138.574** kN

DW = future wearing surface and utilities

DW_G1 = 57.83 kN *(From rxn of DW_DL from bridge StaadPro model)*
DW_G2 = 50.89 kN *(at link below girders)*
DW_G3 = 57.83 kN

DW = **166.544** kN

Superimposed dead loads added to bridge StaadPro model:

SDL for whole bridge: *(See Design of PSCG Girder Calculations)*

2x	w.railings =	1.920	kN/m
2x	w.railpost =	1.060	kN/m
2x	w.sidewalk =	0.000	kN/m
whole	w.fws =	8.800	kN/m
+10%	w.utilities =	1.178	kN/m
	Total SDL =	12.958	kN/m
	SDL per girder =	4.319	kN/m

No. of girders, Ng = 3

Staad model
edge plate width = 1.00 m x2
total inner plates width = 3.50 m
Total super struct. width = 5.50 m

DC_	SDL load to sidewalk edge plate area =	1.490	kPa	say =	*1.50* kPa
	(w.railings+w.railpost+w.sidewalk)/2 * (1/edge plate width) =				
DW_	SDL load to whole plate area =	1.814	kPa	say =	*1.85* kPa
	(w.fws+w.utilties)/total super struct. width =				

b. Substructure

pier cap/coping beam: unit wt. of concrete = 24.00 kN/m3

DC_cap = **269.3** kN

h.cap = 1.20 m
b.cap = 1.70 m
l.cap = 5.50 m

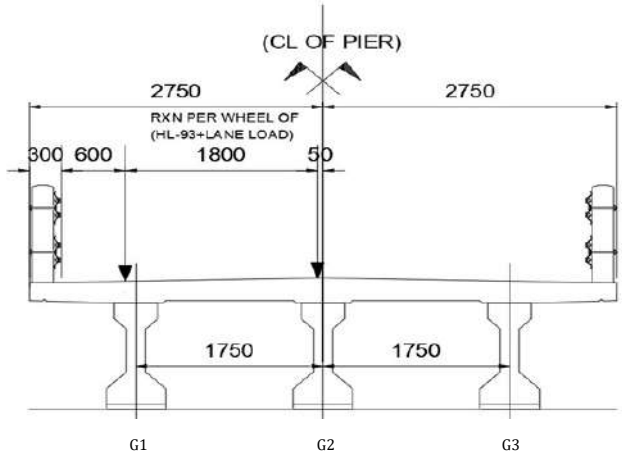
pier column/s:

DC_col = **252.4** kN

D.col = 1.20 m
H.col = 9.30 m
No.col = 1

5.2.3 Vehicular Loads

a. Vehicular Truck + Lane Loads



Live Load Position For Pier 1 Reactions that will produce the worst effect to pier.

From live load (continuous beam) analysis, the reaction of HL93 =	306.08	kN
the reaction of Lane Load =	215.93	kN
Reaction per wheel line HL 93 =	153.0405	kN
Reaction per wheel line Lane load =	107.966	kN
Impact (not to be applied to lane load and footing), IM =	1.33	
Multiple presence factor, m on 1-lane =	1.20	
Multiple presence factor, m on 2-lanes =	1.00	
Number of lanes =	1	
Reaction per wheel line HL 93*IM*m =	**244.25**	kN
Reaction per wheel line Lane load*m =	**129.5592**	kN
	373.81	kN
Number of girders =	3	
Spacing of girders =	1.75	m
Total road width =	5.00	m
distance of second wheel to G2 =	0.05	m

	Reaction to G1 (kN)	Reaction to G2 (kN)	Reaction to G3 (kN)
first wheel	373.81		
second wheel	10.68	363.13	
Total =	**384.49**	**363.13**	**0.00**

b. Vehicular Braking Force, BF

Note: The total braking force is computed based on the number of design lanes in the same direction. Also, braking forces are not increased for dynamic load allowance in accordance with LRFD [3.6.2.1]. The calculation of the braking force for a single traffic lane follows:

Braking force shall be taken as the greater of:
25% (HL93) = 81.25 kN
25% (Tandem) = 55.00 kN

5% (HL93 + Lane Load) = 37.27 kN
5% (Tandem + Lane Load) = 32.02 kN
Lane Load = Design Lane Load x Total Length of Bridge

Therefore, horizontal force BF = 97.50 kN
(max of BF x no. lanes x m)

HL-93 (Design Truck) = 145kN+145kN+35kN = 325 kN
Design Tandem = 110kN+110kN = 220 kN
Design Lane Load = 9.34 kN/m
Total Length of Bridge = 45 m

LRFD 3.6.4 states that the braking force is applied along the longitudinal axis of the bridge at a distance of six feet above the roadway surface. Bearings are assumed incapable of transmitting longitudinal moment, the braking force will be applied at the c.g. of bentcap.

BF along span of bent cap = **17.73** kN/m
Uniform moment along
span of bend cap due to BF = **10.64** kN-m/m
(acting at c.g. of bentcap)
(has bearings so lever arm is upto top of bentcap only)

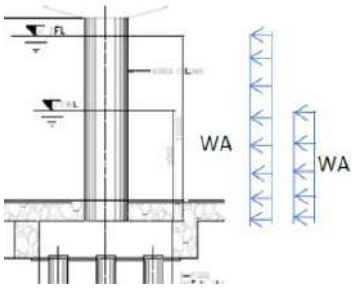
Length of Bent Cap = 5.50 m
Height of Bent Cap = 1.20 m

5.2.4 Stream Flow, WA

a. During Ordinary Water Level (OWL)

Uniform distributed pressure, $P = 5.14 \times 10^{-4} C_d V^2$
Drag coeff., C_d = 0.7 for circular shape
Velocity of water, V = 4.00 m/s *(refer to hydraulic report)*
H_{OWL} = 4.61 m height of ordinary water level
P = 0.00576 Mpa
D_{col} = 1.20 m A.col = 1.13 m^2
WA = P x D_{col} = **6.91** kN/m

H_sub = 4.61 m
V_trans = 31.85 kN
Yw = 10 kN/m^3
Bouyant force, BF = **-52.138** kN *(per column)*
(A.col x H_sub x Yw)



b. During Design Flood Level (DFL) 50 year period

Uniform distributed pressure, $P = 5.14 \times 10^{-4} C_d V^2$
Drag coeff., C_d = 0.7 for circular shape
Velocity of water, V = 4.00 m/s *(refer to hydraulic report)*
H_{DFL} = 9.02 m height of design flood level
P = 0.00576 Mpa
WA = P x D_{col} = **6.91** kN/m

H_sub = 9.02 m
V_trans = 62.31 kN

Bouyant force, BF = **-102.01** kN *(per column)*
(A.col x H_sub x Yw)

5.2.5 Summary of Unfactored Loads/Forces on Column

(From Staad Pier Bent Analysis and Staad Bridge EQ Analysis)
Pier 1 - Column 1 *Shear and Axial Forces unit is kN; Moment unit is kN-m*

Loadings	COLUMN BOTTOM				
	Fx	Fy	Fz	My	Mz
DC_DL	1610.1	0.7	0.0	0.0	7.8
DW_DL	166.0	0.0	0.0	0.0	0.1
LL+IM	747.6	0.0	0.0	672.9	0.0
BF	0.0	97.5	0.0	0.0	1023.9
WA_owl	-52.1	0.0	31.5	71.8	0.0
WA_dfl	-102.0	0.0	62.2	279.9	0.0
EQ1 mom. / R	103.2	544.2	197.5	503.2	1677.8
EQ2 mom. / R	31.0	163.3	658.4	1677.4	503.4

5.2.6 Load combinations and load factors

From DGCS Vol. 5 Bridges

Table 10.3-1 Load Combination and Load Factors

Load Combination / Limit State	DC DD DW EH EV ES EL PS CR SH	LL IM CE BR PL LS	WA	WS	FR	TU	TG	SE	Use one of these at a time: EQ	BL	CT	CV
STRENGTH-I (Unless noted)	γ_p	1.75	1.00	-	1.00	0.50/1.20	0.0	γ_{SE}	-	-	-	-
STRENGTH-II	γ_p	1.35	1.00	-	1.00	0.50/1.20	0.0	γ_{SE}	-	-	-	-
STRENGTH-III	γ_p	1.35	1.00	-	1.00	0.50/1.20	0.0	γ_{SE}	-	-	-	-
STRENGTH-III	γ_p	-	1.00	1.4	1.00	0.50/1.20	0.0	γ_{SE}	-	-	-	-
STRENGTH-IV	γ_p	-	1.00	-	1.00	0.50/1.20	-	-	-	-	-	-
EH, EV,ES,DW,	1.5											
DC ONLY												
STRENGTH-V	γ_p	1.35	1.00	0.40	1.00	0.50/1.20	0.0	γ_{SE}	-	-	-	-
EXTREME EVENT - I	γ_p	γ_{EQ}	1.00	-	1.00	-	-	-	1.00	-	-	-
EVENT - I												
EXTREME EVENT - II	γ_p	0.5	1.00	-	1.00	-	-	-	-	1.00	1.00	1.00
EVENT - II												
SERVICE - I	1.00	1.00	1.00	0.30	1.00	1.00/1.20	0.0	γ_{SE}	-	-	-	-
SERVICE - II	1.00	1.3	1.00	-	1.00	1.00/1.20	-	-	-		-	-
SERVICE - III	1.00	0.8	1.00	-	1.00	1.00/1.20	0.0	γ_{SE}	-	-	-	-
SERVICE - IV	1.00	-	1.00	0.70	1.00	1.00/1.20	-	1.0	-	-	-	-
FATIGUE – I LL, IM, & CE ONLY	-	1.50	-	-	-	-	-	-	-	-	-	-
FATIGUE – II LL, IM, & CE ONLY	-	0.75	-	-	-	-	-	-	-	-	-	-

Table 10.3-2 Load Factors for Permanent Loads, γ_p

Type of Load	Load Factor: Max	Min
DC: Component and Attachments	1.25	0.90
DD: Downdrag	1.80	0.45
DW: Wearing Surfaces and Utilities	1.50	0.65
EH: Horizontal Earth Pressure		
Active	1.50	0.90
At-Rest	1.35	0.90
EL: Locked-in Erection Stresses	1.00	1.00
EV: Vertical Earth Pressure		
Retaining Walls and Abutments	1.35	1.00
Rigid Buried Structure	1.30	0.90
Rigid Frames	1.35	0.90
Flexible Buried Structures other than Metal Box Culverts	1.95	0.90
Flexible Metal Box Culverts	1.50	0.90
ES: Earth Surcharge	1.50	0.75

Seismic load can act in either direction, EQ negative axial force Fx (tension) will produce the critical Pmin among the load combinations, therefore load modifiers / factors that produce Pmin may not be analyzed.

5.2.7 Summary of Factored and Load Combinations on Column

Pier 1 - Column 1 *Shear and Axial Forces unit is kN; Moment unit is kN-m*

Loadings		COLUMN BOTTOM Fx	Fy	Fz	My	Mz
STRENGTH I		3468	172	62	1457	1802
EXTREME Ia (EQ1)	P_{max}	2686	594	229	911	2200
	P_{min}	2480				
EXTREME Ib (EQ2)	P_{max}	2614	213	690	2086	1025
	P_{min}	2552				
SERVICE I		2472	98	32	745	1032

STRENGTH I = 1.25DC + 1.50DW + 1.75[(LL+IM)+BF] + 1.00WA_dfl
EXTREME I = 1.25DC + 1.50DW + 0.50[(LL+IM)+BF] + 1.00WA_owl + 1.00EQ
SERVICE I = 1.00DC + 1.00DW + 1.00[(LL+IM)+BF] + 1.00WA_owl

Governing Column Design Forces (Extreme Event I):

Longitudinal Direction (EQ1)

$P_{max}L$ = 2686 kN
$P_{min}L$ = 2480 kN
VyL = 594 kN
VzL = 229 kN
MyL = 911 kN-m
MzL = 2200 kN-m

Transverse Direction (EQ2)

$P_{max}T$ = 2614 kN
$P_{min}T$ = 2552 kN
VyT = 213 kN
VzT = 690 kN
MyT = 2086 kN-m
MzT = 1025 kN-m

5.2.8 Verification of Slenderness Effect (Moment Magnification Factor)

AASHTO LRFD BRIDGE DESIGN SPECIFICATIONS

Table C4.6.2.5-1—Effective Length Factors, *K*

	(a)	(b)	(c)	(d)	(e)	(f)
Buckled shape of column is shown by dashed line						
Theoretical *K* value	0.5	0.7	1.0	1.0	2.0	2.0
Design value of *K* when ideal conditions are approximated	0.65	0.80	1.0	1.2	2.1	2.0
End condition code		Rotation fixed Rotation free Rotation fixed Rotation free		Translation fixed Translation fixed Translation free Translation free		

if kLu/r>=22 compute magnification factor

if kLu/r < 22 neglect slendersness effects

if kLu/r<=100 proceed to approximate evaluation of slenderness effect

Direction	LONG'L	TRANSV.	
K	2.1	2.1	
Unsupported length, Lu, m.	9.30	9.30	
Diameter, D, m.	1.20		
Mom. Of Inertia, Ig = $\pi D^4/64$, m^4	0.1018		
Area, Ag = $\pi D^2/4$, m^2	1.1310		
Radius of gyration, r = sqrt(Ig/Ag), m	0.30		
kLu/r	65.10	65.10	
Modulus of elasticity, Ec, kPa	27,350,095		
Cm	1		for all cases
Reduction factor, ϕ_k	0.75		for concrete
β_d	0		for EQ loads
EcIg = (EcIg/2.5)/(1+βd), kN-m^2	1113560.23		
Pc = π^2EcIg/$(kLu)^2$, kN	28814.36	28814.36	
Pu, kN	2686	2614	Pmax
Mom. Magnification factor, δ_L & δ_T δ = Cm/(1-(Pu/(ϕkPc)) >= 1	**1.142**	**1.138**	

Modified Column Design Forces:

Longitudinal Direction (EQ1)

$P_{max}L$ =	2686	kN
$P_{min}L$ =	2480	kN
VyL =	594	kN
VzL =	229	kN
MyL(δ_L) =	**1041**	kN-m
MzL(δ_L) =	**2512**	kN-m

Transverse Direction (EQ2)

$P_{max}T$ =	2614	kN
$P_{min}T$ =	2552	kN
VyT =	213	kN
VzT =	690	kN
MyT(δ_T) =	**2373**	kN-m
MzT(δ_T) =	**1166**	kN-m

5.3 VERIFICATION OF COLUMN FLEXURAL RESISTANCE AND DISPLACEMENT

5.3.1 Verification of column resistance

Design Column Forces:

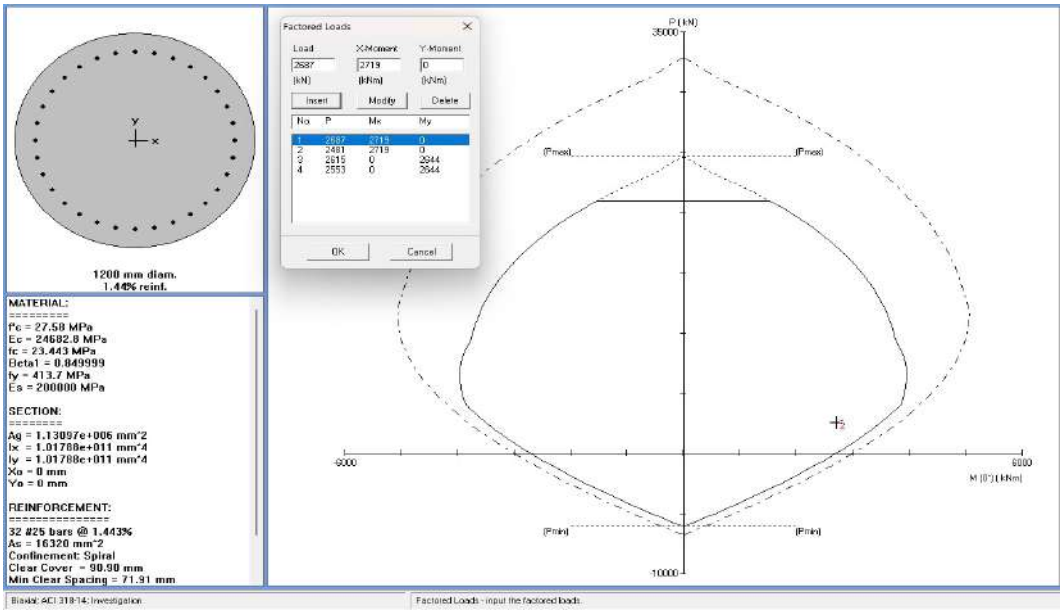
Calculate the combined seismic design forces considering the bi-axial response (for circular column).

$V = sqrt(Vy^2 + Vz^2)$

$M = sqrt(My^2 + Mz^2)$

Longitudinal Direction (EQ1)			**Transverse Direction (EQ2)**		
$P_{max}L$ =	2686	kN	$P_{max}T$ =	2614	kN
$P_{min}L$ =	2480	kN	$P_{min}T$ =	2552	kN
Vdes_L =	637	kN	Vdes_T =	722	kN
Mdes_L =	2719	kN-m	Mdes_T =	2644	kN-m

From spColumn Interaction Diagram Investigation:



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	2687.00	2719.00	0.00	3534.71	0.00	1.300	364	1096	0.00605	0.900
2	2481.00	2719.00	0.00	3481.36	0.00	1.280	357	1096	0.00622	0.900
3	2615.00	0.00	2644.00	0.00	3516.13	1.330	361	1096	0.00611	0.900
4	2553.00	0.00	2644.00	0.00	3500.07	1.324	359	1096	0.00616	0.900

Design Section and Reinforcement:

Dcol =	1200	mm		
Longitudinal Bars =	32	bars -	25	mm Ø
ρs =	1.39%			

Design Flexural Capacities and C/D Ratio:

ϕ_L =	0.9		ϕ_T =	0.9	
ϕMn_L =	3481	kN-m	ϕMn_T =	3500	kN-m
Mn_L =	3868	kN-m	Mn_T =	3889	kN-m
ϕMn_L/Mdes_L =	1.28	**OK!**	ϕMn_T/Mdes_T =	1.32	**OK!**

Verify min. and max longitudinal reinf.

ZONE 1	No seismic requirement
ZONE 2	0.01 < ρs < 0.06
ZONE 3 and 4	0.01 < ρs < 0.04

For Zone 3 and 4:

ρ_{s_min} =	0.01	**OK with min. reinf.!**
ρ_{s_max} =	0.04	**OK with max. reinf.!**

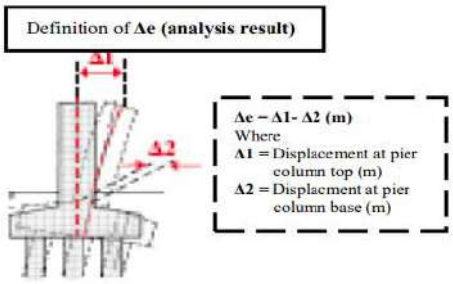
5.3.2 Verification of column displacement

a. Allowable displacement requirement:

$\Delta_{allow} = 0.25\ \varphi Mn / P_{udL}$
where: P_{udL} = factored axial dead load on top of column
$P_{udL} = 1.25DC_DL + 1.5DW_DL$ *(See Unfactored Fx of Column Top)*

P_{udL} = 2261.6 kN

$\Delta_{allow}L$ = 385 mm $\Delta_{allow}T$ = 387 mm



b. Actual design displacement:

$\Delta_{actual} = Rd * \Delta e$

Δe - displacement calculated from elastic seismic analysis, mm

Tlong =	0.68107	sec	*Mode 2 period from Staad EQ Analysis*
Ttrans =	0.69751	sec	*Mode 1 period from Staad EQ Analysis*
Ts =	1.157	sec	*From Design response spectrum analysis*
R =	3		*Modification factor BSDS 3.8*

1.25Ts = 1.446 sec

Rd = (1-1/R)*(1.25*Ts/T)+(1/R), if T<1.25*Ts
Rd = 1 , if T>=1.25*Ts

RdL = 1.75 RdT = 1.72

From bridge seismic analysis due to mass or dead load only the displacement of top and bottom of column are as follows:

Longitudinal Direction (LC1):		
Δtop = Δ1 =	128	mm
Δbot = Δ2 =	10	mm
ΔeL = Δtop - Δbot	118	mm

Transverse Direction (LC2):		
Δtop = Δ1 =	125	mm
Δbot = Δ2 =	11	mm
ΔeT = Δtop - Δbot	114	mm

$\Delta_{actual}L$ = RdL*ΔeL =	206	mm
$\Delta_{allow}L$ =	385	mm
c/d =	1.86	
OK in long'l. displacement!		

$\Delta_{actual}T$ = RdT*ΔeT =	195.852	mm
$\Delta_{allow}T$ =	387	mm
c/d =	1.98	
OK in transv. displacement!		

Therefore use:

Column diameter = 1200 mm
Vertical Bars = 32 bars - 25 mm Ø
ρs = 1.39%

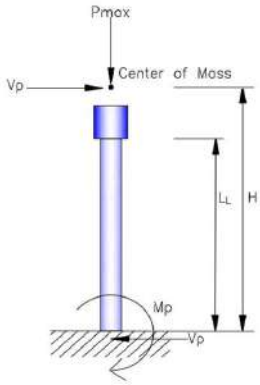
5.4 PIER COLUMN PLASTIC HINGING ANALYSIS

a. Transverse Direction:

$P_{max}T$ =	2614	kN	*at LC2/EQ2*
Mn =	3828	kN-m	*Nominal moment at Pmax*
Mp = ø Mn			
Mp_T =	4976.4	kN-m	*Plastic moment at Pmax*
h =	11.43	m	*Height of column fixity to c.g of superstructure*

Vp =	Mp / h		
Vp_T =	435	kN	*Corresponding plastic shear*

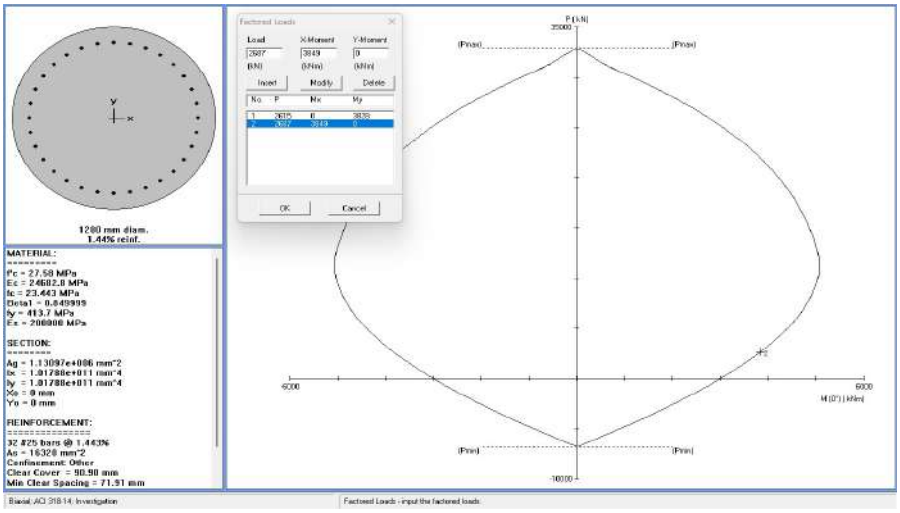
h =	11.427	m (from c.g. of superstructure)
L =	9.300	m (from top of col./bottom of coping)
ø =	1.3	plastic moment factor
M_p =	1.3 M_n	



b. Longitudinal Direction:

$P_{max}L$ =	2686	kN	*at LC1/EQ1*
Mn =	3849	kN-m	*Nominal moment at Pmax*
Mp = ø Mn			
Mp_L =	5003.7	kN-m	*Plastic moment at Pmax*
h =	11.43	m	*Height of column fixity to c.g of superstructure*

Vp =	Mp / h		
Vp_L =	438	kN	*Corresponding plastic shear*



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	2615.00	3828.00	0.00	3828.83	0.00	1.000	352	1096	0.00634	1.000
2	2687.00	0.00	3849.00	0.00	3849.51	1.000	354	1096	0.00628	1.000

5.5 VERIFICATION OF COLUMN SHEAR RESISTANCE

a. Column Design Shear Demand

	Longitudinal Direction:		Transverse Direction:		Design Force:		
Vdes_L =	637	kN	Vdes_T =	722 kN	Vdes =	722 kN	<< Elastic Analysis
Vp_L =	438	kN	Vp_T =	435 kN	Vp =	438 kN	<< Plastic Hinging (Inelastic) Analysis
					Vd_EQ =	**438 kN**	**Plastic Hinging Forces Governs!**

BSDS 5.3.4 ..the lesser of plastic hinging forces and elastic seismic forces governs.

b. Calculate the Shear Resistance

The nominal shear resistance, Vn is lesser of the following:

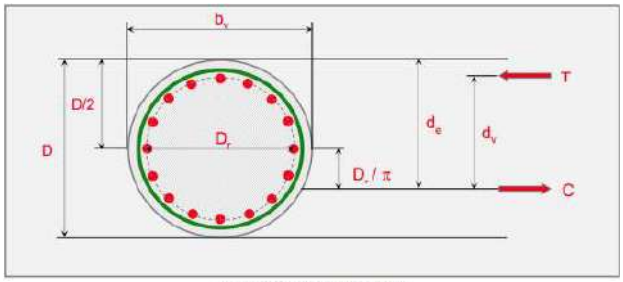
Vn = Vc + Vs + *Vp* OR Vn = 0.25f'c bv dv + *Vp*

where: Vc = 0.083*β*Sqrt(f'c)*bv*dv
Vs = [Av*fy*dv*(cotθ+cotα) sinα] / s
Vp = 0.00 for non-prestressed concrete
Vp = component in the direction of applied shear of effective prestressing force

Ultimate shear resistance, Vr = ØVn: Vr = ØVn > Vd_EQ where Ø= 0.9 for shear

Shear design parameters:

D	diameter of pier column (out to out dimension)	1200	mm
cc	concrete cover (to spiral or hoops)	75	mm
db	main bars or longitudinal bars	25	mm
ds	size of spirals	16	mm
f'c	concrete compressive strength	27.6	MPa
fy	yield strength of reinforcing steel	414	MPa
bv	effective web width, mm = diameter of column	1200	mm
de	de = (D / 2) + (Dr / π)	916	mm
dv	effective shear depth = 0.90 x de	824	mm
β	factor indicating ability of diagonally cracked concrete to transmit tension	2.0	
θ	angle of inclination of diagonal compressive stresses	45	deg
α	angle of inclination of transverse reinforcement	90	deg
Dr	diameter of circle passing thru centers of the longitudinal reinf.	993	mm
Dc	diameter of circle measured ourside the diameter of spiral/hoops	1050	mm
Ac	Area of core measured outside diameter of spiral/hopps	865901	mm^2
Ag	Gross area of section	1130973	mm^2



: For definition of terms:

From above parameters:

ds	size of spirals	16	mm
Asp	area of shear reinf.	201	mm
s	Try spacing	100	mm
No	Try no. of legs	4	No. legs
Total area of shear reinf. Av = Av = No*(Asp)		804	mm^2

Definition of no. of legs

leg ⬭ leg

no. of spirals	No. of legs
single spiral	2 legs
bundle spiral	4 legs

cot θ = 1/tan 45° = 1
cot α = 1/tan 90° = 0
sin α = sin90° = 1

Vc = 0.00 kN *For conservative approach, assume Vc = 0.0, i.e. concrete has no shear strength.; If Pmin = negative value (tension), Vc = 0.0.*
Vp = 0.00 kN *For non prestressed concrete*
Vs = [Av*fy*dv*(cot θ + cot α)sin α]/s = **2745.2** kN

Vn is lesser of the following:

Vn = Vc + Vs + Vp =	2745.2	kN	**Governing shear capacity!**
Vn = 0.25f'cbvdv+Vp=	6826.6	kN	
Vd_EQ =	438	kN	Shear Demand
Vr =	**2471**	**kN**	Ultimate Shear Resistance, ØVn
c/d =	5.64		Shear resistance is OK!

c. Verification of shear/transverse reinforcement for confinement at plastic hinges

DGCS 12.7.11.2

Length of column end region will be the largest of the following :

a.) Max. cross-sectional dimensic 1.20 m

b.) 1/6 clear height = 1.55 m clear height = 9.30 m

c.)450 mm (18 inches) = 0.45 m

therefore, **Length** = **1.55 m**

d. Verification of minimum required transverse reinforcement

1. The greater of the following:

ρs1 = 0.12*(f'c/fy) = 0.00800 *AASHTO 5.11.4.1.4 @ Plastic Hinge Region*

Note: for circular shape:

ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) = 0.00918 **Governs!** *AASHTO 5.6.4.6 @ Column General Region*

2. Checking from provided confinement, where As represent spiral leg on one (1) side or the bars itself only

$$\rho_{sprovided} = \frac{4*As}{Dc*s} = 0.01532$$ **OK!!**

Maximum spacing

1. S should not be greater than 1/4 min dimension of member (=D/4) **OK!!**
2. S should not be greater than 100mm **OK!!**

Minimum clear spacing 68 mm *Actual clear spacing*

1. Sc should not be less than 25mm **OK!!**
2. Sc should not be less than 1.33 x aggregate size (1.33 x 20mm) **OK!!**

e. Transverse reinforcement for outside region

ds	size of spirals	16	mm
Asp	area of shear reinforcement	201	mm^2
s	Try spacing of transverse steel	150	mm
No	Try number of legs	4	No. legs
	Total area of shear reinf. Av = Av = No*(Asp)	804	mm^2

DGCS 12.5.2.4

Min area of transverse reinf. spacing :

Av = 0.083 x sqrt(f'c) x bv x s/fy = 189.585 mm^2 **OK!!**

Note: for circular shape:

ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) = 0.00918 *AASHTO 5.6.4.6 @ Column General Region*

Checking from provided confinement, where As represent spiral leg on one (1) side or the bars itself only

$$\rho_{sprov.Out} = \frac{4*As}{Dc*s} = 0.01021$$ **OK!!**

f. Summary of shear/transverse reinforcement

Location	Range, m	Spacing/Pitch of spirals, mm	Diameter, mm	No. of legs
End regions	1.55	100	16	4
Outside region	6.20	150	16	4

5.6 PIER BENT CAP/COPING BEAM DESIGN

Note: 1. Forces from elastic analysis govern.
2. If forces from plastic hinging govern, revise design calculations accordingly.

hcg =	11.43	m	Height of column fixity to c.g of superstructure
H =	9.90	m	Height of column from c.g. of coping beam
Lt =	5.50	m	Length of coping beam

a. Shear forces (Longitudinal Direction-EQ1):

From summary of unfactored loads at bottom of column.

X-Dir :

Vpier_L = 438 kN Total base shear

Equivalent uniform shear force due elastic base shear:
$V_{E_}L$ = 79.62 kN/m = Vpier_L / Lt

Equivalent uniform moment force due elastic base shear:
$M_{E_}L$ = 121.57 kN-m/m = Vpier_L (hcg - H) / Lt

b. Shear forces (Transverse Direction-EQ2):

Z-Dir :

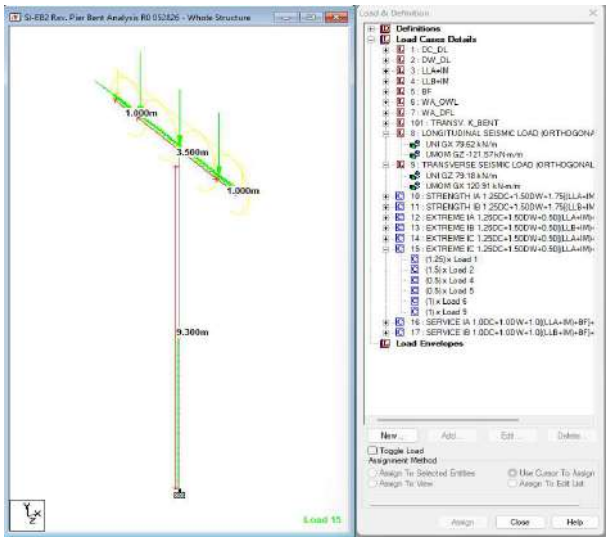
Vpier_T = 435 kN Total base shear

Equivalent uniform shear force due elastic base shear:
$V_{E_}T$ = 79.18 kN/m = Vpier_T / Lt

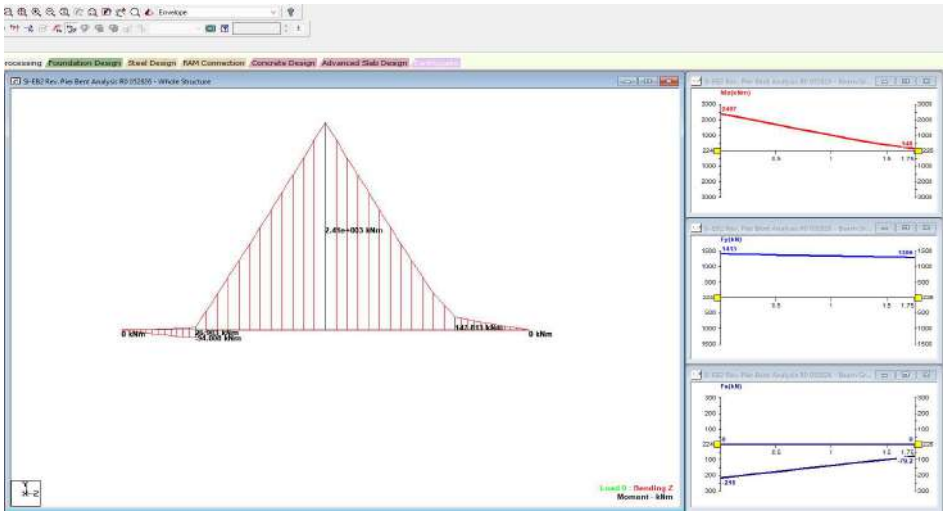
Equivalent uniform moment force due elastic base shear:
$M_{E_}T$ = 120.91 kN-m/m = Vpier_L (hcg - H) / Lt

c. Computer Modeling and Loading (StaadPro)

unit wt. of concrete =	24.00	kN/m3			
h.cap =	1.20	m	D.col =	1.20	m
b.cap =	1.70	m	H.col =	9.30	m
l.cap =	5.50	m	No.col =	1	



d. Design for Top Reinforcement (Negative Moment)





Maximum design negative moment
(at face of column):
M.u_neg = 1638.58 kN-m

Maximum design shear
(at face of column):
V.u = 1379.79 kN

Demand moment, M.u_neg	**1638.58**	**kN-m**	
Height of coping beam, h.cap	1200	mm	
Width of coping beam, b.cap	1700	mm	
Concrete cover, cc	115	mm	*(See Pier Diaphragm Details)*
Diameter of reinforcing bar, db	25	mm	Ab = 490.87 mm2
Diameter of stirrups, ds	16	mm	
Effective depth of concrete, de	1044	mm	
Conc. Compressive strength, f'c	27.6	MPa	
Reinforcing steel, fy	414	MPa	
Flexural Resistance factor, Φ	0.9		

Steel ratio = 0.85

β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ	m1 =	0.0567		m1=0.85*f'c/fy
	m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)
	Rn =	0.983	MPa	$Rn = \frac{Md}{\emptyset bv * (d_e)^2}$
	ρ=	0.0024		ρ=m1*(1-sqrt(1-m2Rn))

Area of main reinforcement required, As = ρbd	As =	4304.47	mm2				
Number of required top bars, As/Ab	n =	8.77		**Say.**	**18**	**bars -**	**25** **mm Ø Top Bars**

Area of main reinforcement provided, Asp = nAb	Asp =	8835.73	mm2	
a = (Asp fy) / (0.85 f'c b)	a =	91.7204	mm2	*Depth of compression block*
Flexural resistance, Mr = Φ [Asp fy (de - a/2)]	**Mr =**	**3286.07**	**kN-m**	
	c/d =	2.01	**OK!**	

Check for maximum reinforcement limit:

"The maximum reinforcement requirements ensure the section has sufficient ductility and is not overreinforced"

Compression fiber to neutral axis, c = a/β	c =	107.906	mm2
Maximum reinforcement ratio c/de =	max c/de =	0.420	
	actual c/de =	0.103	**OK!**

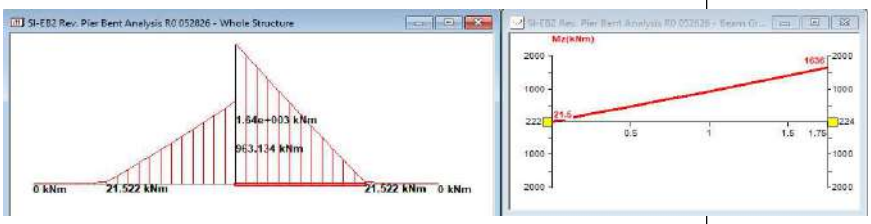
Check for minimum reinforcement limit:

"The minimum reinforcement requirements ensure the moment capacity provided is at least 1.2 times greater than the cracking moment"

Flexural cracking variability factor, *Y1*	1.6	*for all other concrete*		
Ratio of specified min. to ult. tensile strength of steel, *Y3*	0.67	*for A615, 414MPa steel*		
Modulus of rupture, f_r	3.310	MPa	fr=0.63*sqrt(f'c)	
Section modulus, S_c	4.1E+08	mm^3	Sc=I/c	
Cracking Moment, M_{cr}				
$M_{cr}=\gamma_3(\gamma_1*fr)\ S_c$	1447.60	kN-m		
1.20 Mcr =	1737.13	kN-m		
1.33*M.u_neg =	2.2E+03	kN-m		
Minimum design moment, Min(1.2Mcr or 1.33Mu)	1737.13	kN-m		
Flexural resistance provided, Mr	3286.1	kN-m	**OK!**	

Control of cracking by distribution of reinforcement:

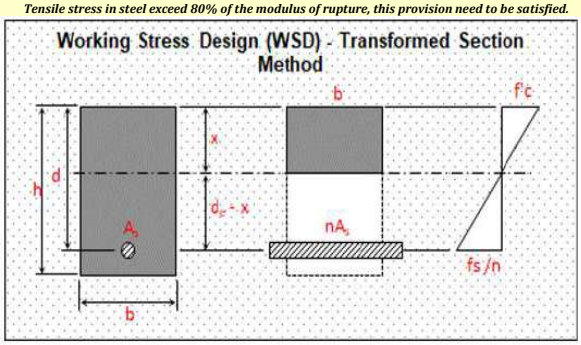
• DGCS 12.4.3.4



Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1 *(adopt max neg or pos moment), Ms=*	1636.00	kN-m	• DGCS 12.1.1.6
80% of Modulus of rupture, fr = 0.52*sqrt(f'c)	2.185	MPa	
Tension in the cross section, fs = Ms/Sc	4.010	MPa	

Tensile stress in steel exceed 80% of the modulus of rupture, this provision need to be satisfied.



b*x*x/2 = nAs*(de-x)
bx^2/2=nAsde-nAsx
bx^2/2+nAsx-nAsde=0
a = b/2
b = nAs
c = -nAs de

Extreme tension fiber to center of flexural reinforcement, d_c	143.5	mm
Overall thickness of component, h	1200	mm
Compression fiber to the centroid of extreme tension steel, d_e	1044	mm
Neutral axis to extreme compression fiber, x	257.36	mm

• by quadratic equation determine x:

a=	850.00
b=	7.2E+04

Modulus elasticity of steel, E_s	200	Gpa	c=	-7.5E+07
Modulus elasticity of concrete, E_c	24.69	GPa		
Modular ratio, n	8.10			
Cracked section moment of inertia of section, I NA = $(bx^3/3) + nAs(de-x)^2$	5.4E+10	mm^4		
Exposure factor, Y_e	1			
Exposure condition: Class 1				
Tensile stress in steel reinforcement at the service limit, fss = nM(de-x)/I	193.23	MPa		
or fss = Ms /(As (de-x/3))				
βs = 1+ dc /(0.7 (*h* −dc))	1.19			
The spacing shall satisfy: s≤123000γe/βsfss −2dc	246	mm		
Provided spacing: **91** mm **SATISFIED!**				
(center to center of main bars)				

e. Design for Shear (Near Support Area)

Ultimate shear based on demand, V_u **1379.8** kN

Effective shear depth, d_v 998.1 mm

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2) 998.1 mm
dv2= $0.9d_e$ 939.6 mm
dv3= 0.72h.cap 864 mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*	
Angle of inclination of diagonal compressive stresses, θ =	45	deg	
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9		
Diameter of stirrups, ds =	16	mm	
Number of legs of stirrups	4		
Total area of shear reinforcement, Av =	804.248	mm2	
Non-prestress, Vp =	0	kN	
Required nominal shear resistance, V_n	1533.1	kN	
**shall be determined as the lesser of:*			
V_n=Vc+Vs+Vp=Vu/Øv	1533.1	kN	
V_n=0.25fc'$b_v d_v$+Vp	1.2E+04	kN	
Shear resistance provided by concrete, V_c	1479.8	kN	Vc= 0.083*β(f'c^0.5)bv dv
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	53.3	kN	**(Near Support Area)**
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	6236	mm	**Say,** 150 **mm spacing,**
			16 **mm Ø stirrups,**
			4 **legs** ***(2 SETS)***

AASHTO and ACI 318-14 Sec. 9.9.1.1

Deep beam classification

h =	1.20	m	Beam depth
L_c =	2.15	m	Clear span between column faces
P_f =	1.20	m	Closest concentrated load (girder) from face of support along clear span

Satisfies a) or b)

a) $L_c/h <= 4$ Lc/h = 1.79
b) $P_f/h <= 2$ Pf/h = 1.00
Check : **Coping is deep beam!**

AASHTO 2020 Sec. 5.8.2.6

Crack control reinforcement - vertical and horizontal bars of deep beams

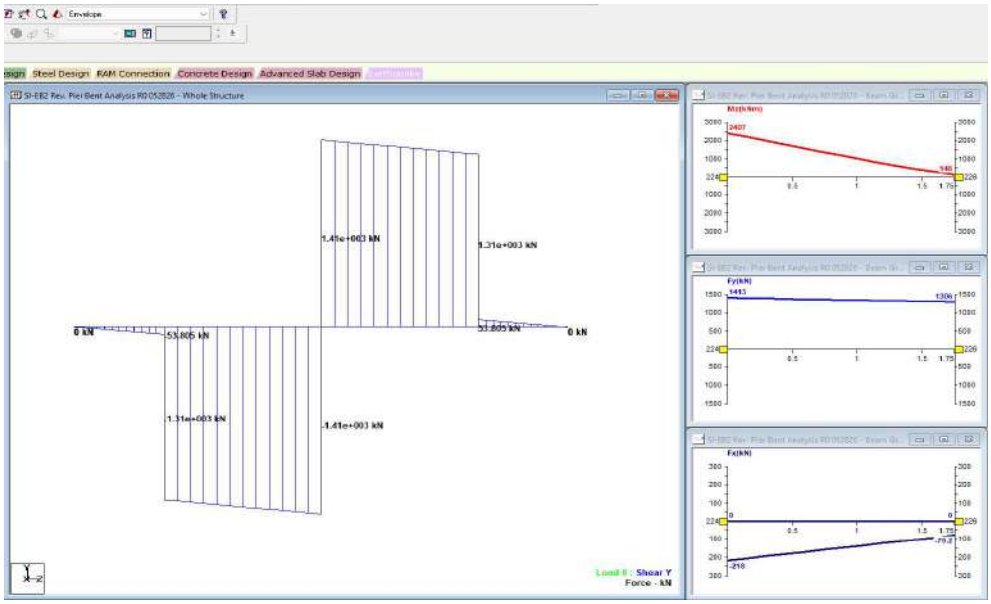
smax = min(d/4 and 300mm)
smax = 261 mm **OK!**
$A_v/b_w S_v \geq 0.003$
$A_v/b_w S_v$ = 0.00315 **SATISFIED!**

f. Design for Bottom Reinforcement (Positive Moment)

Cantilever beam therefore, no design moment - provide same reinforcement as top bars.

Provide: 18 **bars -** 25 **mm Ø Bottom Bars**

g. Design for Shear (Endspan Area)



Ultimate shear based on demand, V_u *mid*	1306	kN
Effective shear depth, d_v	998.1	mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*	
Angle of inclination of diagonal compressive stresses, θ =	45	deg	
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9		
Diameter of stirrups, ds =	16	mm	
Number of legs of stirrups	4		
Total area of shear reinforcement, Av =	804.248	mm2	
Non-prestress, Vp =	0	kN	
Required nominal shear resistance, V_n	1451.1	kN	
**shall be determined as the lesser of:*			
V_n=Vc+Vs+Vp=Vu/Øv	1451.1	kN	
V_n=0.25fc'$b_v d_v$+Vp	1.2E+04	kN	
Shear resistance provided by concrete, V_c	1479.8	kN	Vc= 0.083*β(f'c^0.5)bv dv
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	-28.7	kN	**(Midspan Area)**
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	-11584	mm	**Say, 200 mm spacing,**
			16 mm Ø stirrups,
			4 legs *(2 SETS)*

h. Shrinkage and Temperature Reinforcement (Side Bars):

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete. For the pile cap, bars shall be provided at the perimeter side of the pile cap. *DGCS 12.7.8*

Ast => 0.75 MPa x (bxh)/2(b+h)fy
where:

b = least width of component section, mm	1700	mm
h = least thickness of component section, mm	1200	mm

a) Ast => 0.75 MPa x (bxh)/2(b+h)fy — 0.637 mm^2/mm
b) Limitations: 0.223<=Ast<=1.27

therefore Ast =	0.637	mm^2/mm		
try diameter db =	20	mm		
Ab =	314	mm^2		
spacing, S = Ab/ Ast =	493	mm		
say S_prov =	250	mm -	20 mm Ø Side Bars	***(MINIMUM ONLY)***
Ast_act = Ab/S_prov =	1.257	mm^2/mm	**OK! Satisfied limitations.**	

where:

a) S_prov should not be greater than 3 x thickness, or 450mm	450	mm
b) S_prov should not be greater than 300 for walls and footings > 450mm thk.	300	mm
c) S_prov should be 300mm for other components > 900mm thick.	300	mm
S_max =	300	mm

Therefore, spacing provided for shrinkage and temperature bars are OK!

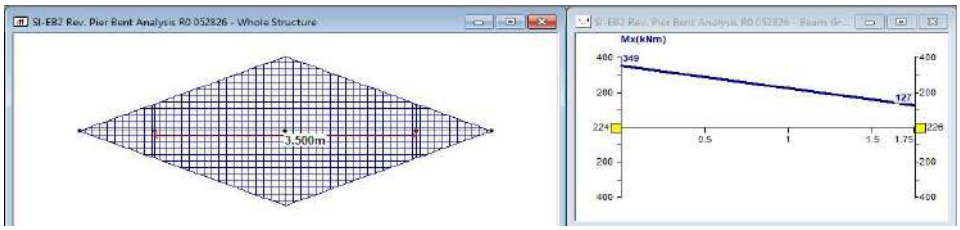
PROVIDE horizontal/side bars: *(1st trial use above as minimum)*

d_h =	25	mm	diameter	
n_h =	1		no. of bundle/layers per side	**Adopt if below is satisfied.**
s_h =	150	mm	spacing	
A_h =	981.748	mm^2	area provided per spacing, both sides	

AASHTO 2020 Sec. 5.8.2.6
Crack control reinforcement - vertical and horizontal bars of deep beams

smax = min(d/4 and 300mm)
smax = 261 mm **OK!**
$A_h/b_w s_h \geq$ 0.0030
$A_h/b_w s_h$ = 0.0038 **SATISFIED!**

i. Torsional Demand and Capacity:



Demand torsion, T.u		279.23	**kN-m**	*(At face of column)*		
Height of coping beam, h.cap		1200	mm			
Width of coping beam, b.cap		1700	mm			
Angle of inclination of diagonal compressive stresses, θ =					45	deg
Diameter of reinforcing bar, db		25	mm	Ab =	490.87	mm2
Diameter of stirrups, ds		16	mm	At =	201.06	mm2
Spacing of stirrups, s		150	mm			
Diameter of side bars, dsb		25	mm	Asb =	490.87	mm2
Conc. Compressive strength, f'c		27.6	MPa			
Reinforcing steel, fy		414	MPa			
Torsion Resistance factor, Φ		0.75				
Area	Acp =	2.04	m^2			
Perimeter	Pcp =	5.8	m			
Threshold Torsion, Tth =		314.127	kN-m			
Tth = 1/12 (f'c^0.5)(Acp^2/Pcp)						
	Φ Tth =	**235.60**	**kN-m**	**<**	**T.u**	
		Therefore, consider torsional effects!				

Torsional Strength, T.n
Lesser of T.n1 and T.n2

T.n1 = 2 Ao At fyt cot θ / s

Aoh =	1.614	m^2	*(area of centerline of outermost closed stirrup)*
Ao = 0.85 Aoh =	1.372	m^2	
At =	402.12	mm^2	*(area of 1/2 of total number of legs of closed stirrups)*
T.n1 =	3045.42	kN-m	

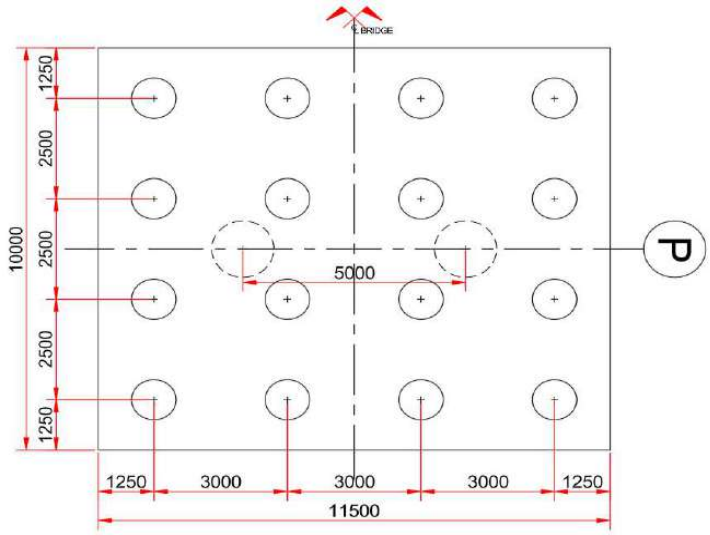
T.n2 = 2 Ao AL fy cot θ / Ph

AL =	24544	mm^2	*(total area of all longitudinal bars: top, bottom, sides)*
Ph =	5.206	m	*(perimeter of centerline of outermost closed stirrup)*
T.n2 =	5355.68	kN-m	

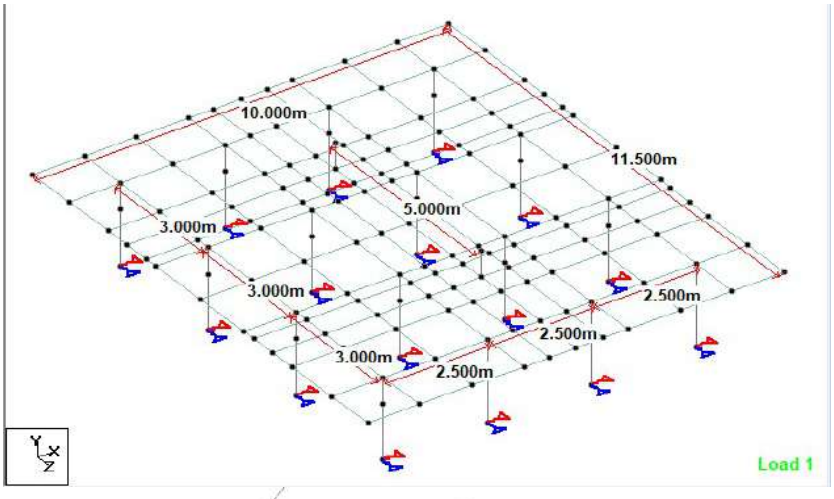
Therefore, T.n =	3045.42	kN-m	
Torsional Resistance, Φ T.n =	**2284.1**	**kN-m**	
c/d =	8.18	**OK!**	

5.7A VERIFICATION OF PILE CAP RESISTANCE - PIER 1

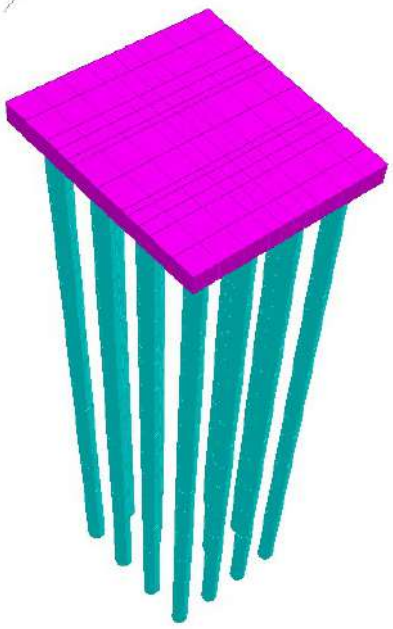
a. Pilecap with Piles Diagram and StaadPro Mathematical Model



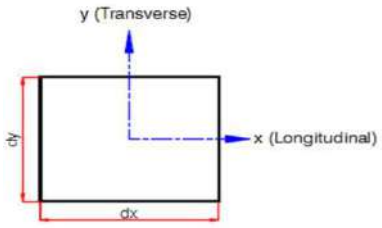
Pilecap with Piles Diagram



Pier Foundation StaadPro Mathematical Model *(Showing Pilecap)*



Pier Foundation StaadPro Mathematical Model *(Showing 3D Render)*



StaadPro Plate Model Orientation

b. Load Combinations / Limit States Column Reactions ***(From StaadPro Pier Bent Analysis)***

SI-EB2 Rev. Pier Bent Analysis R0 052826 - Support Reactions: 232

All / Summary / Envelope /

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
232	10 STRENGTH IA 1.25DC+1.50DW+1.75[(LLA+IM)+BF]+1.0WA_DFL	-170.651	3516.737	-62.190	897.645	-0.000	1791.857
	11 STRENGTH IB 1.25DC+1.50DW+1.75[(LLB+IM)+BF]+1.0WA_DFL	-170.651	3516.737	-62.190	-1457.356	-0.000	1791.857
	12 EXTREME IA 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ1	-486.668	2632.072	-31.510	264.587	0.000	5515.903
	13 EXTREME IB 1.25DC+1.50DW+0.50[(LLB+IM)+BF]+1.0WA_OWL+1.0EQ1	-486.668	2632.072	-31.510	-408.271	0.000	5515.903
	14 EXTREME IC 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2	-48.757	2632.072	-467.000	-4711.770	-0.000	511.959
	15 EXTREME IC 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2	-48.757	2632.072	-467.000	-5384.627	-0.000	511.959
	16 SERVICE IA 1.0DC+1.0DW+1.0[(LLA+IM)+BF]+1.0WA_OWL	-97.515	2510.491	-31.510	601.016	-0.000	1023.918
	17 SERVICE IB 1.0DC+1.0DW+1.0[(LLB+IM)+BF]+1.0WA_OWL	-97.515	2510.491	-31.510	-744.700	-0.000	1023.918

Let My = 0

>>> Reverse signs of column reaction forces and then load to pilecap with piles mathematical model. *(minimal, neglect)*

c. Additional Pilecap Loadings

1. Weight of overlaid rocks above pilecap for scour protection, DC

assumed height of rock, hr = 0.75 m
unit weight of rock, Yr = 20 kN/m^3
pressure load of rock, wr = **15.00** kPa = Yr*hr *(vertical load along surface area of pilecap)*

(Note: Selfweight of pilecap and piles are also included in the analysis.)

2. Horizontal inertial seismic effect of pilecap weight and rock on top of pilecap, EQ

acceleration coeff, As = 0.47 = kho, seismic hori. coeff. *(for foundations)*

height of pilecap, hp = 1.20 m
unit weight of concrete, Yc = 24 kN/m^3
pressure wt. of pilecap, wp = 28.80 kPa = Yc*hp

hori. shear pressure by pilecap, Vp = 13.54 kPa = kho*wp
hori. shear pressure by rock, Vr = 7.05 kPa = kho*wr
total hori. seismic intertial load, V_iEQ = **20.59** kPa *(hori. load along surface area of pilecap: longi. dir. for EQ1, transv. dir. for EQ2)*

3. Buoyancy Force, WA

unit weight of water, Yw = 10 kN/m^3
height of pilecap+rock, hw = 1.95 m *(height of displaced water)*
upward buoyancy pressure, ww = **19.50** kPa = Yw*hw *(vertical load along surface area of pilecap)*

>>> Adopt appropriate load factor for corresponding load combination.

d. Design Forces From StaadPro Analysis, Longitudinal and Transverse Directions

(Note: Critical moment and shear are assumed to act at the face of column/s, so plate reaction stresses under the column are not considered critical; these plates are assumed to have a rigid column to footing connection.)

EXTREME EVENTS

SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Plate Center Stress: GS: _PILECAP

Shear, Membrane and Bending / Summary / Principal and Von Mis / Summary / Global Moments / Combined Stresse

	Plate	L/C	Shear SQX (local) N/mm2	Shear SQY (local) N/mm2	Membrane SX (local) N/mm2	Membrane SY (local) N/mm2	Membrane SXY (local) N/mm2	Bending Moment Mx kNm/m	Bending Moment My kNm/m	Bending Moment Mxy kNm/m
Max Qx	743	16 1.25DC(FTG WT)+EXTREME IB	**0.914**	-0.352	-0.067	0.019	0.033	-680.174	-499.830	94.441
Min Qx	746	18 1.25DC(FTG WT)+EXTREME ID+	**-0.729**	-0.518	-0.015	-0.078	-0.070	-150.423	-336.956	-49.573
Max Qy	731	15 1.25DC(FTG WT)+EXTREME IA+	0.576	**0.762**	-0.080	-0.017	-0.074	-361.435	-203.746	-75.969
Min Qy	748	18 1.25DC(FTG WT)+EXTREME ID+	0.339	**-0.902**	0.017	-0.064	0.033	-494.251	-666.840	94.440
Max Sx	728	16 1.25DC(FTG WT)+EXTREME IB	-0.013	-0.167	**0.070**	0.011	-0.073	199.406	44.836	-224.050
Min Sx	749	16 1.25DC(FTG WT)+EXTREME IB	0.592	-0.773	**-0.081**	-0.023	0.081	-363.938	-209.368	79.543
Max Sy	728	17 1.25DC(FTG WT)+EXTREME IC+	0.118	0.027	0.012	**0.068**	-0.072	33.110	149.195	-203.409
Min Sy	749	17 1.25DC(FTG WT)+EXTREME IC+	0.724	-0.578	-0.024	**-0.080**	0.080	-197.643	-313.728	58.902
Max Sx	749	16 1.25DC(FTG WT)+EXTREME IB	0.592	-0.773	-0.081	-0.023	**0.081**	-363.938	-209.368	79.543
Min Sx	731	15 1.25DC(FTG WT)+EXTREME IA+	0.576	0.762	-0.080	-0.017	**-0.074**	-361.435	-203.746	-75.969
Max Mx	734	16 1.25DC(FTG WT)+EXTREME IB	0.342	-0.180	0.063	-0.029	-0.032	**443.875**	-116.790	-152.177
Min Mx	743	16 1.25DC(FTG WT)+EXTREME IB	0.914	-0.352	-0.067	0.019	0.033	**-680.174**	-499.830	94.441
Max My	729	18 1.25DC(FTG WT)+EXTREME ID+	0.168	-0.330	-0.027	0.060	-0.032	-122.369	**430.543**	-152.178
Min My	748	18 1.25DC(FTG WT)+EXTREME ID+	0.339	-0.902	0.017	-0.064	0.033	-494.251	**-666.840**	94.440
Max Mx	746	15 1.25DC(FTG WT)+EXTREME IA+	-0.029	0.156	0.068	0.005	0.066	196.902	39.213	**220.476**
Min Mx	728	16 1.25DC(FTG WT)+EXTREME IB	-0.013	-0.167	0.070	0.011	-0.073	199.406	44.836	**-224.050**

STRENGTH EVENTS

SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Plate Center Stress: G4: _PILECAP

Shear, Membrane and Bending | Summary | Principal and Von Mis | Summary | Global Moments | Combined Stresses

			Shear		Membrane			Bending Moment		
	Plate	L/C	SQX (local) N/mm2	SQY (local) N/mm2	SX (local) N/mm2	SY (local) N/mm2	SXY (local) N/mm2	Mx kNm/m	My kNm/m	Mxy kNm/m
Max Qx	743	14 1.25DC(FTG WT)+STRENGTH IB+	**0.625**	-0.115	-0.024	0.006	0.014	-369.303	-478.989	33.455
Min Qx	746	14 1.25DC(FTG WT)+STRENGTH IB+	**-0.417**	-0.291	0.013	-0.005	0.010	-53.280	-186.672	98.455
Max Qy	730	13 1.25DC(FTG WT)+STRENGTH IA+	0.049	**0.526**	-0.009	0.009	-0.008	-439.218	-274.857	-22.994
Min Qy	748	14 1.25DC(FTG WT)+STRENGTH IB+	0.081	**-0.594**	-0.002	-0.009	0.015	-460.737	-335.018	35.477
Max Sx	740	13 1.25DC(FTG WT)+STRENGTH IA+	-0.161	-0.079	**0.024**	-0.013	0.005	28.465	-357.992	95.796
Min Sx	737	13 1.25DC(FTG WT)+STRENGTH IA+	0.606	0.143	**-0.025**	0.009	-0.006	-365.602	-464.764	-17.746
Max Sy	736	14 1.25DC(FTG WT)+STRENGTH IB+	0.145	0.096	-0.006	**0.011**	-0.002	-12.522	-304.601	44.549
Min Sy	739	14 1.25DC(FTG WT)+STRENGTH IB+	-0.053	-0.024	0.006	**-0.014**	0.002	44.817	-218.627	-1.903
Max Sx	749	14 1.25DC(FTG WT)+STRENGTH IB+	0.544	-0.556	-0.020	-0.011	**0.026**	-272.454	-265.587	6.094
Min Sx	728	14 1.25DC(FTG WT)+STRENGTH IB+	-0.223	0.210	0.015	0.006	**-0.023**	13.788	6.920	-186.635
Max Mx	739	14 1.25DC(FTG WT)+STRENGTH IB+	-0.053	-0.024	0.006	-0.014	0.002	**44.817**	-218.627	-1.903
Min Mx	747	14 1.25DC(FTG WT)+STRENGTH IB+	-0.284	-0.493	0.004	-0.010	0.008	**-482.207**	-309.364	56.695
Max My	724	14 1.25DC(FTG WT)+STRENGTH IB+	0.033	0.065	-0.007	0.003	0.000	-234.459	**43.410**	-6.390
Min My	737	14 1.25DC(FTG WT)+STRENGTH IB+	0.546	0.291	-0.025	0.009	-0.006	-347.780	**-491.229**	38.685
Max Mx	746	13 1.25DC(FTG WT)+STRENGTH IA+	-0.274	-0.241	0.013	-0.005	0.010	2.801	-20.197	**171.766**
Min Mx	728	14 1.25DC(FTG WT)+STRENGTH IB+	-0.223	0.210	0.015	0.006	-0.023	13.788	6.920	**-186.635**

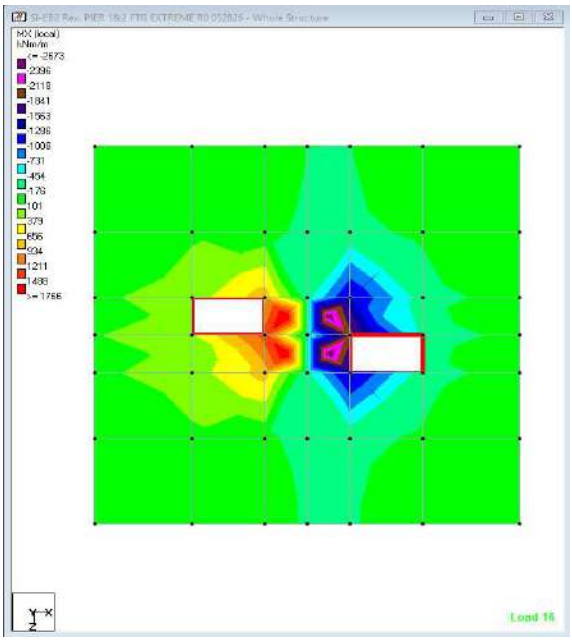
SERVICE EVENT

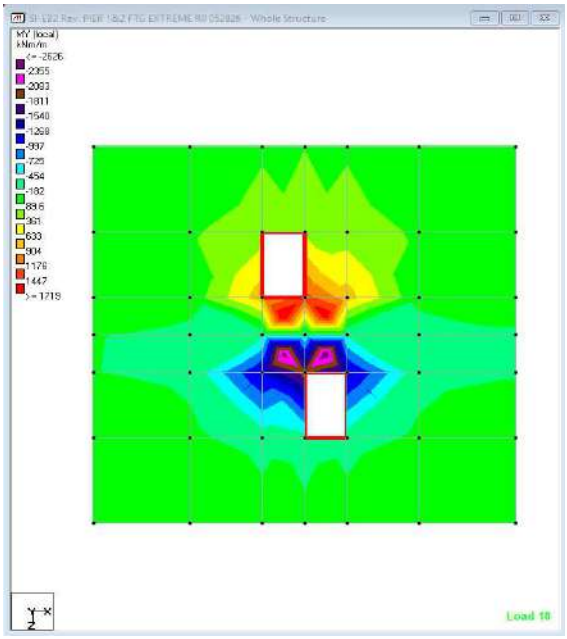
SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Plate Center Stress: G4: _PILECAP

Shear, Membrane and Bending | Summary | Principal and Von Mis | Summary | Global Moments | Combined Stresses

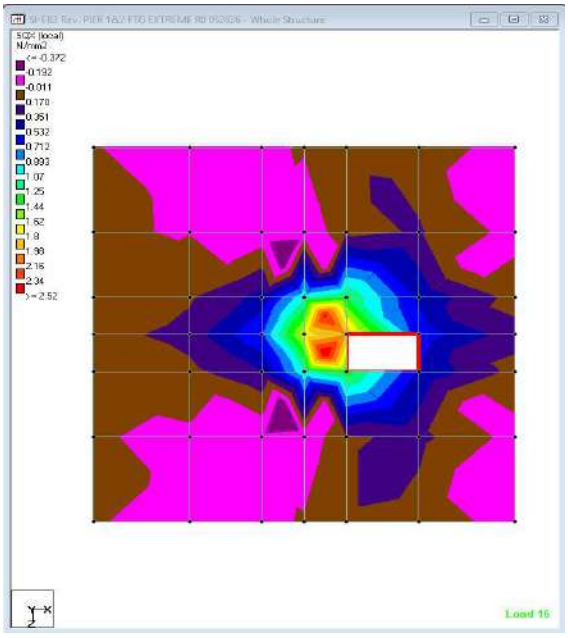
			Shear		Membrane			Bending Moment		
	Plate	L/C	SQX (local) N/mm2	SQY (local) N/mm2	SX (local) N/mm2	SY (local) N/mm2	SXY (local) N/mm2	Mx kNm/m	My kNm/m	Mxy kNm/m
Max Qx	743	20 1.0DC(FTG WT)+SERVICE IB+1.0	**0.409**	-0.087	-0.014	0.003	0.008	-234.500	-333.844	11.339
Min Qx	746	20 1.0DC(FTG WT)+SERVICE IB+1.0	**-0.286**	-0.217	0.007	-0.003	0.006	-47.116	-119.538	71.777
Max Qy	730	19 1.0DC(FTG WT)+SERVICE IA+1.0	0.046	**0.365**	-0.005	0.004	-0.005	-313.772	-190.467	-9.268
Min Qy	748	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.057	**-0.384**	-0.001	-0.004	0.009	-320.002	-206.270	12.958
Max Sx	740	19 1.0DC(FTG WT)+SERVICE IA+1.0	-0.144	-0.066	**0.013**	-0.007	0.003	-7.204	-264.703	82.518
Min Sx	737	19 1.0DC(FTG WT)+SERVICE IA+1.0	0.405	0.092	**-0.014**	0.005	-0.004	-233.683	-336.386	-6.734
Max Sy	736	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.097	0.062	-0.004	**0.006**	-0.001	-4.947	-212.458	27.108
Min Sy	739	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.043	-0.023	0.004	**-0.008**	0.001	27.534	-160.719	3.109
Max Sx	749	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.350	-0.369	-0.012	-0.006	**0.015**	-172.357	-164.832	-12.035
Min Sx	728	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.187	0.176	0.008	0.003	**-0.012**	-12.835	-20.560	-116.847
Max Mx	739	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.043	-0.023	0.004	-0.008	0.001	**27.534**	-160.719	3.109
Min Mx	747	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.174	-0.327	0.002	-0.006	0.005	**-332.270**	-191.610	39.712
Max My	724	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.028	0.054	-0.004	0.002	0.000	-174.050	**25.437**	-0.541
Min My	737	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.369	0.176	-0.014	0.005	-0.003	-223.499	**-340.080**	25.514
Max Mx	746	19 1.0DC(FTG WT)+SERVICE IA+1.0	-0.204	-0.189	0.007	-0.003	0.006	-15.070	-24.409	**113.669**
Min Mx	728	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.187	0.176	0.008	0.003	-0.012	-12.835	-20.560	**-116.847**

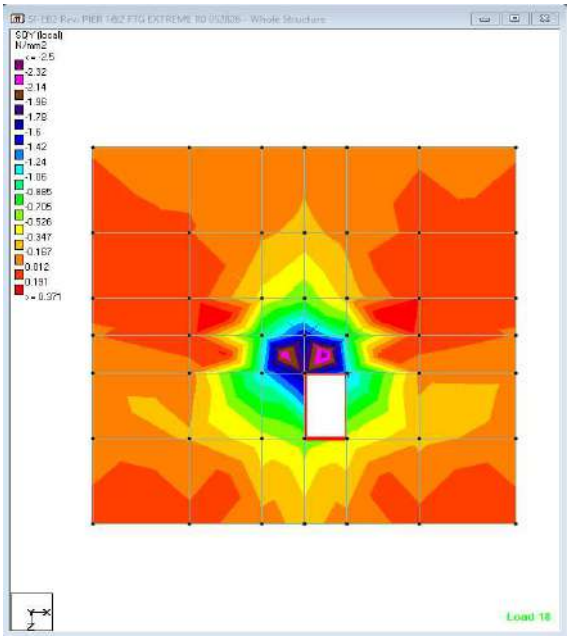
GOVERNING EVENTS





Longitudinal Moment: *(per meter width)*		*Add Torsion Effect Mxy*	Transverse Moment: *(per meter width)*		*Add Torsion Effect Mxy*
Max. +Mx_top =	443.88 kNm/m	152.18 kNm/m	Max. +My_top =	430.54 kNm/m	152.18 kNm/m
Max. -Mx_bot =	680.17 kNm/m	94.44 kNm/m	Max. -My_bot =	666.84 kNm/m	94.44 kNm/m





Longitudinal Shear:
Max. abs. SQx = 0.914 N/mm2
SQx (1000kPa/1MPa)(hp) = 1096.80 kN/m *(per meter width)*
hp = height of pilecap

Transverse Shear:
Max. abs. SQy = 0.902 N/mm2
SQy (1000kPa/1MPa)(hp) = 1082.40 kN/m *(per meter width)*
hp = height of pilecap

d. Pile Cap Resistance

Concrete unit weight,	Yc =	24	kN/m^3
Concrete compressive strength @ 28 days,	f'c =	27.6	MPa
Modulus elasticity of concrete @ 28 days,	Ec =	27350	MPa
Yield strength of steel reinforecement,	fy =	414	MPa
Modulus elasticity of steel reinforcement,	Es =	200000	MPa
Thickness of pile cap,	hp =	1.20	m
Unit width,	b =	1.00	m

1. Longitudinal Direction:

1.a Flexural Requirement for Top Reinforcement:

Design moment per unit width,	Mx_top =	596.05	kN-m	
Flexural cracking variability factor,*Y1*		1.6	**for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, *Y3*		0.67	**for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/*c*
Cracking Moment, M_{cr}				
$M_{cr}=y_3(y_1*fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		792.75	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		792.75	kN-m	
Design moment per unit width,	**Mx_top =**	**792.75**	**kN-m**	
Concrete cover,	cc_top =	75	mm	
Size of reinforcement,	db_top =	32	mm	Area of bar reinf., Ab = 804.248 mm^2
Effective depth of concrete,	de_top =	1093	mm	
Flexural Resistance factor,	Φ =	0.9		

Steel ratio β_1 = 0.85

β_1 Coefficient Criterion:

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

	m1 =	0.0567		m1=0.85*f'c/fy	
For required steel ratio, ρ	m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)	$Rn = \frac{Md}{\phi bv * (d_e)^2}$
	Rn =	0.737	MPa		
	ρ =	0.0018		ρ=m1*(1-sqrt(1-m2Rn))	

Area of main reinforcement required, As = ρbd	As =	1978.17	mm2				
spacing required of top bars, Ab(b)/As = sx_top	sx_top =	406.56	mm	**Say, s =**	**200**	**mm -**	**32** **mm Ø Top Bars**
				Longitudinal TOP:	***USE 32mmØ @ 200mm***		
Area of main reinforcement provided, Ab(b)/s = Asp	Asp =	4021.24	mm2				
a = (Asp fy) / (0.85 f'c b)	a =	70.963	mm2	*Depth of compression block*			
Flexural resistance, Mr = Φ [Asp fy (de - a/2)]	**Mr =**	**1584.49**	**kN-m**				
	c/d =	2.00	**OK!**				

1.b Flexural Requirement for Bottom Reinforcement

Design moment per unit width,	Mx_bot =	774.62	kN-m	
Flexural cracking variability factor, γ_1		1.6	*for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, γ_3		0.67	*for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/c
Cracking Moment, M_{cr}				
$M_{cr}=\gamma_3(\gamma_1 * fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		1030.24	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		1021.84	kN-m	
Design moment per unit width,	**Mx_bot =**	**1021.84**	**kN-m**	

Concrete cover,	cc_bot =	200	mm	*(150mm pile embedment + 50mm clearance)*		
Size of reinforcement,	db_bot =	36	mm	Area of bar reinf., Ab =	1017.88	mm^2
Effective depth of concrete,	de_bot =	964	mm			
Flexural Resistance factor,	Φ =	0.9				

Steel ratio β_1 = 0.85

β_1 Coefficient Criterion:

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

	m1 =	0.0567		m1=0.85*f'c/fy	
For required steel ratio, ρ	m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)	$Rn = \frac{Md}{\phi bv * (d_e)^2}$
	Rn =	1.222	MPa		
	ρ =	0.0030		ρ=m1*(1-sqrt(1-m2Rn))	

Area of main reinforcement required, As = ρbd	As =	2923.08	mm2				
spacing required of bottom bars, Ab(b)/As = sx_bot	sx_bot =	348.22	mm	**Say, s =**	**200**	**mm -**	**36** **mm Ø Bottom Bars**
				Longitudinal BOT.:	***USE 36mmØ @ 200mm***		
Area of main reinforcement provided, Ab(b)/s = Asp	Asp =	5089.38	mm2				
a = (Asp fy) / (0.85 f'c b)	a =	89.8126	mm2	*Depth of compression block*			
Flexural resistance, Mr = Φ [Asp fy (de - a/2)]	**Mr =**	**1742.88**	**kN-m**				
	c/d =	1.71	**OK!**				

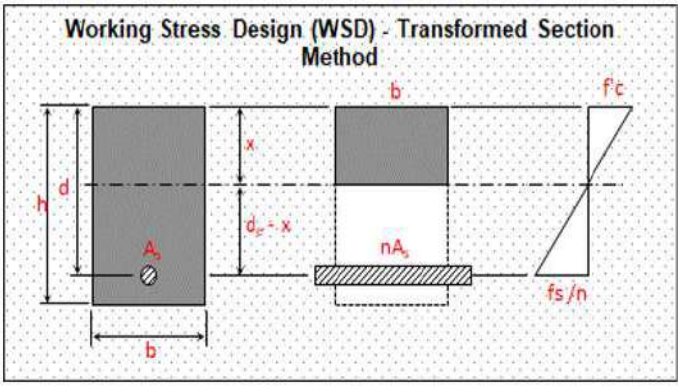
Control of cracking by distribution of reinforcement: • DGCS 12.4.3.4

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1 *(adopt bot., max neg moment), Ms=*	371.98	kN-m	*(plus corresponding Mxy)*
80% of Modulus of rupture, fr = 0.52*sqrt(f'c)	2.185	MPa	
Section modulus, S_c	2.4E+08	mm^3	
Tension in the cross section, fs = Ms/Sc	1.550	MPa	

• DGCS 12.1.1.6

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



b*x*x/2 = nAs*(de-x)
bx^2/2=nAsde-nAsx
bx^2/2+nAsx-nAsde=0
a = b/2
b = nAs
c = -nAs de

Extreme tension fiber to center of flexural reinforcement, d_c	218	mm
Overall thickness of component, h	1200	mm
Compression fiber to the centroid of extreme tension steel, d_e	964	mm

• by quadratic equation determine x:

a= 500.00

Neutral axis to extreme compression fiber, x	233.23	mm	b=	3.7E+04
Modulus elasticity of steel, E_s	200	Gpa	c=	-3.6E+07
Modulus elasticity of concrete, E_c	27.35	GPa		
Modular ratio, n	7.31			
Cracked section moment of inertia of section, I NA = $(bx^3/3) + nAs(de-x)^2$	2.4E+10	mm^4		
Exposure factor, Y_e	0.75			
Exposure condition: Class 2				
Tensile stress in steel reinforcement at the service limit, fss = nM(de-x)/I	82.47	MPa		
or fss = Ms /(As (de-x/3))				
βs = 1+ dc /(0.7 (*h* −dc))	1.32			
The spacing shall satisfy: s≤123000Ye/βsfss −2dc	413	mm		
Provided spacing: **200** mm **SATISFIED!**				

Check for maximum reinforcement limit:

"The maximum reinforcement requirements ensure the section has sufficient ductility and is not overreinforced"

Compression fiber to neutral axis, c = a/β	c_top =	83.4859	mm2	c_bot =	105.662 mm2
Maximum reinforcement ratio c/de =	max c/de_top =	0.420		max c/de_bot =	0.420
	actual c/de_top =	0.076	**OK!**	actual c/de_bot =	0.110 **OK!**

Check for spacing requirements:

•Check for minimum spacing of reinforcement — DGCS 12.7.3.1

For cast in place concrete, clear distance between parallel bars in a layer shall not be less than:

•1.5 x nominal diam of bars	= 54 mm	s_min =	max of 3 values		
•1.5 x max. size of aggregates	= 37.5 mm	s_min =	54 mm		
•38mm	= 38 mm	s_provt =	200 mm	**OK in min. spacing requirement!**	
		s_provb =	200 mm	**OK in min. spacing requirement!**	

•Check for maximum spacing of reinforcement (for walls and slabs) — DGCS 12.7.3.2

•s < 1.5 x hf	= 1800 mm	s_max =	min of 2 values	
•450mm	= 450 mm	s_min =	450 mm	
		s_provt =	200 mm	**OK in max. spacing requirement!**
		s_provb =	200 mm	**OK in max. spacing requirement!**

1.c Shear Requirement in Longitudinal Direction:

Ultimate shear based on demand, *Vxu* — **1096.8** kN *(unit width, b=1.0m)*

Effective shear depth, d_v — 919.1 mm

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2)	919.1	mm
dv2= $0.9d_e$	867.6	mm
dv3= 0.72h.cap	864.0	mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*			
Angle of inclination of diagonal compressive stresses, θ =	45	deg			
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9				
Diameter of cross ties/J-bars, ds =	20	mm			
Number of legs of ties, n = b / spacing along b	2.50	*(@ every no. of top bar)*	s_top =	200 mm	every no. of bars = 2
Total area of shear reinforcement, Av =	785.398	mm2			
Non-prestress, Vp =	0	kN			
Required nominal shear resistance, V_n	1218.7	kN			
**shall be determined as the lesser of:*					
V_n=Vc+Vs+Vp=Vu/Øv	1218.7	kN			
V_n=0.25fc'$b_v d_v$+Vp	6341.7	kN			
Shear resistance provided by concrete, V_c	801.5	kN	Vc= 0.083*β(f'c^0.5)bv dv		
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	417.1	kN			
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	716	mm	**Say,**	**400**	**mm spacing along span,**
				20	**mm Ø ties at**
				400	**mm spacing along width**
				Longitudinal	

2. Transverse Direction:

2.a Flexural Requirement for Top Reinforcement:

Design moment per unit width,	My_top =	582.72	kN-m	
Flexural cracking variability factor, *Y1*		1.6	**for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, *Y3*		0.67	**for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/*c*
Cracking Moment, M_{cr}				
$M_{cr}=y_3(y_1*fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		775.02	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		775.02	kN-m	

Design moment per unit width, **My_top = 775.02 kN-m**

Concrete cover, cc_top = 75 mm
Size of reinforcement, db_top = 32 mm Area of bar reinf., Ab = 804.248 mm^2
Effective depth of concrete, de_top = 1045 mm
Flexural Resistance factor, Φ = 0.9

Steel ratio β_1 = 0.85
β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ
m1 = 0.0567 m1=0.85*f'c/fy
m2 = 0.0853 mm2/N m2=2/(0.85*f'c)
Rn = 0.789 MPa $Rn = \frac{Md}{\phi bv * (d_e)^2}$
ρ = 0.0019 ρ=m1*(1-sqrt(1-m2Rn))

Area of main reinforcement required, As = ρbd As = 2025.09 mm2
spacing required of top bars, Ab(b)/As = sy_top sy_top = 397.14 mm **Say, s = 200 mm - 32 mm Ø Top Bars**
Transverse TOP: *USE 32mmØ @ 200mm*

Area of main reinforcement provided, Ab(b)/s = Asp Asp = 4021.24 mm2
a = (Asp fy) / (0.85 f'c b) a = 70.963 mm2 *Depth of compression block*
Flexural resistance, Mr = Φ [Asp fy (de - a/2)] **Mr = 1512.58 kN-m**
c/d = 1.95 **OK!**

2.b Flexural Requirement for Bottom Reinforcement

Design moment per unit width, My_bot = 761.28 kN-m
Flexural cracking variability factor, *Y1* 1.6 **for all other concrete*
Ratio of specified min. to ult. tensile strength of steel, *Y3* 0.67 **for A615, 414MPa steel*
Modulus of rupture, f_r 3.310 MPa fr=0.63*sqrt(f'c)
Section modulus, S_c 2.4E+08 mm^3 Sc=I/*c*
Cracking Moment, M_{cr}
$M_{cr}=\gamma_3(\gamma_1 * fr)\ S_c$ 851.53 kN-m
1.20 Mcr = 1021.84 kN-m
1.33*M.u_neg = 1012.50 kN-m

Minimum design moment, Min(1.2Mcr or 1.33Mu) 1012.50 kN-m

Design moment per unit width, **My_bot = 1012.50 kN-m**

Concrete cover, cc_bot = 200 mm *(150mm pile embedment + 50mm clearance)*
Size of reinforcement, db_bot = 36 mm Area of bar reinf., Ab = 1017.88 mm^2
Effective depth of concrete, de_bot = 910 mm
Flexural Resistance factor, Φ = 0.9

Steel ratio β_1 = 0.85
β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ
m1 = 0.0567 m1=0.85*f'c/fy
m2 = 0.0853 mm2/N m2=2/(0.85*f'c)
Rn = 1.359 MPa $Rn = \frac{Md}{\phi bv * (d_e)^2}$
ρ = 0.0034 ρ=m1*(1-sqrt(1-m2Rn))

Area of main reinforcement required, As = ρbd As = 3078.01 mm2
spacing required of bottom bars, Ab(b)/As = sy_bot sy_bot = 330.69 mm **Say, s = 200 mm - 36 mm Ø Bottom Bars**
Transverse BOT.: *USE 36mmØ @ 200mm*

Area of main reinforcement provided, Ab(b)/s = Asp Asp = 5089.38 mm2
a = (Asp fy) / (0.85 f'c b) a = 89.8126 mm2 *Depth of compression block*
Flexural resistance, Mr = Φ [Asp fy (de - a/2)] **Mr = 1640.48 kN-m**
c/d = 1.62 **OK!**

Control of cracking by distribution of reinforcement:

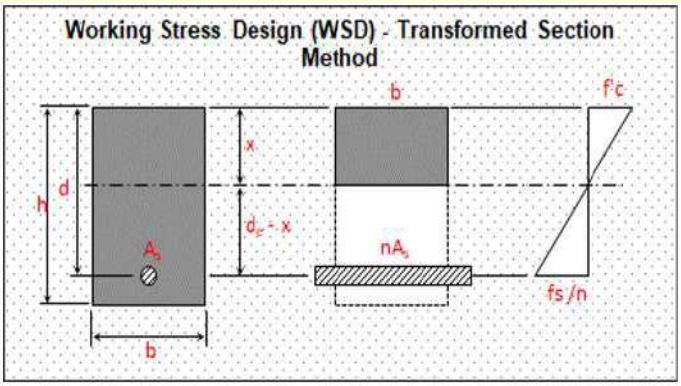
Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

- DGCS 12.4.3.4

Moment demand at Service 1 *(adopt bot., max neg moment), Ms=*	365.59	kN-m	*(plus corresponding Mxy)*
80% of Modulus of rupture, fr = 0.52*sqrt(f'c)	2.185	MPa	
Section modulus, S_c	2.4E+08	mm^3	
Tension in the cross section, fs = Ms/Sc	1.523	MPa	

- DGCS 12.1.1.6

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



b*x*x/2 = nAs*(de-x)
bx^2/2=nAsde-nAsx
bx^2/2+nAsx-nAsde=0
a = b/2
b = nAs
c = -nAs de

Extreme tension fiber to center of flexural reinforcement, d_c	254	mm
Overall thickness of component, h	1200	mm
Compression fiber to the centroid of extreme tension steel, d_e	910	mm
Neutral axis to extreme compression fiber, x	225.69	mm
Modulus elasticity of steel, E_s	200	Gpa
Modulus elasticity of concrete, E_c	27.35	GPa
Modular ratio, n	7.31	
Cracked section moment of inertia of section, I NA = $(bx^3/3) + nAs(de-x)^2$	2.1E+10	mm^4
Exposure factor, Y_e	0.75	
Exposure condition: Class 2		
Tensile stress in steel reinforcement at the service limit, fss = nM(de-x)/I	86.05	MPa
or fss = Ms /(As (de-x/3))		
βs = 1+ dc /(0.7 (*h* −dc))	1.38	
The spacing shall satisfy: s≤123000γe/βsfss −2dc	267	mm

- by quadratic equation

determine x:

a= 500.00
b= 3.7E+04
c= -3.4E+07

Provided spacing: **200** mm **SATISFIED!**

Check for maximum reinforcement limit:

"The maximum reinforcement requirements ensure the section has sufficient ductility and is not overreinforced"

Compression fiber to neutral axis, c = a/β	c_top =	83.4859	mm2	c_bot =	105.662 mm2
Maximum reinforcement ratio c/de =	max c/de_top =	0.420		max c/de_bot =	0.420
	actual c/de_top =	0.080	**OK!**	actual c/de_bot =	0.116 **OK!**

Check for spacing requirements:

•Check for minimum spacing of reinforcement — DGCS 12.7.3.1

For cast in place concrete, clear distance between parallel bars in a layer shall not be less than:

•1.5 x nominal diam of bars	= 54 mm	s_min =	max of 3 values		
•1.5 x max. size of aggregates	= 37.5 mm	s_min =	54	mm	
•38mm	= 38 mm	s_provt =	200	mm	**OK in min. spacing requirement!**
		s_provb =	200	mm	**OK in min. spacing requirement!**

•Check for maximum spacing of reinforcement (for walls and slabs) — DGCS 12.7.3.2

•s < 1.5 x hf	= 1800 mm	s_max =	min of 2 values		
•450mm	= 450 mm	s_min =	450	mm	
		s_provt =	200	mm	**OK in max. spacing requirement!**
		s_provb =	200	mm	**OK in max. spacing requirement!**

2.c Shear Requirement in Transverse Direction:

Ultimate shear based on demand, *Vyu* — **1082.4** kN *(unit width, b=1.0m)*

Effective shear depth, d_v — 865.1 mm

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2)	865.1	mm
dv2= 0.9d_e	819.0	mm
dv3= 0.72h.cap	864.0	mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*
Angle of inclination of diagonal compressive stresses, θ =	45	deg
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9	
Diameter of cross ties/J-bars, ds =	20	mm

Number of legs of ties, n = b / spacing along b	2.50	*(@ every no. of top bar)*	s_top =	200	mm	every no. of bars = 2
Total area of shear reinforcement, Av =	785.398	mm2				
Non-prestress, Vp =	0	kN				
Required nominal shear resistance, V_n	1202.7	kN				
**shall be determined as the lesser of:*						
V_n=Vc+Vs+Vp=Vu/Øv	1202.7	kN				
V_n=0.25fc'$b_v d_v$+Vp	5969.1	kN				
Shear resistance provided by concrete, V_c	754.4	kN	Vc= 0.083*β(f'c^0.5)bv dv			
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	448.2	kN				
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	628	mm	**Say.**	**400**	**mm spacing along span,**	
				20	**mm Ø ties at**	
				400	**mm spacing along width**	
				Transverse		

3. Shrinkage and Temperature Reinforcement (Side Bars):

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete. For the pile cap, bars shall be provided at the perimeter side of the pile cap.

DGCS 12.7.8

Ast => 0.75 MPa x (bxh)/2(b+h)fy
where:

b = least width of component section, mm	5400	mm
h = least thickness of component section, mm	1200	mm
a) Ast => 0.75 MPa x (bxh)/2(b+h)fy	0.889	mm^2/mm
b) Limitations: 0.223<=Ast<=1.27		
therefore Ast =	0.889	mm^2/mm
try diameter db =	20	mm
Ab =	314	mm^2
spacing, S = Ab/ Ast =	353	mm
say S_prov =	**250**	**mm -** **20** **mm Ø Side Bars** ***(MINIMUM Only)***
Ast_act = Ab/S_prov =	1.257	mm^2/mm **OK! Satisfied limitations.**

where:

a) S_prov should not be greater than 3 x thickness, or 450mm	450	mm
b) S_prov should not be greater than 300 for walls and footings > 450mm thk.	300	mm
c) S_prov should be 300mm for other components > 900mm thick.	300	mm
S_max =	300	mm

Therefore, spacing provided for shrinkage and temperature bars are OK!

∴ PROVIDED: 25 mm Ø @ 200 mm Side Bars

4. Verification of Two-Way Shear Action (Punching Shear)

4.a Punching Shear Requirement for Pile:

Maximum reaction of single pile, V_u	**1811.4**	kN	*(From StaadPro Pier Footing Analysis)*
V_n *of the concrete shall be taken as:*			
$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv \le 0.33\sqrt{(fc')}\,bodv$			
Effective shear depth of footing, d_v	865.1	mm	
Dimension of pile, dpile	1000.0	mm	
Ratio of long to short direction of pile, βc	1		*(i.e. for square piles)*
Perimeter of the critical section, b_o	7460	mm	*(dv/2 distance around pile)*
Nominal shear capacity of concrete, V_n	11189.0	kN	
a) $Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv$	1.7E+04	kN	
b) $Vn= 0.33\sqrt{(fc')}\,bodv$	11189.0	kN	
Resistance factor for normal weight concrete, Ø	0.9		
Ultimate shear capacity of section, $ØV_n$	**10070.1**	kN	**OK!**
c/d =	5.56		

Pile cap is adequate for shear due to maximum reaction of pile.

4.b Punching Shear Requirement for Column:

Maximum reaction of single column, P_{max}	**2686**	kN	*(From Design of Column)*
V_n *of the concrete shall be taken as:*			
$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv \le 0.33\sqrt{(fc')}\,bodv$			
Effective shear depth of footing, d_v	865.1	mm	
Diameter of column, dcol	1200.0	mm	
Ratio of long to short direction of pile, βc	1		*(i.e. for square col.)*
Perimeter of the critical section, b_o	6488	mm	*(dv/2 distance around column)*
Nominal shear capacity of concrete, V_c	9730.2	kN	

a) $Vn=(0.17+0.33/\beta c)\ \sqrt{(fc')}\ bodv$	1.5E+04	kN	
b) $Vn= 0.33\ \sqrt{(fc')}\ bodv$	9730.2	kN	
Resistance factor for normal weight concrete, Ø	0.9		
Ultimate shear capacity of section (concrete only), $ØV_c$	**8757**	kN	**OK!** *(Concrete only)*

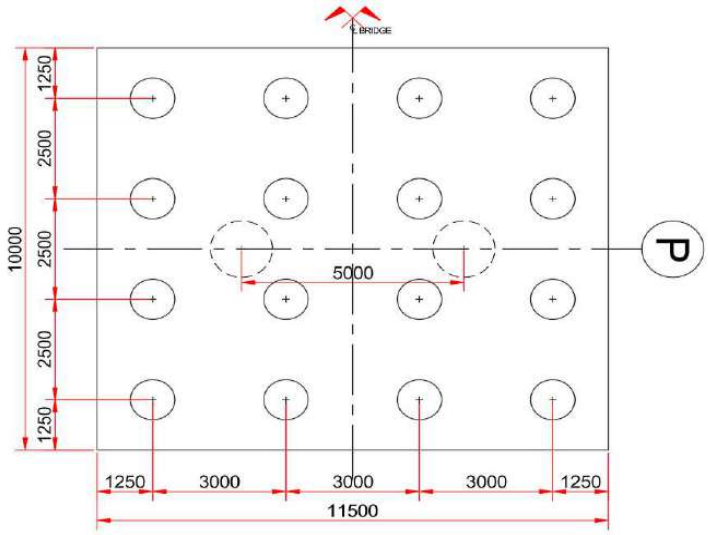
Analyze transverse reinforcement shear capacity on critical section (b_o perimeter):

Approximate spacing of shear reinf. on both directions, s_{ties}	335.41	mm	
$s_{ties} = sqrt(s_{ties_longtl}^2 + s_{ties_transv}^2)$			
Diameter of cross ties/J-bars, ds =	20	mm	
Number of legs of ties, n = bo / s_{ties}	19.34		
Total area of shear reinforcement, Av =	6076.64	mm2	
Shear reinforcement capacity, Vs = ØAv fy dv / s_{ties}	**5840**	kN	
Ultimate shear capacity of section, $ØV_n$ *=Ø(Vc + Vs)*	**14597**	kN	**OK!** *(Concrete and Shear Reinf. Capacities)*
c/d =	5.43		

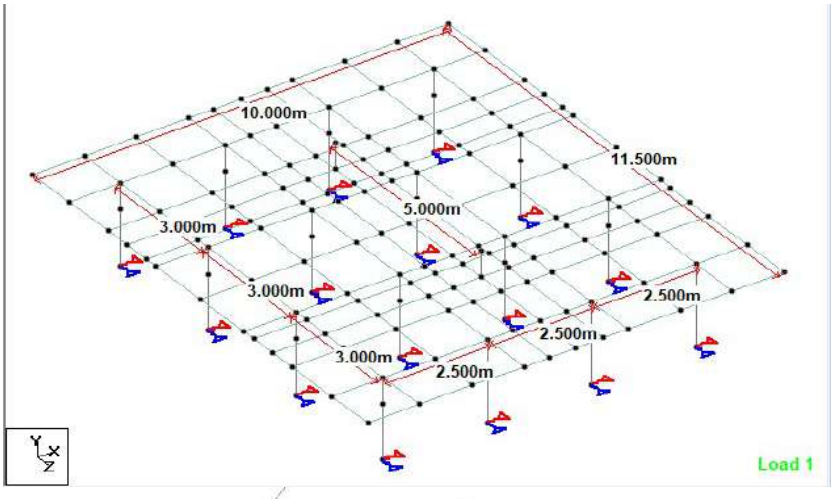
Pile cap is adequate for shear due to maximum reaction of column.

5.7B VERIFICATION OF PILE CAP RESISTANCE - PIER 2

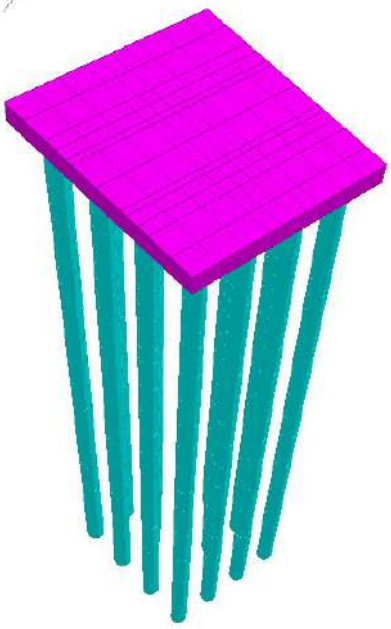
a. Pilecap with Piles Diagram and StaadPro Mathematical Model



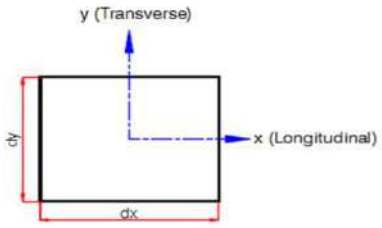
Pilecap with Piles Diagram



Pier Foundation StaadPro Mathematical Model *(Showing Pilecap)*



Pier Foundation StaadPro Mathematical Model *(Showing 3D Render)*



StaadPro Plate Model Orientation

b. Load Combinations / Limit States Column Reactions ***(From StaadPro Pier Bent Analysis)***

SI-EB2 Rev. Pier Bent Analysis R0 052826 - Support Reactions: 232

All / Summary / Envelope /

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
232	10 STRENGTH IA 1.25DC+1.50DW+1.75[(LLA+IM)+BF]+1.0WA_DFL	-170.651	3516.737	-62.190	897.645	-0.000	1791.857
	11 STRENGTH IB 1.25DC+1.50DW+1.75[(LLB+IM)+BF]+1.0WA_DFL	-170.651	3516.737	-62.190	-1457.356	-0.000	1791.857
	12 EXTREME IA 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ1	-486.668	2632.072	-31.510	264.587	0.000	5515.903
	13 EXTREME IB 1.25DC+1.50DW+0.50[(LLB+IM)+BF]+1.0WA_OWL+1.0EQ1	-486.668	2632.072	-31.510	-408.271	0.000	5515.903
	14 EXTREME IC 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2	-48.757	2632.072	-467.000	-4711.770	-0.000	511.959
	15 EXTREME IC 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2	-48.757	2632.072	-467.000	-5384.627	-0.000	511.959
	16 SERVICE IA 1.0DC+1.0DW+1.0[(LLA+IM)+BF]+1.0WA_OWL	-97.515	2510.491	-31.510	601.016	-0.000	1023.918
	17 SERVICE IB 1.0DC+1.0DW+1.0[(LLB+IM)+BF]+1.0WA_OWL	-97.515	2510.491	-31.510	-744.700	-0.000	1023.918

Let My = 0
(minimal, neglect)

>>> Reverse signs of column reaction forces and then load to pilecap with piles mathematical model.

c. Additional Pilecap Loadings

1. Weight of overlaid rocks above pilecap for scour protection, DC

assumed height of rock, hr = 0.75 m
unit weight of rock, Yr = 20 kN/m^3
pressure load of rock, wr = **15.00** kPa = Yr*hr *(vertical load along surface area of pilecap)*

(Note: Selfweight of pilecap and piles are also included in the analysis.)

2. Horizontal inertial seismic effect of pilecap weight and rock on top of pilecap, EQ

acceleration coeff, As = 0.47 = kho, seismic hori. coeff. *(for foundations)*

height of pilecap, hp = 1.20 m
unit weight of concrete, Yc = 24 kN/m^3
pressure wt. of pilecap, wp = 28.80 kPa = Yc*hp

hori. shear pressure by pilecap, Vp = 13.54 kPa = kho*wp
hori. shear pressure by rock, Vr = 7.05 kPa = kho*wr
total hori. seismic intertial load, V_iEQ = **20.59** kPa *(hori. load along surface area of pilecap: longi. dir. for EQ1, transv. dir. for EQ2)*

3. Buoyancy Force, WA

unit weight of water, Yw = 10 kN/m^3
height of pilecap+rock, hw = 1.95 m *(height of displaced water)*
upward buoyancy pressure, ww = **19.50** kPa = Yw*hw *(vertical load along surface area of pilecap)*

>>> Adopt appropriate load factor for corresponding load combination.

d. Design Forces From StaadPro Analysis, Longitudinal and Transverse Directions

(Note: Critical moment and shear are assumed to act at the face of column/s, so plate reaction stresses under the column are not considered critical; these plates are assumed to have a rigid column to footing connection.)

EXTREME EVENTS

SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Plate Center Stress: G5: PILECAP

Shear, Membrane and Bending / Summary / Principal and Von Mis / Summary / Global Moments / Combined Stresse

	Plate	L/C	Shear SQX (local) N/mm2	Shear SQY (local) N/mm2	Membrane SX (local) N/mm2	Membrane SY (local) N/mm2	Membrane SXY (local) N/mm2	Bending Moment Mx kNm/m	Bending Moment My kNm/m	Bending Moment Mxy kNm/m
Max Qx	743	16 1.25DC(FTG WT)+EXTREME IB	**0.914**	-0.352	-0.067	0.019	0.033	-680.174	-499.830	94.441
Min Qx	746	18 1.25DC(FTG WT)+EXTREME ID+	**-0.729**	-0.518	-0.015	-0.078	-0.070	-150.423	-336.956	-49.573
Max Qy	731	15 1.25DC(FTG WT)+EXTREME IA+	0.576	**0.762**	-0.080	-0.017	-0.074	-361.435	-203.746	-75.969
Min Qy	748	18 1.25DC(FTG WT)+EXTREME ID+	0.339	**-0.902**	0.017	-0.064	0.033	-494.251	-666.840	94.440
Max Sx	728	16 1.25DC(FTG WT)+EXTREME IB	-0.013	-0.167	**0.070**	0.011	-0.073	199.406	44.836	-224.050
Min Sx	749	16 1.25DC(FTG WT)+EXTREME IB	0.592	-0.773	**-0.081**	-0.023	0.081	-363.938	-209.368	79.543
Max Sy	728	17 1.25DC(FTG WT)+EXTREME IC+	0.118	0.027	0.012	**0.068**	-0.072	33.110	149.195	-203.409
Min Sy	749	17 1.25DC(FTG WT)+EXTREME IC+	0.724	-0.578	-0.024	**-0.080**	0.080	-197.643	-313.728	58.902
Max Sx	749	16 1.25DC(FTG WT)+EXTREME IB	0.592	-0.773	-0.081	-0.023	**0.081**	-363.938	-209.368	79.543
Min Sx	731	15 1.25DC(FTG WT)+EXTREME IA+	0.576	0.762	-0.080	-0.017	**-0.074**	-361.435	-203.746	-75.969
Max Mx	734	16 1.25DC(FTG WT)+EXTREME IB	0.342	-0.180	0.063	-0.029	-0.032	**443.875**	-116.790	-152.177
Min Mx	743	16 1.25DC(FTG WT)+EXTREME IB	0.914	-0.352	-0.067	0.019	0.033	**-680.174**	-499.830	94.441
Max My	729	18 1.25DC(FTG WT)+EXTREME ID+	0.166	-0.330	-0.027	0.060	-0.032	-122.369	**430.543**	-152.178
Min My	748	18 1.25DC(FTG WT)+EXTREME ID+	0.339	-0.902	0.017	-0.064	0.033	-494.251	**-666.840**	94.440
Max Mx	746	15 1.25DC(FTG WT)+EXTREME IA+	-0.029	0.156	0.068	0.005	0.066	196.902	39.213	**220.476**
Min Mx	728	16 1.25DC(FTG WT)+EXTREME IB	-0.013	-0.167	0.070	0.011	-0.073	199.406	44.836	**-224.050**

STRENGTH EVENTS

SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Plate Center Stress: G4: _PILECAP

Shear, Membrane and Bending | Summary | Principal and Von Mis | Summary | Global Moments | Combined Stresses

	Plate	L/C	Shear		Membrane			Bending Moment		
			SQX (local) N/mm2	SQY (local) N/mm2	SX (local) N/mm2	SY (local) N/mm2	SXY (local) N/mm2	Mx kNm/m	My kNm/m	Mxy kNm/m
Max Qx	743	14 1.25DC(FTG WT)+STRENGTH IB+	**0.625**	-0.115	-0.024	0.006	0.014	-369.303	-478.989	33.455
Min Qx	746	14 1.25DC(FTG WT)+STRENGTH IB+	**-0.417**	-0.291	0.013	-0.005	0.010	-53.280	-186.672	98.455
Max Qy	730	13 1.25DC(FTG WT)+STRENGTH IA+	0.049	**0.526**	-0.009	0.009	-0.008	-439.218	-274.857	-22.994
Min Qy	748	14 1.25DC(FTG WT)+STRENGTH IB+	0.081	**-0.594**	-0.002	-0.009	0.015	-460.737	-335.018	35.477
Max Sx	740	13 1.25DC(FTG WT)+STRENGTH IA+	-0.161	-0.079	**0.024**	-0.013	0.005	28.465	-357.992	95.796
Min Sx	737	13 1.25DC(FTG WT)+STRENGTH IA+	0.668	0.143	**-0.025**	0.009	-0.006	-365.602	-464.764	-17.748
Max Sy	736	14 1.25DC(FTG WT)+STRENGTH IB+	0.145	0.096	-0.006	**0.011**	-0.002	-12.522	-304.831	44.049
Min Sy	739	14 1.25DC(FTG WT)+STRENGTH IB+	-0.053	-0.024	0.006	**-0.014**	0.002	44.817	-218.627	-1.903
Max Sx	749	14 1.25DC(FTG WT)+STRENGTH IB+	0.544	-0.556	-0.020	-0.011	**0.026**	-272.454	-265.587	6.094
Min Sx	728	14 1.25DC(FTG WT)+STRENGTH IB+	-0.223	0.210	0.015	0.006	**-0.023**	13.788	6.920	-186.635
Max Mx	739	14 1.25DC(FTG WT)+STRENGTH IB+	-0.053	-0.024	0.006	-0.014	0.002	**44.817**	-218.627	-1.903
Min Mx	747	14 1.25DC(FTG WT)+STRENGTH IB+	-0.284	-0.493	0.004	-0.010	0.008	**-482.207**	-309.364	56.695
Max My	724	14 1.25DC(FTG WT)+STRENGTH IB+	0.033	0.065	-0.007	0.003	0.000	-234.459	**43.410**	-6.390
Min My	737	14 1.25DC(FTG WT)+STRENGTH IB+	0.546	0.291	-0.025	0.009	-0.006	-347.780	**-491.229**	38.685
Max Mx	746	13 1.25DC(FTG WT)+STRENGTH IA+	-0.274	-0.241	0.013	-0.005	0.010	2.801	-20.197	**171.766**
Min Mx	726	14 1.25DC(FTG WT)+STRENGTH IB+	-0.223	0.210	0.015	0.006	-0.023	13.788	6.920	**-186.635**

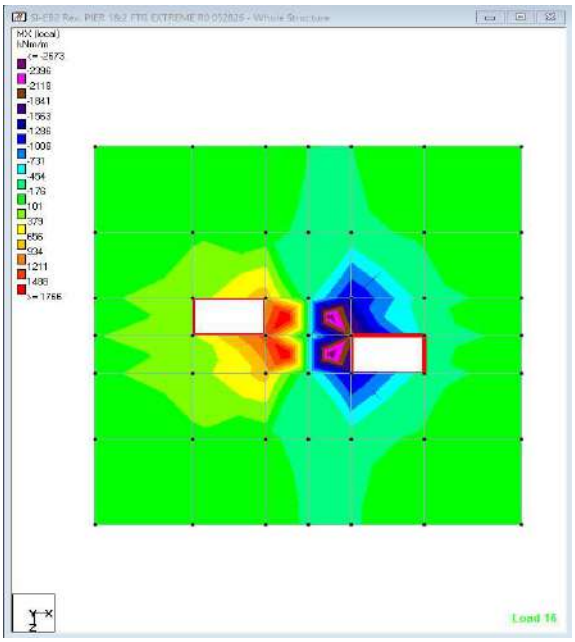
SERVICE EVENT

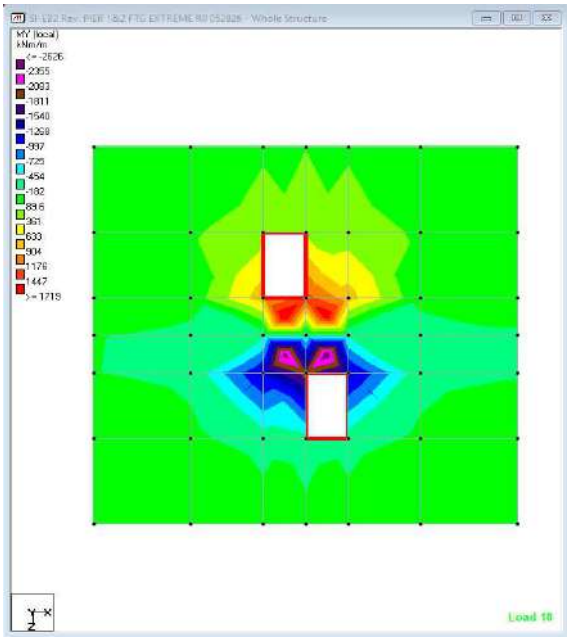
SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Plate Center Stress: G4: _PILECAP

Shear, Membrane and Bending | Summary | Principal and Von Mis | Summary | Global Moments | Combined Stresses

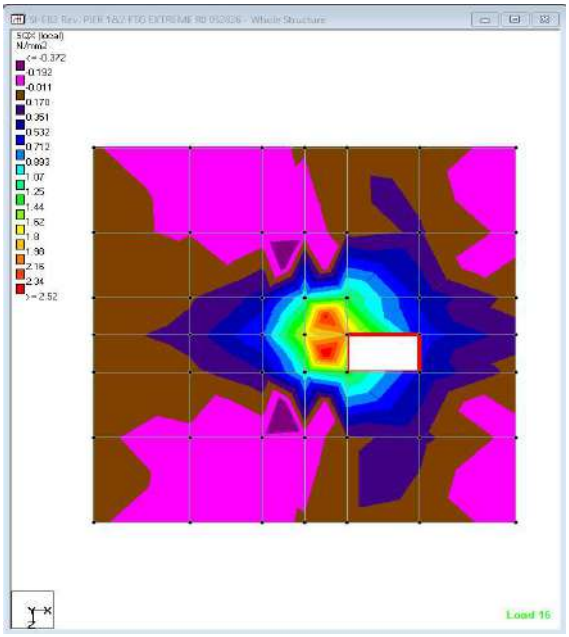
	Plate	L/C	Shear		Membrane			Bending Moment		
			SQX (local) N/mm2	SQY (local) N/mm2	SX (local) N/mm2	SY (local) N/mm2	SXY (local) N/mm2	Mx kNm/m	My kNm/m	Mxy kNm/m
Max Qx	743	20 1.0DC(FTG WT)+SERVICE IB+1.0	**0.409**	-0.087	-0.014	0.003	0.008	-234.500	-333.844	11.339
Min Qx	746	20 1.0DC(FTG WT)+SERVICE IB+1.0	**-0.286**	-0.217	0.007	-0.003	0.006	-47.116	-119.538	71.777
Max Qy	730	19 1.0DC(FTG WT)+SERVICE IA+1.0	0.048	**0.365**	-0.005	0.004	-0.005	-313.772	-190.467	-9.268
Min Qy	748	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.057	**-0.384**	-0.001	-0.004	0.009	-320.602	-206.270	12.958
Max Sx	740	19 1.0DC(FTG WT)+SERVICE IA+1.0	-0.144	-0.066	**0.013**	-0.007	0.003	-7.204	-264.703	82.518
Min Sx	737	19 1.0DC(FTG WT)+SERVICE IA+1.0	0.405	0.092	**-0.014**	0.005	-0.004	-233.683	-336.386	-6.734
Max Sy	736	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.097	0.062	-0.004	**0.006**	-0.001	-4.947	-212.458	27.105
Min Sy	739	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.043	-0.023	0.004	**-0.008**	0.001	27.534	-160.719	3.109
Max Sx	749	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.350	-0.369	-0.012	-0.006	**0.015**	-172.357	-164.632	-12.035
Min Sx	728	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.187	0.176	0.008	0.003	**-0.012**	-12.835	-20.560	-116.847
Max Mx	739	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.043	-0.023	0.004	-0.008	0.001	**27.534**	-160.719	3.109
Min Mx	747	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.174	-0.327	0.002	-0.006	0.005	**-332.270**	-191.810	39.712
Max My	724	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.028	0.054	-0.004	0.002	0.000	-174.050	**25.437**	-0.541
Min My	737	20 1.0DC(FTG WT)+SERVICE IB+1.0	0.369	0.176	-0.014	0.005	-0.003	-223.499	**-340.080**	25.514
Max Mx	746	19 1.0DC(FTG WT)+SERVICE IA+1.0	-0.204	-0.189	0.007	-0.003	0.006	-15.670	-24.409	**113.669**
Min Mx	726	20 1.0DC(FTG WT)+SERVICE IB+1.0	-0.187	0.176	0.008	0.003	-0.012	-12.835	-20.560	**-116.847**

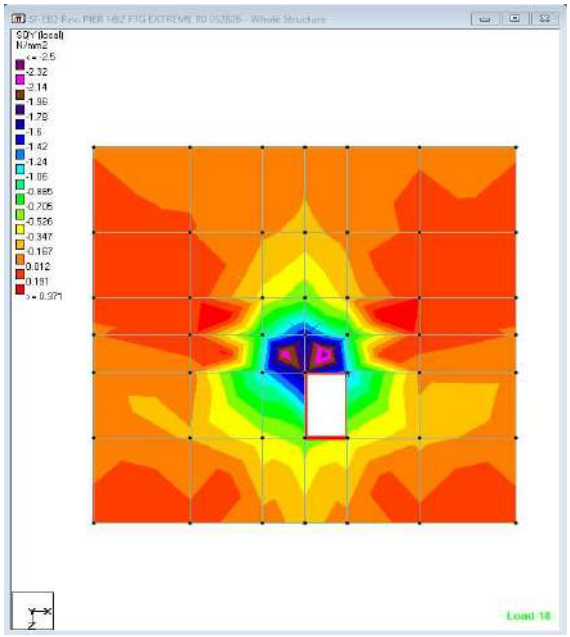
GOVERNING EVENTS





Longitudinal Moment: *(per meter width)*		*Add Torsion Effect Mxy*		Transverse Moment: *(per meter width)*		*Add Torsion Effect Mxy*	
Max. +Mx_top =	443.88 kNm/m		152.18 kNm/m	Max. +My_top =	430.54 kNm/m		152.18 kNm/m
Max. -Mx_bot =	680.17 kNm/m		94.44 kNm/m	Max. -My_bot =	666.84 kNm/m		94.44 kNm/m





Longitudinal Shear:
Max. abs. SQx = 0.914 N/mm2
SQx (1000kPa/1MPa)(hp) = 1096.80 kN/m *(per meter width)*
hp = height of pilecap

Transverse Shear:
Max. abs. SQy = 0.902 N/mm2
SQy (1000kPa/1MPa)(hp) = 1082.40 kN/m *(per meter width)*
hp = height of pilecap

d. Pile Cap Resistance

Concrete unit weight,	Yc =	24	kN/m^3
Concrete compressive strength @ 28 days,	f'c =	27.6	MPa
Modulus elasticity of concrete @ 28 days,	Ec =	27350	MPa
Yield strength of steel reinforecement,	fy =	414	MPa
Modulus elasticity of steel reinforcement,	Es =	200000	MPa
Thickness of pile cap,	hp =	1.20	m
Unit width,	b =	1.00	m

1. Longitudinal Direction:

1.a Flexural Requirement for Top Reinforcement:

Design moment per unit width,	Mx_top =	596.05	kN-m	
Flexural cracking variability factor, $Y1$		1.6	*for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, $Y3$		0.67	*for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/c
Cracking Moment, M_{cr}				
$M_{cr}=Y_3(Y_1*fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		792.75	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		792.75	kN-m	
Design moment per unit width,	**Mx_top =**	**792.75**	**kN-m**	
Concrete cover,	cc_top =	75	mm	
Size of reinforcement,	db_top =	32	mm	Area of bar reinf., Ab = 804.248 mm^2
Effective depth of concrete,	de_top =	1093	mm	
Flexural Resistance factor,	Φ =	0.9		

Steel ratio β_1 = 0.85

β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

	m1 =	0.0567		m1=0.85*f'c/fy	
For required steel ratio, ρ	m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)	$Rn = \frac{Md}{\phi bv * (d_e)^2}$
	Rn =	0.737	MPa		
	ρ =	0.0018		ρ=m1*(1-sqrt(1-m2Rn))	

Area of main reinforcement required, As = ρbd	As =	1978.17	mm2					
spacing required of top bars, Ab(b)/As = sx_top	sx_top =	406.56	mm	**Say, s =**	**200**	**mm -**	**32**	**mm Ø Top Bars**
				Longitudinal TOP:	***USE 32mmØ @ 200mm***			
Area of main reinforcement provided, Ab(b)/s = Asp	Asp =	4021.24	mm2					
a = (Asp fy) / (0.85 f'c b)	a =	70.963	mm2	*Depth of compression block*				
Flexural resistance, Mr = Φ [Asp fy (de - a/2)]	**Mr =**	**1584.49**	**kN-m**					
	c/d =	2.00	**OK!**					

1.b Flexural Requirement for Bottom Reinforcement

Design moment per unit width,	Mx_bot =	774.62	kN-m	
Flexural cracking variability factor, γ_1		1.6	**for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, γ_3		0.67	**for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/c
Cracking Moment, M_{cr}				
$M_{cr}=\gamma_3(\gamma_1 * fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		1030.24	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		1021.84	kN-m	
Design moment per unit width,	**Mx_bot =**	**1021.84**	**kN-m**	

Concrete cover,	cc_bot =	200	mm	*(150mm pile embedment + 50mm clearance)*	
Size of reinforcement,	db_bot =	36	mm	Area of bar reinf., Ab =	1017.88 mm^2
Effective depth of concrete,	de_bot =	964	mm		
Flexural Resistance factor,	Φ =	0.9			

Steel ratio β_1 = 0.85

β_1 Coefficient Criterion:

the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

	m1 =	0.0567		m1=0.85*f'c/fy	
For required steel ratio, ρ	m2 =	0.0853	mm2/N	m2=2/(0.85*f'c)	$Rn = \frac{Md}{\phi bv * (d_e)^2}$
	Rn =	1.222	MPa		
	ρ =	0.0030		ρ=m1*(1-sqrt(1-m2Rn))	

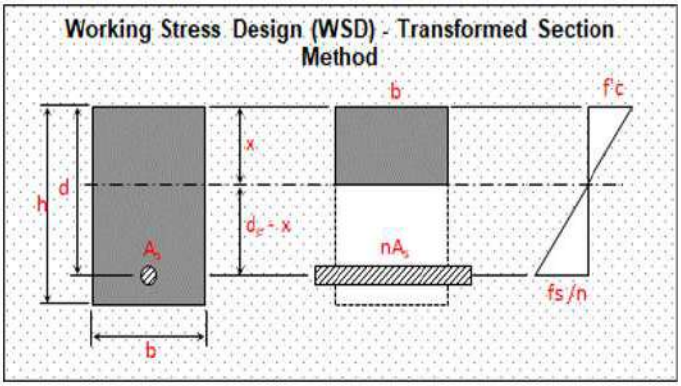
Area of main reinforcement required, As = ρbd	As =	2923.08	mm2					
spacing required of bottom bars, Ab(b)/As = sx_bot	sx_bot =	348.22	mm	**Say, s =**	**200**	**mm -**	**36**	**mm Ø Bottom Bars**
				Longitudinal BOT.:	***USE 36mmØ @ 200mm***			
Area of main reinforcement provided, Ab(b)/s = Asp	Asp =	5089.38	mm2					
a = (Asp fy) / (0.85 f'c b)	a =	89.8126	mm2	*Depth of compression block*				
Flexural resistance, Mr = Φ [Asp fy (de - a/2)]	**Mr =**	**1742.88**	**kN-m**					
	c/d =	1.71	**OK!**					

Control of cracking by distribution of reinforcement: • DGCS 12.4.3.4

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1 *(adopt bot., max neg moment), Ms=*	371.98	kN-m	*(plus corresponding Mxy)* • DGCS 12.1.1.6
80% of Modulus of rupture, fr = 0.52*sqrt(f'c)	2.185	MPa	
Section modulus, S_c	2.4E+08	mm^3	
Tension in the cross section, fs = Ms/Sc	1.550	MPa	

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



b*x*x/2 = nAs*(de-x)
bx^2/2=nAsde-nAsx
bx^2/2+nAsx-nAsde=0
a = b/2
b = nAs
c = -nAs de

Extreme tension fiber to center of flexural reinforcement, d_c	218	mm
Overall thickness of component, h	1200	mm
Compression fiber to the centroid of extreme tension steel, d_e	964	mm

• by quadratic equation
determine x:
a= 500.00

Neutral axis to extreme compression fiber, x	233.23	mm	b=	3.7E+04
Modulus elasticity of steel, E_s	200	Gpa	c=	-3.6E+07
Modulus elasticity of concrete, E_c	27.35	GPa		
Modular ratio, n	7.31			
Cracked section moment of inertia of section, I NA = $(bx^3/3) + nAs(de-x)^2$	2.4E+10	mm^4		
Exposure factor, Y_e	0.75			
Exposure condition: Class 2				
Tensile stress in steel reinforcement at the service limit, fss = nM(de-x)/I	82.47	MPa		
or fss = Ms /(As (de-x/3))				
βs = 1+ dc /(0.7 (*h* −dc))	1.32			
The spacing shall satisfy: s≤123000γe/βsfss −2dc	413	mm		
Provided spacing: **200** mm **SATISFIED!**				

Check for maximum reinforcement limit:

"The maximum reinforcement requirements ensure the section has sufficient ductility and is not overreinforced"

Compression fiber to neutral axis, c = a/β	c_top =	83.4859	mm2	c_bot =	105.662 mm2
Maximum reinforcement ratio c/de =	max c/de_top =	0.420		max c/de_bot =	0.420
	actual c/de_top =	0.076	**OK!**	actual c/de_bot =	0.110 **OK!**

Check for spacing requirements:

•Check for minimum spacing of reinforcement — DGCS 12.7.3.1

For cast in place concrete, clear distance between parallel bars in a layer shall not be less than:

•1.5 x nominal diam of bars	= 54 mm	s_min =	max of 3 values		
•1.5 x max. size of aggregates	= 37.5 mm	s_min =	54 mm		
•38mm	= 38 mm	s_provt =	200 mm	**OK in min. spacing requirement!**	
		s_provb =	200 mm	**OK in min. spacing requirement!**	

•Check for maximum spacing of reinforcement (for walls and slabs) — DGCS 12.7.3.2

•s < 1.5 x hf	= 1800 mm	s_max =	min of 2 values	
•450mm	= 450 mm	s_min =	450 mm	
		s_provt =	200 mm	**OK in max. spacing requirement!**
		s_provb =	200 mm	**OK in max. spacing requirement!**

1.c Shear Requirement in Longitudinal Direction:

Ultimate shear based on demand, *Vxu* — **1096.8** kN *(unit width, b=1.0m)*

Effective shear depth, d_v — 919.1 mm

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2)	919.1	mm
dv2= $0.9d_e$	867.6	mm
dv3= 0.72h.cap	864.0	mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*			
Angle of inclination of diagonal compressive stresses, θ =	45	deg			
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9				
Diameter of cross ties/J-bars, ds =	20	mm			
Number of legs of ties, n = b / spacing along b	2.50	*(@ every no. of top bar)*	s_top =	200 mm	every no. of bars = 2
Total area of shear reinforcement, Av =	785.398	mm2			
Non-prestress, Vp =	0	kN			
Required nominal shear resistance, V_n	1218.7	kN			
**shall be determined as the lesser of:*					
V_n=Vc+Vs+Vp=Vu/Øv	1218.7	kN			
V_n=0.25fc'$b_v d_v$+Vp	6341.7	kN			
Shear resistance provided by concrete, V_c	801.5	kN	Vc= 0.083*β(f'c^0.5)bv dv		
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	417.1	kN			
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	716	mm	**Say,**	**400**	**mm spacing along span,**
				20	**mm Ø ties at**
				400	**mm spacing along width**
				Longitudinal	

2. Transverse Direction:

2.a Flexural Requirement for Top Reinforcement:

Design moment per unit width,	My_top =	582.72	kN-m	
Flexural cracking variability factor,*Y1*		1.6	**for all other concrete*	
Ratio of specified min. to ult. tensile strength of steel, *Y3*		0.67	**for A615, 414MPa steel*	
Modulus of rupture, f_r		3.310	MPa	fr=0.63*sqrt(f'c)
Section modulus, S_c		2.4E+08	mm^3	Sc=I/*c*
Cracking Moment, M_{cr}				
$M_{cr}=y_3(y_1*fr)\ S_c$		851.53	kN-m	
1.20 Mcr =		1021.84	kN-m	
1.33*M.u_neg =		775.02	kN-m	
Minimum design moment, Min(1.2Mcr or 1.33Mu)		775.02	kN-m	

Design moment per unit width, **My_top = 775.02 kN-m**

Concrete cover, cc_top = 75 mm
Size of reinforcement, db_top = 32 mm — Area of bar reinf., Ab = 804.248 mm^2
Effective depth of concrete, de_top = 1045 mm
Flexural Resistance factor, Φ = 0.9

Steel ratio β_1 = 0.85
β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ
m1 = 0.0567 — m1=0.85*f'c/fy
m2 = 0.0853 mm2/N — m2=2/(0.85*f'c)
Rn = 0.789 MPa — $Rn = \frac{Md}{\phi bv * (d_e)^2}$
ρ = 0.0019 — ρ=m1*(1-sqrt(1-m2Rn))

Area of main reinforcement required, As = ρbd — As = 2025.09 mm2
spacing required of top bars, Ab(b)/As = sy_top — sy_top = 397.14 mm — **Say, s = 200 mm - 32 mm Ø Top Bars**
Transverse TOP: *USE 32mmØ @ 200mm*

Area of main reinforcement provided, Ab(b)/s = Asp — Asp = 4021.24 mm2
a = (Asp fy) / (0.85 f'c b) — a = 70.963 mm2 — *Depth of compression block*
Flexural resistance, Mr = Φ [Asp fy (de - a/2)] — **Mr = 1512.58 kN-m**
c/d = 1.95 **OK!**

2.b Flexural Requirement for Bottom Reinforcement

Design moment per unit width, My_bot = 761.28 kN-m
Flexural cracking variability factor, *Y1* = 1.6 **for all other concrete*
Ratio of specified min. to ult. tensile strength of steel, *Y3* = 0.67 **for A615, 414MPa steel*
Modulus of rupture, f_r = 3.310 MPa — fr=0.63*sqrt(f'c)
Section modulus, S_c = 2.4E+08 mm^3 — Sc=I/*c*
Cracking Moment, M_{cr}
$M_{cr}=\gamma_3(\gamma_1 * fr)\ S_c$ = 851.53 kN-m
1.20 Mcr = 1021.84 kN-m
1.33*M.u_neg = 1012.50 kN-m

Minimum design moment, Min(1.2Mcr or 1.33Mu) = 1012.50 kN-m

Design moment per unit width, **My_bot = 1012.50 kN-m**

Concrete cover, cc_bot = 200 mm — *(150mm pile embedment + 50mm clearance)*
Size of reinforcement, db_bot = 36 mm — Area of bar reinf., Ab = 1017.88 mm^2
Effective depth of concrete, de_bot = 910 mm
Flexural Resistance factor, Φ = 0.9

Steel ratio β_1 = 0.85
β_1 Coefficient Criterion:
the factor β_1 shall be taken as 0.85 for concrete strengths not exceeding 28MPa. For concrete strength exceeding 28MPa, β_1 shall be reduced at a rate of 0.05 for each 7MPa strength excess of 28MPa but not less than 0.65

For required steel ratio, ρ
m1 = 0.0567 — m1=0.85*f'c/fy
m2 = 0.0853 mm2/N — m2=2/(0.85*f'c)
Rn = 1.359 MPa — $Rn = \frac{Md}{\phi bv * (d_e)^2}$
ρ = 0.0034 — ρ=m1*(1-sqrt(1-m2Rn))

Area of main reinforcement required, As = ρbd — As = 3078.01 mm2
spacing required of bottom bars, Ab(b)/As = sy_bot — sy_bot = 330.69 mm — **Say, s = 200 mm - 36 mm Ø Bottom Bars**
Transverse BOT.: *USE 36mmØ @ 200mm*

Area of main reinforcement provided, Ab(b)/s = Asp — Asp = 5089.38 mm2
a = (Asp fy) / (0.85 f'c b) — a = 89.8126 mm2 — *Depth of compression block*
Flexural resistance, Mr = Φ [Asp fy (de - a/2)] — **Mr = 1640.48 kN-m**
c/d = 1.62 **OK!**

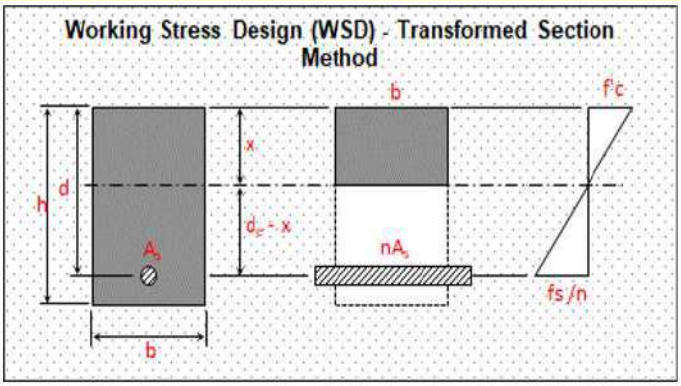
Control of cracking by distribution of reinforcement: • DGCS 12.4.3.4

Applies to all reinforcements of concrete that exceeds 80% of the modulus of rupture, except deck slabs.

Moment demand at Service 1 *(adopt bot., max neg moment), Ms=*	365.59	kN-m	*(plus corresponding Mxy)*
80% of Modulus of rupture, fr = 0.52*sqrt(f'c)	2.185	MPa	
Section modulus, S_c	2.4E+08	mm^3	
Tension in the cross section, fs = Ms/Sc	1.523	MPa	

• DGCS 12.1.1.6

Tensile stress in steel does not exceed 80% of the modulus of rupture, this provision does not need to be satisfied.



b*x*x/2 = nAs*(de-x)
bx^2/2=nAsde-nAsx
bx^2/2+nAsx-nAsde=0
a = b/2
b = nAs
c = -nAs de

Extreme tension fiber to center of flexural reinforcement, d_c	254	mm	
Overall thickness of component, h	1200	mm	
Compression fiber to the centroid of extreme tension steel, d_e	910	mm	
Neutral axis to extreme compression fiber, x	225.69	mm	
Modulus elasticity of steel, E_s	200	Gpa	
Modulus elasticity of concrete, E_c	27.35	GPa	
Modular ratio, n	7.31		
Cracked section moment of inertia of section, I NA = $(bx^3/3) + nAs(de-x)^2$	2.1E+10	mm^4	
Exposure factor, Y_e	0.75		
Exposure condition: Class 2			
Tensile stress in steel reinforcement at the service limit, fss = nM(de-x)/I	86.05	MPa	
or fss = Ms /(As (de-x/3))			
βs = 1+ dc /(0.7 (*h* −dc))	1.38		
The spacing shall satisfy: s≤123000γe/βsfss −2dc	267	mm	

• by quadratic equation determine x:

a= 500.00
b= 3.7E+04
c= -3.4E+07

Provided spacing: **200** mm **SATISFIED!**

Check for maximum reinforcement limit:

"The maximum reinforcement requirements ensure the section has sufficient ductility and is not overreinforced"

Compression fiber to neutral axis, c = a/β	c_top =	83.4859	mm2	c_bot =	105.662	mm2
Maximum reinforcement ratio c/de =	max c/de_top =	0.420		max c/de_bot =	0.420	
	actual c/de_top =	0.080	**OK!**	actual c/de_bot =	0.116	**OK!**

Check for spacing requirements:

•Check for minimum spacing of reinforcement DGCS 12.7.3.1

For cast in place concrete, clear distance between parallel bars in a layer shall not be less than:

•1.5 x nominal diam of bars =	54 mm	s_min =	max of 3 values		
•1.5 x max. size of aggregates =	37.5 mm	s_min =	54 mm		
•38mm =	38 mm	s_provt =	200 mm	**OK in min. spacing requirement!**	
		s_provb =	200 mm	**OK in min. spacing requirement!**	

•Check for maximum spacing of reinforcement (for walls and slabs) DGCS 12.7.3.2

•s < 1.5 x hf =	1800 mm	s_max =	min of 2 values		
•450mm =	450 mm	s_min =	450 mm		
		s_provt =	200 mm	**OK in max. spacing requirement!**	
		s_provb =	200 mm	**OK in max. spacing requirement!**	

2.c Shear Requirement in Transverse Direction:

Ultimate shear based on demand, *Vyu* **1082.4** kN *(unit width, b=1.0m)*

Effective shear depth, d_v 865.1 mm

Taken as the distance measured perpendicular to the neutral axis, between the resultants of the tensile and compressive forces due to flexure; it need not to be taken to be less than the greater of 0.9de or 0.72h

dv1 = (de - a/2)	865.1	mm
dv2= 0.9d_e	819.0	mm
dv3= 0.72h.cap	864.0	mm

Factor relating to longitudinal strain on the shear capacity of conc., β =	2.0	*(for non-prestress)*
Angle of inclination of diagonal compressive stresses, θ =	45	deg
Shear resistance factor (AASHTO LRFD 5.5.4.2.1), Øv =	0.9	
Diameter of cross ties/J-bars, ds =	20	mm

Number of legs of ties, n = b / spacing along b	3.33	*(@ every no. of top bar)*	s_top =	150	mm	every no. of bars = 2
Total area of shear reinforcement, Av =	1047.2	mm2				
Non-prestress, Vp =	0	kN				
Required nominal shear resistance, V_n	1202.7	kN				
**shall be determined as the lesser of:*						
V_n=Vc+Vs+Vp=Vu/Øv	1202.7	kN				
V_n=0.25fc'$b_v d_v$+Vp	5969.1	kN				
Shear resistance provided by concrete, V_c	754.4	kN	Vc= 0.083*β(f'c^0.5)bv dv			
Shear strength required by shear reinforcement, Vs = Vn-Vc-Vp	448.2	kN				
Required shear reinforcement spacing, s = Av fy dv cot(θ) / Vs	837	mm	**Say.**	**400**	**mm spacing along span,**	
				20	**mm Ø ties at**	
				300	**mm spacing along width**	
				Transverse		

3. Shrinkage and Temperature Reinforcement (Side Bars):

Reinforcement for shrinkage and temperature stresses shall be provided near surfaces of concrete exposed to daily temperature changes and in structural mass concrete. For the pile cap, bars shall be provided at the perimeter side of the pile cap.

DGCS 12.7.8

Ast => 0.75 MPa x (bxh)/2(b+h)fy
where:

b = least width of component section, mm	5400	mm
h = least thickness of component section, mm	1200	mm
a) Ast => 0.75 MPa x (bxh)/2(b+h)fy	0.889	mm^2/mm
b) Limitations: 0.223<=Ast<=1.27		
therefore Ast =	0.889	mm^2/mm
try diameter db =	20	mm
Ab =	314	mm^2
spacing, S = Ab/ Ast =	353	mm
say S_prov =	**250**	**mm -** **20** **mm Ø Side Bars** ***(MINIMUM Only)***
Ast_act = Ab/S_prov =	1.257	mm^2/mm **OK! Satisfied limitations.**

where:

a) S_prov should not be greater than 3 x thickness, or 450mm	450	mm
b) S_prov should not be greater than 300 for walls and footings > 450mm thk.	300	mm
c) S_prov should be 300mm for other components > 900mm thick.	300	mm
S_max =	300	mm

Therefore, spacing provided for shrinkage and temperature bars are OK!

∴ PROVIDED: 25 mm Ø @ 200 mm Side Bars

4. Verification of Two-Way Shear Action (Punching Shear)

4.a Punching Shear Requirement for Pile:

Maximum reaction of single pile, V_u	**1811.4**	kN	*(From StaadPro Pier Footing Analysis)*
V_n *of the concrete shall be taken as:*			
$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv \le 0.33\sqrt{(fc')}\,bodv$			
Effective shear depth of footing, d_v	865.1	mm	
Dimension of pile, dpile	1000.0	mm	
Ratio of long to short direction of pile, βc	1		*(i.e. for square piles)*
Perimeter of the critical section, b_o	7460	mm	*(dv/2 distance around pile)*
Nominal shear capacity of concrete, V_n	11189.0	kN	
a)$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv$	1.7E+04	kN	
b) $Vn= 0.33\sqrt{(fc')}\,bodv$	11189.0	kN	
Resistance factor for normal weight concrete, Ø	0.9		
Ultimate shear capacity of section, $ØV_n$	**10070.1**	kN	**OK!**
c/d =	5.56		

Pile cap is adequate for shear due to maximum reaction of pile.

4.b Punching Shear Requirement for Column:

Maximum reaction of single column, P_{max}	**2686**	kN	*(From Design of Column)*
V_n *of the concrete shall be taken as:*			
$Vn=(0.17+0.33/\beta c)\sqrt{(fc')}\,bodv \le 0.33\sqrt{(fc')}\,bodv$			
Effective shear depth of footing, d_v	865.1	mm	
Diameter of column, dcol	1200.0	mm	
Ratio of long to short direction of pile, βc	1		*(i.e. for square col.)*
Perimeter of the critical section, b_o	6488	mm	*(dv/2 distance around column)*
Nominal shear capacity of concrete, V_c	9730.2	kN	

a) $Vn=(0.17+0.33/\beta c)\ \sqrt{(fc')}\ bodv$	1.5E+04	kN	
b) $Vn= 0.33\ \sqrt{(fc')}\ bodv$	9730.2	kN	
Resistance factor for normal weight concrete, Ø	0.9		
Ultimate shear capacity of section (concrete only), $ØV_c$	**8757**	kN	**OK!** *(Concrete only)*

Analyze transverse reinforcement shear capacity on critical section (b_o perimeter):

Approximate spacing of shear reinf. on both directions, s_{ties}	335.41	mm	
$s_{ties} = sqrt(s_{ties_longtl}^2 + s_{ties_transv}^2)$			
Diameter of cross ties/J-bars, ds =	20	mm	
Number of legs of ties, n = bo / s_{ties}	19.34		
Total area of shear reinforcement, Av =	6076.64	mm2	
Shear reinforcement capacity, Vs = ØAv fy dv / s_{ties}	**5840**	kN	
Ultimate shear capacity of section, $ØV_n$ =Ø(Vc + Vs)	**14597**	kN	**OK!** *(Concrete and Shear Reinf. Capacities)*
c/d =	5.43		

Pile cap is adequate for shear due to maximum reaction of column.

5.8A DESIGN OF PILES - PIER 1

5.8.1 Determine the pile springs and geometric properties

a. Horizontal pile spring constant (K_h)

Horizontal Modulus of Subgrade Reaction
(See New Geotechnical Report)

Borehole No., **BH-34** (P1)
Pile width, D = 1.000 m

STRENGTH EVENT				ks = Kh*D*z	EXTEREME EVENT				ks = Kh*D*z
BH depth (m)	Thickness (m)	Depth of pile z, (m)	Kh (kN/m^3)	Soil Spring Stiffness, ks (kN/m)	BH depth (m)	Thickness (m)	Depth of pile z, (m)	Kh (kN/m^3)	Soil Spring Stiffness, ks (kN/m)
0.0				*NGL*	0.0				*NGL*
1.5					1.5				
3.0		EMBED and PILECAP		EMBED and PILECAP	3.0		EMBED and PILECAP		EMBED and PILECAP
4.5					4.5				
6.0	0.0				6.0	0.0			
7.5	1.5	1.5	4,956	7434	7.5	1.5	1.5	59,450	89175
9.0	1.5	3.0	4,956	7434	9.0	1.5	3.0	59,450	89175
10.5	1.5	4.5	4,956	7434	10.5	1.5	4.5	36,331	54497
12.0	1.5	6.0	1,803	2705	12.0	1.5	6.0	26,785	40178
13.5	1.5	7.5	2,821	4232	13.5	1.5	7.5	38,029	57044
15.0	1.5	9.0	839	1259	15.0	1.5	9.0	9,830	14745
16.5	1.5	10.5	8,325	12488	16.5	1.5	10.5	51,338	77007
18.0	1.5	12.0	11,831	17747	18.0	1.5	12.0	67,807	101711
19.5	1.5	13.5	15,438	34736	19.5	1.5	13.5	132,492	298107
22.5	3.0	16.5	16,659	49977	22.5	3.0	16.5	139,390	418170
25.5	3.0	19.5	38,437	115311	25.5	3.0	19.5	152,163	456489
28.5	3.0	22.5	55,993	167979	28.5	3.0	22.5	204,362	613086
31.5	3.0	25.5	65,678	197034	31.5	3.0	25.5	227,296	681888
34.5	3.0	28.5	69,871	262016	34.5	3.0	28.5	236,871	888266
39.0	4.5	33.0	89,737	403817	39.0	4.5	33.0	279,873	1259429
43.5	4.5	37.5	96,939	436226	43.5	4.5	37.5	294,653	1325939
48.0	4.5	42.0	104,191	703289	48.0	4.5	42.0	309,172	2086911
57.0	9.0	51.0	110,027	495122	57.0	9.0	51.0	320,612	1442754

Pile end vertical spring	
Kv (kN/m^3)	Soil Axial Spring, kv (kN/m)
157,009	123315

Pile end vertical spring	
Kh (kN/m^3)	Soil Axial Spring, kv (kN/m)
413,867	325050

b. Geometric properties of piles

Select Pile Section :
-- Circular Section
Square Section

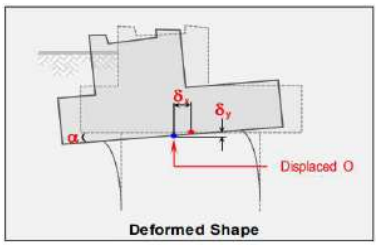
Select Pile Installation Method :
Driven Piles (Blow Method)
Driven Piles (Vibro-Hammer Method)
Cast-in-place RC Piles
-- Bored Piles
Pre-Boring Piles
Steel Pile Soil Cement Piles

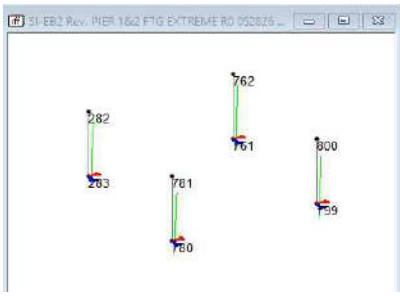
Input Pile Dimension : Diameter	1.000	m
Input Number of Piles	8	piles
Input Pile Length :	51.0	m
Calculate Section Properties :		
Cross-section Area	0.785	m^2
Perimeter of Pile :	3.142	m
Pile Moment of Inertia :	0.049	m^4
Pile Flexural Stiffness :	1.34E+06	kN-m^2
Concrete Material Properties :		
Design Compressive Strength at 28th day	27.6	N/mm^2
Unit Density for Concrete	2400	kg/m^3
Unit weight for Reinforced Concrete	24	kN/m^3
Young's Modulus of Elasticity	2.74E+07	kN/m^2
Reinforcement Material Properties :		
Minimum Yield Strength	414	N/mm^2
Ultimate Tensile Strength	620	N/mm^2
Young's Modulus of Elasticity	2.00E+08	kN/m^2

5.8.2 Determine the pile displacement, reaction forces, and critical forces along piles

a. Displacement of pile heads:

Displacement			
For Governing Load Case			
Extreme Event I			
Location	Displacement (Max. Values)		
	Horizontal [Resultant, sqrt(dx2+dz2)]	Vertical	Rotational
	d_x	d_y	a
	mm	mm	rad
Pile Head	**7.711**	**14.804**	**0.003**





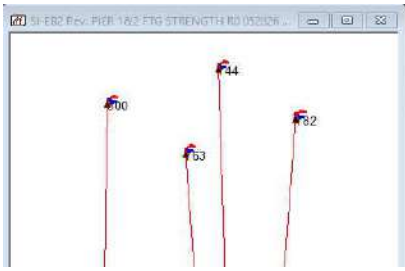
SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Node Displacements: G3: _PILEHEAD

All | Summary

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	rY rad	rZ rad
Max X	800	16 1.25DC(F	**7.711**	-14.804	0.434	16.698	0.000	-0.000	-0.003
Min X	781	18 1.25DC(F	**0.472**	-14.106	7.546	16.035	0.003	-0.000	-0.000
Max Y	282	16 1.25DC(F	7.856	**-5.522**	0.379	9.447	0.000	0.000	-0.003
Min Y	800	16 1.25DC(F	7.711	**-14.804**	0.434	16.698	0.000	-0.000	-0.003
Max Z	800	18 1.25DC(F	0.568	-14.786	**7.550**	16.612	0.003	0.000	-0.000
Min Z	782	15 1.25DC(F	7.716	-14.632	**-0.092**	16.539	0.000	0.000	-0.003
Max rX	782	18 1.25DC(F	0.526	-6.220	7.491	9.751	**0.003**	0.000	-0.000
Min rX	800	15 1.25DC(F	7.709	-14.457	0.005	16.384	**-0.000**	-0.000	-0.003
Max rY	782	16 1.25DC(F	7.708	-14.286	0.337	16.235	0.000	**0.000**	-0.003
Min rY	781	17 1.25DC(F	0.474	-13.760	7.117	15.498	0.003	**-0.000**	-0.000
Max rZ	800	18 1.25DC(F	0.568	-14.786	7.550	16.612	0.003	0.000	**-0.000**
Min rZ	781	16 1.25DC(F	7.853	-6.041	0.391	9.758	0.000	-0.000	**-0.003**
Max Rs	800	16 1.25DC(F	7.711	-14.804	0.434	**16.698**	0.000	-0.000	-0.003

b. Reaction forces at bottom of piles:

at SERVICE:

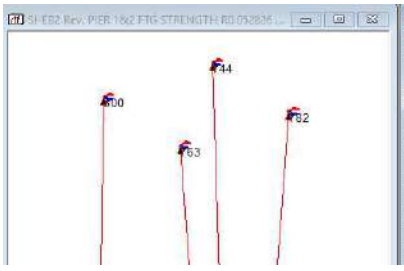


SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Support Reactions: G3: _PILEBOTTOM

All | Summary | Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	20 1.0DC(FTG WT)+SERVICE IB-	**0.000**	1724.333	0.000	0.000	0.000	0.000
Min Fx	782	20 1.0DC(FTG WT)+SERVICE IB-	**-0.000**	1972.136	-0.000	0.000	0.000	0.000
Max Fy	782	20 1.0DC(FTG WT)+SERVICE IB+	-0.000	**1972.136**	-0.000	0.000	0.000	0.000
Min Fy	300	20 1.0DC(FTG WT)+SERVICE IB+	0.000	**1573.307**	0.000	0.000	0.000	0.000
Max Fz	744	20 1.0DC(FTG WT)+SERVICE IB-	-0.000	1821.105	**0.000**	0.000	0.000	0.000
Min Fz	782	19 1.0DC(FTG WT)+SERVICE IA+	-0.000	1858.784	**-0.000**	0.000	0.000	0.000
Max Mx	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	**0.000**	0.000	0.000
Min Mx	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	**0.000**	0.000	0.000
Max My	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	0.000	**0.000**	0.000
Min My	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	0.000	**0.000**	0.000
Max Mz	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	0.000	0.000	**0.000**
Min Mz	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1688.674	-0.000	0.000	0.000	**0.000**

at STRENGTH:

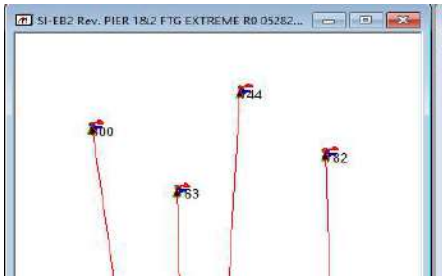


SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Support Reactions: G3: _PILEBOTTOM

All | Summary | Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	14 1.25DC(FTG WT)+STRENGTH IB+	**0.000**	2278.625	0.000	0.000	0.000	0.000
Min Fx	782	14 1.25DC(FTG WT)+STRENGTH IB+	**-0.000**	2712.263	0.000	0.000	0.000	0.000
Max Fy	782	14 1.25DC(FTG WT)+STRENGTH IB+	-0.000	**2712.263**	0.000	0.000	0.000	0.000
Min Fy	300	14 1.25DC(FTG WT)+STRENGTH IB+	0.000	**1982.682**	0.000	0.000	0.000	0.000
Max Fz	744	14 1.25DC(FTG WT)+STRENGTH IB+	-0.000	2416.320	**0.000**	0.000	0.000	0.000
Min Fz	782	13 1.25DC(FTG WT)+STRENGTH IA+	-0.000	2510.371	**-0.000**	0.000	0.000	0.000
Max Mx	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	**0.000**	0.000	0.000
Min Mx	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	**0.000**	0.000	0.000
Max My	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	**0.000**	0.000
Min My	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	**0.000**	0.000
Max Mz	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	0.000	**0.000**
Min Mz	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	0.000	**0.000**

at EXTREME:



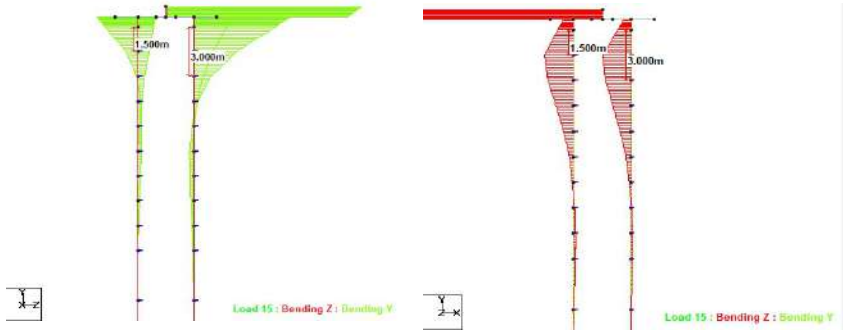
SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Support Reactions: G4: _PILEBOTTOM

All | Summary | Envelope

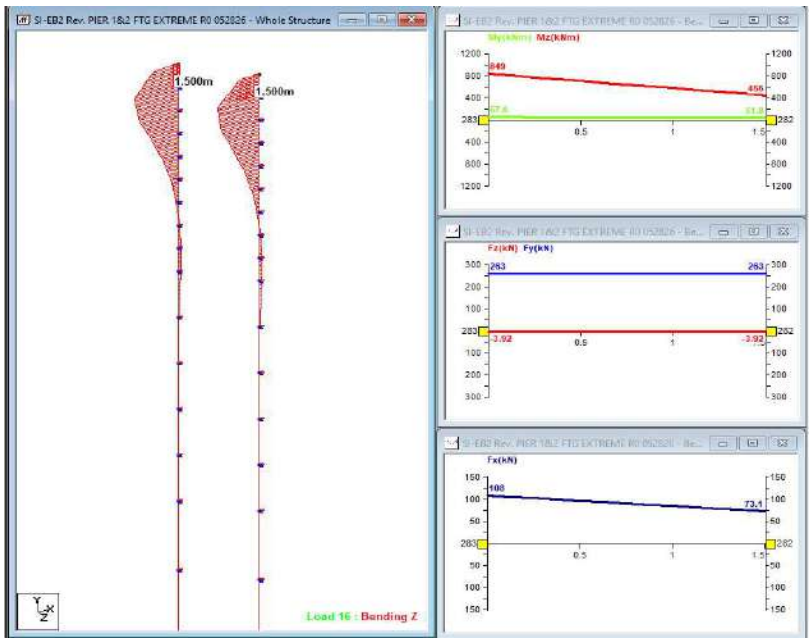
	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	18 1.25DC(F	-0.000	2849.714	-0.000	0.000	0.000	0.000
Min Fx	782	18 1.25DC(F	**-0.000**	2977.757	-0.000	0.000	0.000	0.000
Max Fy	782	18 1.25DC(F	-0.000	**2977.757**	-0.000	0.000	0.000	0.000
Min Fy	300	18 1.25DC(F	-0.000	**1274.754**	-0.000	0.000	0.000	0.000
Max Fz	744	15 1.25DC(F	-0.000	2946.130	**0.000**	0.000	0.000	0.000
Min Fz	782	18 1.25DC(F	-0.000	2974.456	**-0.000**	0.000	0.000	0.000
Max Mx	300	15 1.25DC(F	-0.000	1338.381	0.000	**0.000**	0.000	0.000
Min Mx	300	15 1.25DC(F	-0.000	1338.381	0.000	**0.000**	0.000	0.000
Max My	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	**0.000**	0.000
Min My	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	**0.000**	0.000
Max Mz	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	0.000	**0.000**
Min Mz	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	0.000	**0.000**

c. Critical axial with flexure forces:

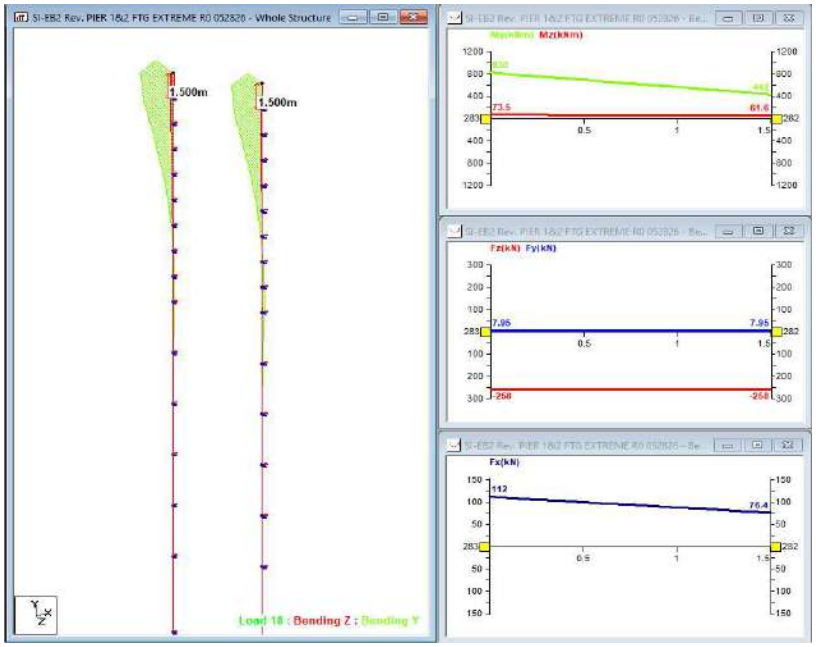
Moment Envelope



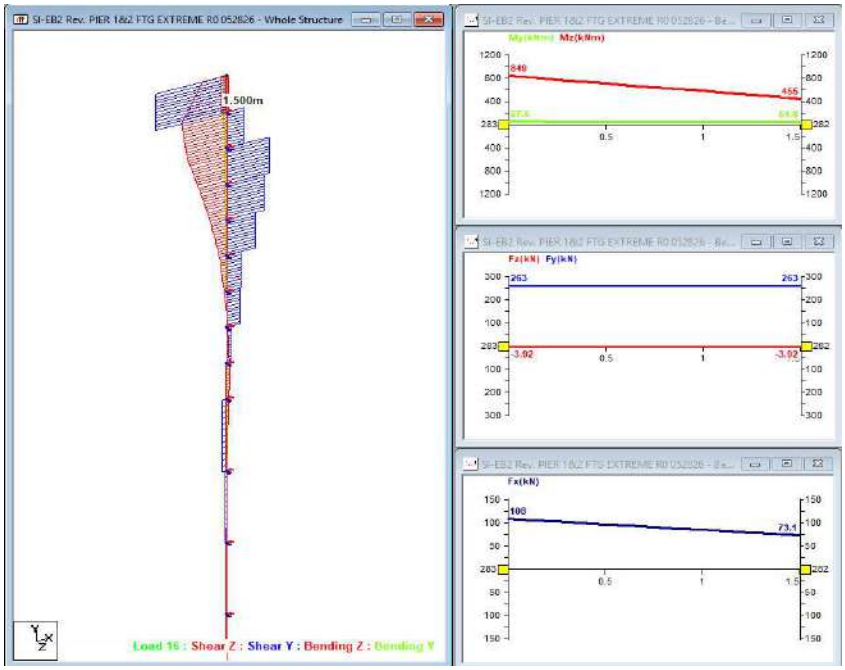
Longitudinal (LC1)



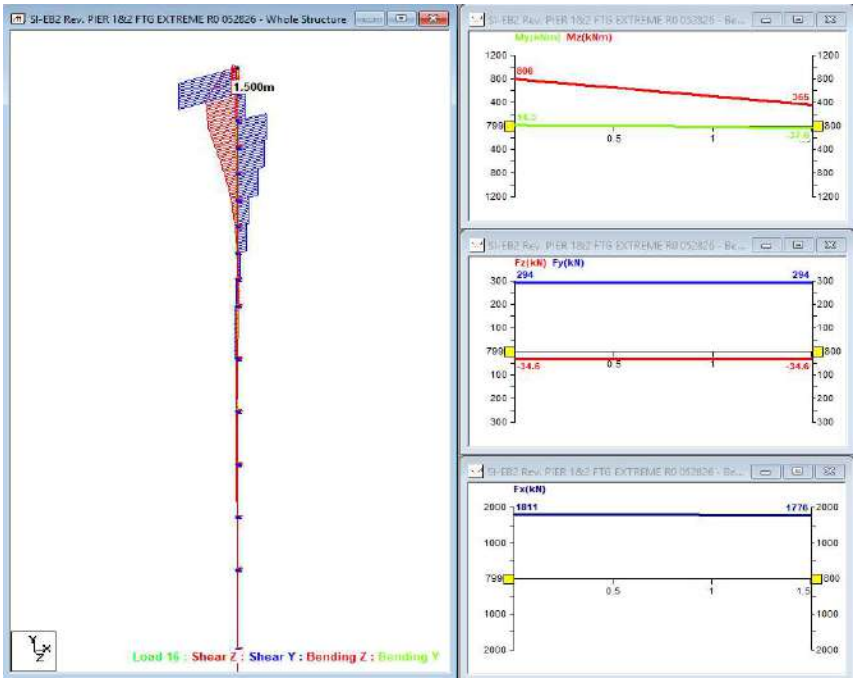
Transverse (LC2)



1) Max Negative Axial (Tension) with Max Moment



2) Max Positive Axial with Max Moment



5.8.3 Verification of Pile Stability

a. The factored resistance of piles shall be taken as : *(If comparing demand at top of pile)*

$$R_R = \Upsilon(\emptyset R_n - Ws) + W_s - W$$

where:

R_R	factored resitance of pile, kN	
R_n	nominal resistance of pile, kN	
W_s	effective weight of soil replaced by pile, kN	
W	effective weight of pile and soil inside pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state -BSDS Article 5.4.1(5)	0.65
Υ	modification coefficient depending on nominal bearing resistance 1.00-BSDS Table 5.4.3.3-1	

b. The nominal bearing capacity can be obtained from the empirical bearing capacity estimation formula:

$$R_n = q_d A_p + U \Sigma L_i f_i$$

where:

R_n	nominal bearing capacity of pile, kN
A_p	area of pile tip
q_d	nominal end bearing resistance intensity per unit area, kN/m^2
U	perimeter of pile
L_i	thickness of soil layer considering shaft resistance, m
f_i	maximum shaft resistance of soil layer considering pile shaft resistance, kN/m^2

c. The factored axial pull-out resistance of a single pile shall be obtained considering soil conditions and construction methods: *(If comparing demand at top of pile)*

$$P_R = \emptyset P_n + W$$

where:

P_R	factored axial pull-out resistance of pile, kN	
P_n	nominal axial pull-out resistance, kN	
W	effective weight of pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state -BSDS Article 5.4.1(5)	0.50

Pile Bearing and Pull-Out Capacity for D=1.0m BORED PILE (From New Geotechnical Report)

BH-34 L = 51.0 m *(Adopted BH-34 for Bearing Capacity)*

Allowable Pile Bearing Capacity of Pile, Ra :	**(vs Service)**	9435.0 kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Bearing Capacity of Pile, Rn:	**(vs Strength)**	9435.0 kN	
Extreme (Factored) Resistance of Pile, RR :	**(vs Extreme)**	8344.0 kN	
Allowable Uplift Capacity of Pile, Pa :	**(vs Service)**	5442.0 kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Pull-out Capacity of Pile, Pn:	**(vs Strength)**	5442.0 kN	
Extreme (Factored) Pull-out Resistance of Pile, PR	**(vs Extreme)**	4946.0 kN	

Verification for Max. Axial Resistance (Service)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1972.13	9435	4.78	OK

Verification for Max. Axial Pull-out Resistance (Service)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1573.31	-5442	OK	OK	*(No pull-out demand at bottom)*

Verification for Max. Axial Resistance (Strength)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
2712.26	9435	3.48	OK

Verification for Max. Axial Pull-out Resistance (Strength)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1982.68	-5442	OK	OK	*(No pull-out demand at bottom)*

Verification for Maximum Axial Resistance (Extreme)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
2977.76	8344	2.80	OK

(See Reaction at Bottom of Pile)

Verification for Maximum Axial Pull-out Resistance (Extreme)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1274.75	-4946	OK	OK	*(No pull-out demand at bottom)*

(See Reaction at Bottom of Pile)

Verification for Lateral Displacement

Displacement		C/D	Verification
Demand	Capacity		
mm	mm		
7.71	30	3.89	OK

Capacity = 3% x Dpile. (See commentary BSDS pp 4-15)

(Use average of 1 and 5% <= 50mm)

Verification for Tip Displacement/Settlement

Displacement		C/D	Verification
Demand	Allowable		
mm	mm		
9.16	25	2.73	OK

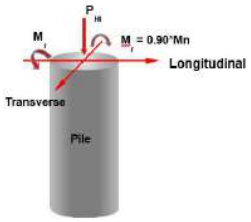
5.8.4 Verification of Pile Structural Resistance

Maximum Neg P-M		
Rigid Pile Head		
P_u	M_{trans}	M_{long}
kN	kN-m	kN-m
108	57.6	849

Maximum Pos P-M		
Rigid Pile Head		
P_u	M_{trans}	M_{long}
kN	kN-m	kN-m
1811	14.3	806

Note: The connection is rigid.

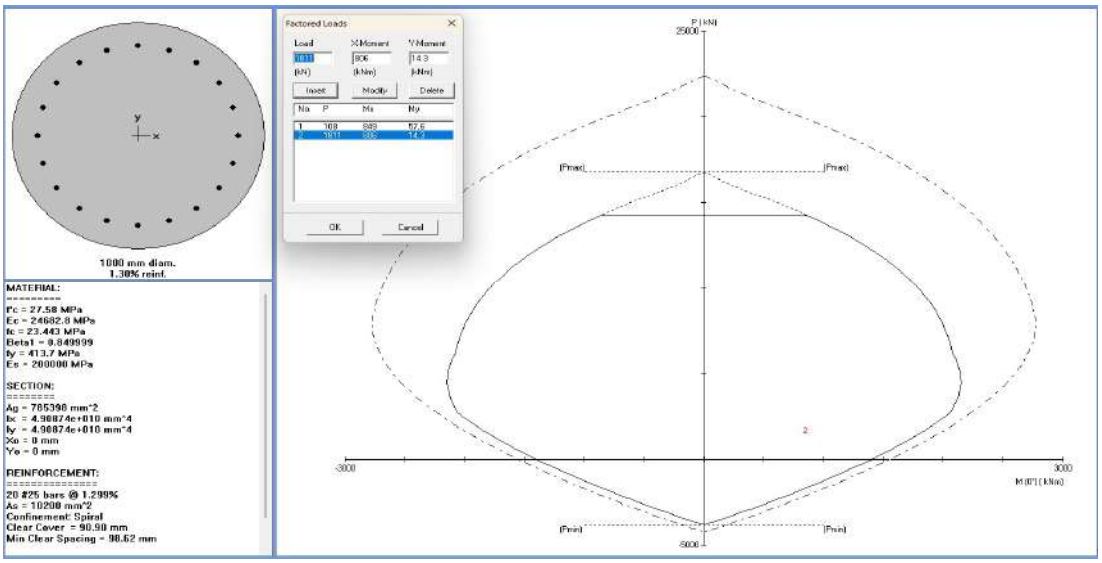
(See Critical Axial with Flexure Forces)



Define Reinforcement of Pile :

Diameter of Longitudinal Bars	25	mm
Diameter of Hoops/Spiral	16	mm
Reinforcement concrete cover to spiral/hoops	75	mm

SPColumn Axial and Flexure (P-M) Design of Pile



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	108.00	849.00	57.60	1421.35	96.43	1.674	224	895	0.00896	0.900
2	1811.00	806.00	14.30	1871.37	33.20	2.322	299	896	0.00600	0.900

Verification of Flexure with Axial Load Resistance

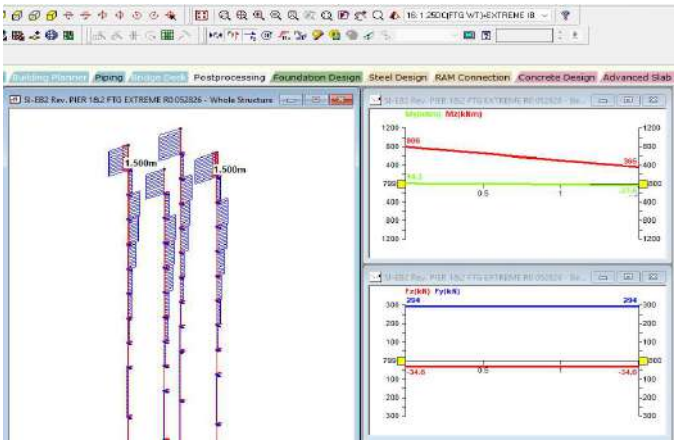
P = 108 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
850.95	1424.62	1.67	OK

(Note: Resultant Moments)

P = 1811 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
806.13	1871.66	2.32	OK

(Note: Resultant Moments)

Calculate Pile Shear Design and Resistance



Shear Demand, Vu = 294 kN *(Critical Resultant Shear)*

The nominal shear resistance, Vn is lesser of the following:

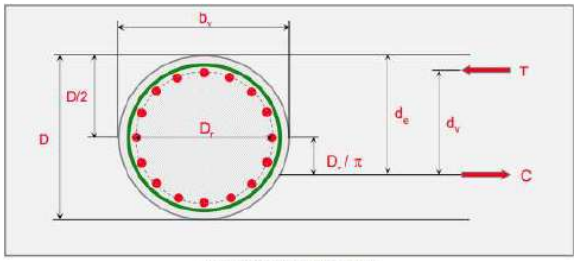
Vn = Vc + Vs + *Vp* OR Vn = 0.25f'c bv dv + *Vp*

where: Vc = 0.083*β*Sqrt(f'c)*bv*dv
Vs = [Av*fy*dv*(cotθ+cotα) sinα] / s
Vp = 0.00 for non-prestressed concrete
Vp = component in the direction of applied shear of effective prestressing force

Ultimate shear resistance, Vr = ØVn: Vr = ØVn > Vd_EQ where Ø= 0.9 for shear

Shear design parameters:

D	diameter of pier column (out to out dimension)	1000	mm
cc	concrete cover (to spiral or hoops)	75	mm
db	main bars or longitudinal bars	25	mm
ds	size of spirals	16	mm
f'c	concrete compressive strength	27.6	MPa
fy	yield strength of reinforcing steel	414	MPa
bv	effective web width, mm = diameter of column	1000	mm
de	de = (D / 2) + (Dr / π)	752	mm
dv	effective shear depth = 0.90 x de	677	mm
β	factor indicating ability of diagonally cracked concrete to transmit tension	2.0	
θ	angle of inclination of diagonal compressive stresses	45	deg
α	angle of inclination of transverse reinforcement	90	deg
Dr	diameter of circle passing thru centers of the longitudinal reinf.	793	mm
Dc	diameter of circle measured ourside the diameter of spiral/hoops	850	mm
Ac	Area of core measured outside diameter of spiral/hopps	567450	mm^2
Ag	Gross area of section	785398.16	mm^2



: For definition of terms:

From above parameters:

ds	size of spirals	16	mm
Asp	area of shear reinf.	201	mm
s	Try spacing	75	mm
No	Try no. of legs	2	No. legs
Total area of shear reinf. Av = Av = No*(Asp)		402	mm^2

Definition of no. of legs

leg eg

no. of spirals	No. of legs
single spiral	2 legs
bundle spiral	4 legs

cot θ = 1/tan 45° = 1
cot α = 1/tan 90° = 0
sin α = sin90° = 1

Vc = 0.00 kN *For conservative approach, assume Vc = 0.0, i.e. concrete has no shear strength.; If Pmin = negative value (tension), Vc = 0.0.*
Vp = 0.00 kN *For non prestressed concrete*
Vs = [Av*fy*dv*(cot θ + cot α)sin α]/s = **1503.1** kN

Vn is lesser of the following:

Vn = Vc + Vs + Vp =	1503.1	kN	**Governing shear capacity!**
Vn = 0.25f'cbvdv+Vp=	4672.5	kN	
Vd_EQ =	294	kN	Shear Demand
Vr =	**1353**	**kN**	Ultimate Shear Resistance, ØVn
c/d =	4.60		Shear resistance is OK!

Verification of shear/transverse reinforcement for confinement at plastic hinges

DGCS 12.7.11.2

Length of pile end region will be the larger of the following :

a.) 3D + Max M Loc. *(assume 5D)* =	8.00	m	*Conservative, see pile moment diagram above*
b.)450 mm (18 inches) =	0.45	m	
therefore, **Length =**	**8.00**	**m**	

Verification of minimum required transverse reinforcement

1. The greater of the following:

ρs1 = 0.12*(f'c/fy) =	0.00800		*AASHTO 5.11.4.1.4 @ Plastic Hinge Region*
Note: for circular shape:			
ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) =	0.01152	**Governs!**	*AASHTO 5.6.4.6 @ Column General Region*

2. Checking from provided confinement, where As represent spiral leg on one (1) side or the bars itself only

$$\rho_{sprovided} = \frac{4*As}{Dc*s} = 0.01262$$ **OK!!**

Maximum spacing

1. S should not be greater than 1/4 min dimension of member (=D/4)	**OK!!**
2. S should not be greater than 100mm	**OK!!**

Minimum clear spacing 59 mm *Actual clear spacing*

1. Sc should not be less than 25mm	**OK!!**
2. Sc should not be less than 1.33 x aggregate size (1.33 x 25mm)	**OK!!**

Transverse reinforcement for outside region *(not considered as column but as foundation element, minimal to zero shear demand region)*

ds	size of spirals	16	mm
Asp	area of shear reinforcement	201	mm^2
s	Try spacing of transverse steel	150	mm
No	Try number of legs	2	No. legs
	Total area of shear reinf. Av = Av = No*(Asp)	402	mm^2

DGCS 12.5.2.4

Min area of transverse reinf. spacing :

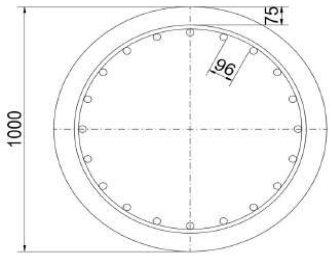
Av = 0.083 x sqrt(f'c) x bv x s/fy = 157.9878 mm^2 **OK!!**

Summary of shear/transverse reinforcement

Location	Range, m	Spacing/Pitch of spirals, mm	Diameter, mm	No. of legs	Layer	Note
From Pile Cap	8.00	75	16	2	1st	
The Rest Upto Bottom	43.00	150	16	2	1st	

Pile Details:

Use 4 - 1.0m Dia. Bored Piles, Total Length=51.0m ea., with 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 75mm for the first 15.0m from the pile cap, and then for the rest of the length, use 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 150mm.



5.8B DESIGN OF PILES - PIER 2

5.8.1 Determine the pile springs and geometric properties

a. Horizontal pile spring constant (K_h)

Horizontal Modulus of Subgrade Reaction
(See New Geotechnical Report)

Borehole No., **BH-34** (P2)
Pile width, D = 1.000 m

STRENGTH EVENT				ks = Kh*D*z	EXTEREME EVENT				ks = Kh*D*z
BH depth (m)	Thickness (m)	Depth of pile z, (m)	Kh (kN/m^3)	Soil Spring Stiffness, ks (kN/m)	BH depth (m)	Thickness (m)	Depth of pile z, (m)	Kh (kN/m^3)	Soil Spring Stiffness, ks (kN/m)
0.0				*NGL*	0.0				*NGL*
1.5					1.5				
3.0		EMBED and PILECAP		EMBED and PILECAP	3.0		EMBED and PILECAP		EMBED and PILECAP
4.5					4.5				
6.0	0.0				6.0	0.0			
7.5	1.5	1.5	4,956	7434	7.5	1.5	1.5	59,450	89175
9.0	1.5	3.0	4,956	7434	9.0	1.5	3.0	59,450	89175
10.5	1.5	4.5	4,956	7434	10.5	1.5	4.5	36,331	54497
12.0	1.5	6.0	1,803	2705	12.0	1.5	6.0	26,785	40178
13.5	1.5	7.5	2,821	4232	13.5	1.5	7.5	38,029	57044
15.0	1.5	9.0	839	1259	15.0	1.5	9.0	9,830	14745
16.5	1.5	10.5	8,325	12488	16.5	1.5	10.5	51,338	77007
18.0	1.5	12.0	11,831	17747	18.0	1.5	12.0	67,807	101711
19.5	1.5	13.5	15,438	34736	19.5	1.5	13.5	132,492	298107
22.5	3.0	16.5	16,659	49977	22.5	3.0	16.5	139,390	418170
25.5	3.0	19.5	38,437	115311	25.5	3.0	19.5	152,163	456489
28.5	3.0	22.5	55,993	167979	28.5	3.0	22.5	204,362	613086
31.5	3.0	25.5	65,678	197034	31.5	3.0	25.5	227,296	681888
34.5	3.0	28.5	69,871	262016	34.5	3.0	28.5	236,871	888266
39.0	4.5	33.0	89,737	403817	39.0	4.5	33.0	279,873	1259429
43.5	4.5	37.5	96,939	436226	43.5	4.5	37.5	294,653	1325939
48.0	4.5	42.0	104,191	703289	48.0	4.5	42.0	309,172	2086911
57.0	9.0	51.0	110,027	495122	57.0	9.0	51.0	320,612	1442754

Pile end vertical spring	
Kv (kN/m^3)	Soil Axial Spring, kv (kN/m)
157,009	123315

Pile end vertical spring	
Kh (kN/m^3)	Soil Axial Spring, kv (kN/m)
413,867	325050

b. Geometric properties of piles

Select Pile Section :
-- Circular Section
Square Section

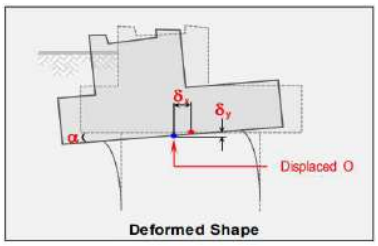
Select Pile Installation Method :
Driven Piles (Blow Method)
Driven Piles (Vibro-Hammer Method)
Cast-in-place RC Piles
-- Bored Piles
Pre-Boring Piles
Steel Pile Soil Cement Piles

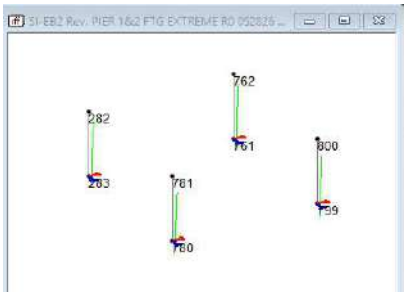
Input Pile Dimension : Diameter	1.000	m
Input Number of Piles	8	piles
Input Pile Length :	51.0	m
Calculate Section Properties :		
Cross-section Area	0.785	m^2
Perimeter of Pile :	3.142	m
Pile Moment of Inertia :	0.049	m^4
Pile Flexural Stiffness :	1.34E+06	kN-m^2
Concrete Material Properties :		
Design Compressive Strength at 28th day	27.6	N/mm^2
Unit Density for Concrete	2400	kg/m^3
Unit weight for Reinforced Concrete	24	kN/m^3
Young's Modulus of Elasticity	2.74E+07	kN/m^2
Reinforcement Material Properties :		
Minimum Yield Strength	414	N/mm^2
Ultimate Tensile Strength	620	N/mm^2
Young's Modulus of Elasticity	2.00E+08	kN/m^2

5.8.2 Determine the pile displacement, reaction forces, and critical forces along piles

a. Displacement of pile heads:

Displacement			
For Governing Load Case			
Extreme Event I			
Location	Displacement (Max. Values)		
	Horizontal [Resultant, sqrt(dx2+dz2)]	Vertical	Rotational
	d_x	d_y	a
	mm	mm	rad
Pile Head	**7.711**	**14.804**	**0.003**





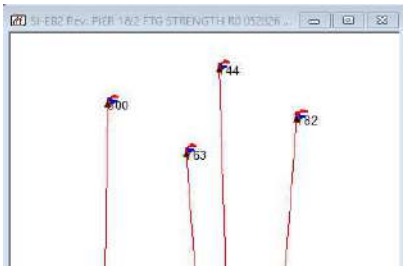
SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Node Displacements: G3: _PILEHEAD

All / Summary

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	rY rad	rZ rad
Max X	800	16 1.25DC(F	**7.711**	-14.804	0.434	16.698	0.000	-0.000	-0.003
Min X	791	18 1.25DC(F	**0.472**	-14.106	7.546	16.005	0.003	-0.000	-0.000
Max Y	282	16 1.25DC(F	7.856	**-5.522**	0.379	9.447	0.000	0.000	-0.003
Min Y	800	16 1.25DC(F	7.711	**-14.804**	0.434	16.698	0.000	-0.000	-0.003
Max Z	800	16 1.25DC(F	0.568	-14.786	**7.550**	16.612	0.003	0.000	-0.000
Min Z	782	15 1.25DC(F	7.710	-14.632	**-0.092**	16.539	0.000	0.000	-0.003
Max rX	782	16 1.25DC(F	0.526	-0.220	7.491	9.751	**0.003**	0.000	-0.000
Min rX	800	15 1.25DC(F	7.709	-14.457	0.005	16.384	**-0.000**	-0.000	-0.003
Max rY	782	16 1.25DC(F	7.708	-14.288	0.337	16.235	0.000	**0.000**	-0.003
Min rY	791	17 1.25DC(F	0.474	-13.769	7.117	15.493	0.003	**-0.000**	-0.000
Max rZ	800	16 1.25DC(F	0.568	-14.786	7.550	16.612	0.003	0.000	**-0.000**
Min rZ	781	16 1.25DC(F	7.653	-6.041	0.391	9.758	0.000	-0.000	**-0.003**
Max Rs	800	16 1.25DC(F	7.711	-14.804	0.434	**16.698**	0.000	-0.000	-0.003

b. Reaction forces at bottom of piles:

at SERVICE:

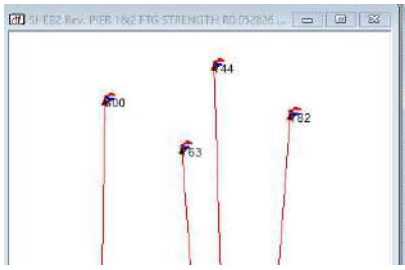


SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Support Reactions: G3: _PILEBOTTOM

All / Summary / Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	20 1.0DC(FTG WT)+SERVICE IB+	**0.000**	1724.333	0.000	0.000	0.000	0.000
Min Fx	782	20 1.0DC(FTG WT)+SERVICE IB+	**-0.000**	1972.136	-0.000	0.000	0.000	0.000
Max Fy	782	20 1.0DC(FTG WT)+SERVICE IB+	-0.000	**1972.136**	-0.000	0.000	0.000	0.000
Min Fy	300	20 1.0DC(FTG WT)+SERVICE IB+	0.000	**1573.307**	0.000	0.000	0.000	0.000
Max Fz	744	20 1.0DC(FTG WT)+SERVICE IB+	-0.000	1821.105	**0.000**	0.000	0.000	0.000
Min Fz	782	18 1.0DC(FTG WT)+SERVICE IA+	-0.000	1856.764	**-0.000**	0.000	0.000	0.000
Max Mx	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	**0.000**	0.000	0.000
Min Mx	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	**0.000**	0.000	0.000
Max My	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	0.000	**0.000**	0.000
Min My	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	0.000	**0.000**	0.000
Max Mz	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	0.000	0.000	**0.000**
Min Mz	300	19 1.0DC(FTG WT)+SERVICE IA+	0.000	1868.674	-0.000	0.000	0.000	**0.000**

at STRENGTH:

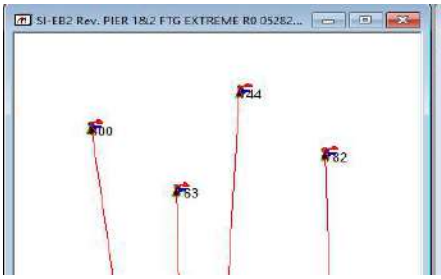


SI-EB2 Rev. PIER 1&2 FTG STRENGTH R0 052826 - Support Reactions: G3: _PILEBOTTOM

All / Summary / Envelope

	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	14 1.25DC(FTG WT)+STRENGTH IB+	**0.000**	2278.625	0.000	0.000	0.000	0.000
Min Fx	782	14 1.25DC(FTG WT)+STRENGTH IB+	**-0.000**	2712.263	0.000	0.000	0.000	0.000
Max Fy	782	14 1.25DC(FTG WT)+STRENGTH IB+	-0.000	**2712.263**	0.000	0.000	0.000	0.000
Min Fy	300	14 1.25DC(FTG WT)+STRENGTH IB+	0.000	**1982.682**	0.000	0.000	0.000	0.000
Max Fz	744	14 1.25DC(FTG WT)+STRENGTH IB+	-0.000	2416.320	**0.000**	0.000	0.000	0.000
Min Fz	782	13 1.25DC(FTG WT)+STRENGTH IA+	-0.000	2510.371	**-0.000**	0.000	0.000	0.000
Max Mx	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	**0.000**	0.000	0.000
Min Mx	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	**0.000**	0.000	0.000
Max My	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	**0.000**	0.000
Min My	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	**0.000**	0.000
Max Mz	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	0.000	**0.000**
Min Mz	300	13 1.25DC(FTG WT)+STRENGTH IA+	0.000	2184.574	-0.000	0.000	0.000	**0.000**

at EXTREME:



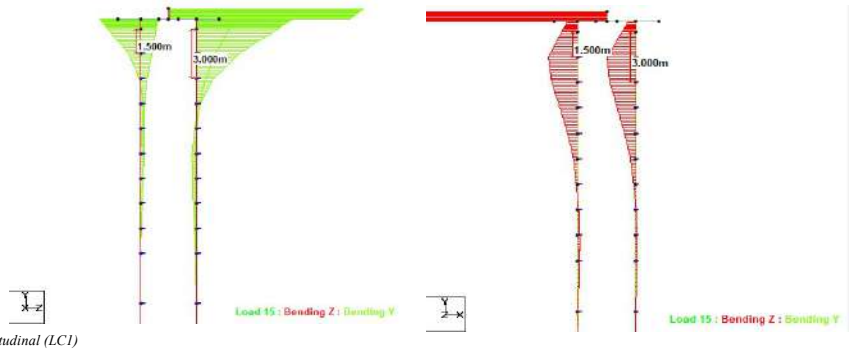
SI-EB2 Rev. PIER 1&2 FTG EXTREME R0 052826 - Support Reactions: G4: _PILEBOTTOM

All / Summary / Envelope

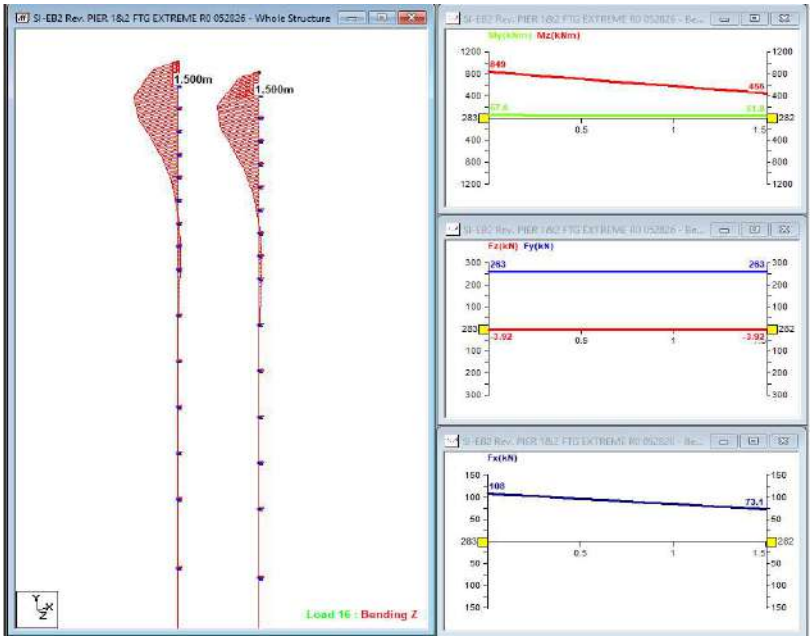
	Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	Moment Mx kNm	My kNm	Mz kNm
Max Fx	783	18 1.25DC(F	**-0.000**	2649.714	-0.000	0.000	0.000	0.000
Min Fx	782	18 1.25DC(F	**-0.000**	2977.757	-0.000	0.000	0.000	0.000
Max Fy	782	18 1.25DC(F	-0.000	**2977.757**	-0.000	0.000	0.000	0.000
Min Fy	300	18 1.25DC(F	-0.000	**1274.754**	-0.000	0.000	0.000	0.000
Max Fz	744	15 1.25DC(F	-0.000	2946.130	**0.000**	0.000	0.000	0.000
Min Fz	782	18 1.25DC(F	-0.000	2974.456	**-0.000**	0.000	0.000	0.000
Max Mx	300	15 1.25DC(F	-0.000	1338.381	0.000	**0.000**	0.000	0.000
Min Mx	300	15 1.25DC(F	-0.000	1338.381	0.000	**0.000**	0.000	0.000
Max My	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	**0.000**	0.000
Min My	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	**0.000**	0.000
Max Mz	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	0.000	**0.000**
Min Mz	300	15 1.25DC(F	-0.000	1338.381	0.000	0.000	0.000	**0.000**

c. Critical axial with flexure forces:

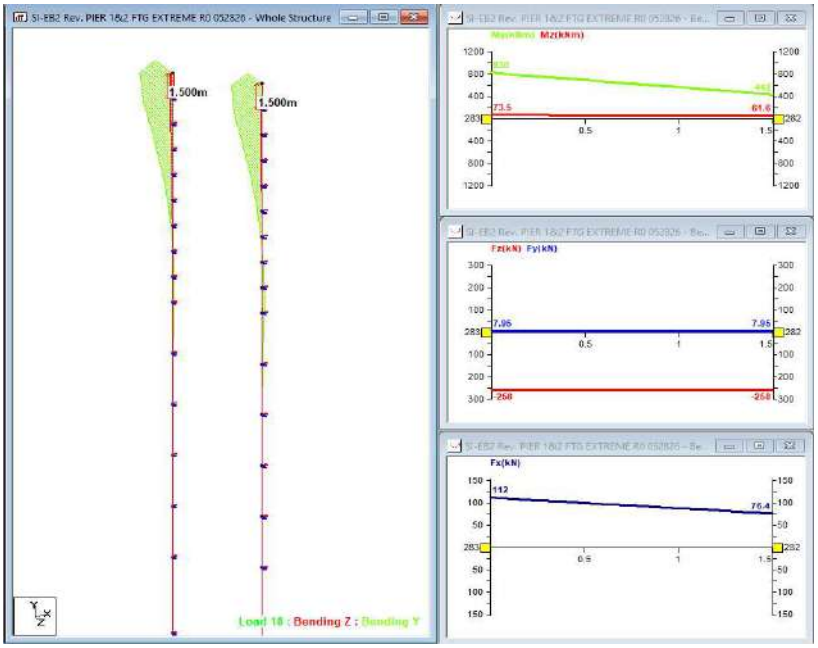
Moment Envelope



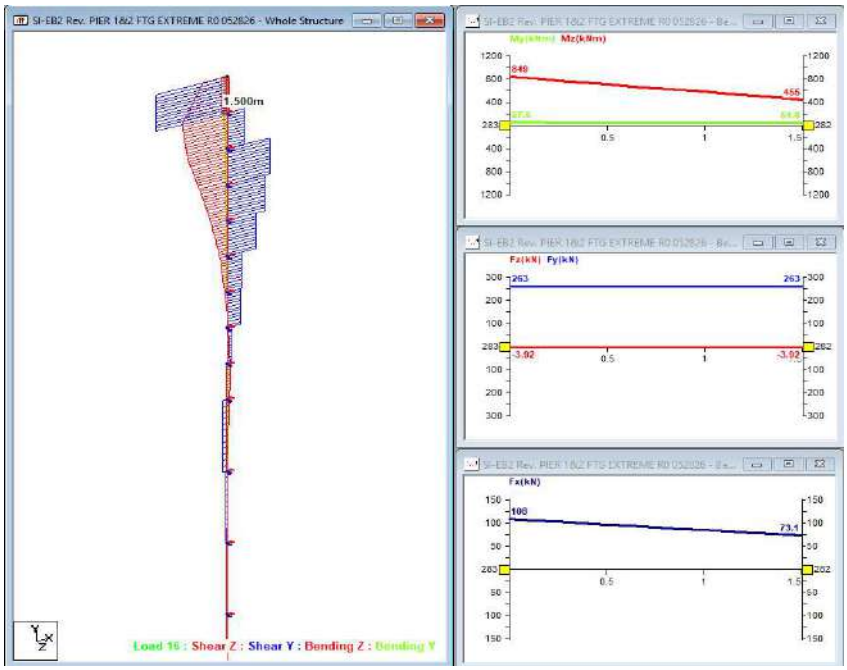
Longitudinal (LC1)



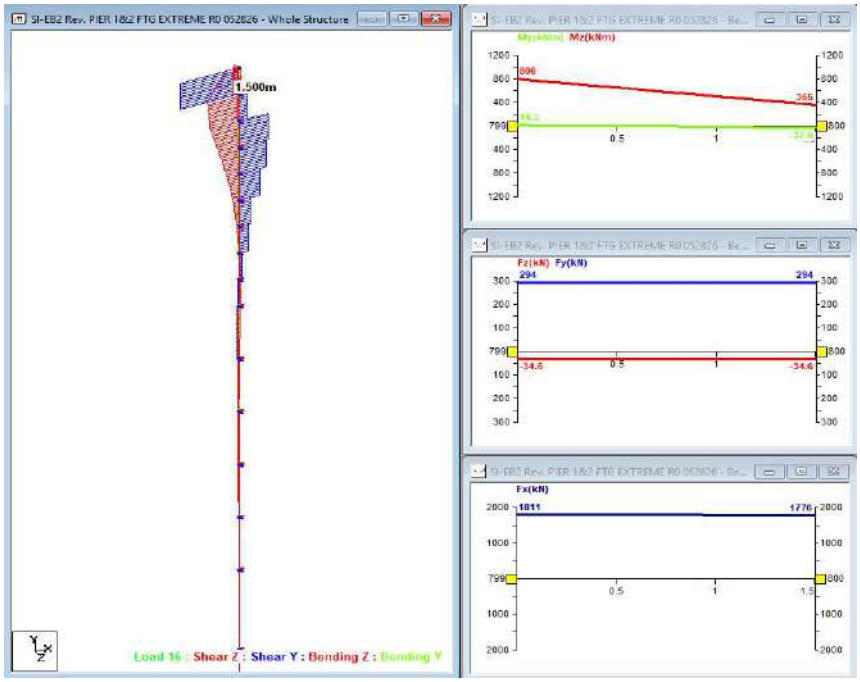
Transverse (LC2)



1) Max Negative Axial (Tension) with Max Moment



2) Max Positive Axial with Max Moment



5.8.3 Verification of Pile Stability

a. The factored resistance of piles shall be taken as : *(If comparing demand at top of pile)*

$$R_R = \Upsilon(\emptyset R_n - Ws) + W_s - W$$

where:

R_R	factored resitance of pile, kN	
R_n	nominal resistance of pile, kN	
W_s	effective weight of soil replaced by pile, kN	
W	effective weight of pile and soil inside pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state -BSDS Article 5.4.1(5)	0.65
Υ	modification coefficient depending on nominal bearing resistance 1.00-BSDS Table 5.4.3.3-1	

b. The nominal bearing capacity can be obtained from the empirical bearing capacity estimation formula:

$$R_n = q_d A_p + U \Sigma L_i f_i$$

where:

R_n	nominal bearing capacity of pile, kN
A_p	area of pile tip
q_d	nominal end bearing resistance intensity per unit area, kN/m^2
U	perimeter of pile
L_i	thickness of soil layer considering shaft resistance, m
f_i	maximum shaft resistance of soil layer considering pile shaft resistance, kN/m^2

c. The factored axial pull-out resistance of a single pile shall be obtained considering soil conditions and construction methods: *(If comparing demand at top of pile)*

$$P_R = \emptyset P_n + W$$

where:

P_R	factored axial pull-out resistance of pile, kN	
P_n	nominal axial pull-out resistance, kN	
W	effective weight of pile, kN	
$\emptyset$	resistance factor for pile under extreme event limit state -BSDS Article 5.4.1(5)	0.50

Pile Bearing and Pull-Out Capacity for D=1.0m BORED PILE (From New Geotechnical Report)

BH-34 L = 51.0 m

Allowable Pile Bearing Capacity of Pile, Ra :	**(vs Service)**	9435.0 kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Bearing Capacity of Pile, Rn:	**(vs Strength)**	9435.0 kN	
Extreme (Factored) Resistance of Pile, RR :	**(vs Extreme)**	8344.0 kN	
Allowable Uplift Capacity of Pile, Pa :	**(vs Service)**	5442.0 kN	*(If comparing demand at bottom of pile, where pile weight is already considered)*
Ultimate Strength Pull-out Capacity of Pile, Pn:	**(vs Strength)**	5442.0 kN	
Extreme (Factored) Pull-out Resistance of Pile, PR	**(vs Extreme)**	4946.0 kN	

Verification for Max. Axial Resistance (Service)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
1972.13	9435	4.78	OK

Verification for Max. Axial Pull-out Resistance (Service)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1573.31	-5442	OK	OK	*(No pull-out demand at bottom)*

Verification for Max. Axial Resistance (Strength)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
2712.26	9435	3.48	OK

Verification for Max. Axial Pull-out Resistance (Strength)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1982.68	-5442	OK	OK	*(No pull-out demand at bottom)*

Verification for Maximum Axial Resistance (Extreme)

Axial Load		C/D	Verification
Demand	Capacity		
kN	kN		
2977.76	8344	2.80	OK

(See Reaction at Bottom of Pile)

Verification for Maximum Axial Pull-out Resistance (Extreme)

Axial Pull-out		C/D	Verification	
Demand	Capacity			
kN	kN			
1274.75	-4946	OK	OK	*(No pull-out demand at bottom)*

(See Reaction at Bottom of Pile)

Verification for Lateral Displacement

Displacement		C/D	Verification
Demand	Capacity		
mm	mm		
7.71	30	3.89	OK

Capacity = 3% x Dpile. (See commentary BSDS pp 4-15)

(Use average of 1 and 5% <= 50mm)

Verification for Tip Displacement/Settlement

Displacement		C/D	Verification
Demand	Allowable		
mm	mm		
9.16	25	2.73	OK

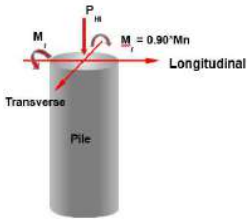
5.8.4 Verification of Pile Structural Resistance

Maximum Neg P-M		
Rigid Pile Head		
P_u	M_{trans}	M_{long}
kN	kN-m	kN-m
108	57.6	849

Maximum Pos P-M		
Rigid Pile Head		
P_u	M_{trans}	M_{long}
kN	kN-m	kN-m
1811	14.3	806

Note: The connection is rigid.

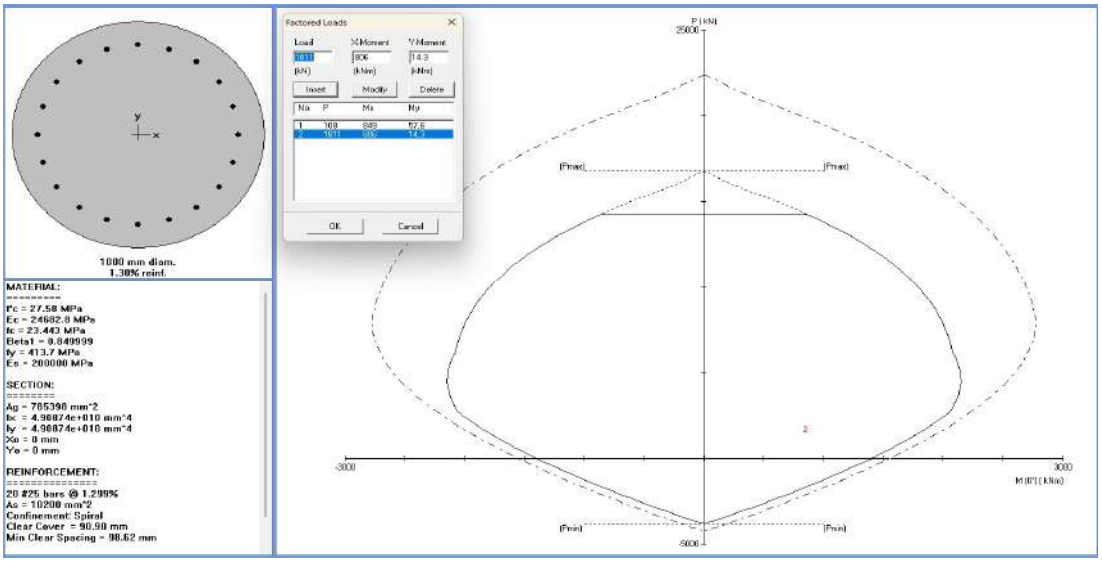
(See Critical Axial with Flexure Forces)



Define Reinforcement of Pile :

Diameter of Longitudinal Bars	25	mm
Diameter of Hoops/Spiral	16	mm
Reinforcement concrete cover to spiral/hoops	75	mm

SPColumn Axial and Flexure (P-M) Design of Pile



Factored Loads and Moments with Corresponding Capacities

No	Pu	Mux	Muy	ϕMnx	ϕMny	ϕMn/Mu	NA Depth	dt Depth	εt	ϕ
	kN	kNm	kNm	kNm	kNm		mm	mm		
1	108.00	849.00	57.60	1421.35	96.43	1.674	224	895	0.00896	0.900
2	1811.00	806.00	14.30	1871.37	33.20	2.322	299	896	0.00600	0.900

Verification of Flexure with Axial Load Resistance

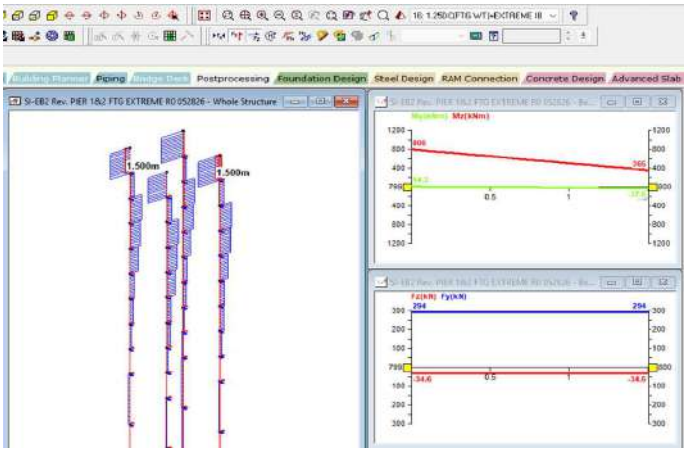
P = 108 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
850.95	1424.62	1.67	OK

(Note: Resultant Moments)

P = 1811 kN		C/D	Verification
M-Demand	M-Capacity		
kN-m	kN-m		
806.13	1871.66	2.32	OK

(Note: Resultant Moments)

Calculate Pile Shear Design and Resistance



Shear Demand, Vu = 294 kN *(Critical Resultant Shear)*

The nominal shear resistance, Vn is lesser of the following:

$Vn = Vc + Vs + Vp$ OR $Vn = 0.25f'c\ bv\ dv + Vp$

where: $Vc = 0.083*\beta*Sqrt(f'c)*bv*dv$

$Vs = [\ Av*fy*dv*(cot\theta+cot\alpha)\ sin\alpha\ ] / s$

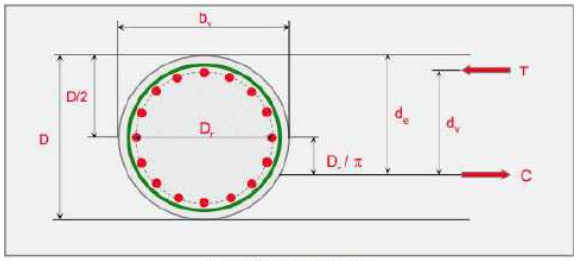
Vp = 0.00 for non-prestressed concrete

Vp = component in the direction of applied shear of effective prestressing force

Ultimate shear resistance, Vr = ØVn: Vr = ØVn > Vd_EQ where Ø= 0.9 for shear

Shear design parameters:

D	diameter of pier column (out to out dimension)	1000	mm
cc	concrete cover (to spiral or hoops)	75	mm
db	main bars or longitudinal bars	25	mm
ds	size of spirals	16	mm
f'c	concrete compressive strength	27.6	MPa
fy	yield strength of reinforcing steel	414	MPa
bv	effective web width, mm = diameter of column	1000	mm
de	de = (D / 2) + (Dr / π)	752	mm
dv	effective shear depth = 0.90 x de	677	mm
β	factor indicating ability of diagonally cracked concrete to transmit tension	2.0	
θ	angle of inclination of diagonal compressive stresses	45	deg
α	angle of inclination of transverse reinforcement	90	deg
Dr	diameter of circle passing thru centers of the longitudinal reinf.	793	mm
Dc	diameter of circle measured ourside the diameter of spiral/hoops	850	mm
Ac	Area of core measured outside diameter of spiral/hopps	567450	mm^2
Ag	Gross area of section	785398.16	mm^2



: For definition of terms:

From above parameters:

ds	size of spirals	16	mm
Asp	area of shear reinf.	201	mm
s	Try spacing	75	mm
No	Try no. of legs	2	No. legs
Total area of shear reinf. Av = Av = No*(Asp)		402	mm^2

Definition of no. of legs

leg eg

no. of spirals	No. of legs
single spiral	2 legs
bundle spiral	4 legs

cot θ = 1/tan 45° = 1

cot α = 1/tan 90° = 0

sin α = sin90° = 1

Vc = 0.00 kN *For conservative approach, assume Vc = 0.0, i.e. concrete has no shear strength.; If Pmin = negative value (tension), Vc = 0.0.*

Vp = 0.00 kN *For non prestressed concrete*

$Vs = [\ Av*fy*dv*(cot\ \theta + cot\ \alpha)sin\ \alpha\ ]/s$ = **1503.1** kN

Vn is lesser of the following:

Vn = Vc + Vs + Vp =	1503.1	kN	**Governing shear capacity!**
Vn = 0.25f'cbvdv+Vp=	4672.5	kN	
Vd_EQ =	294	kN	Shear Demand
Vr =	**1353**	**kN**	Ultimate Shear Resistance, ØVn
c/d =	4.60		Shear resistance is OK!

Verification of shear/transverse reinforcement for confinement at plastic hinges

DGCS 12.7.11.2

Length of pile end region will be the larger of the following :

a.) 3D + Max M Loc. *(assume 5D)* =	8.00	m	*Conservative, see pile moment diagram above*
b.)450 mm (18 inches) =	0.45	m	
therefore, **Length =**	**8.00**	**m**	

Verification of minimum required transverse reinforcement

1. The greater of the following:

ρs1 = 0.12*(f'c/fy) =	0.00800		*AASHTO 5.11.4.1.4 @ Plastic Hinge Region*
Note: for circular shape:			
ρs2 = 0.45*[(Ag/Ac) - 1]*(f'c/fy) =	0.01152	**Governs!**	*AASHTO 5.6.4.6 @ Column General Region*

2. Checking from provided confinement, where As represent spiral leg on one (1) side or the bars itself only

$$\rho_{sprovided} = \frac{4*As}{Dc*s} = 0.01262$$ **OK!!**

Maximum spacing

1. S should not be greater than 1/4 min dimension of member (=D/4)	**OK!!**
2. S should not be greater than 100mm	**OK!!**

Minimum clear spacing 59 mm *Actual clear spacing*

1. Sc should not be less than 25mm	**OK!!**
2. Sc should not be less than 1.33 x aggregate size (1.33 x 25mm)	**OK!!**

Transverse reinforcement for outside region *(not considered as column but as foundation element, minimal to zero shear demand region)*

ds	size of spirals	16	mm
Asp	area of shear reinforcement	201	mm^2
s	Try spacing of transverse steel	150	mm
No	Try number of legs	2	No. legs
	Total area of shear reinf. Av = Av = No*(Asp)	402	mm^2

DGCS 12.5.2.4

Min area of transverse reinf. spacing :

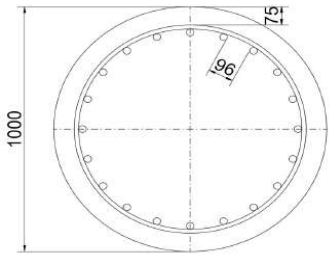
Av = 0.083 x sqrt(f'c) x bv x s/fy = 157.9878 mm^2 **OK!!**

Summary of shear/transverse reinforcement

Location	Range, m	Spacing/Pitch of spirals, mm	Diameter, mm	No. of legs	Layer	Note
From Pile Cap	8.00	75	16	2	1st	
The Rest Upto Bottom	43.00	150	16	2	1st	

Pile Details:

Use 4 - 1.0m Dia. Bored Piles, Total Length=51.0m ea., with 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 75mm for the first 15.0m from the pile cap, and then for the rest of the length, use 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 150mm.



DESIGN SUMMARY

DESIGN OF PIER 1 & 2

Project: Prop San Isidro - Eastern Branch 2 (SI-EB2) Bridge Proj. Location: Minalin, Pampanga
Designed by/Date: KH-RCDG 05/28/26

5.9 DESIGN SUMMARY

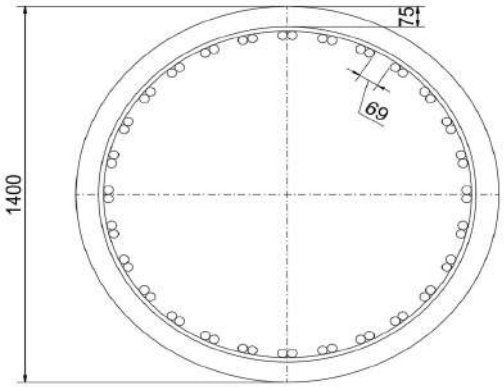
5.9.1 COLUMN/S

D.col =	1.20	m	diameter
H.col =	9.30	m	height
No.col =	1		number

A. Axial & Flexure

Design Reinforcement:

Longi. Bars:	32	bars -	25	mm Ø



< Column Diagram >

Longitudinal Direction (EQ1)

DEMAND			CAPACITY			C/D RATIO	
PmaxL =	2686	kN					
PminL =	2480	kN	PminL =	2480	kN		
Vdes_L =	637	kN					
Mdes_L =	2719	kN-m	φMn_L =	3481	kN-m	C/D =	1.28

Transverse Direction (EQ2)

DEMAND			CAPACITY			C/D RATIO	
PmaxT =	2614	kN					
PminT =	2552	kN	PminT =	2552	kN		
Vdes_T =	722	kN					
Mdes_T =	2644	kN-m	φMn_T =	3500	kN-m	C/D =	1.32

B. Shear

Design Reinforcement:

Location	Range, m	Spacing/Pitch of Spirals, mm	Diameter, mm	No. of Legs	Layer	Note
End regions	1.55	100	16	4	1st	
Outside region	6.20	150	16	4	1st	

At End Regions:

DEMAND			CAPACITY			C/D RATIO	
Vd_EQ =	438	kN	Vr =	2471	kN	C/D =	5.64

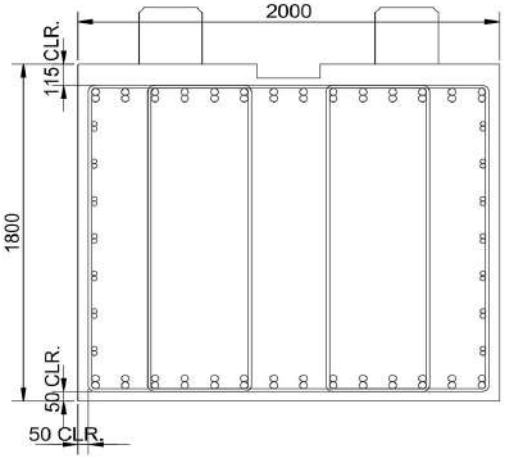
5.9.2 COPING BEAM

h.cap =	1.20	m	height
b.cap =	1.70	m	width
l.cap =	5.50	m	length

A. Flexure

Design Reinforcement:

Top Bars:	18	bars -	25	mm Ø
	0			
Bottom Bars:	18	bars -	25	mm Ø
	0			



< Coping Beam Diagram >

	DEMAND			CAPACITY			C/D RATIO	
At face of Col.	M.u_neg =	1639	kN-m	Mr =	3286	kN-m	C/D =	2.01
At face of Col.	M.u_pos =	#REF!	kN-m	Mr =	3286	kN-m	C/D =	#REF!

B. Shear

Design Reinforcement:

Location	Range, m	Spacing of **Stirrups**, mm	Diameter, mm	No. of Legs	
L/4 Ends	0.95	150	16	4	(2 SETS)
Middle L/2	1.90	200	16	4	(2 SETS)

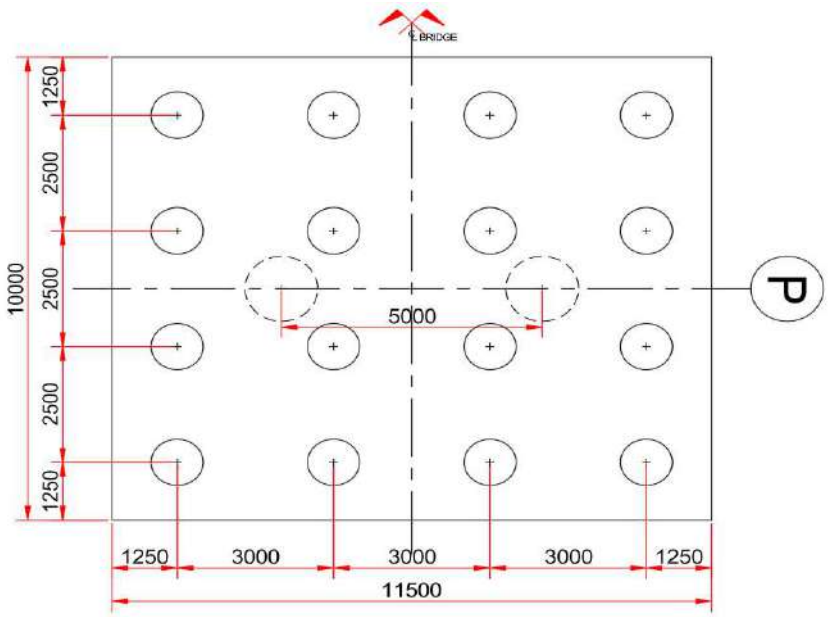
	DEMAND			CAPACITY			C/D RATIO	
At face of Col.	V.u =	1380	kN	Vr =	3326	kN	C/D =	2.41
Middle L/2	V.umid =	1306	kN	Vrmid =	2827	kN	C/D =	2.16

C. Side Bars

Design Reinforcement:

Spacing @	150	mm -	25	mm Ø Side Bars	1	No. of Bundle

5.9.3 PILE CAP



< Pile Cap Diagram >

height of pilecap, hp = 1.20 m

A. Resulting Flexure

Design Reinforcement:

Longitudinal TOP:	USE 32mmØ @ 200mm 0	**Transverse TOP:**	USE 32mmØ @ 200mm
Longitudinal BOT.:	USE 36mmØ @ 200mm *0*	**Transverse BOT.:**	USE 36mmØ @ 200mm

Longitudinal TOP:

DEMAND		CAPACITY		C/D R.
Mx_top =	793 kN-m	Mrx_t =	1584 kN-m	2.00

Transverse TOP:

DEMAND		CAPACITY		C/D R.
My_top =	775 kN-m	Mry_t =	1513 kN-m	1.95

Longitudinal BOT.:

DEMAND		CAPACITY		C/D R.
Mx_bot =	1022 kN-m	Mrx_b =	1743 kN-m	1.71

Transverse BOT.:

DEMAND		CAPACITY		C/D R.
My_bot =	1013 kN-m	Mry_b =	1640 kN-m	1.62

B. Shear

Design Reinforcement:

Diameter of cross ties/J-bars, ds = 20 mm

Use spacing @ 400mm x 400mm O.C.

Longitudinal Shear:

DEMAND		CAPACITY		C/D R.
Vxu =	1097 kN	Vxr =	1394 kN	1.27

Transverse Shear:

DEMAND		CAPACITY		C/D R.
Vyu =	1082 kN	Vyr =	1523 kN	1.41

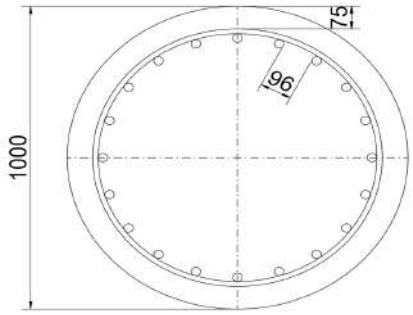
C. Side Bars

Design Reinforcement:

Spacing =	250 mm -	20 mm Ø	Side Bars	(MINIMUM Only)
USE Spacing =	200 mm -	25 mm Ø	Side Bars	**∴ PROVIDED:**

5.9.4 PILES

Pile Diameter =	1.00	m
Number of Piles =	8	piles
Pile Length =	51.00	m



< Bored Pile Diagram >

A. Axial & Flexure

Design Reinforcement:

Longi. Bars:	20	bars -	25	mm Ø

1) Max Negative Axial (Tension) with Max Moment

DEMAND			CAPACITY			C/D RATIO	
Pu =	108	kN	Pu =	108	kN		
Mu =	851	kN-m	φMn =	1425	kN-m	C/D =	1.67

2) Max Positive Axial with Max Moment

DEMAND			CAPACITY			C/D RATIO	
Pu =	1811	kN	Pu =	1811	kN		
Mu =	806	kN-m	φMn =	1872	kN-m	C/D =	2.32

B. Shear

Design Reinforcement:

Location	Range, m	Spacing/Pitch of **Spirals**, mm	Diameter, mm	No. of legs	Layer	Note
From Pile Cap	15.00	75	16	2	1st	
The Rest Upto Bot.	36.00	150	16	2	1st	

At Start Region:

DEMAND			CAPACITY			C/D RATIO	
Vd_EQ =	294	kN	Vr =	1353	kN	C/D =	4.60

Use 4 - 1.0m Dia. Bored Piles, Total Length=51.0m ea., with 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 75mm for the first 15.0m from the pile cap, and then for the rest of the length, use 20-25mm dia. main bars and 16mm dia. spiral hoops spaced @ 150mm.

Piers 1&2 Bent and Footing Analysis Staad Output/Code

1. Piers 1&2 Bent Analysis Staad Output/Code


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        *        STAAD.Pro V8i SELECTseries6             *
        *        Version  20.07.11.33                    *
        *        Proprietary Program of                  *
        *        Bentley Systems, Inc.                   *
        *        Date=    JUN  4, 2026                   *
        *        Time=    15:37: 7                       *
        *                                                *
        *    USER ID:                                    *
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   1. STAAD SPACE
INPUT FILE: D:\SA3\Bridge\JBJ\BP11_SI-EB2\Pier Design\STAAD R0\SI-EB2 Rev. Pier
Bent Analysis R0 05282...STD
   2. START JOB INFORMATION
   3. ENGINEER DATE 15-APR-26
   4. END JOB INFORMATION
   5. INPUT WIDTH 79
   6. UNIT METER KN
   7. JOINT COORDINATES
   8. 221 15 -1.933 0; 222 15 -1.933 1; 224 15 -1.933 2.75; 226 15 -1.933 4.5
   9. 227 15 -1.933 5.5; 228 15 -2.533 2.75; 229 15 -4.858 2.75; 230 15 -7.183 2.75
  10. 231 15 -9.508 2.75; 232 15 -11.833 2.75
  11. MEMBER INCIDENCES
  12. 187 221 222; 188 222 224; 189 224 226; 190 226 227; 192 232 231; 193 231 230
  13. 194 230 229; 195 229 228; 196 228 224
  14. START GROUP DEFINITION
  15. MEMBER
  16. JOINT
  17. END GROUP DEFINITION

 ** WARNING: NO VALID GROUPS IN GROUP DEFINITION DATA **
  18. START USER TABLE
  19. TABLE 1
  20. UNIT METER KN
  21. GENERAL
  22. TYPEIII
  23. 0.360685 1.143 0.001 0.56 0.001 0.0521963 0.00509875 0.00710869 0.0830343 -
  24. 0.0181982 0.214693 0.200946 0.123389 0.0346505 0.00061616 1.143
  26. END
  27. DEFINE MATERIAL START
  28. ISOTROPIC CONCRETE0
  29. E 2.735E+007
  30. POISSON 0.17
  31. ALPHA 1E-005
  32. DAMP 0.05
  33. ISOTROPIC CONCRETE1
  34. E 2.735E+007
  35. POISSON 0.17
  36. DENSITY 24
  37. ALPHA 1E-005
STAAD SPACE                                -- PAGE NO.  2

  38. DAMP 0.05
  39. ISOTROPIC RIGID_LINK
  40. E 2.735E+007
  41. POISSON 0.17
  42. ALPHA 1E-005
  43. DAMP 0.05
  44. ISOTROPIC PCONCRETE1
  45. E 3.3497E+007
  46. POISSON 0.17
  47. DENSITY 24
  48. ALPHA 1E-005
  49. DAMP 0.05
  50. END DEFINE MATERIAL
  51. MEMBER PROPERTY AMERICAN
  52. 188 189 PRIS YD 1.2 ZD 1.7
  53. 187 190 PRIS YD 1.055 ZD 1.7
  54. 192 TO 195 PRIS YD 1.2
  55. 196 PRIS AX 100 IX 1000 IY 1000 IZ 1000
  56. CONSTANTS
  57. MATERIAL CONCRETE1 MEMB 187 TO 190 192 TO 195
  58. MATERIAL RIGID_LINK MEMB 196
  59. SUPPORTS
  60. 232 FIXED
  61. LOAD 1 LOADTYPE NONE  TITLE DC_DL
  62. SELFWEIGHT Y -1
  63. JOINT LOAD
  64. 222 226 FY -394.35
  65. 224 FY -349.88
  66. LOAD 2 LOADTYPE NONE  TITLE DW_DL
  67. JOINT LOAD
  68. 222 226 FY -57.83
  69. 224 FY -50.89
  70. LOAD 3 LOADTYPE NONE  TITLE LLA+IM
  71. JOINT LOAD
  72. 222 FY -384.49
  73. 224 FY -363.13
  74. LOAD 4 LOADTYPE NONE  TITLE LLB+IM
  75. JOINT LOAD
  76. 226 FY -384.49
  77. 224 FY -363.13
  78. LOAD 5 LOADTYPE NONE  TITLE BF
  79. MEMBER LOAD
  80. 187 TO 190 UNI GX 17.73
  81. 187 TO 190 UMOM GZ -10.64
  82. LOAD 6 LOADTYPE NONE  TITLE WA_OWL
  83. MEMBER LOAD
  84. 192 UNI GZ 6.91
  85. 193 UNI GZ 6.91 0 2.235
  86. 192 CON GY 52.14 0.001
  87. LOAD 7 LOADTYPE NONE  TITLE WA_DFL
  88. MEMBER LOAD
  89. 192 TO 194 UNI GZ 6.91
  90. 195 UNI GZ 6.91 0 2.025
  91. 192 CON GY 102 0.001
  92. LOAD 101 LOADTYPE NONE  TITLE TRANSV. K_BENT
  93. MEMBER LOAD
STAAD SPACE                                -- PAGE NO.  3

  94. 187 TO 190 UNI GZ 18.18
  95. LOAD 8 LOADTYPE NONE  TITLE LONGITUDINAL SEISMIC LOAD
(ORTHOGONAL)
  96. MEMBER LOAD
  97. 187 TO 190 UNI GX 79.62
  98. 187 TO 190 UMOM GZ -121.57
  99. LOAD 9 LOADTYPE NONE  TITLE TRANSVERSE SEISMIC LOAD (ORTHOGONAL)
 100. MEMBER LOAD
 101. 187 TO 190 UNI GZ 79.18
 102. 187 TO 190 UMOM GX 120.91
 103. LOAD COMB 10 STRENGTH IA 1.25DC+1.50DW+1.75[(LLA+IM)+BF]+1.0WA_DFL
 104. 1 1.25 2 1.5 3 1.75 5 1.75 7 1.0
 105. LOAD COMB 11 STRENGTH IB 1.25DC+1.50DW+1.75[(LLB+IM)+BF]+1.0WA_DFL
 106. 1 1.25 2 1.5 4 1.75 5 1.75 7 1.0
 107. LOAD COMB 12 EXTREME IA
1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ1
 108. 1 1.25 2 1.5 3 0.5 5 0.5 6 1.0 8 1.0
 109. LOAD COMB 13 EXTREME IB
1.25DC+1.50DW+0.50[(LLB+IM)+BF]+1.0WA_OWL+1.0EQ1
 110. 1 1.25 2 1.5 4 0.5 5 0.5 6 1.0 8 1.0
 111. LOAD COMB 14 EXTREME IC
1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2
 112. 1 1.25 2 1.5 3 0.5 5 0.5 6 1.0 9 1.0
 113. LOAD COMB 15 EXTREME IC
1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2
 114. 1 1.25 2 1.5 4 0.5 5 0.5 6 1.0 9 1.0
 115. LOAD COMB 16 SERVICE IA 1.0DC+1.0DW+1.0[(LLA+IM)+BF]+1.0WA_OWL
 116. 1 1.0 2 1.0 3 1.0 5 1.0 6 1.0
 117. LOAD COMB 17 SERVICE IB 1.0DC+1.0DW+1.0[(LLB+IM)+BF]+1.0WA_OWL
 118. 1 1.0 2 1.0 4 1.0 5 1.0 6 1.0
 119. PERFORM ANALYSIS


      P R O B L E M   S T A T I S T I C S
      -----------------------------------

 NUMBER OF JOINTS        10  NUMBER OF MEMBERS      9
 NUMBER OF PLATES         0  NUMBER OF SOLIDS       0
 NUMBER OF SURFACES       0  NUMBER OF SUPPORTS     1


      SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

 ORIGINAL/FINAL BAND-WIDTH=    3/    3/    24 DOF
TOTAL     PRIMARY LOAD CASES =    10, TOTAL DEGREES OF FREEDOM =     54
TOTAL LOAD COMBINATION  CASES =     8  SO FAR.
 SIZE OF STIFFNESS MATRIX =        2 DOUBLE  KILO-WORDS
 REQRD/AVAIL. DISK SPACE  =     12.0/1679851.6 MB


 120. FINISH
STAAD SPACE                                -- PAGE NO.  4


          *********** END OF THE STAAD.Pro RUN ***********

           **** DATE= JUN  4,2026   TIME= 15:37: 8 ****

        ************************************************************
        *  For technical assistance on STAAD.Pro, please visit     *
        *  http://selectservices.bentley.com/en-US/                *
        *                                                          *
        *  Details about additional assistance from                *
        *  Bentley and Partners can be found at program menu       *
        *  Help->Technical Support                                 *
        *                                                          *
        *     Copyright (c) 1997-2015 Bentley Systems, Inc.        *
        *             http://www.bentley.com                       *
        ************************************************************
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2. Pier 1&2 Footing Extreme Analysis Staad Output/Code


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 *        Bentley Systems, Inc.               *
 *        Date=    JUN  4, 2026               *
 *        Time=    17:32: 8                   *
 *                                            *
 *    USER ID:                                *
 ****************************************************
```

```
   1. STAAD SPACE
INPUT FILE: D:\SA3\Bridge\JBJ\BP11_SI-EB2\Pier Design\STAAD R0\SI-EB2 Rev. PIER
1&2 FTG EXTREME R0 052.. .STD
   2. START JOB INFORMATION
   3. ENGINEER DATE 15-APR-26
   4. END JOB INFORMATION
   5. INPUT WIDTH 79
   6. UNIT METER KN
   7. JOINT COORDINATES
   8. 232 15 -11.833 2.75; 233 15 -12.433 2.75; 282 13.5 -13.033 1.25
   9. 283 13.5 -14.533 1.25; 284 13.5 -16.033 1.25; 285 13.5 -17.533 1.25
  10. 286 13.5 -19.033 1.25; 287 13.5 -20.533 1.25; 288 13.5 -22.033 1.25
  11. 289 13.5 -23.533 1.25; 290 13.5 -25.033 1.25; 291 13.5 -26.533 1.25
  12. 292 13.5 -29.533 1.25; 293 13.5 -32.533 1.25; 294 13.5 -35.533 1.25
  13. 295 13.5 -38.533 1.25; 296 13.5 -41.533 1.25; 297 13.5 -46.033 1.25
  14. 298 13.5 -50.533 1.25; 299 13.5 -55.033 1.25; 300 13.5 -64.033 1.25
  15. 648 12.25 -12.433 0; 649 13.5 -12.433 0; 650 14.45 -12.433 0; 651 15 -12.433 0
  16. 652 15.55 -12.433 0; 653 16.5 -12.433 0; 654 17.75 -12.433 0
  17. 655 12.25 -12.433 1.25; 656 13.5 -12.433 1.25; 657 14.45 -12.433 1.25
  18. 658 15 -12.433 1.25; 659 15.55 -12.433 1.25; 660 16.5 -12.433 1.25
  19. 661 17.75 -12.433 1.25; 662 12.25 -12.433 2.2; 663 13.5 -12.433 2.2
  20. 664 14.45 -12.433 2.2; 665 15 -12.433 2.2; 666 15.55 -12.433 2.2
  21. 667 16.5 -12.433 2.2; 668 17.75 -12.433 2.2; 669 12.25 -12.433 2.75
  22. 670 13.5 -12.433 2.75; 671 14.45 -12.433 2.75; 672 15.55 -12.433 2.75
  23. 673 16.5 -12.433 2.75; 674 17.75 -12.433 2.75; 675 15 -12.433 3.3
  24. 676 12.25 -12.433 3.3; 677 13.5 -12.433 3.3; 678 14.45 -12.433 3.3
  25. 679 15.55 -12.433 3.3; 680 16.5 -12.433 3.3; 681 17.75 -12.433 3.3
  26. 682 15 -12.433 4.25; 683 12.25 -12.433 4.25; 684 13.5 -12.433 4.25
  27. 685 14.45 -12.433 4.25; 686 15.55 -12.433 4.25; 687 16.5 -12.433 4.25
  28. 688 17.75 -12.433 4.25; 689 15 -12.433 5.5; 690 12.25 -12.433 5.5
  29. 691 13.5 -12.433 5.5; 692 14.45 -12.433 5.5; 693 15.55 -12.433 5.5
  30. 694 16.5 -12.433 5.5; 695 17.75 -12.433 5.5; 744 16.5 -64.033 1.25
  31. 745 16.5 -55.033 1.25; 746 16.5 -50.533 1.25; 747 16.5 -46.033 1.25
  32. 748 16.5 -41.533 1.25; 749 16.5 -38.533 1.25; 750 16.5 -35.533 1.25
  33. 751 16.5 -32.533 1.25; 752 16.5 -29.533 1.25; 753 16.5 -26.533 1.25
  34. 754 16.5 -25.033 1.25; 755 16.5 -23.533 1.25; 756 16.5 -22.033 1.25
  35. 757 16.5 -20.533 1.25; 758 16.5 -19.033 1.25; 759 16.5 -17.533 1.25
  36. 760 16.5 -16.033 1.25; 761 16.5 -14.533 1.25; 762 16.5 -13.033 1.25
  37. 763 13.5 -64.033 4.25; 764 13.5 -55.033 4.25; 765 13.5 -50.533 4.25
  38. 766 13.5 -46.033 4.25; 767 13.5 -41.533 4.25; 768 13.5 -38.533 4.25
  STAAD SPACE                              -- PAGE NO.  2

  39. 769 13.5 -35.533 4.25; 770 13.5 -32.533 4.25; 771 13.5 -29.533 4.25
  40. 772 13.5 -26.533 4.25; 773 13.5 -25.033 4.25; 774 13.5 -23.533 4.25
  41. 775 13.5 -22.033 4.25; 776 13.5 -20.533 4.25; 777 13.5 -19.033 4.25
  42. 778 13.5 -17.533 4.25; 779 13.5 -16.033 4.25; 780 13.5 -14.533 4.25
  43. 781 13.5 -13.033 4.25; 782 16.5 -64.033 4.25; 783 16.5 -55.033 4.25
  44. 784 16.5 -50.533 4.25; 785 16.5 -46.033 4.25; 786 16.5 -41.533 4.25
  45. 787 16.5 -38.533 4.25; 788 16.5 -35.533 4.25; 789 16.5 -32.533 4.25
  46. 790 16.5 -29.533 4.25; 791 16.5 -26.533 4.25; 792 16.5 -25.033 4.25
  47. 793 16.5 -23.533 4.25; 794 16.5 -22.033 4.25; 795 16.5 -20.533 4.25
  48. 796 16.5 -19.033 4.25; 797 16.5 -17.533 4.25; 798 16.5 -16.033 4.25
  49. 799 16.5 -14.533 4.25; 800 16.5 -13.033 4.25
  50. MEMBER INCIDENCES
  51. 191 233 232; 233 300 299; 234 299 298; 235 298 297; 236 297 296; 237 296 295
  52. 238 295 294; 239 294 293; 240 293 292; 241 292 291; 242 291 290; 243 290 289
  53. 244 289 288; 245 288 287; 246 287 286; 247 286 285; 248 285 284; 249 284 283
  54. 250 283 282; 251 282 656; 793 744 745; 794 745 746; 795 746 747; 796 747 748
  55. 797 748 749; 798 749 750; 799 750 751; 800 751 752; 801 752 753; 802 753 754
  56. 803 754 755; 804 755 756; 805 756 757; 806 757 758; 807 758 759; 808 759 760
  57. 809 760 761; 810 761 762; 811 762 660; 812 763 764; 813 764 765; 814 765 766
  58. 815 766 767; 816 767 768; 817 768 769; 818 769 770; 819 770 771; 820 771 772
  59. 821 772 773; 822 773 774; 823 774 775; 824 775 776; 825 776 777; 826 777 778
  60. 827 778 779; 828 779 780; 829 780 781; 830 781 684; 831 782 783; 832 783 784
  61. 833 784 785; 834 785 786; 835 786 787; 836 787 788; 837 788 789; 838 789 790
  62. 839 790 791; 840 791 792; 841 792 793; 842 793 794; 843 794 795; 844 795 796
  63. 845 796 797; 846 797 798; 847 798 799; 848 799 800; 849 800 687
  64. ELEMENT INCIDENCES SHELL
  65. 721 655 656 649 648; 722 656 657 650 649; 723 657 658 651 650
  66. 724 658 659 652 651; 725 659 660 653 652; 726 660 661 654 653
  67. 727 662 663 656 655; 728 663 664 657 656; 729 664 665 658 657
  68. 730 665 666 659 658; 731 666 667 660 659; 732 667 668 661 660
  69. 733 669 670 663 662; 734 670 671 664 663; 735 671 233 665 664
  70. 736 233 672 666 665; 737 672 673 667 666; 738 673 674 668 667
  71. 739 676 677 670 669; 740 677 678 671 670; 741 678 675 233 671
  72. 742 675 679 672 233; 743 679 680 673 672; 744 680 681 674 673
  73. 745 683 684 677 676; 746 684 685 678 677; 747 685 682 675 678
  74. 748 682 686 679 675; 749 686 687 680 679; 750 687 688 681 680
  75. 751 690 691 684 683; 752 691 692 685 684; 753 692 689 682 685
  76. 754 689 693 686 682; 755 693 694 687 686; 756 694 695 688 687
  77. START GROUP DEFINITION
  78. MEMBER
  79. _BOREDPILE 233 TO 250 793 TO 810 812 TO 829 831 TO 848
  80. _PILETOP_BEAM 250 810 829 848
  81. JOINT
  82. _PILEHEAD 282 762 781 800
  83. _PILEBOTTOM 300 744 763 782
  84. ELEMENT
  85. _PILECAP 721 TO 734 737 TO 740 743 TO 756
  86. END GROUP DEFINITION
  87. START USER TABLE
  88. TABLE 1
  89. UNIT METER KN
  90. GENERAL
  91. TYPEIII
  92. 0.360685 1.143 0.001 0.56 0.001 0.0521963 0.00509875 0.00710869 0.0830343 -
  93. 0.0181982 0.214693 0.200946 0.123389 0.0346505 0.00061616 1.143
  95. END
  STAAD SPACE                              -- PAGE NO.  3

  96. ELEMENT PROPERTY
  97. 721 TO 756 THICKNESS 1.2
  98. SUPPORTS
  99. 283 761 780 799 FIXED BUT FY MX MY MZ KFX 89175 KFZ 89175
 100. 284 760 779 798 FIXED BUT FY MX MY MZ KFX 89175 KFZ 89175
 101. 285 759 778 797 FIXED BUT FY MX MY MZ KFX 54497 KFZ 54497
 102. 286 758 777 796 FIXED BUT FY MX MY MZ KFX 40178 KFZ 40178
 103. 287 757 776 795 FIXED BUT FY MX MY MZ KFX 57044 KFZ 57044
 104. 288 756 775 794 FIXED BUT FY MX MY MZ KFX 14745 KFZ 14745
 105. 289 755 774 793 FIXED BUT FY MX MY MZ KFX 77007 KFZ 77007
 106. 290 754 773 792 FIXED BUT FY MX MY MZ KFX 101711 KFZ 101711
 107. 291 753 772 791 FIXED BUT FY MX MY MZ KFX 298107 KFZ 298107
 108. 292 752 771 790 FIXED BUT FY MX MY MZ KFX 418170 KFZ 418170
 109. 293 751 770 789 FIXED BUT FY MX MY MZ KFX 456489 KFZ 456489
 110. 294 750 769 788 FIXED BUT FY MX MY MZ KFX 613086 KFZ 613086
 111. 295 749 768 787 FIXED BUT FY MX MY MZ KFX 681888 KFZ 681888
 112. 296 748 767 786 FIXED BUT FY MX MY MZ KFX 888266 KFZ 888266
 113. 297 747 766 785 FIXED BUT FY MX MY MZ KFX 1.25943E+006 KFZ 1.25943E+006
 114. 298 746 765 784 FIXED BUT FY MX MY MZ KFX 1.32594E+006 KFZ 1.32594E+006
 115. 299 745 764 783 FIXED BUT FY MX MY MZ KFX 2.08691E+006 KFZ 2.08691E+006
 116. 300 744 763 782 FIXED BUT MX MY MZ KFX 1.44275E+006 KFY 325050 KFZ
1.44275E+006
 117. DEFINE MATERIAL START
 118. ISOTROPIC CONCRETE0
 119. E 2.735E+007
 120. POISSON 0.17
 121. ALPHA 1E-005
 122. DAMP 0.05
 123. ISOTROPIC CONCRETE1
 124. E 2.735E+007
 125. POISSON 0.17
 126. DENSITY 24
 127. ALPHA 1E-005
 128. DAMP 0.05
 129. ISOTROPIC RIGID_LINK
 130. E 2.735E+007
 131. POISSON 0.17
 132. ALPHA 1E-005
 133. DAMP 0.05
 134. ISOTROPIC PCONCRETE1
 135. E 3.3497E+007
 136. POISSON 0.17
 137. DENSITY 24
 138. ALPHA 1E-005
 139. DAMP 0.05
 140. END DEFINE MATERIAL
 141. MEMBER PROPERTY AMERICAN
 142. 191 251 811 830 849 PRIS AX 100 IX 1000 IY 1000 IZ 1000
 143. 233 TO 250 793 TO 810 812 TO 829 831 TO 848 PRIS YD 1
 144. CONSTANTS
 145. MATERIAL CONCRETE1 MEMB 233 TO 250 721 TO 756 793 TO 810 812 TO 829
831 TO 848
 146. MATERIAL RIGID_LINK MEMB 191 251 811 830 849
 147. LOAD 1 DC_DL (PILECAP + PILES & SCOUR PROTECTION WT.)
 148. SELFWEIGHT Y -1
 149. ELEMENT LOAD
 150. 721 TO 756 PR GY -15
 151. LOAD 2 WA (BUOYANCY FROM PILECAP + ROCK)
  STAAD SPACE                             -- PAGE NO.  4

 152. ELEMENT LOAD
 153. 721 TO 756 PR GY 19.5
 154. LOAD 3 EQ1 (LONGI. INERTIAL SEISMIC EFFECT OF PILECAP + ROCK)
 155. ELEMENT LOAD
 156. 721 TO 756 PR GX 20.59
 157. LOAD 4 EQ2 (TRANSV. INERTIAL SEISMIC EFFECT OF PILECAP + ROCK)
 158. ELEMENT LOAD
 159. 721 TO 756 PR GZ 20.59
 160. **** LOADS FROM PIER BENT ANALYSIS
 161. LOAD 7 EXTREME IA 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ1
 162. JOINT LOAD
 163. 232 FX 486.67 FY -2632.07 FZ 31.51 MX -264.59 MZ -5515.9
 164. LOAD 8 EXTREME IB 1.25DC+1.50DW+0.50[(LLB+IM)+BF]+1.0WA_OWL+1.0EQ1
 165. JOINT LOAD
 166. 232 FX 486.67 FY -2632.07 FZ 31.51 MX 408.27 MZ -5515.9
 167. LOAD 9 EXTREME IC 1.25DC+1.50DW+0.50[(LLA+IM)+BF]+1.0WA_OWL+1.0EQ2
 168. JOINT LOAD
 169. 232 FX 48.76 FY -2632.07 FZ 467 MX 4711.77 MZ -511.96
 170. LOAD 10 EXTREME ID 1.25DC+1.50DW+0.50[(LLB+IM)+BF]+1.0WA_OWL+1.0EQ2
 171. JOINT LOAD
```

172. 232 FX 48.76 FY -2632.07 FZ 467 MX 5384.63 MZ -511.96
173. LOAD COMB 15 1.25DC(FTG WT)+EXTREME IA+1.0WA(BUOYANCY)+1.0EQ1(L.INERTIA)
174. 1 1.25 7 1.0 2 1.0 3 1.0
175. LOAD COMB 16 1.25DC(FTG WT)+EXTREME IB +1.0WA(BUOYANCY)+1.0EQ1(L.INERTIA)
176. 1 1.25 8 1.0 2 1.0 3 1.0
177. LOAD COMB 17 1.25DC(FTG WT)+EXTREME IC+1.0WA(BUOYANCY)+1.0EQ2(T.INERTIA)
178. 1 1.25 9 1.0 2 1.0 4 1.0
179. LOAD COMB 18 1.25DC(FTG WT)+EXTREME ID+1.0WA(BUOYANCY)+1.0EQ2(T.INERTIA)
180. 1 1.25 10 1.0 2 1.0 4 1.0
181. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 126 NUMBER OF MEMBERS 77
NUMBER OF PLATES 36 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 72

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 65/ 15/ 96 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 756
TOTAL LOAD COMBINATION CASES = 4 SO FAR.
SIZE OF STIFFNESS MATRIX = 73 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.2/1679821.0 MB

STAAD SPACE -- PAGE NO. 5

***WARNING - INSTABILITY AT JOINT 782 DIRECTION = MY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1284402E+06 L-MATRIX DIAG= 1.0401406E-04 EQN NO 755
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

182. FINISH
STAAD SPACE -- PAGE NO. 6

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 4,2026 TIME= 17:32: 9 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2015 Bentley Systems, Inc.
* http://www.bentley.com *
****

3. Pier 1&2 Footing Strength Analysis Staad Output/Code

```
                                    PAGE NO.    1




        **************************************************
        *                                                *
        *        STAAD.Pro V8i SELECTseries6             *
        *        Version  20.07.11.33                    *
        *        Proprietary Program of                  *
        *        Bentley Systems, Inc.                   *
        *        Date=    JUN  4, 2026                   *
        *        Time=    17:33: 2                       *
        *                                                *
        *    USER ID:                                    *
        **************************************************



    1. STAAD SPACE
INPUT FILE: D:\SA3\Bridge\JBJ\BP11_SI-EB2\Pier Design\STAAD R0\SI-EB2 Rev. PIER
1&2 FTG STRENGTH R0 05.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 15-APR-26
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 232 15 -11.833 2.75; 233 15 -12.433 2.75; 282 13.5 -13.033 1.25
    9. 283 13.5 -14.533 1.25; 284 13.5 -16.033 1.25; 285 13.5 -17.533 1.25
   10. 286 13.5 -19.033 1.25; 287 13.5 -20.533 1.25; 288 13.5 -22.033 1.25
   11. 289 13.5 -23.533 1.25; 290 13.5 -25.033 1.25; 291 13.5 -26.533 1.25
   12. 292 13.5 -29.533 1.25; 293 13.5 -32.533 1.25; 294 13.5 -35.533 1.25
   13. 295 13.5 -38.533 1.25; 296 13.5 -41.533 1.25; 297 13.5 -46.033 1.25
   14. 298 13.5 -50.533 1.25; 299 13.5 -55.033 1.25; 300 13.5 -64.033 1.25
   15. 648 12.25 -12.433 0; 649 13.5 -12.433 0; 650 14.45 -12.433 0; 651 15 -12.433 0
   16. 652 15.55 -12.433 0; 653 16.5 -12.433 0; 654 17.75 -12.433 0
   17. 655 12.25 -12.433 1.25; 656 13.5 -12.433 1.25; 657 14.45 -12.433 1.25
   18. 658 15 -12.433 1.25; 659 15.55 -12.433 1.25; 660 16.5 -12.433 1.25
   19. 661 17.75 -12.433 1.25; 662 12.25 -12.433 2.2; 663 13.5 -12.433 2.2
   20. 664 14.45 -12.433 2.2; 665 15 -12.433 2.2; 666 15.55 -12.433 2.2
   21. 667 16.5 -12.433 2.2; 668 17.75 -12.433 2.2; 669 12.25 -12.433 2.75
   22. 670 13.5 -12.433 2.75; 671 14.45 -12.433 2.75; 672 15.55 -12.433 2.75
   23. 673 16.5 -12.433 2.75; 674 17.75 -12.433 2.75; 675 15 -12.433 3.3
   24. 676 12.25 -12.433 3.3; 677 13.5 -12.433 3.3; 678 14.45 -12.433 3.3
   25. 679 15.55 -12.433 3.3; 680 16.5 -12.433 3.3; 681 17.75 -12.433 3.3
   26. 682 15 -12.433 4.25; 683 12.25 -12.433 4.25; 684 13.5 -12.433 4.25
   27. 685 14.45 -12.433 4.25; 686 15.55 -12.433 4.25; 687 16.5 -12.433 4.25
   28. 688 17.75 -12.433 4.25; 689 15 -12.433 5.5; 690 12.25 -12.433 5.5
   29. 691 13.5 -12.433 5.5; 692 14.45 -12.433 5.5; 693 15.55 -12.433 5.5
   30. 694 16.5 -12.433 5.5; 695 17.75 -12.433 5.5; 744 16.5 -64.033 1.25
   31. 745 16.5 -55.033 1.25; 746 16.5 -50.533 1.25; 747 16.5 -46.033 1.25
   32. 748 16.5 -41.533 1.25; 749 16.5 -38.533 1.25; 750 16.5 -35.533 1.25
   33. 751 16.5 -32.533 1.25; 752 16.5 -29.533 1.25; 753 16.5 -26.533 1.25
   34. 754 16.5 -25.033 1.25; 755 16.5 -23.533 1.25; 756 16.5 -22.033 1.25
   35. 757 16.5 -20.533 1.25; 758 16.5 -19.033 1.25; 759 16.5 -17.533 1.25
   36. 760 16.5 -16.033 1.25; 761 16.5 -14.533 1.25; 762 16.5 -13.033 1.25
   37. 763 13.5 -64.033 4.25; 764 13.5 -55.033 4.25; 765 13.5 -50.533 4.25
   38. 766 13.5 -46.033 4.25; 767 13.5 -41.533 4.25; 768 13.5 -38.533 4.25
   STAAD SPACE                              -- PAGE NO.    2

   39. 769 13.5 -35.533 4.25; 770 13.5 -32.533 4.25; 771 13.5 -29.533 4.25
   40. 772 13.5 -26.533 4.25; 773 13.5 -25.033 4.25; 774 13.5 -23.533 4.25
   41. 775 13.5 -22.033 4.25; 776 13.5 -20.533 4.25; 777 13.5 -19.033 4.25
   42. 778 13.5 -17.533 4.25; 779 13.5 -16.033 4.25; 780 13.5 -14.533 4.25
   43. 781 13.5 -13.033 4.25; 782 16.5 -64.033 4.25; 783 16.5 -55.033 4.25
   44. 784 16.5 -50.533 4.25; 785 16.5 -46.033 4.25; 786 16.5 -41.533 4.25
   45. 787 16.5 -38.533 4.25; 788 16.5 -35.533 4.25; 789 16.5 -32.533 4.25
   46. 790 16.5 -29.533 4.25; 791 16.5 -26.533 4.25; 792 16.5 -25.033 4.25
   47. 793 16.5 -23.533 4.25; 794 16.5 -22.033 4.25; 795 16.5 -20.533 4.25
   48. 796 16.5 -19.033 4.25; 797 16.5 -17.533 4.25; 798 16.5 -16.033 4.25
   49. 799 16.5 -14.533 4.25; 800 16.5 -13.033 4.25
   50. MEMBER INCIDENCES
   51. 191 233 232; 233 300 299; 234 299 298; 235 298 297; 236 297 296; 237 296 295
   52. 238 295 294; 239 294 293; 240 293 292; 241 292 291; 242 291 290; 243 290 289
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   54. 250 283 282; 251 282 656; 793 744 745; 794 745 746; 795 746 747; 796 747 748
   55. 797 748 749; 798 749 750; 799 750 751; 800 751 752; 801 752 753; 802 753 754
   56. 803 754 755; 804 755 756; 805 756 757; 806 757 758; 807 758 759; 808 759 760
   57. 809 760 761; 810 761 762; 811 762 660; 812 763 764; 813 764 765; 814 765 766
   58. 815 766 767; 816 767 768; 817 768 769; 818 769 770; 819 770 771; 820 771 772
   59. 821 772 773; 822 773 774; 823 774 775; 824 775 776; 825 776 777; 826 777 778
   60. 827 778 779; 828 779 780; 829 780 781; 830 781 684; 831 782 783; 832 783 784
   61. 833 784 785; 834 785 786; 835 786 787; 836 787 788; 837 788 789; 838 789 790
   62. 839 790 791; 840 791 792; 841 792 793; 842 793 794; 843 794 795; 844 795 796
   63. 845 796 797; 846 797 798; 847 798 799; 848 799 800; 849 800 687
   64. ELEMENT INCIDENCES SHELL
   65. 721 655 656 649 648; 722 656 657 650 649; 723 657 658 651 650
   66. 724 658 659 652 651; 725 659 660 653 652; 726 660 661 654 653
   67. 727 662 663 656 655; 728 663 664 657 656; 729 664 665 658 657
   68. 730 665 666 659 658; 731 666 667 660 659; 732 667 668 661 660
   69. 733 669 670 663 662; 734 670 671 664 663; 735 671 233 665 664
   70. 736 233 672 666 665; 737 672 673 667 666; 738 673 674 668 667
   71. 739 676 677 670 669; 740 677 678 671 670; 741 678 675 233 671
   72. 742 675 679 672 233; 743 679 680 673 672; 744 680 681 674 673
   73. 745 683 684 677 676; 746 684 685 678 677; 747 685 682 675 678
   74. 748 682 686 679 675; 749 686 687 680 679; 750 687 688 681 680
   75. 751 690 691 684 683; 752 691 692 685 684; 753 692 689 682 685
   76. 754 689 693 686 682; 755 693 694 687 686; 756 694 695 688 687
   77. START GROUP DEFINITION
   78. MEMBER
   79. _BOREDPILE 233 TO 250 793 TO 810 812 TO 829 831 TO 848
   80. JOINT
   81. _PILEHEAD 282 762 781 800
   82. _PILEBOTTOM 300 744 763 782
   83. ELEMENT
   84. _PILECAP 721 TO 734 737 TO 740 743 TO 756
   85. END GROUP DEFINITION
   86. START USER TABLE
   87. TABLE 1
   88. UNIT METER KN
   89. GENERAL
   90. TYPEIII
   91. 0.360685 1.143 0.001 0.56 0.001 0.0521963 0.00509875 0.00710869 0.0830343 -
   92. 0.0181982 0.214693 0.200946 0.123389 0.0346505 0.00061616 1.143
   94. END
   95. ELEMENT PROPERTY
   STAAD SPACE                              -- PAGE NO.    3

   96. 721 TO 756 THICKNESS 1.2
   97. SUPPORTS
   98. 283 761 780 799 FIXED BUT FY MX MY MZ KFX 7434 KFZ 7434
   99. 284 760 779 798 FIXED BUT FY MX MY MZ KFX 7434 KFZ 7434
  100. 285 759 778 797 FIXED BUT FY MX MY MZ KFX 7434 KFZ 7434
  101. 286 758 777 796 FIXED BUT FY MX MY MZ KFX 2705 KFZ 2705
  102. 287 757 776 795 FIXED BUT FY MX MY MZ KFX 4232 KFZ 4232
  103. 288 756 775 794 FIXED BUT FY MX MY MZ KFX 1259 KFZ 1259
  104. 289 755 774 793 FIXED BUT FY MX MY MZ KFX 12488 KFZ 12488
  105. 290 754 773 792 FIXED BUT FY MX MY MZ KFX 17747 KFZ 17747
  106. 291 753 772 791 FIXED BUT FY MX MY MZ KFX 34736 KFZ 34736
  107. 292 752 771 790 FIXED BUT FY MX MY MZ KFX 49977 KFZ 49977
  108. 293 751 770 789 FIXED BUT FY MX MY MZ KFX 115311 KFZ 115311
  109. 294 750 769 788 FIXED BUT FY MX MY MZ KFX 167979 KFZ 167979
  110. 295 749 768 787 FIXED BUT FY MX MY MZ KFX 197034 KFZ 197034
  111. 296 748 767 786 FIXED BUT FY MX MY MZ KFX 262016 KFZ 262016
  112. 297 747 766 785 FIXED BUT FY MX MY MZ KFX 403817 KFZ 403817
  113. 298 746 765 784 FIXED BUT FY MX MY MZ KFX 436226 KFZ 436226
  114. 299 745 764 783 FIXED BUT FY MX MY MZ KFX 703289 KFZ 703289
  115. 300 744 763 782 FIXED BUT MX MY MZ KFX 495122 KFY 123315 KFZ 495122
  116. DEFINE MATERIAL START
  117. ISOTROPIC CONCRETE0
  118. E 2.735E+007
  119. POISSON 0.17
  120. ALPHA 1E-005
  121. DAMP 0.05
  122. ISOTROPIC CONCRETE1
  123. E 2.735E+007
  124. POISSON 0.17
  125. DENSITY 24
  126. ALPHA 1E-005
  127. DAMP 0.05
  128. ISOTROPIC RIGID_LINK
  129. E 2.735E+007
  130. POISSON 0.17
  131. ALPHA 1E-005
  132. DAMP 0.05
  133. ISOTROPIC PCONCRETE1
  134. E 3.3497E+007
  135. POISSON 0.17
  136. DENSITY 24
  137. ALPHA 1E-005
  138. DAMP 0.05
  139. END DEFINE MATERIAL
  140. MEMBER PROPERTY AMERICAN
  141. 191 251 811 830 849 PRIS AX 100 IX 1000 IY 1000 IZ 1000
  142. 233 TO 250 793 TO 810 812 TO 829 831 TO 848 PRIS YD 1
  143. CONSTANTS
  144. MATERIAL CONCRETE1 MEMB 233 TO 250 721 TO 756 793 TO 810 812 TO 829
831 TO 848
  145. MATERIAL RIGID_LINK MEMB 191 251 811 830 849
  146. LOAD 1 DC_DL (PILECAP + PILES & SCOUR PROTECTION WT.)
  147. SELFWEIGHT Y -1
  148. ELEMENT LOAD
  149. 721 TO 756 PR GY -15
  150. LOAD 2 WA (BUOYANCY FROM PILECAP + ROCK)
  151. ELEMENT LOAD
   STAAD SPACE                              -- PAGE NO.    4

  152. 721 TO 756 PR GY 19.5
  153. LOAD 5 STRENGTH IA 1.25DC+1.50DW+1.75[(LLA+IM)+BF]+1.0WA_DFL
  154. JOINT LOAD
  155. 232 FX 170.65 FY -3516.94 FZ 62.19 MX -897.65 MZ -1791.86
  156. LOAD 6 STRENGTH IB 1.25DC+1.50DW+1.75[(LLB+IM)+BF]+1.0WA_DFL
  157. JOINT LOAD
  158. 232 FX 170.65 FY -3516.94 FZ 62.19 MX 1457.36 MZ -1791.86
  159. LOAD 11 SERVICE IA 1.0DC+1.0DW+1.0[(LLA+IM)+BF]+1.0WA_OWL
  160. JOINT LOAD
  161. 232 FX 97.52 FY -2510.49 FZ 31.51 MX -601.02 MZ -1023.92
  162. LOAD 12 SERVICE IB 1.0DC+1.0DW+1.0[(LLB+IM)+BF]+1.0WA_OWL
  163. JOINT LOAD
  164. 232 FX 97.52 FY -2510.49 FZ 31.51 MX 744.7 MZ -1023.92
  165. LOAD COMB 13 1.25DC(FTG WT)+STRENGTH IA+1.0WA(BUOYANCY)
  166. 1 1.25 5 1.0 2 1.0
  167. LOAD COMB 14 1.25DC(FTG WT)+STRENGTH IB+1.0WA(BUOYANCY)
  168. 1 1.25 6 1.0 2 1.0
  169. LOAD COMB 19 1.0DC(FTG WT)+SERVICE IA+1.0WA(BUOYANCY)
  170. 1 1.0 11 1.0 2 1.0
  171. LOAD COMB 20 1.0DC(FTG WT)+SERVICE IB+1.0WA(BUOYANCY)
  172. 1 1.0 12 1.0 2 1.0
```

173. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 126 NUMBER OF MEMBERS 77
NUMBER OF PLATES 36 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 72

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 65/ 15/ 96 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 756
TOTAL LOAD COMBINATION CASES = 4 SO FAR.
SIZE OF STIFFNESS MATRIX = 73 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.1/1679820.2 MB

***WARNING - INSTABILITY AT JOINT 782 DIRECTION = MY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1284402E+06 L-MATRIX DIAG= 1.0888674E-04 EQN NO 755
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
STAAD SPACE -- PAGE NO. 5

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

174. FINISH
STAAD SPACE -- PAGE NO. 6

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 4,2026 TIME= 17:33: 3 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

San Isidro -EB2 Bridge

Design of Anchor Bars at Pier and Abutment

DESIGN OF ANCHOR BARS AT ABUTMENT A/B

Conversion factor:

$kN \equiv 1000 \cdot newton \quad kPa := \frac{kN}{m^2} \quad MPa := \frac{newton}{mm^2} \quad ksi := 1000 \cdot psi \quad kip := 4.448 \cdot kN$

Input Data:

$V_z := [(51.97 \cdot 2) + 74.76] \cdot kN = 178.70 \cdot kN$ Shear z from STAAD EQ analysis of beams for LC1 load

$V_y := [(246.39 \cdot 2) + 174.44] \cdot kN = 667.22 \cdot kN$ Axial x from STAAD EQ analysis of beams for LC1 load

$f'c := 27.6 \cdot MPa$ Concrete Strength

$fy := 414 \cdot MPa$ Yield Strength of Steel

$:= 1.0$ Correction Factor for Concrete

$_1 := 28mm$ Diameter of Anchor Bars

$n_1 := 6$ Number of Anchor Bars

$A_b := \frac{\pi}{4} \cdot {}_1^2 = 615.75 \cdot mm^2$ Area of one anchor bar

$R := 0.8$ Response modification factor

Design Shear

$V_u := \frac{\sqrt{V_y^2 + V_z^2}}{R}$ $\quad V_u = 863.42 \cdot kN$

Shear Capacity

$Avf := n_1 \cdot A_b \quad Avf = 3694.51 \cdot mm^2$

$Vn := 0.90 Avf \cdot fy \cdot 1.0 \quad Vn = 1376.58 \cdot kN$

Evaluation

$Evaluation := \begin{cases} \text{"ADEQUATE"} & \text{if } Vn > V_u \\ \text{"Revise Reinforcement"} & \text{otherwise} \end{cases}$ $\quad Evaluation = \text{"ADEQUATE"}$

Check the minimum development length required

Basic development length

For l_{dh}: AASHTO 8.29.2

$l_{hb1} := \frac{1200 \cdot psi}{\sqrt{f'c \cdot psi}} \cdot {}_1 \quad l_{hb1} = 531.06 \cdot mm$

$l_{hb2} := \begin{cases} 8 \cdot {}_1 & \text{if } 8 \cdot {}_1 > 150 \cdot mm \\ 150 \cdot mm & \text{otherwise} \end{cases}$

$l_{hb} := \begin{cases} l_{hb1} & \text{if } l_{hb1} > l_{hb2} \\ l_{hb2} & \text{otherwise} \end{cases}$

$l_{dh_min} := 0.7 \cdot l_{hb} \quad l_{dh_min} = 371.74 \cdot mm$

For l_{dh}: AASHTO 8.25.1

$l_{d_min} := \begin{cases} \frac{0.04}{in} \cdot A_b \cdot fy & \text{if } \frac{0.04}{in} \cdot A_b \cdot fy \geq 0.0004 \cdot {}_1 \cdot \frac{fy}{psi} \\ 0.0004 \cdot {}_1 \cdot \frac{fy}{psi} & \text{otherwise} \end{cases}$ $\quad l_{d_min} = 920.28 \cdot mm$

DESIGN OF ANCHOR BARS AT PIER 1/2

Conversion factor:

$kN \equiv 1000 \cdot newton$ $\quad kPa := \frac{kN}{m^2}$ $\quad MPa := \frac{newton}{mm^2}$ $\quad ksi := 1000 \cdot psi$ $\quad kip := 4.448 \cdot kN$

Input Data:

$V_z := 435.49 \cdot kN$ Shear z from Total Shear at C.G. of Superstructure @ Design of Pier, LC2 load

$V_y := 0 \cdot kN$ Shear y from Total Shear at C.G. of Superstructure @ Design of Pier, LC2 load

$f'c := 27.6 \cdot MPa$ Concrete Strength

$fy := 414 \cdot MPa$ Yield Strength of Steel

$\lambda := 1.0$ Correction Factor for Concrete

$d_1 := 28mm$ Diameter of Dowel

$n_1 := 13$ Number of Dowels

$A_b := \frac{\pi}{4} \cdot d_1^2 = 615.75 \cdot mm^2$ Area of one dowel bar

$R := 1.0$ Response modification factor

Design Shear

$V_u := \frac{\sqrt{V_y^2 + V_z^2}}{R}$ $\quad V_u = 435.49 \cdot kN$

Shear Capacity

$Avf := n_1 \cdot A_b$ $\quad Avf = 8004.78 \cdot mm^2$

$Vn := 0.90 Avf \cdot fy \cdot 1.0$ $\quad Vn = 2982.58 \cdot kN$

Evaluation

$$Evaluation := \begin{cases} \text{"ADEQUATE"} & \text{if } Vn > V_u \\ \text{"Revise Reinforcement"} & \text{otherwise} \end{cases}$$

Evaluation = "ADEQUATE"

Static Friction Force From Bearing Pads Will Also Help Resist the EQ Shear Forces
(μ = 0.20 assumption)
Plus Interface Shear Resistance of Concrete Diaphragm.

Check the minimum development length required

Basic development length

For l_{dh}:
AASHTO 8.29.2

$l_{hb1} := \frac{1200 \cdot psi}{\sqrt{f'c \cdot psi}} \cdot d_1$ $\quad l_{hb1} = 531.06 \cdot mm$

$$l_{hb2} := \begin{cases} 8 \cdot d_1 & \text{if } 8 \cdot d_1 > 150 \cdot mm \\ 150 \cdot mm & \text{otherwise} \end{cases}$$

$$l_{hb} := \begin{cases} l_{hb1} & \text{if } l_{hb1} > l_{hb2} \\ l_{hb2} & \text{otherwise} \end{cases}$$

$l_{dh_min} := 0.7 \cdot l_{hb}$ $\quad l_{dh_min} = 371.74 \cdot mm$

For l_{dh}:
AASHTO 8.25.1

$$l_{d_min} := \begin{cases} \frac{0.04}{in} \cdot A_b \cdot fy \Big/ \sqrt{f'c \cdot psi} & \text{if } \frac{0.04}{in} \cdot A_b \cdot fy \Big/ \sqrt{f'c \cdot psi} \geq 0.0004 \cdot d_1 \cdot \frac{fy}{psi} \\ 0.0004 \cdot d_1 \cdot \frac{fy}{psi} & \text{otherwise} \end{cases}$$

$l_{d_min} = 920.28 \cdot mm$

San Isidro-EB2 Bridge

Design of Shear Keys at Pier and Abutment

DESIGN OF SHEAR KEY AT ABUTMENT A/B (Redundant)

Conversion factor:

$kN \equiv 1000 \cdot newton$ $\quad kPa := \frac{kN}{m^2}$ $\quad MPa := \frac{newton}{mm^2}$ $\quad ksi := 1000 \cdot psi$ $\quad kip := 4.448 \cdot kN$

Input Data:

$L := 1.17m \cdot 2 + 0.56m$ total length of shear blocks (2mid,1end)

$L = 2.90\,m$

$w := 280 \cdot mm$ width of shear block

$h := 300 \cdot mm$ height of shear block

$n := 3$ total no. of keys on design

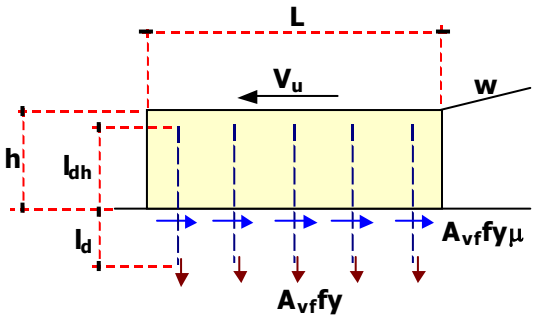
$d_b := 20 \cdot mm$ diameter of shear reinf.

$R := 0.8$ Response modification factor

$:= 0.80$ Shear capacity reduction factor (Enter 0.80)

$V_z := 178.70\ kN$ Shear z from STAAD EQ analysis of beams for LC1 load

$V_y := 667.22\ kN$ Axial x from STAAD EQ analysis of beams for LC1 load



$f_y := 414 \cdot MPa$ Yield strength of steel anchors

$f'c := 27.6 \cdot MPa$ Concrete strength

$:= 1.0$ correction factor (1.0 for normal wt. of conc.)

$:= 1.0 \cdot$ coeff. of friction (AASHTO 8.16.6.4.4)

$A_b := \frac{\pi}{4} \cdot d_b^2$ area of one shear reinf. steel

$A_b = 314.16 \cdot mm^2$

Shear demand per shear key

Longitudinal $\quad V_{d_long} := \frac{V_y}{R}$ $\quad V_{d_long} = 834.025 \cdot kN$

Transverse $\quad V_{d_transv} := \frac{V_z}{R}$ $\quad V_{d_transv} = 223.375 \cdot kN$

$V_u := 2\sqrt{V_{d_long}^2 + V_{d_transv}^2}$ $\quad V_u = 1726.84 \cdot kN$ (double load, only one abutment will trigger at a time)

Area of steel required for shear friction

$A_{vf} := \frac{V_u}{\phi \cdot f_y \cdot \mu}$ $\quad A_{vf} = 5213.889 \cdot mm^2$

Let spacing of bars: $\quad s_p := 200mm$

Est. number of bars: $\quad n_b := \frac{L}{s_p} \cdot 2 = 29.00$

C/D Ratio:

$CD := \frac{A_{total}}{A_{vf}} = 1.75$

Area of bars provided:

$A_{total} := n_b \cdot A_b$

$A_{total} = 9110.62 \cdot mm^2$

Check the minimum development length required

Basic development length

For l_{dh}:
AASHTO 8.29.2

$$l_{hb1} := \frac{1200 \cdot psi}{\sqrt{f'c \cdot psi}} \cdot d_b \qquad l_{hb1} = 379.33 \cdot mm$$

$$l_{hb2} := \begin{cases} 8 \cdot d_b & \text{if } 8 \cdot d_b > 150 \cdot mm \\ 150 \cdot mm & \text{otherwise} \end{cases} \qquad l_{hb} := \begin{cases} l_{hb1} & \text{if } l_{hb1} > l_{hb2} \\ l_{hb2} & \text{otherwise} \end{cases}$$

$$l_{dh_min} := 0.7 \cdot l_{hb} \qquad l_{dh_min} = 265.53 \cdot mm$$

For l_{dh}:
AASHTO8.25.1

$$l_{d_min} := \begin{cases} \frac{0.04}{in} \cdot A_b \cdot f_y \Big/ \sqrt{f'c \cdot psi} & \text{if } \frac{0.04}{in} \cdot A_b \cdot f_y \Big/ \sqrt{f'c \cdot psi} \geq 0.0004 \cdot d_b \cdot \frac{f_y}{psi} \\ 0.0004 \cdot d_b \cdot \frac{f_y}{psi} & \text{otherwise} \end{cases}$$

$$l_{d_min} = 480.36 \cdot mm$$

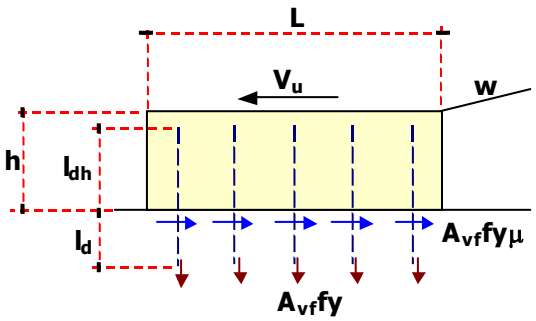
DESIGN OF SHEAR KEY AT PIER 1/2 (Redundant)

Input Data:

total length of shear blocks (4mids,1end)

$L := 1.17m \cdot 4 + 1.5 \cdot m$

$L = 6.18\,m$



$w := 300 \cdot mm$	width of shear block
$h := 300 \cdot mm$	height of shear block
$n := 5$	total no. of keys on design
$d_b := 20 \cdot mm$	diameter of shear reinf.
$R := 1$	Response modification factor
$:= 0.80$	Shear capacity reduction factor (Enter 0.80)
$V_z := 435.49 \cdot kN$	Shear z from Total Shear at C.G. of Superstructure @ Design of Pier, LC2 load
$V_y := 0 \cdot kN$	Shear y from Total Shear at C.G. of Superstructure @ Design of Pier, LC2 load

$f_y := 414 \cdot MPa$	Yield strength of steel anchors
$f'c := 27.6 \cdot MPa$	Concrete strength
$:= 1.0$	correction factor (1.0 for normal wt. of conc.)
$:= 1.0 \cdot$	coeff. of friction (AASHTO 8.16.6.4.4)
$A_b := \frac{}{4} \cdot d_b^2$	area of one shear reinf. steel

$A_b = 314.16 \cdot mm^2$

Shear demand per shear key

Longitudinal $V_{d_long} := \frac{V_y}{R}$ $V_{d_long} = 0 \cdot kN$

Transverse $V_{d_transv} := \frac{V_z}{R}$ $V_{d_transv} = 435.49 \cdot kN$

$V_u := \sqrt{V_{d_long}^2 + V_{d_transv}^2}$ $V_u = 435.49 \cdot kN$

Area of steel required for shear friction

$A_{vf} := \frac{V_u}{\cdot f_y \cdot}$ $A_{vf} = 1314.885 \cdot mm^2$

Let spacing of bars: $s_p := 200.mm$

Est. number of bars: $n_b := \frac{L}{s_p} \cdot 2 = 61.80$

Area of bars provided: $A_{total} := n_b \cdot A_b$ $A_{total} = 19415.04 \cdot mm^2$

C/D Ratio: $CD := \frac{A_{total}}{A_{vf}} = 14.77$

Check the minimum development length required

Basic development length

For l_{dh}: AASHTO 8.29.2

$l_{hb1} := \frac{1200 \cdot psi}{\sqrt{f'c \cdot psi}} \cdot d_b$ $l_{hb1} = 379.33 \cdot mm$

$l_{hb2} := \begin{cases} 8 \cdot d_b & \text{if } 8 \cdot d_b > 150 \cdot mm \\ 150 \cdot mm & \text{otherwise} \end{cases}$

$l_{hb} := \begin{cases} l_{hb1} & \text{if } l_{hb1} > l_{hb2} \\ l_{hb2} & \text{otherwise} \end{cases}$

$l_{dh_min} := 0.7 \cdot l_{hb}$ $l_{dh_min} = 265.53 \cdot mm$

For l_{dh}: AASHTO8.25.1

$l_{d_min} := \begin{cases} \frac{0.04}{in} \cdot A_b \cdot f_y \Big/ \sqrt{f'c \cdot psi} & \text{if } \frac{0.04}{in} \cdot A_b \cdot f_y \Big/ \sqrt{f'c \cdot psi} \geq 0.0004 \cdot d_b \cdot \frac{f_y}{psi} \\ 0.0004 \cdot d_b \cdot \frac{f_y}{psi} & \text{otherwise} \end{cases}$

$l_{d_min} = 480.36 \cdot mm$

Design of Approach Slab

Conversion Factors:

$kN := 1000 \cdot newton$ $\quad kPa := \frac{kN}{m^2}$ $\quad MPa := 1000 \cdot kPa$

Design slab reinforcement is computed considering 1-m strip of approach slab

1. INPUT DATA

Geometry

$S := 4000 \cdot mm$ clear span between backwall corbel and slab end support

$S_d := 90\% \cdot S = 3.6\,m$ design span as simply supported for design forces

$t_s := 270 \cdot mm$ approach slab thickness (average depth)

$t_{pave} := 80 \cdot mm$ thickness of pavement

$b := 1000 \cdot mm$ unit width considered in design

$20 \cdot mm$

$Cover := 50 \cdot mm$ concrete cover

Material Properties

$conc := 24 \cdot \frac{kN}{m^3}$ unit weight of concrete

$pave := 22 \cdot \frac{kN}{m^3}$ unit weight of pavement

$f_y := 414 \cdot MPa$ yield strength of reinforcement

$f_{c'} := 27.6 \cdot MPa$ concrete strength

$:= 0.90$ flexural strength reduction factor

$d_b := 25 \cdot mm$ diameter of main reinforcement

$d_s := 20 \cdot mm$ diameter of distribution reinforcement

2. BENDING MOMENTS AT MIDSPAN

2.1 Due to Dead Load

a. Slab (DC-S)

$W_{SLAB} := t_s \cdot b \cdot conc$ $\quad W_{SLAB} = 6.48 \cdot \frac{kN}{m}$ << load

$M_{slab} := \frac{W_{SLAB} \cdot S_d^2}{8}$ $\quad M_{slab} = 10.498 \cdot kN \cdot m$ << moment

b. Pavement (DW-P)

$W_{PAVE} := t_{pave} \cdot b \cdot pave$ $\quad W_{PAVE} = 1.76 \cdot \frac{kN}{m}$ << load

$M_{pave} := \frac{W_{PAVE} \cdot S_d^2}{8}$ $\quad M_{pave} = 2.851 \cdot kN \cdot m$ << moment

c. Utilities (DW-U)

$W_{UTILITIES} := 1.0 \cdot \frac{kN}{m}$ << load

$M_{utilities} := \frac{W_{UTILITIES} \cdot S_d^2}{8}$ $\quad M_{utilities} = 1.62 \cdot kN \cdot m$ << moment

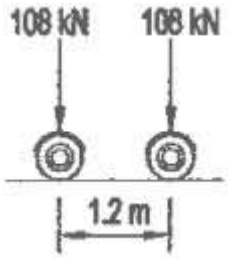
2.2 Due to Live Load

Distribution Width of Axle Load for Main Reinforcement Perpendicular to Span:
(DGCS Vol.5 Sec 11.3.2.10-1)

$E := 2440mm + 0.12 \cdot (S_d)$ $E = 2872 \cdot mm$

$mp := 1.20$ multiple presence factor single lane

a1. Design Tandem at Midspan Will Govern



One Axle Load $P_T := 108 \cdot kN$

Effective Axle Load per unit width of the slab with multiple presence factor

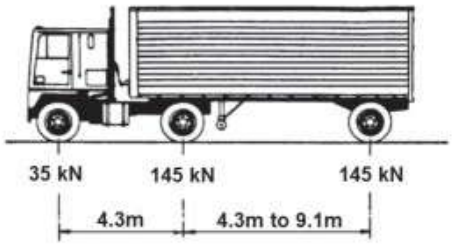
$b = 1\ m$

$$P_{Teff} := \frac{P_T \cdot b \cdot mp}{E} = 45.125 \cdot kN$$

Moment due to Live load *(Design Tandem at Midspan)*

$$M_{LL_pos1} := P_{Teff} \cdot \frac{(S_d - 1.2m)}{2} \qquad M_{LL_pos1} = 54.15 \cdot kN \cdot m$$

a2. Design Truck, 2 Axles of 145kN at Midspan



One Axle Load $P_{T2} := 145 \cdot kN$

Effective Axle Load per unit width of the slab with multiple presence factor

$$P_{Teff2} := \frac{P_{T2} \cdot b \cdot mp}{E} = 60.585 \cdot kN$$

Moment due to Live load *(Design Truck - 2 145kN Axle at Midspan)*

$$M_{LL_pos2} := P_{Teff2} \cdot \frac{(S_d - 4.3m)}{2} \qquad M_{LL_pos2} = -21.20 \cdot kN \cdot m$$

a3. Design Moment due to Live load *(Max. of Tandem or Truck)*

$M_{LL_pos} := \max(M_{LL_pos1}, M_{LL_pos2})$ $M_{LL_pos} = 54.15 \cdot kN \cdot m$

b. Design Lane load $w_{lane} := 9.34 \cdot \frac{kN}{m}$

Effective Lane Load per unit width of the slab with multiple presence factor

$$w_{laneeff} := \frac{w_{lane} \cdot b \cdot mp}{E} = 3.903 \cdot \frac{kN}{m}$$

Moment due to Live load *(Design Lane Load at Midspan)*

$$M_{lane} := \frac{w_{laneeff} \cdot S_d^{\,2}}{8} \qquad M_{lane} = 6.322 \cdot kN \cdot m$$

c. Total Design Live Load

Dynamic Load Allowance Factor $IM := 1.33$

$$M_{LL_I} := \left(M_{LL_pos} \cdot IM\right) + M_{lane} \qquad M_{LL_I} = 78.342 \cdot kN \cdot m$$

2.3 Ultimate Design Moment

$$M_u := 1.25 \cdot \left(M_{slab}\right) + 1.50 \cdot \left(M_{pave} + M_{utilities}\right) + 1.75 \cdot M_{LL_I} \qquad M_u = 156.93 \cdot kN \cdot m$$

3. MAIN FLEXURAL REINFORCEMENT

3.1 Minimum Flexural Moment (Minimum of 1.2 x Cracking Moment and 1.33 x Design Moment)

$b_{eff} := b = 1\,m$

$f_r := 0.63 \cdot \sqrt{f_c' \cdot MPa} \qquad f_r = 3.31 \cdot MPa$

$I_r := \frac{b_{eff} \cdot t_s^{\,3}}{12} \qquad I_r = 0.0016\,m^4$

$c_r := \frac{t_s}{2} \qquad c_r = 0.135\,m$

$M_{cr} := \frac{f_r \cdot I_r}{c_r} \qquad M_{cr} = 40.213 \cdot kN \cdot m \qquad 1.2 \cdot M_{cr} = 48.256 \cdot kN \cdot m$

$M_{u_min} := \min\left(1.2 \cdot M_{cr}, 1.33 \cdot M_u\right) \qquad M_{u_min} = 48.256 \cdot kN \cdot m$

3.2 Design Flexural Reinforcement

$Mu := \max(M_{u_min}, M_u)$ $\quad Mu = 156.928 \cdot kN \cdot m$

$d := t_s - Cover - \frac{d_b}{2}$ $\quad d = 0.208\,m$

$:= if\left[f_{c'} < 30 \cdot MPa, 0.85, 0.85 - 0.008 \cdot \left(\frac{f_{c'}}{MPa} - 30\right)\right]$

$= 0.85$

$m1 := 0.85 \cdot \frac{f_{c'}}{f_y} = 0.057$ $\quad m2 := \frac{2}{0.85 \cdot f_{c'}} = 0.085 \cdot \frac{mm^2}{N}$

$Rn := \frac{Mu}{\left(\cdot b_{eff} \cdot d^2\right)}$ $\quad Rn = 4.05 \cdot MPa$

Required steel ratio:

$:= m1 \cdot \left(1 - \sqrt{1 - m2 \cdot Rn}\right)$ $\quad = 0.011$

Required area of main reinf.:

$A_s := \cdot b_{eff} \cdot d$ $\quad A_s = 2243.821 \cdot mm^2$

Area of one reinforcing bar:

$A_b := \frac{\cdot d_b^2}{4}$ $\quad A_b = 490.874 \cdot mm^2$

Required spacing of main reinf.:

$sp := \frac{A_b \cdot b_{eff}}{A_s}$ $\quad sp = 218.8 \cdot mm$

Let: $space := 150 \cdot mm$ *(provided actual spacing)*

Area of main reinf. provided:

$A_{sp} := \frac{A_b \cdot b_{eff}}{space}$ $\quad A_{sp} = 3272.492 \cdot mm^2$

Depth of comression block:

$a := \frac{A_{sp} \cdot f_y}{0.85 \cdot f_{c'} \cdot b_{eff}}$ $\quad a = 57.75 \cdot mm$

Flexural Resistance:

$Mr := \cdot\left[A_{sp} \cdot f_y \cdot \left(d - \frac{a}{2}\right)\right]$ $\quad Mr = 217.803 \cdot kN \cdot m$

C/D Ratio:

$CD := \frac{Mr}{Mu} = 1.388$

USE: 25mmϕ @ 150mm O.C. for bottom bars

3.3 Check for Maximum Reinforcement Limit

Compression fiber to neutral axis:

$c := \frac{a}{\quad}$ $\quad c = 67.941 \cdot mm$

Maximum reinforcement ratio c/d_{eff}:

$maxratio := 0.420$

$actualratio := \frac{c}{d} = 0.327$

$Check_maxreinf := if(actualratio \le maxratio, "OK does not exceed max. limit!" , "NG exceeded max. reinf.!!")$

$Check_maxreinf = "OK does not exceed max. limit!"$

4. DISTRIBUTION REINFORCEMENT

Distribution reinforcement for main reinforcement parallel to traffic,
(DGCS Vol.5 Sec 14.4.3.1)

$S_d = 3600 \cdot mm$

$Percentage := \frac{1750}{\sqrt{\frac{S_d}{mm}}}$ $\quad Percentage = 29$

$Percentage := if(Percentage \le 50, Percentage, 50)$ $\quad Percentage = 29$

$A_{dist_rebars} := \frac{Percentage}{100} \cdot A_{sp}$ $\quad A_{dist_rebars} = 954 \cdot mm^2$

$A_{bs} := \frac{\cdot d_s^{\,2}}{4}$ $\quad A_{bs} = 314.159 \cdot mm^2$

$spacing_{dist} := \frac{A_{bs} \cdot b}{A_{dist_rebars}}$ $\quad spacing_{dist} = 329 \cdot mm$

USE: 20 mm ϕ @ 300 mm O.C. for Distribution Reinforcement

5. SHEAR DEMAND AND CAPACITY CHECK

5.1 Shear Demand at Support

Slab (DC)

$V_{slab} := \frac{W_{SLAB} \cdot S_d}{2}$ $\quad V_{slab} = 11.664 \cdot kN$

Pavement (DW)

$V_{sup} := \frac{W_{PAVE} \cdot S_d}{2}$ $\quad V_{sup} = 3.168 \cdot kN$

LL Due to Design Tandem

$V_T := P_{Teff}$ $\quad V_T = 45.125 \cdot kN$

LL Due to Design Lane Load

$$V_{lane} := \frac{w_{laneeff} \cdot S_d}{2} \qquad V_{lane} = 7.025 \cdot kN$$

Total Design Live Load (LL)

$$V_{LL_I} := \left(P_{Teff} \cdot IM\right) + V_{lane} \qquad V_{LL_I} = 67.04 \cdot kN$$

Ultimate Design Shear

$$V_u := 1.25 \cdot V_{slab} + 1.50 \cdot V_{sup} + 1.75 \cdot V_{LL_I} \qquad V_u = 136.65 \cdot kN$$

5.2 Shear Capacity

$$0.72 \cdot t_s = 0.194\,m$$

Factor relating to longitudinal strain on the shear capacity of conc.

$v := 2.0$ *(for non-prestress)*

$$0.9 \cdot d = 0.187\,m$$

Shear resistance factor

$v := 0.9$

$$d - \frac{a}{2} = 0.179\,m$$

Effective shear depth

$$d_v := \max\left(d - \frac{a}{2}, 0.9 \cdot d, 0.72 \cdot t_s\right) \qquad d_v = 194.4 \cdot mm$$

Shear resistance provided by concrete

$$V_c := 0.083 \cdot {}_v\sqrt{f_c' \cdot MPa} \cdot b_{eff} \cdot d_v \qquad V_c = 169.53 \cdot kN$$

$$V_r := {}_v \cdot V_c \qquad V_r = 152.58 \cdot kN$$

5.3 Shear Capacity / Demand Ratio

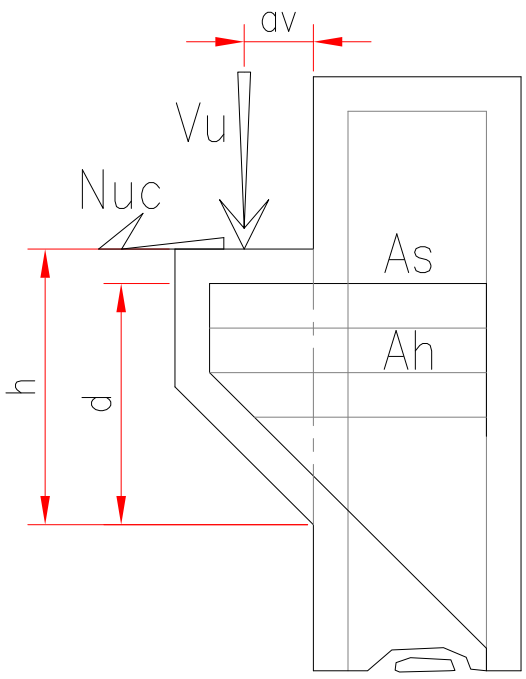
$$CD_v := \frac{V_r}{V_u} = 1.117$$

$$Check_shear := \text{if}\left(CD_v \geq 1.0, \text{"Section is OK in shear!"}, \text{"NG need shear reinf.!!"}\right)$$

Check_shear = "Section is OK in shear!"

Design of Corbel @ Backwall of Abutment

A. Conversion: $kN := 1000 \cdot newton$ $kPa := \frac{kN}{m^2}$ $MPa := 1000 \cdot kPa$



B. Input Data:

Dimensions:

$h := 400 \cdot mm$	height of corbel at face of backwall
$a_v := 100 \cdot mm$	lever arm of Vu from face of backwall
$b := 1000 \cdot mm$	unit width considered in design
$c_c := 50 \cdot mm$	clear cover

Material Data:

$c := 24 \cdot \frac{kN}{m^3}$	unit weight of concrete
$f'c := 27.6 \cdot MPa$	specified concrete strength
$f_y := 414 \cdot MPa$	specified yield strength of reinforcement
$d_b := 16 \cdot mm$	assumed diameter of main rebar As

Design data:

$d := h - c_c - 0.5 \cdot d_b$ effective depth of As

$d = 342 \cdot mm$

$:= 0.85$ corbel design reduction factor

$:= 1.0$ correction factor (1.0 for normal wt. of conc.)

$:= 1.0 \cdot$ coeff. of friction (AASHTO 8.16.6.4.4)

C. Design Loads

Ultimate Load:

$V_u = 136.654 \cdot kN$ shear support reaction from approach slab

(design shear per unit meter)

$N_{uc} := 1.75 \cdot (12.0 \cdot kN)$ from horizontal breaking force at approach slab, BF per meter

(calc. see in abutment design)

$N_{uc} = 21 \cdot kN$

D. Required Main Reinforcement (As):

Shear Friction Requirement (Avf):

$V_{nmax1} := 0.2 \cdot f'c \cdot b \cdot d$

$V_{nmax2} := 800 \cdot \frac{b \cdot d}{in^2} \cdot lb \cdot 9.81 \frac{m}{s^2}$

$V_{nmax} := \min(V_{nmax1}, V_{nmax2})$

$V_{nmax} = 1887.050 \cdot kN$

$V_u := \begin{cases} V_u & \text{if } \frac{V_u}{} \le V_{nmax} \\ V_{nmax} & \text{otherwise} \end{cases}$ $V_u = 136.654 \cdot kN$

$A_{vf} := \frac{V_u}{\cdot f_y}$ $A_{vf} = 388.332 \cdot mm^2$

Tensile Requirement (An):

$N_{uc} := \max(N_{uc}, 0.2 \cdot V_u)$

$N_{uc} = 27.331 \cdot kN$

$A_n := \frac{N_{uc}}{\cdot f_y}$ $A_n = 77.666 \cdot mm^2$

Flexural Requirement (Af):

$M_u := \left[V_u \cdot a_v + N_{uc} \cdot (h - d)\right]$

$M_u = 15.251 \cdot kN \cdot m$

Modulus of rupture of conrete: $f_r := 0.63 \cdot \sqrt{f'c \cdot MPa}$ $f_r = 3.31 \cdot MPa$

$I'_g := \frac{b \cdot h^3}{12}$ $I'_g = 0.0053\ m^4$ $C' := \frac{h}{2}$

Cracking Moment: $M_{CR'} := \frac{f_r \cdot I'_g}{C'}$ $M_{CR'} = 88.26 \cdot kN \cdot m$

Minimum Design Moment: $M_{u_min} := \min(1.2 \cdot M_{CR'}, 1.33 \cdot M_u)$ $M_{u_min} = 20.283 \cdot kN \cdot m$

Design Moment: $Mu := \max(M_{u_min}, M_u)$ $Mu = 20.283 \cdot kN \cdot m$

$m1 := 0.85 \cdot \frac{f'c}{f_y} = 0.057$ $m2 := \frac{2}{0.85 \cdot f'c} = 0.085 \cdot \frac{mm^2}{N}$

$Rn := \frac{Mu}{(\cdot b \cdot d^2)}$ $Rn = 0.204 \cdot MPa$

Required steel ratio: $des := m1 \cdot (1 - \sqrt{1 - m2 \cdot Rn})$ $= 0.011$

Required flexural reinf.: $A_f := des \cdot b \cdot d$ $A_f = 169.276 \cdot mm^2$

Main Reinforcement (As):

$A_{min} := 0.04 \cdot \frac{f'c}{f_y} \cdot b \cdot d$ $A_{min} = 912 \cdot mm^2$

$A_{s1} := A_f + A_n$ $A_{s1} = 246.942 \cdot mm^2$

$A_{s2} := \frac{2 \cdot A_{vf}}{3} + A_n$ $A_{s2} = 336.555 \cdot mm^2$

$A_s := \max(A_{min}, A_{s1}, A_{s2})$ $A_s = 912 \cdot mm^2$

Area of one bar: $A_b := \frac{\cdot d_b^2}{4}$ $A_b = 201.062 \cdot mm^2$

Required spacing of bars: $s_s := \frac{A_b \cdot b}{A_s}$ $s_s = 220.463 \cdot mm$

Adopted: 16 mm dia. @ 200 mm main reinforcement

E. Closed Hoop Stirrups (Ah):

$d_h := 12mm$ diameter of hoop bars Ah

$A_{dh} := \frac{\cdot d_h^2}{4}$ $A_{dh} = 113.097 \cdot mm^2$

$A_h := \max\left(0.5 \cdot A_f, \frac{A_{vf}}{3}\right)$ $0.5 \cdot A_f = 84.638 \cdot mm^2$

$\frac{A_{vf}}{3} = 129.444 \cdot mm^2$

$A_h = 129.444 \cdot mm^2$

$n_h := \frac{A_h}{A_{dh}}$ $n_h = 1.145$

NEGLECTED, Corbel load is continuous

Revetment structure design

Input data

Project

Task : Sheet Pile design
Part : Cross-section & Material Properties
Description :
Customer : DPWH
Author : Kyong-Ho
Date :
Project ID : IDRR-CCA Stage 2
Section TRL2-63
Station :2+850 TO 3+500

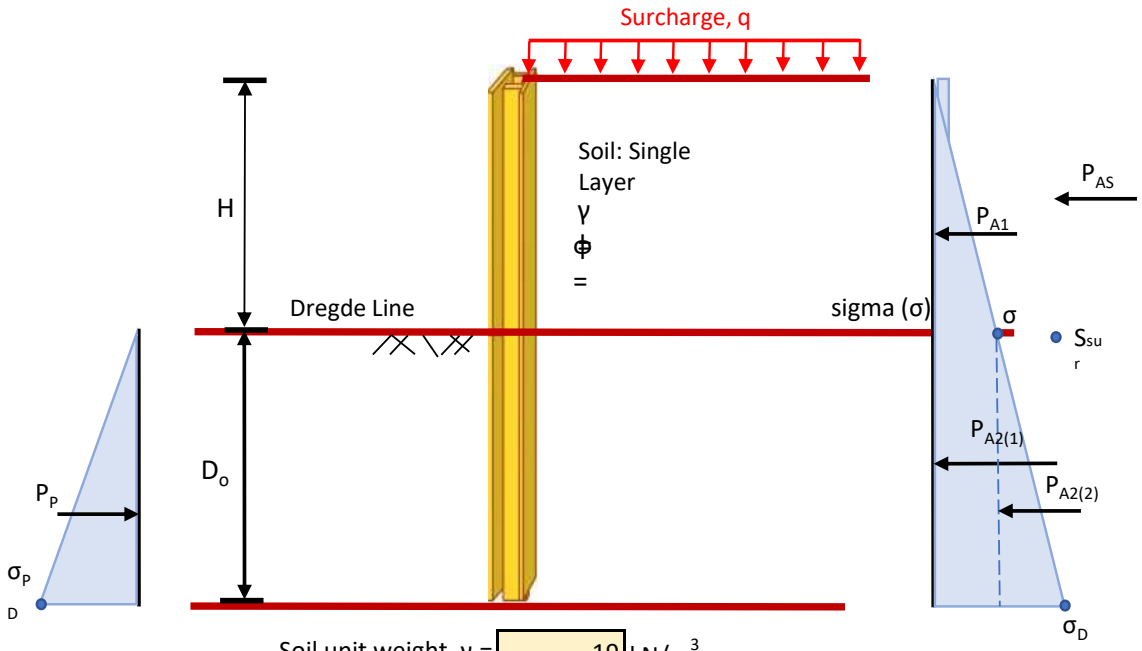
Settings

Philippines - DPWH Settings

Pressure analysis

Active earth pressure calculation: Rankine

Input Data / Material Properties



Soil unit weight, γ_s=	19	kN/m^3
Internal Angle of Friction, ϕ=	19	deg
Height above dredge line, H=	2	m
Height of surcharge, H_{sur}=	8.152	m
Highway load, LL=	12	kPa

A. SERVICE LIMIT STATE

LRFD Factors due to Service Limit:

Active Pressure=	1
Passive Earth Pressure=	1
Live Load Surcharge=	1

A.1. Active and Passive Earth Pressure Coefficients

$$K_a = tan^2(45 - \frac{\phi}{2})$$

K_a= 0.509

$$K_p = tan^2(45 + \frac{\phi}{2})$$

K_p= 1.965

A.2. Earth Pressure Distribution

$Total_{sur}$= LL + ($\gamma s * H_{sur}$)

$Total_{sur}$= 166.888 kPa

$$S_{sur} = Total_{sur}\ x\ K_a\ x\ LRFD\ Factor$$

S_{sur}= 84.91 kPa

$$\sigma_D = \sigma + (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)$$

σ_D= 19.33+9.67Do kPa

$$\sigma = \gamma_S\ x\ K_a\ x\ H\ x\ LRFD\ Factor$$

σ= 19.33 kPa

$$\sigma_{PD} = \gamma_S\ x\ K_p\ x\ LRFD\ Factor\ x\ D_o$$

σ_{PD}= 37.34Do kPa

A.3. Resultant Earth Forces

$$P_{AS} = H\ x\ S_{SUR}$$

P_{AS}= 169.82 kN

$$P_{A2(1)} = \sigma\ x\ D_O$$

$P_{A2(1)}$= 19.33Do kN

$$P_{A1} = \frac{H}{2}\ x\ \sigma$$

P_{A1}= 19.33 kN

$$P_{A2(2)} = (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)x\ \frac{D_O}{2}$$

$P_{A2(2)}$= 4.83Do^2 kN

$$P_P = P_D\ x\ \frac{D_O}{2}$$

P_P= 18.67Do^2 kN

A.4. Driving and Resisting Moments

Driving force (kN)	Arm_{DR} (m)	Driving Moment, M_{DR}
169.82	1+Do	.69.82+169.82Do
19.33	0.67+Do	12.89+19.33Do
19.33Do	Do/2	9.67Do^2
4.83Do^2	Do/3	1.61Do^3
Resisting Force (kN)	**Arm_{RS} (m)**	**Resisting Moment, M_{RS}**
18.67Do^2	Do/3	6.22Do^3

A.5. Length of Sheetpile

5.1 DEPTH OF PENETRATION

$M_{DR}=M_{RS}$: D_o= 7.91 m

5.2 ACTUAL DEPTH OF EMBEDMENT

$D= 1.3D_o$

D= 10.29 m

5.3 LENGTH OF SHEETPILE

L= H+D

L= 13 m

A.6. Maximum Moment

6.1 LOCATION OF MAXIMUM MOMENT BELOW DREDGE LINE, Y

$$\sum F_x = 0$$

Y^2 -1.397 Y -13.67 =0

→ Y= 4.46 m (below the dredge line)

6.2 MAXIMUM MOMENT

Summation of moment about depth Y

M_{max}= 809.40 kN-m

A.7. Section Modulus

Rebar yield strength, fy= 325 MPa

Allowable flexural stress, $\sigma_{allowable}$= 0.6fy

$\sigma_{allowable}$= 195 MPa

Section Modulus, $S_m= S_x= M_{max}/\sigma_{allowable}$

S_m= 4150.77 cm^3

B. STRENGTH LIMIT STATE

LRFD Factors due to Strength Limit:

Active Pressure=	1.5
Passive Earth Pressure=	1
Live Load Surcharge=	1.75

B.1. Active and Passive Earth Pressure Coefficients

$$K_a = tan^2(45 - \frac{\phi}{2})$$

K_a= 0.509

$$K_p = tan^2(45 + \frac{\phi}{2})$$

K_p= 1.965

B.2. Earth Pressure Distribution

$Total_{sur}$= LL + ($\gamma s * H_{sur}$)

$Total_{sur}$= 166.888 kPa

$$S_{sur} = Total_{sur}\ x\ K_a\ x\ LRFD\ Factor$$

S_{sur}= 148.59 kPa

$$\sigma_D = \sigma + (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)$$

σ_D= 29+14.5Do kPa

$$\sigma = \gamma_S\ x\ K_a\ x\ H\ x\ LRFD\ Factor$$

σ= 29.001 kPa

$$\sigma_{PD} = \gamma_S\ x\ K_p\ x\ LRFD\ Factor\ x\ D_o$$

σ_{PD}= 37.34Do kPa

B.3. Resultant Earth Forces

$$P_{AS} = H\ x\ S_{SUR}$$

P_{AS}= 297.19 kN

$$P_{A2(1)} = \sigma\ x\ D_O$$

$P_{A2(1)}$= 29Do kN

$$P_{A1} = \frac{H}{2}\ x\ \sigma$$

P_{A1}= 29.00 kN

$$P_{A2(2)} = (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)x\ \frac{D_O}{2}$$

$P_{A2(2)}$= 7.25Do^2 kN

$$P_P = P_D\ x\ \frac{D_O}{2}$$

P_P= 18.67Do^2 kN

B.4. Driving and Resisting Moments

Driving force (kN)	Arm_{DR} (m)	Driving Moment, M_{DR}
297.19	1+Do	297.19+297.19Do
29.00	0.67+Do	19.33+29Do
29Do	Do/2	14.5Do^2
7.25Do^2	Do/3	2.42Do^3
Resisting Force (kN)	**Arm_{RS} (m)**	**Resisting Moment, M_{RS}**
18.67Do^2	Do/3	6.22Do^3

B.5. Length of Sheetpile

5.1 DEPTH OF PENETRATION

M_{DR}=M_{RS}: D_o= 11.72 m

5.2 ACTUAL DEPTH OF EMBEDMENT

D= 1.3D_o

D= 15.24 m

5.3 LENGTH OF SHEETPILE

L= H+D

L= 18 m

B.6. Maximum Moment

6.1 LOCATION OF MAXIMUM MOMENT BELOW DREDGE LINE, Y

$$\sum F_x = 0$$

Y^2 -1.397 Y -15.71 =0

→ Y= 4.72 m (below the dredge line)

6.2 MAXIMUM MOMENT

Summation of moment about depth Y

M_{max}= 1451.59 kN-m

B.7. Section Modulus

Rebar yield strength, fy= 325 MPa

Allowable flexural stress, $\sigma_{allowable}$= 0.6fy

$\sigma_{allowable}$= 292.5 MPa

Section Modulus, S_m= S_x= $M_{max}/\sigma_{allowable}$

S_m= 4962.71 cm^3

C. EXTREME EVENT LIMIT STATE

LRFD Factors due to Seismic Event:

Active Pressure=	1
Passive Earth Pressure=	1
Live Load Surcharge=	0

C.1. Active and Passive Earth Pressure Coefficients

other parameters:

k_h= 0.15

k_v= 0

$$\theta = tan^{-1}\left(\frac{k_h}{1-k_v}\right)$$

θ= 8.531

δ= 2ø/3

δ= 12.67

$$K_{ae} = \frac{\cos^2(\emptyset - \theta - \omega)}{\cos\theta \cos^2\omega \cos(\theta + \omega + \delta)\left(1 + \sqrt{\frac{\sin(\emptyset + \delta)\sin(\emptyset - \theta - \beta)}{\cos(\delta + \theta + \omega)\cos(\beta - \omega)}}\right)^2}$$

K_{ae}= 0.602

$$K_{pe} = \frac{\cos^2(\emptyset - \theta + \omega)}{\cos\theta \cos^2\omega \cos(\theta - \omega + \delta)\left(1 - \sqrt{\frac{\sin(\emptyset + \delta)\sin(\emptyset - \theta + \beta)}{\cos(\delta + \theta - \omega)\cos(\beta - \omega)}}\right)^2}$$

K_{pe}= 2.268

C.2. Earth Pressure Distribution

$Total_{sur}$= LL + ($\gamma s * H_{sur}$)

$Total_{sur}$= 0 kPa

$$S_{sur} = Total_{sur}\ x\ K_a\ x\ LRFD\ Factor$$

S_{sur}= 0 kPa

$$\sigma_D = \sigma + (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)$$

σ_D= 22.88+11.44Do kPa

$$\sigma = \gamma_S\ x\ K_a\ x\ H\ x\ LRFD\ Factor$$

σ= 22.882 kPa

$$\sigma_{PD} = \gamma_S\ x\ K_p\ x\ LRFD\ Factor\ x\ D_o$$

σ_{PD}= 43.1Do kPa

C.3. Resultant Earth Forces

$$P_{AS} = H\ x\ S_{SUR}$$

P_{AS}= 0 kN

$$P_{A1} = \frac{H}{2}\ x\ \sigma$$

P_{A1}= 22.88 kN

$$P_{A2(1)} = \sigma\ x\ D_O$$

$P_{A2(1)}$= 22.88Do kN

$$P_{A2(2)} = (\gamma_S\ x\ K_a\ x\ LRFD\ Factor\ x\ D_o)x\ \frac{D_O}{2}$$

$P_{A2(2)}$= 5.72Do^2 kN

$$P_P = P_D\ x\ \frac{D_O}{2}$$

P_P= 21.55Do^2 kN

C.4. Driving and Resisting Moments

Driving force (kN)	Arm_{DR} (m)	Driving Moment, M_{DR}
22.88	1+Do	0+0Do
22.88Do	0.67+Do	15.25+22.88Do
5.72Do^2	Do/2	11.44Do^2
21.55Do^2	Do/3	1.91Do^3
Resisting Force (kN)	**Arm_{RS} (m)**	**Resisting Moment, M_{RS}**
21.55Do^2	Do/3	7.18Do^3

C.5. Length of Sheetpile

5.1 DEPTH OF PENETRATION

$M_{DR}=M_{RS}$: D_o= 3.60 m

5.2 ACTUAL DEPTH OF EMBEDMENT

$D= 1.3D_o$

D= 4.68 m

5.3 LENGTH OF SHEETPILE

L= H+D

L= 7 m

C.6. Maximum Moment

6.1 LOCATION OF MAXIMUM MOMENT BELOW DREDGE LINE, Y

$$\sum F_x = 0$$

Y^2 -1.446 Y -1.45 =0

→ Y= 2.13 m (below the dredge line)

6.2 MAXIMUM MOMENT

Summation of moment about depth Y

M_{max}= 64.92 kN-m

C.7. Section Modulus

Rebar yield strength, fy= 325 MPa

Allowable flexural stress, $\sigma_{allowable}$= 0.6fy

$\sigma_{allowable}$= 195 MPa

Section Modulus, $S_m= S_x= M_{max}/\sigma_{allowable}$

S_m= 332.92 cm^3

CHOOSE GOVERNING LENGTH: **18 m**

Hard Strata Height= 27.0 m

CHOOSE LARGEST SECTION MODU S_m= 4962.71 cm^3

Based on available Z type sheetpiles:

Larger S_m: 5070 cm^3

Smaller S_m: 3455 cm^3

Trial S_m: 5070 cm^3

(Sm/Smaller Sm)>1.05, ∴Choose Larger Sm as Trial Sm!

Weight of Larger S_m: 216.09 kg/m^2

Weight of Trial S_m: 216.09 kg/m^2

*Use smaller weight: 216.09 kg/m^2

Use: **Section Modulus: 5,070 cm^3**

Area of Sheetpile: 275.28 cm^2

8. Maximum Shear

	Vmax	
SERVICE	524.07	kN
STRENGTH	2186.26	kN
SIESMIC	99.65	kN

CHOOSE GOVERNING SHEAR: Vr 2186.26 kN

$$V_{max} = \sum F_x \ @ \ D_o$$

V_{max}= 2186.26 kN

$q_v = V_{max}/A$

q_v= 79419.37 kPa

$q_{vallowable}$= 0.6fy

$q_{vallowable}$= 195 MPa

Check Shear:

qvallowable>qv, ∴OK!